MOLECULAR BIOLOGY 1

Lectures

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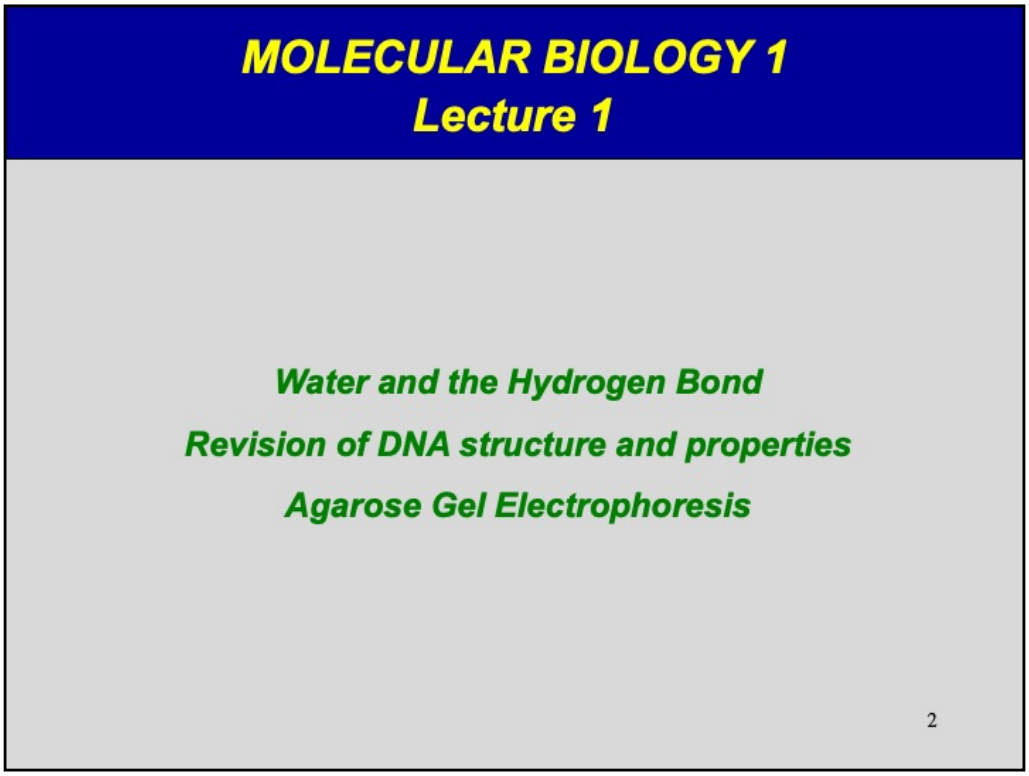
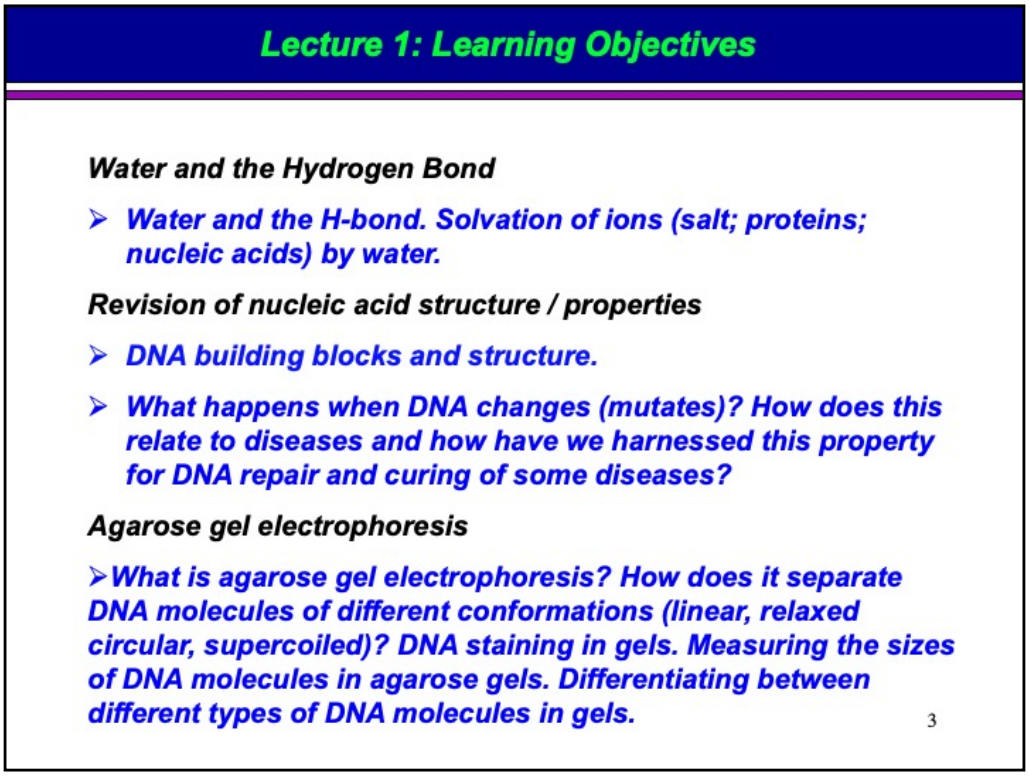


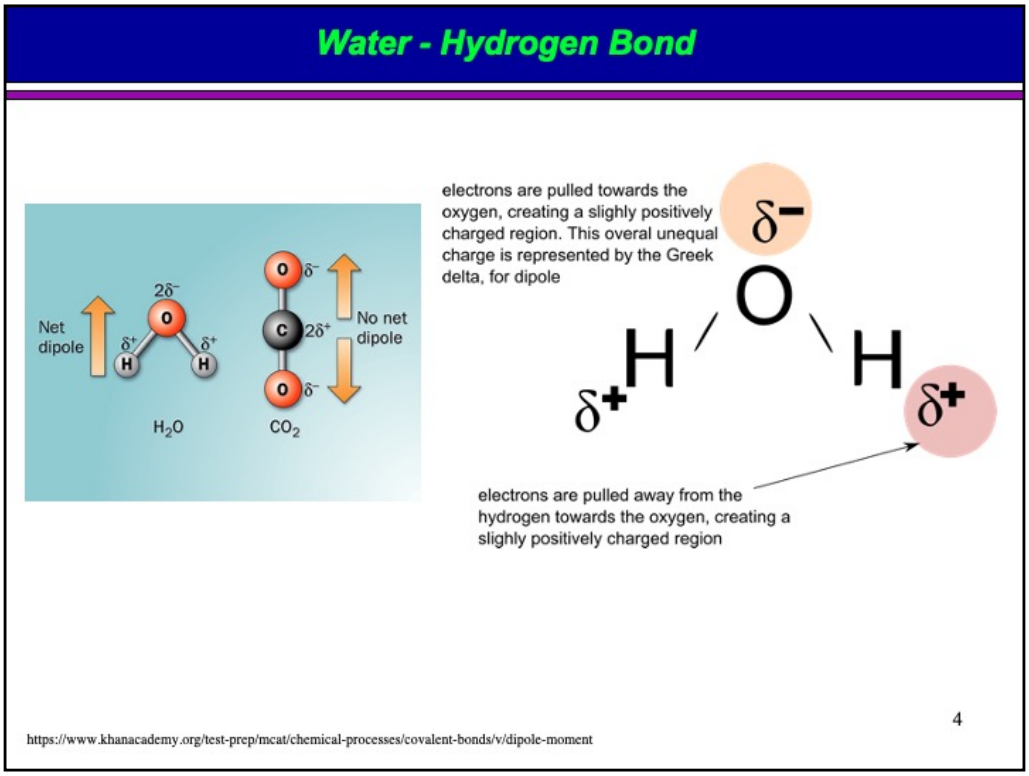
Here we go. Hang on to your hat!



This lecture begins by reacquainting us with water and the hydrogen bond, which are fundamental to most biochemistry and molecular biology.

The middle section of this lecture will have historical and revision content on the discovery and properties of DNA.

This last section of this lecture introduces a fundamental method in molecular biology, namely, agarose gel electrophoresis, which you will first encounter in Experiment 1, and thereafter in several other experiments in the practical content of Molecular Biology 1.

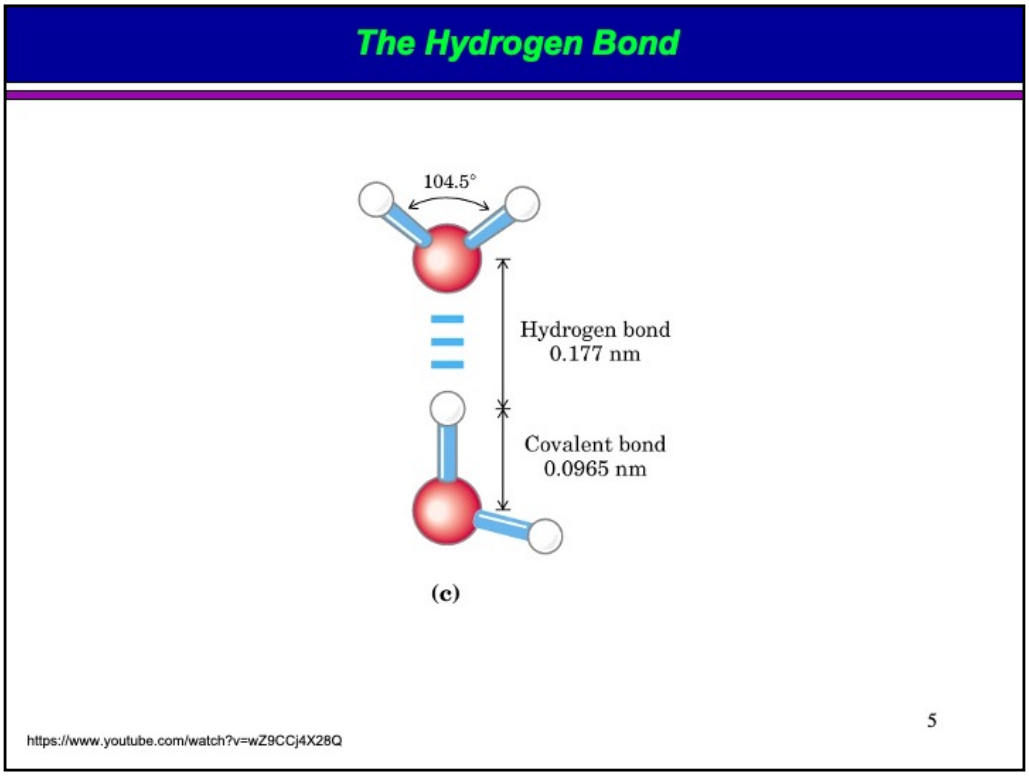


Water is the most important molecule on the Earth. Note in the left panel how it is capable of bearing positive and negative diploes with a **net negative dipole**, but that the linear molecule CO_2 (carbon dioxide) does not have the same dipole (opposite arrowed dipoles cancel each other out). Importantly, **carbon dioxide doesn't have any lone pair electrons on its two oxygens, as the extra electrons are engaged in double bonds to the central carbon atom**.

Oxygen is much more **electronegative** than hydrogen, due to the lone pair electrons, which attract positive charge, hence the dipole effect (right panel). **As a result of the charged dipole, water is capable of forming very strong hydrogen bonds (H-bonds) with itself and with other polar or charged molecules**.

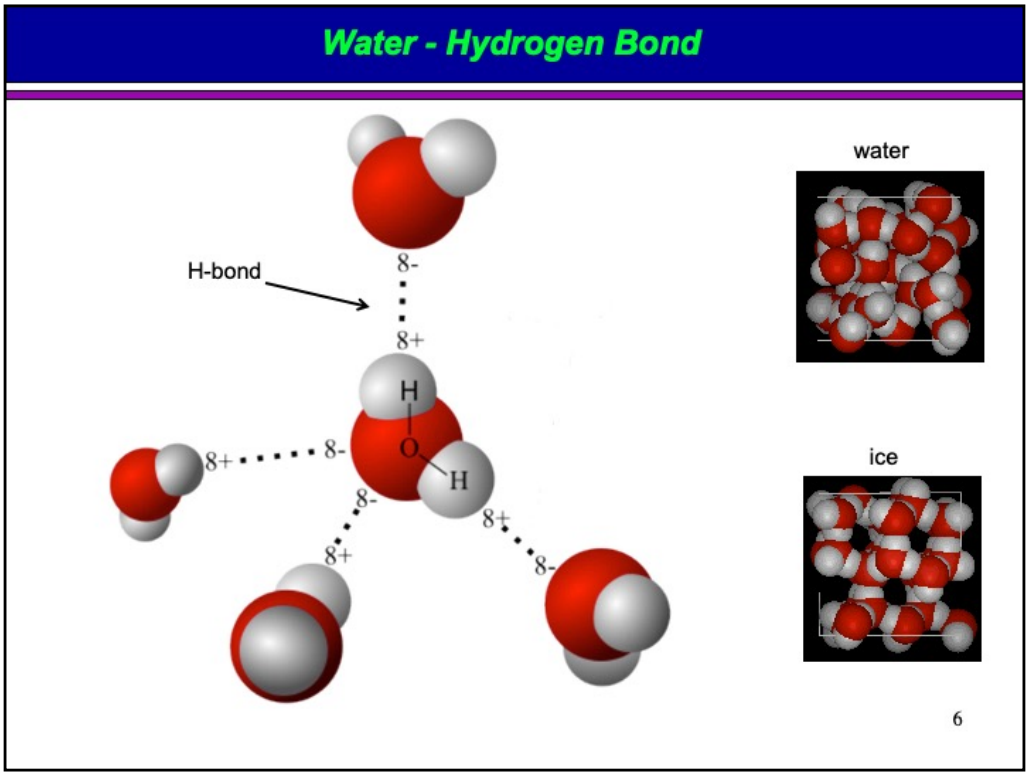
Delta dipole charges are usually small, but for water there is a large molecular dipole towards the oxygen atom, due to the extra negativity imparted by the two unshared electron pairs on the oxygen atom; and the gathering positive dipoles of the two hydrogens.

Much of what happens to nucleic acids and, for that matter, among all macromolecules, and in the working of cells and organisms generally, depends completely on water, the hydrogen bond, and the ways in which these two items interact with all other molecules. The link (bottom left) is a very good You Tube video on dipoles.



The direction and strength of a **hydrogen bond** (often simply written as 'H-bond') in comparison to a covalent bond. H-bonds are the most important in biology (as we'll see throughout these lectures). They control the way that water is used by organisms and within cells. Though the covalent bond between atoms is stronger than the H-bond, the latter exerts its attraction over a greater distance.

H-bonds are responsible for the stability and structure of nucleic acids and proteins and they are used during most enzymatic reactions in which H-bonds are broken and reformed, or new ones are formed, during the transfer of electrons, chemical groups, and substrates.

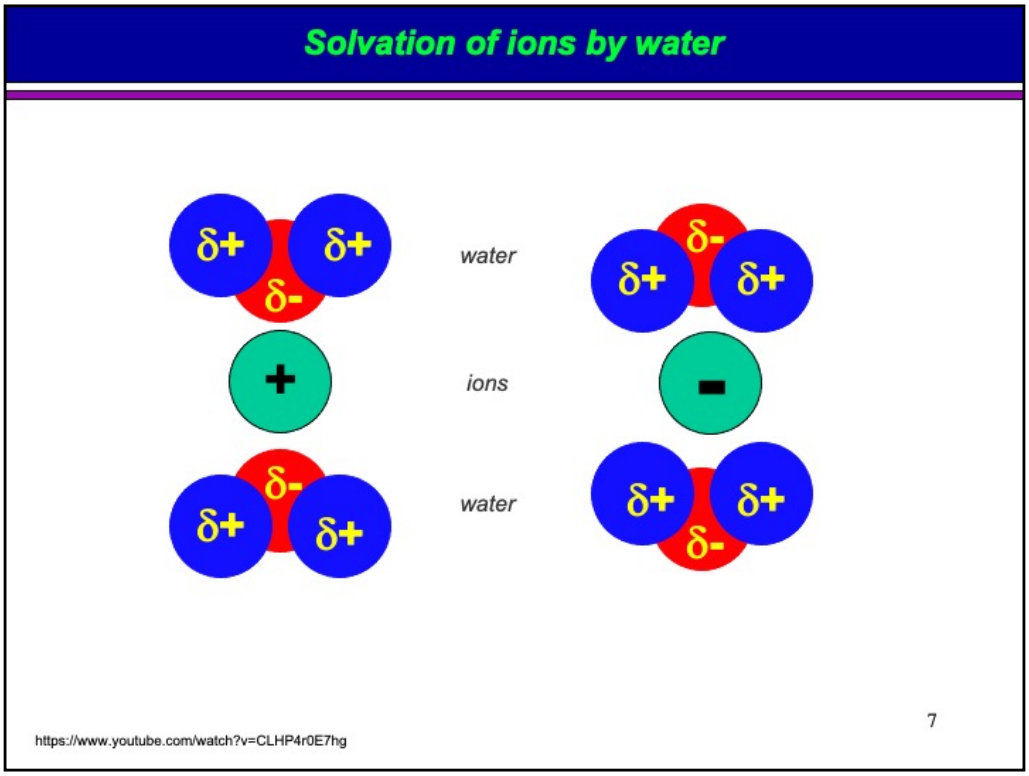


Water is a liquid at room temperature chiefly due to the extensive network of H-bonds between the water molecules. Similar molecules (e.g. CO_2, SO_2) are gases because there are no bonds to hold the molecules together. In this diagram there are four H-bonds between the central water molecule and four others, alternately to hydrogens (lower left) and lone electron pairs of oxygens, top and right. You can see how this simple representation can build to a networked continuum of H-bonded waters in liquid water (top right panel) and ice (bottom right panel).

When water freezes into ice, the ordered structure is less dense, because water expands slightly as it freezes, which you can see by the spaces between the water molecules in ice. Therefore, ice is less dense than water, which is why ice floats on the surface of water.

Collectively, the H-bonds in liquid water impart a strong stability to the liquid that resists becoming gaseous as steam until the H-bonds are broken by energy (heat). The surface of liquid water has a **surface tension** due to the 'skin' of H-bonded water, and it supports small insects and other objects that are heavier than water, but cannot break the H-bond skin. In fact, the molecules of water in the liquid state are constantly breaking and reforming the H-bonds at a slow but continuous rate, occasionally escaping by evaporation, but most remain in the liquid state at room temperature.

As water is heated toward boiling (100^{o}C) the molecules move much more rapidly, the H-bonds can no longer be maintained, and the now separated water molecules are free to enter the vapour or gaseous state. However, whereas CO_2 and SO_2 are gases at room temperature, water can only be so at its boiling point.

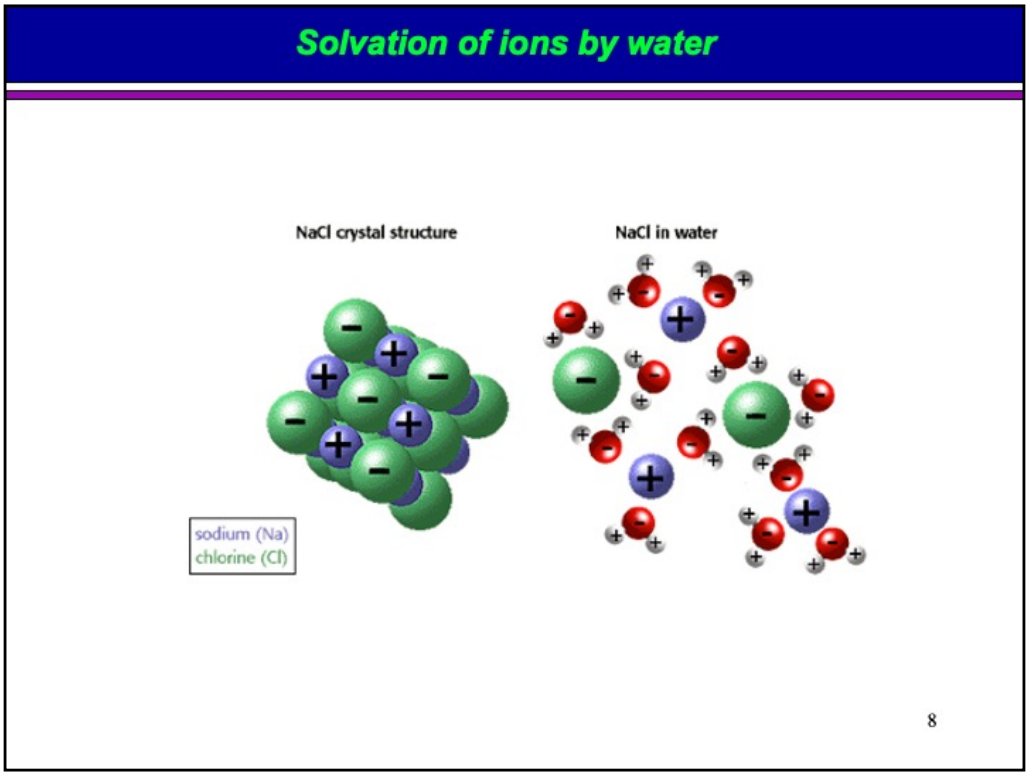


Salts, indeed any compounds with charges or polar groups (e.g. amino acids, sugars, proteins, nucleic acids), will dissolve in water by the process of **solvation**. Water molecules literally pull salt ions from the solid crystal with the water hydrogens gathering the negative chloride ions and the water oxygens the positive sodium ions. Water molecules then surround and set up strong dipolar hydrogen bonds with the solubilised ions, or with larger charged or polar molecules like proteins or nucleic acids. This process of solvation also results in the disruption of some of the water hydrogen bonded to other water molecules.

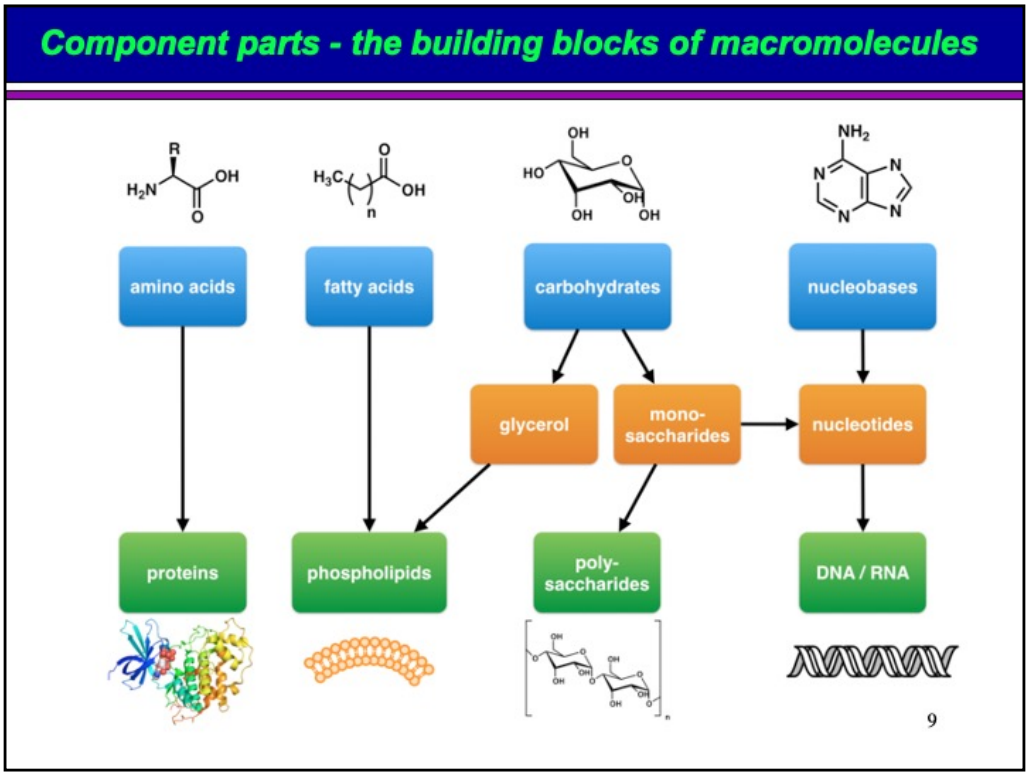
Proteins, nucleic acids, sugars etc are dissolved (solvated) in aqueous cellular fluid in exactly this way. The link at the bottom of the slide is very useful. [As you probably know, any such You Tube teaching video comes with others on the right side panel of the site page and some of these are worth exploring.]

Note! In addition to the strong ionic charge interactions depicted here between positive or negative ions and water molecules, the actual interaction is a combination of ionic and H-bonds. The two types of bonds, acting together, are stronger than either alone; and have about one third the strength of a covalent bond.

The lower alcohols, methanol, ethanol, and propanol (also iso-propanol) all have terminal OH groups and the lone paired electrons in these groups are able to engage in H-bonding in water, which is why they are soluble and have relatively high boiling points. As the non-polar carbon chain increases, such as in the next member, butanol ($CH_3CH_2CH_2CH_2OH$), the long aliphatic chain now dominates butanol's physical properties such that it is much less soluble in water than the simpler alcohols.

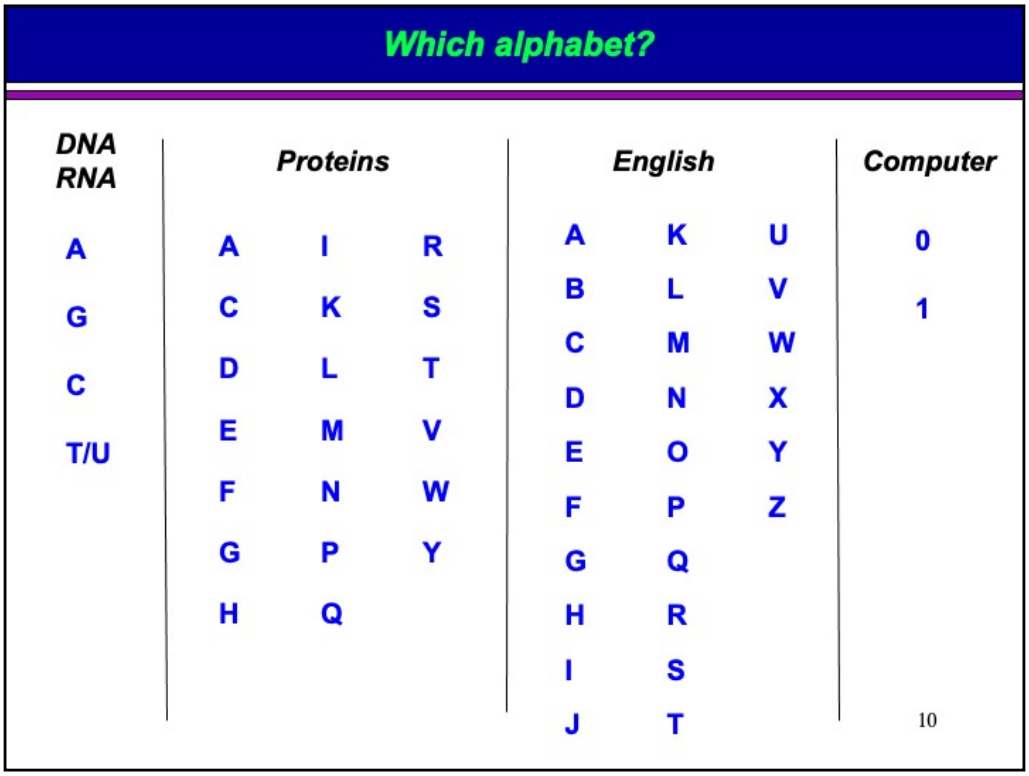


Related to the previous slide, here showing how common salt (sodium chloride, NaCl) dissolves (is solvated) in water.

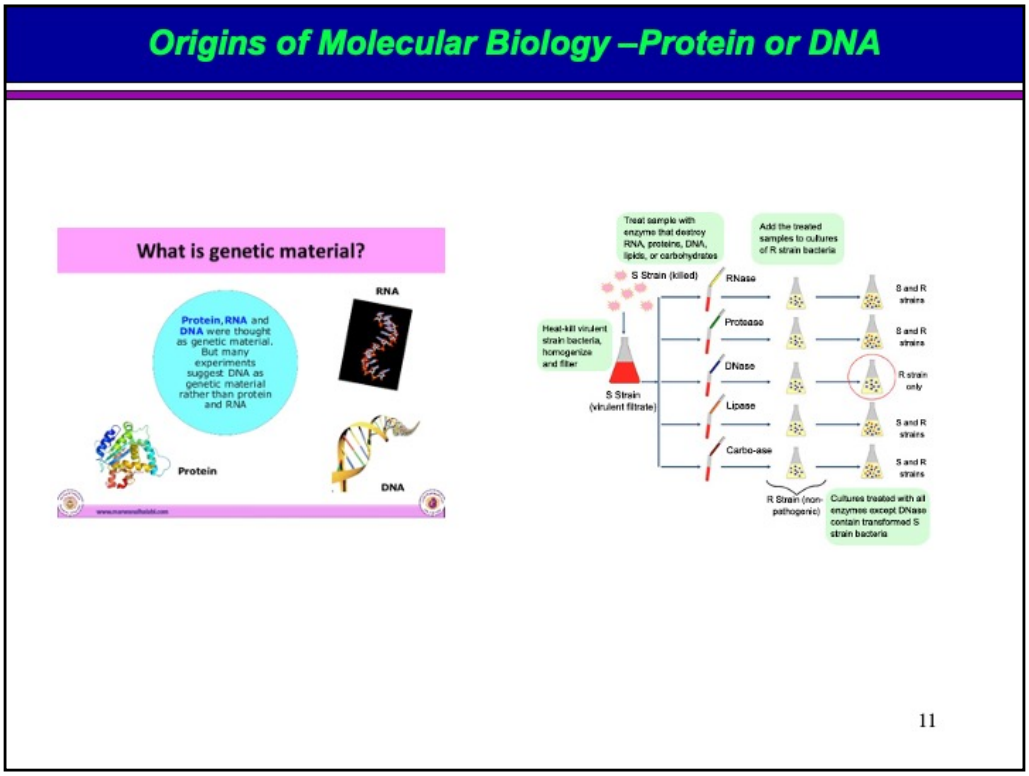


A simple revision of the major components of macromolecules. The 4 types or classes of macromolecules are shown in the green boxes at bottom, and the component 'building blocks' are in the blue boxes at the top.

Molecular Biology 1 will cover mostly nucleic acids, but proteins (and enzymes) will loom large within many aspects of the subject. Carbohydrates, in particular the ribose and deoxyribose 5-carbon sugars will show up as important cogs in the structures of RNA and DNA, respectively. Lipids will get a mention as they constitute most of the cell membrane bilayer structure.

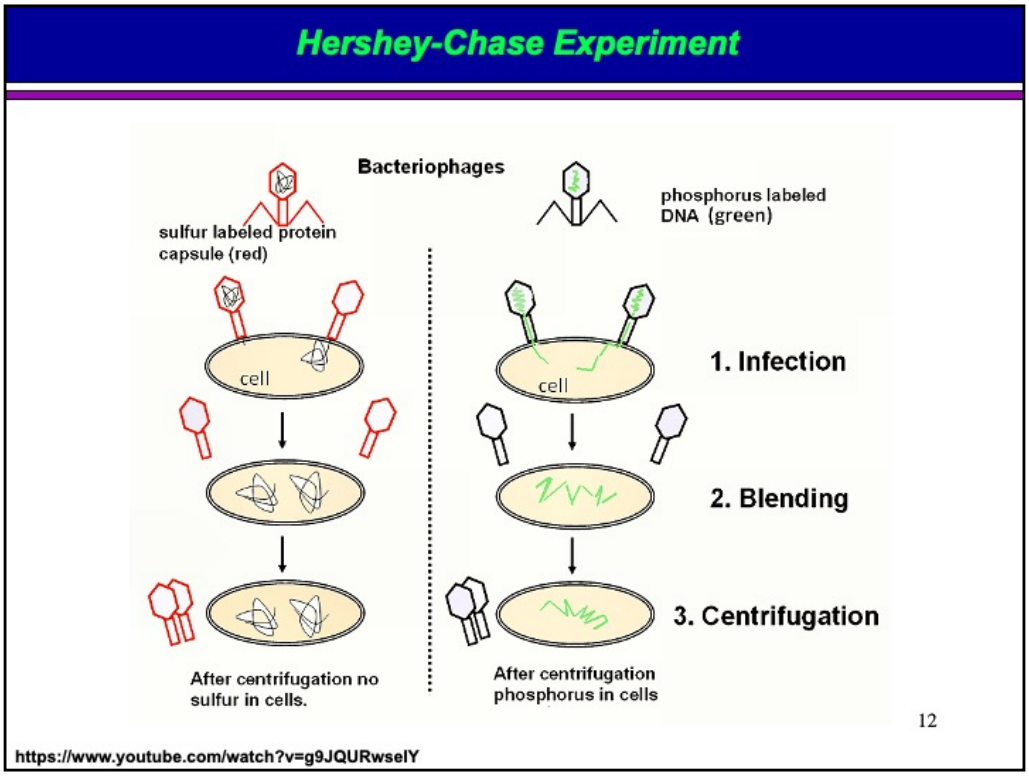


Four alphabets. The complexity of the protein 'alphabet', that is 20 amino acids, versus only 4 for DNA or RNA, led many early geneticists and biochemists to propose that proteins had to be origin of the genetic material due to its much larger alphabet. Moreover, as we will see shortly, the two DNA strands were believed to be intertwined with the bases on the outside, as this was the only way by which a 'complexity' of sequence could be attained, but even then there was not enough diversity in the 'alphabet'.. We know now that that was an incorrect supposition. There were several other such misconceptions in the gradual evaluation of the structure of DNA.



Until the late 1940s, most researchers though that the genetic material was encoded in proteins. Confusion reigned in the scientific community at that time (left panel). Why? There was a enormous library of different proteins with different functional specificities; and proteins had a 20 letter alphabet of amino acids. On the other hand, little was known about nucleic acids, except for the fact that in eukaryotes, DNA was only found in the nucleus. In addition, DNA only had a 4-letter alphabet and most of its physical and chemical properties remained a mystery. Some geneticists and biochemists even gave RNA a run as the genetic candidate molecule.

In 1944, Ostwald Avery and colleagues at the Rockefeller Institute NY concluded and reported a brilliant experiment to try to work out which type of macromolecule was the genetic material. They know that a strain of bacterial with a rough outer membrane (R pneumococci) was avirulent (non0infectious) and the corresponding smooth form of pneumococci (S), with a polysaccharide outer coat was virulent (infectious). They took a heat killed cell-free filtrate of the S strain and mixed it with the non-infectious R strain. Then they treated 5 identical flasks of living R bacteria plus S filtrate with enzymes that would destroy proteins, lipids, RNA, carbohydrates, or DNA. All samples bar one returned both S and R living forms of the bacteria, indicating that the R form could be transformed to the S form by some of the R bacteria taking up the S ‘genetic principle’ and producing S bacteria. Only the sample treated with deoxyribonuclease didn’t ‘work’, with only R bacteria growing. This was a clear indication that the infectious material was DNA, and not any of the other macromolecules.



DNA had been known to biologists since 1869, but many scientists still assumed at the time and almost to as recently as 1950 that proteins carried the information for inheritance, because DNA appeared to be an inert molecule, and was located in the nucleus, rather than in the more active cytoplasm.

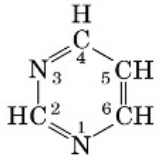
This clever experiment by Alfred Hershey and Martha Chase in 1952 presented (almost) incontrovertible evidence for DNA rather than proteins as the genetic material. Quite simply, one bacteriophage virus culture was labelled with a sulfur isotope in the coat proteins and another virus culture was labelled with a phosphorus isotope in the DNA.

E. coli cells were infected with each virus type and the cells centrifuged to remove the attached virus 'ghosts'. The cells were found to contain only phosphorus labelled DNA and no sulfur labelled proteins. The infecting agent therefore had to be DNA. Mind you, Avery and his team had found a similar result six years earlier, but for reasons unknown, were not accorded the same degree of acknowledgement among their peers. Indeed, back then, in a slightly misogynist world, yes even among scientists(!), Men like Maurice Wilkens and Alfred Hershey won Nobel prizes, but female scientists of equivalent noteworthiness such as Rosalind Franklin and Martha Chase did not win the Nobel prizes their work deserved. On the other hand, neither did Ostwald Avery, a man!

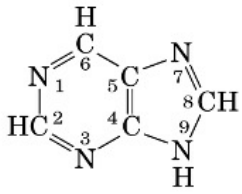
A You Tube video link is at bottom left of the slide and is worth a look.

Nucleotide Bases

Pyrimidines	Purines
Cytosine Thymine (DNA only) Uracil (RNA only)	Guanine Adenine



Pyrimidine



Purine

DNA contains (weak) bases

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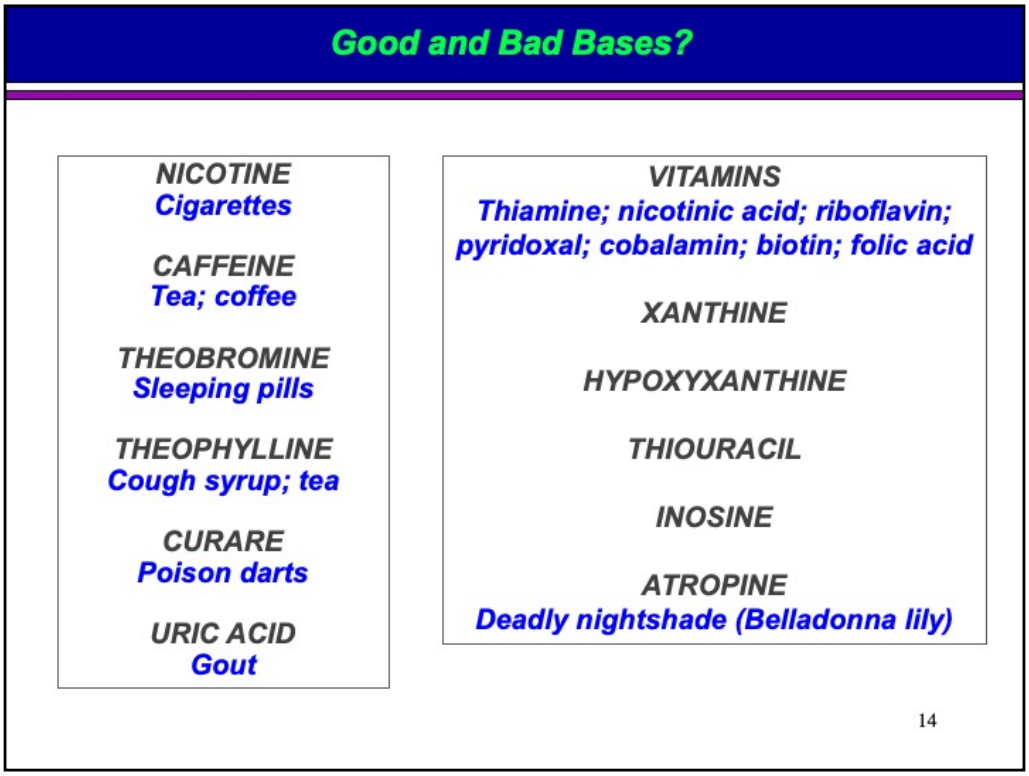
The five major bases in nucleic acids.

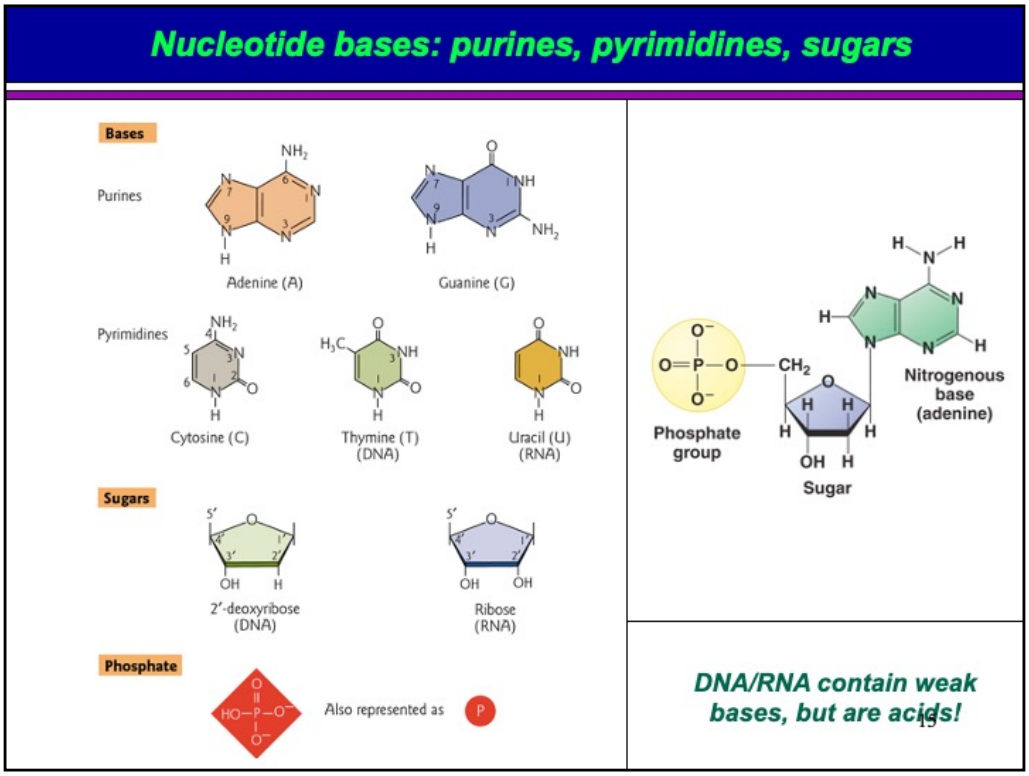
The pyrimidines have a single six-membered aromatic ring. Purines have a double ring with the larger one being the same as the pyrimidine ring.

Thymine is only found in DNA. Uracil is the Thymine equivalent in RNA.

The ‘bases’ are very weakly alkaline, non-polar, and absorb UV light at a maximum wavelength of 260 nm.



In addition to the five well know bases in DNA and RNA (adenine, cytosine, guanine, thymine, uracil), there are many others in nature that are involved in metabolism (as vitamins) and in nucleic acid biosynthesis. There are also a number of so-called 'bad' bases, such as in the first panel above.



The two purine and three pyrimidine bases that give DNA and RNA their uniqueness (Uracil is only found in RNA; Thymine in DNA).

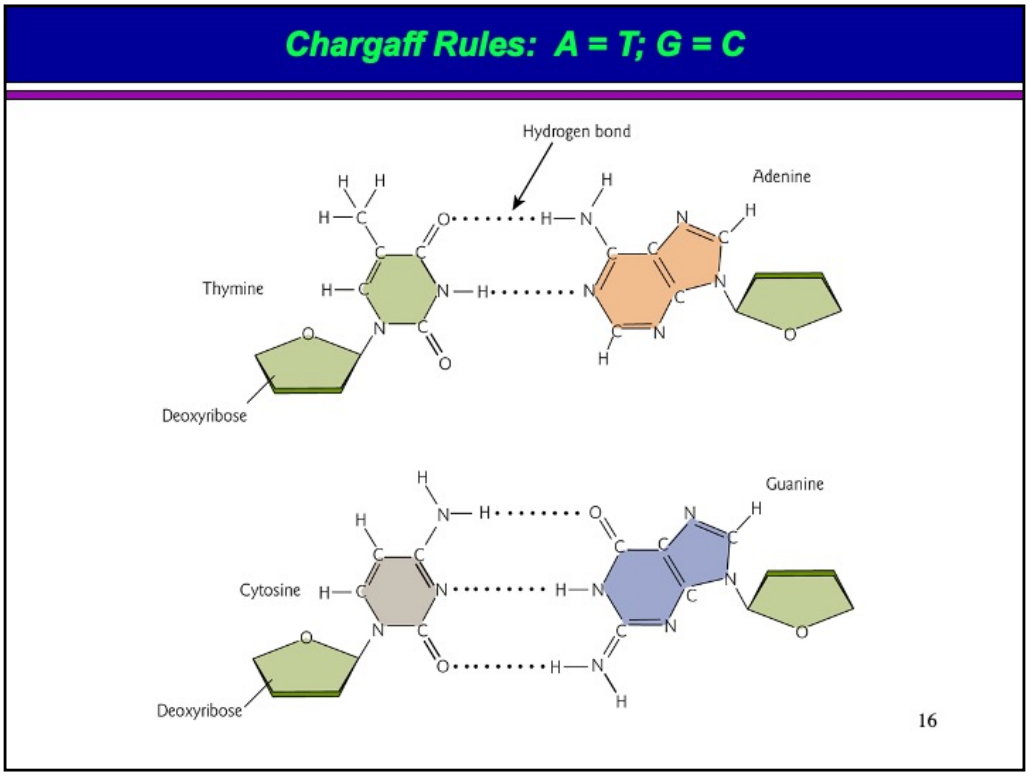
The contributing sugars and phosphate are also shown. Note that DNA has deoxyribose and RNA ribose. These sugars differ in the 2-prime hydroxyl position. The positions are numbered according to the sugar carbons, as indicated in the structures.

Deoxyribose is really 2-deoxyribose, but is usually abbreviated as 'deoxyribose'. 'Deoxy' means 'lacking oxygen' and ribose is converted to deoxyribose when the 2'-carbon of ribose loses its hydroxyl group (-OH), leaving a hydrogen (H) in that position.

Finally, nucleic acids have phosphoric acid (often abbreviated as just 'P') attached the to the sugar and base, generating nucleotides, an example of which is shown on the right.

Note that though nucleotides contain weak bases, the phosphoric acid moiety dominates, and nucleotides polynucleotides are known as acids.

All of this information was known to chemists, biochemists, and geneticists since the early 1900s, but how this jigsaw puzzle of nucleotides fitted into a DNA structure was a mystery, until three independent groups began trying to solve the structre in the late 1940s and early 1950s. The three teams were: Linus Pauling and co. (Californian Institute of Technology); Maurice Wilkens / Rosalind Franklin (Kings College London); and James Watson and Francis Crick (Cavendish Laboratory, Cambridge).

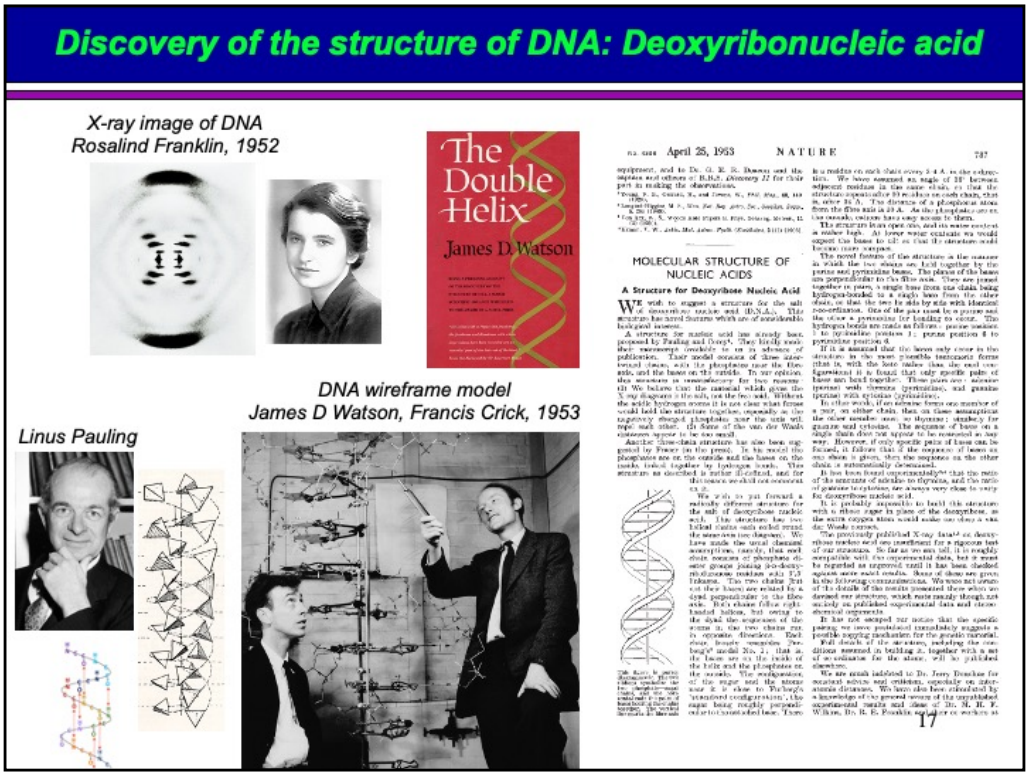


Erwin Chargaff performed experiments in the 1930s that showed DNA to have equal amounts of A and T; and of G and C. These equalities became known as 'Chargaff's Rules'. You can see from the diagram how the base pairings obey these rules.

Note that the A-T base pair (upper) has two H-bonds and the C-G pair three H-bonds.

A third H-bond between the H and CO in the upper pair is not formed due to the distance being too great for the interaction to occur. The limit of an H-bond is about 3.5 angstroms (Å), which is approximately three and one half covalent bond distances. For simplicity, this diagram does not show the phosphates that would be connected to deoxyribose on the outside.

These 'rules' were crucial in the solution of the structure of DNA by Watson and Crick a few years later.



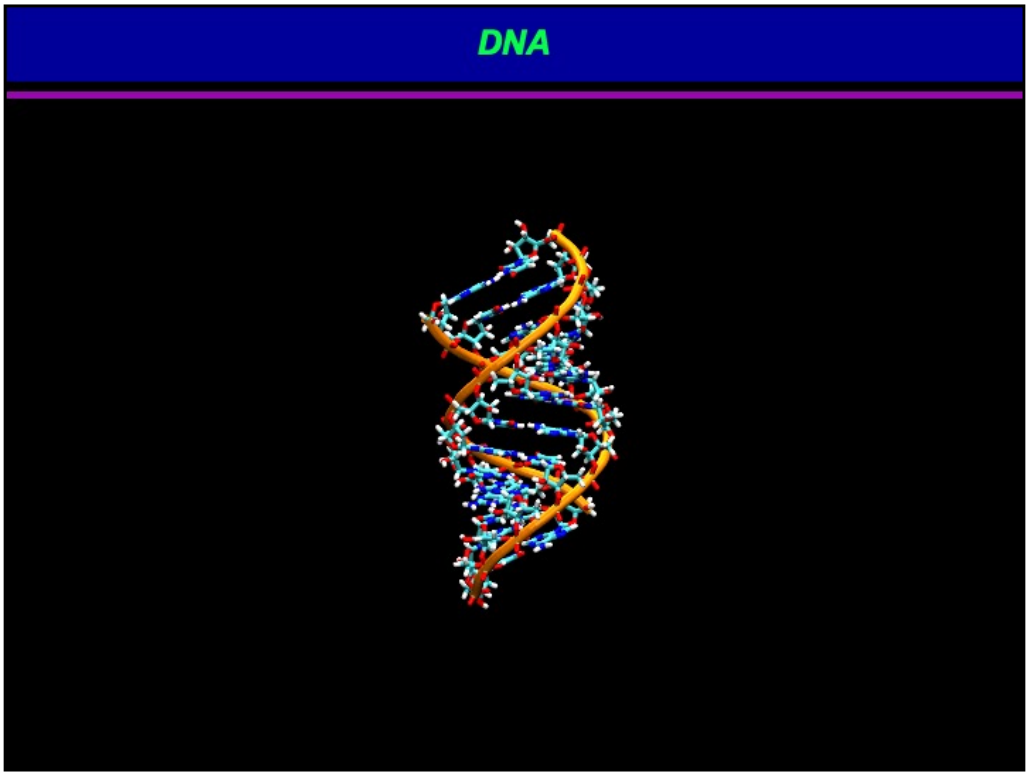
The discovery of the structure of DNA in 1953. James Watson and Francis Crick decided in 1951 that DNA was almost certainly the genetic molecule, from the results of the Avery and Hersey-Chase experiments. But the nature of DNA (and RNA) was still mostly unknown. When they entered the race to build the structure of DNA, they were fighting against time and one of the most brilliant scientists of the Twentieth Century, Linus Pauling (bottom left). Pauling had already taken out a Nobel Prize for Chemistry for his work on the nature of the chemical bond. He also worked out the alpha-helical structure of polypeptides. Pauling's alpha-helix is at bottom left and his 3-stranded DNA original drawing is shown to the right of his photo.

Top left, the original X-ray image of a double-stranded DNA helix with the even diameter 'X' depicted. The dark shading top and bottom represents the sugar phosphates on the outside of the double helix, and the crosses in the centre are the stacked H-bonded base pairs in a double helical formation. Rosalind Franklin, who took this first ever clearest image of double-stranded DNA is shown to the right of the X-ray image.

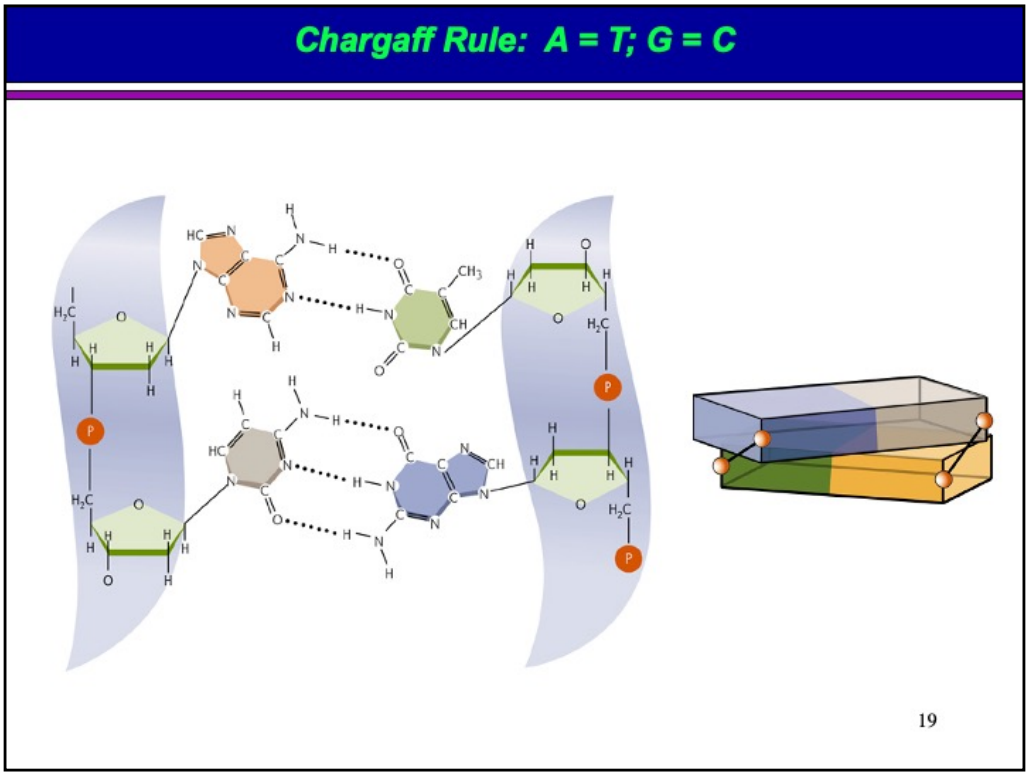
Bottom middle shows Watson and Crick with their wire-frame model. The analogy of a winding staircase is often cited, with the bannister rails representing the sugar phosphate backbone and the steps as the base pairs in the core of the molecule (see the stairwell in the new UTS building going up alongside the Tower!).

An narrative account of the race to discover the structure of DNA was published by James Watson in 1968 as 'The Double Helix' , and it makes compelling reading (top centre). A copy of the first page of the *Nature* report of the structure is shown at right.

Women hard done by in Science!! Be careful if you make a woman cry, because God counts her tears.



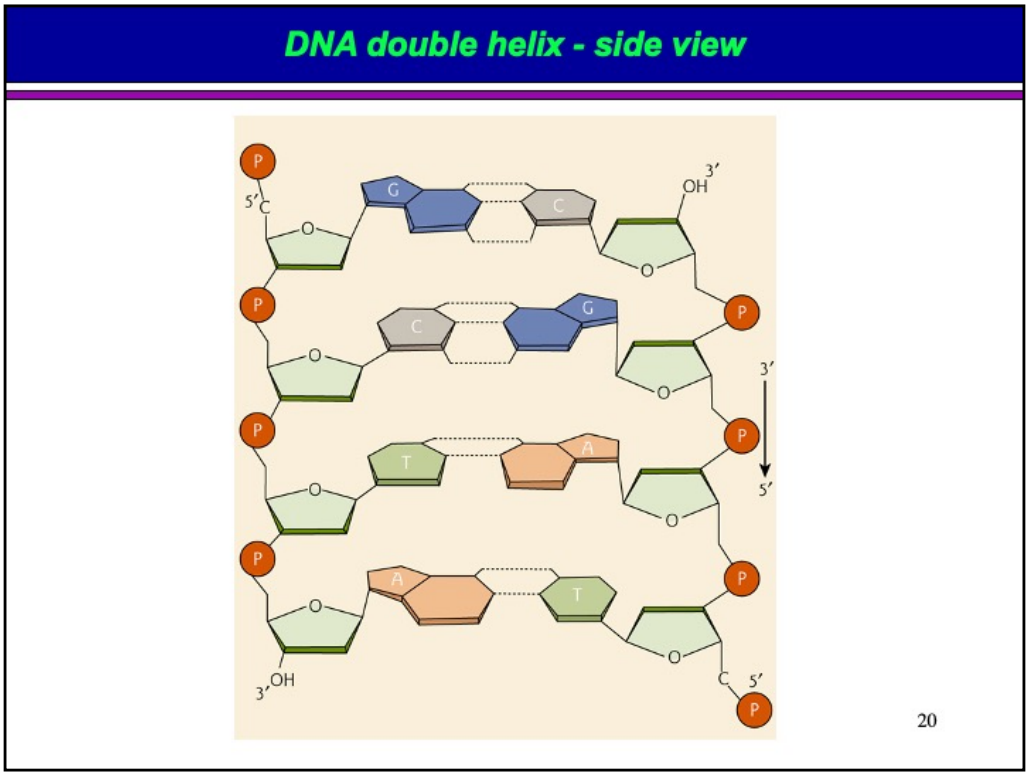
A movie showing the 3-D structure of DNA.



The left panel diagram shows how nucleic acid (DNA or RNA) bases are paired by the complementarity rule (Chargaff Rules) and this takes place across the double helix and within its hydrophobic core. The sugar phosphate hydrophilic "backbone" of each nucleotide is positioned on the outside of each helix strand facing the aqueous surroundings, making the whole molecule soluble and acidic. Note the covalent attachment of the nucleotides (by phosphodiester bonds) in the vertical direction only (on the outside of the structure).

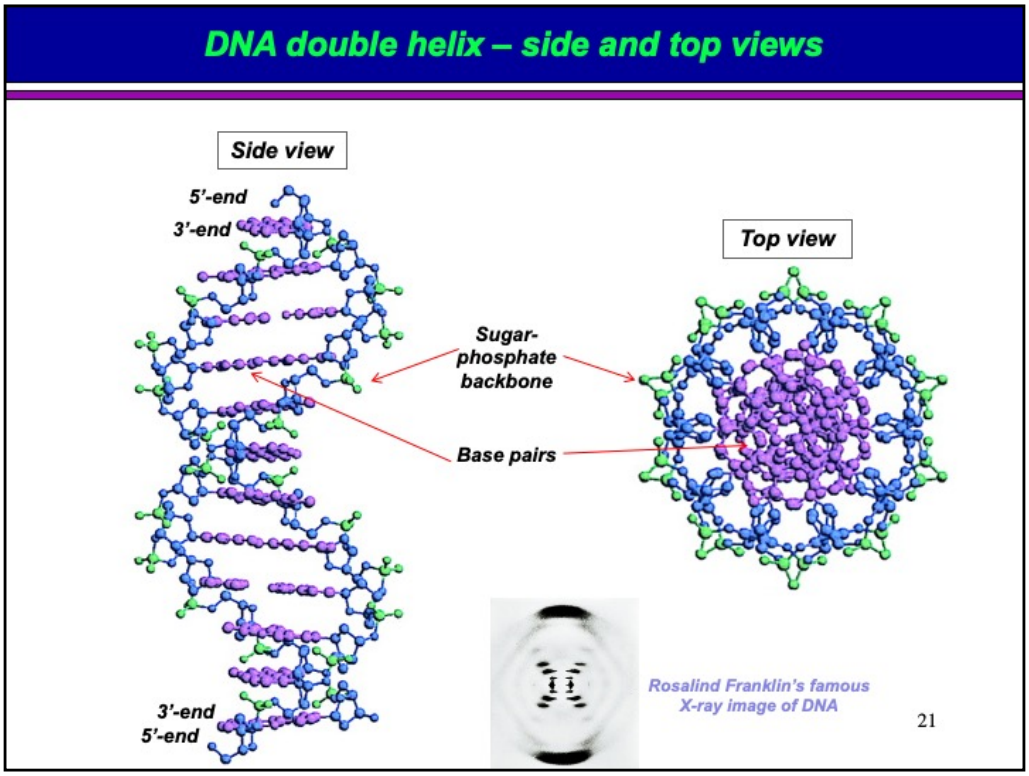
The small cartoon (right panel) shows how the stacked base pairs gradually twist to give the nucleic acid its helical shape. Each DNA or RNA nucleotide consists of base, sugar, and phosphate.

If you haven't noticed yet, you will find throughout the Molecular Biology world, and not just in this subject, that the nucleic acid nucleotides, which comprise a base, sugar, and phosphate, are often abbreviated to the nomenclature 'base' for simplification. So, the two are used almost interchangeably, but when a distinction is required, the specific and proper name is used (base or nucleotide).



Base-pairing in nucleic acids. These pairings are always made, with each larger purine base opposite a smaller pyrimidine one so that the DNA double-stranded molecule has an even diameter. The base-pairing (dashed lines) occurs due to spontaneous formation of H-bonds across the gap. Note the evenness of the diameter of the paired bases that form the double helix. Note also the **opposite polarity** of the strands, with each strand running in opposite directions (left strand 5’ to 3’; right strand 3’ to 5’). This polarity is always opposite and the double stranded nucleic acid molecule will only form and remain stable in this opposite arrangement of the strands.

The 5’ designation refers to a free 5’-carbon on the first nucleotide (5’-P at the top of the left strand). The 3’ designation refers to a free 3’-carbon on the last nucleotide of the strand (3’-OH at the bottom of the left strand). Every nucleotide is joined covalently to the one above and below by **phosphodiester bonds**.

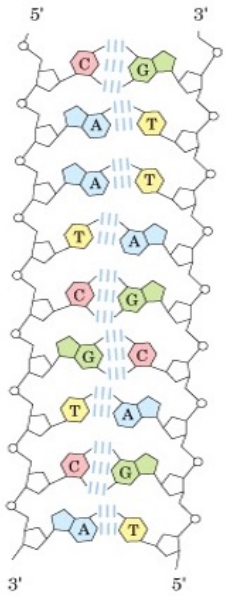


A double-stranded 'spiral staircase'. In these views of double-stranded DNA, note that there are H-bonds between the nucleotide bases in the centre of the molecule and it is predominantly **these non-covalent forces that hold the two strands together**. The outside of the structure is very polar due to the presence of the sugar (deoxyribose) and phosphates (negatively charged) that surround the base pairs in the centre.

DNA "dissolves" by solvation into water. Note also that the two strands are antiparallel. The two strands of DNA or RNA are held together by the collective action of three types of non-covalent interacting bonds: (i) H-bonds between the horizontally stacked base pairs in the core of the molecule; (ii) hydrophobic interactions between the vertically stacked base rings in the core; and (iii) sodium ions that neutralise the strong oxygen negative charges on the acidic phosphates on the outside of the molecules.

Note in the 'top view' diagram on the right that the non-polar bases are in the centre and the polar sugar phosphates surround the non-polar core and are exposed to water. Unsurprisingly, soluble proteins are configured in exactly the same way, but in this case, with the non-polar amino acids within the core and the polar or charged amino acids on the outside in contact with water.

DNA double helix is stabilised by three forces



- *Horizontal hydrogen-bonding between base pairs.*
- *Vertical hydrophobic interactions between the rings of the stacked bases.*
- *Sodium charge clouds that neutralize the negative charges on the phosphate groups.*

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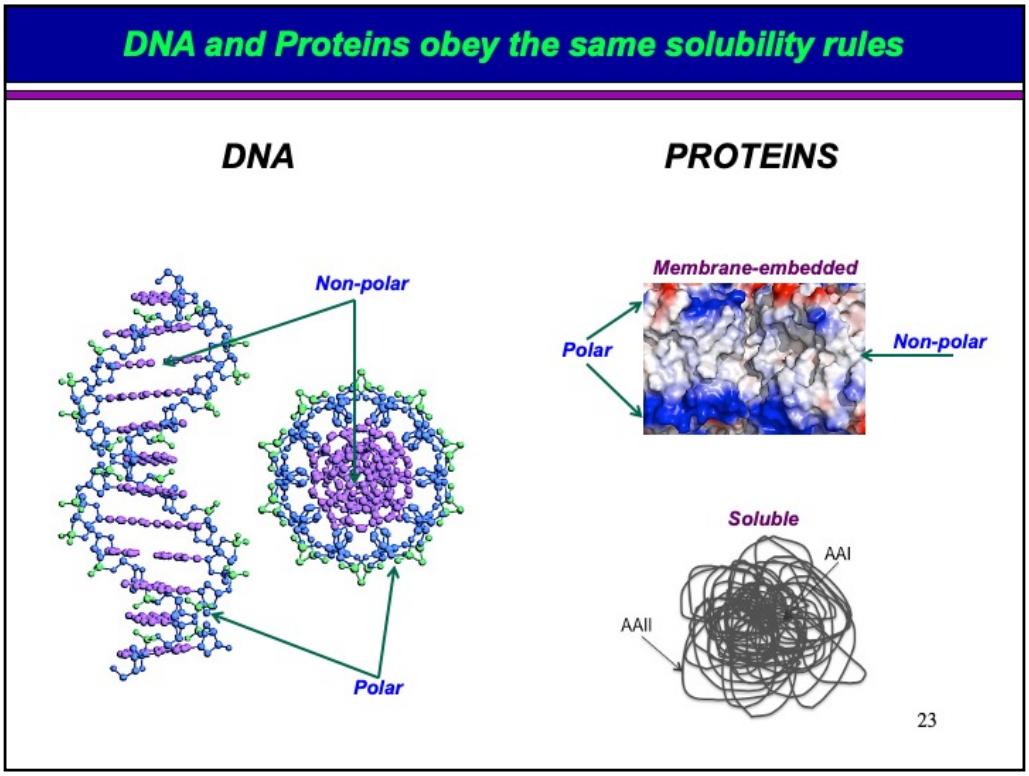
A summary of the forces contributing to the stability of the DNA double helix, but bear in mind that the H-bonded bases in the centre are by far the most important in stabilising the molecule.

On the other hand, the sodium clouds around the negative phosphates are also important in that they neutralize the acidic negative phosphates and enable the two strands to associate.

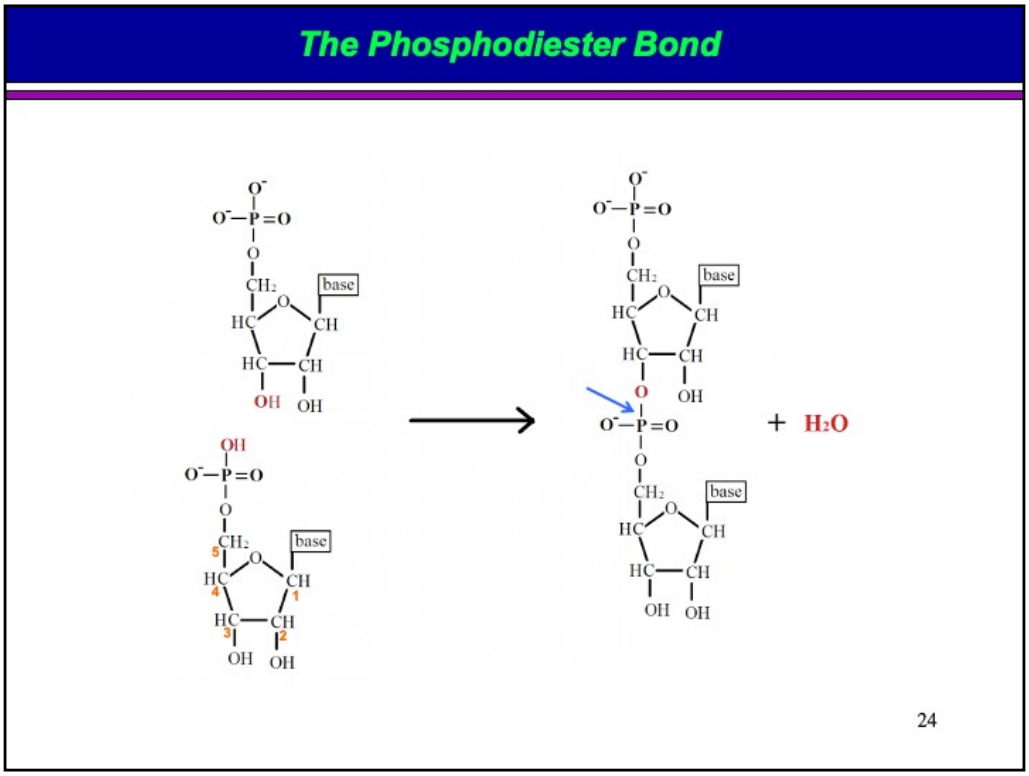
In fact, if the two strands were separated in pure water, with no sodium ions around the negatively charged phosphates, the two strands would not associate to form hydrogen-bonded double helix, even though the strands are perfectly complementary.



DNA and proteins obey the same solubility rules.

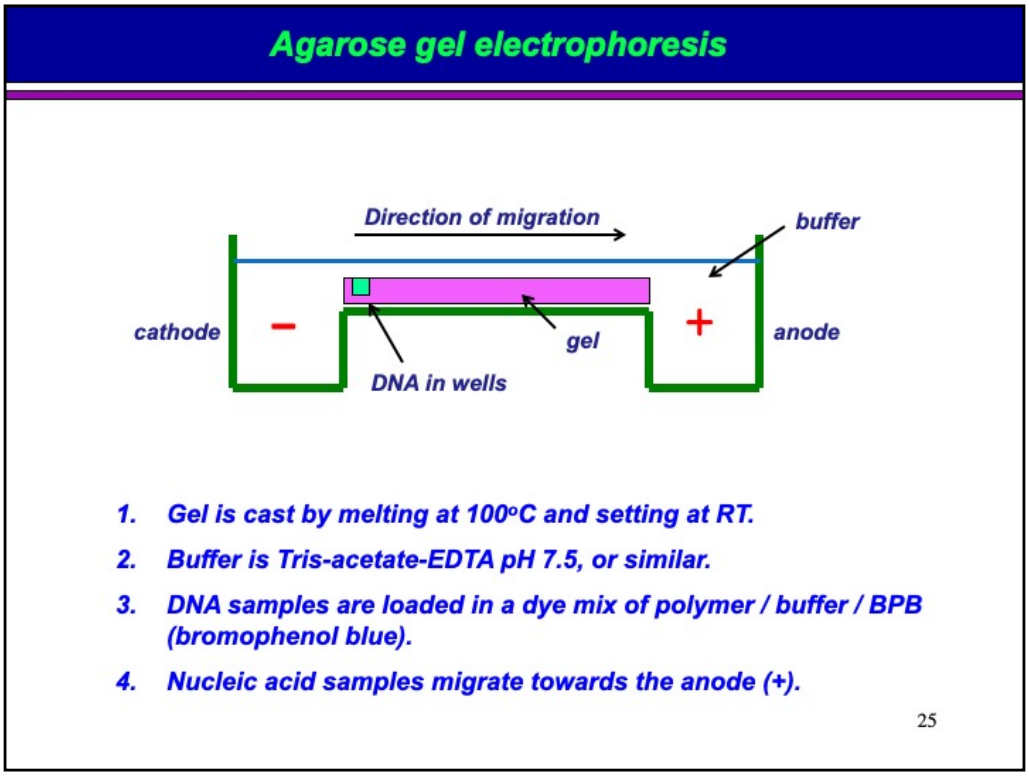
On the left are two images of double-stranded DNA (from the previous slide). The purple spheres are the nitrogenous weak bases within the core or interior of the structure and are in a water-free, hydrophobic, non-polar environment. The blue and green spheres are the sugars and phosphates that protect or cover the bases within the core. The sugar-phosphates are in contact with water and are polar and acidic.

On the right are two images of different proteins. The version at the top is a membrane-embedded protein and site within the hydrophobic, non-polar lipid bilayer (white, space-filled non-polar amino acids); and on the inner and outer membrane surfaces in contact with water (blue, red space-filled amino acids), where red is for acidic amino acid side chains and blue for alkaline amino acid side chains. The grey soluble protein drawing at the bottom is a folded soluble protein. Two amino acids, designated AAI and AAII, are depicted. AAI is a core or buried amino acid and therefore has a non-polar, hydrophobic sides chain; and AAII is a surface amino acid with a polar side chain.



Two monophosphate nucleotides joined with a 3’ to 5’ phosphodiester bond (**water is eliminated across the bond, just as it is in the peptide bond between two amino acids**). The phosphodiester bond is formed by the elimination of the 3’-Carbon hydrogen, leaving the oxygen (in red on the right figure) and elimination of an hydroxyl group from the 5’-phosphate group (OH in red, left figure). These covalently joined nucleotides are extended at either end with new phosphodiester bonds between added nucleotides, generating a polynucleotide strand.

If another antiparallel strand is added, then a double-stranded DNA or RNA molecule is formed. The 5-prime and 3-prime nomenclature derives from the sugar moiety (deoxyribose in DNA; ribose in RNA). The carbon numbering is depicted on the ribose ring at bottom left.

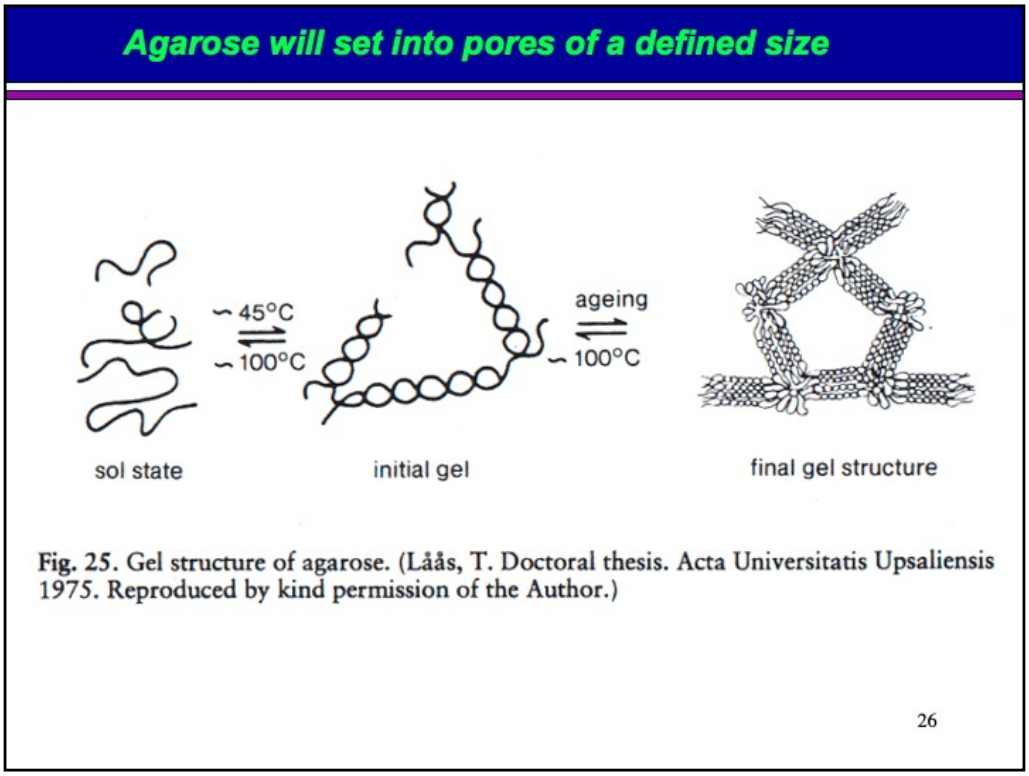


This is the standard method for identifying and analysing nucleic acids (DNA, RNA). It is also known as submarine gel electrophoresis, as the gel is submerged beneath the running buffer.

The nucleic acid samples are loaded in special dye mix that contains marker dyes to indicate the extent of the run – these dyes run ahead of (bromophenol blue, BPB) or with (xylene cyanol) nucleic acids of various sizes. The sample loading dye mix also contains a nuclease inhibitor (EDTA) to prevent digestion of DNA during the run, and ~30% sucrose or a high MW polymer to give the sample dye mix viscosity, so that it will settle and remain in the well until the run is started.

Gels are run until the BPB dye reaches at least halfway or near the end of the gel. Note that the DNA is added at the cathode (negative) end, as DNA has a net negative charge at the pH of the buffer (~7.5) and will therefore be conducted towards the anode by the negatively charged buffer ions (and by the DNA's own –ve charges).

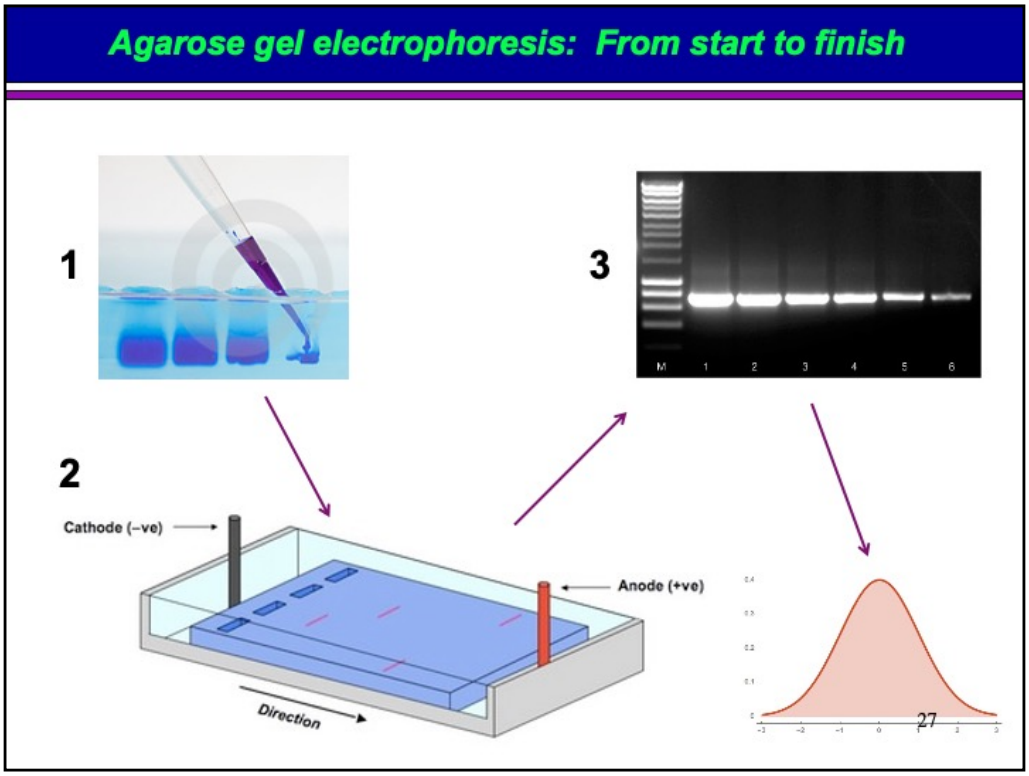
The DNA molecules are conducted through the regularly sized pores of the gel. The migration rates of nucleic acid molecules is determined by their size and conformation (see next slides).



Agarose is a purified component of agar that melts (100°C) and sets (~45°C) in the same way as agar.

The pore size (on the right, in the final gel structure) is determined by the agarose gel percentage. Most gels are cast in the range of 0.5 to 3.0%.

Higher gel percentages produce smaller pore sizes and will only allow ready migration of small fragments. Lower percentage gels will have larger pores and allow larger nucleic acids to migrate with relatively minimal frictional resistance. In your MB1 practical experiments, all gels will be around 0.7%.

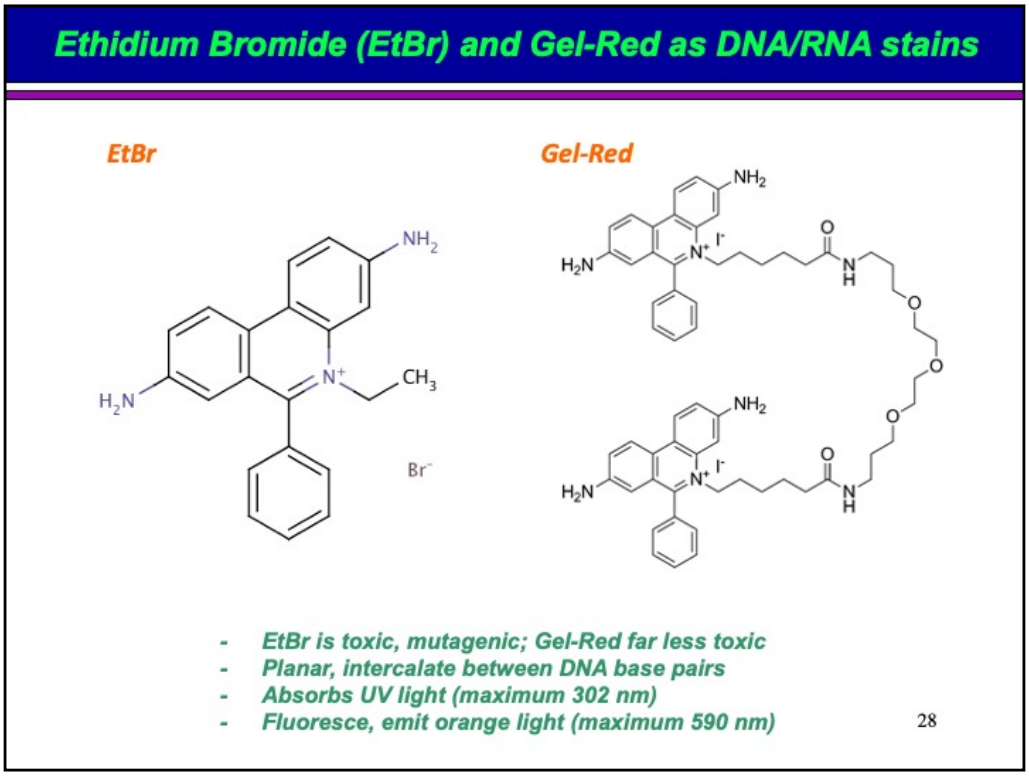


Small aliquots of DNA or RNA samples are mixed with loading/running dye. Typically, about 5 µl dye to 15 µl sample. The loading dye mixture is supplied or made at ~5x strength so that only a few µls are required per sample. It contains buffer, EDTA, bromophenol blue dye (and often a second dye xylene cyanol, which is green and runs slower than bromophenol blue), and something to make the dye mix viscous, usually 30% sucrose, glycerol or a high MW polysaccharide polymer such as Ficoll.

Figure 1. When the sample is added to the well via a pipette tip, it drops into the well, as it is more dense than the running buffer; and it remains there until the current is applied.

Figure 2 depicts the resolving of the DNA fragment bands in the gel as they migrate from cathode (-) to anode (+), and the completed run with the fragments arranged in maximum to minimum sizes from top to bottom (left to right in the figure, as feint orange bands).

Figure 3 depicts linear molecular weight markers in the left lane; and decreasing amounts of the same DNA sample in lanes 1 to 6. The point to notice about these samples is the way the broadness of the bands spreads from the centre outwards, like a normal curve (bottom right), with the highest concentration of sample at the centre of the band in the gel. Although not a precise issue, the actual size of the sample band is best taken from the lane 6 band, or from the middle of any of the other bands.

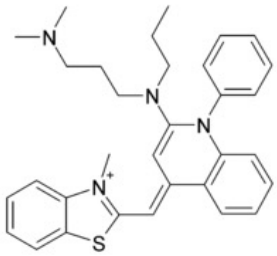


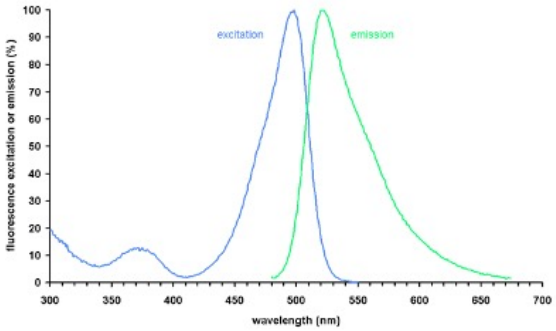
Structures of the two DNA/RNA stains for agarose gels. Ethidium bromide (EtBr; left) was the original stain used until several years ago (but is still used in some places). It has been superseded by Gel-Red (right), which has the same degree of binding and staining sensitivity as EtBr, but is far less toxic and is now widely used.

You can see from the structures above that they are comparatively similar, with Ethidium ionised with Bromine (Br-) versus Gel-Red ionised with Iodine (I-). Gel-Red is virtually a duplication of the EtBr multiple ring structure, with a long aliphatic connecting chain between the aromatic ring clusters.

Both dyes are planar (flat) and are able to **intercalate** (see next slides) between the stacked base pairs of double-stranded DNA or RNA, or against single-stranded RNA. The positive charge on the nitrogen (N^+) allows the dye to bind to the nucleic acid's opposite charged (-ve) phosphates.

Sybr Green as a DNA stain





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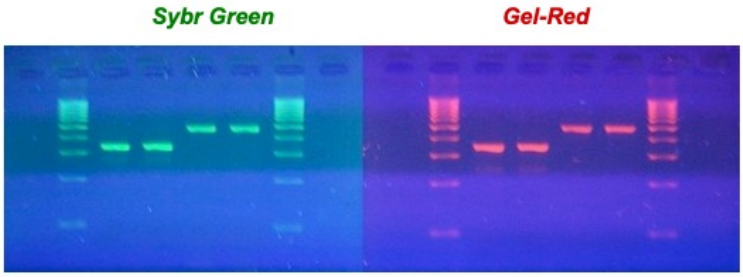
A third type of DNA staining dye, Sybr Green, which as you can see from the structure on the left, will bind to DNA more or less in the same way as Gel-Red and ethidium bromide (small positive charge; intercalating aromatic rings).

DNA stained with Sybr Green can be visualised by UV excitation in the same way as the other dyes, though Sybr Green works at different wavelengths, absorbing blue light at ~495 nm and emitting green light at around 520-540 nm, as depicted in the spectrum above.

The one property of Sybr Green that differs from the other dyes is that it does not bind as well to single stranded DNA or RNA (fluorescence is 11x weaker).

DNA staining in agarose gels



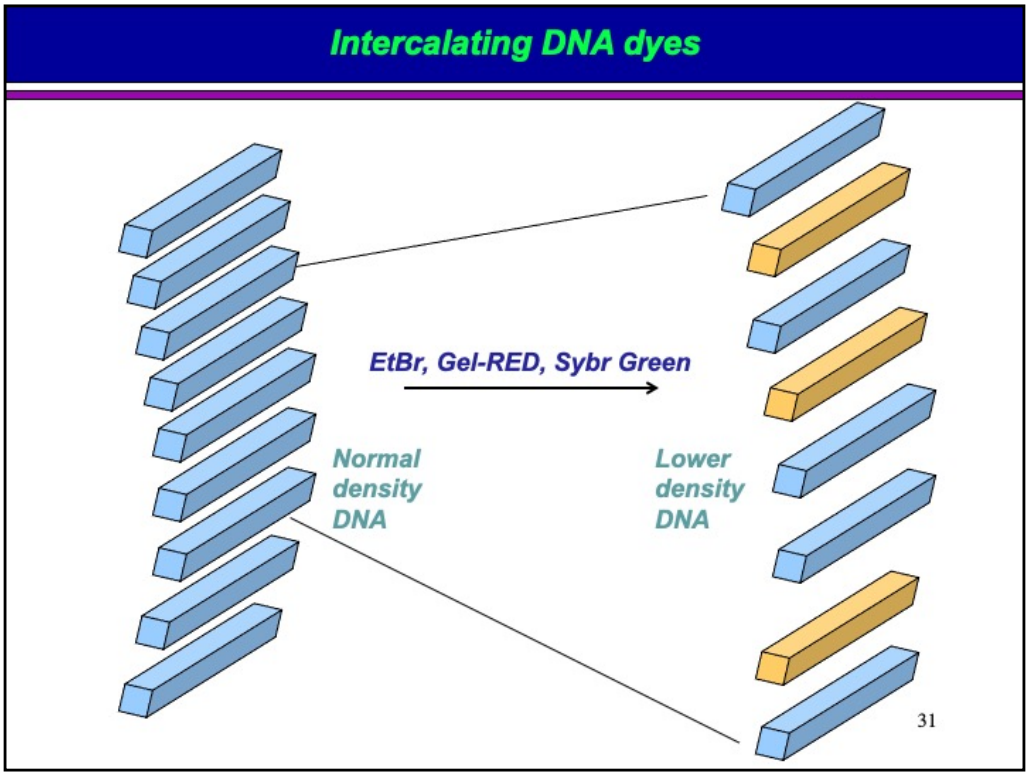
Each DNA band contains many hundreds or thousands of exactly the same DNA molecule – same mass, same conformation.

30

DNA "bands" in an agarose gel after electrophoresis and stained with Sybr Green or Gel-Red. The molecules of DNA in the original sample added to the origin wells migrate in the electric current. They are sorted by the current into bands according to their mass and conformation so that all molecules of the nucleic acid in any one band are the same.

For Gel-Red, the stained gel is viewed on an ultraviolet (UV) light box with the UV lamp set to irradiate the nucleic acids at a wavelength of 302 nm, whereupon the nucleic acids absorb the UV light, fluoresce, and re-emit in the red-orange region of the light spectrum. Photographic records of the gel are made and saved as jpg or tif files.

Note that each DNA 'band' contains hundreds or thousands of exactly the same DNA molecule, same mass, same conformation.



A cartoon representation of the **intercalating dye** binding of EtBr, Gel-Red, or Sybr Green in double-stranded DNA.

The planar rings of the dye molecules 'stack' within the core of the structure like extra 'stairs' in between the base pairs and this stacking is aided by the similarity of the hydrophobic planar rings of the dyes to those of the DNA or RNA bases.

Note that the DNA '**swells**' when the dye is bound, with the entire dye-bound DNA molecule becoming **less dense**. Dye binding is easier in linear or relaxed circular molecules, more difficult in supercoiled DNA due to the tightly wound DNA strands, making dye access harder.

Separation range for agarose and polyacrylamide gels

Gel type	Separation range (bp)
0.3% agarose	50000 to 1000
0.7% agarose	20000 to 300
1.5% agarose	6000 to 300
3% agarose	3000 to 200
4% acrylamide	1000 to 100
10% acrylamide	500 to 25
20% acrylamide	50 to 1

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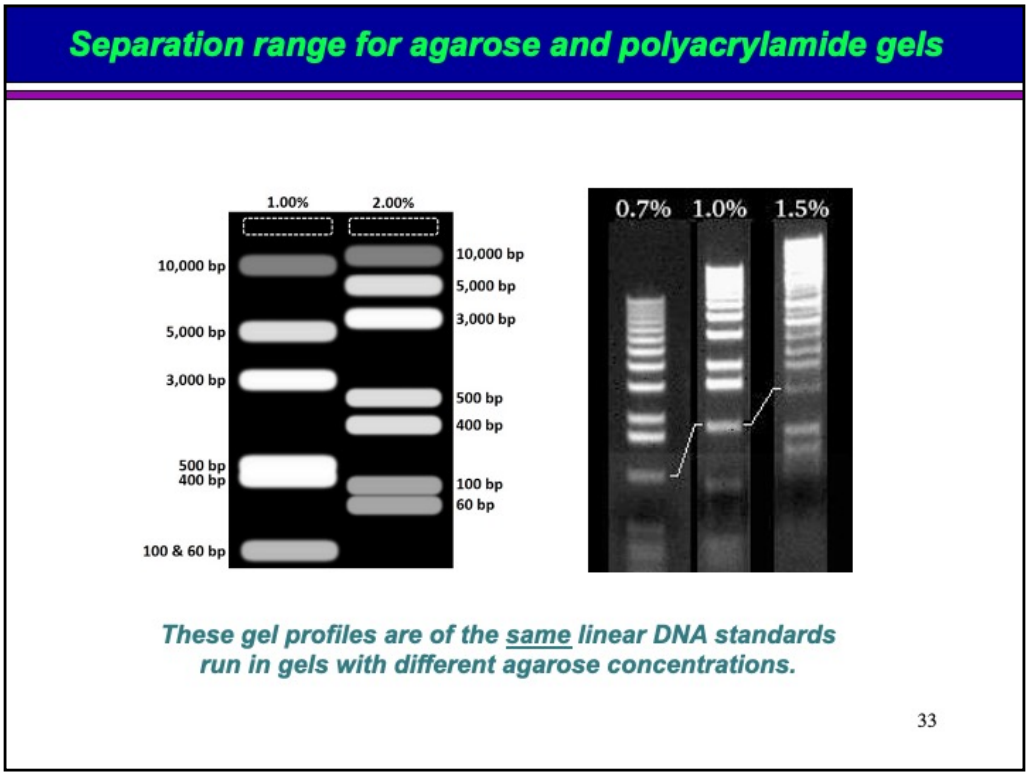
The table gives approximate nucleic acid separation and resolution ranges for various percentages of agarose or polyacrylamide gels. Note that for smaller DNA molecules, gels must be cast with polyacrylamide to ensure the pores of the gel are small enough to resolve the population of small DNA molecules. Agarose gels may be cast up to 3% to resolve small DNA fragments, but these high % gels are brittle and produce too much heat (as resistance), which may distort the DNA bands, and so these gels must be run at lower voltages for longer times.

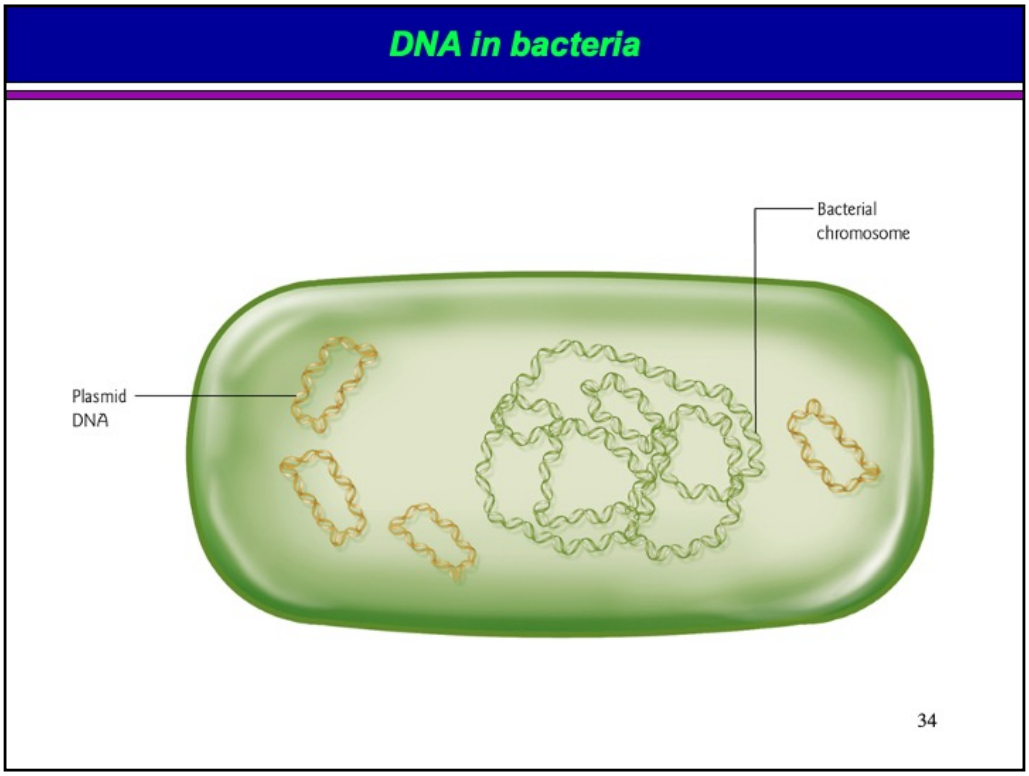


Note the different positional profiles of the same set of linear standards in increasing percentage agarose gels depicted in the gel photos above.

The gel figure at left is a drawing only of the same set of linear DNA bands running in a 1% gel (left lane) and a 2% gel (right lane). Note how the positions and resolution of the bands change depending on the gel percentage.

The gel figure at right is a real representation of the same linear DNA bands running in gels of three different percentages, namely 0.7%, 1.0%, and 1.5%. The difference is best illustrated by looking at the top few standards in the different % gels/lanes. See how they are spread out in the 0.7% gel, whose larger pores allow bigger DNA molecules to migrate through the pores with less frictional resistance. The 1.5% gel shows the same high mass DNA bands 'compressed', that is less spread out, unresolved, and much closer to the origin, because they are too large to migrate freely, encountering frictional resistance through the smaller pores.

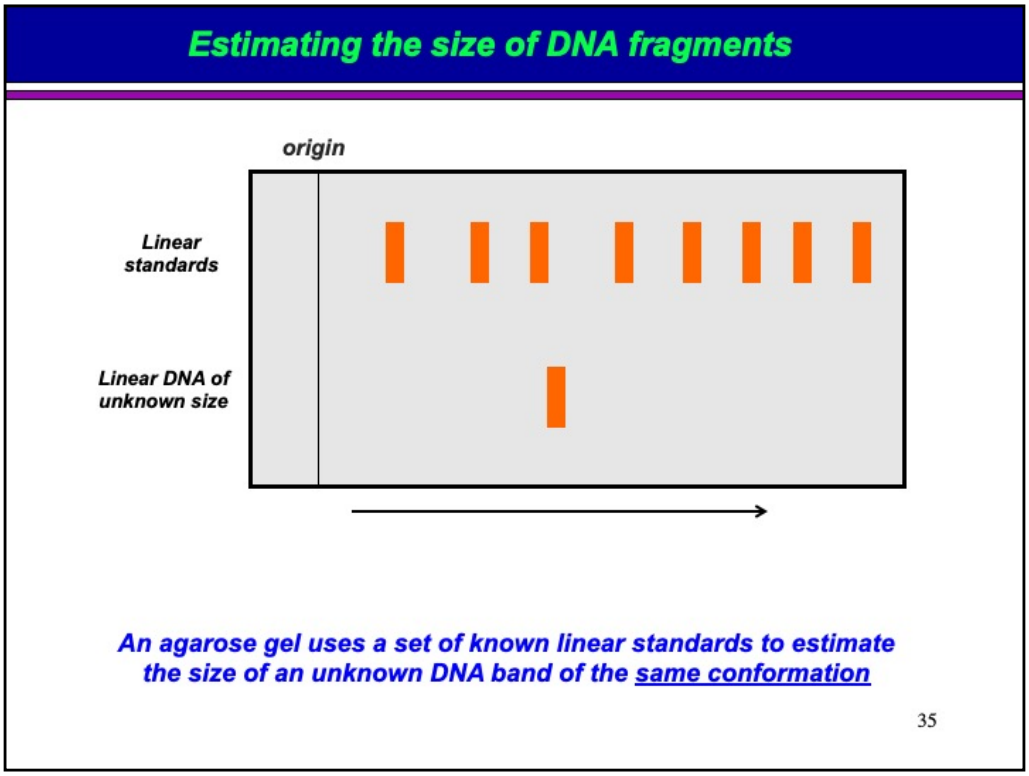
Note also that linear DNA molecules migrate end-on-end, but their lengths still prevent them from fitting easily through small pores where their flexibility and length make them bump continuous into the walls of the pores. More on this in a few slides ahead, when we look at different DNA conformations being run in agarose gels; and the effect of conformational geometry on migration rate.



This cartoon diagram contains an exaggerated representation of circular plasmid DNA molecules that are only present in some bacteria and contain genes that give an advantage to the host bacterium. The plasmid is replicated synchronously with the chromosome and "inherited" by daughter cells along with the customary single chromosome copy.

Native plasmids may occur in single to several copies per cell and range in size from as small as 2 kilobases (kb) to perhaps over 400 kb. The *E. coli* chromosome is ~4600 kb in length and very large plasmids may reach almost 1/10th the size of the chromosome.

When we come to look at plasmids used in Molecular Biology and Genetic Engineering (Molecular Cloning; Recombinant DNA Technology), we will see that only small hybrid plasmids are used and these have been engineered to contain special features for molecular cloning.

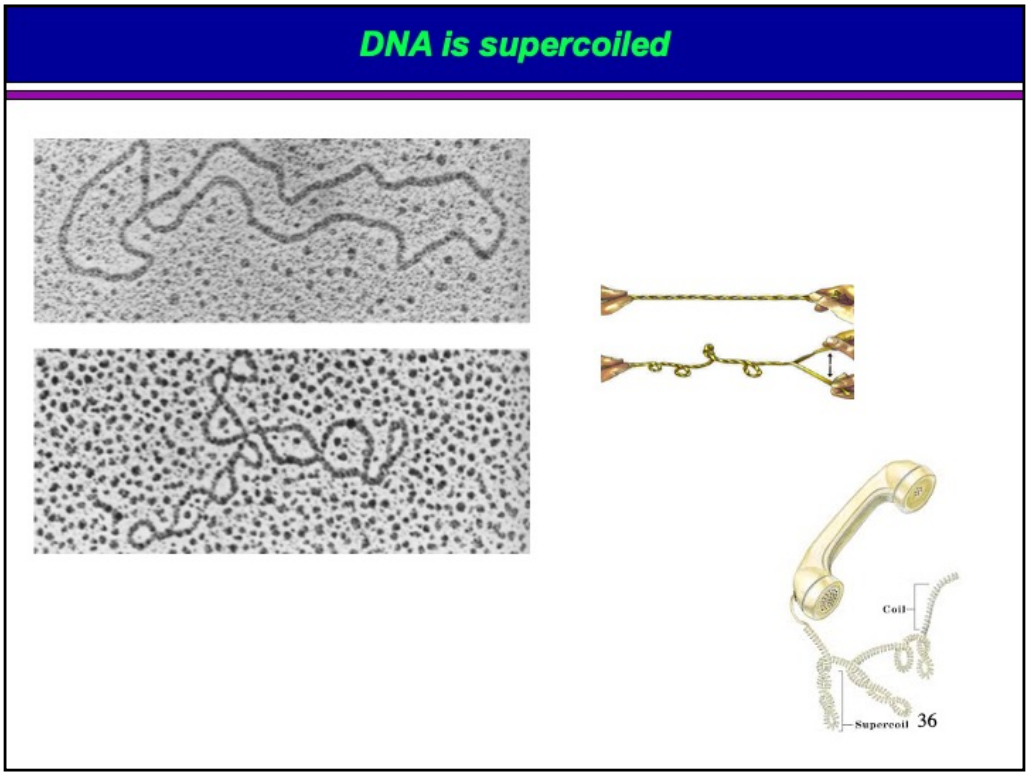


Estimating the sizes of DNA fragments typically involves running the unknown DNA against a set of standards of known sizes.

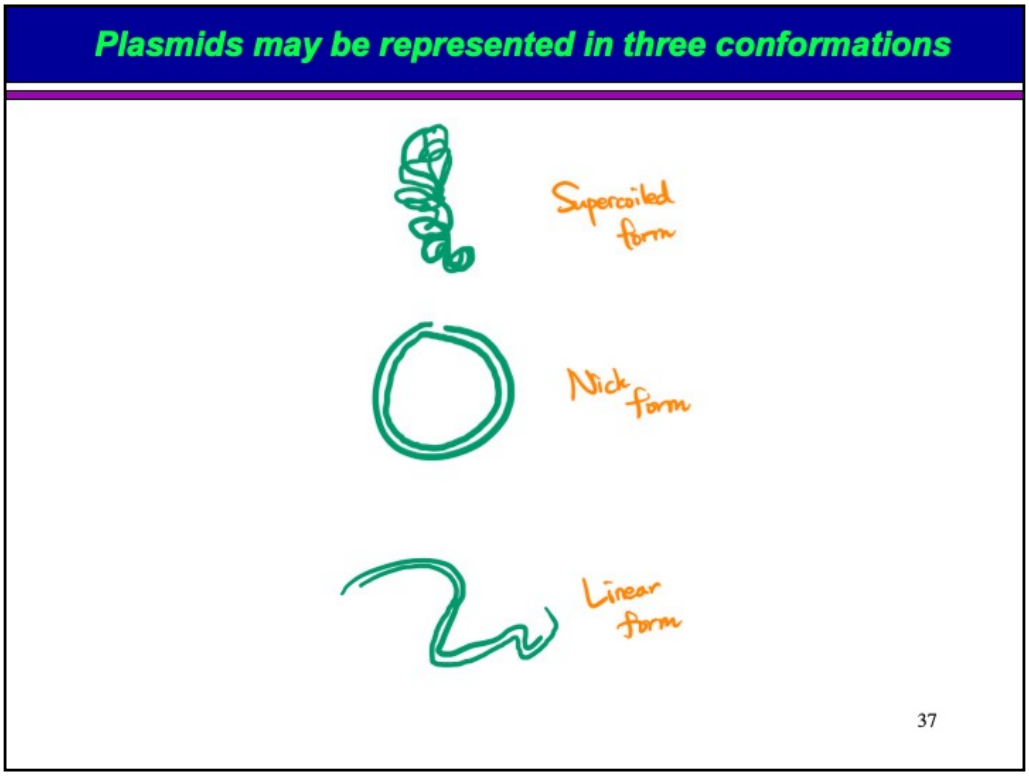
In the diagram above, the top lane has a set of linear standards running according to their mass/length sizes and will migrate according to the inverse log10 of their masses (see the next slide). A DNA band of unknown size**, but of the same conformation as the set of standards** is run in the bottom lane.

It is easy to make a good guess of the approximate size of the unknown fragment, by eye, as the unknown band lies between two standards of known sizes, bands 3 and 4. If nothing else, this at least will tell you if your more accurate value from the standard curve is correct.

Note: unless otherwise stated, when reference is made to circular or linear DNA, the molecules are deemed to be **double-stranded** by implication, as this is the 'normal' state for most such molecules. Single stranded DNA or RNA will be indicated as such. There is also the convention of abbreviating the type of nucleic acid, for example, as ds DNA or ss DNA, where 'ds' and 'ss' are double-stranded and single-stranded, respectively.



Different views of ‘supercoils’. The electron micrograph (top) is of a normal ‘relaxed’ circular DNA molecule, whilst the lower electron micrograph is a supercoiled circular molecule. A comparison is depicted in the two diagrams to the right of the EMs and in the telephone cable at lower right.



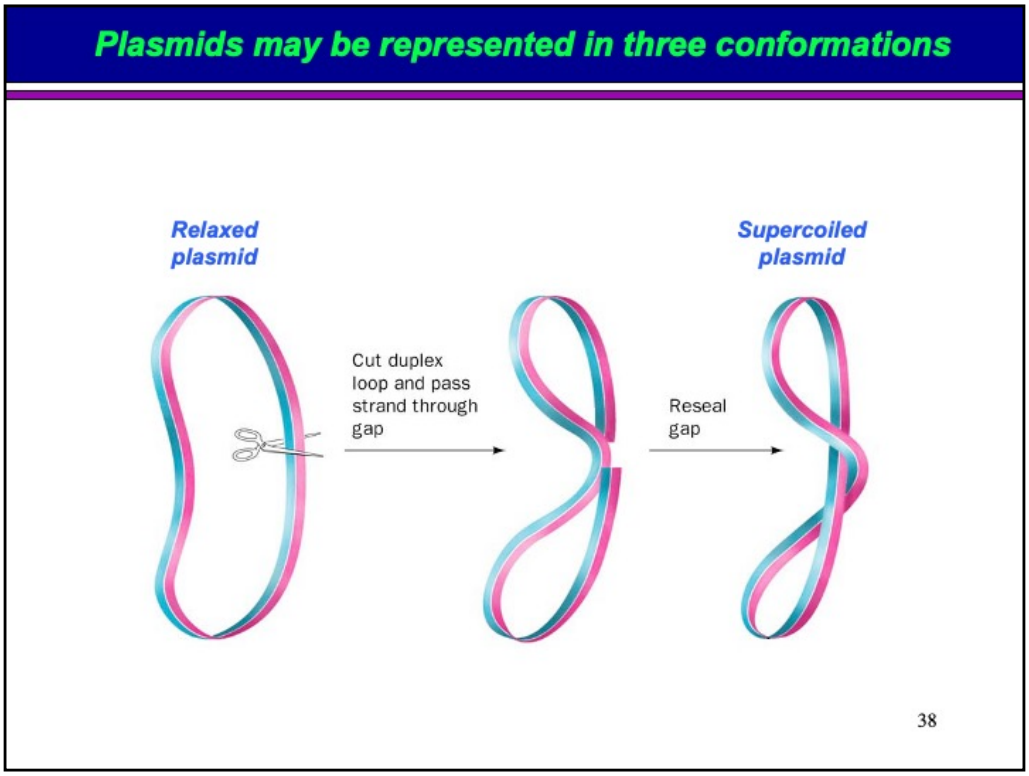
Typically, plasmids in bacterial cells are in the supercoiled configuration (top cartoon drawing), just as chromosomes are in all cells (bacteria to human). This is the so-called resting state and is a 'packaging' device that cells use to store DNA molecules (plasmids and chromosomes).

The nicked configuration, also known as relaxed or open circle (middle cartoon), is the unwound version of the supercoiled plasmid. Nicks come about through enzymatic activity, irradiation, chemical scission, or accidental breakages from time to time.

The relaxed configuration is also 'forced' during normal cellular function, so that transcription of a gene or many genes is made possible. It also comes about during the replication cycle of the DNA molecule, as the plasmid or chromosome is replicated progressively.

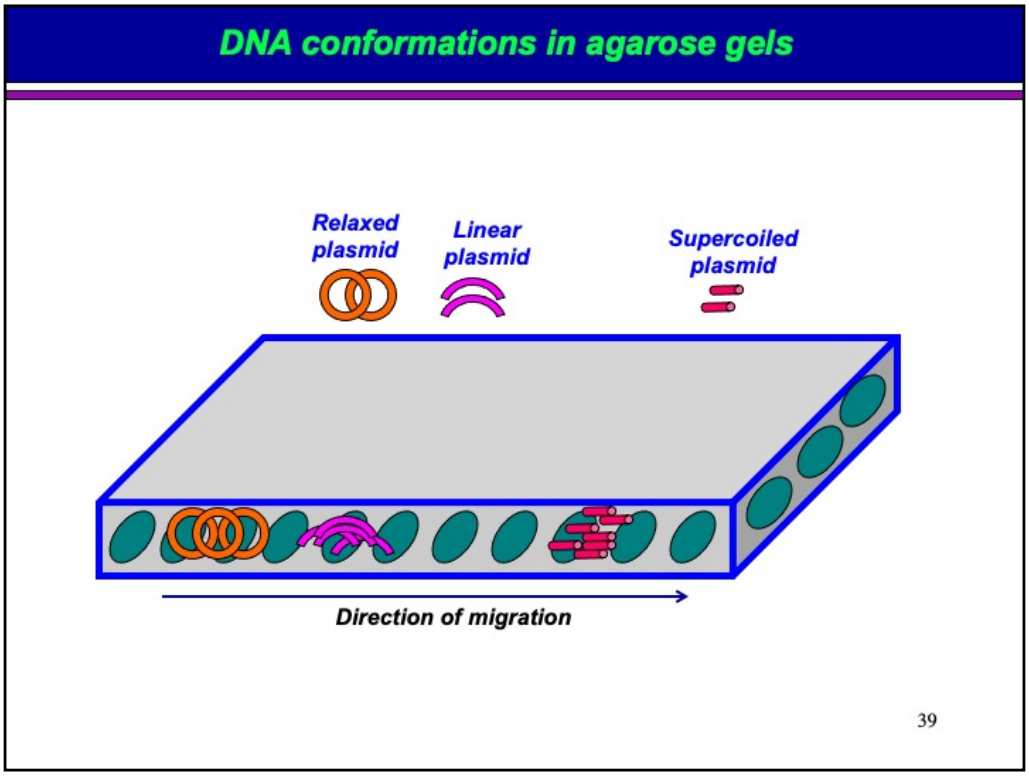
Plasmids and chromosomes within cells are interconverted into supercoiled or relaxed configurations by a family of enzymes known as gyrases or topoisomerases.

The linear configuration of the plasmid shown here is artificial (bottom cartoon), and is not found in cells. It is generated by a double nick across both strands, either by physical breakage or enzyme activity.



How the first supercoil is introduced into a relaxed, open circular plasmid. The scissors in this diagram is substituted in cells by a gyrase or topoisomerase.

There are two types of these 'winding/unwinding' enzymes, type I and type II. One type introduces supercoils, the other removes them. In this way, cells are able to fully supercoil or relax DNA molecules (plasmids or chromosomes) on demand, which is chiefly for transcription, DNA repair, or genome replication.



The previous two slides were a forerunner to this slide, which depicts the way different conformations (spatial geometric shapes) of the same DNA molecule migrate in an agarose gel.

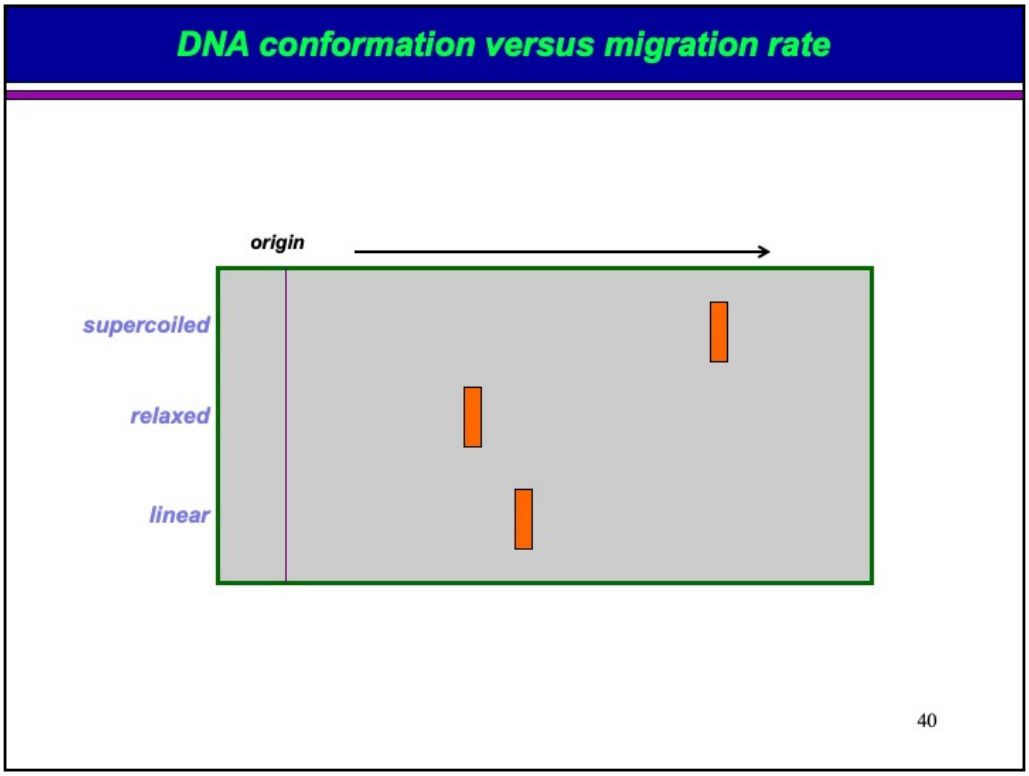
Note, these three DNA types (supercoiled plasmid, relaxed plasmid, linear plasmid) are **exactly the same molecular mass size**, but migrate differently due to shape/conformation differences alone; and this is caused by resistance to movement through the pores of the gel.

The floppier relaxed molecules (orange), show the greatest resistance to movement through the restrictive pores (shown as even-sized green circles in the gel cross-section) and migrate slowest of the three types.

The drawing of the supercoiled plasmids (red) makes them look 'linear', but they are actually 'squashed' into that apparent shape by the supercoiling effect on their conformation. They migrate fastest as they are rigid, narrow, and migrate end-on-end.

The linear version (pink) also migrates end-on-end, but they are more flexible than the supercoiled molecules and bump into the sides of the pores, and this slows them down, but not as much as the relaxed molecules.

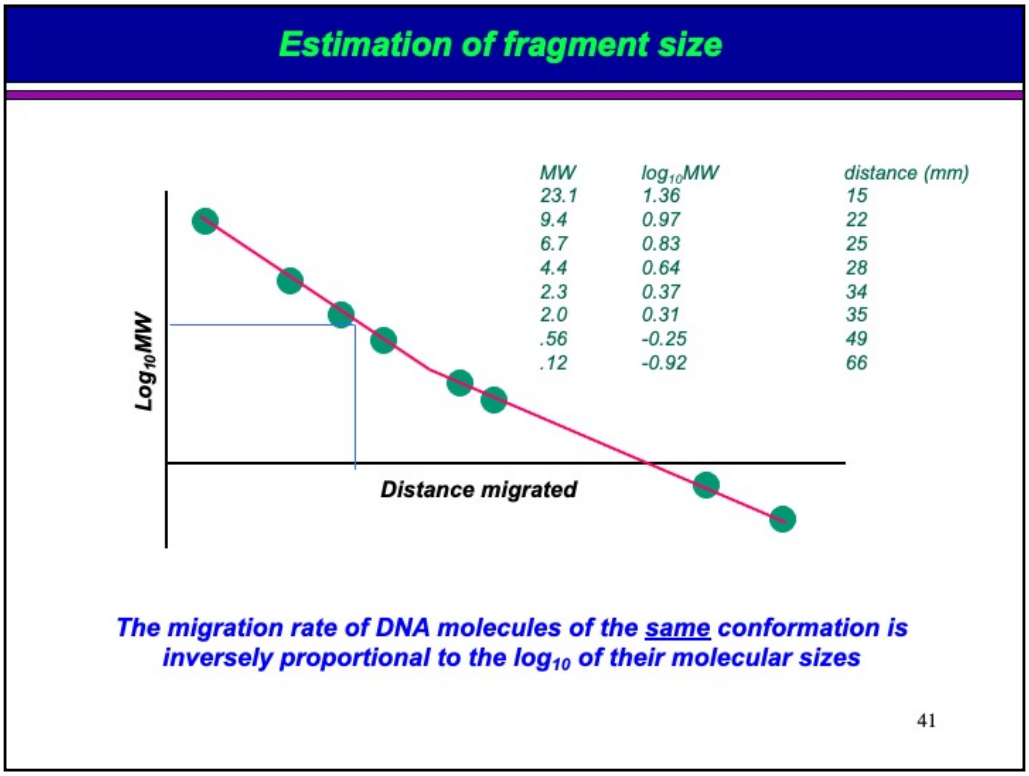
Agarose gel electrophoresis that displays profiles of three plasmid conformations and the estimation of the size of the linear version, is Experiment 1 in the laboratory course.



This slide is a follow up (top down view) of the previous one and emphasizes the point about conformation versus migration rate. Three conformations of the **same** DNA plasmid are shown and labelled back at the origin at left.

Only two of these (relaxed, supercoiled) are seen in bacterial cell lysate preparations; and there is much more supercoiled (~95%) than relaxed (5%). The linear version is provided artificially, in this case to signpost the three different mobilities.

You will see all three of these bands in Experiment 1, but all in the same lane, and the relaxed band could be very feint and possibly not even visible.



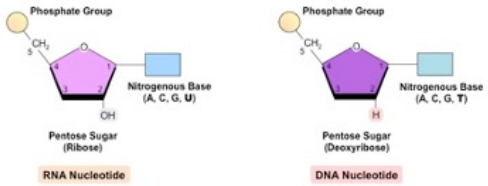
A standard curve drawn by plotting the distances migrated from the origin to the DNA bands in an agarose gel (abscissa or x-axis) versus the $\log_{10}$ of molecular masses (ordinate or y-axis). Alternatively, this plot could have been drawn on semi-log paper by plotting the MW values directly on the log scale and the distances on the linear scale.

The size of the unknown DNA molecule (from the previous slide) is estimated from the standard curve by drawing a vertical line from its distance migrated (bottom axis) up to the standard line and across to the ordinate axis from which its molecular mass is read directly or calculated from the log10 value.

An important point about standard curves is that because the pore sizes in the gel are the same throughout, as determined by the agarose concentration used, a broad range of standards will migrate in two (or more) modes as shown above. That is, the larger DNA bands run a little slower than the smaller bands – due to the resistance caused by the pore size on the larger molecules; and this is reflected in the standard line with two slightly different gradients.

Take home message: **The standard 'curve' or line is always drawn through every point and NOT as a line of best fit**.

MCQ L1



This drawing of RNA and DNA nucleotides was taken from on Online website. What is wrong with it?

(A) The RNA is an incorrect structure.

(B) Nothing is wrong with it.

(C) Both are missing the 3' hydroxyl group on the sugar.

(D) Only RNA has an hydroxyl group on the 2' sugar.

(E) The DNA is an incorrect structure.

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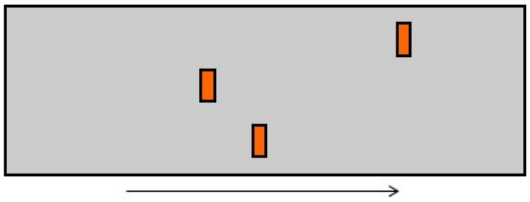
A simple representation of the two types of nucleotides in RNA and DNA.

MCQ L1



This gel profile depicts different conformations of a plasmid (not in any order as: linear, relaxed, supercoiled) migrating from left to right. Which one would I use with linear DNA standards (not shown here) to estimate the size of the plasmid?

A. The top band because it is migrating the fastest.

B. The bottom band because it is migrating in between the other two bands.

C. The bottom band because it is the linear conformation of the plasmid.

D. The middle band because it is migrating the slowest.

E. The top band because it is the linear conformation of the plasmid.

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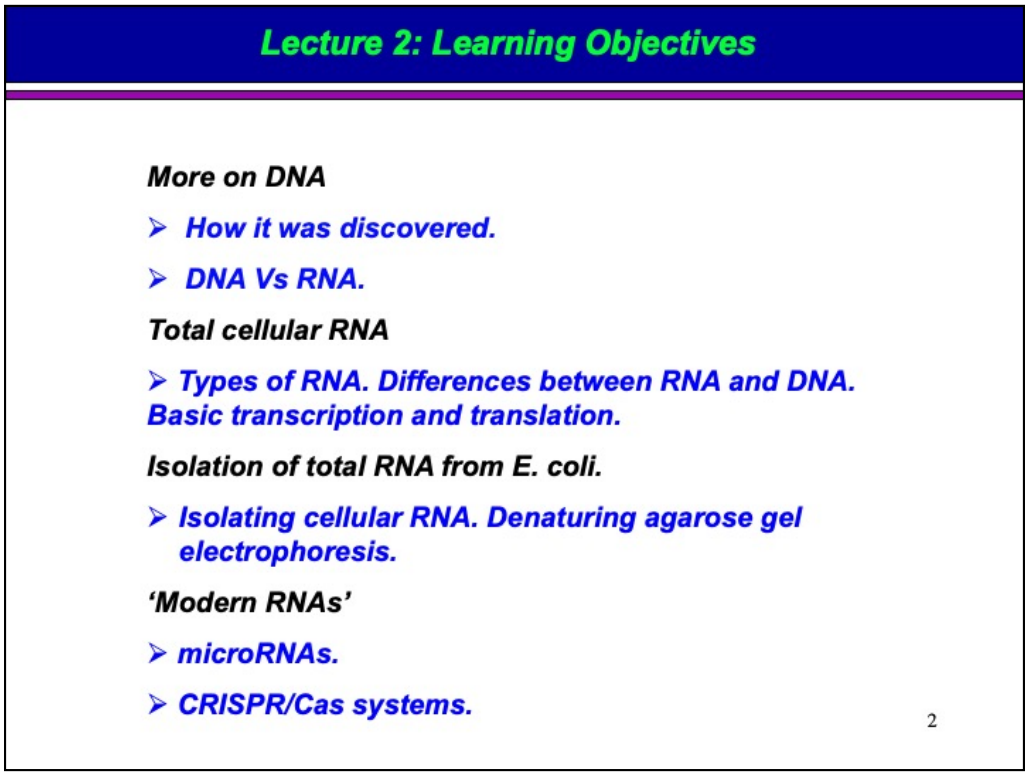
MOLECULAR BIOLOGY 1
Lecture 2

Deoxyribonucleic Acid (DNA)

Ribonucleic Acid (RNA)

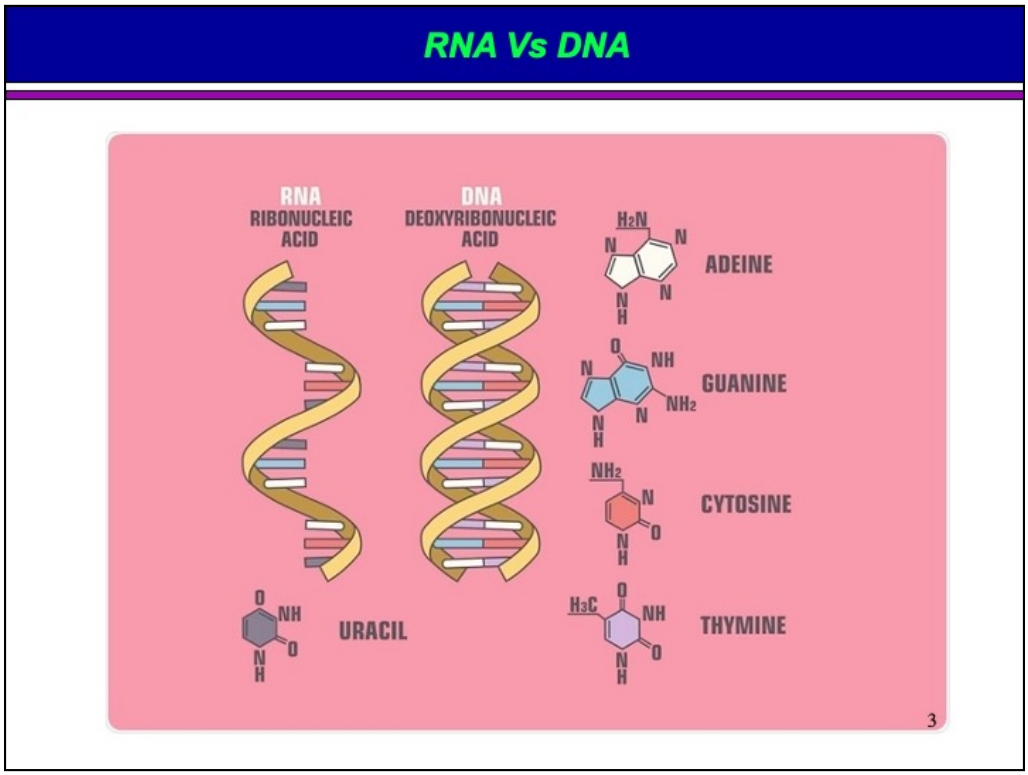
1



This lecture will introduce us to the world of RNA, the 'poor relative' of its more well-known sibling, DNA. In this first section, we will 'join the dots' of comparing DNA and RNA as well as looking briefly at transcription and translation, which involves all three types of RNA and DNA genes.

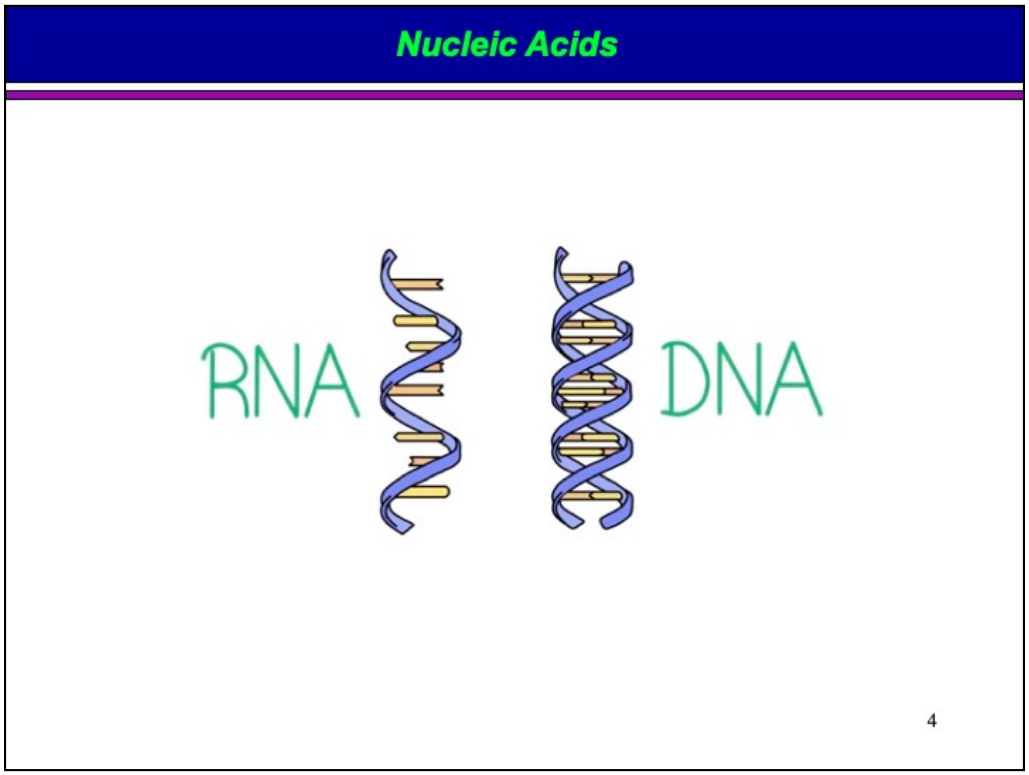
In the secod section, we will focus on a method for isolating total RNA from an *Escherichia coli* culture and resolving the total RNA into its component parts in denaturing gels. Only RNA is run in denaturing gels, which maintains all types of RNA as single-stranded linear molecules so that the separate molecular types migrate in the gel according to their lengths. The denaturing conditions are brought about by the use of chaotropic agents that destroy H-bonds. Expt. 2 covers this section.

In the last section, we will see an overview of 'modern RNAs', which in truth are of ancient origin, but it is only in the past few years that we have discovered and come to realize the importance and potential of these types of RNAs, which in a nutshell, work as switches and regulators (microRNAs) or gene editors (CrispR-caspase).



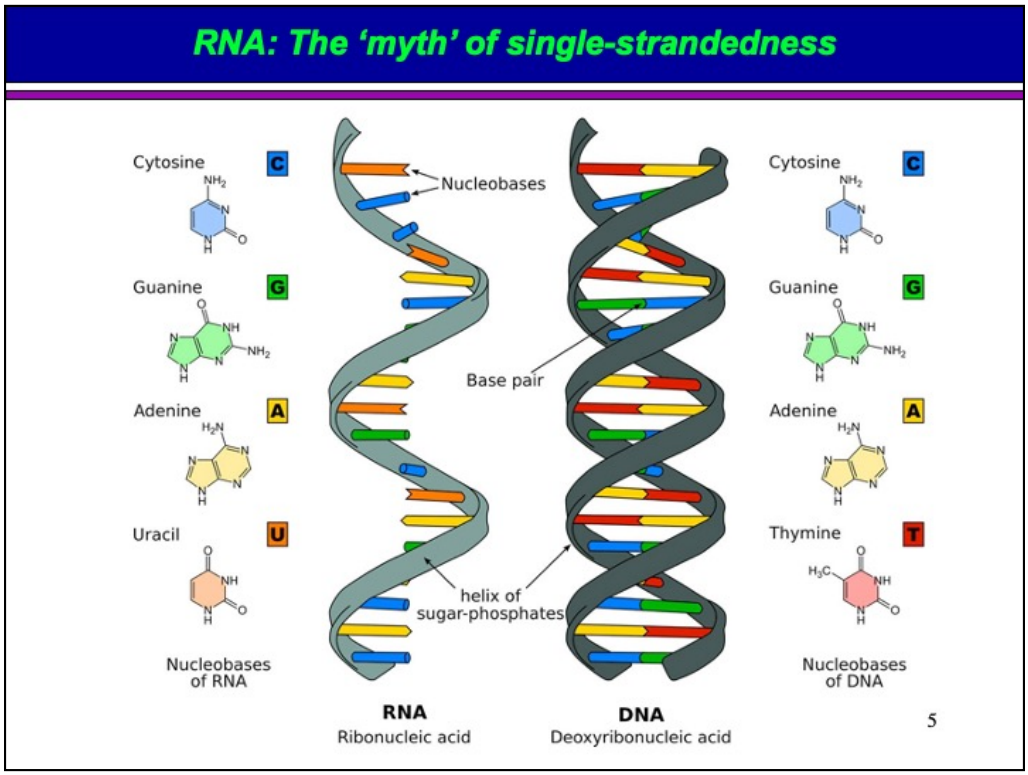
There are three major differences between RNA and DNA.

1. RNA occurs as single strands, whereas DNA is double-stranded).
2. RNA contains Uracil in place of Thymine (in DNA). The other three bases are common to both nucleic acids.
3. RNA contains the sugar Ribose and DNA has 2-Deoxyribose (neither show here).



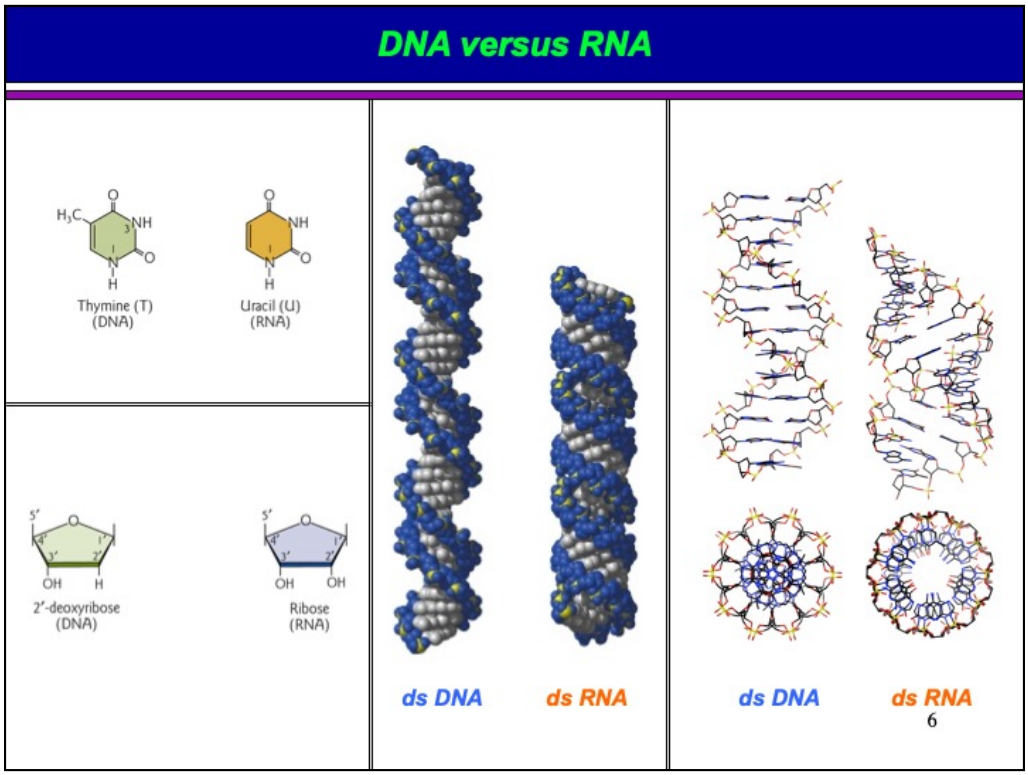
A cartoon of basic structures of the nucleic acids DNA and RNA.

We know from this simple diagram, and from the basic building blocks of both structures that these two types of nucleic acids work 'lock-in-step' in all cells to convert the genetic material, via transcription and translation, into 'workhorse' proteins, lipids, and polysaccharides, for cellular structure and function. However, we now know that there are many types of RNAs (see following slides), and that the single-stranded RNA helix depicted in this cartoon is a rare member of the RNA family, which is found as messenger RNA and as some single strand viral chrososomes.



RNA chromosomes may be single-stranded (ss) or double-stranded (ds), and these genomes reside in many types of viruses. RNA viruses use bacteria, fungi, plants, and animal as hosts. Human diseases caused by RNA viruses include Ebola, the common cold, influenza, polio, and measles. Retroviruses are a separate class of RNA viruses that use DNA intermediates in their replication cycles, such as the HIV-I and HIV-2 human AIDS viruses.

In terms of the generic structure of RNAs, this slide depicts ss-RNA (left) and ds-DNA (right), and their respective base or nucleotide compositions, but ds-RNA is nearly identical to ds-DNA, except for two differences (see next slide).

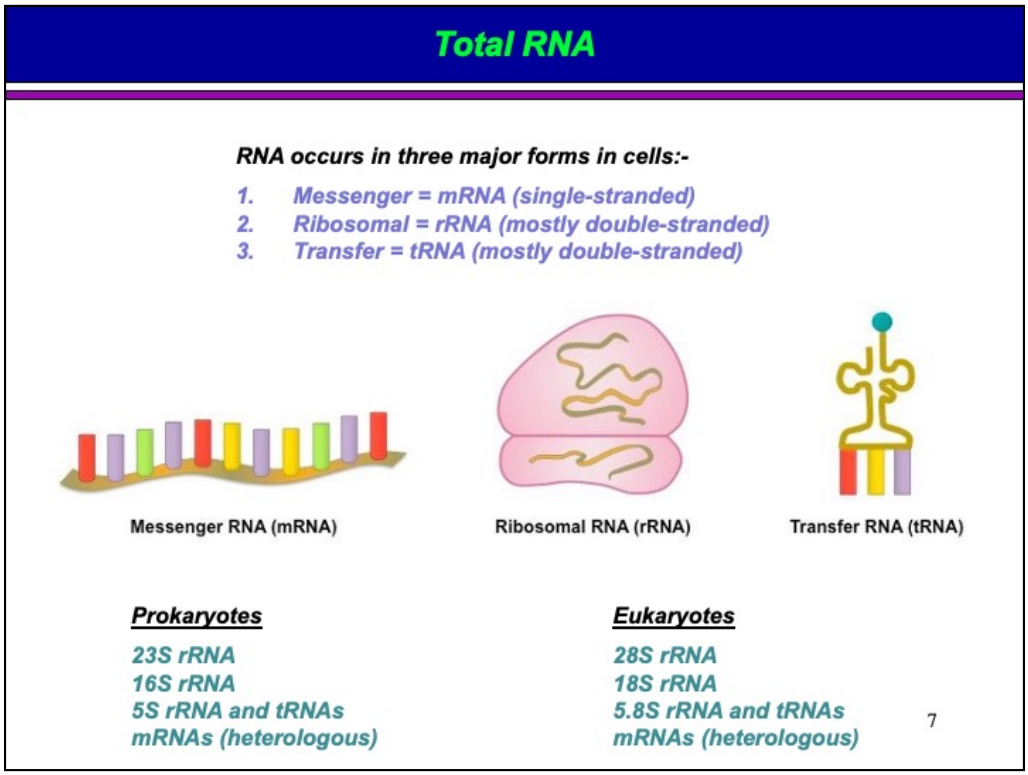


Differences between DNA and RNA. One is the RNA nucleotide base uracil substituting for the DNA base thymine (top left).

The second difference is that in DNA, the sugar-phosphate backbone utilises the sugar 2'-deoxyribose (hence the name deoxyribonucleic acid) (bottom left), whereas RNA utilises ribose (hence the name ribonucleic acid) (bottom right).

Double-stranded nucleic acids assume different double-stranded configurations (middle and right panels). The striking difference between ds-DNA and ds-RNA here is that ds-RNA is about 20% wider and shorter than ds-DNA. Why?

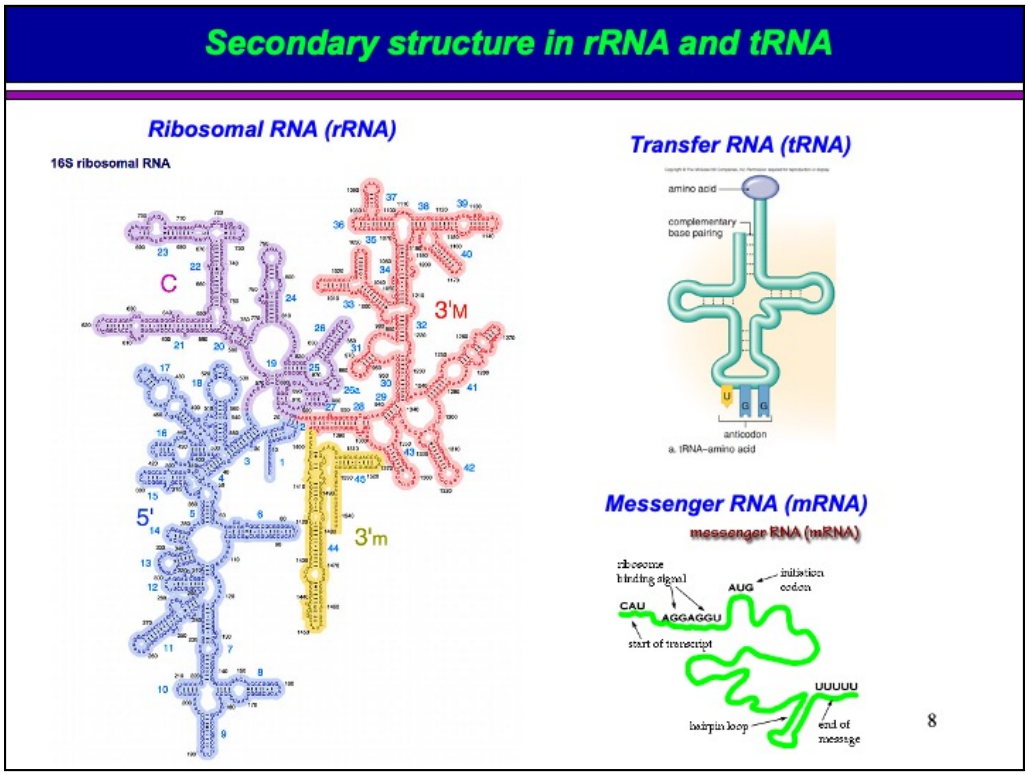
The middle panel depicts space-filled models of DNA and RNA, while the right panel has ball and stick side and top views of both types of structures.



This is a summary slide of the respective types of RNAs (rRNA; tRNA, mRNA) in prokaryotic and eukaryotic cells. Apart from small differences in the sizes of the three types of rRNAs, the RNA populations are essentially similar in both dominions (parokarya and eukarya); and this points to conservation during evolution for the task of translation, which is the transfer of the gene codes into proteins and other cellular molecules.

The 'S' values refer to Svedberg units, which were given to the sedimentation rates of the RNA types in high-speed centrifugation. The simple numeric values (23S, 16S, 5S etc) are indicative of their respective sizes.

On the other hand, mRNA is '**heterologous**', which is a word coined in this case to mean of different lengths. Since each mRNA transcript represents a different gene or genomic DNA sequence (with associated ORF), the family of mRNAs from any species will be of different lengths, with no single mRNA size dominating, hence the family of mRNAs from a given cell or species is said to be heterologous.

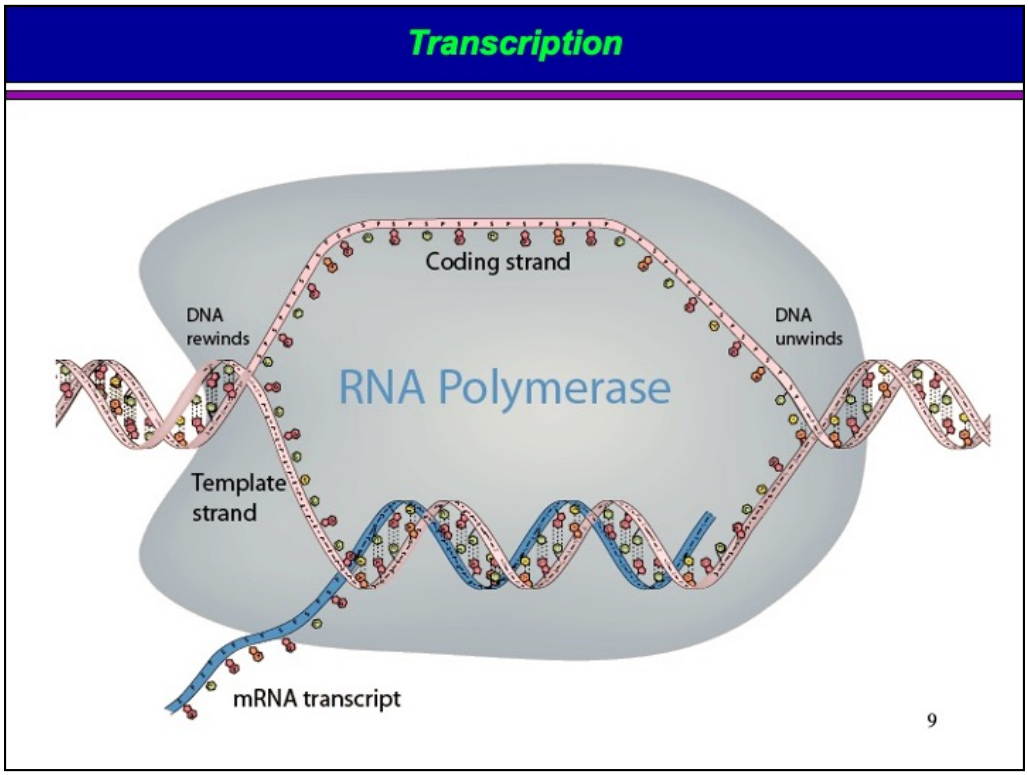


Schematic or cartoon drawings of ribosomal RNA (rRNA; left) and transfer RNA (tRNA; top right), which both have extensive regions of double-strandedness, which is achieved - as for DNA – by H-bonded base pairing (G with C; A with U). This pairing gives the RNAs distinctive conformational shapes that are essential for their functions. You can see from these diagrams that secondary structure predominates in these RNAs, with single-stranded very short regions occurring only at junctions or turns in the scaffolds. rRNAs combine with many proteins that together are assembled into ribosomes (riboproteins).

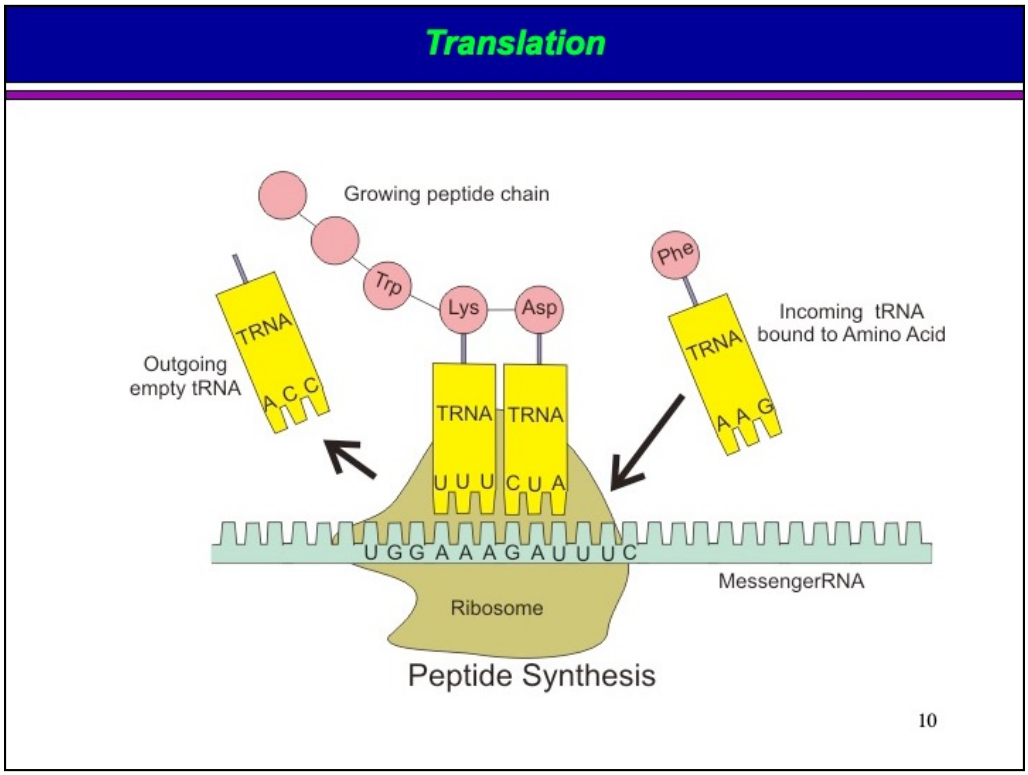
tRNAs actually have a roughly L-shape, but are drawn as a '**clover leaf**' with arms at either end (amino acid arm and **anticodon loop**) that bind, respectively, to a specific amino acid, and to the specific triplet codon in the **Open Reading Frame** (**ORF**) of a **messenger RNA (mRNA) transcript.** mRNAs are of various lengths, determined by the lengths of the encoding genes.

An mRNA is depicted at bottom right with distinctive features: ribosome binding signal; start codon; occasional short double-stranded hairpin loops; and an end of message or termination signal, for detachment from the ribosomal cleft at the end of translation. Note that – unlike rRNAs and tRNAs that have extensive double-strandedness, mRNAs are single-stranded.

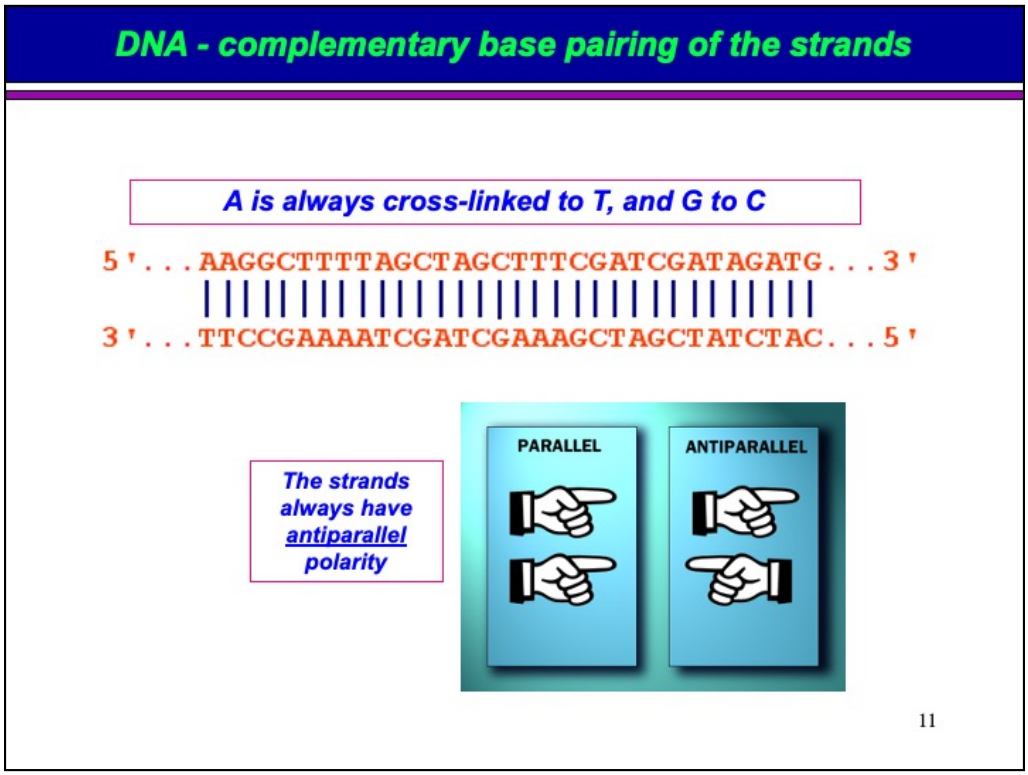
In Experiment 2, total RNA will be isolated from a bacterial cell culture and subjected to electrophoresis in a **denaturing agarose gel**. The notion of using such a gel is to observe the three types of denatured RNA (with all H-bonds broken) migrating according to their respective single-stranded lengths or masses, rather than their structural conformations, as illustrated above.



Transcription is the process whereby RNA Polymerase (grey blob above) synthesizes a complementary copy of one of the DNA strands (template strand at bottom) that is actually an exact copy of the coding strand (top), or non-template strand.



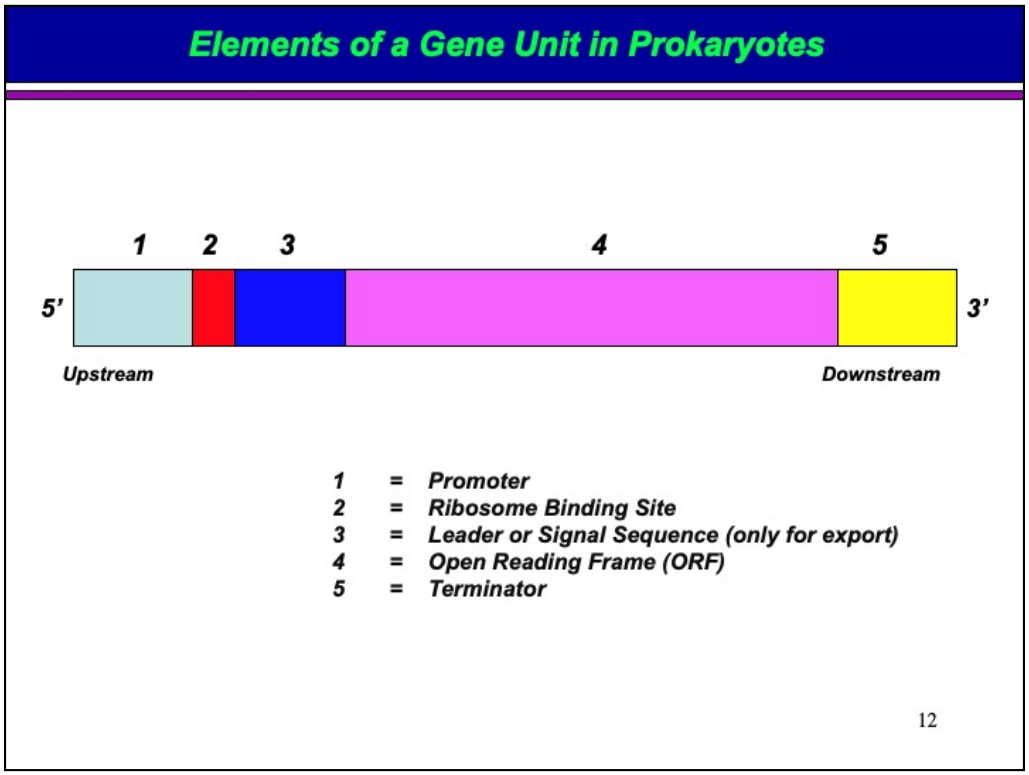
Translation is the process of 'reading' the mRNA codons using tRNAs with complementary anticodons (bottom) that attach amino acids to the growing polypeptide chain (top).



A simple representation of base-pairing showing the precise (Chargaff Rules) complementarities of A with T and G with C in any double-stranded DNA molecule. In RNA, it is A with U (substituting for 'T') and G with C.

Note the antiparallel (5' to 3' and 3' to 5') arrangement of the two strands. The 5' ends carry free phosphate groups on the 5' carbons of deoxyribose, and 3' ends carry free hydroxyl groups on the 3' carbons of deoxyribose.

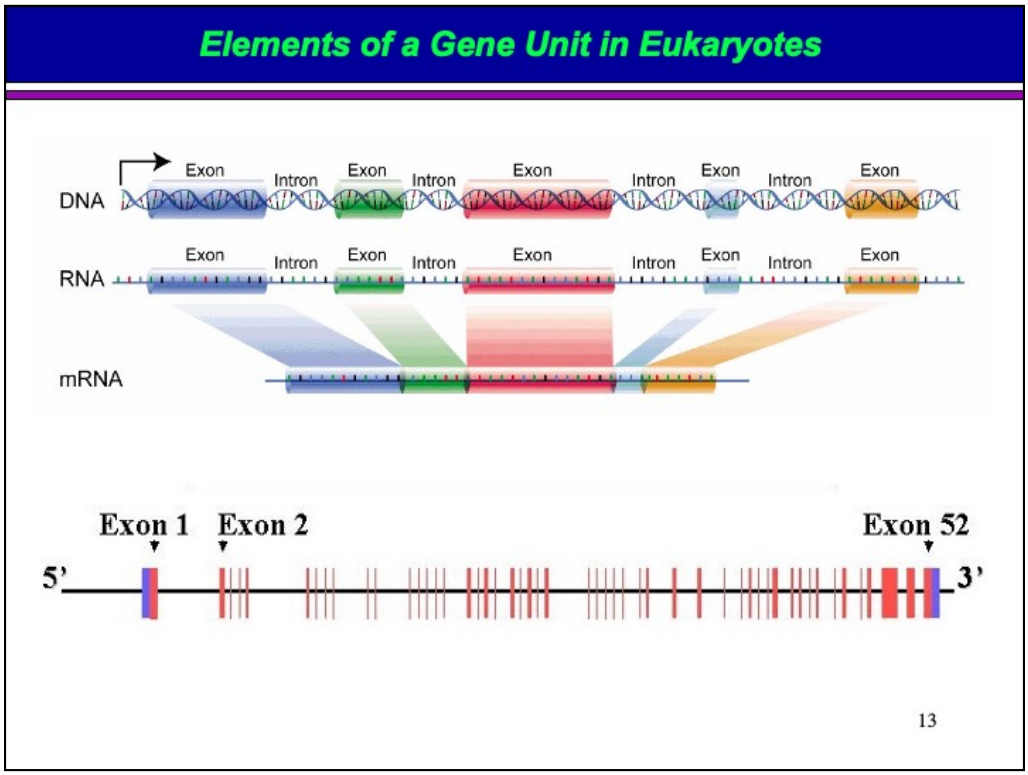
If the sequence of one strand is known, then the complementary sequence of the partner strand is easy to deduce, using the Chargaff Rules.



We think of a gene in terms of the translated protein product, but all genes have defined elements that control the expression of the gene, namely:

1.The initiation of transcription, by the binding of RNA polymerase to the Promoter site;
2.The ribosome binding sequence that directs the RNA transcript to the protein translation site on the ribosome;
3.A signal or leader sequence, which is only a feature of proteins that need to be exported out of the cytoplasm;
4.The Open Reading Frame – the set of triplet nucleotide codons that specify the sequence of amino acids for the translated polypeptide.
5.A termination signal that enables the RNA polymerase to disengage from the mRNA transcript.

In the illustration above, note the polarity of the sequence, which is 5' to 3' left to right. The second non-coding strand (not shown) will have the opposite polarity. Note also how the region to the left of the 5'-end of the gene sequence is referred to as 'upstream'; and the region to the right of the 3'-end is 'downstream'.



In eukaryotes, the genes in chromosomes are arranged differently to those in prokaryotes. The Open Reading Frame (ORF) is non-contiguous (see upper diagram) and consists of a series of alternating exons and introns. The exons are sections of the ORF and comprise some of the amino acid codons. The introns, also known as intervening sequences, are segments of innocuous DNA that interrupt and act as bridges between successive exons.

The primary or pre-mRNA transcript made from these genes comprises the entire exon-intron DNA gene sequence. This pre-mRNA is then spliced within the nucleus to yield a mature mRNA transcript (bottom of top diagram). This mRNA is now similar to those from prokaryotes, in that it now comprises only the conjoined, uninterrupted exons with the amino acid codons now contiguous in the manner of a normal ORF.

Note, all mRNAs (from whichever source) do not contain the promoter sequence, which is back upstream (in the 5-prime direction) from the ORF. It is not transcribed because its only purpose is in providing a 'landing spot' for RNA polymerase to begin transcribing the gene.

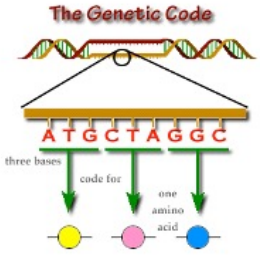
The bottom diagram is an example of a collagen gene with extensive exon-intron structure. We shall see in a later lecture how molecular cloning gets around the problem of eukaryotic gene structure and enables a copy of the gene to be cloned and expressed in bacteria such as *E. coli*.

DNA Alphabet – The Genetic Code

20 Amino Acids

↓

Type of Code	Number of Bases in a Codon	Number of Codons
Singlet	1	4
Doublet	2	16
Triplet	3	64



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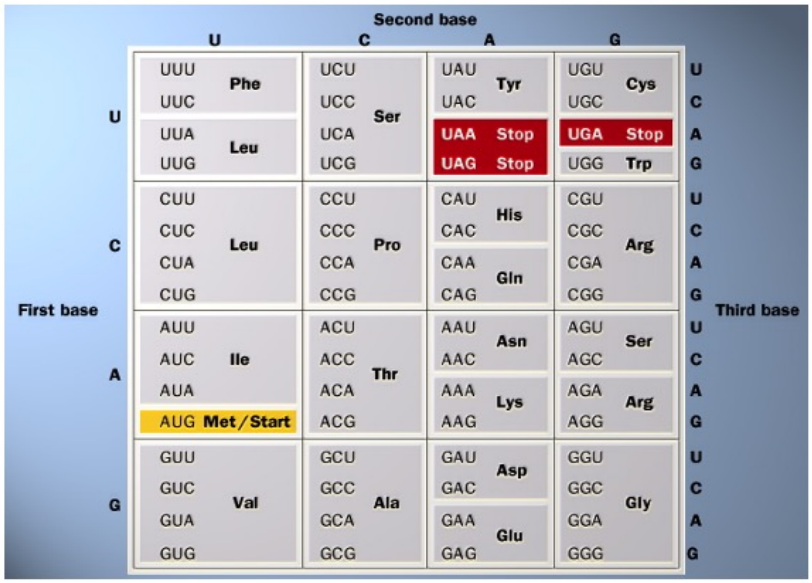
The next problem was sorting out how an alphabet of 4 letters (A, C, G, T) encodes the 20 amino acids? Looking at the table above, it turns out that 64 code words are needed from an 'alphabet' of four letters to encode 20 amino acids.

One of these code words, the 'start signal' begins all the sequences that code for amino acid chains. Three of these code words act as 'stop signals' that indicate that the message is over. All the other sequences code for specific amino acids. Some amino acids are only coded for by a single 'word', while some others are coded for by up to six 'words'. The genetic code is redundant (see next slide).

The complete Genetic Code was not worked out until 1966 by Marshal Nirenberg and his team. However, it began back in the late 1950s when Francis Crick designed experiments that showed that the genetic code was non-overlapping and had to have three letters in frame. Nirenberg then performed an experiment with a poly-U RNA and found that only phenylalanine was produced in a cell free protein synthesising system. The number of phenylalainines was equal to 1/3rd of the number of uracils. Eventually all 64 codons were ‘cracked’.

The Genetic Code



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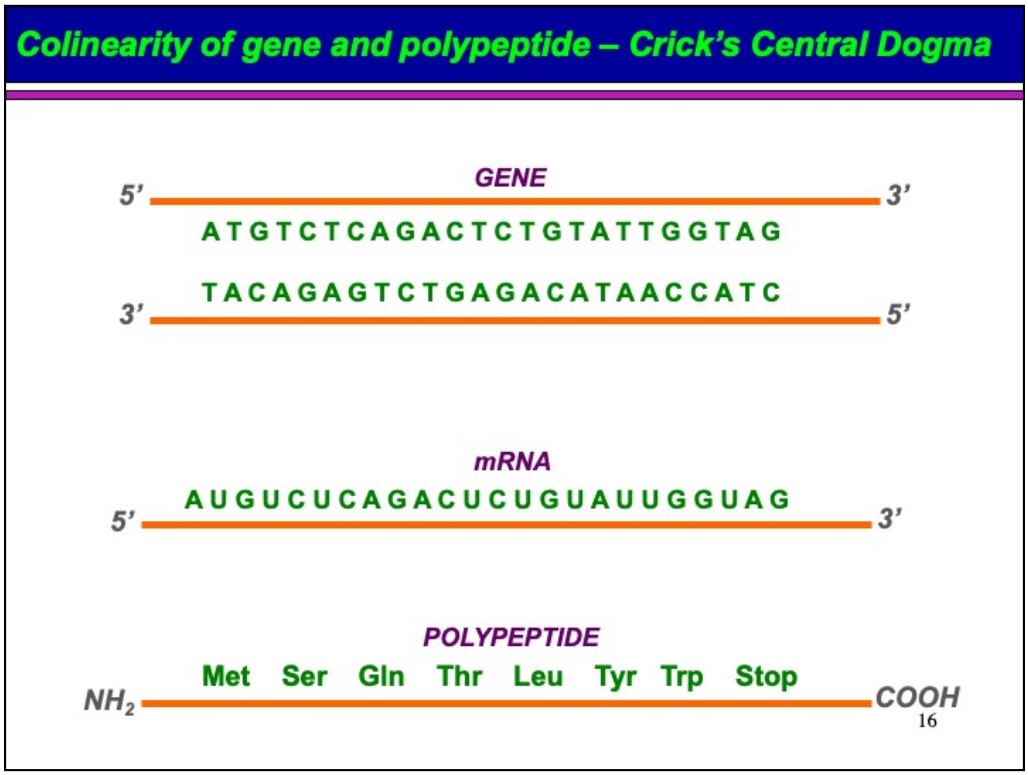
The 'Genetic Code' of triplet codons of which there are a total of 64 (4x4x4). As you can see, there are three stop or nonsense codons (UAA; UAG; UGA) and a single start codon (Met). Tryptophan also has a single codon (UGG). However, the remaining 59 available codons are distributed among the other 18 amino acids, resulting in a range of codons for each amino acid (from two to six), making the Code 'degenerate', meaning, in this case, more than one amino acid among the extra codon choices.

How this Code is used and "adjusted" in molecular cloning will be made apparent on the following slide and in more detail in a later lecture. Note that this table of codons is written for the mRNA transcript, but in the original gene ORFs, the DNA triplet codons have 'T' in place of 'U'.



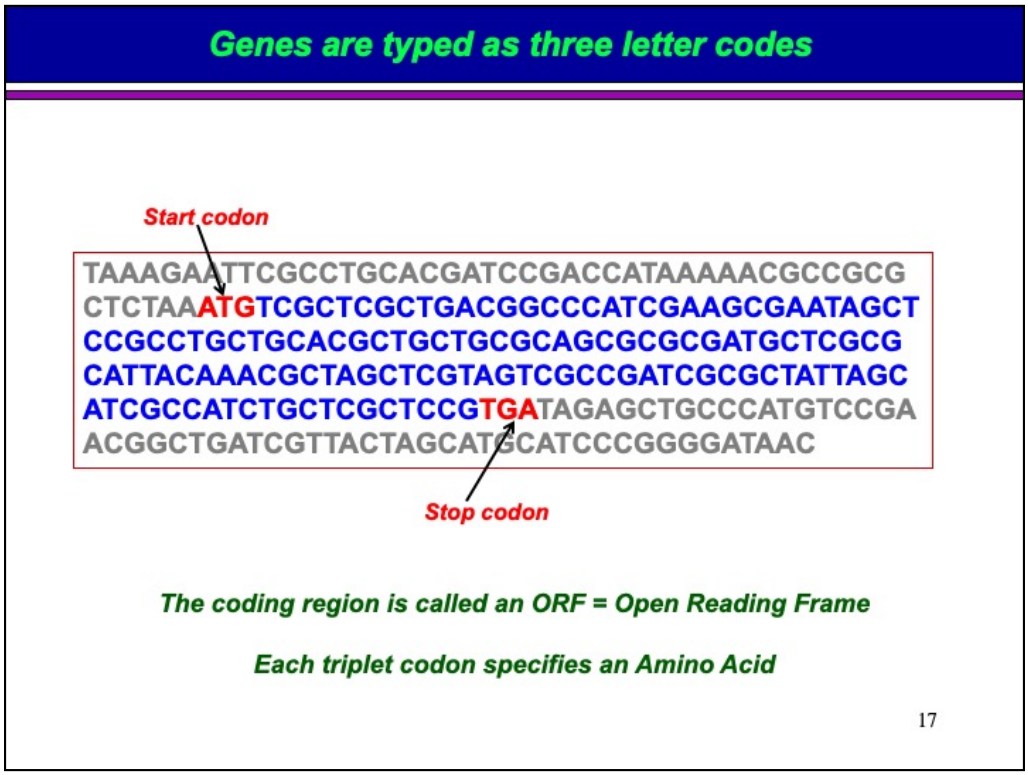
Francis Crick figured early on after the DNA structure was devised that the gene 'code' had to be read in one direction, from the DNA to an intermediate RNA molecule and then into protein. mRNA as the genetic 'messenger' was solved by Sydney Brenner and a host of others, including our ubiquitous friend Francis Crick, in 1960. Crick also came up with the concept of 20 adaptor molecules (tRNAs as it turned out) to join the mRNA-specified amino acids into polypeptides.

A gene (top) is transcribed (by RNA polymerase) into mRNA (middle) and this transcript is copied into a polypeptide (bottom). Within the Open Reading Frame (ORF) of a gene, each triplet codon specifies an amino acid in the translated polypeptide and the two – triplet codons and amino acids – are collinear or contiguous, that is, they are aligned precisely. ['Collinear' and especially 'contiguous' are terrible words I know, but we have to put up with the English language and its seemingly infinite terminology.]

The messenger RNA transcript triplets are identical to the triplets of the gene's ORF coding strand (upper sequence), but complementary to the triplets of the template strand (lower sequence). The template strand is the "go between" strand that enables the gene sequence to be copied exactly by the principle of complementarity. Note that the gene sequence and mRNA strands also have the same polarity (5' - 3'), which is opposite to the template strand polarity (3' – 5').

Lastly, notice that in RNA, the base uracil (U) substitutes for thymine (T). Note that each triplet codon specifies an amino acid, but that the redundancy of the Genetic Code for most amino acids means that changes in the third position, and sometimes the second position may specify the same amino acid. Having said that, problems may arise when the DNA (in

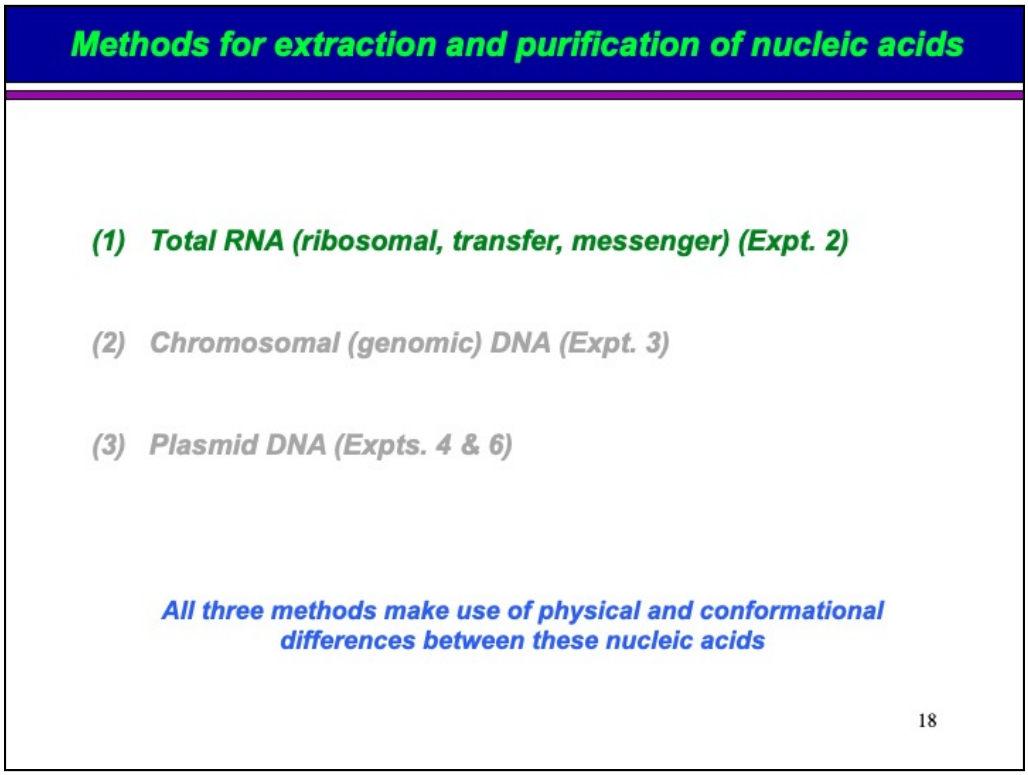
particular, the gene sequence or ORF, but sometimes other crucial regions such as the promoter or other regulatory sequences) is mutated, spontaneously, or by errors in replication, or by the action of radiation or toxins, or free radicals.



A DNA sequence begins and ends with non-coding nucleotide bases (in light grey type above). An Open Reading Frame (ORF) for a gene encodes a polypeptide, which is bookended by Start and Stop codons (in red), between which are the triplet codons (3 bases each, shown in blue) for the linear sequence (string) of amino acids that will make up the polypeptide or protein for this gene.

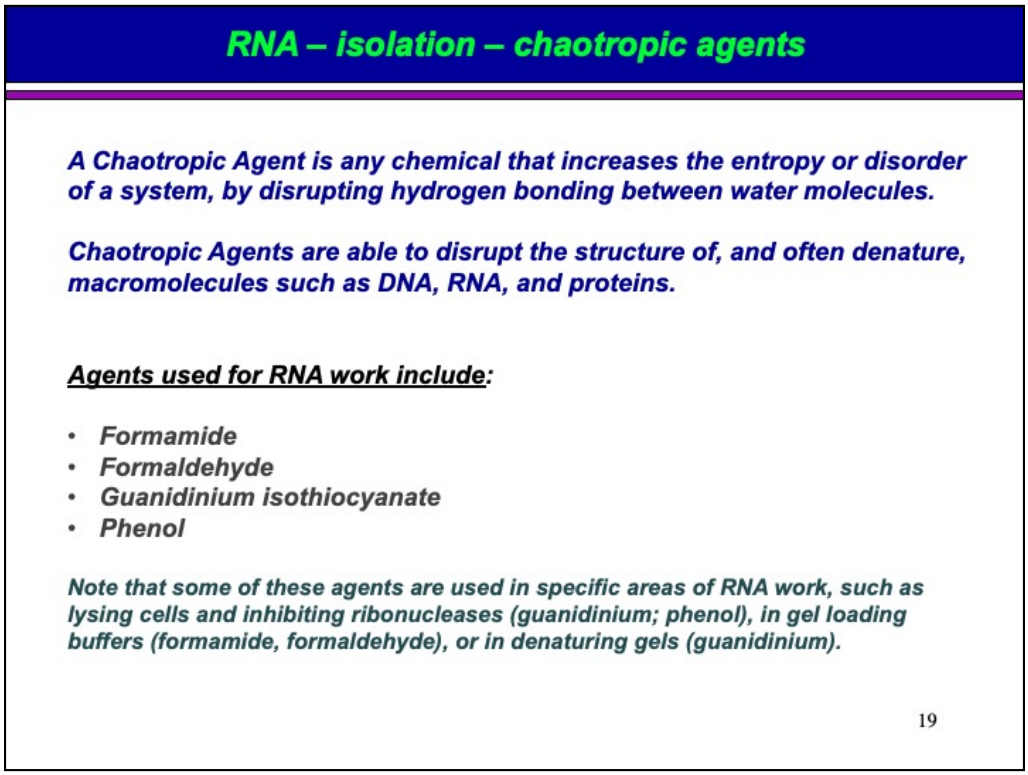
One thing to note, but which is not shown here, is that any changes in the ORF will almost always be catastrophic in terms of the expression of the protein. Deleting or inserting one or two bases will alter the ORF and the protein will not be expressed (translated). One or two bases added or subtracted anywhere will ruin (change) the reading frame (the correct types of amino acids in the correct order) downstream from the addition or deletion.

Insert or delete three bases and one complete amino acid will be removed or added. While not always deleterious, this type of alteration will almost always lead to the protein not folding correctly, or not working as it should.



In the experimental section of this subject, you will isolate each of these three types of nucleic acids from *E. coli* bacterial cells. The isolation and purification method for each nucleic acid differs on the basis of physical and conformational features of the DNA or RNA. The remainder of this lecture will cover RNA only. The other two will be dealt with in upcoming lectures.

The overarching themes for isolating these nucleic acids broaches the fundamental methods (technical, scientific) of the isolation processes themselves; and the broader downstream goals of studying the structure and function of these macromolecules and why they are crucial in the way cells, organs, and organisms operate, or fail to operate.

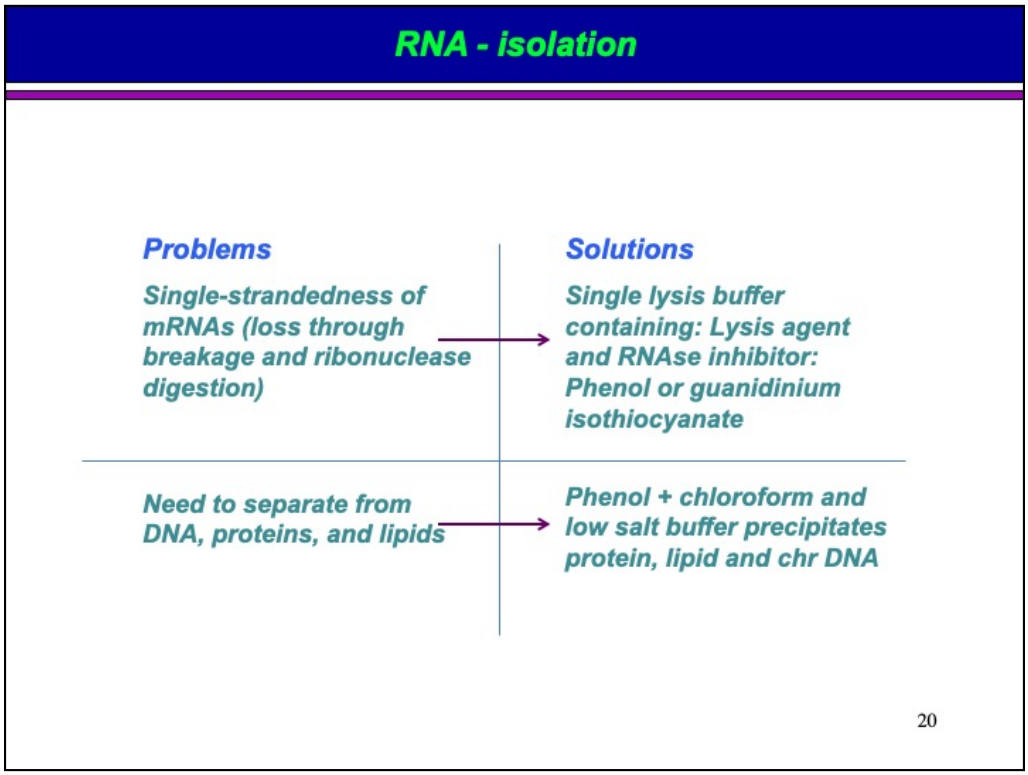


Chaotropic Agents (CAs) exert their effect by disrupting hydrogen bonds in water and between and within water and macromolecules. The list above is far from exhaustive, but these are the major CAs used in molecular biology for the purpose in assisting in the isolation of RNA or DNA. They have roles as denaturants/inhibitors and as H-bond breakers.

Once H-bonds are broken, such as in the case of RNA that is heated, the bonds are prevented from reforming if a CA is present. In a similar sense, guanidinium isothiocyanate is typically added to RNA agarose gels, hence the name 'denaturing gel'. This CA maintains the single stranded conformation of RNA molecules in denaturing gels – it prevents H-bonds from reforming within rRNAs and tRNAs. In this way, we are able to measure the profile of RNA molecules and their respective lengths or masses, in a denaturing agarose gel.

Phenol and guanidinium isothiocyanate are also powerful inhibitors of ribonucleases (RNases) and are common components of lysing buffers used in RNA isolations from cells.

Urea serves the same role as formamide, formaldehyde, and guanidinium isothiocyanate, but only in special polyacrylamide DNA sequencing gels, in which it is necessary to run many different sized single-stranded DNA fragments (see DNA sequencing lecture, further on in the semester).



Isolating total RNA from a bacterial culture will constitute Experiment 2 and will be dealt with in more detail therein, and in slides further on in this lecture. This slide gives only a summary of some of the technical considerations in the isolation procedure.

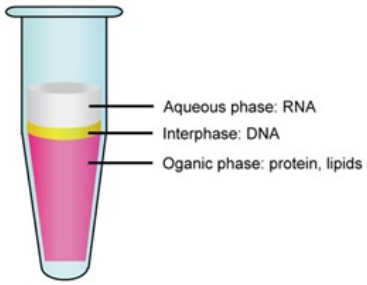
Above the line, the two main features to note are the single-stranded nature of mRNAs that are very easily fragmented by ribonuclease enzymatic digestion. This will greatly affect the yield of full length heterologous mRNAs. Is can be a major problem because many mRNAs are present in cells and organisms in varied abundance, from very small to very large copies. The solution to this problem is to use a cell lysis buffer that contains a ribonuclease inhibitor (typically, guanidinium isothiocyanate, or phenol). On the other hand, physical breakage during subsequent isolation is not really a problem as the number of handling operations are minor.

The second consideration (shown below the line) is the need to separate the RNA fraction from all other macromolecules. This is done in a simple two-step isolation process – phenol-chloroform removes membrane fragments, proteins, and lipids, and the low salt buffer causes most to all of the DNA (which is larger than RNAs and is still connected to membrane fragments) to be removed with the proteins/lipids.

RNA - isolation

RNA isolation methods use a lysis/RNase inhibitor 'cocktail' containing buffered phenol/chloroform, and/or guanidinium isothiocyanate. A commercial preparation containing these ingredients plus a phenol red indicator is called TRIzol™

In Expt 2, we will use a phenol/chloroform lysing/inhibitor mix.



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This slide indicates the main commercial RNA lysing buffer, TRizol, shown in the bottle at left. Its formulation comprises buffered phenol, chloroform, guanidium isothiocyanate, and a phenol red indicator.

The tube at right shows the result of a Trizol lysis of cells and subsequent high-speed centrifugation, which separates the layers into the components indicated.

As effective as this method is, you will see (in a couple of slides further on, and in Expt 2) that we can achieve the same separation/purification shown in the tube above using a simpler lysing cocktail of just phenol/chloroform and a separation device.

RNA isolation and purification kit

22

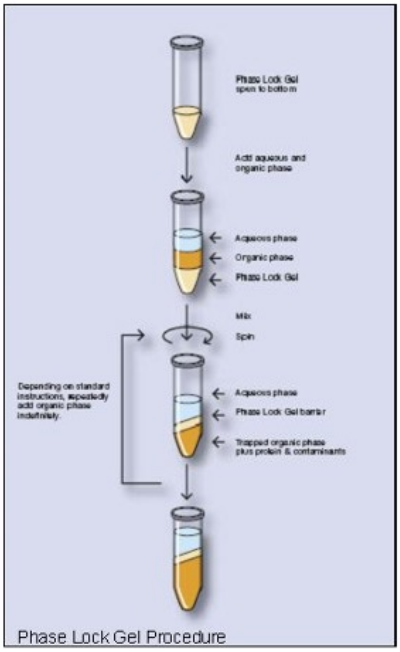
Commercial kits for the isolation of nucleic acids are now commonplace. This kit has all the requirements for isolating and purifying total RNA from a bacterial culture. Unlike the very simple and effective scheme shown on the previous and next slides, kits like these give a sense of quality control and ensure good yields of mRNA that may be used in downstream experiments.

The contents of this Ambion kit are: Lysis/binding buffer (with guanidinium isothacyanate), spin cartridge columns (with hydroxylated silica RNA binding disc and collection tubes), column wash buffers I and II, and RNase-free water for elution. Commercial kits are far from essential lab tools and basic home lab recipe methods work nearly or as well in most cases.

The total RNA extraction method we will use in Experiment 2 is simple, effective, and inexpensive!

RNA - isolation

A simple one-step isolation procedure for recovering total RNA from a bacterial culture



Phase Lock Gel Procedure

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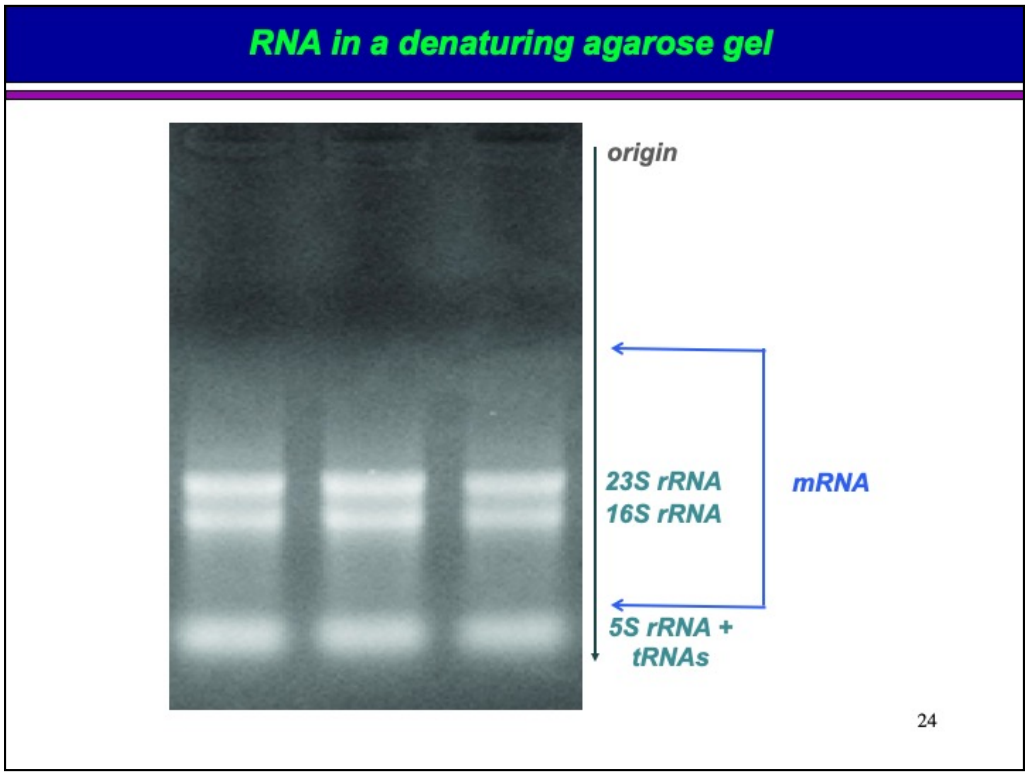
This slide summarises the essence of Experiment 2. The phenol-chloroform used to lyse the bacterial culture and liberate nucleic acids, also inactivates host ribonucleases (in the absence of guanidinium isothiocyanate).

The phenol/chloroform lyses the cells and precipitates membrane protein/lipid and chromosomal DNA into an insoluble mass. This lysate is added to a Phase-Lock Gel (PLG) tube that contains an intermediate density "gel" (pale lemon in the diagram above). Under high-speed centrifugation, the phenol/chloroform organic phase moves below and under the gel, which has an intermediate density, lower than phenol/chloroform and higher than aqueous, and the insoluble protein/lipid/DNA white flocculent pellet is centrifuged just into the gel at the interface of the aqueous and organic phases.

The RNA is recovered in the buffered upper aqueous layer of the PLG tube.

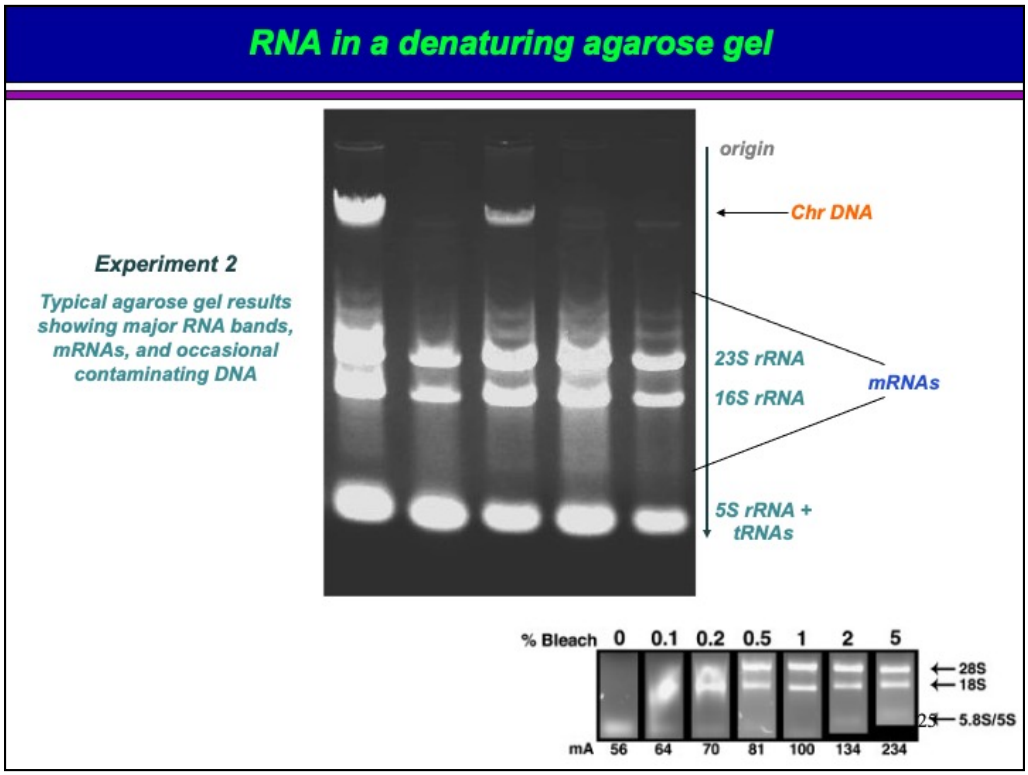
The low salt buffer aqueous phase (containing soluble total RNA) is of lower density than the gel and remains above it, as indicated in the final profile above. The so-called 'phase lock' gel thus achieves the effect of separating (locking) the two phases from one another, making the whole but short process very efficient and simple.

This single slide summarises the essence of Expt 2. All that is required after this extraction protocol is to add dye mix to the aqueous total RNA sample in the top layer of the centrifuged tube, then denature and run the RNA in a denaturing agarose gel.



This profile of total RNA in a denaturing agarose gel depicts the three major RNA bands and a 'smear' of heterologous messenger RNA molecules, within the bracketed arrowed section of the gel.

All of the RNA is migrating as single-stranded molecules, including the previously extensively double-stranded ribosomal and transfer RNAs, because the RNA samples were denatured in a special dye mix buffer (+formamide/formaldehyde) at 90^{o}C before being added to the gel; and this state is maintained in the 'denaturing' gel by the presence of guanidinium isothiocyanate. Hence, the distinctive bright sharply focussed bands at the respective positions of the RNA molecules of similar masses.

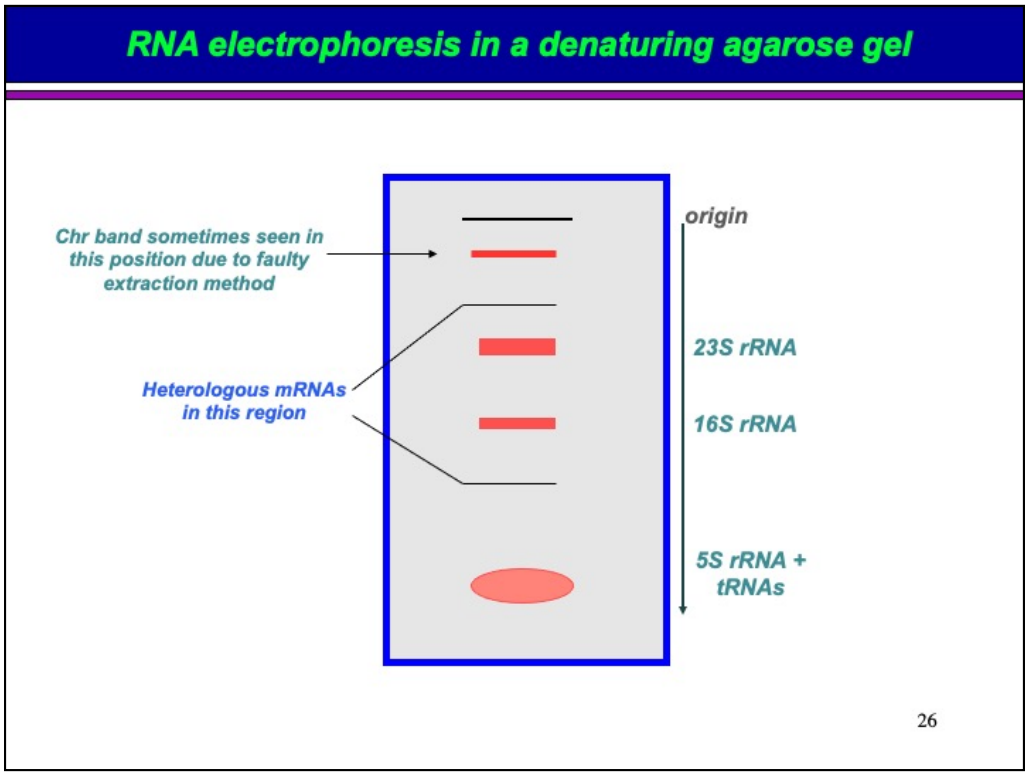


The main central gel panel depicts a profile of total RNA in a denaturing agarose gel. The three sharply focused, brightly Gel-Red stained major RNA bands and a 'smear' of heterologous messenger RNA molecules are depicted. Note, all of the RNA is migrating as single-stranded molecules, including the previously extensively double-stranded ribosomal and transfer RNAs.

This gel depicts five different student preparations from a previous year in MB1. Notice that two of them have contaminating genomic DNA, which would have occurred during the phase-lock gel step by pipetting too much of the upper soluble RNA layer and drawing off some of the precipitated genomic DNA from the interface.

As mentioned on the previous slide, the tight, discrete banding pattern of the main RNAs is indicative of a good isolation procedure, with minimal degradation of the RNAs. Although not visible in the gel (except for a smearing pattern), you can rely on the mRNAs being mostly preserved as full length transcripts on the basis of the rRNA bands being largely intact, as evidenced by the discretely tight banding pattern.

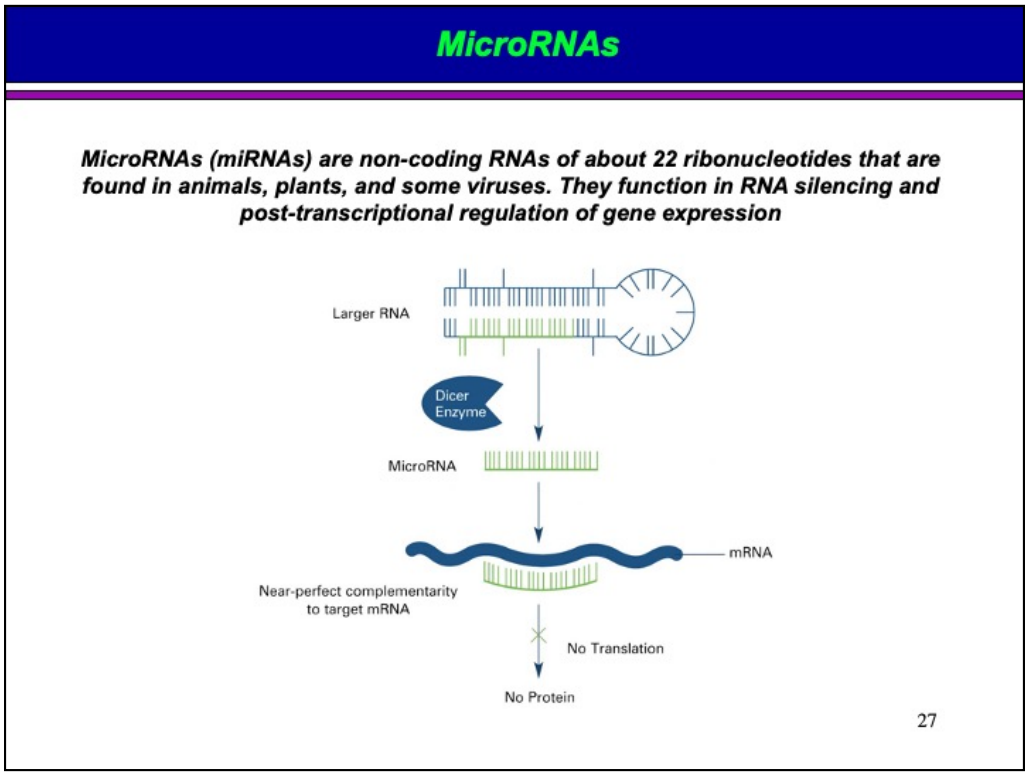
The gel diagram profile at bottom left, depicts total RNA in a denaturing agarose gel, with the three major eukaryotic RNA bands shown as signposted on the right. The variable conditions are percentages of common household bleach (top) and current in mA (bottom), Bleach is able to both denature double-stranded RNA (as you can see from left to right) and inhibit ribonucleases. This was taken from a recent publication that highlighted the nature of RNA isolations in 'quirky' substances such as using household bleach instead of proprietary chaotropic agents.



A cartoon of agarose gel electrophoresis of total RNA extracted from a bacterial culture. Denaturation is achieved by heating the RNA sample in a special buffer cocktail (containing formaldehyde, formamide, EDTA, sucrose, and bromophenol blue running dye) and heating at 90^{o}C for a couple of minutes to break all hydrogen bonding and render all rRNA and tRNA molecules single-stranded (note, the 'cocktail' is a combination of denaturing agents and buffer dye mix). This, together with the guanidinium isothiocyanate in the denaturing gel) ensures that the RNAs will migrate according to their respective molecular sizes, generating the profile seen above.

The heterologous mRNAs run over a wide tract of the lane due to their different sizes (within the bracketed region above), and unless the preparation is a large one, individual mRNAs cannot be seen, but a 'smear' of mRNAs is always visible.

RNA agarose gels are 'denaturing' gels because they are cast with the inclusion of 5 mM guanidinium isothiocyanate in the gel mixture. This compound is required to **maintain** the (already) denatured state of the RNAs during the gel run; and has the added benefit of being a ribonuclease inhibitor. Note, the integrity (sharpness) of the main rRNA bands is indicative of a good preparation (with minimal degradation of RNA). Most of the points made in these notes were stressed on the previous slide as well and therefore could be important to note!

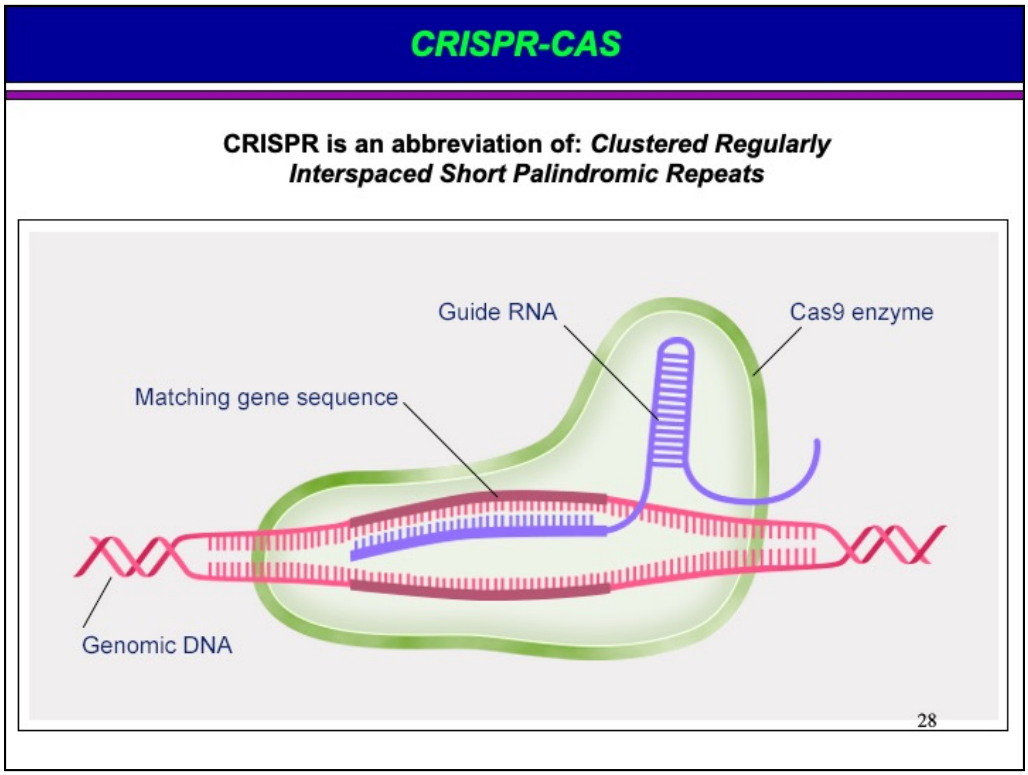


MicroRNAs (miRNAs) are short non-coding RNAs of about 22 ribonucleotides that are found in animals, plants, and some viruses. MiRNAs are encoded either within introns, exons, or in intergenic regions. Some microRNAs are in close proximity to each other and are transcribed from a common promoter.

They function in RNA silencing and post-transcriptional regulation of gene expression. Each miRNA may target numerous mRNAs and each mRNA is targeted by many miRNAs. It has been estimated that a single miRNA may target hundreds of mRNAs, but to a varying level of potency. They were only discovered in the 1990s to early 2000s and their roles in regulation are still being evaluated.

A pri-miRNA (primary miRNA) is cleaved within the nucleus by Drosha, a dsRNA-dependent ribonuclease, then processed further when exported by another ribonuclease called Dicer into the mature miRNA (shown in the middle of the diagram above). This miRNA can undergo two potential fates, namely as near-perfect, or partial complementation to mRNAs – in complex with the RISC protein (RNA-Induced Silencing Complex; not shown in this figure) – that results in either mRNA degradation or translational repression, both being forms of post-transcriptional gene regulation (also known as gene silencing).

The human genome may encode up to 1000 miRNAs and they appear to target up to 60% of human genes. miRNAs are involved in a wide range of biological processes such as, cell cycle control, apoptosis, stem cell differentiation, hematopoiesis, hypoxia, cardiac and skeletal muscle development, neurogenesis, insulin secretion, cholesterol metabolism, aging, and immune responses. Changes in expression levels of microRNAs have been implicated in a number of diseases including a broad range of cancers, heart and neurological diseases.

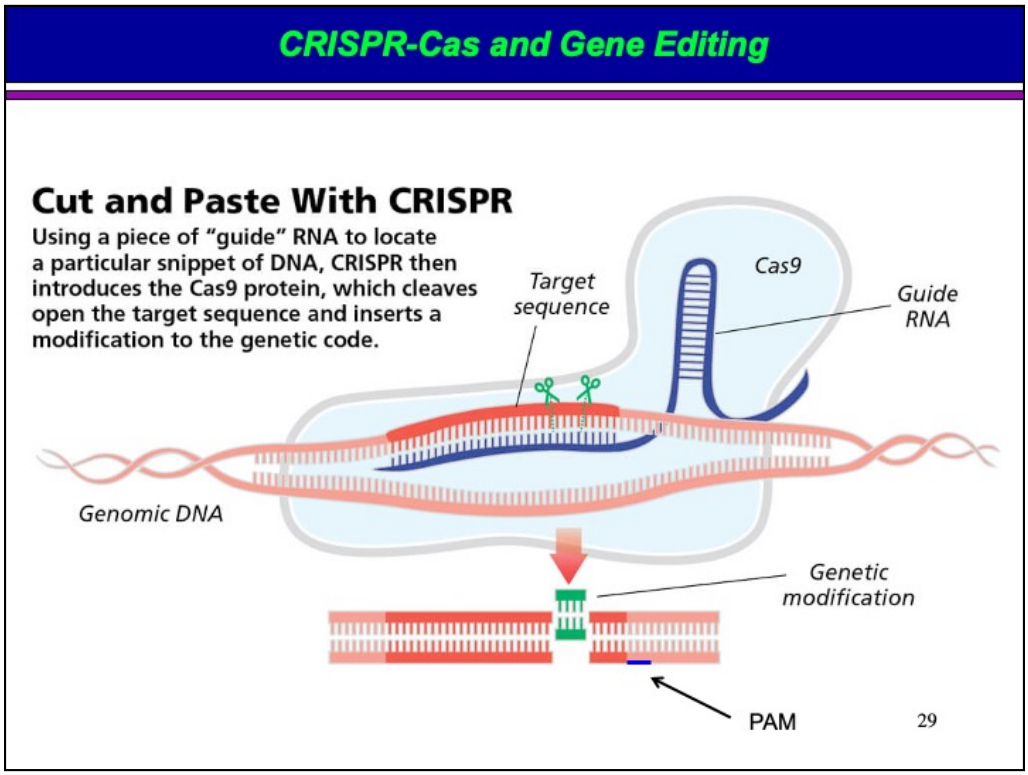


CRISPR is an abbreviation of: *Clustered Regularly Interspaced Short Palindromic Repeats.* CRISPRs are found in approximately 50% of bacterial genomes; and they contain short snippets of viral **RNAs** that have infected the bacteria at some past event.

Acting like a barcode, short crispr RNAs (crRNAs) that match part of the foreign viral sequences are loaded onto members of the CRISPR family of caspase (CAS) enzymes, which perform the function of sentries that roam the bacteria and monitor for foreign code. If these CRISPR-CAS riboprotein complexes encounter genetic material that matches its crRNA, they chop up that DNA to render it harmless to the bacteria. As such, the CRISPR system is a form of prokaryotic immune or defence mechanism to protect the bacterial genome from invasive viral RNA or DNA that may be incorporated into the bacterial chromosome and may inactive vital genes and cause the death of the host bacterium.

The crRNA target regions contain short palindromic repeats separated by short spacer sequences, hence CRISPR = Clustered Regularly Interspaced Short Palindromic Repeats. In this slide, the Cas enzyme helps guide a crRNA to a target region in the human genome that has a defective gene/sequence.

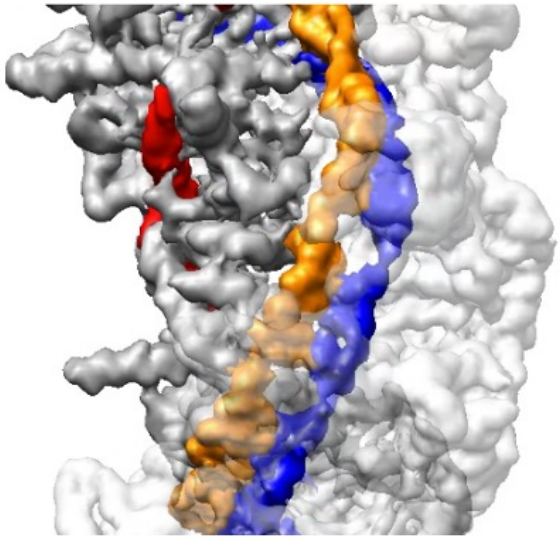
In 2005, three independent research groups showed that some CRISPR spacers are fragments of DNA gathered from viruses that previously tried to attack the bacterial cell. The source of the spacers was a sign that the CRISPR/*cas* system could have a role in adaptive immunity in bacteria. All three studies proposing this idea were initially rejected by high-profile journals, but eventually appeared in other lower echelon journals.



How the CRISPR-Cas editing system works. CRISPR was originally designed to 'knock-out' or silence defective target genes in cell types and organisms, but it has now been applied in human medicine to silence, activate, or repair genes by replacing, inserting, or deleting DNA.

The short RNA sequence (in blue) of approximately 20 ribonucleotides is known as the Guide RNA (gRNA). The associated caspase (usually Cas3 or Cas9) recognises a very short upstream sequence called PAM (Protospacer Adjacent Motif: e.g. NGG etc), shown underlined in the lower diagram. A segment of the genomic DNA is excised and replaced with DNA that will correct a mutation, or silence a gene.

CRISPR-Cas and Gene Editing



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Cryo-electron microscope image of CRISPR-Cas binding to a target site in DNA (orange and red). The DNA is unwound by the Cas3 enzyme (space-filled grey and white atoms) in this example and the DNA (orange) compared to CRISPR RNA (blue) to determine if the target DNA should be cut by the Cas3 enzyme.

MCQ L2

RNA is denatured by heating at 90°C in a special dye mix that breaks all of the H-bonds. The now linearlised RNAs are then subjected to electrophoresis in a denaturing agarose gel containing guanidinium isothiocyanate. The guanidinium isothiocyanate is in the gel because…

A. The RNAs might not have been completely denatured in the heating step.

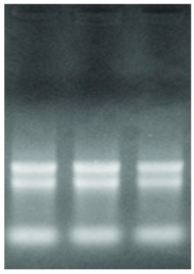
B. RNAs need to be protected from ribonucleases in the gel and buffer.

C. The denatured status of the RNAs needs to be maintained during the gel run.

D. Formamide and formaldehyde will not break H-bonds in the agarose gel.

E. RNAs migrate faster in gels containing guanidinium isothiocyanate.

MCQ L2



The agarose gel profile on the left is of…

A. Chromosomal DNA that has been degraded into fragments of much smaller sizes.

B. Three conformations of the same plasmid, relaxed, linear, supercoiled.

C. The major ribosomal bands, tRNAs and a smear of different sized mRNAs.

D. The three ribosomal RNA types (23S, 16S, 5S) from bacterial cells.

E. Chromosomal DNA (top), plasmid DNA (middle), and total RNA (bottom) bands.

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MOLECULAR BIOLOGY 1
Lecture 3

Structure, characterisation, extraction and purification of genomic (chromosomal) DNA

1

Lecture 3: Learning Objectives

Genomic DNA structure

- ***Structure of genomic DNA. Replication. Transcription. Supercoiling***
- ***The Human Genome.***

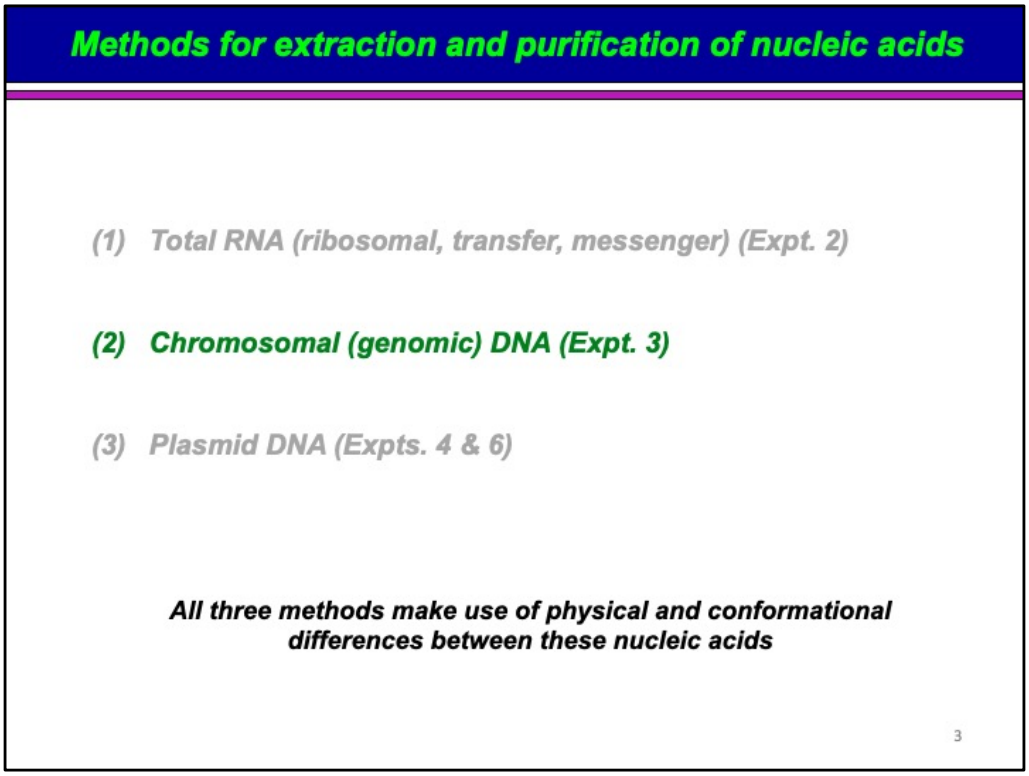
Genomic DNA isolation

- ***The isolation of chromosomal (genomic) DNA from E. coli. Detergents. Salt-alcohol precipitation. Agarose gel electrophoresis of genomic DNA.***

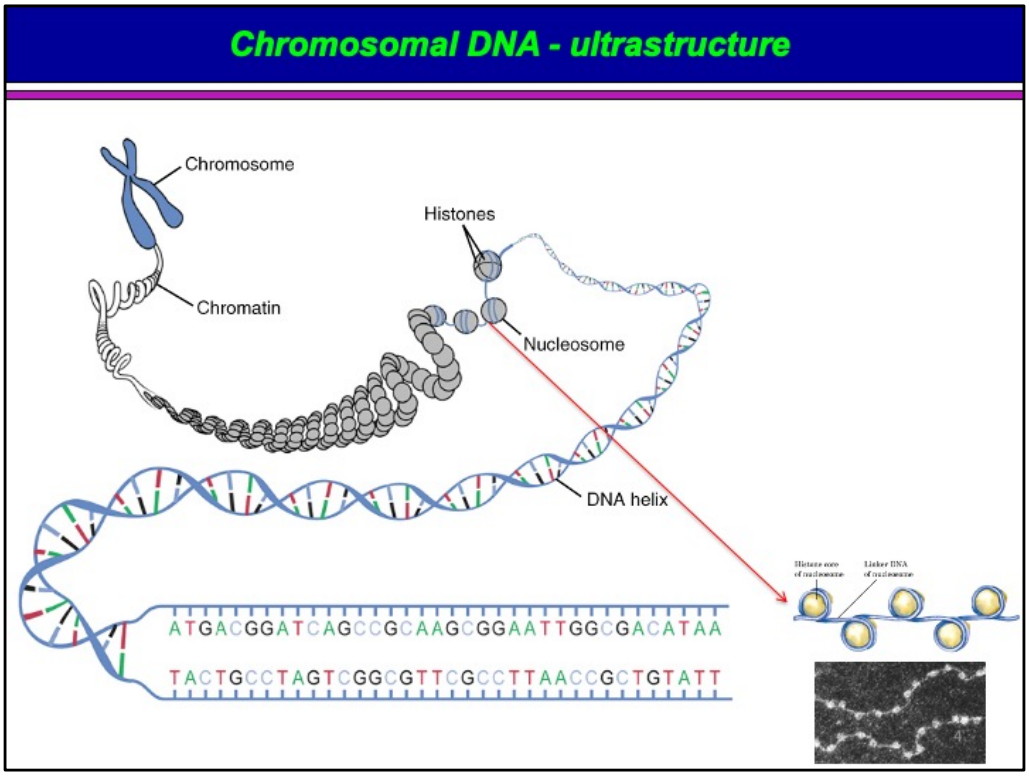
Epigenetics

- ***A brief introduction to non-genetic changes in behaviour.***
- ***Epigenetics***

2



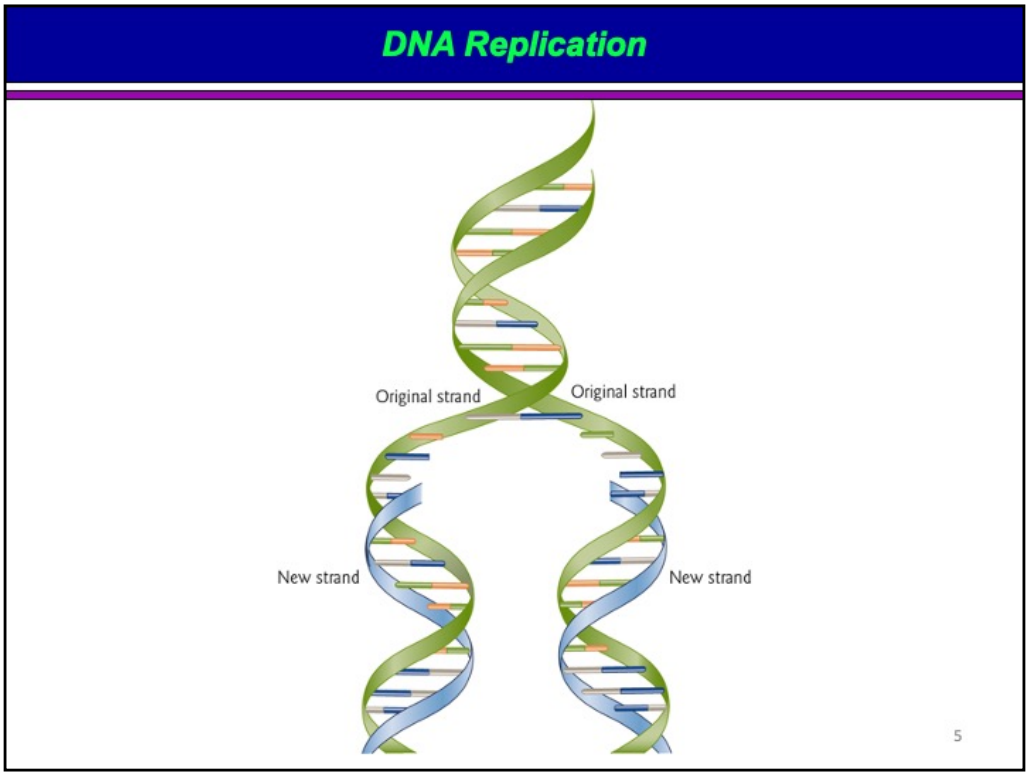
In the experimental section of this subject, you will isolate each of these three types of nucleic acids from *E. coli* bacterial cells. The isolation and purification method for each nucleic acid differs on the basis of physical and conformational features of the DNA or RNA. The remainder of this lecture will cover chromosomal DNA only. The other two are dealt with in other lectures. The overarching themes for isolating these nucleic acids broaches the fundamental methods (technical, scientific) of the isolation processes themselves; and the broader downstream goals of studying the structure and function of these macromolecules and why they are crucial in the way cells, organs, and organisms operate, or fail to operate.



Eukaryotic DNA is packaged as chromosomes into the nuclei of cells. There are 23 pairs of chromosomes, or 46 in all of which 44 are autosomal and 2 constitute the sex chromosome pair. One example of an eukaryotic chromosome is shown here.

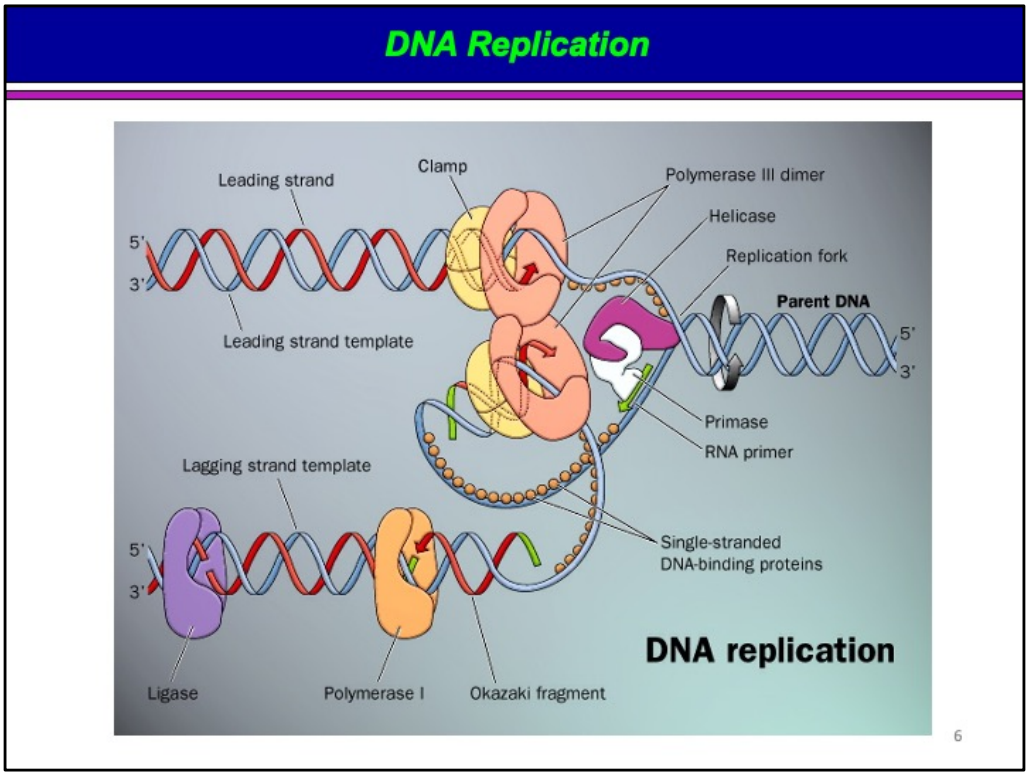
Chromatin is DNA plus histones, positively charged proteins organised into octameric units, around which the DNA is wound 2.5 times constituting a nucleosome. These are supercoiled and packaged into the nucleus, which is part the way of enabling such large molecules to fit into cells (the other crucial requirement is that the DNA be supercoiled). The two smaller diagrams at bottom right are an exploded drawn view of nucleosomes, and an electron micrograph of chromatin as nucleosome 'beads'.

In prokaryotes, the situation is much simpler, with most bacteria having a single naked chromosome, that is without histones, but still supercoiled. The bacterial chromosome resides in the cytoplasm and is attached to the cell membrane.



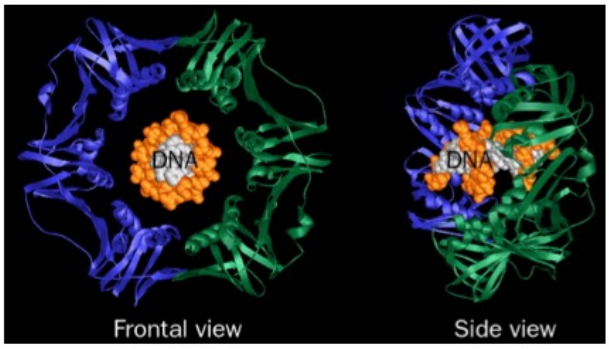
DNA replication is a simple concept, but a complex process involving a 'choreographic performance' by a number of enzymes and enzyme complexes (see the next slide for a detailed summary). In this simple representation, the parental DNA strands (in green) are relaxed and unwound to provide separate template strands for the replication process. In bacteria, the multisubunit enzyme DNA Polymerase III catalyses the addition and linking of nucleotides into new polynucleotide strands (in blue) that are complementary to the template strands against which they are copied.

The net result will be two daughter molecules with one parental strand (green above) and one newly synthesized strand (blue above). Each new 'daughter' double-stranded molecule (green plus blue) is both a 'hybrid' and an exact copy of the original molecule.



A summary slide of the DNA replication process, depicting the different assembly and mechanism on the separated ‘leading’ and ‘lagging’ DNA strands.

DNA polymerase 'sliding clamp'



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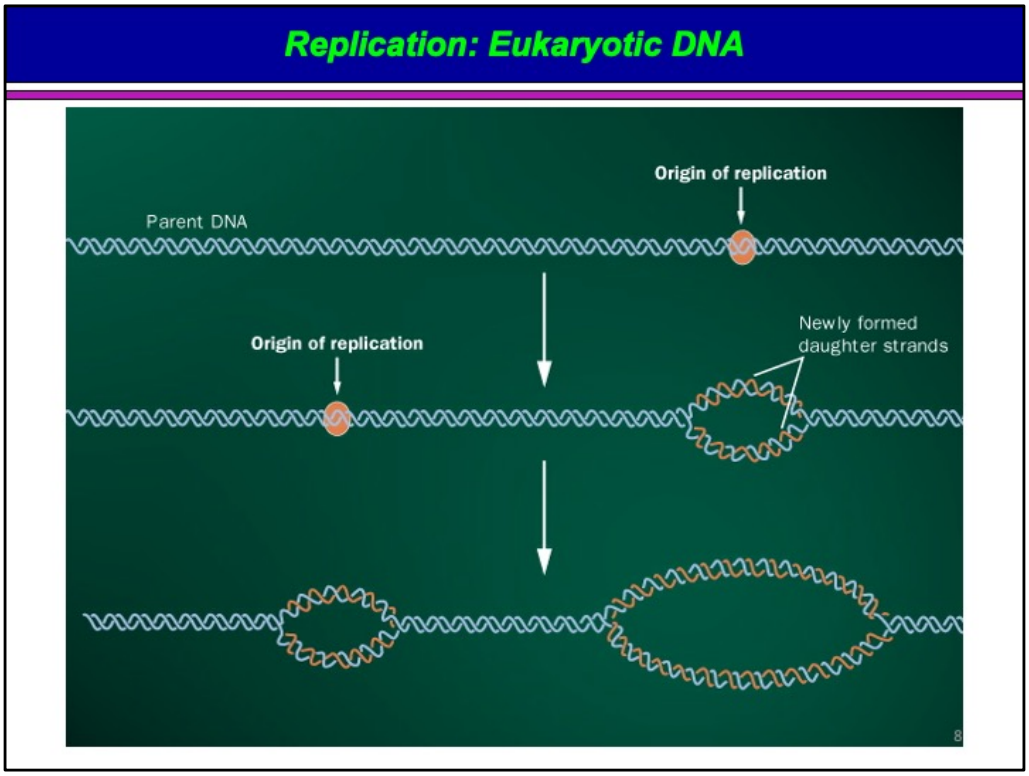
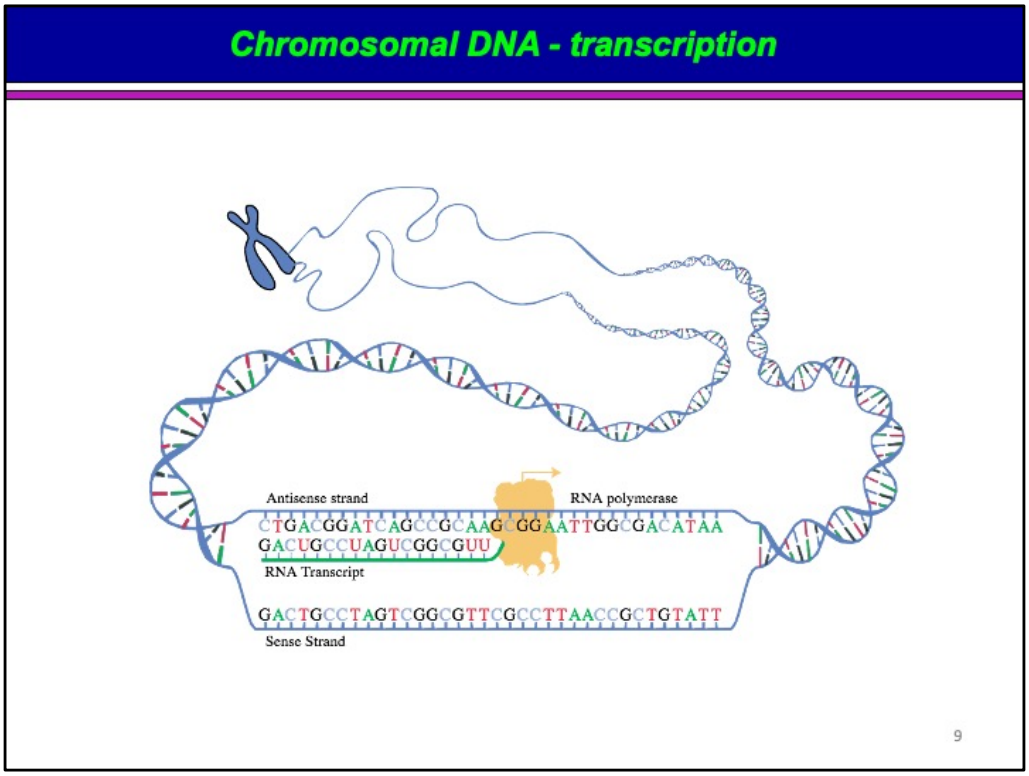
The DNA Pol III holoenzyme includes two b subunits (purple and green in this diagram) that form the 'sliding clamp' assembly and gives Pol III the required processivity, or accuracy of replicating DNA.



This slide depicts the replication of an eukaryotic chromosome, at different origins of replication, of which there are up to 50,000 in a human cell's complement of chromosomes. The origins are specific sequences that bind the replication enzyme machinery. Note how each origin creates a replicating 'bubble' that extends from the origin bidirectionally to the next 'bubble'.

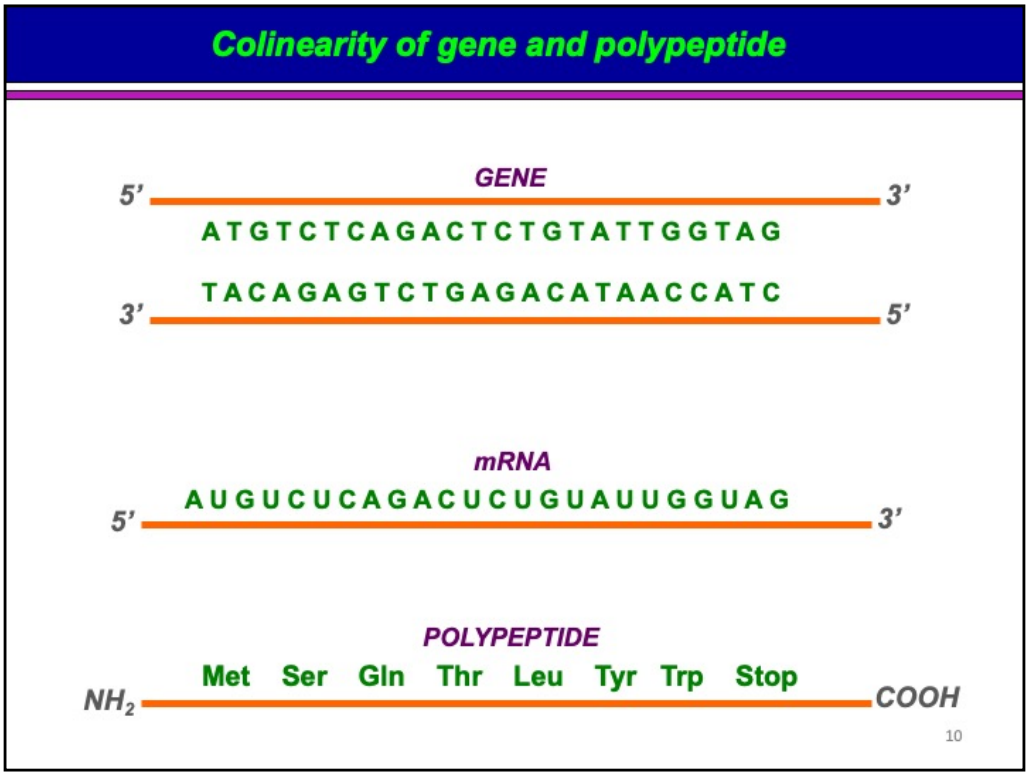
DNA molecules, being very long, require rapid replication speeds by DNA polymerases. In bacteria, this is about 1000 nucleotides per second, but in eukaryotes the process is much slower, and more complex. For instance the histone scaffolding needs to be removed and placed back as the sliding replication fork moves along the DNA strands. Eukaryote replication runs at about 50 nucleotides per second. Hence the large number of replication origins spaced at regular intervals.



At the bottom of the diagram you can see that the two DNA strands are prised apart (by an enzyme called helicase, but not shown here) and a gene sequence read/copied by RNA polymerase into an mRNA transcript.

Note that the strands are labelled ‘antisense’ and ‘sense’, which could also be ‘non-coding’ and ‘coding’, ‘minus’ and ‘plus’, or ‘template’ and ‘non-template’. Unfortunately, all of these terms have been devised over decades and have been used interchangeably in textbooks etc. Whichever the label, the important point to note is that the strand that is copied/transcribed by RNA polymerase is the first of all these names, thus, ‘antisense’, ‘non-coding’, ‘minus’, or ‘template’.

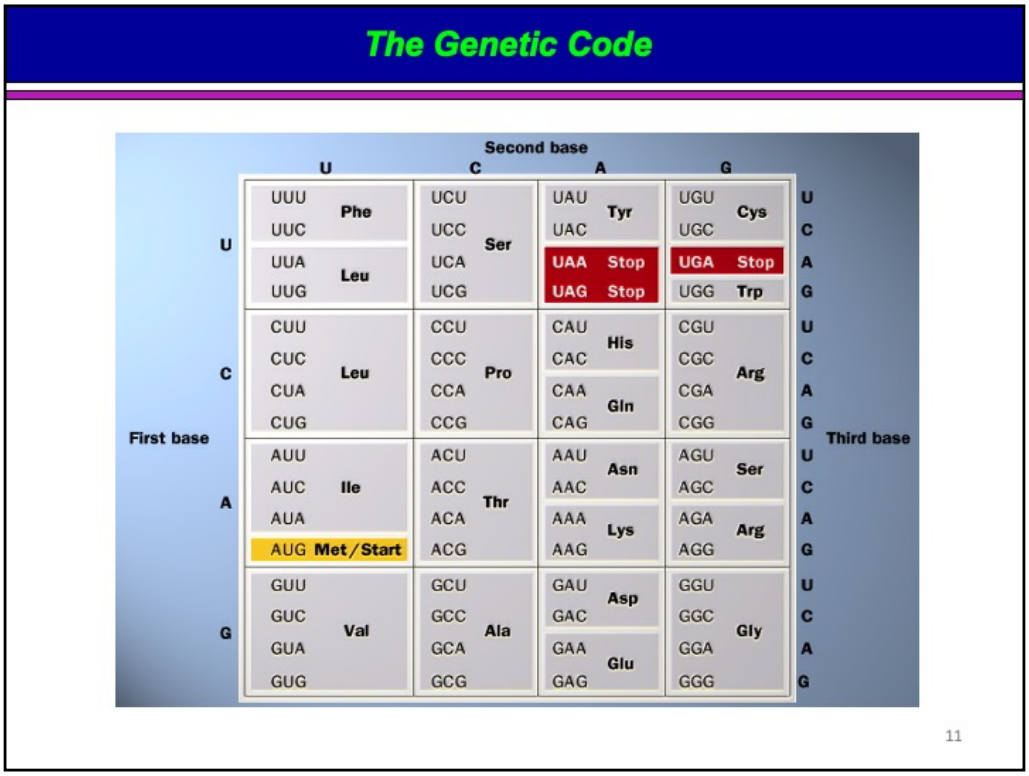
The copied strand (antisense) is the complementary strand of the actual DNA gene sequence (sense). Thus, the mRNA transcript is a complementary copy of the **gene sequence on the opposite strand**. In this way, the strand used by RNA polymerase is a template to enable the mRNA to be processed as an exact copy of the gene sequence on the opposite strand.



A gene (top) is transcribed (by RNA polymerase) into mRNA (middle) and this transcript is copied into a polypeptide (bottom). Within the Open Reading Frame (ORF) of a gene, each triplet codon specifies an amino acid in the translated polypeptide and the two – triplet codons and amino acids – are collinear or contiguous, that is, they are aligned precisely. ['Collinear' and especially 'contiguous' are terrible words I know, but we have to put up with the English language and its seemingly infinite terminology.]

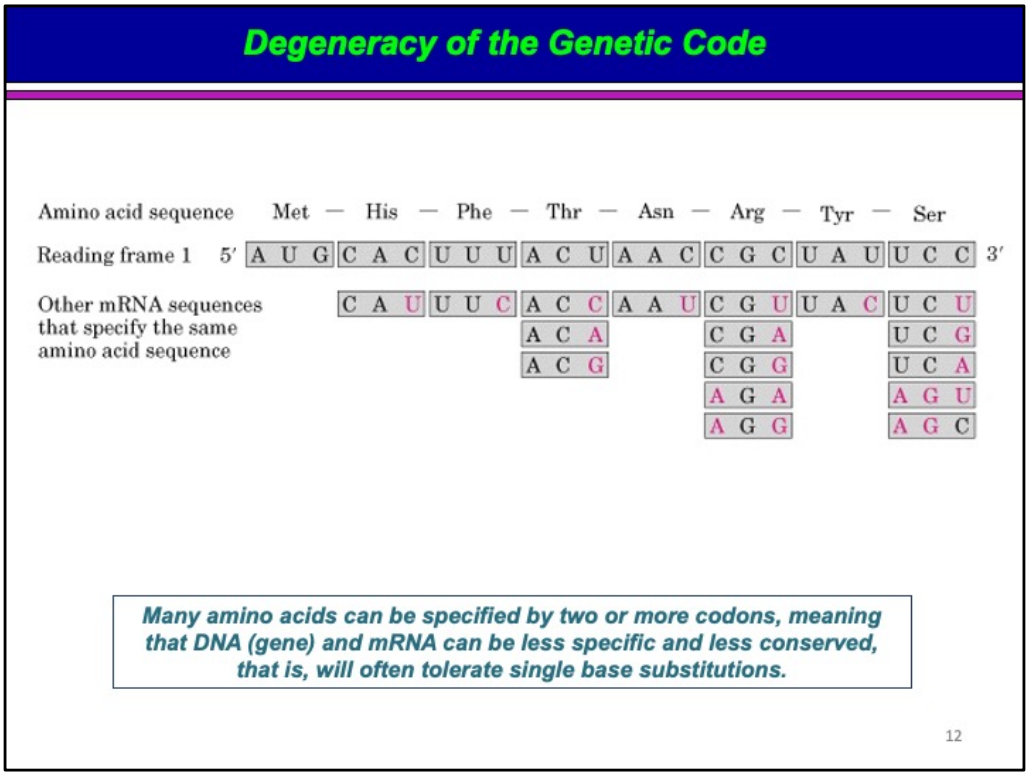
The messenger RNA transcript triplets are identical to the triplets of the gene's ORF coding strand (upper sequence), but complementary to the triplets of the template strand (lower sequence). The template strand is the "go between" strand that enables the gene sequence to be copied exactly by the principle of complementarity. Note that the gene sequence and mRNA strands also have the same polarity (5' - 3'), which is opposite to the template strand polarity (3' – 5').

Lastly, notice that in RNA, the base uracil (U) substitutes for thymine (T). Note that each triplet codon specifies an amino acid, but that the redundancy of the Genetic Code for most amino acids means that changes in the third position, and sometimes the second position may specify the same amino acid. Having said that, problems may arise when the DNA (in particular, the gene sequence or ORF, but sometimes other crucial regions such as the promoter or other regulatory sequences) is mutated, often spontaneously, or by errors in replication, or by the action of radiation or toxins, or free radicals.



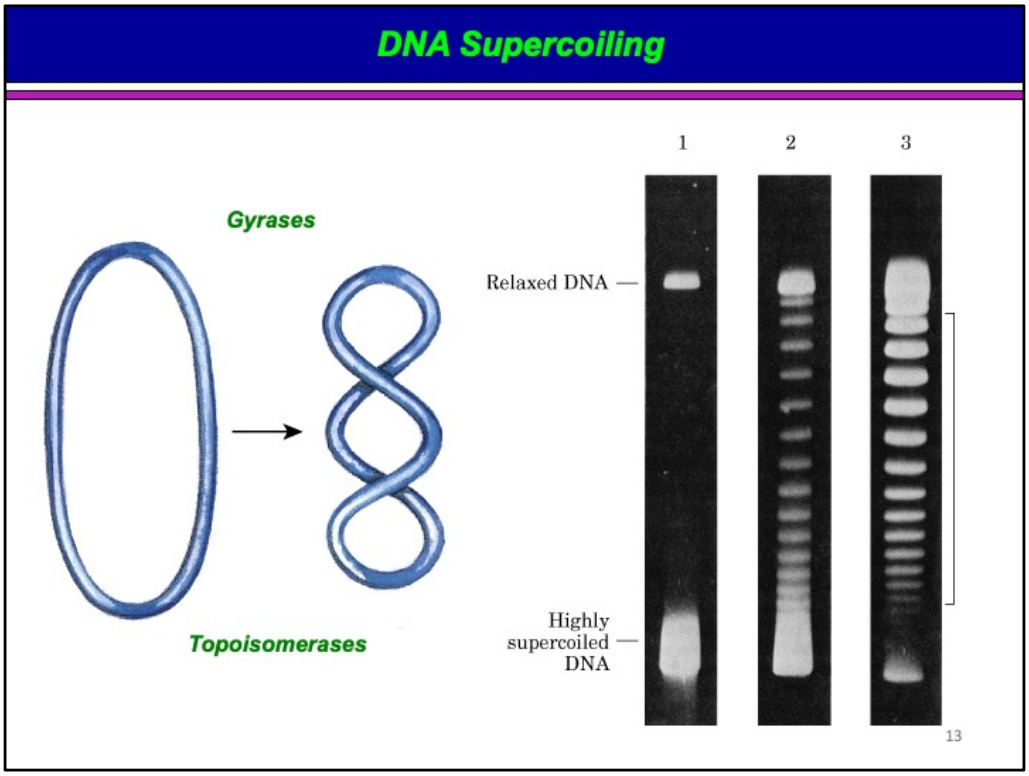
The ‘Genetic Code’ of triplet codons of which there are a total of 64 (4x4x4). As you can see, there are three stop or nonsense codons (UAA; UAG; UGA) and a single start codon (Met). Tryptophan also has a single codon (UGG). However, the remaining 59 available codons are distributed among the other 18 amino acids, resulting in a range of codons for each amino acid (from two to six), making the Code ‘redundant’, meaning, in this case, repeats of the same amino acids among the extra codon choices.

How this Code is used and “adjusted” in molecular cloning will be made apparent on the following slide and in more detail in a later lecture. Note that this table of codons is written for the mRNA transcript, but in the original gene ORFs, the DNA triplet codons have ‘T’ in place of ‘U’.



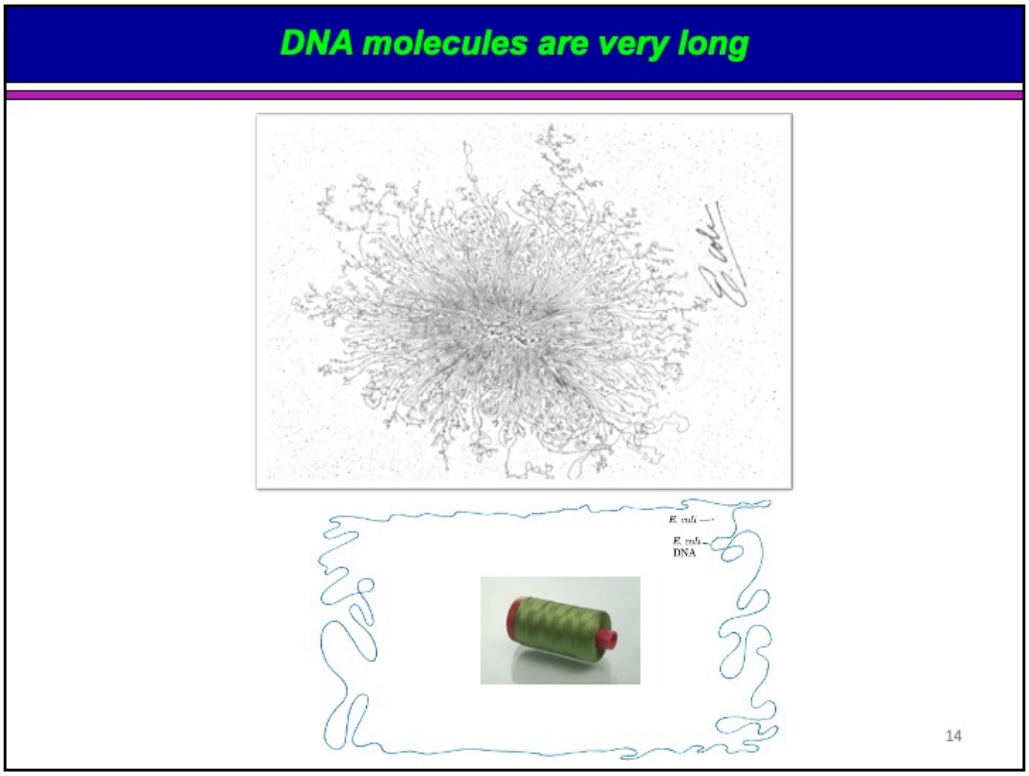
This simple example shows how several amino acids may be specified by more than one codon, and up to as many as six. In one sense, we can look at this as a simple mathematical exercise in that there are four DNA nucleotide bases, but only three positions in any codon. This means that there are 4x4x4 = 64 possible three base combinations that need to be assorted among 20 amino acids and stop codons (of which there are three). You can see the full assignment of the 64 codons on the previous slide.

The text in the box at the bottom of this slide is particularly important! Different codons for the same amino acid may be used in different parts of the polypeptide sequence; or in different species. It also signifies that some amino acids in proteins are more conserved than the DNA.



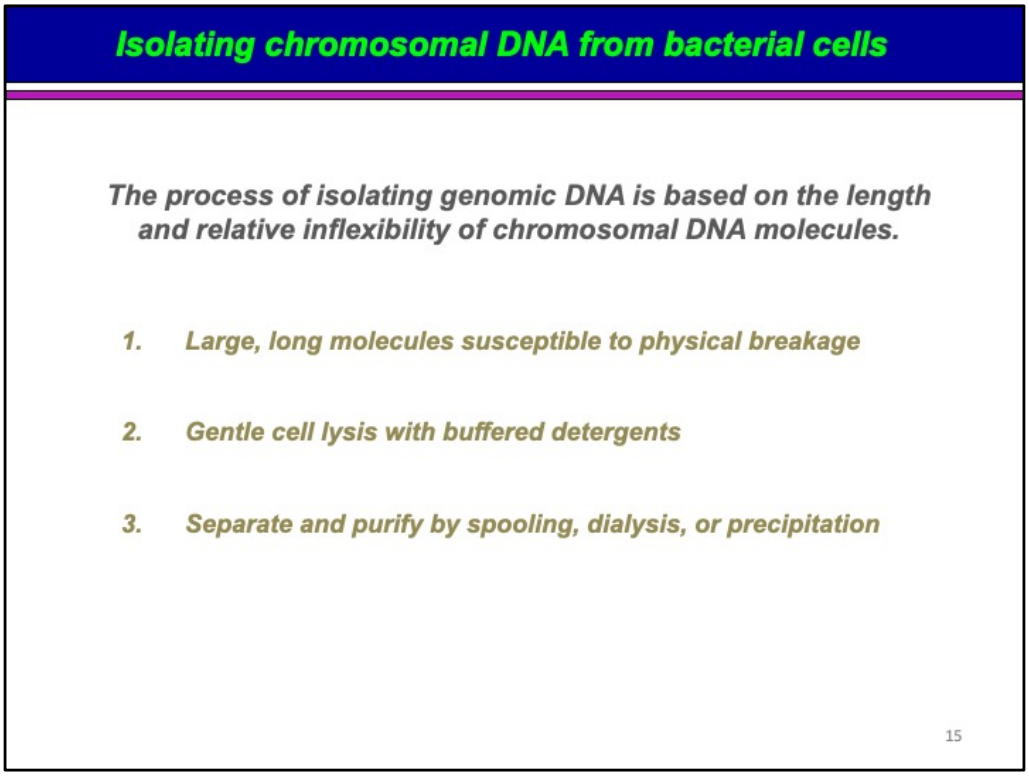
On the left, are depictions of the conversion of relaxed to supercoiled DNA circle. This function is carried out by DNA gyrases (in eukaryotes) and topoisomerases (in prokaryotes). Though this conversion is shown in one direction only, partner gyrases/topoisomerases (I; II) effect the conversion in either direction, as DNA needs to be supercoiled in the resting state, and relaxed when genes are transcribed or DNA is being repaired or replicated.

The gel photo on the right shows relaxed and highly (completely) supercoiled DNA bands at the top and bottom of gel lane 1; and the time course of the interconversion of relaxed to supercoiled DNA (lane 2) and back again (lane 3), by gyrases. The stepladder effect in lanes 2 and 3 are time points in the interconversions, with different stages of supercoiling and relaxing within the DNA molecules.



The upper figure is an electron micrograph showing a single chromosome ‘escaping’ a lysed bacterial cell, which is at the centre of the figure.

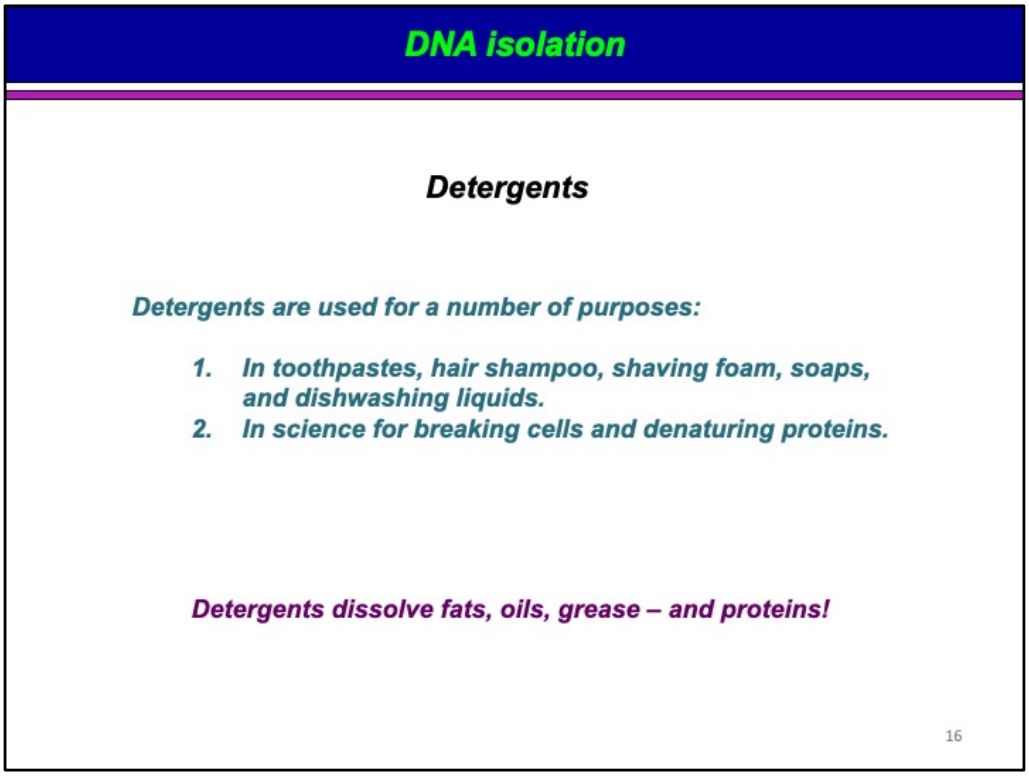
The simple line drawing at bottom, while not precisely to scale, shows the length of the bacterial *E. coli* chromosome in relation to the size of the entire *E. coli* cell (represented by a dot). The chromosome can only “fit” into the cell by being supercoiled so that it occupies a much smaller space. The very thin diameter of the double-stranded DNA chromosome helps the supercoils to be wound into a much smaller bundle (for a comparison, consider the size of a cotton reel, which is about the size of a golf ball, with approx. 120 metres of cotton spooled onto the reel.



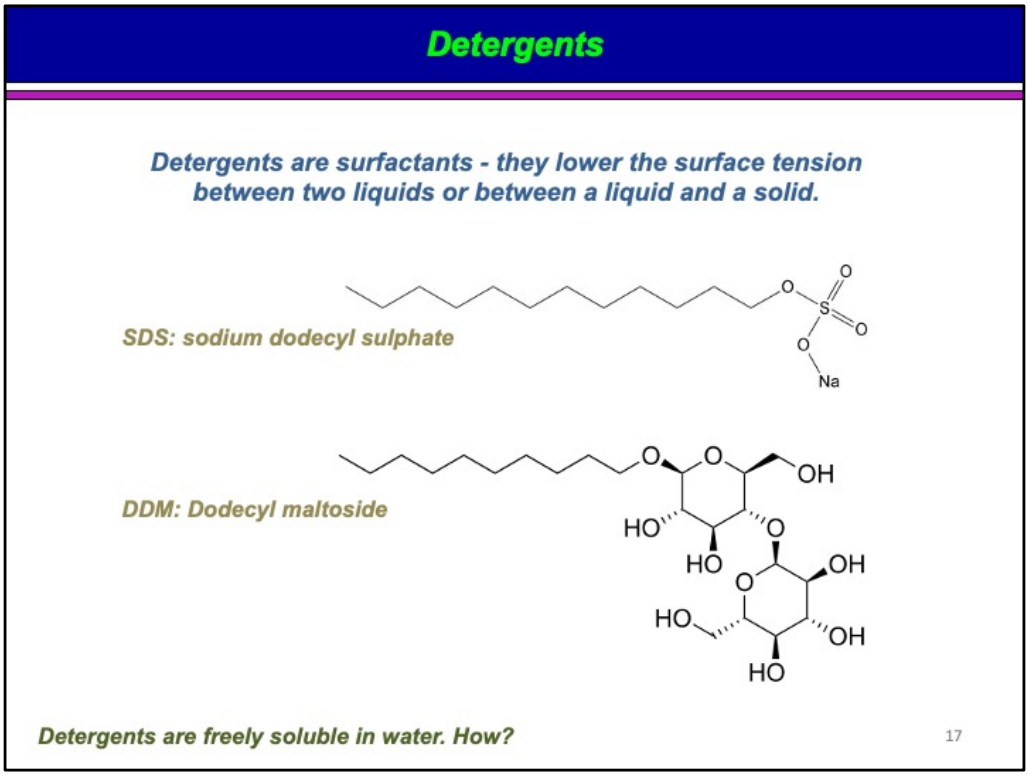
Chromosomes are very long and their double-strandedness renders them relatively inflexible, making them susceptible to physical breakage into much shorter fragments. For instance, the *E. coli* chromosome is 4600 kilobase pairs (kb) in length, but no matter how gentle the isolation procedure, an average fragment size of around 100-300 kb is the best that can be achieved, due to the physical strain on the molecules, from shaking, pipetting, vortexing etc. Special conditions are required to ensure a minimum of physical shearing. These conditions are explored in the next slides that cover the isolation of genomic DNA from a bacterial cell culture.

Note, unlike the isolation of RNA and plasmid DNA (next lecture), chromosomal DNA is not isolated free of RNA (or plasmids if they are present in the bacterial cells). Since most bacteria do not normally harbour plasmids, their absence is almost always guaranteed. That still leaves the RNA, but there is a simple way of dealing with RNA, namely, by adding ribonuclease to digest all of the RNA into ribonucleotides that are far too small to be precipitated later in salt/alcohol. Ribonuclease may be added at the cell membrane lysis stage or later when the cell debris is removed by high speed centrifugation and before the final precipitation of the chromosomal DNA.

In Expt 3, we will leave the RNA in the cleared lysate, as it combines with the chromosomal DNA to form a more visible pellet and assist in the precipitation step.



Understanding the principle of detergent action on DNA, lipids, and proteins is important. See the next two slides.

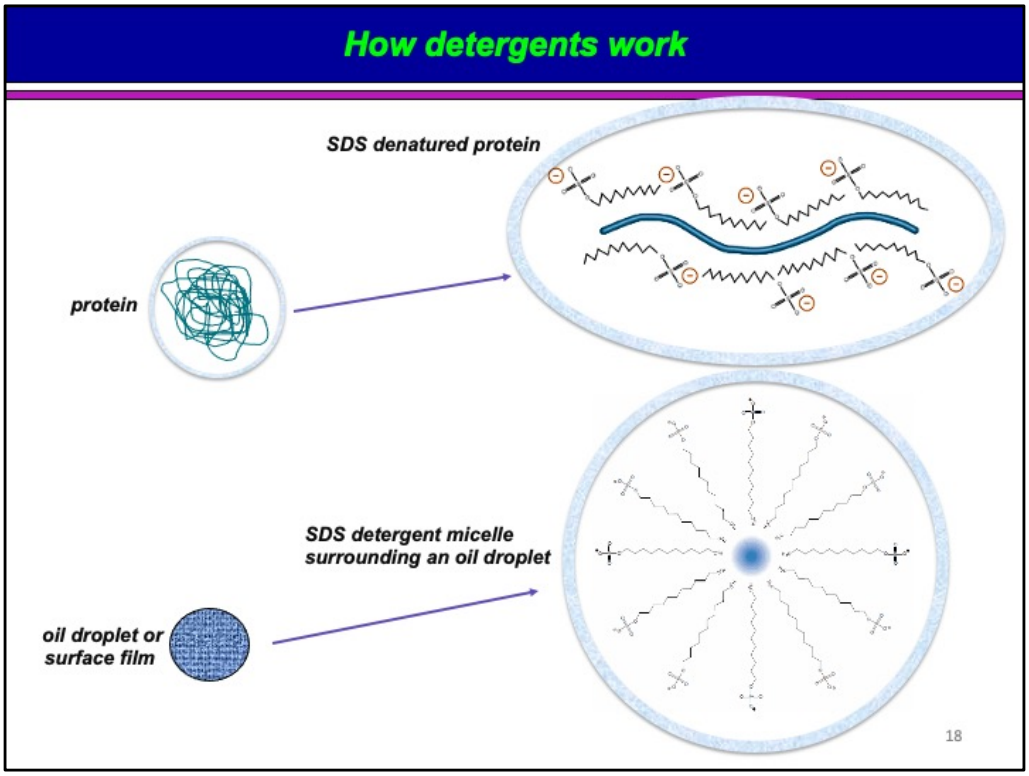


Detergents are surfactants – they lower the surface tension between two liquids or between a liquid and a solid. They act as wetting agents, emulsifiers, foaming agents, and dispersants.

Two common detergents are used in biochemistry and molecular biology. SDS is the detergent of choice for bacterial lysis; and DDM the choice for eukaryotic cell lysis. Their effect is the same, that is, to provide a long aliphatic hydrophobic chain beginning with a methyl group at one end and capped at the other end by a charged (anionic) group (SDS), or polar group (DDM). In this way, they are able to associate with, and dissolve, lipids and proteins and disperse them in aqueous buffers.

SDS (also called SLS, sodium lauryl sulfate) has many common domestic and industrial uses for lathering and dissolving oils and grease, for example in soaps, shampoos, shaving cream, toothpaste, dishwashing and laundry liquids, floor cleaners, and engine degreasers.

SDS is an anionic detergent because it has a negative charge on the terminal polar sulfoxide group (bound to a sodium ion in the diagram above).



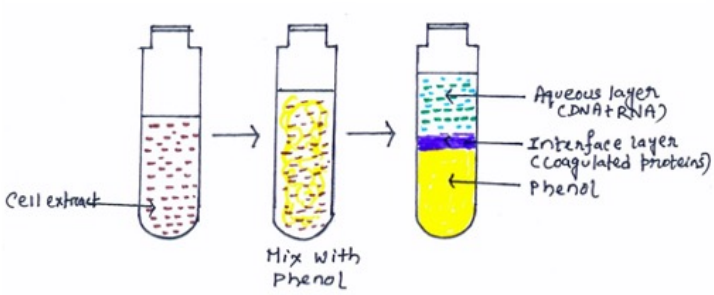
Schematic look at how the detergent SDS (sodium dodecyl sulfate) “dissolves” proteins, or fats, oils, and lipids. The two processes are very different for proteins versus lipids!

SDS denatures proteins by firstly using its charged polar end to interrupt protein surface polar and ionic interactions, including surrounding water that is hydrogen bonded to the protein surface Blue halo; top left), thus allowing the long aliphatic chains of the SDS to gain access to the interior core of the folded protein where the aliphatic chains are able to unravel and bind to the inner non-polar amino acid side chains of the protein, while the polar amino acids and surrounding water shell are dissociated (top panel). Note how the SDS molecules coat and line the now random coil polypeptide by alternately remaining bound to respective polar and non-polar amino acids in the unravelled polypeptide chain, which itself is surrounded by a newly formed H-bonded shell of water (blue halo; top right), therefore completing the dissolution/denaturation effect of the SDS.

The lower panel depicts the SDS dissolution effect on lipids, oils, or fats. The oil droplet (blue circle, left) becomes surrounded within a **detergent micelle** (right) with the aliphatic SDS non-polar lipidic chains forming a “bicycle wheel” of spokes, with the polar heads on the outer rim in contact with water (blue halo; bottom right) . In this way, the oil droplet is “dissolved” in that it is removed from other droplets and is dispersed in the aqueous medium. This can easily be demonstrated by adding SDS to water that has an oil layer on its surface, shaking vigorously for a few seconds then allowing the liquid to settle, whereupon the oil layer has seemingly disappeared (dispersed) within the dissolved SDS micelles.

In molecular biology we use SDS to dissolve and disperse the lipid bilayer and its associated proteins in bacterial membranes; and denature and remove soluble proteins from the cytoplasm. This is part of the process of removing unwanted macromolecules (lipids; proteins; polysaccharides) during the purification of nucleic acids from cells.

Isolating DNA from E. coli (phenol method)



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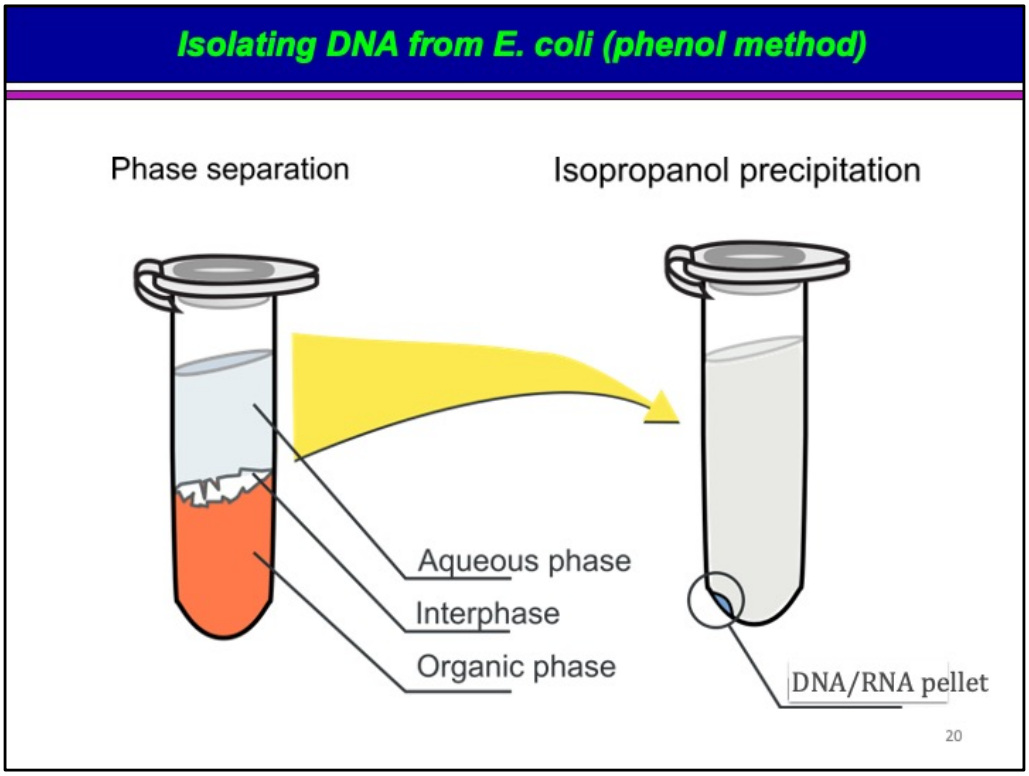
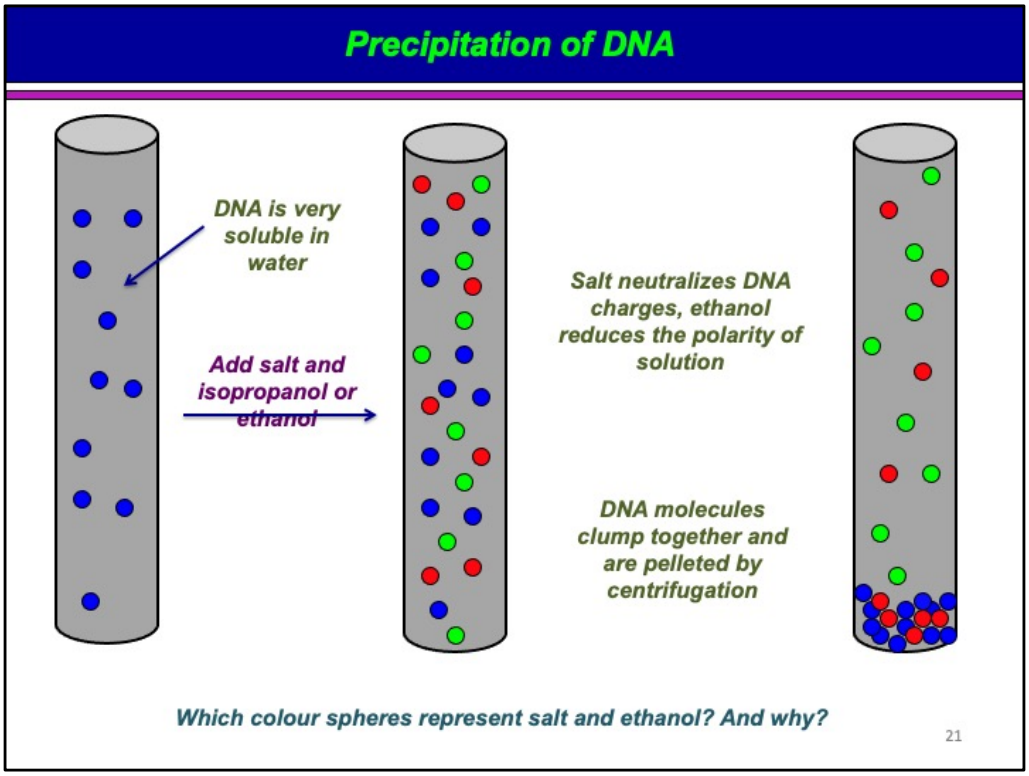
This panel illustrate the basic idea behind the isolation of chromosomal DNA away from cellular proteins, polysaccharides, lipids, and membrane debris. This cartoon only indicates the use of phenol as the sole organic solvent and proteins as the chief contaminants that 'precipitate' at the interface during high-speed centrifugation. More details are given in the next two slides.



This slide illustrates the basic ideas behind the isolation of chromosomal DNA away from cellular proteins, polysaccharides, lipids, and membrane debris. A protocol designed around these ideas is the basis of Experiment 3: Isolation of genomic DNA from an *E. coli* culture.

Following a gentle lysis method, to minimise breakages along the relatively inflexible double-stranded chromosome, detergents (SDS; CTAB), a general protein lysing enzyme (proteinase K) and high salt (1M) are added to facilitate the precipitation and removal of unwanted materials (protein, lipid, membrane fragments, polysaccharides) by phenol/chloroform extractions and centrifugation, with the precipitated debris at the interface (tube at left). A new organic solvent mix is used in a second extraction, namely phenol/chloroform/isoamyl alcohol. Isoamyl alcohol serves the purpose of sharpening the demarcation boundary/interface between the bottom organic and upper aqueous phases – this second extraction illustration isn't shown because it is essentially the same as the tube at left, but with very little debris at the interface and a different solvent mix now in the bottom layer after high speed centrifugation. At this stage, the upper aqueous phase contains genomic DNA (and RNA, if still undigested).

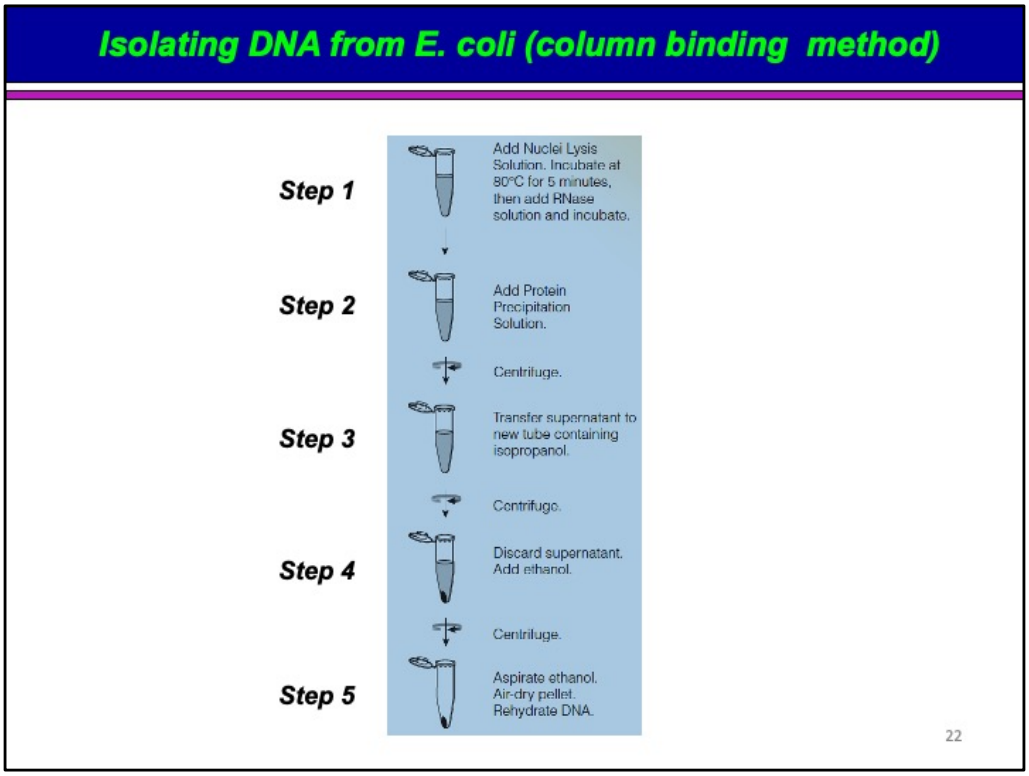
The upper aqueous phase is removed to a new tube and isopropanol (or ethanol) is added. In the presence of the alcohol and salt, the nucleic acids aggregate, then precipitate into a pellet under high-speed centrifugation. The pellet is washed with 70% ethanol to remove residual salts and isopropanol, dried briefly, then resuspended in a small volume of buffer in preparation for agarose gel electrophoresis – voila! Expt 3 done and dusted, or home and hosed, as they say!



Nucleic acids are precipitated by high-speed centrifugation in buffered salt-alcohol (ethanol or isopropanol).

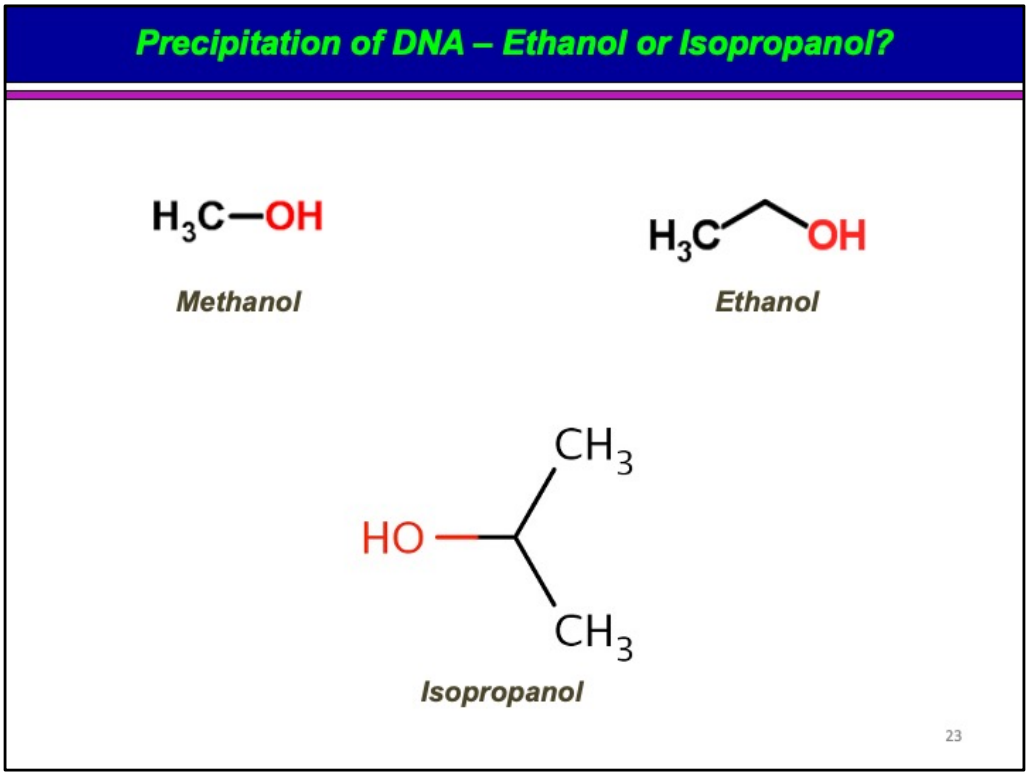
DNA is very soluble in water because it is an acid (negative phosphate charges on each strand). The addition of NaCl brings about the neutralization of the charges, because (+ve) Na ions bind to the (-ve) phosphates.

Ethanol or isopropanol lower the dielectric constant of water, that is, reduce water's polar solubilizing effect by breaking the H-bonding network between water molecules and between water and the DNA. This enables the neutralized DNA molecules to clump together or aggregate and then be collected into a pellet following high-speed centrifugation.



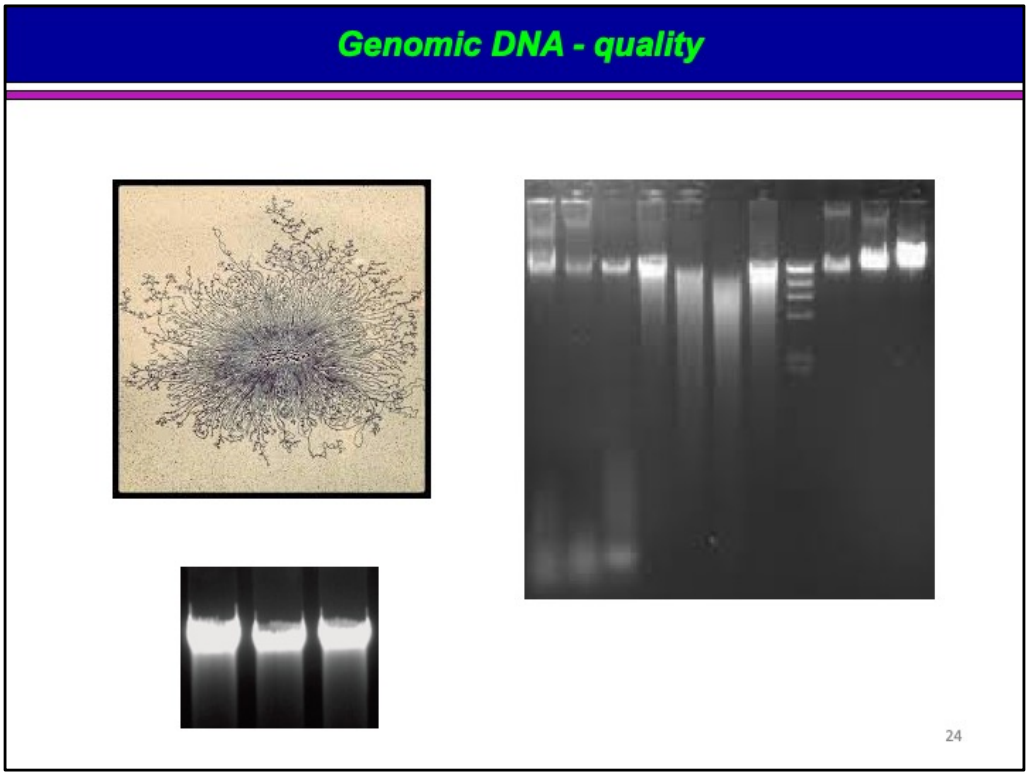
This slide a scheme for the isolation of chromosomal/genomic DNA from an *E. coli* culture. This /scheme/protocol is the basis of Experiment 3: Isolation of genomic DNA from an *E. coli* culture.

To carry out the process, a Promega commercial kit is used. Each step or stage is illustrated in the scheme on this slide and explained in detail in the Lab Manual. Briefly, in the first step, an *E. coli* culture is centrifuged and resuspended in lysis solution and heated. RNA is digested with ribonuclease. Protein precipitation solution (step 2) is added and all of the proteinaceous material (including membrane fragments) is pelleted by high-speed centrifugation. The supernatant (containing the semi-purified, soluble, genomic DNA is transferred to a new tube and isopropanol is added (step 3). The presence of the isopropanol and buffer/salt permits the DNA to aggregate, then precipitate into a pellet under high-speed centrifugation. In step 4, the pellet is washed with 70% ethanol to remove residual salts and isopropanol, dried briefly, then resuspended/rehydrated in a small volume of Tris-EDTA (TE) buffer in preparation for agarose gel electrophoresis. Voila! Expt 3 done and dusted, or home and hosed, as they say.



Isopropanol may be substituted for ethanol. Isopropanol and ethanol are water miscible (soluble) aliphatic alcohols. The first alcohol in this series, namely methanol, is not used, as it is far more toxic than ethanol or isopropanol. The next up in the series, butanol, is only partially soluble in water and is therefore not useful. Ethanol and isopropanol do not have long enough aliphatic methyl chains to overcome the polar solubilising effect of the hydroxyl group, but butanol with four methyl groups reaches the crossover point of solubility.

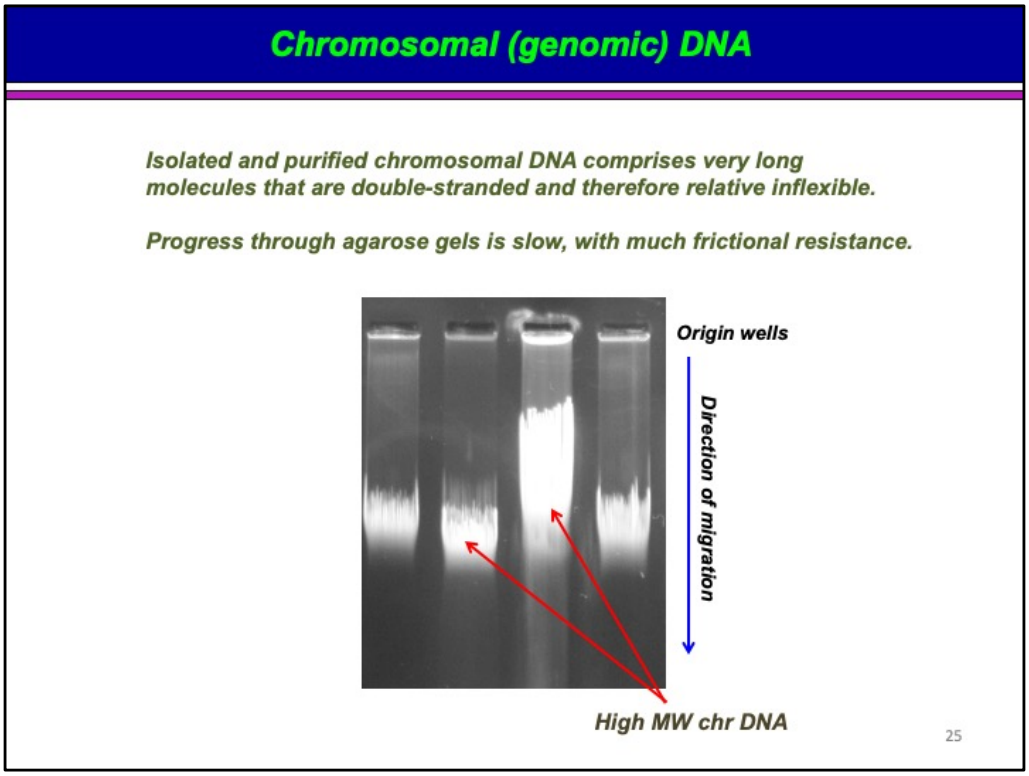
Ethanol is used to precipitate nucleic acids in the ration 2.5:1 (EtOH:water), whereas isopropanol only needs to be used at a 1:1 ratio (or even less). Thus, not as much isopropanol is needed than ethanol to achieve the same outcome. Why is isopropanol more efficient than ethanol in precipitating nucleic acids?



The left upper panel depicts the single bacterial chromosome being released from a lysed bacterial cell. It is very large and very long, and because its double-strandedness makes it relatively inflexible, it fractures or fragments quite easily, especially with vigorous isolation methods (sloshing, pipetting, vortexing).

The gel on the right of this slide depicts a series of lanes with genomic DNA in various degrees of purity and integrity. The three lanes on the left are reasonably good, with RNA contamination (lower in the gel). The middle four lanes are only of fair quality, with some streaking/streaming from too much pipetting and vortexing during the isolation procedure, where shearing forces break the DNA into smaller fragments, which can be seen lower in the gel lanes. The last three lanes depict excellent high molecular weight chromosomal DNA, with virtually no telltale fragmentation ahead of the major bands.

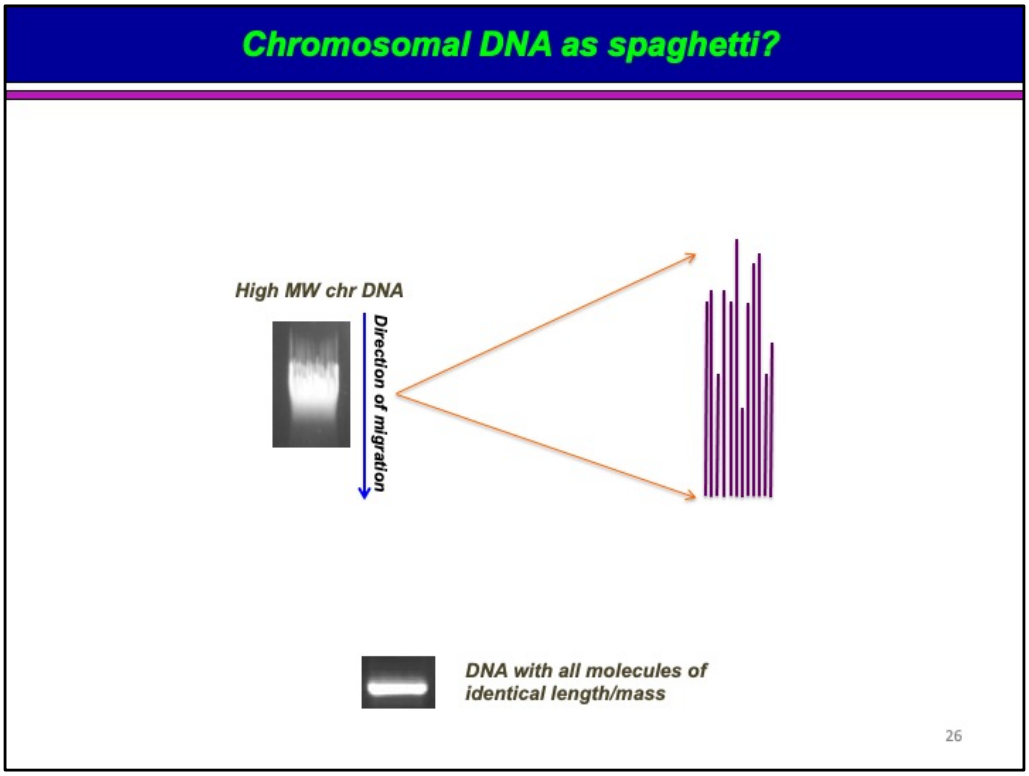
The gel profile at lower left depicts a close up view of three lanes of an agarose gel with excellent high MW chromosomal DNA. Note the leading edges are flush or evenly straight, but the trailing edges are streaky or striated. This is because the randomly broken molecules of DNA migrate in the current with leading edges aligned in the current and the trailing edges indicating the differences in lengths. Linear DNA molecules migrate end on end, or top to tail. Note also that no matter how careful and gentle the isolation process, there is always a faint smear of shorter fragments ahead of the main band.



A typical chromosomal DNA gel profile such as we will see in Experiment 3. The migration rate is governed by the voltage/current and agarose gel percentage. Too high a voltage and/or too high an agarose concentration will result in high frictional resistance, and in 'streaming' and 'dragging' of the DNA fragments, some of the larger fragments of which might not even migrate beyond the origin well, as depicted in the gel diagram above.

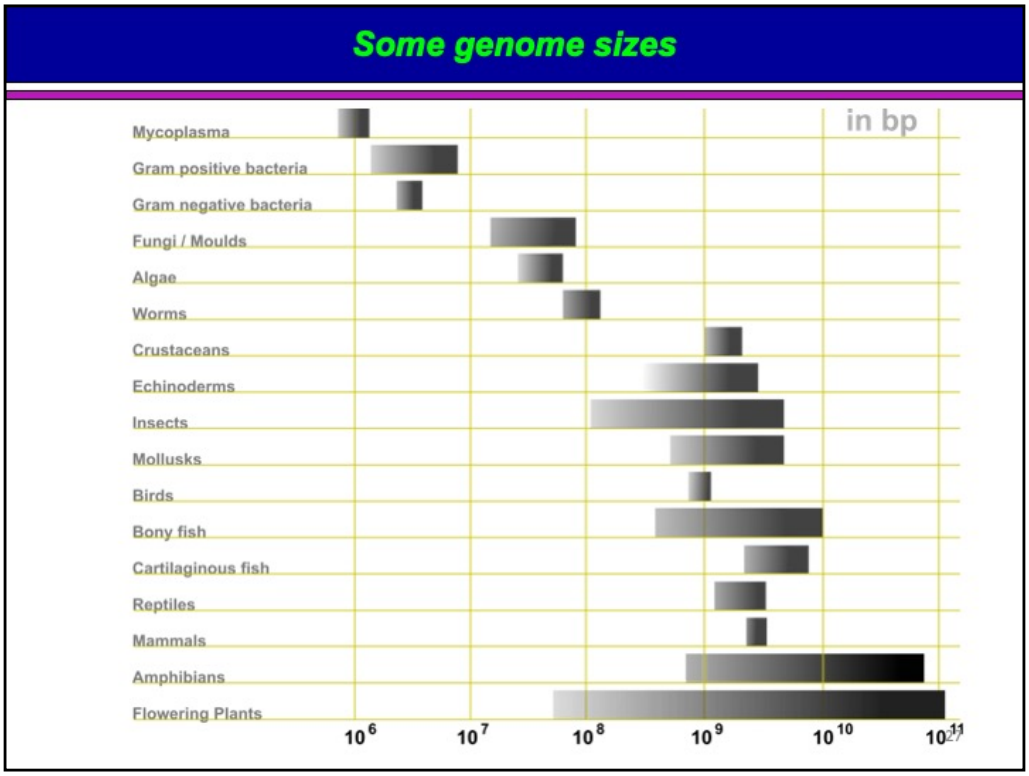
The characteristic striated or streaky pattern seen in the lanes is typical of chromosomal DNA running in an agarose gel. The molecules are of different lengths and migrate end on end, and the ends of the longer molecules poke out and trail behind the main band in this streaky pattern. Only the top of the gel is shown, with the high MW DNAs near the origin wells, even after a longish gel run.

You might also notice the difference between the band in lane 3 with the others. Quite simply, this difference is due to the chr DNA sample in lane 3 being of greater mass and quantity than the samples in the other lanes and it 'stacks' up behind the leading front of the molecules.

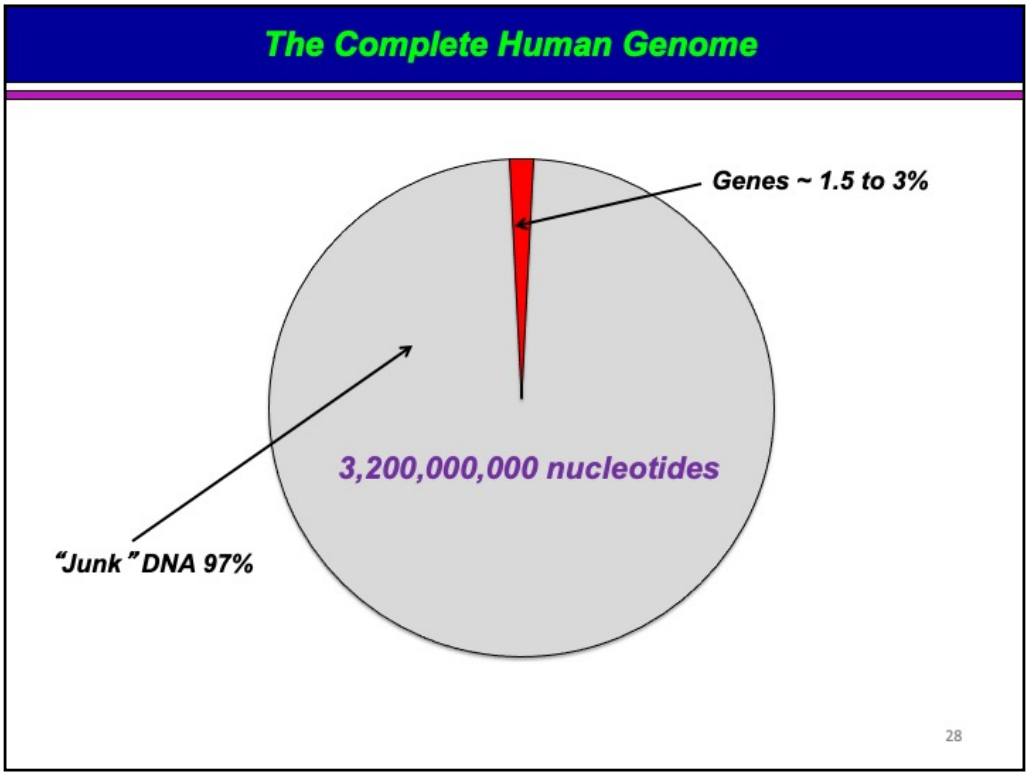


One lane of a gel depicting a migrating band of chromosomal DNA is shown on the left, with the direction of migration top to bottom. The drawing on the right is a stylised drawn version of the band with the different length DNA fragment molecules ("spaghetti", as it were) shown aligned at the bottom edges as they move in the leading edge of the buffer ions current, but tailing at the top due to their different lengths, hence the striated or streaky effect of the migration of genomic DNA in an agarose gel.

In contrast, any band of DNA such as a linear DNA fragment or a plasmid band of identical length/mass, migrate as a more discrete all over smooth-edged band, as seen in the small diagram at the bottom of this slide.



A short table of several genomes for comparison. Note that many rows have genomes much larger than those of mammals. Why? We are not sure, but some of it is due to complex processes such as metamorphosis in amphibians, for instance. It is certainly not due to evolutionary lineage and the number of genes.

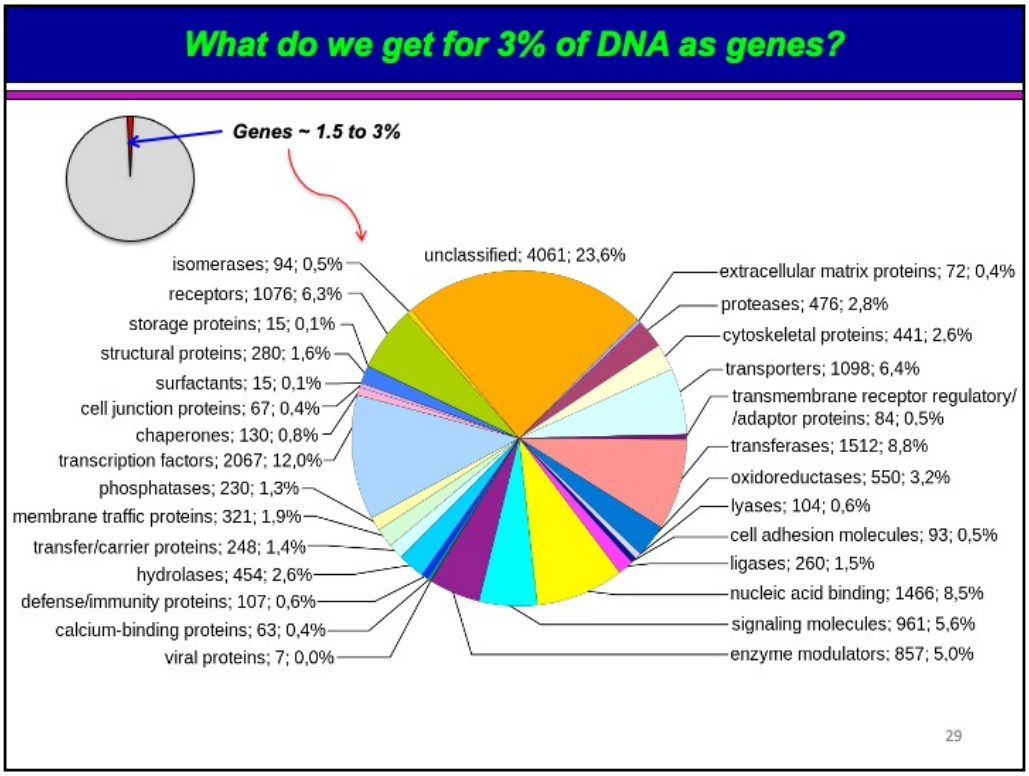


The complete human genome (23 chromosome pairs) has been sequenced many times now, beginning with the first complete human genome in April 2003. The project began in 1990 and was a massive task, undertaken by several major labs and using batch automation to speed up the process of obtaining the precise order of every nucleotide in the haploid human genome that comprises approx. 3.2 billion bases (3.2×10^9).

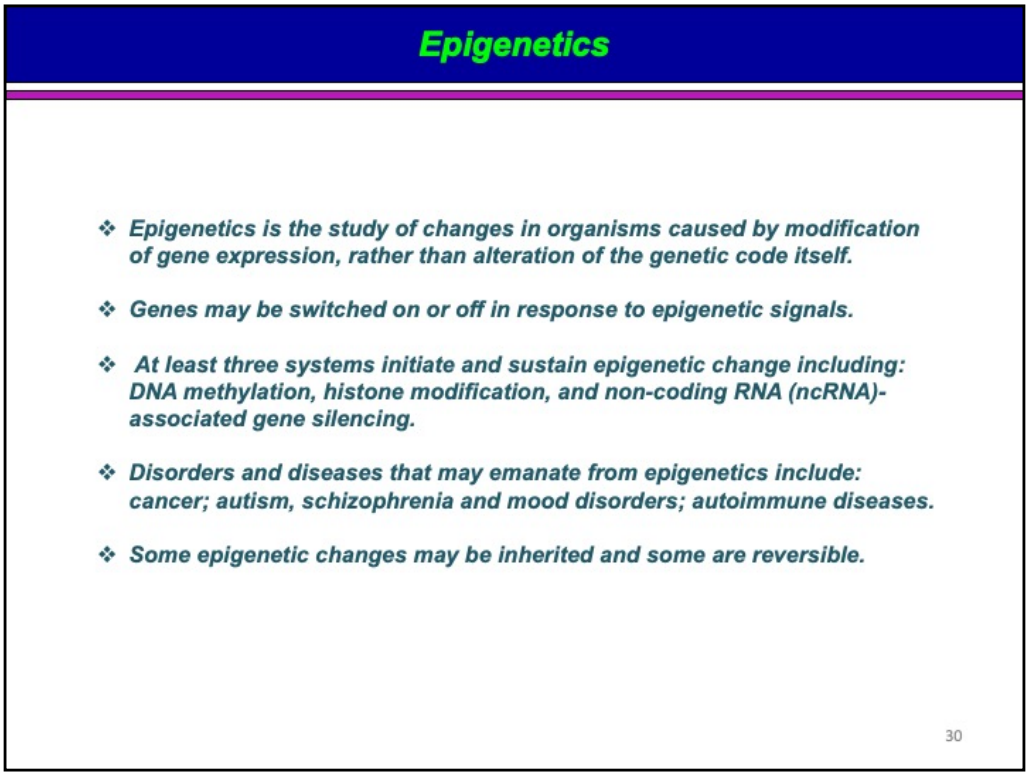
As you can see from the pie chart in this slide, most of the DNA is so-called "junk" or non-coding "silent" DNA; and only about 1.5 to 3% of the genome actually encodes about 24,000 proteins. This compares with only about 20% "junk" DNA in a prokaryote, and ~80% encoding genes. Why the difference? We now know that much of the human junk DNA is important and includes such features as, centromeres and telomeres, multiple, regular origins of replication, transcription factors, numerous CG 'islands', ribosomal and transfer RNA genes, introns within genes that separate exons, and an enormous number of "silent" RNAs, mostly small, micro-RNAs (miRNAs) that are known to be regulatory.

A large amount of junk DNA is also involved in epigenetic activity, a process that leads to variations in traits, chiefly caused by environmental stimuli and that are mostly not inherited (see paragraph below and slides further on). "Junk" DNA, a pejorative term applied before the function of this DNA was known, is therefore important!

When the final tally is made, it turns out that nearly 80% of human DNA contains repetitive sequences, or is involved in biochemical or genetic processes, so not so silent or junky after all. Much of the "junk" DNA is transcribed at different times into RNA. These RNAs are the "messengers" of the brain, for instance – not true mRNA that transcribes genes, but regulatory miRNAs that switch genes on and off. Over 4 million sites are involved in regulation – not unlike electrical switches, turning genes on and off in different tissues and cells and coordinating the activities of these genes. Potentially an infinite number of circuits may be formed and unformed by this regulatory network.



Coding sequences have been assigned to about 24,000 human genes, distributed into families as indicated on the pie chart above, but remember that this represents only ~3% of the human genome (top left, and previous slide).



Two reproductive cells combine and divide to form every type of cell in the adult organism. In order for this process to occur in a way that preserves the exact genomic code, the epigenome 'imprint' must be erased through a process called 'reprogramming'. Reprogramming is important because eggs and sperm are 'marked' with epigenetic tags e.g. methylation patterns in cytosines, histone tagging and non-coding RNAs – see the next slide). Before the new organism can grow into a healthy embryo, the epigenetic tags must be erased.

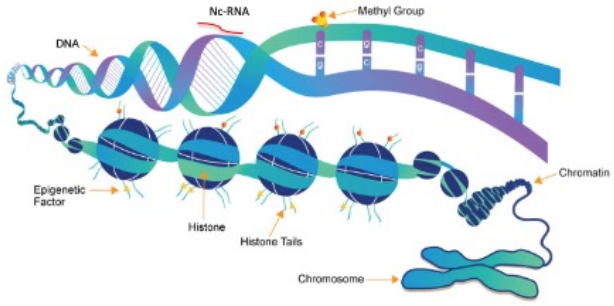
At certain times during development, specialized cellular 'machinery' (aka enzymes) scours the genome and erases its epigenetic tags in order to return the cells to a genetic 'blank slate'. Yet, for a small minority of genes, epigenetic tags make it through this process and pass unchanged from parent to offspring. In mammals, about 1% of genes escape epigenetic reprogramming through a process called Imprinting.

Epigenetic inheritance adds another dimension to the modern picture of evolution. The genome changes slowly, through the processes of random mutation and natural selection. It takes many generations for a genetic trait to become common in a population. The epigenome, on the other hand, can change rapidly in response to signals from the environment. Through epigenetic inheritance, some of the experiences of the parents may pass to future generations. At the same time, the epigenome remains flexible as environmental conditions continue to change. Epigenetic inheritance may allow an organism to continually adjust or adapt its gene expression to fit its environment, without changing its DNA code.

Epigenetics

Epigenetic changes are triggered by factors such as:

- *Diet*
- *Exercise*
- *Drugs, alcohol; smoking*
- *Aging*
- *Pollutants, toxins*
- *Changes to sleep patterns*
- *Any severe environmental stress*
- *Cancer and probably other human disorders and fatal diseases*



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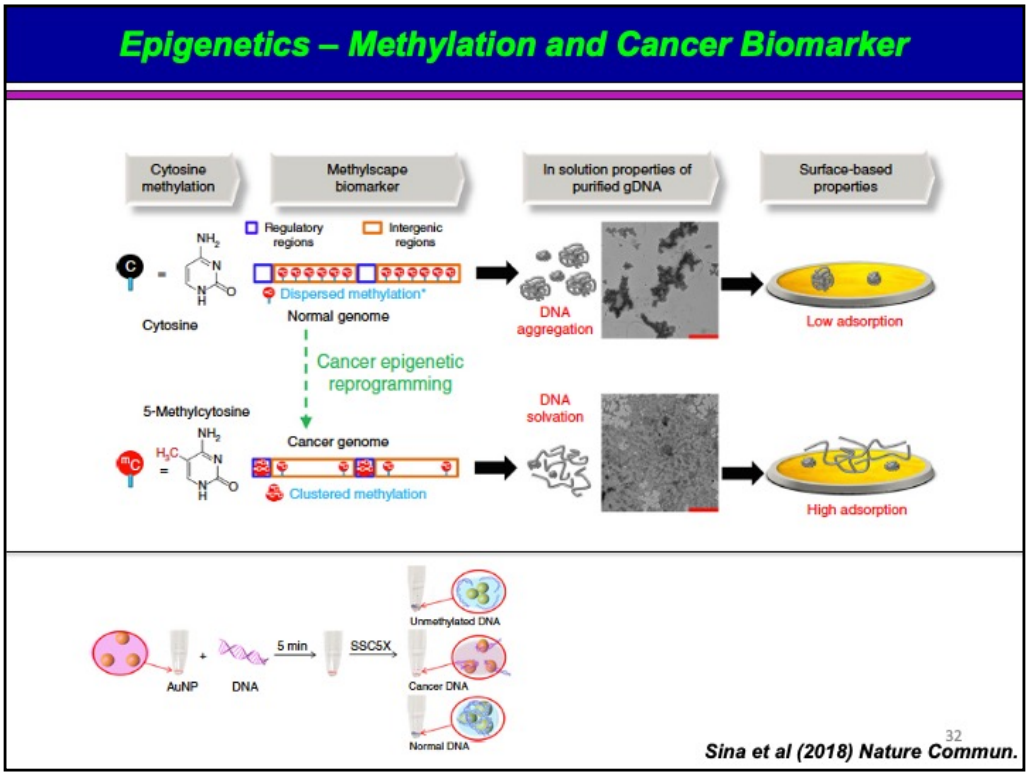
A list of some environmental factors that can trigger epigenetic changes in cells, chiefly through changes in the methylation pattern of cytosine bases in DNA, but not the DNA sequence itself.

The figure indicates the three possible types of changes triggered by external factors. Changes are indicated to the histones, to the methylation of cytosines, or via non-coding RNAs (Nc-RNAs).



So, what has all this epigentic stuff been leading up to? Recent research (QLD, Aust) has produced a potential universal cancer biomarker that detects epigenetic changes in DNA, as depicted in this slide.

In the upper panel, epigenetic reprogramming is shown as different cytosine methylation patterns in normal DNA (dispersed) and cancer DNA (clustered). The clustered methylation pattern occurs in CG rich regulatory islands that determine the genome-wide cancer DNA signature. The different DNAs (normal, cancer) behave differently when solvated and in the way in which they adsorb colloidal gold nanoparticles (AuNPs).

The lower panel depicts a simple 10 min colour test for the different DNA-AuNP (where 'AuNP' = gold nanoparticle) aggregates in high salt solutions, as reddish (cancer DNA methylation) to blue (normal DNA methylation). To date, this scheme has been shown to work on cancer cell lines, tissues and plasma, but is yet to be instituted in clinical trials, which are perhaps a year or two away?

MCQ L3

The drawing above is of a folded protein being denatured by the detergent SDS. The denaturation process occurs because….

A. The polar heads of the SDS attack the outer polar shell of the protein and pull the outside apart, making the interior non-polar amino acids accessible to the long non-polar lipid chains in SDS.

B. The SDS non-polar lipid chains invade the protein core, unravel it, and allow the polar SDS heads to interact with the polar surface amino acids of the protein.

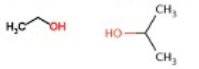
C. The two actions described in (A) or (B) occur simultaneously.

D. The negative charges on SDS dissociate the electrostatic and polar interactions on the protein surface so that the SDS non-polar lipid chains can invade and unravel the protein non-polar core.

33

MCQ L3

Ethanol *Isopropanol*



Ethanol is used to precipitate nucleic acids in the ratio 2.5:1 (EtOH:water). Isopropanol is used at a 1:1 ratio. Not as much isopropanol is needed than ethanol to achieve the same outcome. Why is isopropanol more effective than ethanol in precipitating nucleic acids?

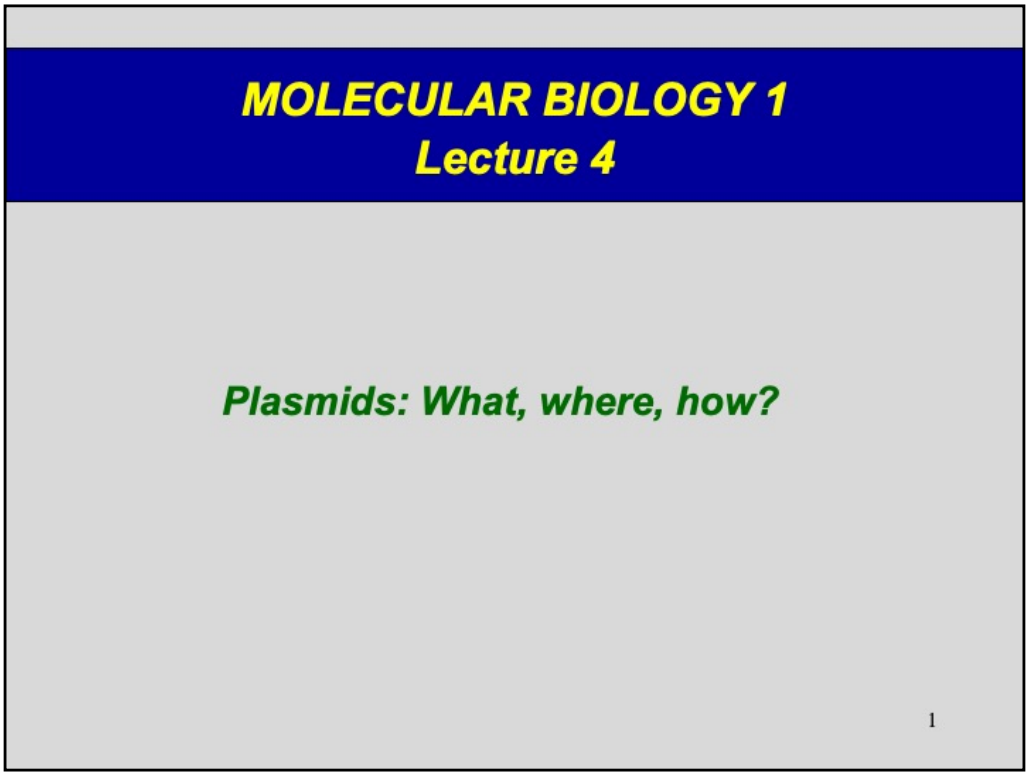
A. Isopropanol is more effective because it is more non-polar and the non-polar chains penetrate the hydrophobic core of nucleic acids more easily than ethanol.

B. Ethanol has a lower molecular weight than isopropanol and therefore more of it is needed to achieve the same outcome of precipitating nucleic acids.

C. Isopropanol has a lower dielectric constant than ethanol, making the water/isopropanol mix less polar than an ethanol/water mix, allowing nucleic acids to aggregate more readily.

D. Water associates more readily with isopropanol than ethanol because isopropanol, though less polar than ethanol, has a stronger acting OH group that interacts with water more easily.

E. Ethanol is more polar than isopropanol and associates more readily with water, which makes an ethanol/water mix more polar than an isoproanol/water mix.



This lecture introduces us to plasmids, and to plasmids as cloning vectors in molecular biology. Much of the information is not only central to this lecture and others later in the series, but covers two of the seven laboratory experiments, namely Expts 4 & 6.

Lecture 4: Learning Objectives

Plasmids and molecular cloning

- ***What are plasmids? Features of genetically engineered plasmids. Plasmid vectors. Transformation. Molecular cloning of genes to generate recombinant plasmids. Plasmids in E. coli – growth, amplification, isolation.***

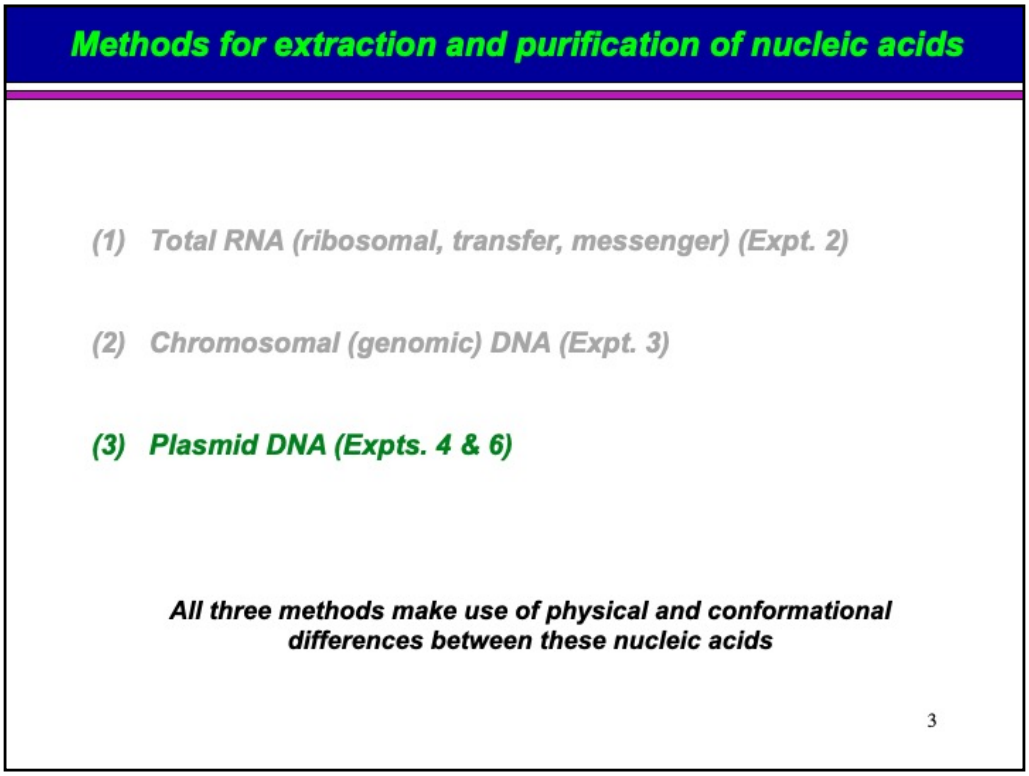
Alkaline lysis method of plasmid isolation. Purification methods

- ***Plasmids are semi-purified from bacterial cell lysates using detergent and alkali to remove cellular debris. Precipitation using salt-alcohol and centrifugation. Binding and elution in spin columns using salt and water.***

Plasmids; and agarose gel electrophoresis revisited

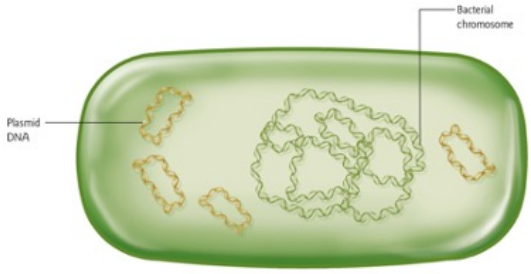
- ***How circular and linear plasmids are detected in agarose gels. Gel-red staining clues to the banding patterns.***

2



In the experimental section of this subject, you will isolate each of these three types of nucleic acids from *E. coli* bacterial cells. The isolation and purification method for each nucleic acid differs on the basis of physical and conformational features of the DNA or RNA. The remainder of this lecture will cover plasmid DNA only. The overarching themes for isolating these nucleic acids broaches the fundamental methods (technical, scientific) of the isolation processes themselves; and the broader downstream goals of studying the structure and function of these macromolecules and why they are crucial in the way cells, organs, and organisms operate, or fail to operate.

Bacterial Plasmids

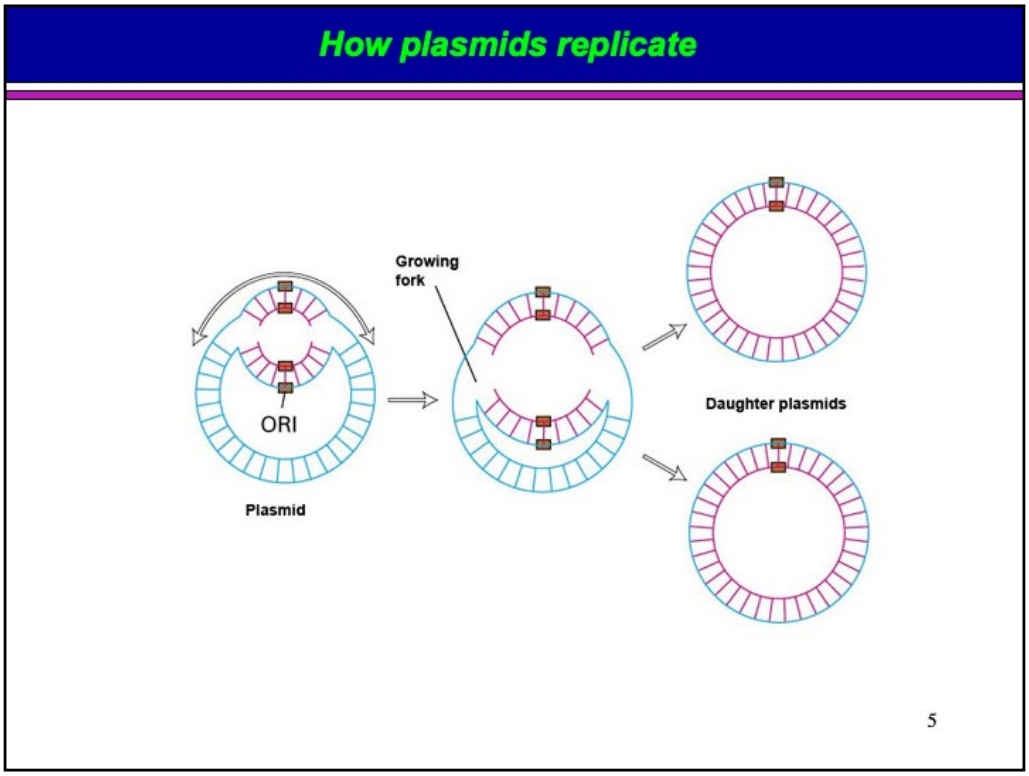


- ***Loops of DNA found in some bacteria.***
- ***Contain 'special' genes usually not related to the cell's function – for example, antibiotic resistance.***
- ***Can be transferred from the host cell to another related, but plasmid-less cell.***
- ***Are 'inherited' by stringent replication control.***

4

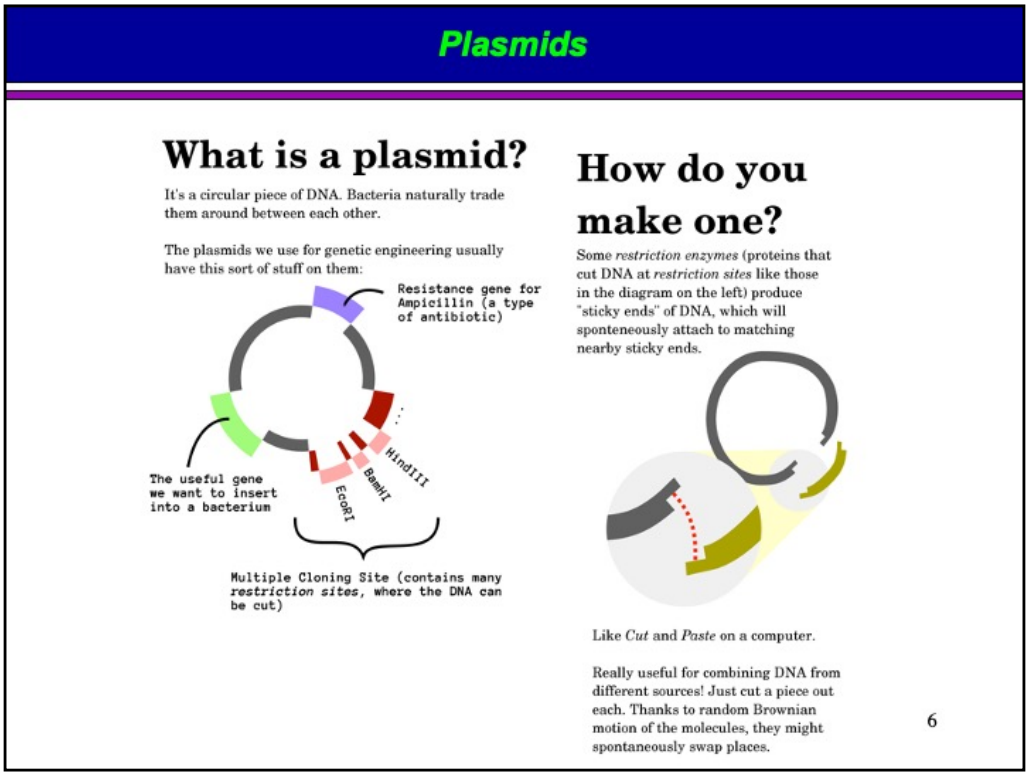
Plasmids are extrachromosomal elements that exist outside the chromosome, but within the bacterial cytoplasm. Many plasmids are naturally occurring and have roles inside their respective host cells. They replicate syncronously with the chromosome (termed stringent replication) and so are partitioned to, and therefore inherited by, all daughter cells. They may be transferred to the same or closely related bacterial species, by the process of conjugation (by way of a mating tube that connects two cells). They may also be transformed into bacterial cells, artificially, as naked DNA, as we do in molecular biology in the process termed 'transformation', which we will look at in the next lecture.

As you will see in the slides that follow, hybrid plasmids have been 'fashioned' (aka genetically engineered) from natural plasmids in order to facilitate and effect DNA recombinant technology, in which genes from any source may be 'cloned' into these hybrid plasmids, acting, and often referred to, as vectors. Once a gene is cloned into a plasmid vector, it may be subjected to any of a number of diverse molecular biology techniques, many of which will be visited in these lectures and in the lab course. Such techniques include: molecular characterisation of cloned DNA and genes; DNA sequencing; site-directed mutagenesis of cloned genes; and expression of proteins from cloned genes.

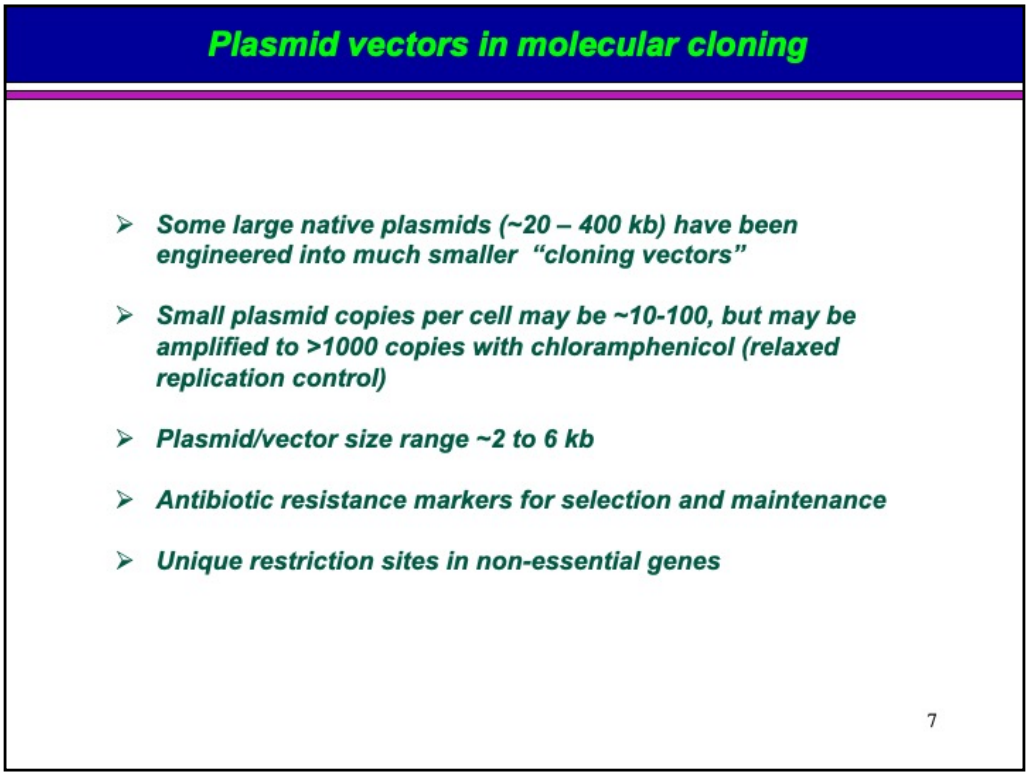


Plasmids replicate from a fixed origin of replication, with bidirectional 'rolling circle' forks moving clockwise and anticlockwise around the double-stranded plasmid until they meet on the other side, whereupon the replicating enzyme complexes disengage at the replication terminus, enabling the new daughter plasmids to separate.

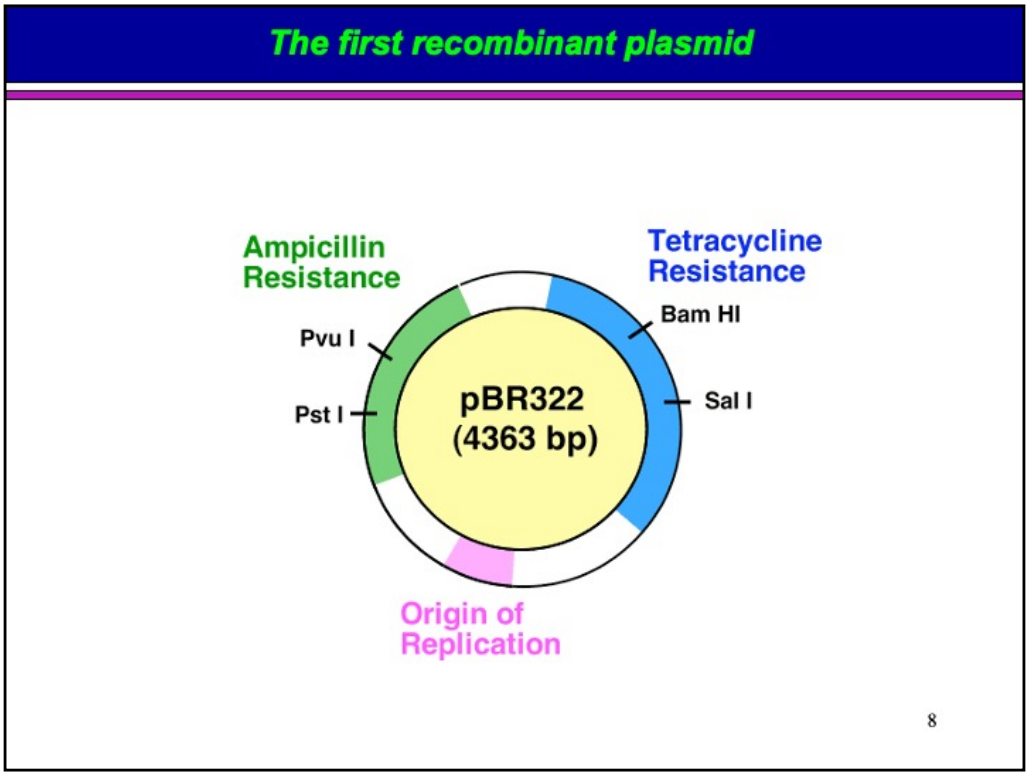
The "new" plasmids comprise one original parental strand (the template) and one newly copied complementary daughter strand. This type of replication (bidirectional fork) is similar to that of the bacterial host chromosome. In fact, both chromosome and host (natural) plasmid are replicated simultaneously, in the process known as stringent replication control.



Notice the essential elements of an engineered (synthetic, hybrid) plasmid as a cloning vector in the left panel: resistance gene for maintenance and selection; multiple cloning region with unique restriction sites; and the gene of interest to be cloned/inserted. This shows us how plasmids as cloning vectors may be "engineered" artificially, by a cut-and-paste method (right panel), to introduce foreign genes, which will be explained in other slides and lectures, and performed by yourselves in Expt 6.



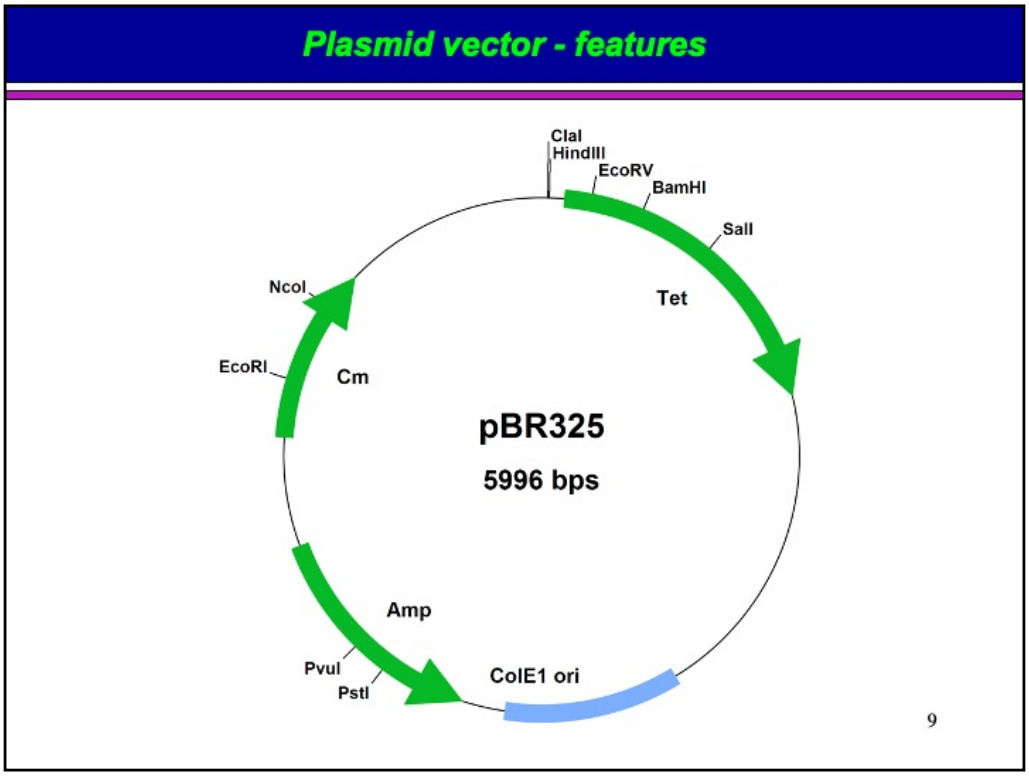
A list of features of engineered small plasmids as cloning vectors. These points are elaborated in the next several slides.



pBR322 was the first engineered (artificial) plasmid to be employed as a cloning vector in molecular biology. This first plasmid, coming soon after the discovery and use of restriction endonucleases, formed the beginnings of modern molecular biology from the late 1970s. pBR322 was engineered in 1977, in the laboratory of HW Boyer (U Cal), and was named for the post-doc "creators" or "molecular engineers" of this plasmid (Bolivar and Rodriguez).

Names were registered at Harvard University, but self-lettering choices (such as pBR, pUC, pAMG etc) were soon exhausted. Recent commercially derived names can be in any script: pYES; pZero; pHAGE-CMV-MCS-IRES-ZsGreen etc).

Take note of the important essential regions of pBR322. The origin of replication allows these engineered plasmids to be copied independently of the bacterial host chromosome (relaxed replication control), as we will see later in this lecture. Such a plasmid will replicate out of phase with the stringent replication cycle of the chromosome and will be copied into higher numbers (~ 10 to 100) than the usual one or two copies of naturally occurring plasmids. The low copy number of natural or host plasmids is due to the physiological burden to maintain the (usually large) plasmids in the host bacterium.



One of the first engineered cloning vectors (circa 1978); a sister plasmid to pBR322, in fact near identical except that pBR325 has a third resistance gene Cm (for chloramphenicol resistance). Note the defined parameters: Origin of replication, antibiotic resistance marker genes, unique restriction enzyme cloning sites.

In this earliest iteration of molecular cloning and recombinant DNA strategy, unique restriction sites were positioned within resistance genes. The cloning vector plasmid (pBR322, pBR325 etc) therefore required at least two resistance genes, with one being inactivated by the cut-and-paste method of DNA insertion, and interruption (discontinuation) of the resistance gene open reading frame (ORF). The second resistance gene remained intact and acted as the selection agent on agar plates and in growth medium containing that antibiotic.

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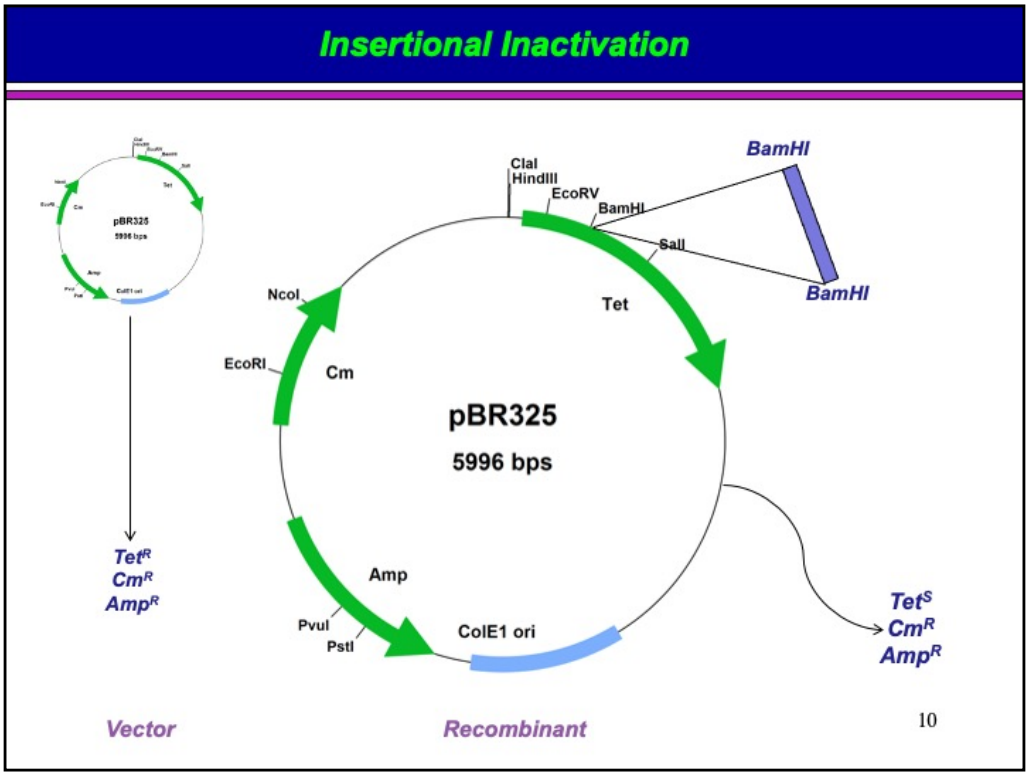
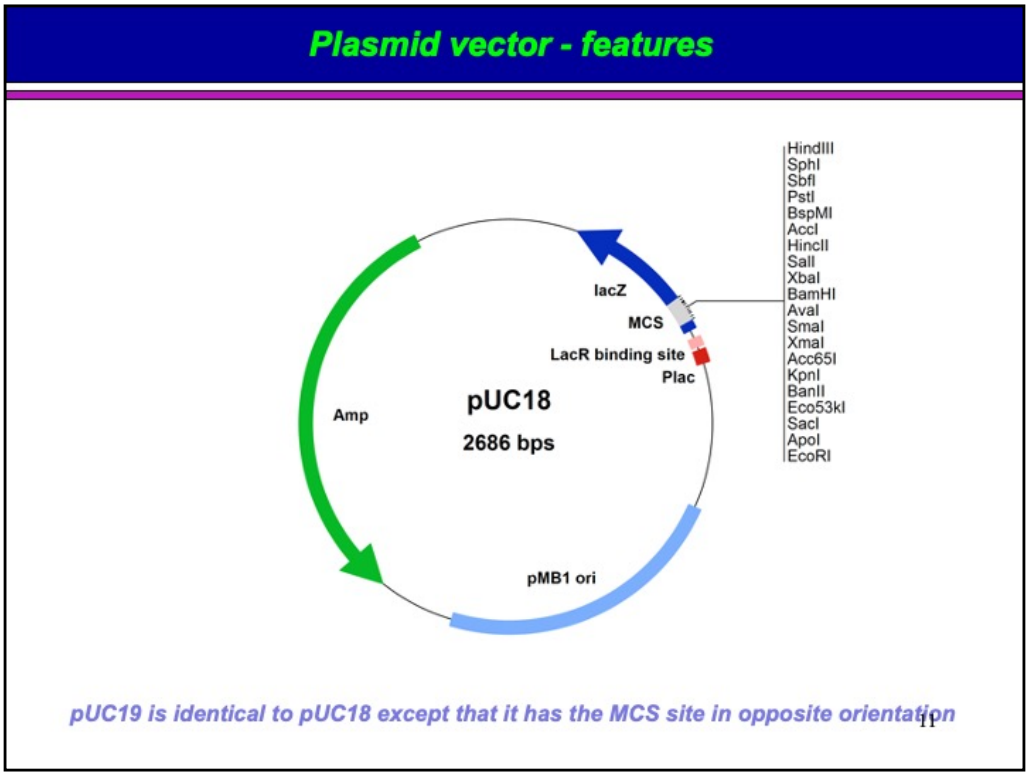


The cloning vector pBR325 has three resistance genes, namely chloramphenicol, ampicillin, tetracycline. with one being inactivated by the cut-and-paste method of gene insertion, and interruption (discontinuation) of the Open Reading Frame (ORF) for the resistance gene. The other two resistance genes remaining intact, with either acting as selection agent on agar plates containing that antibiotic.

E. coli cells containing pBR325 will grow on agar plates containing any of the three antibiotics. However, if we insert a foreign DNA fragment, say within the Tet (tetracycline) gene, at the BamHI site, as depicted in the diagram above, and the recombinant plasmid is introduced/transformed into *E. coli*, those cells with the recombinant plasmid will now be unable to grow on tetracycline-agar plates, but will grow on chloramphenicol-agar or ampicillin-agar.

This constitutes the original method of insertional inactivation and selection of recombinant plasmids. *E. coli* cells with the original vector will grow on tetracycline-agar plates; and hence, this is one simple way of distinguishing between vector (left) and recombinant clone (right) in a mixed population of plasmids in different cells. Note how the insertion of a DNA fragment at a cloning site creates an additional or duplicate site at the insertion point. At the insertion junction of the BamHI fragment, the recombinant plasmid will now have two BamHI sites.

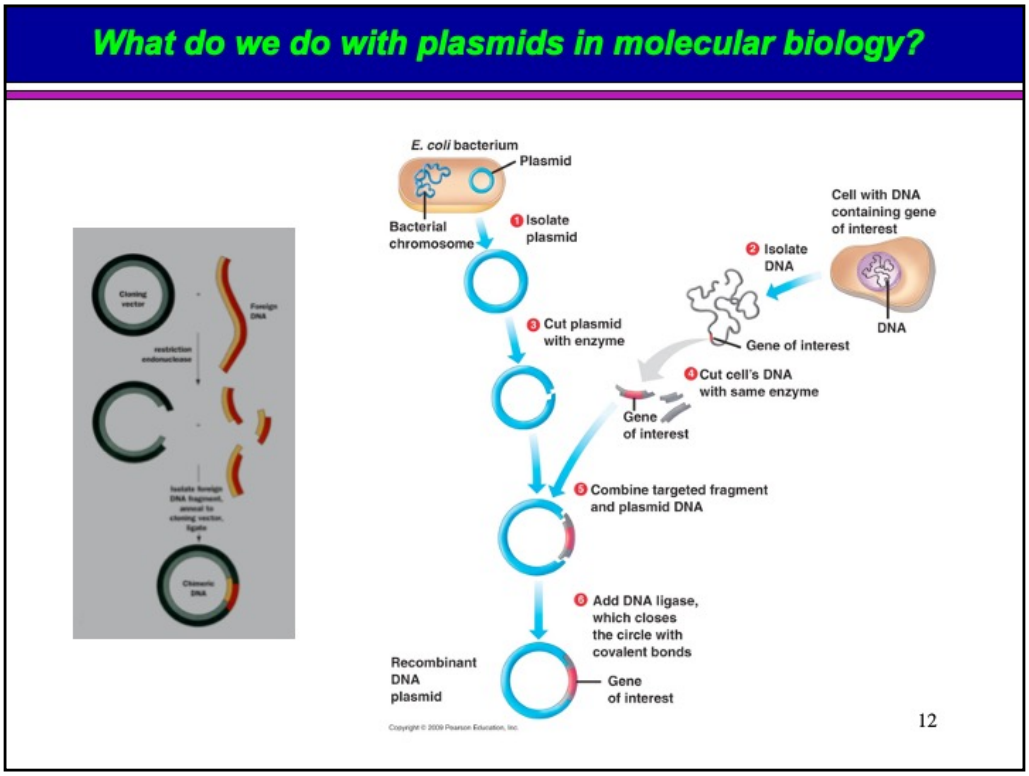


A very small plasmid vector, developed in the mid-1980s and still used as a universal cloning vector. Note that it is smaller (~2.7 kb) than pBR325 (6 kb) and pBR322 (4.3 kb), meaning that it will generate a higher copy number per bacterial cell – and therefore a higher yield of plasmid DNA. It has only one antibiotic resistance gene (beta-lactamase, which hydrolyses and hence inactivates ampicillin) for selection and maintenance in a bacterial culture.

LacZ is the first portion of the lacZ gene, comprising the promoter/operator region and the first 30 amino acid codons of beta-galactosidase, which runs out to 1021 amino acids in total. The lacZ amino acid codons (dark blue arrow) overlap the MCS (Multi-Cloning Site). The LacR repressor binding site (orange) is just "upstream" (as we say) from the beginning of the promoter site (red) and it also contains the operator site, which is part of the LacZ gene's promoter (in red). This expression system allows a cloned gene to be held 'silent' (not expressed at all), or to be switched on (expressed) at different rates (like water from a tap). The overall intention is to produce a protein product from the cloned gene, but under controlled conditions (more on this in later lectures).

The MCS region contains an engineered short sequence of DNA with many single restriction enzyme sites in tandem as indicated. The restriction enzyme sites are unique to the MCS and thus do not occur again anywhere else in the plasmid. The partner plasmid to pUC18, namely pUC19, is identical in all respects except for the opposite orientation of the MCS. We will be using pUC18 in several of the practical classes.

This and all subsequent plasmid cloning vectors do not use the antibiotic insertional inactivation system, and instead use a different type of insertional inactivation, hence the need for only a single resistance gene (most usually ampicillin resistance, as shown here for pUC18). We will be looking at the pUC18 method of insertional activation in a later lecture.



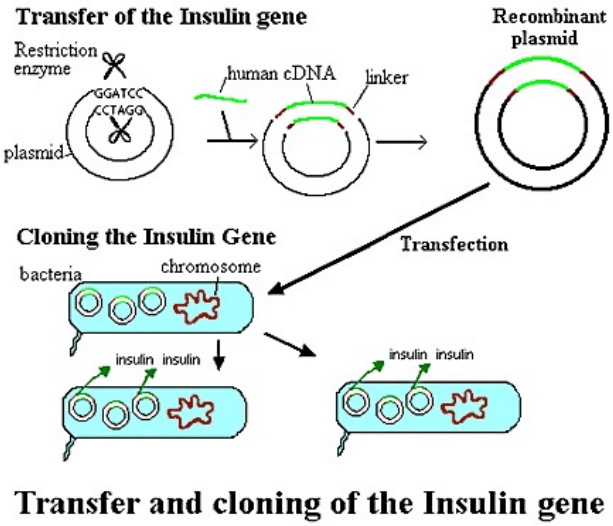
The simple scheme on the left illustrates the way in which cloning vectors (and restriction endonuclease enzymes) are used to insert foreign DNA, the 'cut-and-paste' method, resulting in the formation of a recombinant (hybrid, or chimeric) plasmid.

The scheme on the right depicts the steps involved in cloning experiments, in its simplest iteration using plasmid vector and foreign DNA fragments prepared with enzyme-generated compatible ends. In this way, the gene of interest will be cloned into the vector.

The crucial point of cloning in both these representations is the insertion of foreign DNA into a cloning vector. In this way, the resulting recombinant plasmid "perpetuates" the gene, or any DNA insert, independently of its original host and enables a gene to be processed and manipulated and studied by a number of methods and technologies, including: gene mapping, DNA sequencing, sub-cloning, amplification of copy number, site-directed mutagenesis, to name a few that we will be looking at in later lectures.

Although not obvious in these simple examples, you will note that, in each case, a number of different potential insert fragments are generated and therefore a family of recombinant plasmids will ensue. Only one example is depicted in the schemes above, but cloning often involves a so-called 'shot-gun' approach, in which many inserts result in many recombinant plasmids that will at some point, and in some way, will need to be sorted/selected for the recombinant of interest – as we will see in a later lecture.

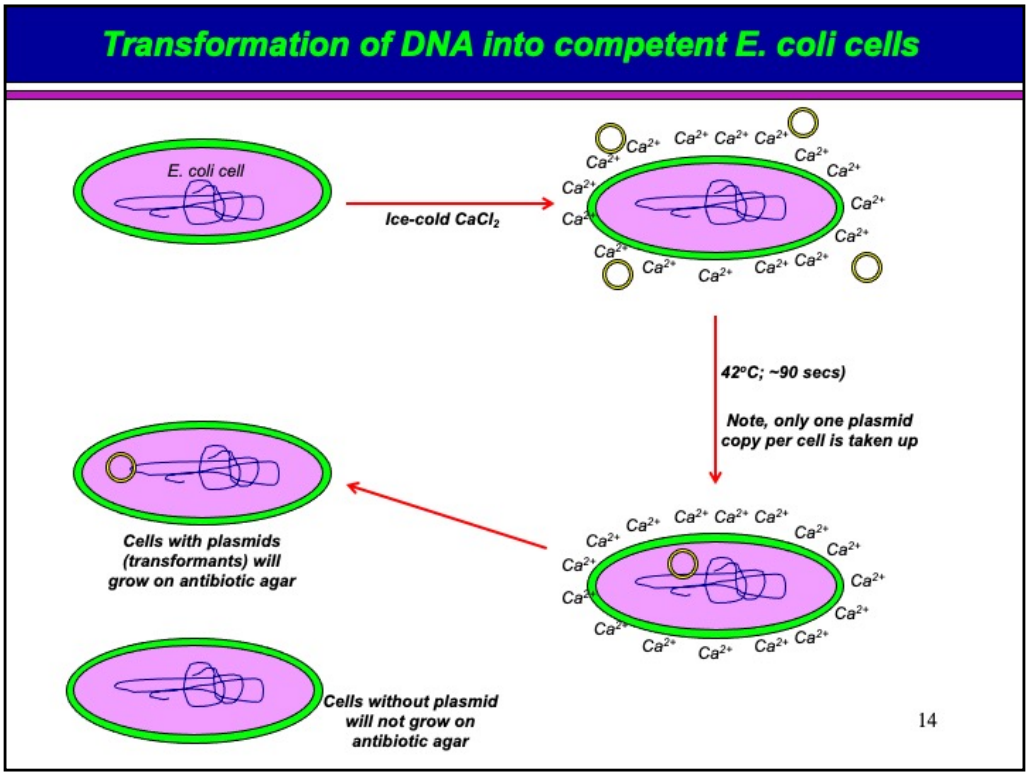
The first human genetic engineering experiment (1979)



Transfer and cloning of the Insulin gene

13

How insulin became available using genetic engineering in 1979, which superseded the older method of obtaining insulin from the pancreases of cattle. However, the production needed scaling up some 50-fold, so the replacement of cattle-derived insulin with bacterial recombinant insulin was not achieved until 1983. This slide is only an interlude to show how recombinant plasmids were first employed in the fight against diseases. We will look at other examples in more detail in a later lecture.



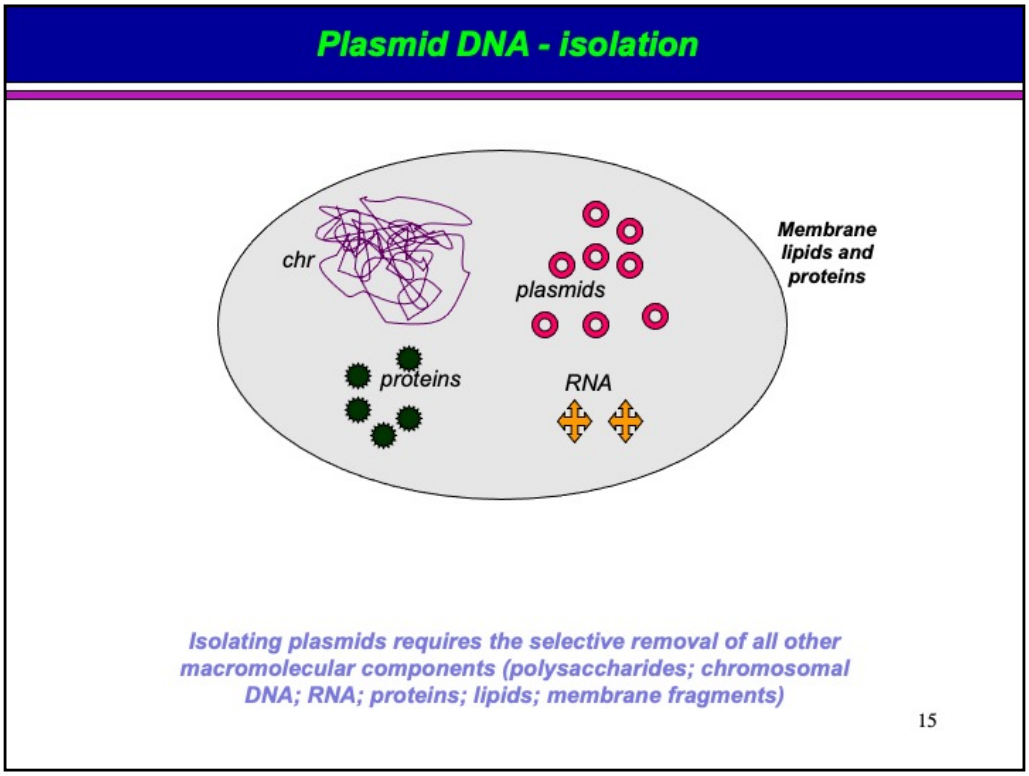
The process of **transformation** in steps that outline: The ice-cold calcium chloride treatment; the addition of plasmids; the transient passage of a plasmid into a 'competent' *E. coli* cell; and the survival of the transformed cells on antibiotic-containing agar plates.

Most "competent" cells do not receive plasmids (only about 3% of cells are transformed) and do not grow on the agar plates, as they are killed by the antibiotic in the agar (bottom left). Only transformed cells are able to grow in the presence of antibiotic, as this property is conferred by the plasmid (middle left). Separate colonies will grow from each plasmid-containing transformed cell. Transformation efficiency can be enumerated by calculating the number of transformed cells (transformants) per µg of plasmid DNA. The blue squiggly line in each cell is the chromosome.

The essential features of this process are: *E. coli* cells are initially made 'competent' for transformation by washing them in several changes of ice-cold calcium chloride solutions. This results in the adsorption of many calcium ions (Ca^{2+}) to the outer cell membrane (top right). The cells are now **competent** for transformation. The cells need to be kept ice-cold, to allow the calcium ions and DNA to remain bound (adsorbed) to the cell envelope. The plasmid is adsorbed to the calcium ions by the electrostatic interaction of the Ca positive and DNA phosphate negative charges. [What happens to the sodium ions that usually form charge neutralizing clouds around the negative phosphates of the nucleotide bases?].

A short high (42°C) temperature pulse results in the cells becoming transiently (temporarily) permeable to the plasmid DNA, with the high temperature also dislodging the Ca-DNA interaction, thus enabling the plasmid DNA to cross the cell envelope. Once the plasmid enters the cell, it is primed to replicate and be inherited by successively dividing *E. coli* cells, which grow happily on the antibiotic containing agar and form colonies when the plates are left overnight at 37°C (middle left).

Each cell, though only transformed by a single plasmid, will rapidly replicate many copies of the plasmid in each of the dividing progeny within the colony, or in broth culture.



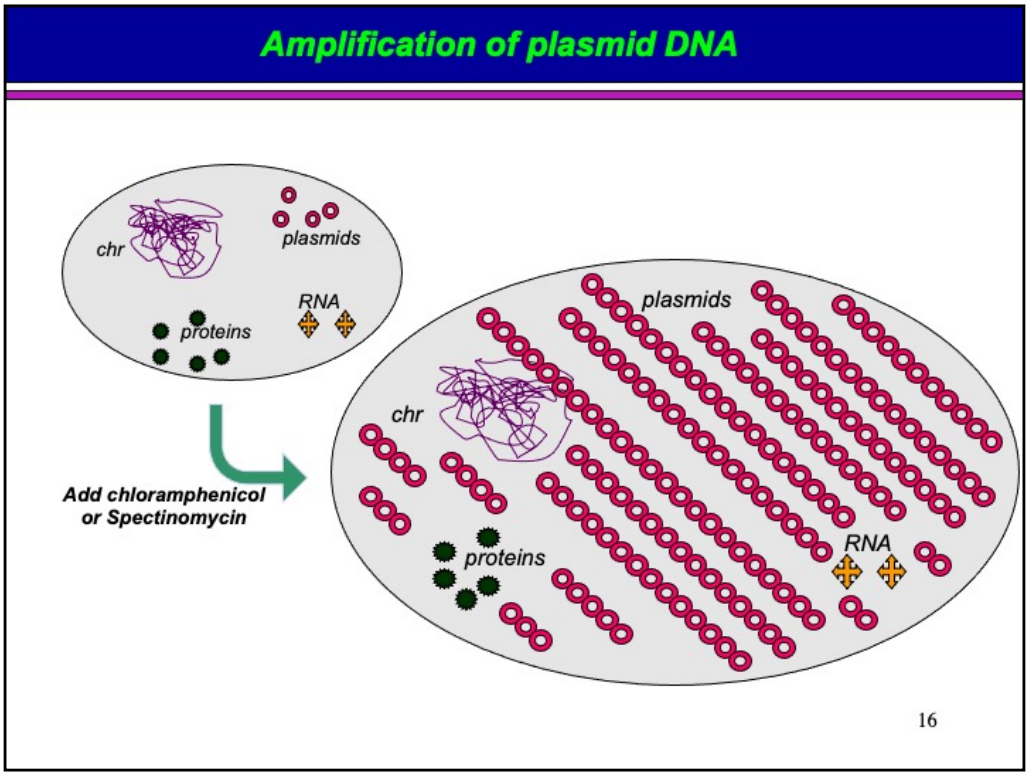
This cartoon of a bacterial cell shows the major macromolecular types, of which there are four in the cell cytoplasm and two in the cell membrane. In some bacteria, polysaccharides are also present in significant quantity.

There are three major considerations in nucleic acid isolation and purification procedures:

(1) How to maximize the yield of plasmid DNA (by antibiotic amplification);

(2) How to isolate and purify just the nucleic acid of interest, that is, only chromosomal DNA, or only RNA, or in this instance, only plasmid DNA; and

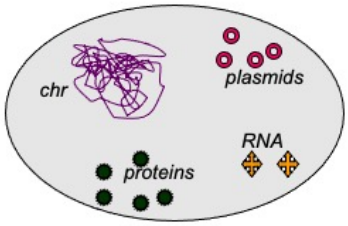
(3) How to remove cellular debris (protein; lipid; polysaccharide) during the isolation procedure. These processes are explained in detail in the next slides and in the lab manual.



Amplification of plasmid DNA is achieved by adding chloramphenicol or spectinomycin to the early-log phase of the vigorously aerated culture growing at 37°C. Chloramphenicol and spectinomycin are protein synthesis inhibitors and when either is added, the culture immediately ceases to grow and divide, that is, there is no further increase in the number of bacterial cells or in the cellular mass of undesirable macromolecules (protein, lipid, polysaccharide, or chromosomal DNA).

Although protein synthesis is stopped, the plasmid DNA, which is under relaxed replication control, continues to be copied to very large numbers as the culture is aerated overnight. The relaxed replication of the plasmid is not linked to the replication of the bacterial chromosome or division of the cell, both of which are inhibited by the presence of the antibiotic. The net result next morning is a "cleaner", smaller mass of culture (~1 x 10^6 vs ~2 x 10^9) with not a lot of contributing unwanted macromolecules, but with a very large number of plasmid molecules (many hundreds of copies versus around 20 to 50 or so in a fully saturated overnight cell culture.

Plasmid DNA - isolation



STEPS

1. Alkaline lysis (SDS buffered at pH 12.6).
2. Neutralize with potassium acetate (KOAc) (pH 4.5).
3. Centrifuge to remove denatured chr DNA, protein, lipid, SDS, KOAc.
4. Digest RNA with ribonuclease enzyme.
5. Purify plasmid DNA by precipitation in salt/alcohol; or by silica column binding and elution.

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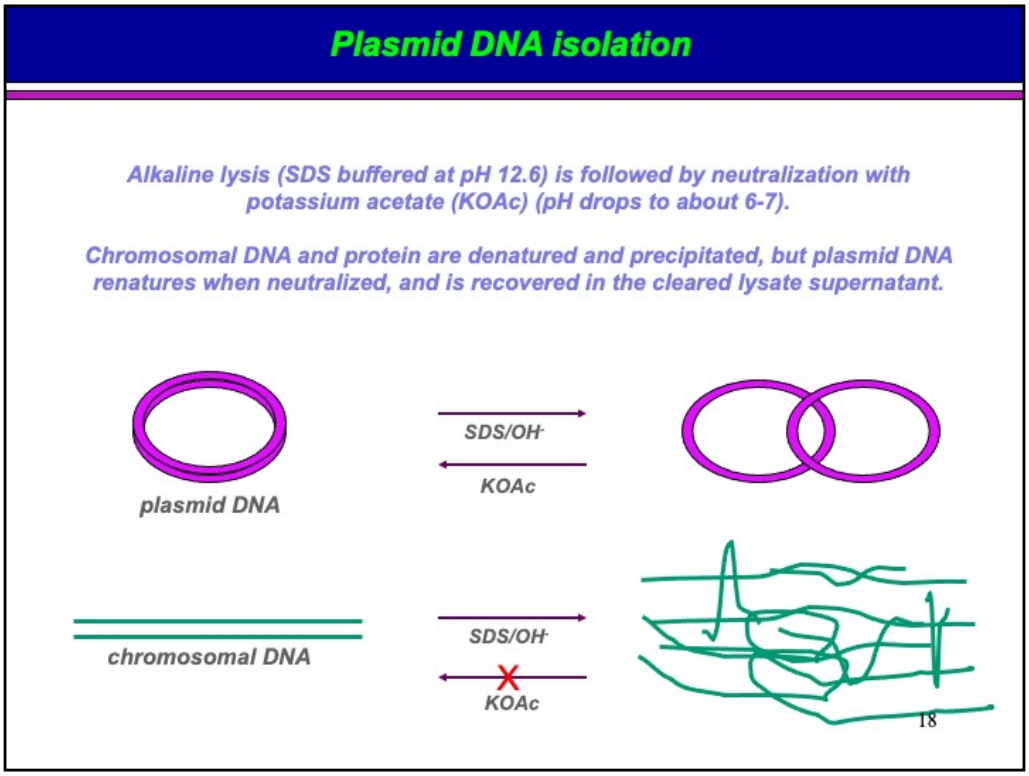
This slide and the next two summarise the 'alkaline lysis' 'method of choice' for the isolation of plasmid DNA from bacterial cultures. It works very well except for large plasmids (> 100 kb) and is the basis of all commercially available plasmid isolation and purification kits – one of which we will be using in Experiments 4 and 6 (Promega SV Minipreps Kit).

The fundamentals of the alkaline lysis method are explained briefly here and the next slides, but is more fully covered in the laboratory manual. Essentially, the purpose of the method is to obtain a cleared cell lysate that purifies the plasmid DNA away from the major contaminating macromolecules (chromosomal DNA, proteins, membranes), which are all precipitated during the lysis procedure. Note, although we have broached amplification of plasmids in the previous slide, it is not always necessary, especially if the plasmid vector is known to be present in high copy number, which is so for pUC18 and many related vectors.

The technical nuances of the steps listed above will be explained on the next slides and in the lab manual. Recall, as we saw for the isolation of genomic DNA, that the RNA may be retained in the cleared lysate along with the plasmid, but most kits include ribonuclease A in the cell pellet resuspension buffer and RNA is digested away before the column purification step.



The basis of the alkaline lysis method. Chromosomal AND plasmid DNA are denatured in alkali (left to right), but only the chr DNA, which is linear and fragmented, is irreversibly denatured, becoming largely single-stranded, and entangled with membrane fragments, proteins, lipids, polysaccharides, SDS, and KOAc (pH 4.5), which collectively forms into a precipitate (aka, "snow") and is easily removed (pelleted) by high-speed centrifugation, leaving a cleared lysate.

On the other hand, the much smaller, unbroken, but now single-stranded and denatured plasmid DNA remains linked (as above), as it is in a closed circle configuration. The KOAc-neutralised cleared lysate supernatant contains the renatured plasmid, as double-stranded closed circles (back on the left). RNA, if present, is also rendered single-stranded, but consists of much smaller molecules and is not significantly entrapped with the "snow". It survives the process and will carry through to the purification stage, which is why is is often removed br prior digestion in the resuspension step.

Plasmid DNA isolation

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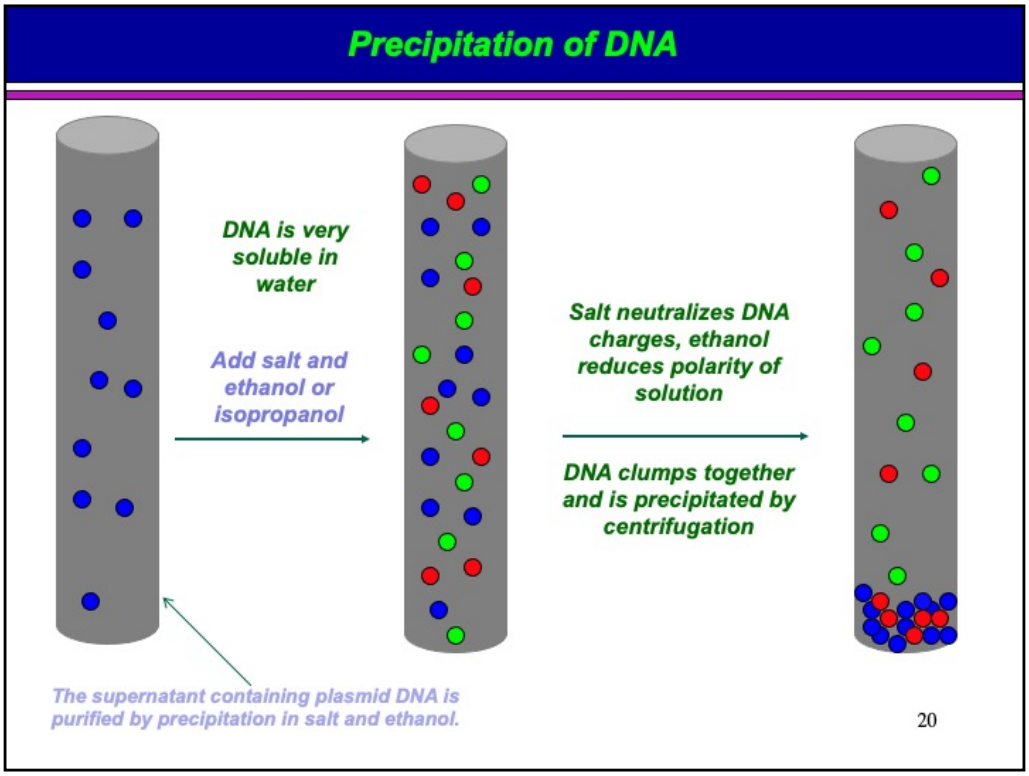
The “snow”, shown in the tube on the right, is a precipitate of aggregated chromosomal DNA, membrane fragments, proteins, lipids, SDS, and potassium acetate. This aggregated snow is removed as a pellet in high-speed centrifugation, leaving a cleared lysate supernatant, which contains plasmid DNA.

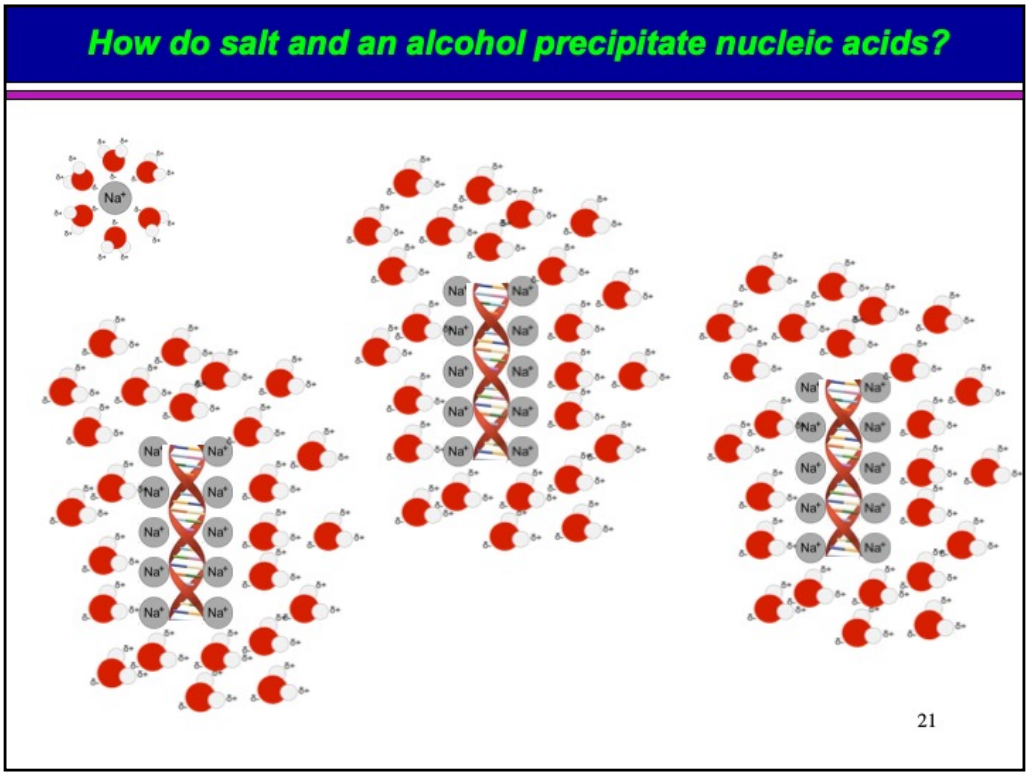


Nucleic acids may be precipitated by high-speed centrifugation in buffered salt-ethanol. DNA is very soluble in water because it is an acid (negative phosphate charges on each strand). The addition of NaCl brings about the neutralization of the charges because Na ions bind to the phosphates.

Ethanol, or isopropanol, lowers the dielectric constant of water, that is, it reduces its polar solubilizing effect by breaking the H-bond network between water molecules themselves, and between water and the DNA, with the alcohol sequestering (partnering) much of the released water in a solvated network of hydrogen-bonds. This enables the neutralized DNA molecules to clump or aggregate and then be collected into a pellet following high-speed centrifugation.

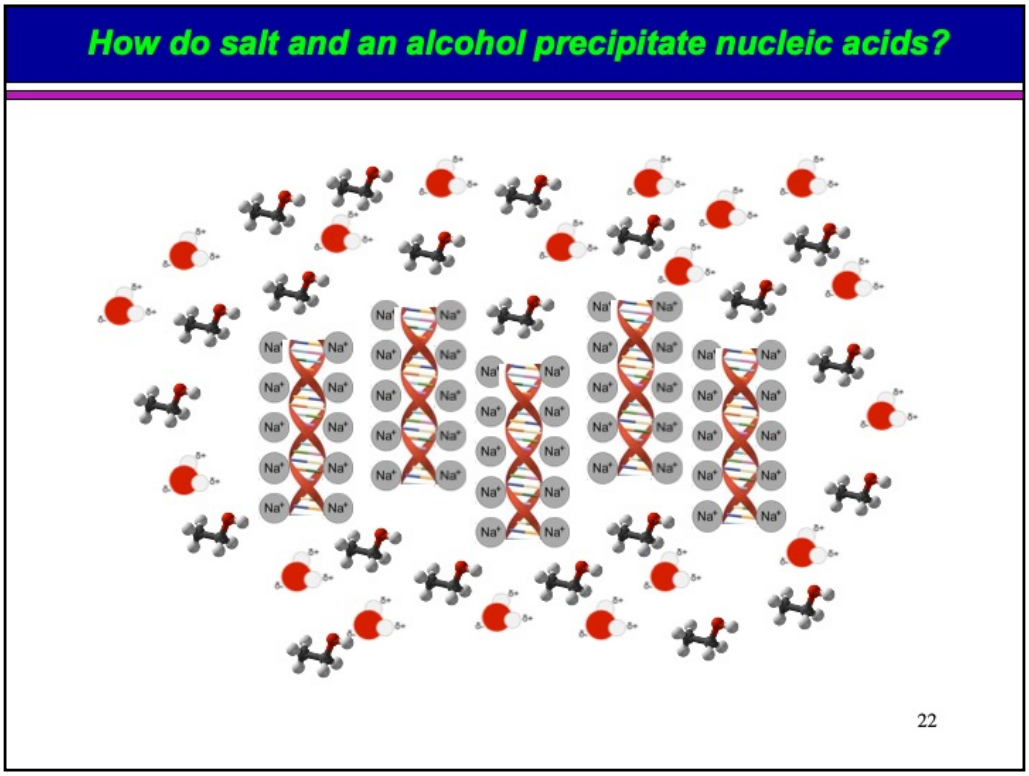
This slide is repeated from Lecture 3, which demonstrated the same process of collecting/purifying chromosomal DNA in salt-isopropanol.

Although we will be using a column purification method for the plasmids, the principle of the different DNA precipitation/binding methods is illustrated on the next few slides.



This slide is the first in the two-stage process of DNA precipitation in salt-alcohol. At top left, you can see the solvation of sodium ions by water.

After cells are lysed, the DNA, which was 'in solution' within the cell cytoplasm, continues to be solvated in the cleared lysate, which contains a buffer with a high salt content. The sodium ions are in excess and continue to be solvated by water, as shown at top left. However, an extensive secondary solvation occurs, in which sodium ions form ionic interactions with the negative phosphates on the DNA, as indicated in three representations in this slide. The sodium ions are themselves surrounded by many water molecules that set up a H-bonded network with the sodium ions. In this situation, the DNA molecules are still solvated/dissolved in the solvent.



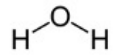
The second stage of the salt-alcohol precipitation process.

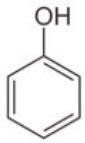
When ethanol (or isopropanol) is added to the solvent/solute mixture (water + sodium salt + DNA), the water is dispersed by the ethanol and the hydrogen-bonding network is broken. Now, the DNA molecules (with phosphate charges still neutralised by sodium) are able to associate and aggregate in this far less polar environment and precipitate into a pellet during high-speed centrifugation.

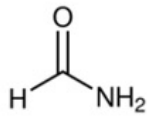
Alcohols are less polar than water and are said to decrease the dielectric constant of the water/alcohol mixture, that is, to make the combined solvent mixture of alcohol plus water less polar and less likely to allow extensive solvent H-bonding and therefore less able to hydrate or solvate the DNA. This is manifested by a new network of hydrogen bonds between the polar ends of the alcohol (-OH) and water molecules. Dielectric constants are explained on the next slide.

Dielectric Constant = Polarity of a Solvent

Solvent	Miscible	Dielectric Constant
Water	✔	80.4
Ethanol	✔	24.3
Isopropanol	✔	17.9
Phenol	✗	4.3
Chloroform	✗	4.8
Glycerol	✔	42.5
Diethyl ether	✗	4.3
Benzene	✗	2.3
Acetaldehyde	✔	21.1
Acetone	✔	20.7
Acetic acid	✔	6.2
Formamide	✔	111







23

A table of some solvent Dielectric Constants (DC). The higher the value, the more soluble (miscible) the solvent is in water. Note that formamide has an even higher DC than water! How is this possible? In fact, they stand first and second among solvents. Check out the formulae on the right. Formamide must have stronger dipoles than water.

Ethanol is more soluble than isopropanol (the latter has an extra methylene group making it less polar than ethanol). EtOH is more soluble because it has a higher H-bonding (solvation) propensity in water than isopropanol.

Note that phenol (benzene with an hydroxyl group) is only slightly more soluble than benzene itself. Phenol has a lower DC than chloroform ($CHCl_3$), which might be surprising except for the fact that phenol is 'burdened' with a very non-polar aromatic ring. In general, aromatics are less soluble than aliphatics. Miscibility, and therefore solubility, in water requires a DC of greater than 15.

Lastly, note the related solvents acetaldehyde (an aldehyde), acetone (a ketone) and acetic acid (a carboxylic acid). Both acetaldehyde and acetone are freely soluble in water and have relatively high DC values. Yet acetic acid, which you would expect to be even more soluble than the other two is in fact less soluble! Acetic acid (and carboxylic acids generally) are weak acids in that the hydroxyl proton does not dissociate very much (~4%) due to the propensity of the carboxyl group (-COOH) to readily form H-bonds with other molecules of acetic acid and less so with water, hence less solvation/ionisation. Its pKa is about 4.75. Strong acids have pKas < 1.

Plasmid DNA – isolation using spin columns

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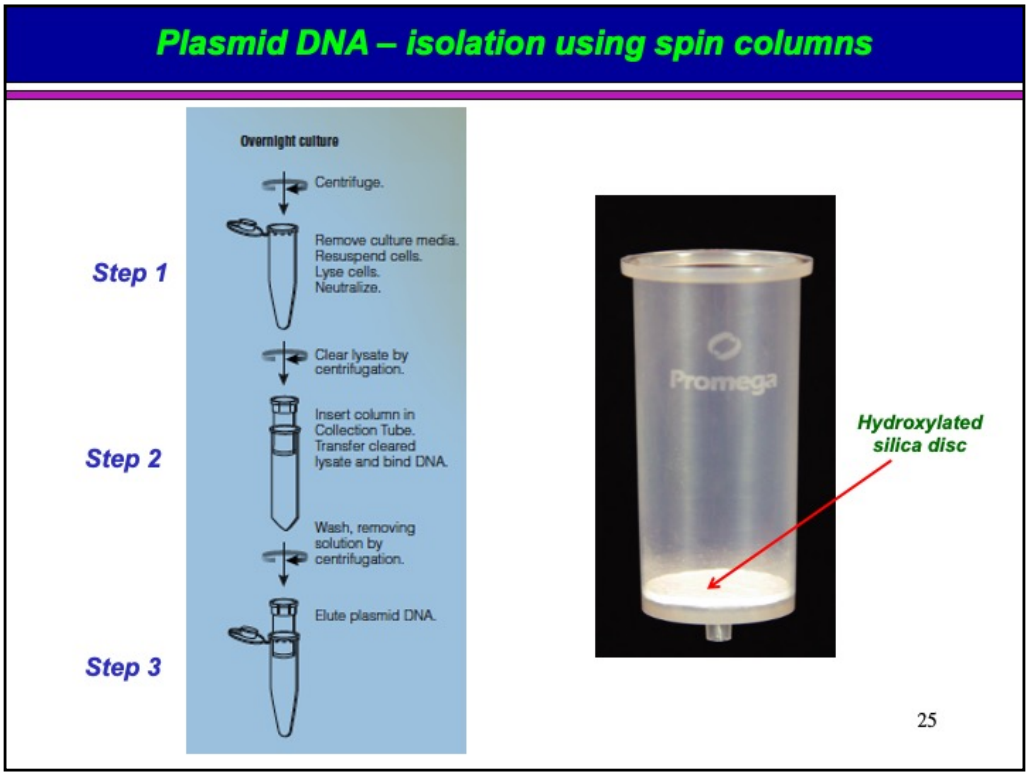
Promega SV Minipreps Kit for isolating plasmid DNA using spin columns. The collection tubes and silica disc columns are in the bags (bottom).

The bottles contain the proprietary alkaline lysis solutions. These are: the cell pellet resuspension buffer (Solution 1), containing RNaseA to digest retained RNA in the cleared lysate; Solution 2, the alkali/SDS mixture (pH 12.6); and Solution 3, the potassium acetate (KOAc) neutralising solution (pH 4.6). The ethanol/salt wash buffer and the eluting pure water solvent are also supplied in bottles. You will be isolating plasmid DNA using this kit in Expts 4 and 6.



The Promega SV Mini-plasmid isolation method: A column-based scheme of plasmid alkaline lysis isolation and purification scheme from bacterial culture to purified plasmid DNA.

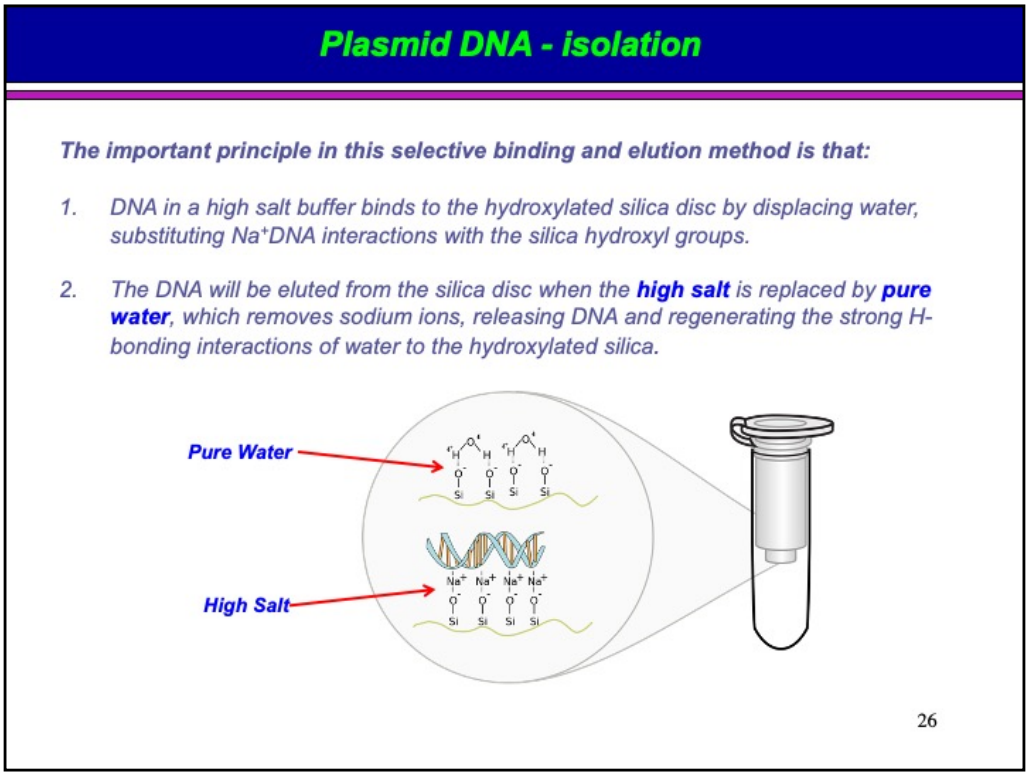
The cupule with an hydroxylated silica disc (at right) is inserted into a collection tube (left panel, step 2). The cell resuspension, alkaline lysis, and neutralising steps are not depicted here. They were shown earlier, with the resulting 'snow' removed by high-speed centrifugation (Step 1), yielding a cleared supernatant, which is added to the silica-disc column inserted into a collection tube (Step 2).

The hydroxylated silica disc (arrowed in the panel at right) binds DNA in a high salt buffer, but allows all other cleared lysate material to pass through in a brief centrifugation (Step 2). Residual adsorbed contaminants are removed in successive 70% ethanol/salt washes.

Plasmid DNA is released in a small aliquot (~100 µl) of pure water (Step 3).

Each step (cell lysis; binding; ethanol wash; water) involves brief high-speed centrifugations.

The 'secret' to how the silica-OH disc binds DNA in high salt, but releases it in pure water, is explained on the next two slides.

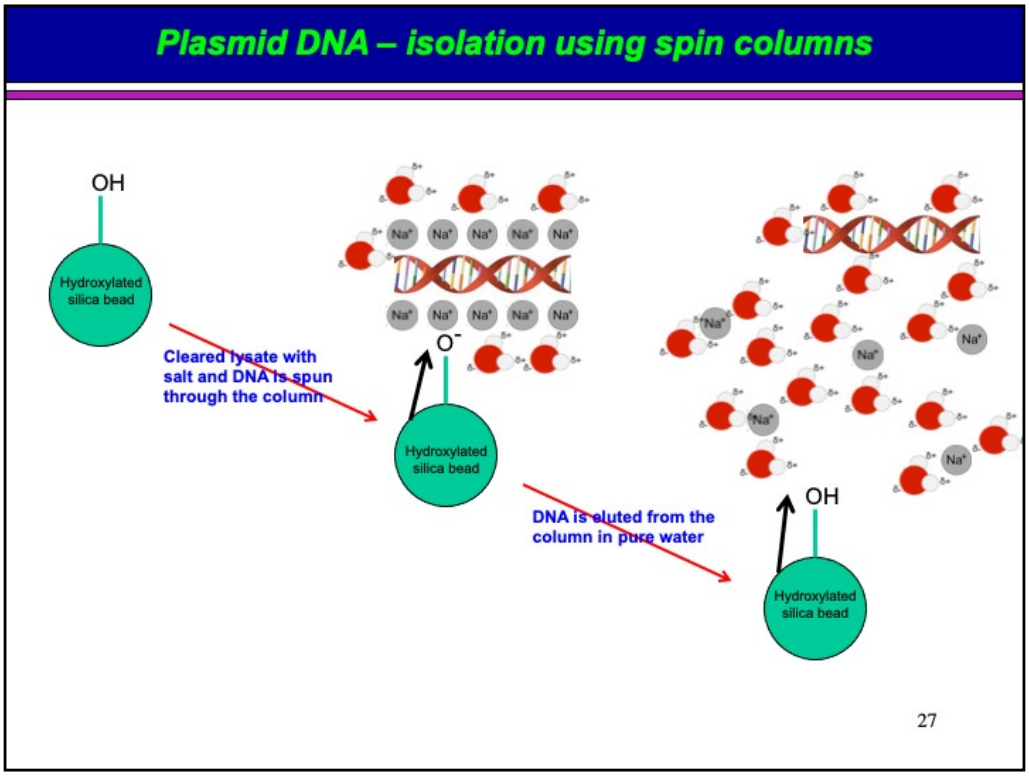


Modern alternatives to salt-alcohol precipitation use (negatively charged) hydroxylated silica columns to bind DNA from cell lysates, then elute it later in pure water.

The principle of binding and elution of plasmid DNA in silica columns is illustrated in the diagram and described in points 1 and 2.

DNA is negatively charged, and so is the silica disc, so ordinarily they would repel each other. However, the use of high salt buffer disrupts the shell of H-bonded water around the DNA molecules, effectively dehydrating it, and the positively charged sodium ions then act as bridges between the negatively charged silica and the negatively charged DNA (bottom half of diagram). DNA binds to silica gel under high salt conditions, while proteins and extraneous neutral materials do not.

After washing the column-DNA complex in a high salt + ethanol solution, a small shot (aliquot) of pure water will overcome the affinity of the DNA for the silica and the DNA will elute, because the water easily 'invades' the complex and disrupts (pulls away) sodium from the silica. As the sodium is removed the negative charges of the hydroxylated silica and DNA repel one another, and presto, the DNA runs free. The Hydrogen Bond (surprise!) plays a big role in this process (next slide). Recall the slide in Lecture 1 that showed how water "invade" and solvates common salt (sodium chloride).



This slide shows the principle of DNA binding in high salt to hydroxylated silica beads (first black arrow; middle green bead) and its elution in pure water (second black arrow; last green bead).

Normally, the negative oxygens on the silica beads and the negative phosphates on DNA will repel one another, but in high salt, the positively charged sodium ions 'intercede', and form a neutralising 'sandwich' between the silica and DNA, which displaces water and allows DNA-silica binding/adsorption.

When excess pure water is added, sodium ions are pulled away by water, and the negative repelling forces of the silica (O^-) and DNA (PO_4^-) are renewed and result in the DNA becoming detached from the silica and eluted.

How does water bring about this detachment? It occurs because the water molecules form hydrogen bonds readily to the sodium ions and to themselves, thereby disrupting the non-covalent attachment of the DNA to the sodium ions and to the hydroxyl groups on the silica disc. In the diagram above, only a single hydroxyl OH group is depicted for simplicity.

Plasmid-Safe™ ATP-Dependent DNase

Lane 1. SmaI digest of chromosomal DNA.

Lane 2. Purified plasmid DNA.

Lane 3. Lanes 1 and 2 combined.

Lane 4. Lane 3 after a short digest with a plasmid-safe DNase.

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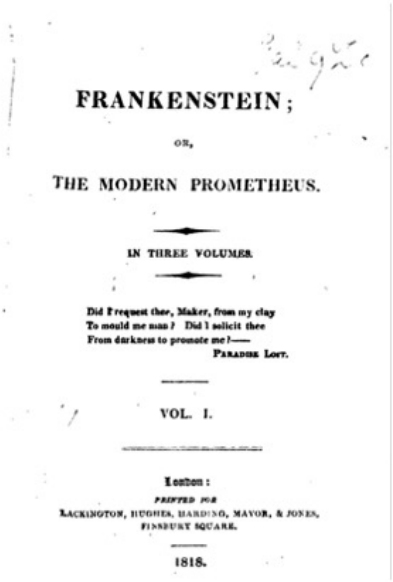
Even the best plasmid isolation procedures and kits may leave traces of contaminating chromosomal DNA. Normally, this might not be a problem, but some techniques require absolute purity so that the wrong DNA source is not cloned accidentally.

This slide demonstrates the effectiveness of a so-called 'plasmid-safe DNase'. It works at slightly alkaline pH and preferentially digests open-ended single or double-stranded DNA (= denatured genomic DNA), but not covalently closed circular DNA (= plasmids).

The Power of Impression and Inference



FRANKENSTEIN;

OR,

THE MODERN PROMETHEUS.

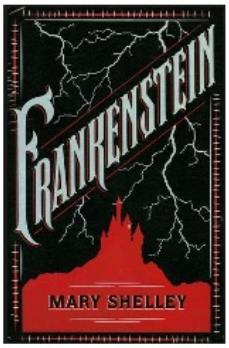
IN THREE VOLUMES.

Did I request thee, Maker, from my clay
To mould me man? Did I solicit thee
From darkness to promote me?—
PARADISE LOST.

VOL. I.

London:
PRINTED FOR
LACKINGTON, HUGHES, HARDING, MAVOR, & JONES,
FINSBURY SQUARE.

1818.



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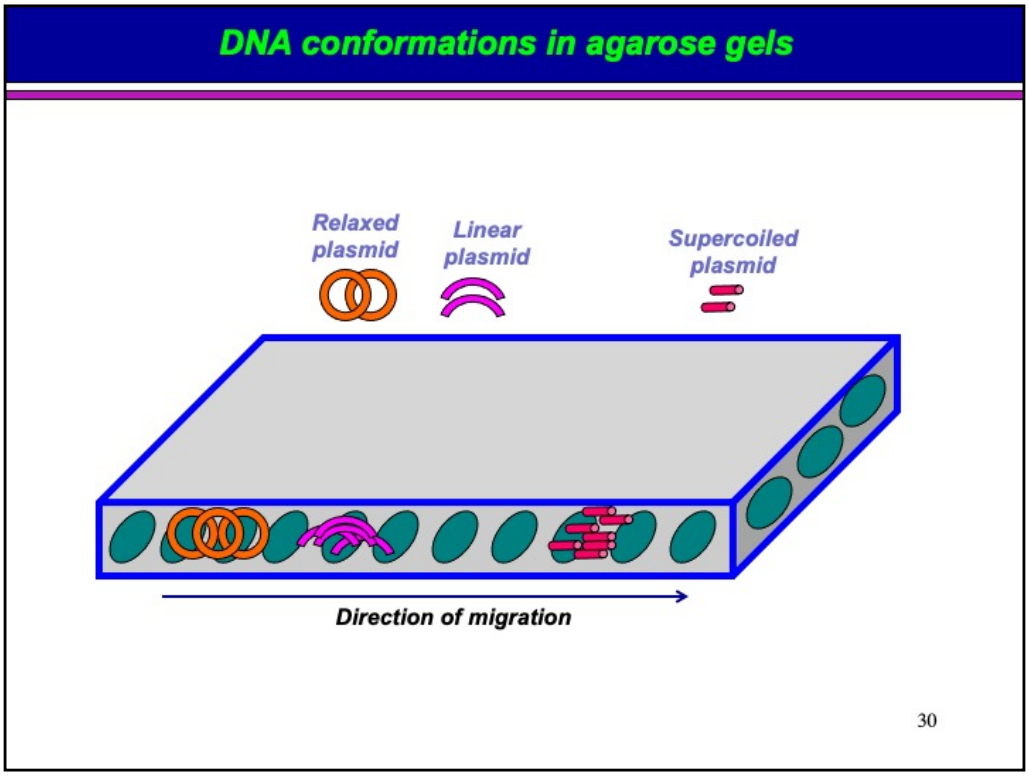
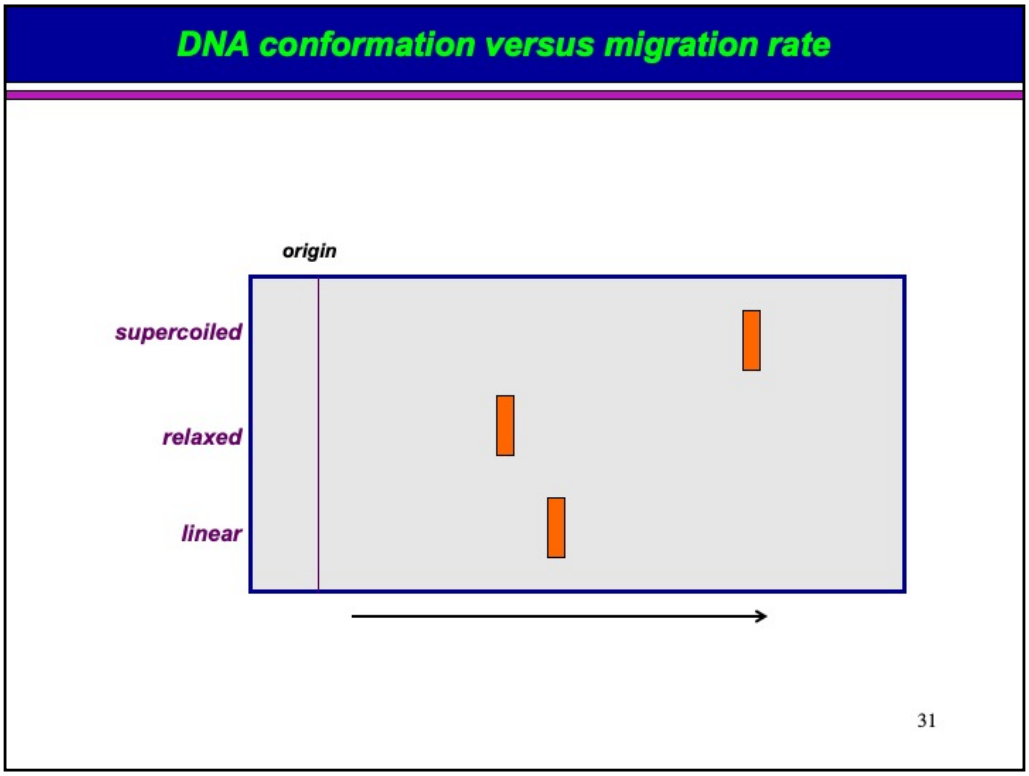
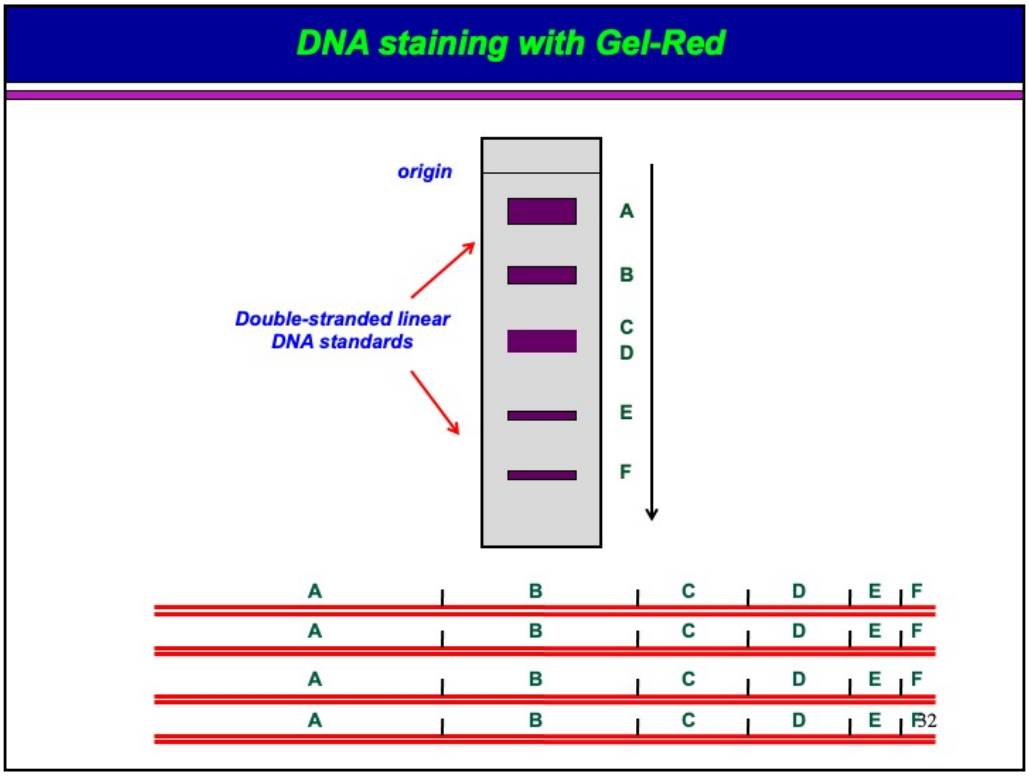


This revision slide indicates how different conformations (geometric shapes) of DNA molecules can affect their migration rates in an agarose gel, even if they are of exactly the same mass (as we saw with pUC18 in an earlier lecture and in Expt 1).

Note, these three DNA types (supercoiled plasmid, relaxed plasmid, linear plasmid) are **exactly the same molecular mass size**, but migrate differently due to conformational differences alone; and this is caused by frictional resistance to movement through the gel pores. The floppier the molecules, the higher is the resistance to movement through the restrictive pores (shown as even-sized green circles in the gel cross-section above). Note that the drawing of the supercoiled plasmids makes them look "linear", but they are actually "squashed" into that apparent shape by the supercoiling effect.



This slide is a follow up (top down view) of the previous one and emphasizes the point about conformation versus migration rate. Three conformations of the **same** DNA plasmid are shown. Only two of these (relaxed, supercoiled) are seen in bacterial cell lysate preparations; and there is much more supercoiled (~95-99%) than relaxed (1-5%) plasmid DNA.



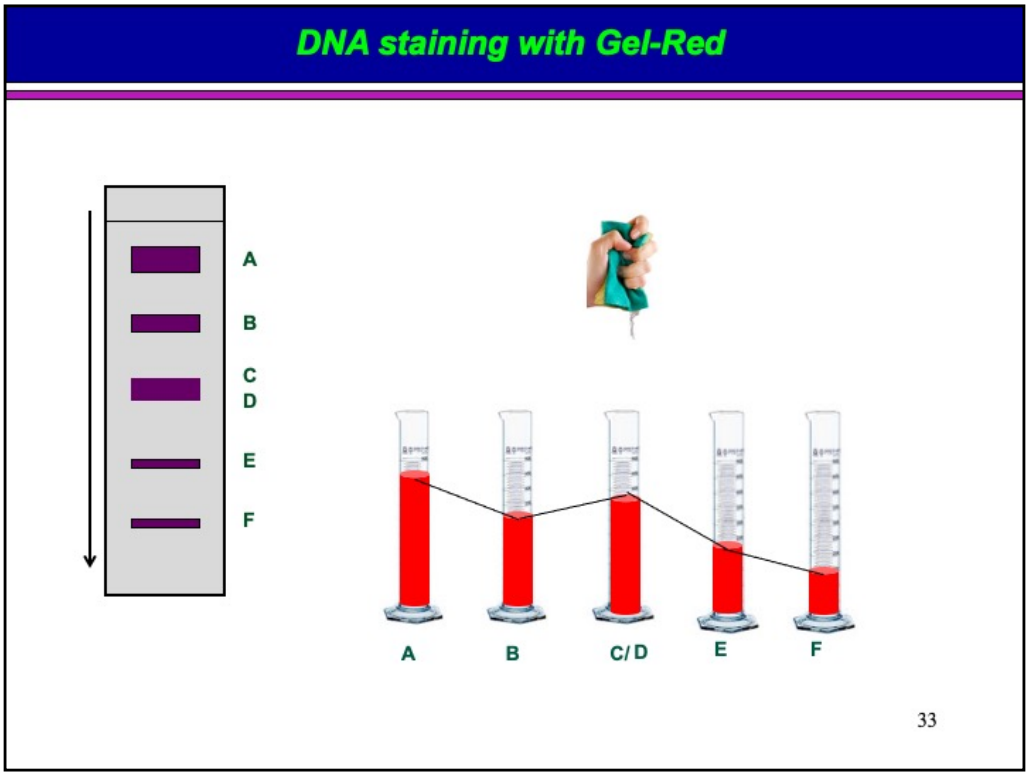
Two simple running and staining principles are illustrated in this slide.

A general rule is that all DNA molecules of the same conformation migrate in agarose gels to positions that are inversely related to the log10 of their molecular masses. The corollary of this principle (see the previous two slides) is that molecules of the same molecular mass but of different conformations, migrate at different rates that are determined by their relative densities (not masses!).

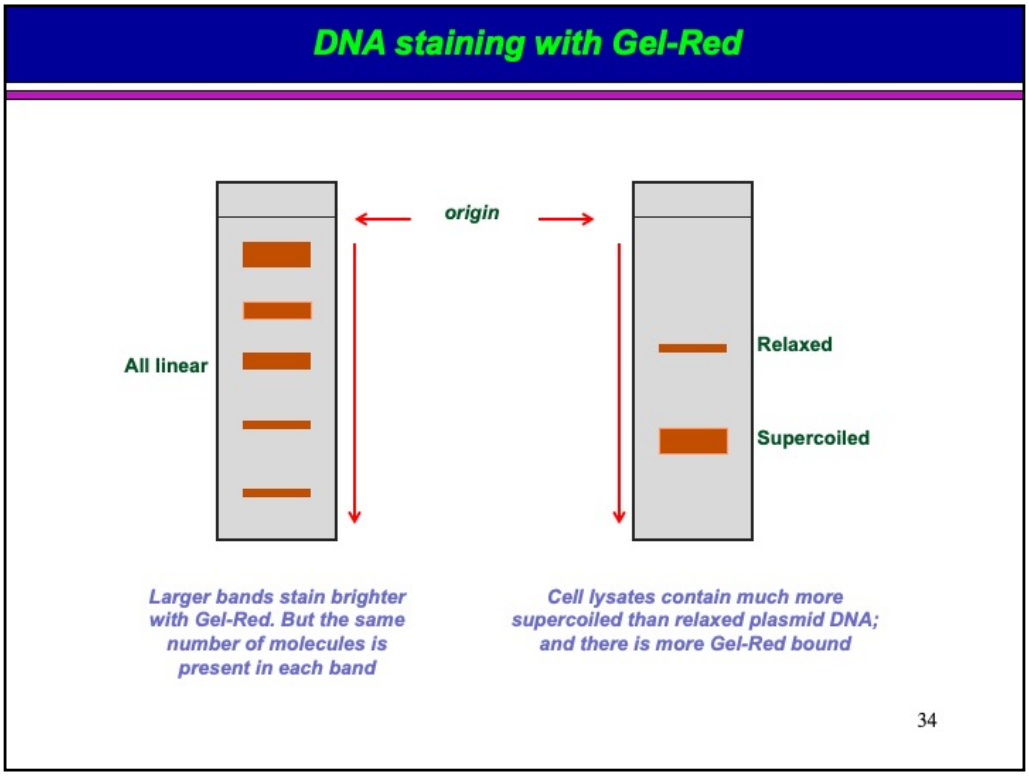
The bands in the gel drawing above show the **degree of binding of dye** (Gel-Red). If all bands are of the same conformation and are, for example, a set of linear standards derived from the same source DNA molecule by being fragmented into standard sizes (as depicted on the labelled DNA molecules at bottom), then there is the **SAME** number of molecules in each band. The upper bands appear broader than the lower bands because they are longer molecules and occupy more space in the gel; and they are brighter because, in being longer, they bind more of the dye.

The equivalent lines at the foot of the slide give the fragment lengths scale for these linear standards, but remember that there are thousands of these same molecules fragmented into segments A-F. Note the anomaly for the 'C' and 'D' fragments. These two fragments do not run separately, but are virtually merged together, as they are nearly identical in size/length. This effect is easily detected in the gel profile because the combined C/D band is about double the width (and Gel-Red stained brightness) than it would be as single C and D bands. So the otherwise smooth transitional reduction in band widths and staining intensity from top to bottom (A.....F) is "disrupted" by the CD **doublet** band.

The amount of dye bound increases with the length/mass of the DNA molecules. A good analogy for this is a long rectangular sponge that you cut into sections like the DNA in the slide (A to F). If you now dip the sections of sponge into a large bath filled with dye then wring out the dye into measuring cylinders, you will see that the amount of dye absorbed by the sponges reflect their sizes, with the volume decreasing as the sponge sizes decrease.



A 'sponge-cylinder' analogy to illustrate the amount of Gel-Red bound to each DNA band in the gel at left.



The first gel panel is a reminder of the principle seen in the previous slide, but this time without the doublet effect. The second panel shows the two major circular plasmid bands obtained from bacterial cell lysates, namely, relaxed and supercoiled.

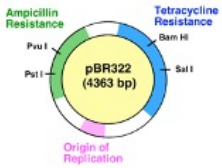
Plasmid DNA is present in cells in the supercoiled conformation (vegetative or resting state). When plasmid DNA is isolate from cell lysates, some of the supercoils become "nicked" by the physical stresses of the isolation procedure (shaking, vortexing, pipetting etc) and this causes breaks in some of the strands - usually only one strand in the doubled-stranded supercoiled molecule so that the plasmid remains circular, but now assumes the relaxed conformation. The amount of relaxed plasmid DNA from cell lysates seen in agarose gels depends on the severity of the isolation procedure and on the size of the plasmid (larger plasmids will be more likely to be nicked) and can vary from a few % to about 25% of the plasmid yield.

The footnotes to both gel profiles above are very important! The number of DNA molecules of DNA in each band in the left gel drawing of linear standards is the same, because they were derived from the same original longer molecule. By analogy, say you take a bundle of very long spaghetti (say 100 strands) and you break it progressively into shorter and shorter sections (like the 5 standards above). The bundles of spaghetti – though of different lengths have the **same** number of strands, but they differ in length, hence the 'sponge' analogy in the previous slide to see how much dye is bound by each band.

In the right gel drawing, the wider and brighter (lower) supercoiled band has many more molecules of the plasmid and binds much more dye than does the upper relaxed band.

Take home message from this slide: The "upside down effect", where linear bands are larger and bind more dye higher up in the gel than the lower bands, whereas circular plasmid bands have the much more abundant and more brightly stained broader supercoiled band lower in the gel than the relaxed band higher up.

MCQ L4



In a molecular cloning experiment using the cloning vector pBR322 above, a foreign DNA insert is cloned into the BamHI site of pBR322. The resulting recombinant plasmid will be….

(A) Ampicillin resistant and tetracycline resistant.

(B) Ampicillin susceptible and tetracycline susceptible.

(C) Ampicillin susceptible and tetracycline resistant.

(D)Ampicillin resistant and tetracycline susceptible.

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MCQ L4

We have seen how plasmid vectors may be amplified to large numbers due to their relaxed origin of replication. However, naturally occurring single copy plasmids are under stringent replication control that is synchronised to the replication of the bacterial chromosome. Stringent replication control is necessary for host single copy plasmids because….

(A) It is the only way that the plasmid is able to be replicated in the host cell.

(B) Natural plasmids are too large to be replicated independently of the host chromosome.

(C) Plasmid vectors, unlike natural plasmids, are not found in nature and they have been designed to replicate independently of the host chromosome.

(D) Natural plasmids are associated with the chromosome and are replicated synchronously with the chromosome, but are not always inherited.

(E) It's all about inheritance. It is the only way that the host natural plasmid may be replicated and inherited by the dividing bacterial cells.

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MOLECULAR BIOLOGY 1
Lecture 5

Restriction Endonucleases

DNA Ligation

1

This lecture introduces two of the most important methods in molecular biology; namely, (i) the use of restriction endonucleases to cut double-stranded DNA; and (ii) DNA ligation, to paste it back together. These are also used in the practical course in Experiments 5 and 6. This lecture therefore, like several others, blends learning from lectures and practicals in a way that reinforces and underpins both arms of the subject. These methods and themes also blend into the two assignments. You may decide to write an MCQ on a topic from this lecture in Assignment 1A. The material will certainly be relevant to Assignment 1B that you will attempt next.

Lecture 5: Learning Objectives

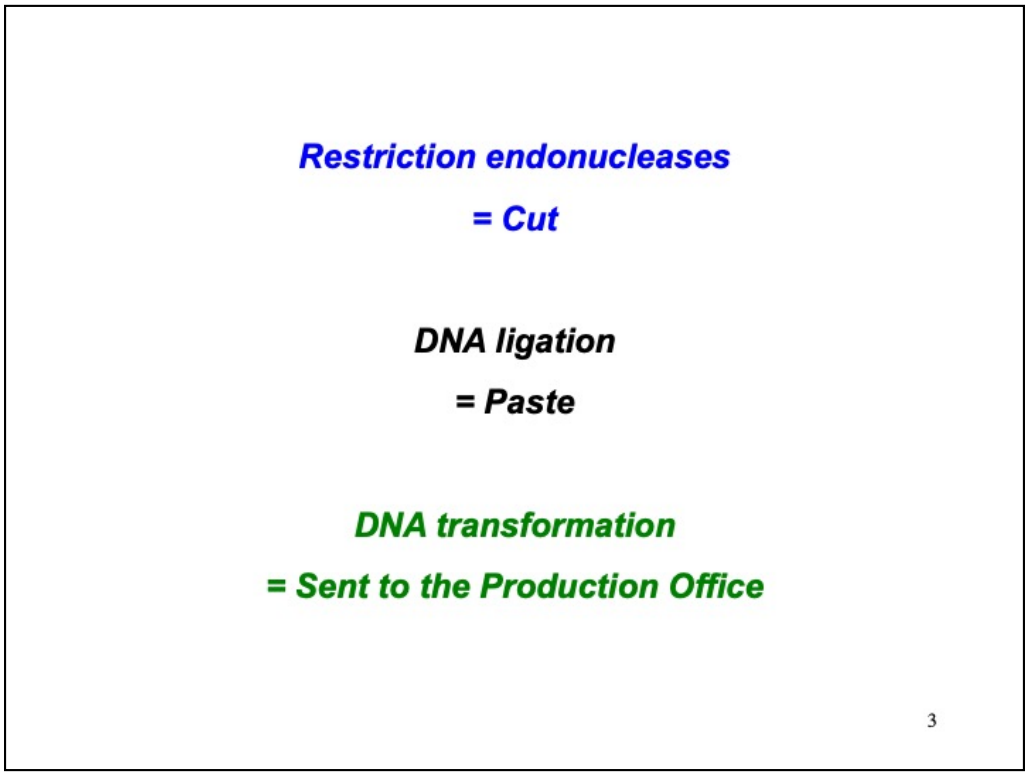
Restriction endonucleases

- ***What are restriction enzymes? How do they work? How are they used in molecular biology?***

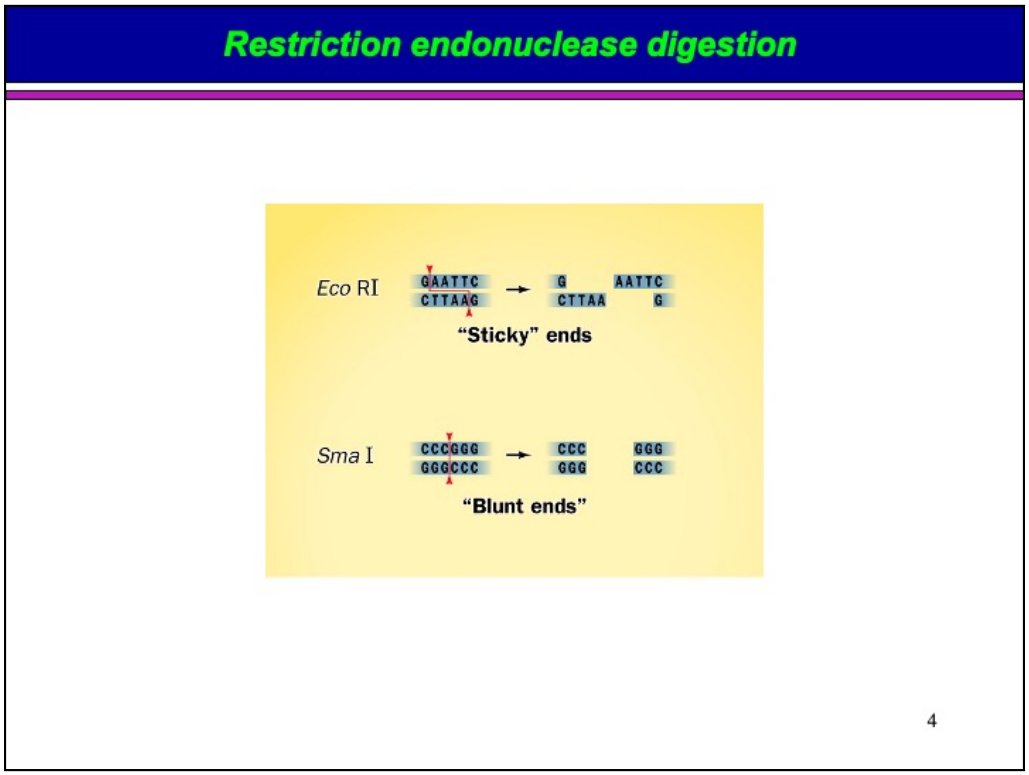
DNA ligation

- ***DNA ligase is the enzyme that joins ('ligates') DNA ends together. How does it work? Its role in molecular cloning.***

2



A simple but effective way of looking at the three processes that govern the molecular cloning of DNA. The first two (restriction, ligation) are covered substantially in this lecture and in the upcoming lecture on Molecular Cloning Strategies; and transformation, which has been touched on in the previous lecture, will be broached in more detail in Experiment 4 and in the next lecture.



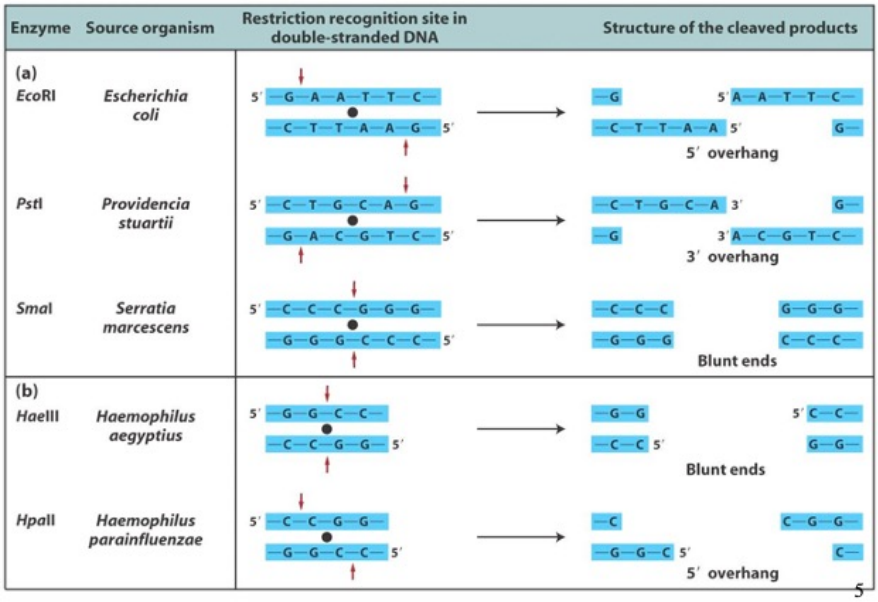
The basics about how restriction enzymes work, using the two examples here that depict the two major forms of DNA cutting, namely, staggered or blunt-end cutting. The resulting free ends are referred to as: 'sticky', 'staggered', or 'cohesive' ends for the example at top; and 'blunt' or 'flush' ends for the example at bottom.

Whichever the type of restricting/cutting, the cuts (shown by the red arrowed lines) are across the twofold or dyad axis of symmetry of the substrate sites. Note, examples such as these are typically drawn with only the substrate or cutting sites shown in nucleotide base lettering. You need to remember that the bases represent purely the specificity of the substrate site, but that each base is of course a nucleotide and it is the 3'-5' phosphodiester bonds (invisible here, but approximately at the arrowhead points) that are actually cut/restricted by the endonuclease.

Other points worth noting are that: (i) The DNA molecules must be double stranded; (ii) The substrate site is a 'palindrome' (reverse repeat), with some exceptions; and (iii) The sites are almost always internal, within longer DNA molecules, with the two strands extending in either direction beyond the sites, as depicted for EcoRI above.

The first discovered restriction endonuclease was HindII in 1970. Many others followed. Werner Arber, Hamilton O. Smith and Daniel Nathans shared the 1978 Nobel Prize for Medicine and Physiology for their discovery of restriction enzymes and their application to molecular genetics.

Restriction endonucleases

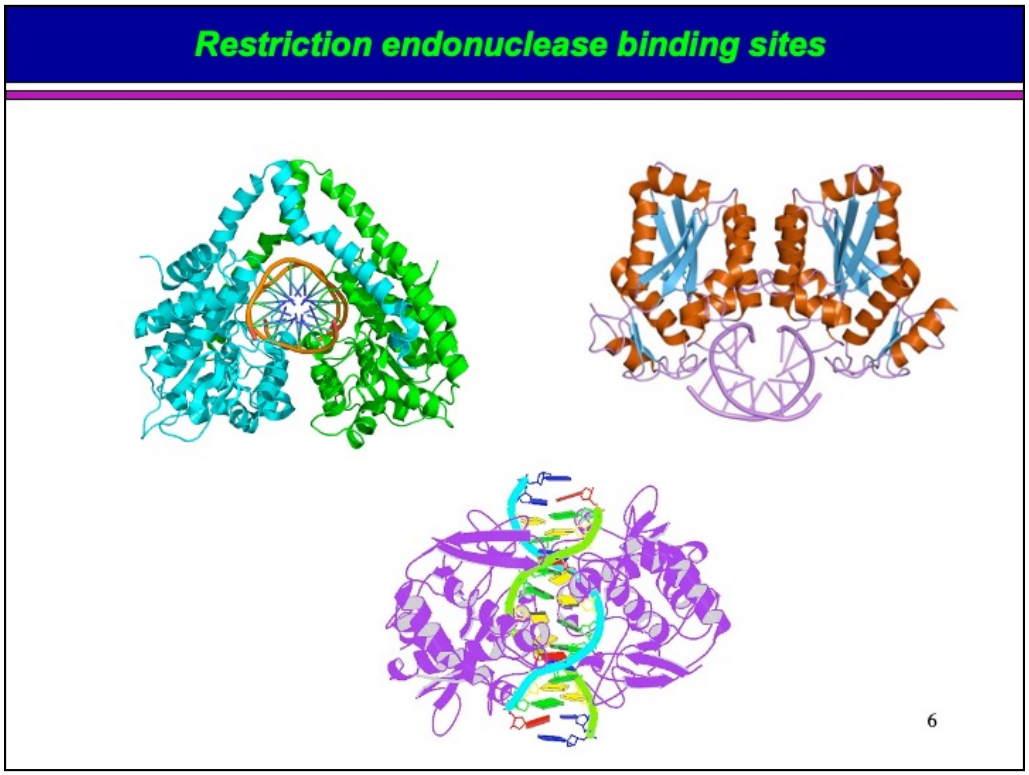


5

More examples of restriction enzyme substrate sites, including EcoRI and SmaI from the previous slide. Here, newer enzymes are listed together with their cutting sites (red arrows) about the axes of dyad (twofold) symmetry (shown here as black dots).

Additional points of note are: The opposite stagger for PstI (Vs EcoRI); and the bottom two examples showing staggered and blunt cuts for two restriction endonucleases with four base recognition palindrome sites. In fact, most restriction endonucleases have either six or four base recognition substrate palindromes. There are a sizable number with palindromes of 5, 7, 8, 9, 10, 11, and 12 bases. Some have their sites spanning over as many as 20 nucleotides, with many of the bases non-specific with 'N' for any base.

A last point to note is that each restriction digest generates new 5-prime (5') phosphate and 3-prime (3') hydroxyl ends, either overhanging (from staggered cuts) or flush/blunt, as shown on the right of the table. When a restriction site is cut, the reaction is actually an ATP-dependent hydrolysis reaction, in which water is involved in ATP catalysis and the addition of OH^- and H across the cut/broken phosphodiester bonds on each strand of the site.

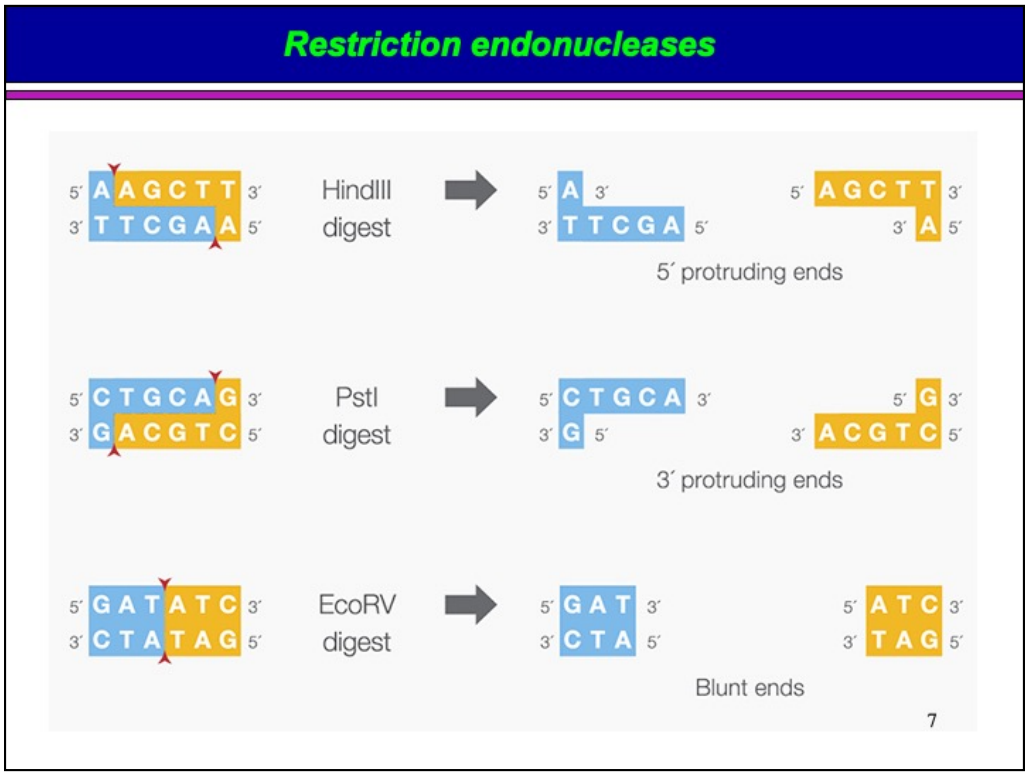


This slide depicts HindIII (top left) and BamHI (top right) bound to their respective DNA palindromic substrate sequences.

Each restriction enzyme is a dimer, with two identical catalytic sites in each monomer binding across or around each DNA strand of the substrate site, illustrated here as the circular 'wheels' at the centre of each dimer. The two monomers are identical and have identical catalytic arms that embrace the palindromic DNA site. Note, this dimeric similarity occurs for restriction enzymes that perform staggered or flush-ended cuts. The image at the botom of this slide is a side view of the two halves of a restriction enzyme dimer (purple) binding to its palindromic substrate recognition site in a DNA molecule (blue and green strands).

After binding at a non-cognate (non-site) sequence, several enzymes have been shown to locate their targets through linear diffusion. For example, *Eco*RI diffuses along linear DNA at a rate of approximately 7×10^6 bp s^{-1} and *Eco*RV diffuses at approximately 1.7×10^6 bp s^{-1}. During this process a large number of water molecules appear to fill the spaces between the enzyme and the DNA. Once the cognate (recognition) sequence is found, most of the water is excluded as a number of contacts evolve between the enzyme and the bases and phosphodiester backbone of the DNA. In the case of *Eco*RI, 50 water molecules are excluded at its recognition site.

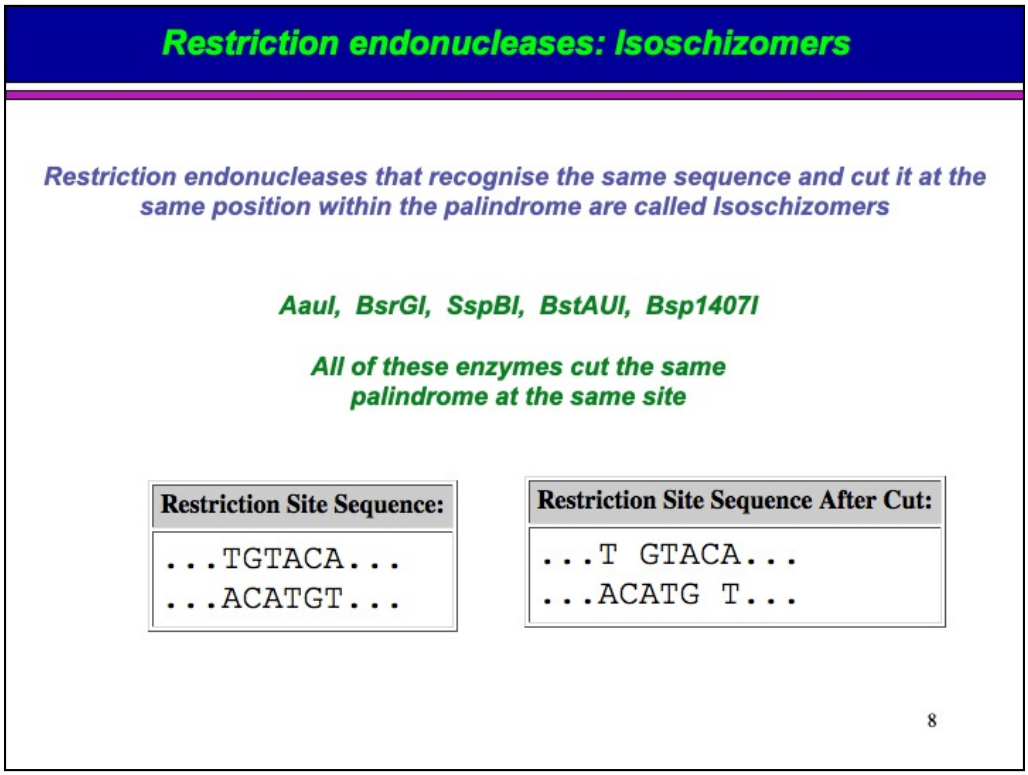
It s worth noting that most soluble enzymes and binding proteins do much the same in terms of binding to their substrate sites. Water surrounding the site becomes excluded as the protein sets up a number of contacts between amino acid side chains in the protein and the substrate.



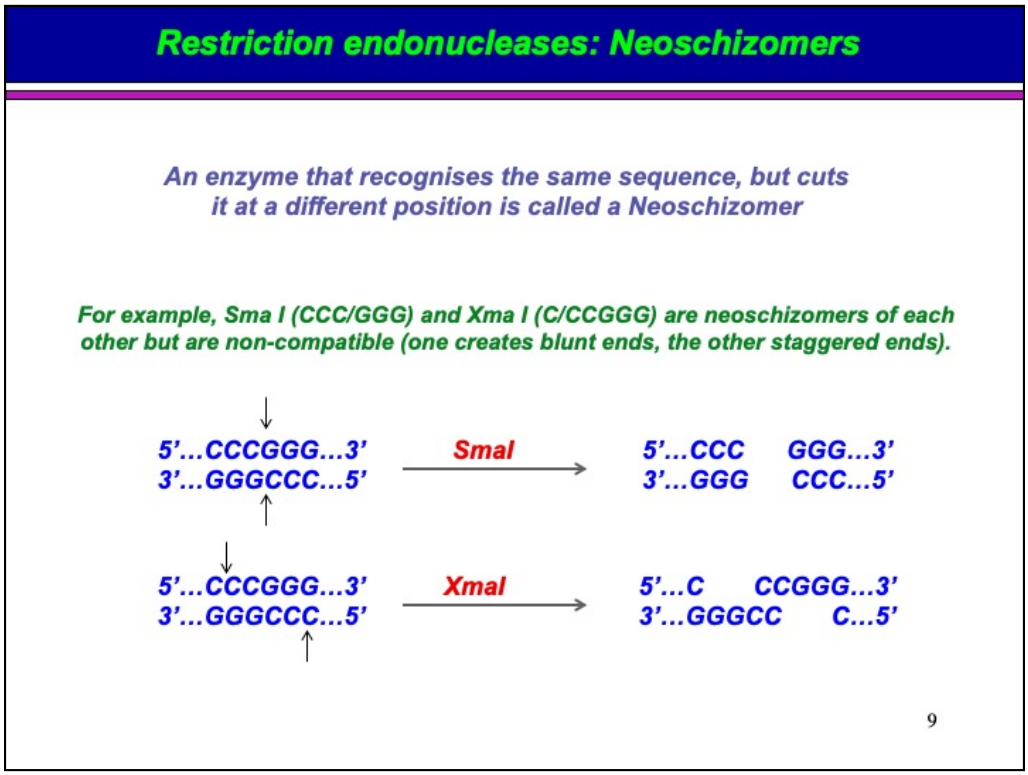
The three common types of restriction endonuclease digestions, with staggered or flush cuts in the double-stranded DNA palindrome recognition sites. Staggered cuts can be right to left, or left to right, across the axis of dyad symmetry, which is in the centre of the palindrome.

Only these three types of cuts occur in all palindromes, irrespective of the length of the site. Though the most common sites are of 4 or 6 bases, other rarer sites may contain any number of bases from 4 to 12.

Of the 3000 characterized restriction endonucleases, about 75 of them recognize non-palindromic sites. This simply means that the enzyme is not a perfect dimer and there is a small degree of asymmetry in the substrate site and enzyme dimer structure. Commercial supplier companies such as New England Biolabs or Promega sell about 300 different types of restriction enzymes.

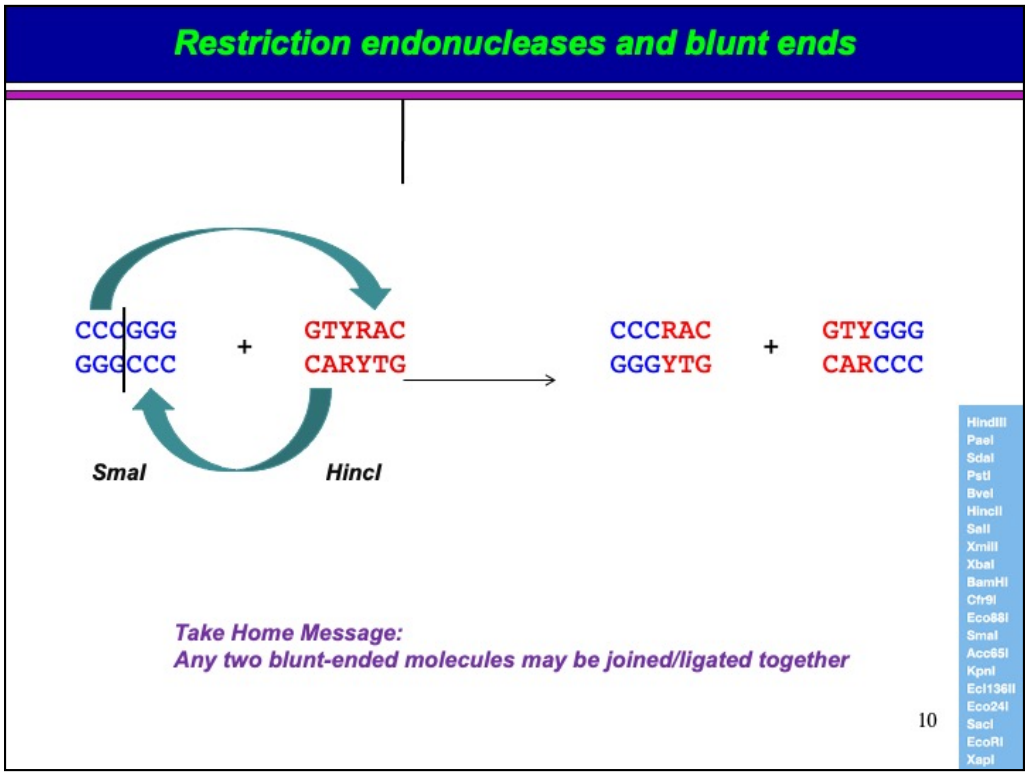


Many restriction enzymes are **isoschizomers** – they come from different bacterial hosts, are different enzymes, but recognize and cut exactly the same site at the same position across the palindrome. Naturally enough, the ends generated by isoschizomers may be religated/rejoined.



A subset of isoschizomers is called **neoschizomers**, and these enzymes recognize the same site, but cut in different positions across the site.

Over 3000 restriction endonucleases have been characterised, and they recognize about 230 different sites. This immediately tells us that most enzymes must be isoschizomers or neoschizomers. They occurmostly in bacteria, where they carry out a denfinsive function (see slide later on), but have also been isolated from viruses, archaea and eukaryotes. About 25% of all bacteria contain at least one restriction enzyme and as many as 7 have been found in a single species.



The blunt-ended enzymes SmaI (CCC/GGG) and StuI (AGG/CCT) cut at their respective sites, then ligated, will create new sites (see above), which cannot be cut by EITHER original enzyme.

In cloning experiments and cloning strategies, there is flexibility in being able to use isoschizomers or neoschizomers, as we saw on the previous two slides, or by using the same enzyme for cloning a DNA fragment into a vector, which will enable the cloned fragment to be recovered, or able to be subcloned at any time in downstream work.

Another possibility is to carry out cloning in the way depicted in this slide, that is, by using blunt-end enzymes that generate new sites that cannot be cut by either original enzyme. Though the recovery of the original cut sites is lost, an advantage of this type of cloning strategy is that **any two blunt-ended molecules may be joined/ligated together**. There is no requirement for specific and restrictive (pun intended!) compatibility in the vector insert sites, as is the case when using enzymes that generate staggered ends. The set of sites at bottom right is the MCS from pUC18. All cloning vectors will have an MCS with similar or different 'libraries' of sites.

There is an 'in-built' advantage in joining incompatible blunt-ended DNA molecules, as illustrated in the set of sites in the panel on the right of the slide (taken from a multicloning site in a plasmid vector, bottom right). The cloning 'trick' here is to insert/ligate the blunt-ended fragment into a blunt-end site, or any staggered site made blunt-ended. It is then an easy task to subclone out the fragment by cutting around it, that is, outside the edges of the fragmentnow with fused and unusable sites, by using any two restriction sites that bracket the insert.

Restriction Endonucleases - Properties

1. *Occur in bacteria, where they act as protectors against foreign, invasive DNA*
2. *Cut double-stranded DNA palindromes.*
3. *Most are specific, but some enzymes cut exactly the same site (isoschizomers). Presently > 500 enzymes with >300 specificities*
4. *Four-base recognition enzymes cut on average every 4^4 = 256 base pairs*
5. *Six-base recognition enzymes cut on average every 4^6 = 4096 base pairs*
6. *Sites are distributed in DNA on the basis of the frequency of the palindromes and not at random or average*
7. *Digestions generate new 5'P and 3'OH ends*
8. *Enzymes have specific reaction requirements of buffer pH, low, medium or high ionic strength, and occasional cation and reaction temperature specificities*
9. *By definition, one enzyme unit will digest 1 µg of DNA at 37°C in 15 mins*
10. *New "FastDigest™" enzymes have been released that use a universal buffer and complete the unit reaction in just 5 mins*

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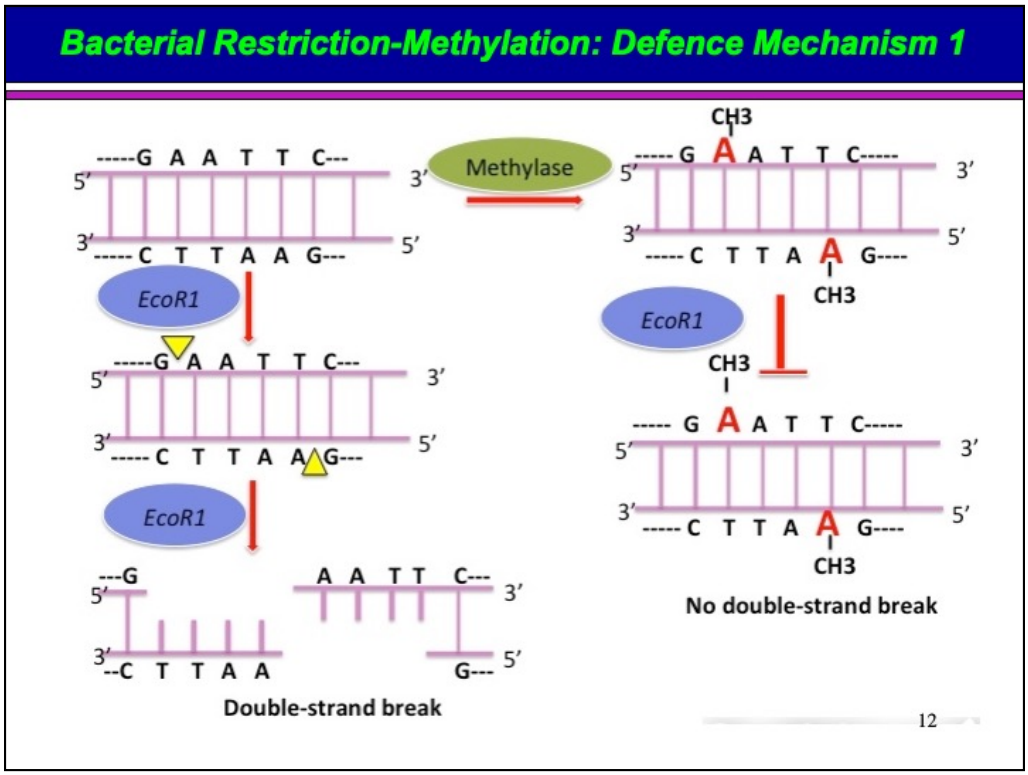
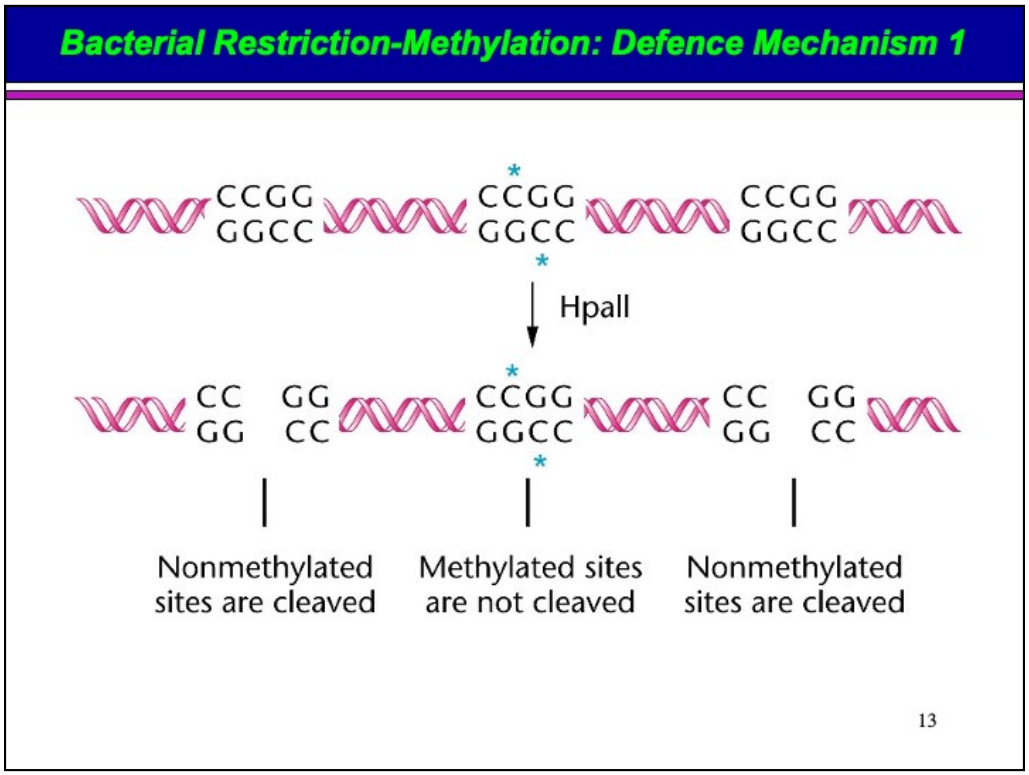
Summary of characteristics of restriction enzymes. These will be explained in more detail in following slides.



First off, why do these methylase enzymes exist? They are part of the bacterial host restriction-methylation system and are protective of the integrity of the bacterial genome. The host bacterium 'screens' invasive foreign DNA, and if not identified as 'self' by the methylation pattern, will be restricted/cut into many small pieces so that the DNA cannot recombine with the host genome and change nature of the species over time, or cause the cell to cease functioning.

The bacterial host has a methylase partner enzyme for its specific restriction endonuclease. The methylase adds bulky methyl groups to an adenine or cytosine to every recognition site for the host restriction enzyme (top, left to right), the result being that the host bacterial chromosome is protected from self-digestion, as the bulky methyl groups prevent the enzyme (EcoRI, in this example) from binding across its site (bottom right).

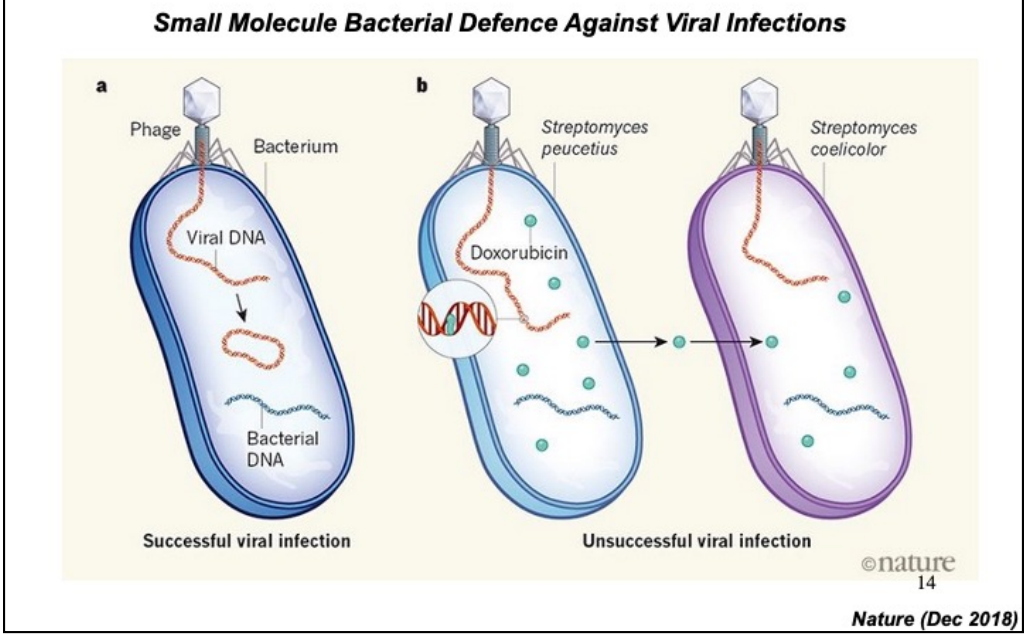
On the other hand, if foreign incoming DNA is anything other than that of the same bacterial species, it will not have the host methylation pattern, and will be chopped up by the host bacterium's restriction endonuclease, as depicted at bottom left.



Another slide showing how the restriction-methylation system works. The central HpaII site is methylated, as indicated by the asterisks and cannot be cut by the enzyme. However, HpaII will cut the (unmethylated) sites either side of the central site, as depicted at top and bottom.

The point to remember about the host restriction-methylation scheme is that a specific bacterial strain, say *Escherichia coli* RI, will have both the EcoRI restriction enzyme and the EcoRI methylase.

Bacterial Restriction-Methylation: Defence Mechanism 2

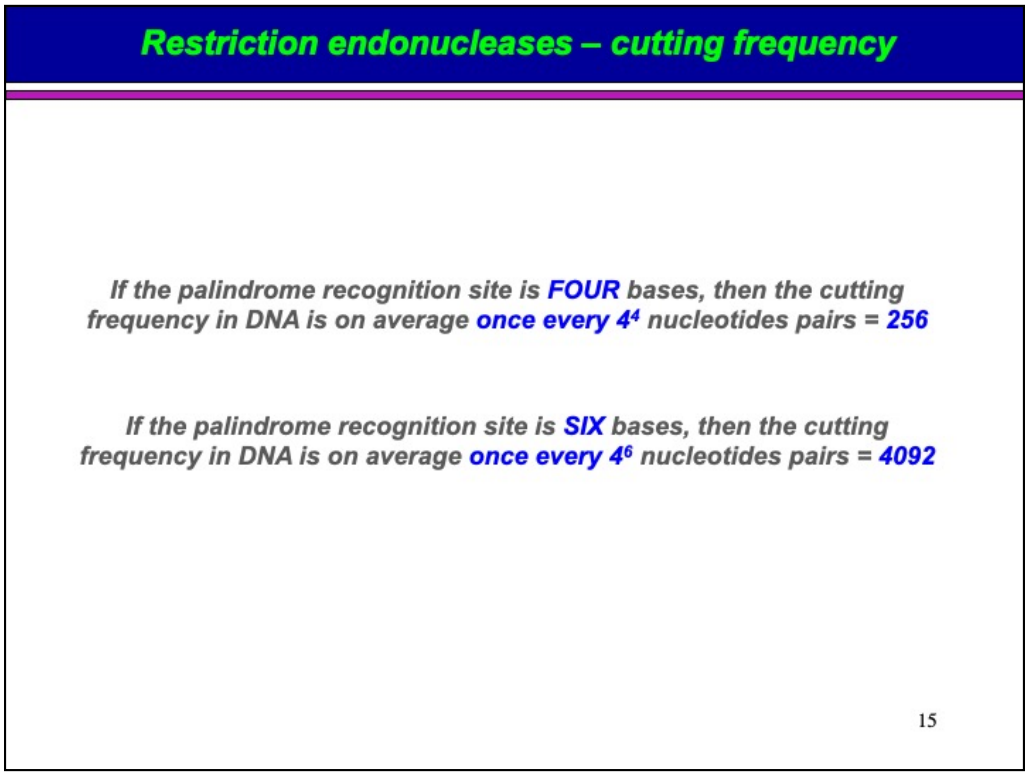


Nature (Dec 2018)

On the left is depicted a normal viral infection of a bacterium, with the injection of the viral genome from the attached phage particle, circularisation of the linear viral chromosome, followed by multiple replication events, packaging of new viral particles, and cell lysis.

Streptomyces peucetius (centre) is infected with a viral particle as on the left, but this time the invading viral DNA is subjected to doxorubicin binding, as shown in the inset. [*S. peuceticus* synthesises doxirubicin, which is an anticancer drug used in humans.] Doxorubicin in its host cell is an intercalating molecule that becomes embedded in the viral DNA in a manner akin to Gel-Red. The still linear viral DNA now cannot circularise due to the presence of bound doxorubicin; and therefore the virus cannot replicate within the host bacterium. The mechanism of this inhibition is not yet known, nor is it known how the bacterial chromosome is protected from doxorubicin, but it works as a bacterial defence mechanism against viral infection.

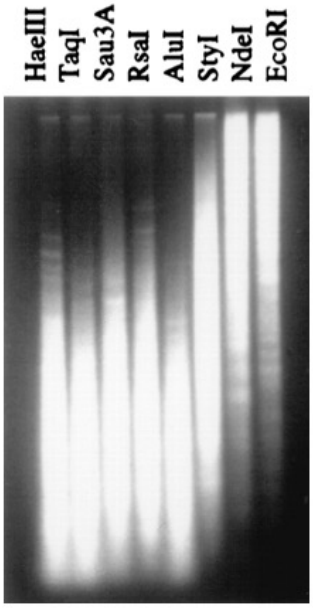
The third panel depicts *Streptomyces coelicolor,* a related species to *S. peucetius*, that does not produce doxorubicin. However, if the drug is introduced, as shown in the diagram, then this cell will be protected from viral infection in then same way as was seen in *S. peucetius*.



Simple cutting rules of combinations of the four-bases ('ACGT') to the index power of the number of bases in the recognition site. Some enzymes have recognition sites of 5, 8, 10, 12 and more and, in some cases, the sites are more complex with any base (signified by 'N') in between or at the ends of some recognition sites, but most sites are palindromes and the enzymes still cut the strands in staggered or blunt positions across the site.

The numbers in these two examples are theoretical cutting frequencies for restriction endonucleases that recognise 4 or 6 base palindromes.

Genomic DNA – restriction digests



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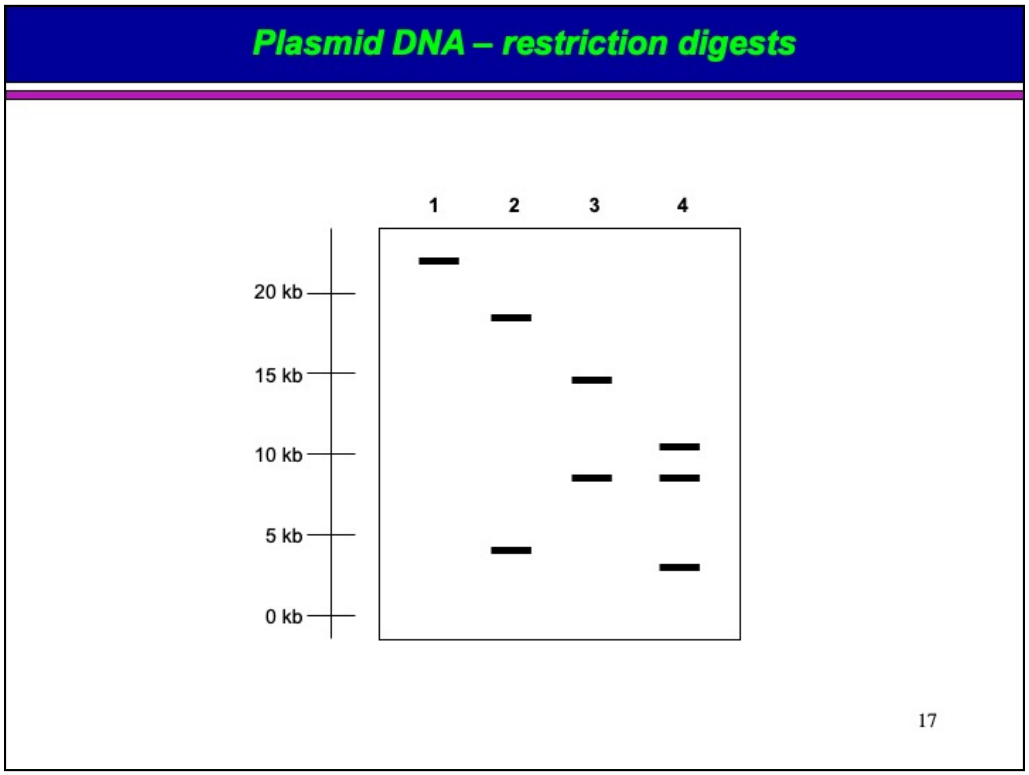
The gel depicts the same amount of chromosomal DNA cut with different restriction endonucleases, as indicated in the enzyme names at the top of each lane.

The first 5 lanes used enzymes that have 4-base recognition site palindromes and the last 3 lanes uses enzymes with 6-base recognition site palindromes. The most obvious difference between the two sets of digests is that the first 5 are distributed further from the origin, whereas the last 3 digests are closer to the origin.

The distributed positions of the many fragments from each set of digests (lanes 1-5 Vs 6-8) reflect both the number and lengths of the fragments. Though not obvious in the gel lanes, there are many more fragments in lanes 1-5 than 6-8 and they are much shorter than the longer fragments in lanes 6-8. This would be much more obvious if we could see all of the fragments, but there are too many for them to be resolved into individual bands along these gel lanes.



This drawing depicts a gel profile of DNA linear fragment bands from the same plasmid, but digested with different restriction enzymes. Lane 1 is the plasmid cut at a single site and its size is approximately 23 kb, according to the scale bar on the left. The other three lanes have the same plasmid cut with different enzymes that generate two (lanes 2 and 3) or three (lane 4) fragments.

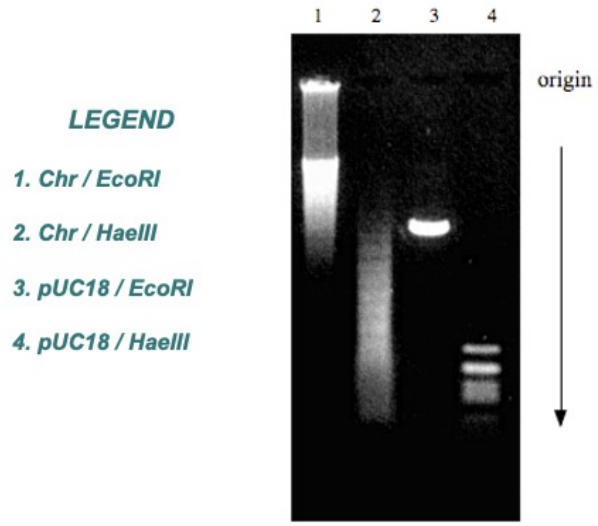
If you do the sums, you will see that the fragment bands in lanes 2-4 all add up to the same value as the single band in lane 1, namely 23 kb.

However, two things that are not immediately obvious from this drawing are:

(i)The number of molecules in all fragment bands is exactly the same, though their lengths are clearly longer or shorter; and

(ii)The Gel-Red staining intensity is reduced proportionally to length in each band in lanes 2, 3 and 4, compared to that of the single band in lane 1. It also means that the amount of Gel-Red in the multiple bands in lanes 2 to 4 must add to the same amount of Gel-Red bound to the single band in Lane 1. This feature will be more obvious in the next slide.

Restriction digestion profiles of genomic and plasmid DNAs

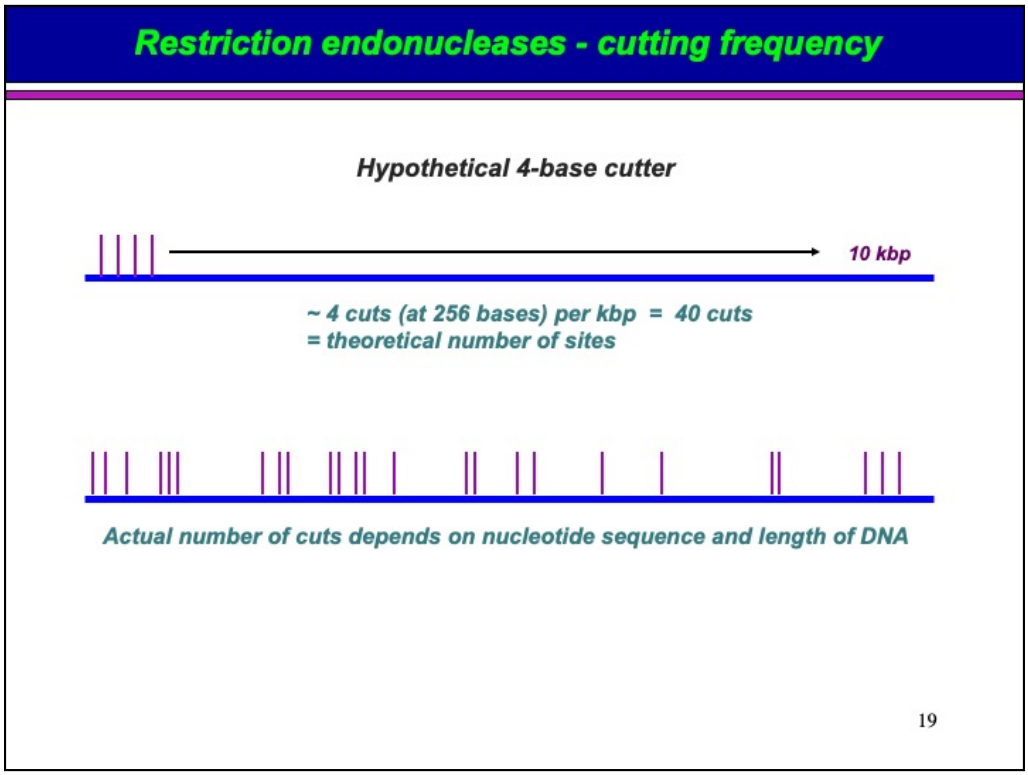


18

Following on from the last two slides, this actual gel photo represents a typical result for the distribution of restriction fragments of *E. coli* chromosomal and plasmid DNAs, treated with 4-base (lanes 2 and 4) or 6-base (lanes 1 and 3) restriction enzyme cutters. This gel photo is the one that will be seen in Experiment 5.

A point to note is that the number of fragments for EcoRI (6-base site) is far fewer than for HaeIII (4-base site) for both genomic and plasmid DNA targets. Note that the positions of the fragments confirm the cutting frequency. The 4-base cutter HaeIII generates smaller and more numerous fragments, as is evident by their positions lower in the gel (compare lanes 2 and 4 to lanes 1 and 3).

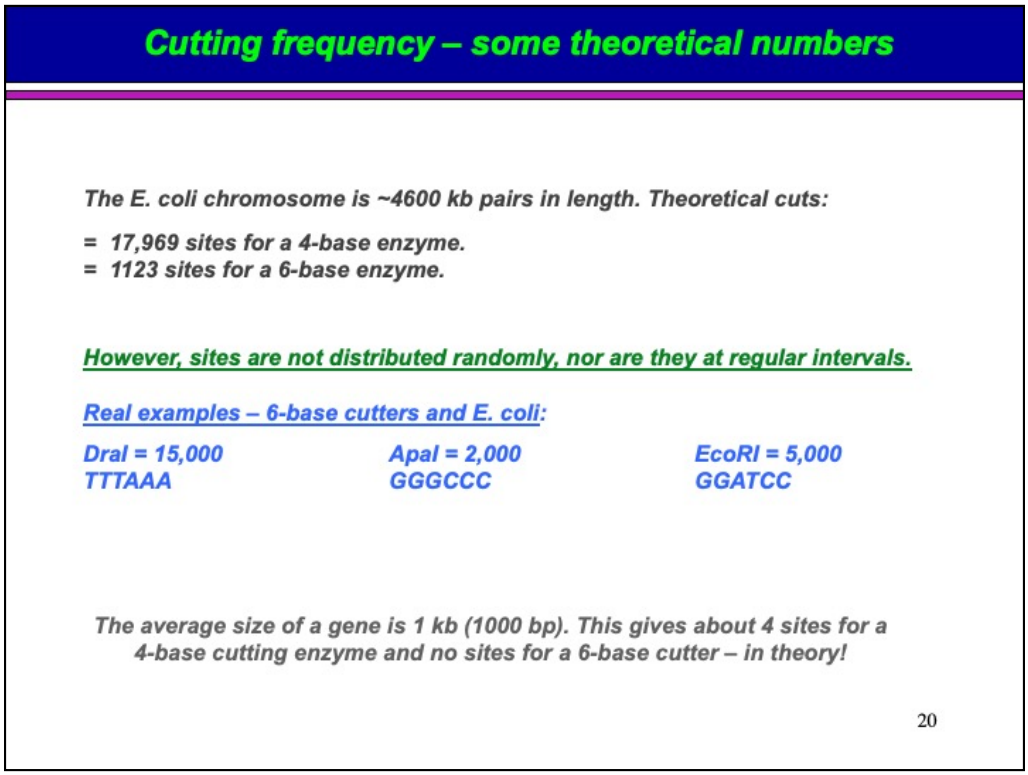
Also, if you look closely enough, you will see that Gel-Red staining is less intense in the more numerous HaeIII fragments, which is more apparent in the plasmid digests in lanes 3 and 4. In a theoretical sense, for pUC18, the number of HaeIII fragments and the amount of Gel-Red bound, should both add up to the length of the single EcoRI pUC18 fragment, and the amount of bound Gel-Red.



Top panel shows that on average you might expect ~40 restriction sites for a 4-base enzyme in a linear fragment of 10 kbp (10,000/256 = 39), but in reality, the actual number of cuts is more random in being dependent **solely on base composition of the DNA**. If the length is vast, you might expect the number of cuts to be closer to the average, but again this is still dependent on base composition,

DNA can show biases across different species, usually manifested in different AT/GC composition; and if you look at many restriction endonuclease palindrome sites, you will see how site numbers per species might vary with higher AT or GC content. For example, if a species has 65% GC content, you would expect more SmaI (CCCGGG) than DraI (TTTAAA) sites.

Nevertheless, the take home message is that the upper theoretical calculation is just that, whereas the lower one, or any variation of it, is more typically based on DNA base composition.



From the previous slide and the question of cutting frequency. These examples show that the number of actual sites for a 6-base enzyme on the *E. coli* chromosome may be close to (*ApaI*, about twice average) or very different (*EcoRI*, *DraI*) to the theoretical average of 1123 sites. Note how the site sequence composition contributes to the differences in cutting frequencies for these three enzymes.

Take home message: It's easy to calculate the theoretical number of restriction sites in any given DNA sequence, but the ACTUAL number is often different due to the uniqueness of different DNA sequences and an asymmetric distribution of any given restriction site within the sequence.

Restriction may be near or at the ends of DNA molecules

Enzyme	Sequence	% DNA cleaved 2 hr	% DNA cleaved 20 hr
AccI	CGTCGACC	0	0
	CCCGTCGACCGG	0	0
AscI	GGCGCGCC	>90	>90
EcoRI	GGAATTCC	>90	>90
NotI	CCCATGGG	0	0
	CATCCCCATGGGCATC	50	75
PstI	GCTGCAGC	0	0
	TGCACTGCAGTGCA	10	10

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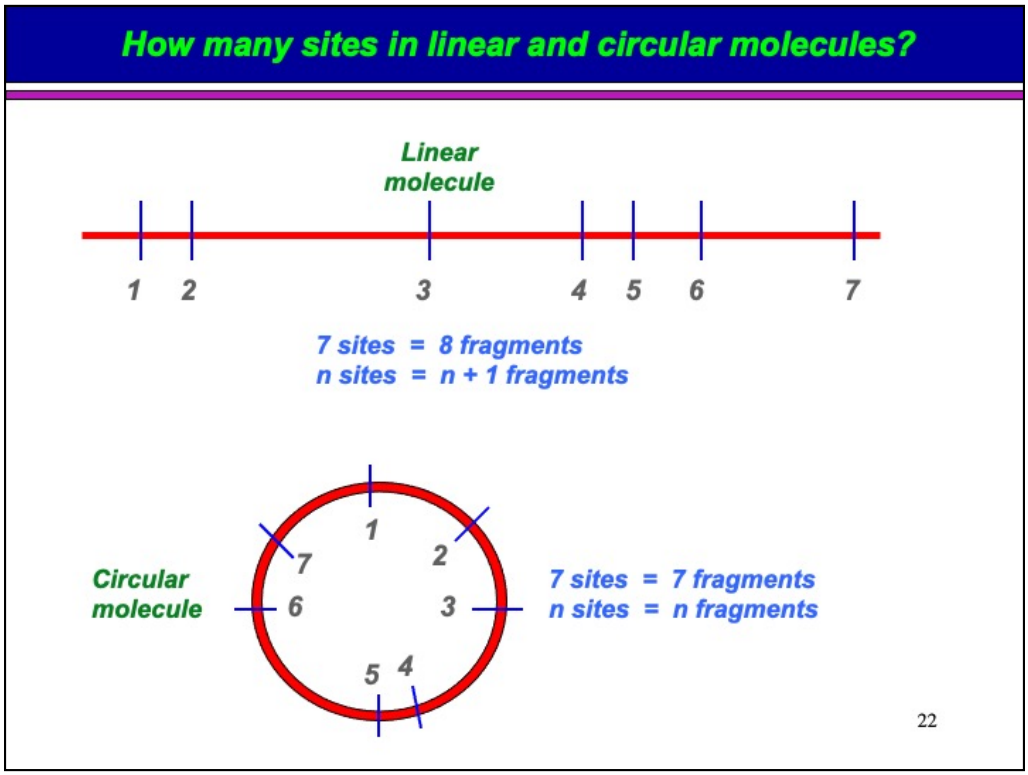
Endonucleases cut within DNA molecules, exonucleases cut inwards from the open ends, base by base.

Despite the fact that restriction enzymes are called 'endonucleases', and though most of them are true endonucleases (e.g. AccI, NotI, PstI, above), some can cut efficiently very near or at the end of DNA fragments (e.g. AscI, EcoRI). Most cut at internal sites, but many will cut at high efficiency if their palindrome sites are very near or at the end of a DNA molecule.

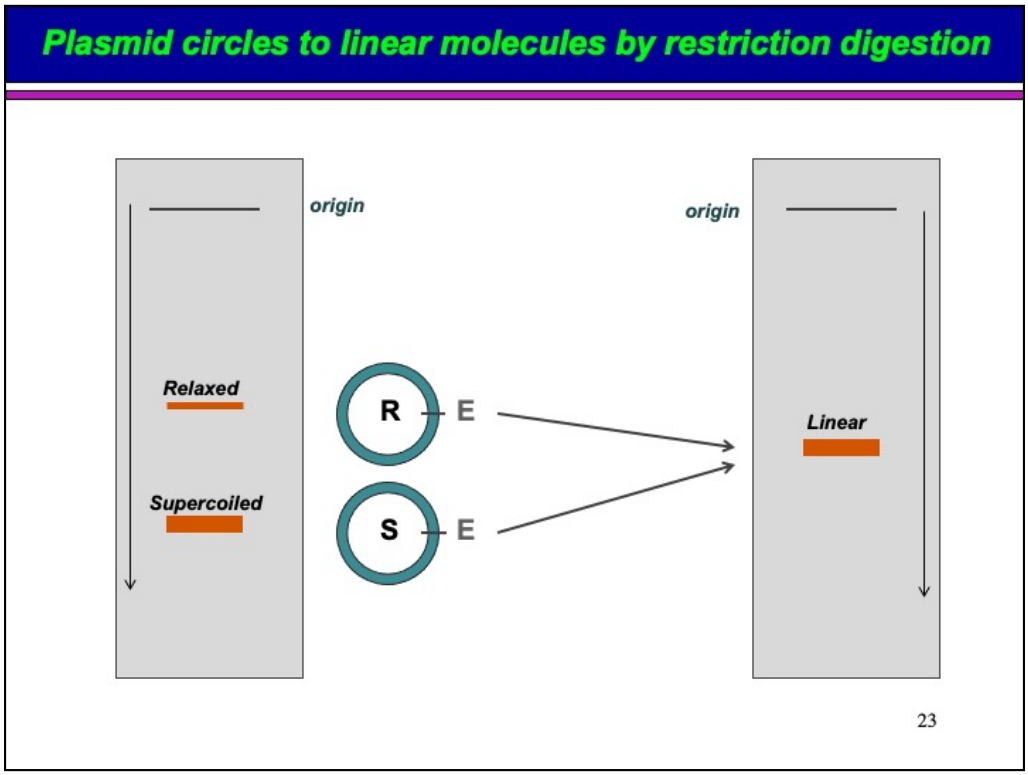
So, why is this bit of news important? In molecular cloning, often a plasmid vector, which is circular, needs to be double-digested, that is, cut sequentially or simultaneously with two different restriction enzymes. The first cut will linearise the circular vector, and the second site is likely to be near or at the open ends. If the enzyme is one that cannot cut near DNA ends (such as AccI, NotI, PstI etc), then the second cut will not happen, or will happen so inefficently as to make the cloning scheme unworkable.

By reversing the enzyme cutting order, the desired double cut will more than likely be achieved, unless both are true endonucleases, in which case another enzyme needs to be chosen, or a different strategy.



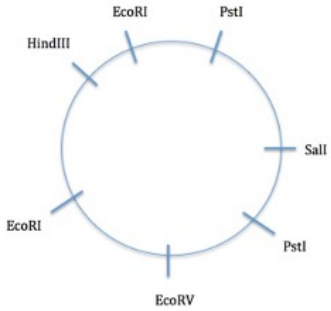
A simple rule-of-thumb to calculate the number of restriction sites in linear and circular DNAs.

Quite simply, for linear DNA the number of fragments generated is one more than the number of sites (top) , but is the same number of fragments as there are sites in the same closed, circular molecule of DNA (bottom).



The two plasmid bands in the gel at left – relaxed and supercoiled – are shown to be identical molecules by doing a restriction enzyme digestion at the unique restriction site 'E' for the two plasmid conformations; and then running the products in an agarose gel where only a single linear band is seen. You might recall Experiment 1 used all three versions of the same plasmid in the way depicted above, but with all three in the same lane.

MCQ L5



A plasmid with restriction sites shown in the figure above is double-digested with the enzymes PstI and EcoRI. How many fragments will be generated?

(A) Seven.
(B) Six.
(C) Four.
(D) Three.
(E) Two.

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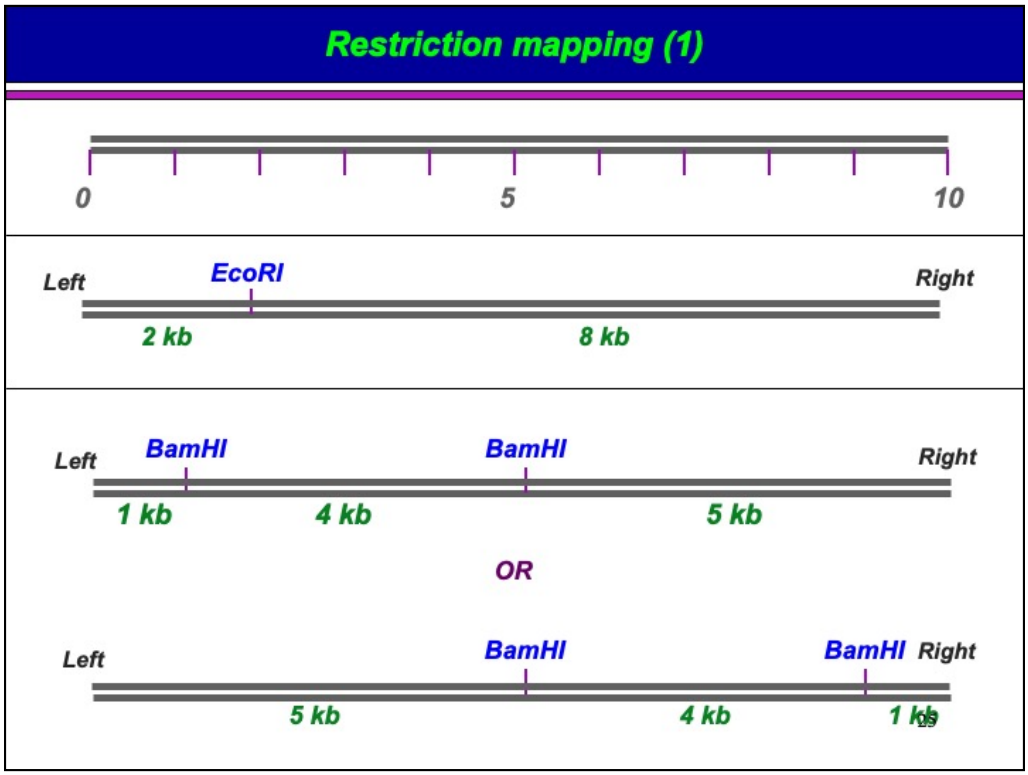
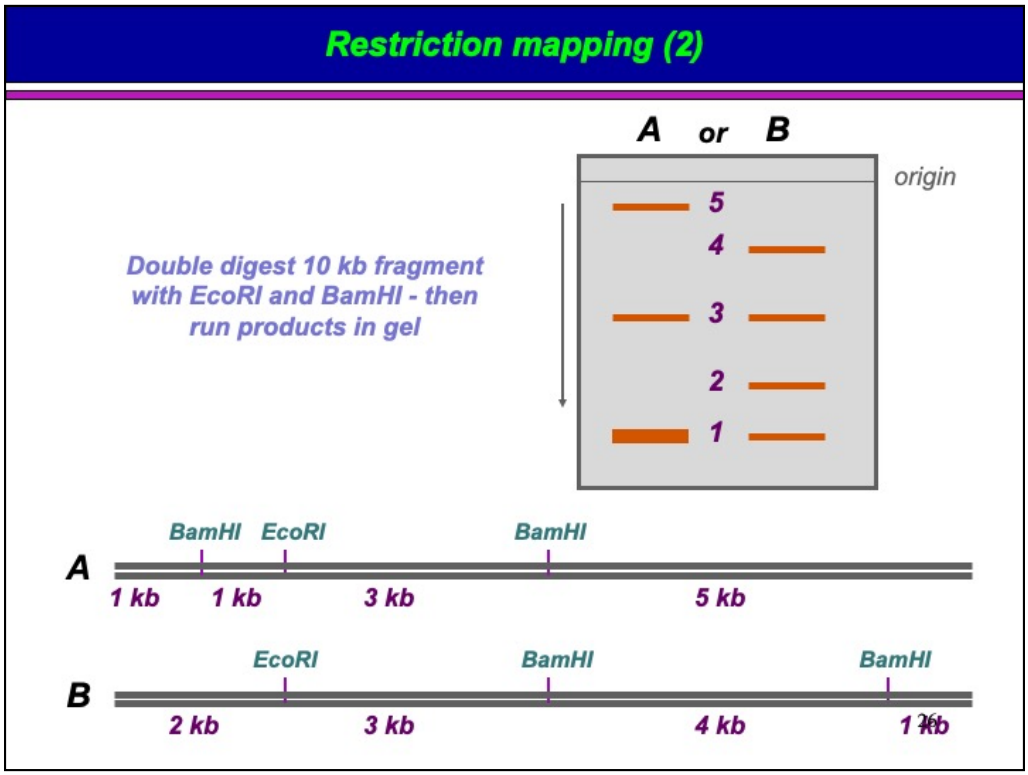


This slide needs to be considered along with the next slide. However, at this point the problem or question is that we have a piece of DNA 10 kb in length and we are told that it has one EcoRI site and two BamHI sites, but given that the DNA molecule has a specific base sequence, or as indicated in the figure, left and right ends, we need to place the EcoRI and BamHI sites **relative** to one another.

When we digest the DNA with EcoRI, we get 2 kb and 8 kb fragments, and when digested with BamHI, there are three fragments of 1, 4, and 5 kb, respectively. However, what we need to know is where the BamHI sites are in relation to the single EcoRI site that we have positioned 2 kb in from the left.

Looking at the bottom two maps, the BamHI sites can be in either of two orientations relative to the EcoRI site. How do we discern the true map for both enzymes? Easy, we do a **double** digest with both enzymes and run the fragments in an agarose gel (go to the next slide).

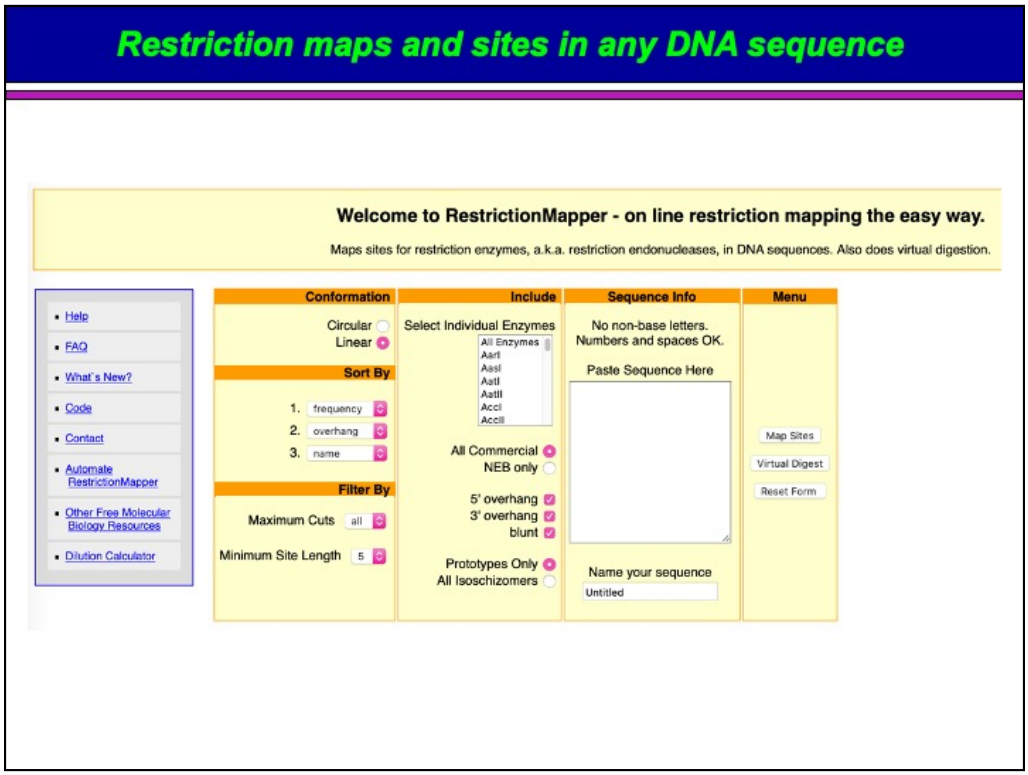


From the previous slide, we have two possible maps indicated as either ‘A’ or ‘B’.

If the true map was A, then the gel profile on the left is obtained (gel lane’A’), with a doublet of 1kb (note thicker band) and 3kb and 5kb bands, adding to 10kb.

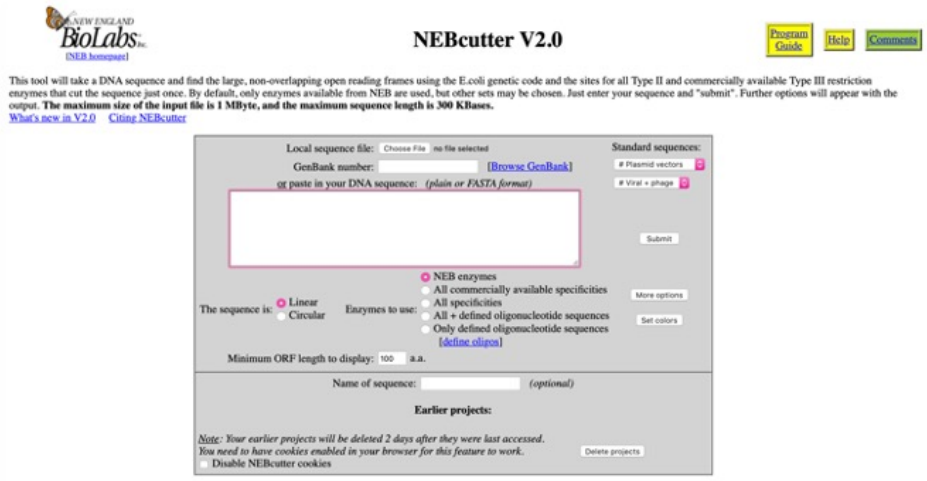
If the map is ‘B’, then the gel profile on the right is obtained (gel lane ‘B’), with four bands of 1, 2, 3, and 4kb which adds to 10kb.

Note only one outcome is possible, but both are shown here for clarity.

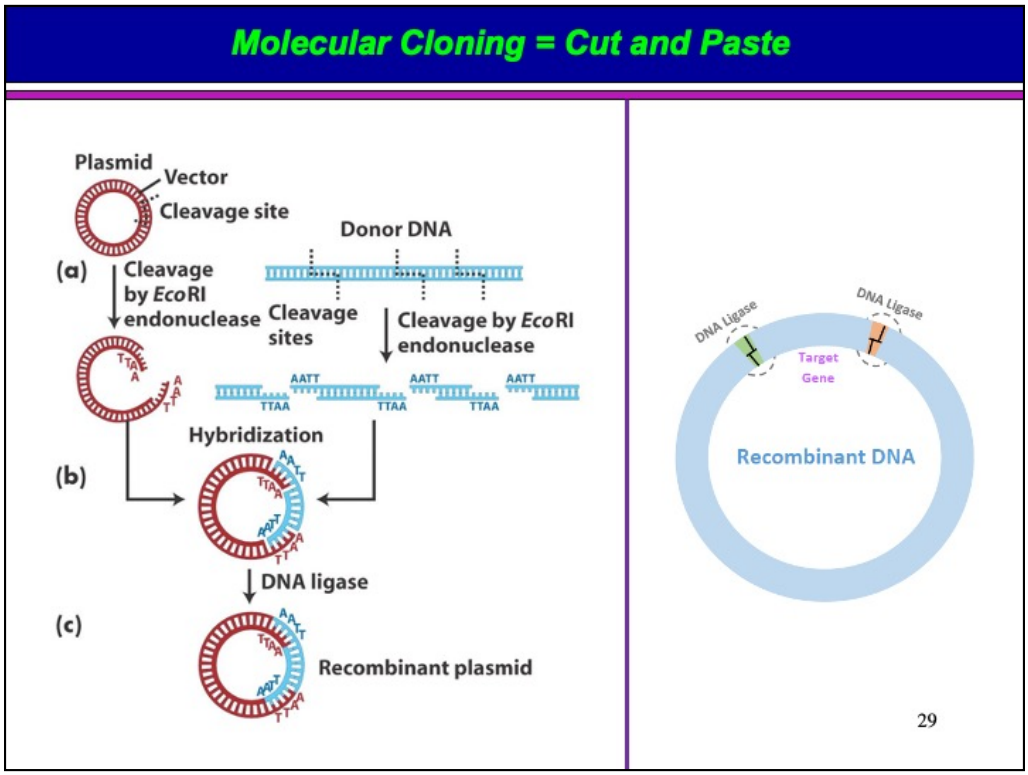


Online RestrictionMapper. Using this tool allows any DNA sequence to be analysed for any number of restriction enzyme cuts and sites. All it requires is to import the sequence (cut and paste) then begin the selection of enzyme(s) and readout options.

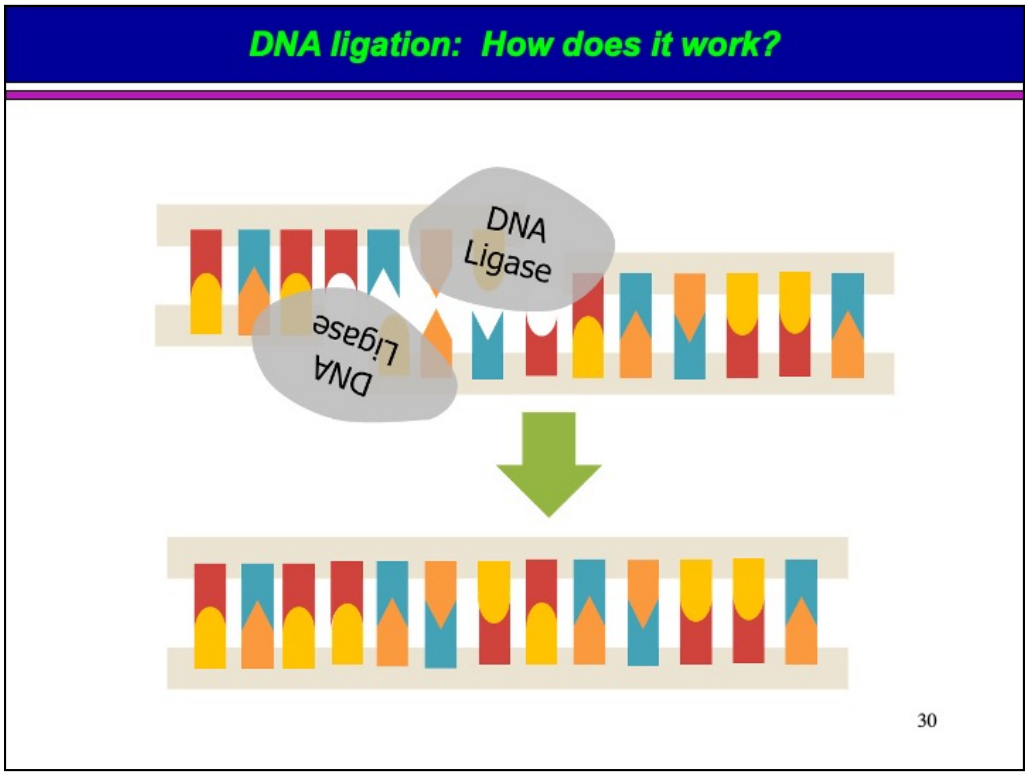
Restriction maps and sites in any DNA sequence



Ditto previous slide, using the NEBcutter.



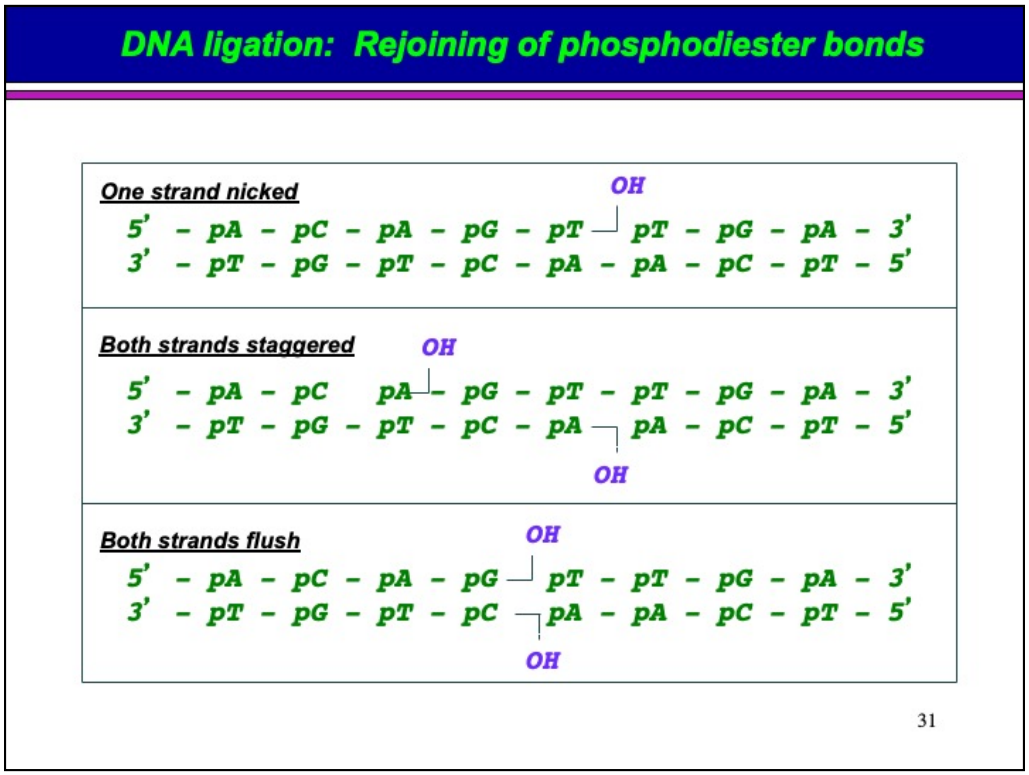
One of the major uses of restriction enzymes is in molecular cloning of recombinant DNA, the simplest iteration of genetic engineering. Looking at the panel on the left: (a) Restriction enzymes cut the plasmid vector and insert foreign (donor) DNA (hydrolysis); (b) The DNA vector and inserts anneal to one another by hydrogen bonding of their compatible ends; (c) The recombinant plasmid is generated after DNA ligase pastes the insert and vector compatible EcoRI ends (phosphorylation). The panel on the right shows a simplified version of cloning, with the ligation sites signposted. DNA ligase will be looked at in detail in the second section of this lecture.



Topic two for today is DNA ligation, catalysed by the enzyme T4 DNA ligase, which you can see in this simple illustration works as a non-covalent dimer, with each monomer binding across the gaps in the DNA strands.

It calalyses the formation of phosphodiester bonds (paste) between adjacent 3'OH and 5'-PO_4 ends; and is opposite to the action of restriction endonucleases (cut). DNA ligase thus catalyses an esterification reaction, with the elimination of OH^- and H from across the 5' and 3' ends on each strand.

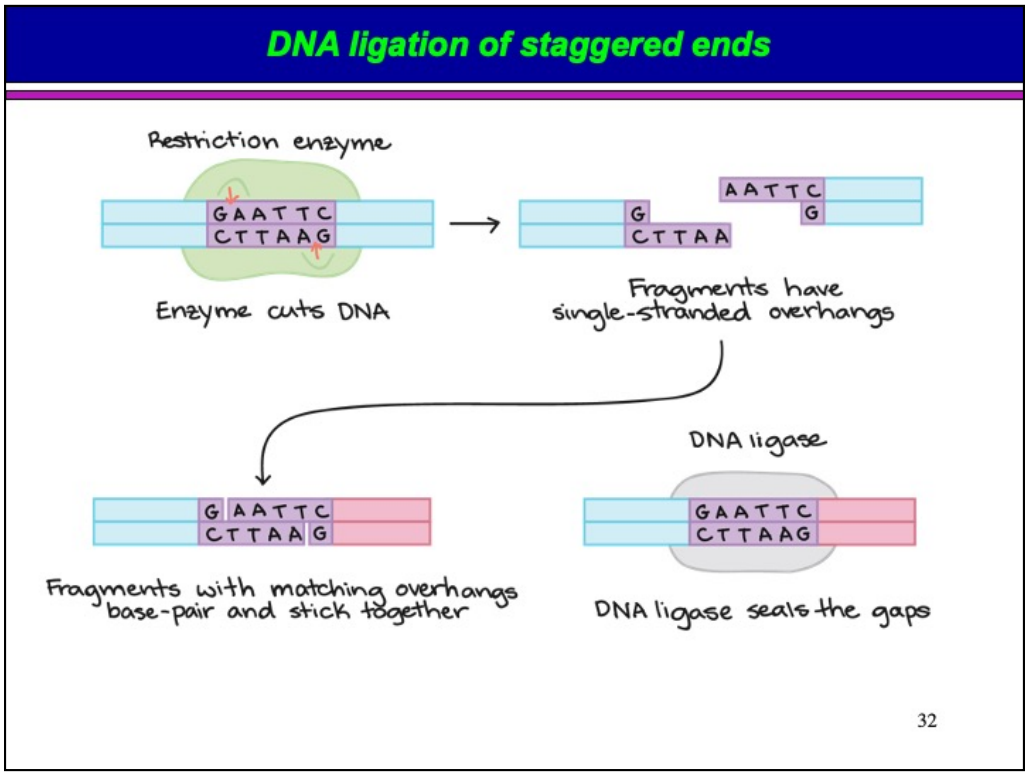
T4 DNA ligase requires ATP and Mg and works at 4° to 37°C; but most ligations are done at around 8° to 16°C.



DNA ligase catalyses the reverse reaction to restriction endonuclease, that is, it esterifies new 5'-3' (phosphodiester) bonds (eliminating water) in single or double stranded DNA molecules.

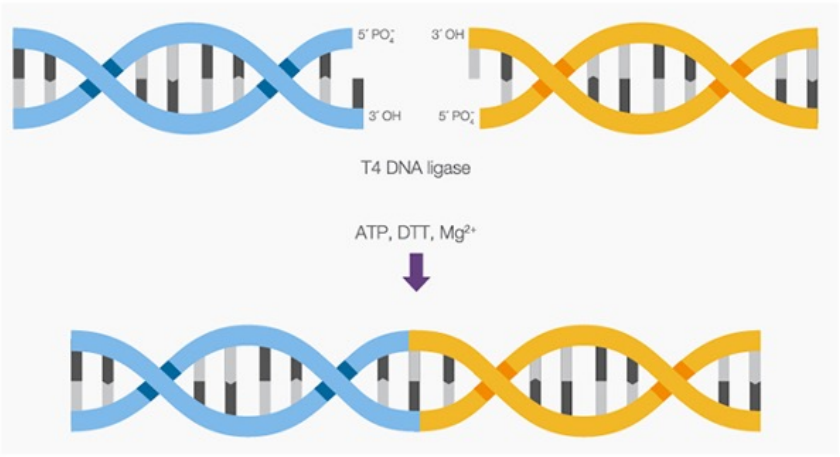
The top panel shows the repair function of ligase in a single strand.

The bottom two panels show how the enzyme ligates restriction sites (staggered or blunt). In simple terms, restriction and ligation may be compared to cutting and pasting. In the figure above, the 'OHs' and gaps indicate the unligated phosphodiester bonds that will be joined by DNA ligase.



The ligation of staggered, sticky, or cohesive ends is made much simpler by the fact that when the ends collide at low temperature, they stick together by virtue of precise complementary H-bonding, whereupon DNA ligase binds to the (annealed) site and joins the ends in new covalent phosphodiester bonds.

DNA ligation of blunt ends



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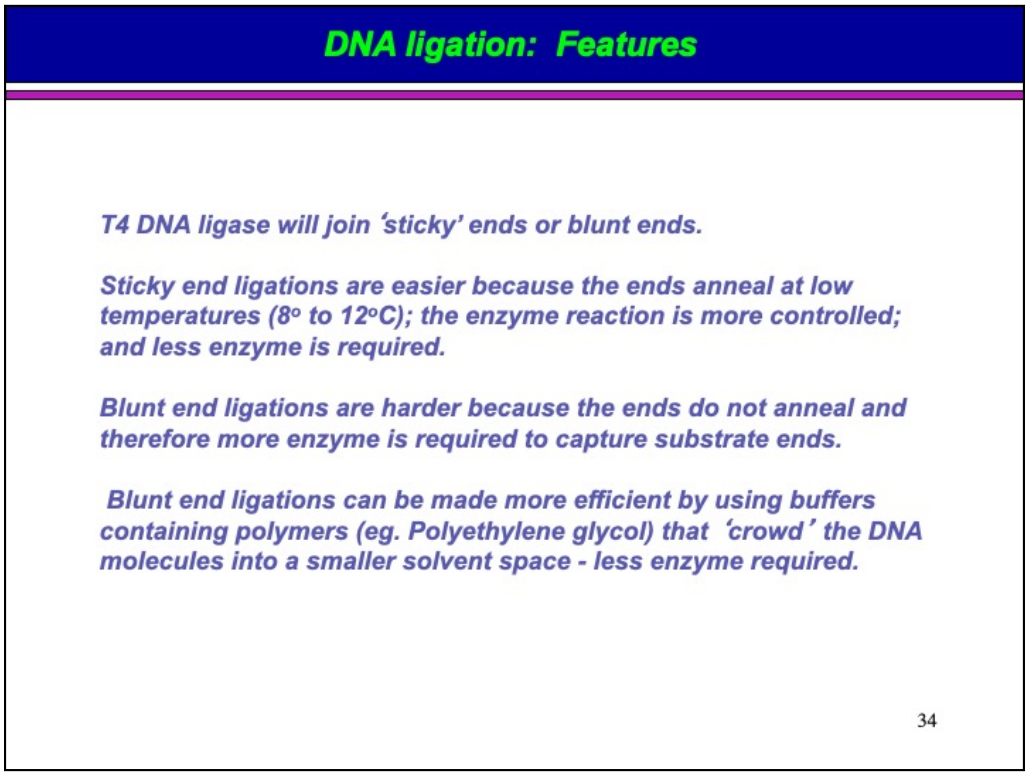
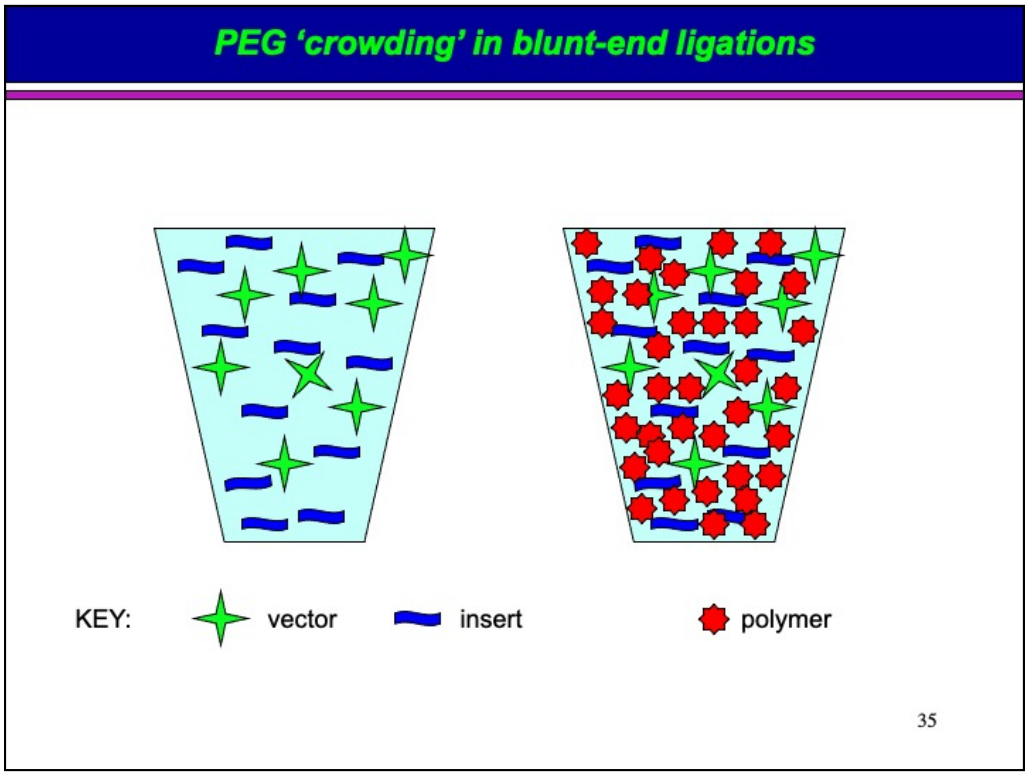
The ligation of blunt DNA ends is quite different from that of sticky ends (previous slide). Though the chemistry is the same, the ends are blunt or flush and do not stick together when they collide.

The 'tricks' that make DNA ligation of blunt ends more efficient are that: (i) They are carried out at low temperature (<10°C), which slows the movement of the DNA molecules in the reaction mixture, and (ii) The reaction is assisted by macromolecular crowding, which is achieved by adding a soluble polymer, such as polyethylene glycol, to the reaction mixture (explained in the slide further on).



Some simple properties and technical details about DNA ligation reactions that are self-explanatory, but the next few slides show how complicated such a simple reaction can be in practice.

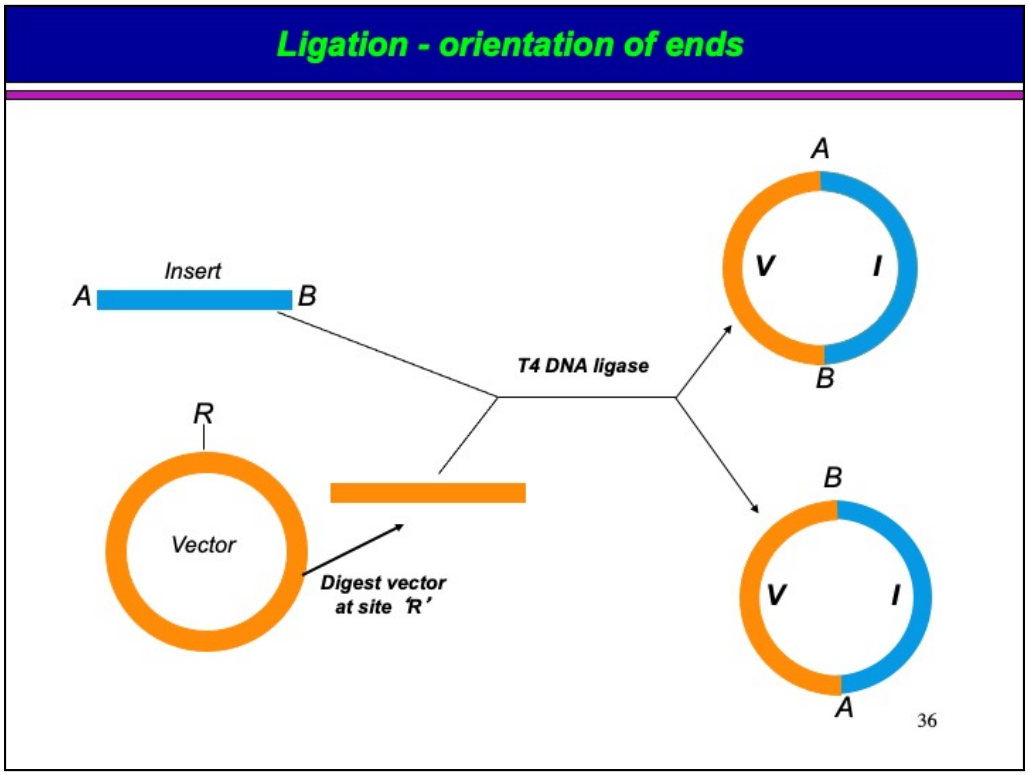


The principle of macromolecular crowding by including a polymer in ligation reactions involving blunt ends.

The volume occupied by the insoluble polymer molecules (such as polyethylene glycol, PEG) is unavailable to the solvent, vector, and insert molecules. The DNA is therefore forced or crowded into a smaller volume/space and this enables the reaction to be more efficient by increasing contact/collisions between the vector and insert fragments. Compare left and right vessels, noting that Archimedes' Principle applies in that the polymer in the right vessel displaces its equivalent volume of water.

By analogy, you fill a bucket of water and then add say 20 golf balls into the bucket. An equivalent volume of water is forced to spill over the side. Anything now dissolved in the remaining water will be more concentrated, that is, will occupy a more confined space in the lesser volume of water among the gold balls.

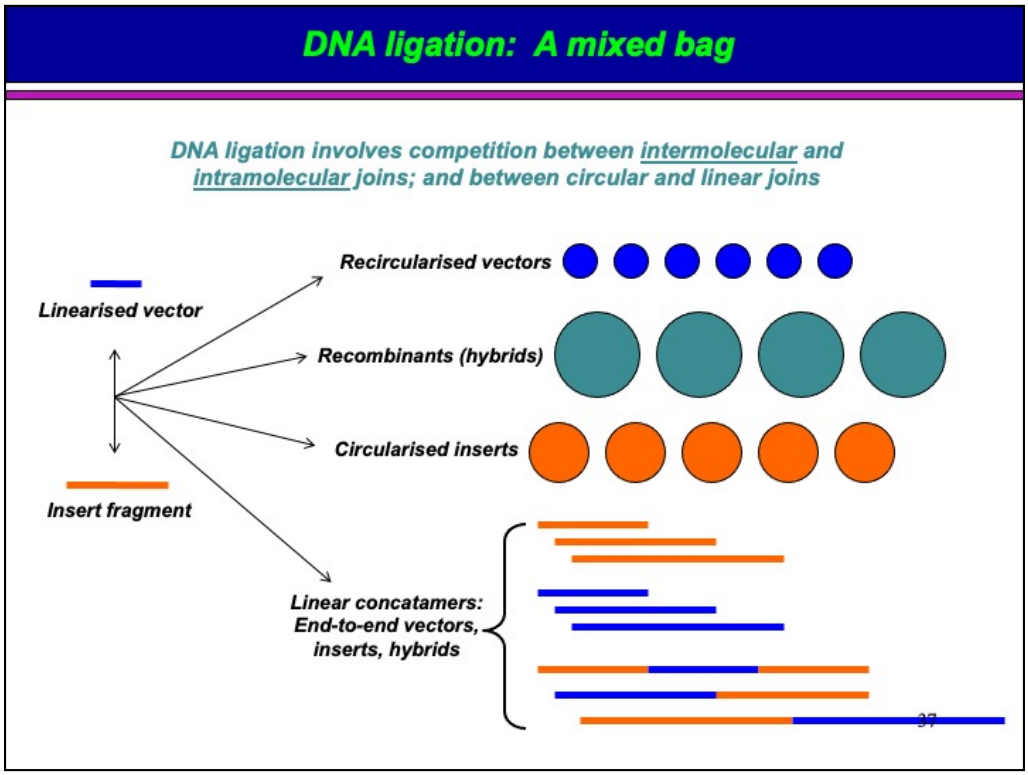
Although this ligation method was developed with blunt end ligations in mind, since collisions of blunt ends are only fleeting, it may be used in sticky end ligations as well, and will improve the yield of recombinant plasmids and transformants.



Molecular cloning or genetic engineering, in which a new 'recombinant' DNA plasmid is formed.

The linear insert (A-B; blue) is to be ligated to the linearised plasmid vector (orange). If the ends of the insert (A and B) have the same restriction site 'R' as that of the plasmid (i.e. all ends are compatible), or if all molecules have blunt ends, then there are two possible products, shown at right, with the orientation of the insert opposite in each case to the fixed position of the vector, as shown by the A/B and B/A orientations in the two recombinant plasmids.

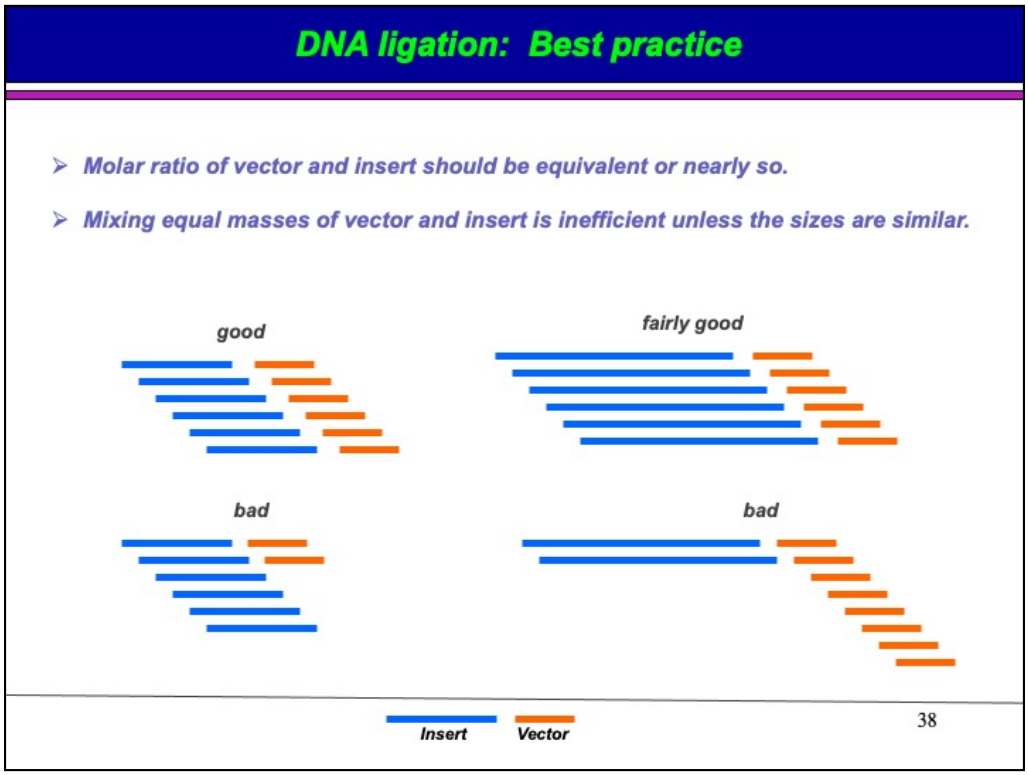
Note that in all other respects the two recombinant plasmids 'look' the same, as they are the same size, but the orientation of the insert is of course different (an important detail to be discussed in a later lecture on the expression of cloned genes).



In actual experiments, DNA ligations are – for want of a better phrase – a bit of a dog's breakfast, with a number of competing reactions happening at the same time. The ligating of compatible ends can set up a number of composite outcomes, as shown above (three circular; three linear).

What we are actually trying to do in this ligation reaction is generate only recombinant circular molecules (like that shown in the previous slide), as shown here as large green circles, but in reality, we also finish up with a number of byproduct circles (orange; blue), and linear molecule composites (bottom of slide). The net result is that the efficiency and yield of the main and desired reaction is reduced.

How can we make the reaction more efficient and reduce the byproducts? The easiest way is shown on the next slide – another method will be dealt with in a later lecture.



Typical stoichiometric and length/ratio considerations in ligation reactions.

The most efficient reaction giving the highest yield of product is that in which the ratio of reactants and their respective lengths are the same or very close to one another ('good' in the example above).

Mind you, all four examples above will work in practice, but efficiency and recovery are less successful with non-ideal fragment/vector numbers and lengths. In some instances we might need to work with poor matching of DNA molecules, as shown in the two examples at the bottom.

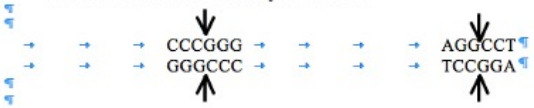
Take home messages are:

✧DNA ligation reactions generate a maximum number of circular recombinant plasmids when the molar ratio of vector and insert are equivalent or nearly so.

✧Mixing equal masses of vector and insert is inefficient unless the sizes are similar.

MCQ L5

The two sequences shown below are palindromes for restriction enzymes that make blunt-ended cuts as shown by the arrows.



If composite hybrid sites were ligated from these blunt cuts, the new sites would be...

(A) The same as the original sites.

(B) Different to either of the original sites and unable to be cut with either enzyme.

(C) Different to either of the original sites and able to be cut with one but not both enzymes.

(D) Different to either of the original sites and able to be cut with both enzymes.

(E) Composites of the original sites and able to be cut with both enzymes.

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MCQ L5

The gel depicts chromosomal DNA cut with different restriction endonucleases. Lanes 1-5 have the fragments distributed in the lower half of the gel, while lanes 6-8 have the fragments in the top half. The source chromosomal DNA is the same for all digests. If you were able to calculate the amount of Gel-Red bound to all fragments in say lanes 1 and 8, what do you think would be the result?

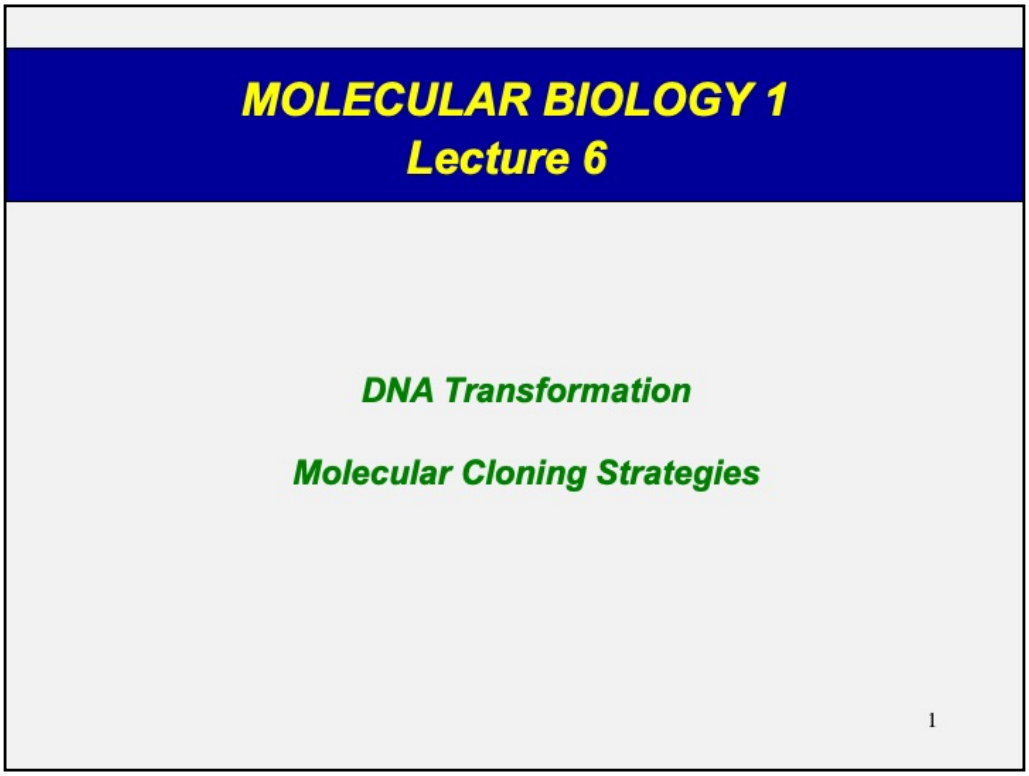
A. There would be more Gel-Red bound in Lane 1.

B. It is not really possible to estimate the amounts of Gel-Red in each lane.

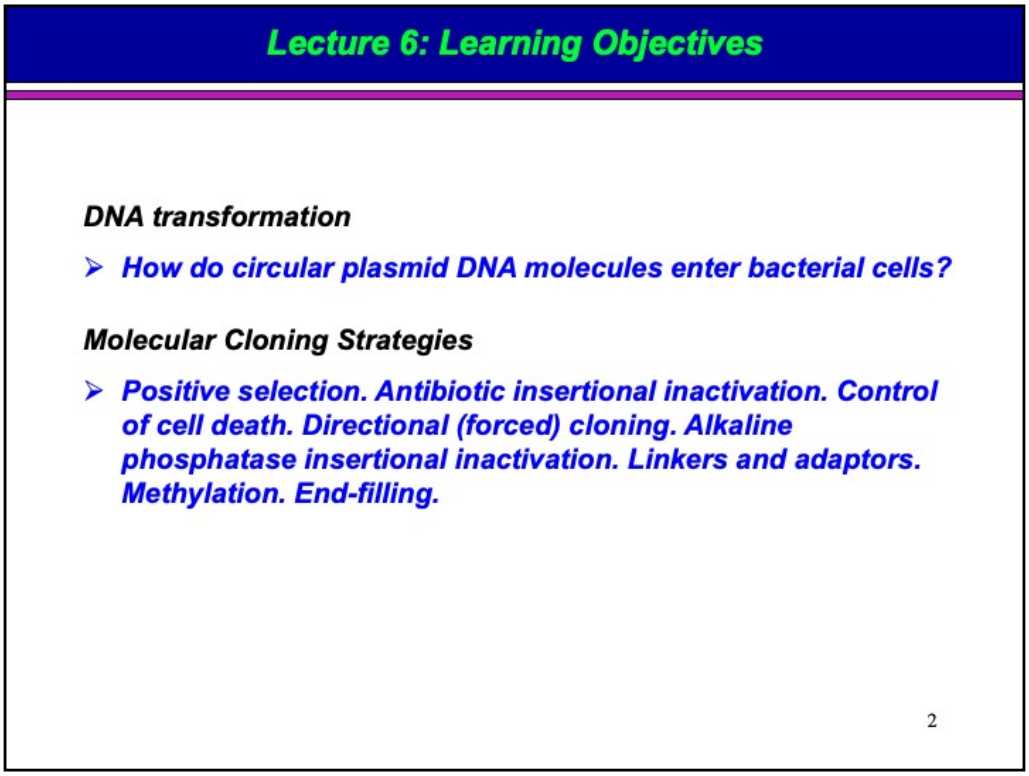
C. There would be the same amount of Gel-Red bound in both lanes.

D. The amount of Gel-Red bound is not linearly related to the amount of DNA present.

E. There would be more Gel-Red bound in Lane 8.



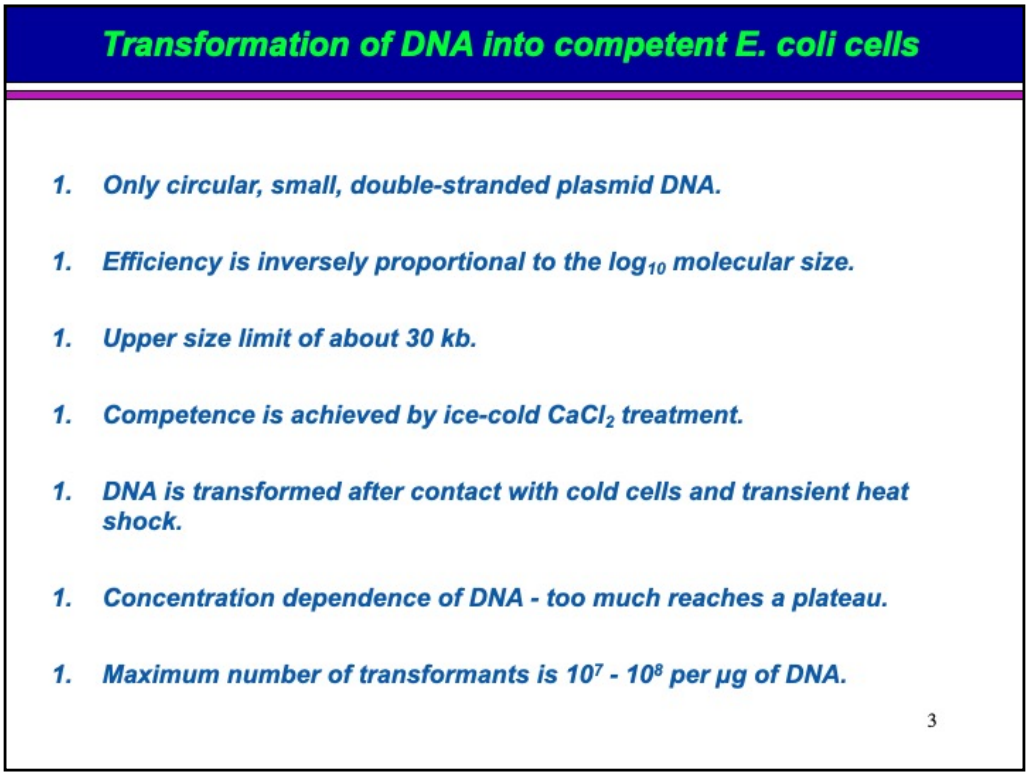
A detailed account of plasmid transformation; and a sampling of mini-strategies topics to assist in molecular cloning.



The first topic, transformation is intended to expand on the limited introduction in Lecture 4, to enable a fuller background for Expts 4 and 6.

The second topic explores a few simple strategies used in molecular cloning. As always, the science behind these ideas/methods are crucially important in understanding and learning in this subject.

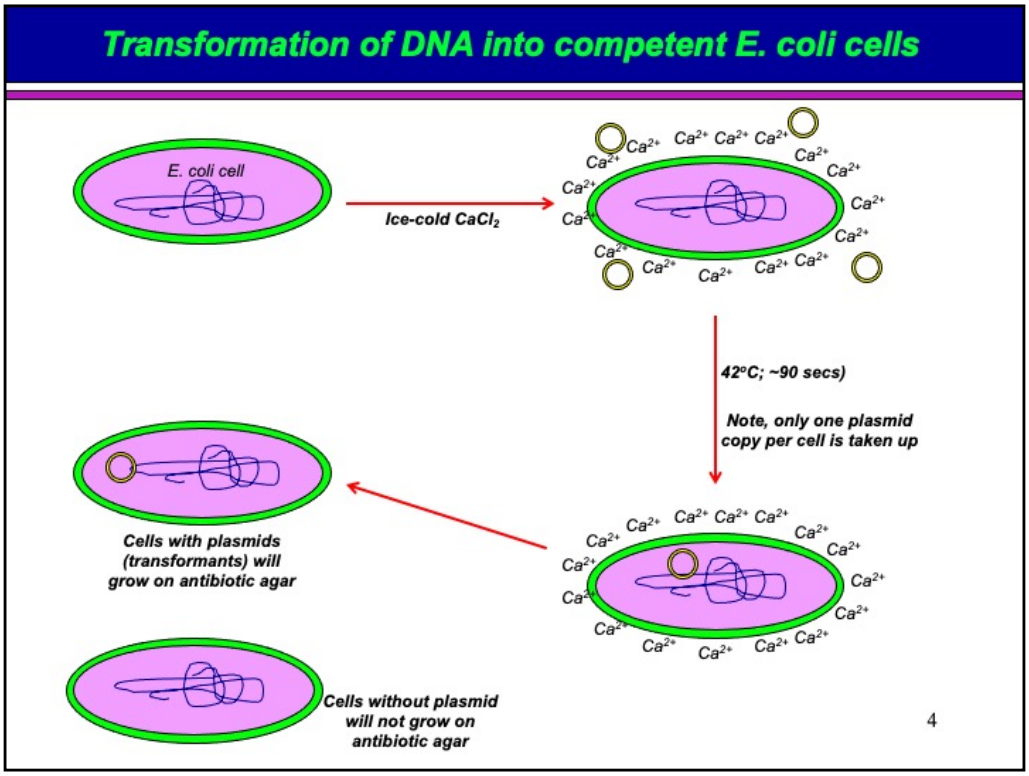
We will be looking at a few **insertional activation** methods, but far and away the most important will be the one that involves the LacZ complementation system. Its importance lies in several directions, including: How it has been 'captured' from the 1960s Operon Theory; how cloning vectors work; and how we use this method for our own cloning experiment (Expt 6).



Topic one for this lecture. How do we put plasmids into bacterial cells? Easy, and we'll be doing this in a couple of experiments.

The main conditions are that the *E. coli* cells must be COMPETENT for transformation, in an ice-cold calcium chloride suspension, and the DNA must be circular. There is an upper plasmid size limit, governed by the size of a circular molecule that can cross the cell envelop. In addition, too much DNA actually lowers, or at least does not improve, the yield of transformants, due to 'clogging' of the available uptake pores in the cell membrane.

Transformation is a passive chemically-induced process, as opposed to 'electroporation', which makes use of a powerful electric pulse to induce the DNA to enter the cell (a few slides on).



This slide was used in Lecture 4. The process of **transformation** in steps that outline: The ice-cold calcium chloride treatment; the addition of plasmids; the transient passage of a plasmid into a 'competent' *E. coli* cell; and the survival of the transformed cells on antibiotic-containing agar plates.

Most "competent" cells do not receive plasmids (~3% of cells are transformed) and do not grow on the agar plates, as they are killed by the antibiotic in the agar (bottom left). Only transformed cells are able to grow in the presence of antibiotic, as this property is conferred by the plasmid (middle left). Separate colonies will grow from each plasmid-containing transformed cell. Transformation efficiency can be enumerated by calculating the number of transformed cells (transformants) per µg of plasmid DNA. The blue squiggly line in each cell is the chromosome.

The essential features of this process are: *E. coli* cells are made 'competent' for transformation by washing them in several changes of ice-cold calcium chloride solutions. This results in the adsorption of calcium ions (Ca^{2+}) to the outer cell membrane (top right). The cells are now **competent** for transformation. The cells need to be kept ice-cold, to allow the calcium ions and DNA to remain bound to the cell envelope. The plasmid is adsorbed to the calcium ions by the electrostatic interaction of the Ca positive and DNA phosphate negative charges. [What happens to the sodium ions that usually form charge neutralizing clouds around the negative phosphates of the nucleotide bases?].

A short high (42°C) temperature pulse results in the cells becoming transiently (temporarily) permeable to the plasmid DNA, with the high temperature also dislodging the Ca-DNA interaction, thus enabling the plasmid DNA to cross the cell envelope. Once the plasmid enters the cell, it is primed to replicate and be inherited by successively dividing *E. coli* cells, which grow happily on the antibiotic containing agar and form colonies when the plates are left overnight at 37°C (middle left).

Each cell, though only transformed by a single plasmid, will rapidly replicate many copies of the plasmid in each of the dividing progeny within the colony, or in broth culture.

Electroporation of DNA into competent E. coli cells

1. Circular DNA, but can be used to transfer any DNA or even across cell membranes in bacteria and eukaryotic cell lines.

1. Electrocompetence by washing in ice-cold cold water.

1. Size dependence has much higher upper limit (< 100 kb).

1. Linear uptake dependence on DNA concentration.

1. DNA is transformed after contact with cold cells and high voltage electric pulse, creating transient pores.

1. Immediate recovery in growth medium is essential.

1. Maximum number of transformants is about 10^{10} per µg of DNA.

An Electroporator or Gene Pulser

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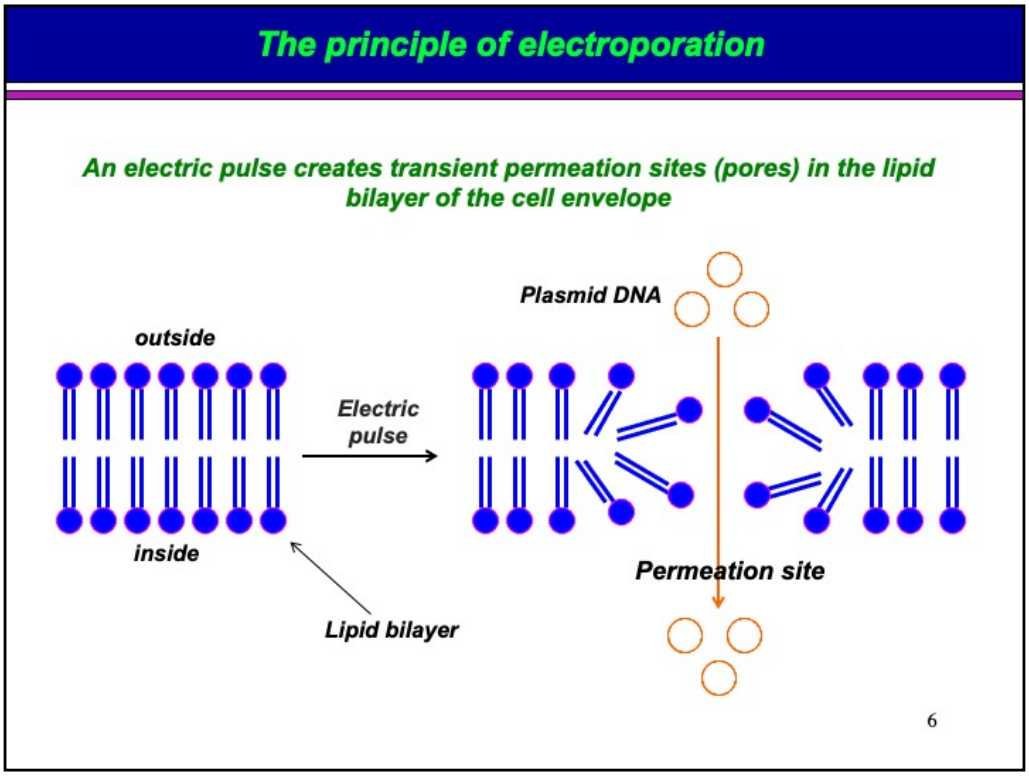
Electroporation is a process that uses an instrument (shown on right) that delivers a very high electric pulse of only picoseconds in duration. The electrodes are attached to a probe that fits over a special type of cuvette with +ve and –ve terminals.

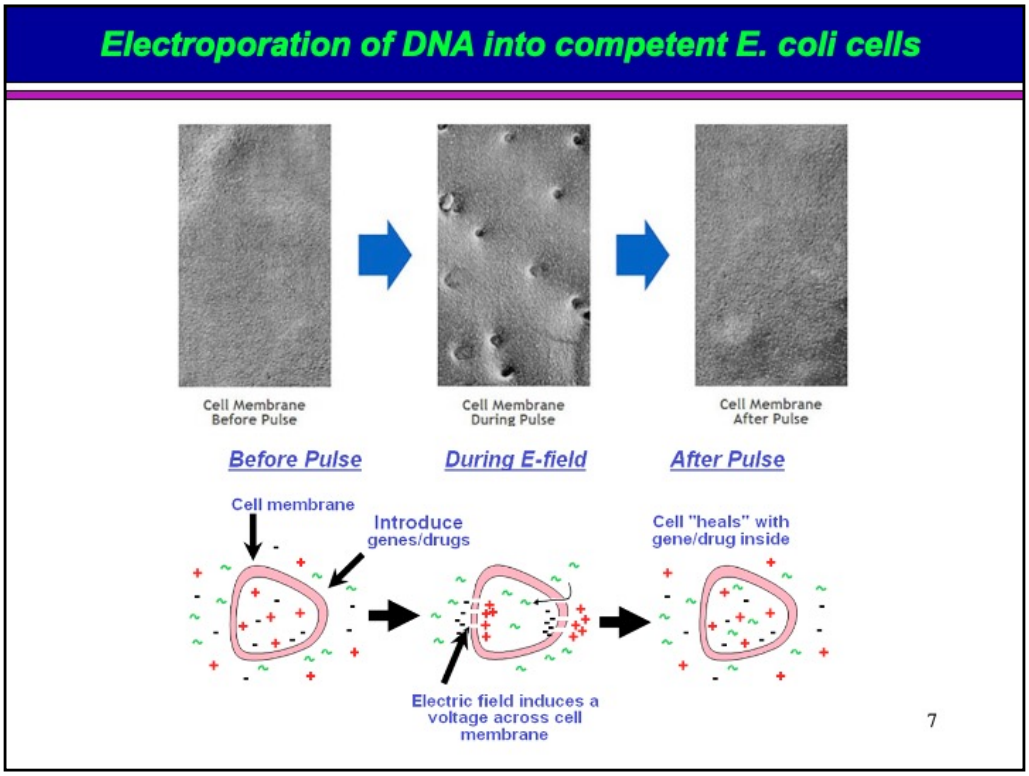
The ice-cold 'electrocompetent' cell suspension is placed in the cuvette, the pulse is delivered and the suspension is quickly added to pre-warmed growth medium to enable physiological recovery. The transient pulse punches small pores in the bacterial cell membrane through which the DNA is able to pass into the cell. The pores close just as rapidly, trapping the DNA inside the cell.

Electroporation has one important difference to transformation – the ice-cold bacterial cells are prepared in pure water (rather than calcium chloride) as ions will dissipate the electric charge and 'short-circuit' the pulse so that the pores do not form. Other differences are marginal and relate to quantity and size of DNA that can be transferred.

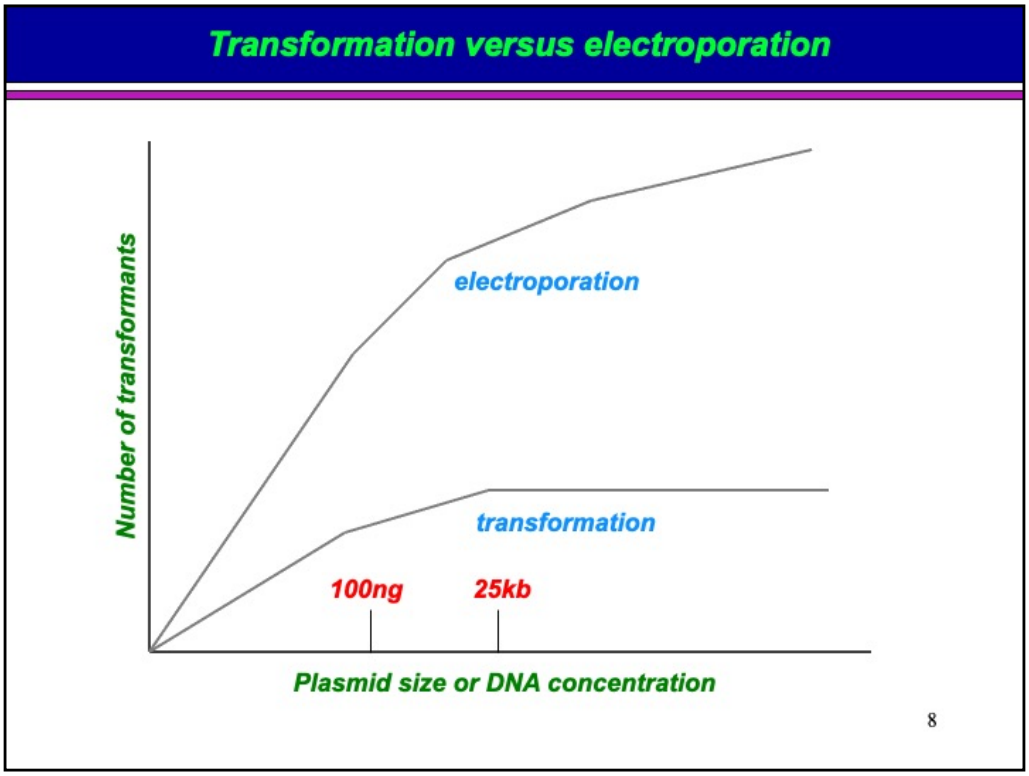
Linear DNA and proteins (e.g. antibodies) can be electroporated and the method is applicable to most types of cells (bacteria, yeasts, eukaryotic cells).



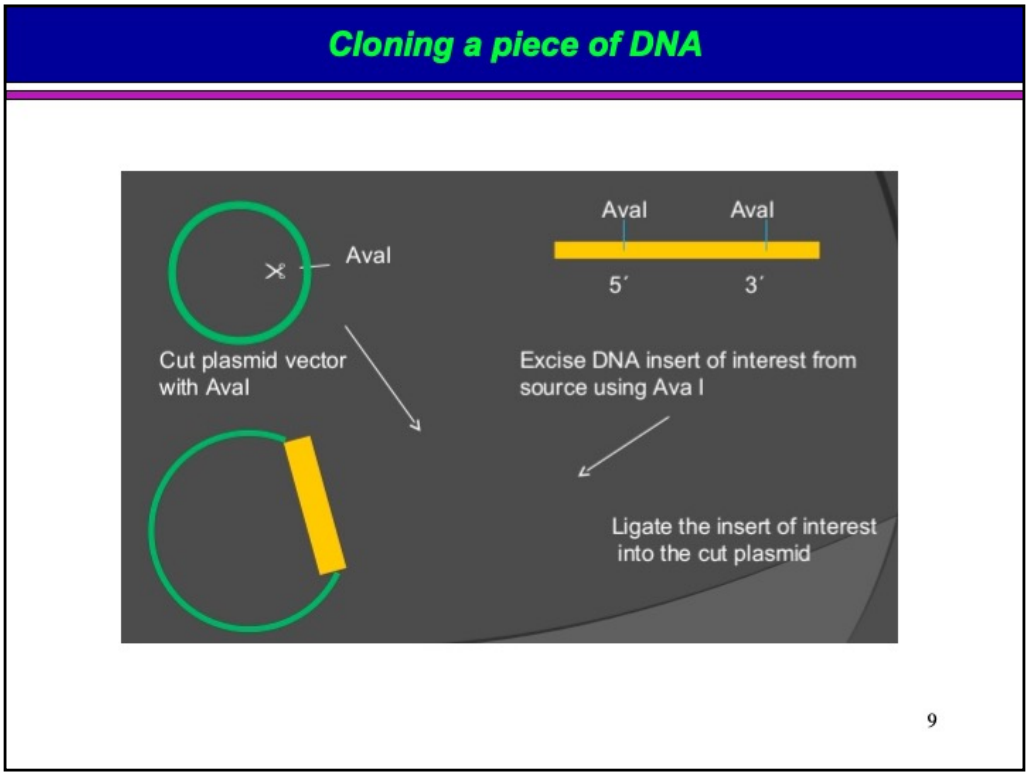
A illustration showing how a high voltage electric pulse creates a transient pore through which the plasmids can enter the cell. The permeation pore is created by the electric current and lasts for only a few picoseconds during which the DNA crosses the membrane; and becomes 'inherited' by the electrocompetent cell, as the pore closes as rapidly as it opened.



EM images of transient micropores in the 'electrocompetent' cell membrane (top) and a schematic of how the process works (bottom).



An arbitrary composite graph showing the differences in transformation and electroporation efficiency and dependency on plasmid size and concentration. In all other respects, both processes achieve the same result – that is, to enable bacterial cells to take up circular plasmids.



A simple cloning scheme to engineer a recombinant plasmid. There are a large number of molecular cloning strategies. Many of them are too advanced, or limited in their usage, and will not be dealt with in these lectures. The strategies that teach us about the science and methodology of fundamental molecular cloning will be described in this lecture.

In the simplest iteration of a molecular cloning experiment, a vector is linearised by digesting it at a unique restriction site, whilst preparing a source of linear insert DNA with the same or compatible restriction ends. In the example above, the vector plasmid is cut with the restriction enzyme, then ligated to the target gene, both with compatible AvaI sites.

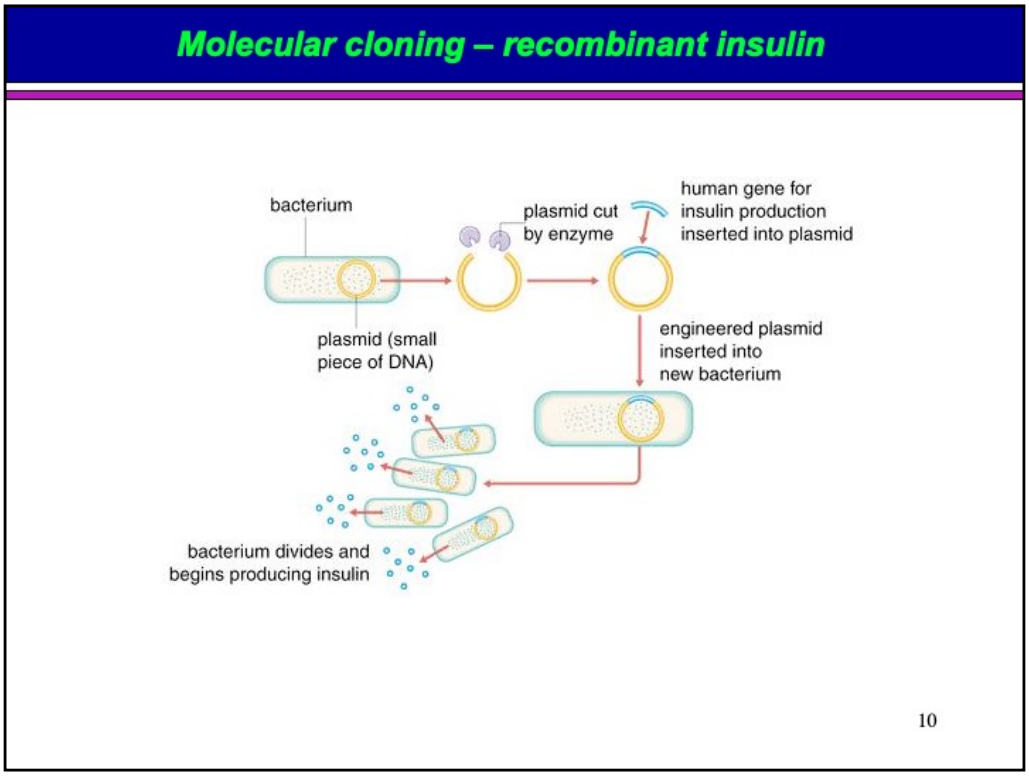
Note though, there are at least four ways of doing this type of cloning experiment, namely:

(a)Using a single site compatible enzyme, such as in the example above;

(b)Using enzymes that are isoschizomers of one another (they cut at the same palindrome thereby generating the same ends);

(c)Using any blunt-ended restriction enzyme, and

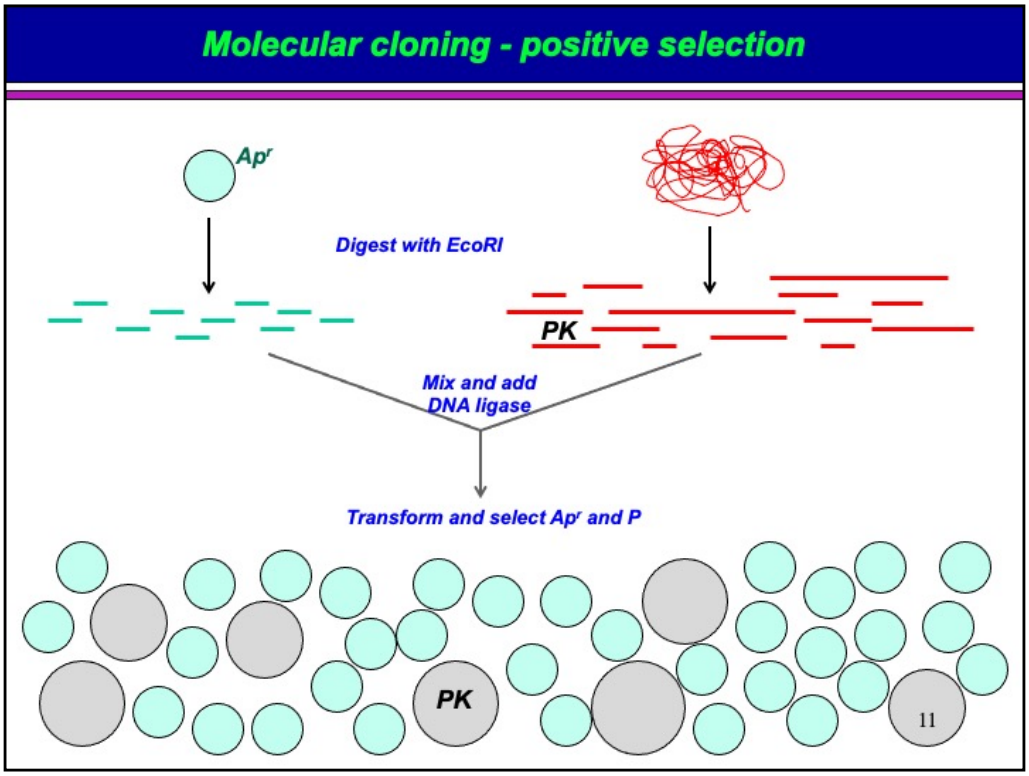
(d)Using two different, incompatible enzymes to cut the vector and insert.



The full process of cloning and production of insulin, for instance. This and the previous slide serve to depict the simplest and most direct route for cloning a gene into a vector plasmid and generating a recombinant (hybrid, chimeric, engineered) plasmid. This may work in many cloning strategies, but in practice, there are issues to be solved. Chief among these is the eventual successful expression of the cloned gene product, the protein, in the bacterial host, from which it can be purified and used downstream for its purpose, such as the insulin example shown here.

Gene expression is a dual problem of regulated or rampart expression, the latter of which could kill the host cell. These problems will be teased out in a later lecture.

This lecture continues, by looking at other problems in molecular cloning strategies, namely, ensuring a good yield of transformants/recombinant plasmids, and some ideas that will provide other cloning choices such as modifying the vector or inserts before the ligation reaction.

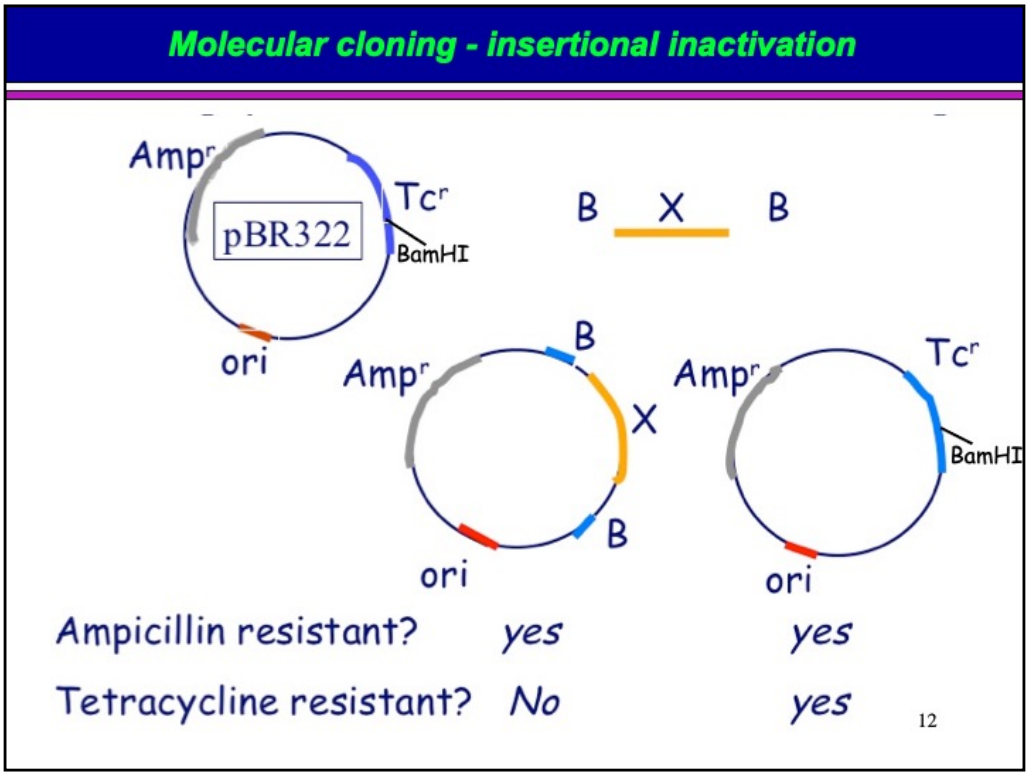


For the exercise of molecular cloning, only two types of molecules are important and these are recircularised vectors (small light green circles above) and true recombinant plasmids (grey circles). In the example above, the "background" of vector and various recombinants is unimportant because we have a telltale selectable phenotype 'PK' for the clone we are seeking; and that phenotype is easily identified by say antibiotic resistance, or a colour pigment in the colony. If the phenotype 'PK' is kanamycin resistance for example, we only need to use agar plates with ampicillin (vector signature resistance gene) and kanamycin to obtain the required recombinant clones, which will be the only ones that grow.

In practice, most molecular cloning experiments do not have such 'signature' phenotypes that are easily selected or recognized and we would then need to trawl through many dozens, hundreds, or even thousands of transformants to find the recombinant clone of interest. Luckily, there are ways of reducing or eliminating the vast background of recircularised vectors, so that mostly the desired recombinant clones will be present.

The most important consideration for the purpose of a successful molecular cloning experiment are to generate ONLY recombinant plasmids that may be screened subsequently for the clone of interest. Linear concatamers (not shown here) are rendered essentially redundant by using ideal ligation conditions of molar equivalents of vector/insert (see earlier lecture); and by noting that transformation will only enable circular plasmids to enter the competent bacterial cells.

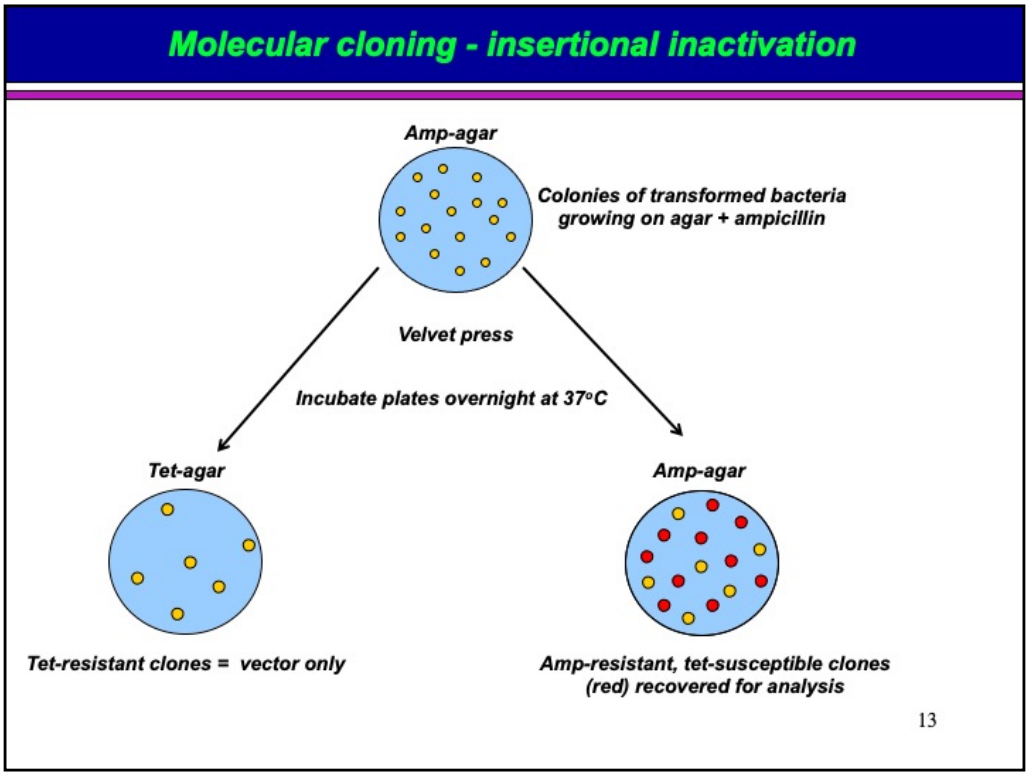
The chief problem then is the background of original vector plasmids that will represent the majority of transformants (~60-95%). This is because small plasmids transform much more efficiently than larger ones – at the rate of about one log less per doubling in plasmid size. There are a number of ways of identifying recombinants, or of reducing or eliminating vector background entirely and the next few slides will illustrate several ways of doing this.



Insertional inactivation of an antibiotic resistance gene is one way of discriminating between religated vectors and recombinant plasmids.

In this example, which we have looked at in an earlier lecture, using the vector pBR322, a DNA linear fragment ('X') with BamHI ends is ligated into the unique BamHI site within the vector's tetracycline resistance gene. The resulting recombinant plasmid (middle) is now susceptible to tetracycline, as the tet gene ORF sequence has been interrupted (by the orange insert fragment) and the Tet protein cannot be made and therefore cannot elicit tetracycline resistance. However, the original vectors that religated without an insert (at right) will still express resistance to tetracycline.

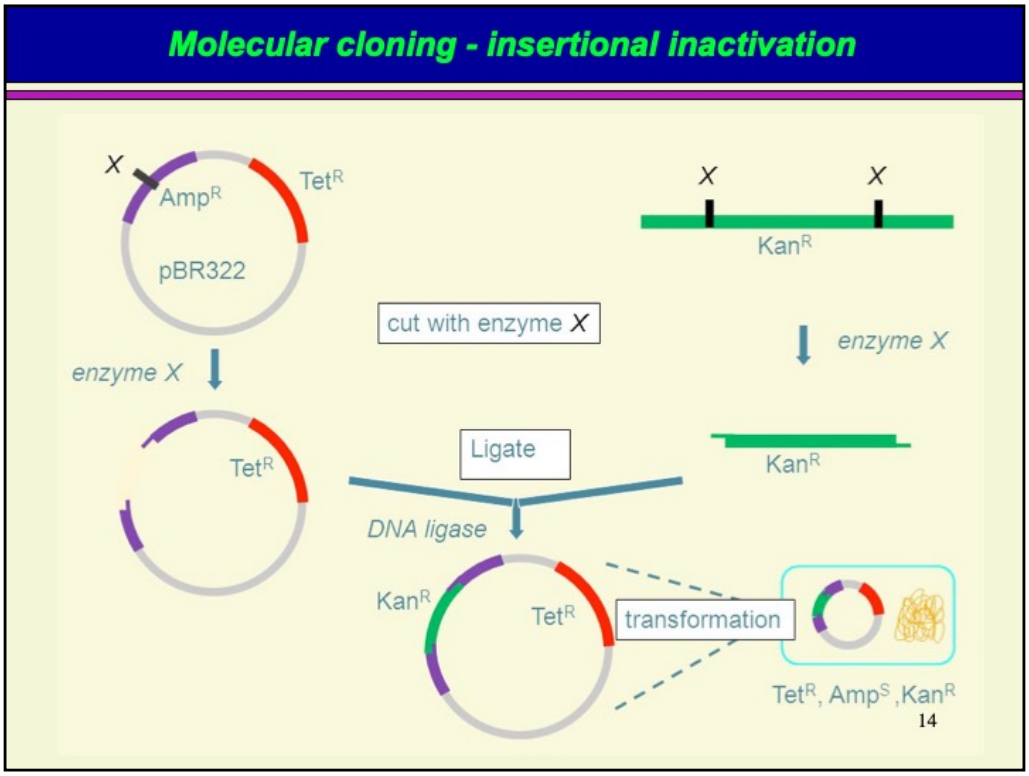
Thus, a transformed ligation mixture will yield two types of clones. Those with vectors, which will be resistant to ampicillin and tetracycline; and those with recombinants, which will be resistant to ampicillin but susceptible to tetracycline.



A simple grid plating method to screen clones from the indirect (difference) cloning method (insertional inactivation) shown in the previous slide.

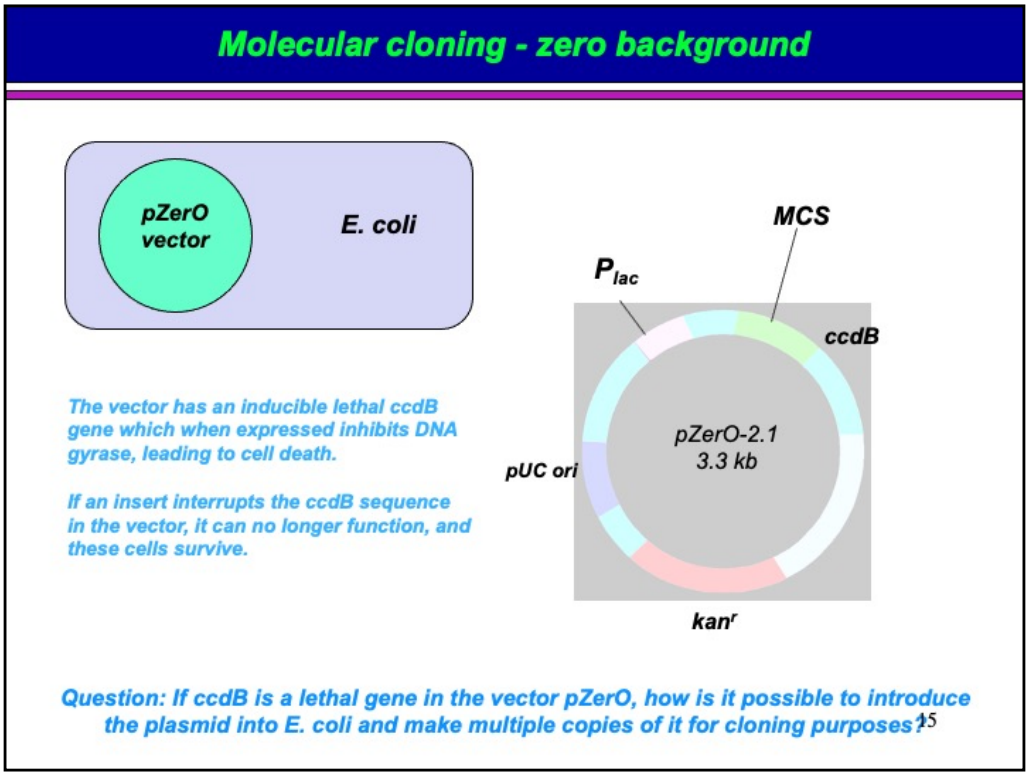
A master grid of transformants (top) is patched (velvet press) onto Tet-only (left) and Amp-only (right) agar plates. Only vector transformants (background) grow on the Tet plate (bottom left) and so the actual recombinants are those that did not grow, and these may be recovered from the master patched plate (red clones, bottom right).

In practice, this method is effective, but requires a special vector (with two antibiotic markers), and is labour intensive, in that many grids would need to be screened to obtain a sizable number of recombinants. The grid result illustrated in this slide is exaggerated to show the potential result, but in practice, a large percentage of transformants (60% to >90%) will be vectors (yellow clones; Tet^R), requiring extensive screening.



Another example of insertional inactivation using pBR322. In this case, the restriction site 'X' within the Amp^R gene is used to clone/ligate a linear fragment bearing a gene for kanamycin resistance (Kan^R). The recombinant plasmid and resultant phenotype (TetR, AmpS, KanR) in transformed *E. coli* cells are shown at the bottom of this slide.

Of interest here is the fact that one antibiotic resistance gene (Amp^R) has been inactivated due to its ORF being split, and another antibiotic resistance factor (Kan^R) is now expressed in the recombinant. The contiguous Kanamycin resistance ORF now sits within the interrupted Ampicillin resistance ORF, which generates the required $Tet^RAmp^SKan^R$ phenotype in the recombinants.

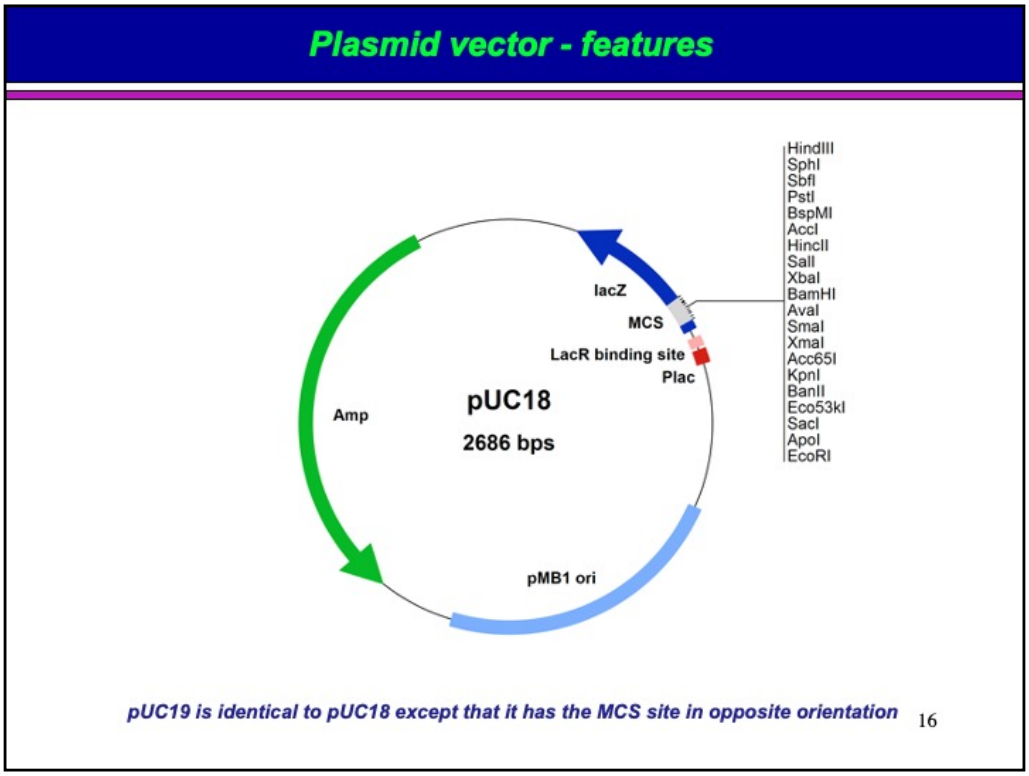


As we have seen already, a large background of religated vectors is the bane of recombinant DNA technology.

There are two ways of controlling vector background**:** By preventing the vector being regenerated, or by preventing its survival in transformants. The latter method is illustrated in this slide.

The pZerO cloning vector makes use of an *E. coli* "Death Gene", *ccdB*. The CcdB protein is lethal, as it is a powerful inhibitor of DNA gyrase, without which the cell cannot repair or replicate its chromosome. The pZerO-2.1 vector has a composite ccdB sequence and MCS cloning site. The ccdB gene is expressed (transcribed and translated) only when the Plac promoter is activated.

A DNA fragment is inserted at the MCS and within the ccdB sequence, which is thus interrupted by insertional inactivation (as we saw in the antibiotic gene examples in pBR322). Recombinant plasmids will now survive in transformed cells, whereas religated vector will result in the death of the transformed cells, because the still intact CcdB protein inhibits DNA gyrase. The vector background is absent completely, and **all** surviving colonies contain recombinant plasmids.



pUC18 is a small vector developed in the mid-1980s and still used as an effective cloning vector. Note that it is smaller than pBR325 and pBR322, meaning that it will generate a higher copy number per bacterial cell – and therefore a higher yield of plasmid DNA.

Features:

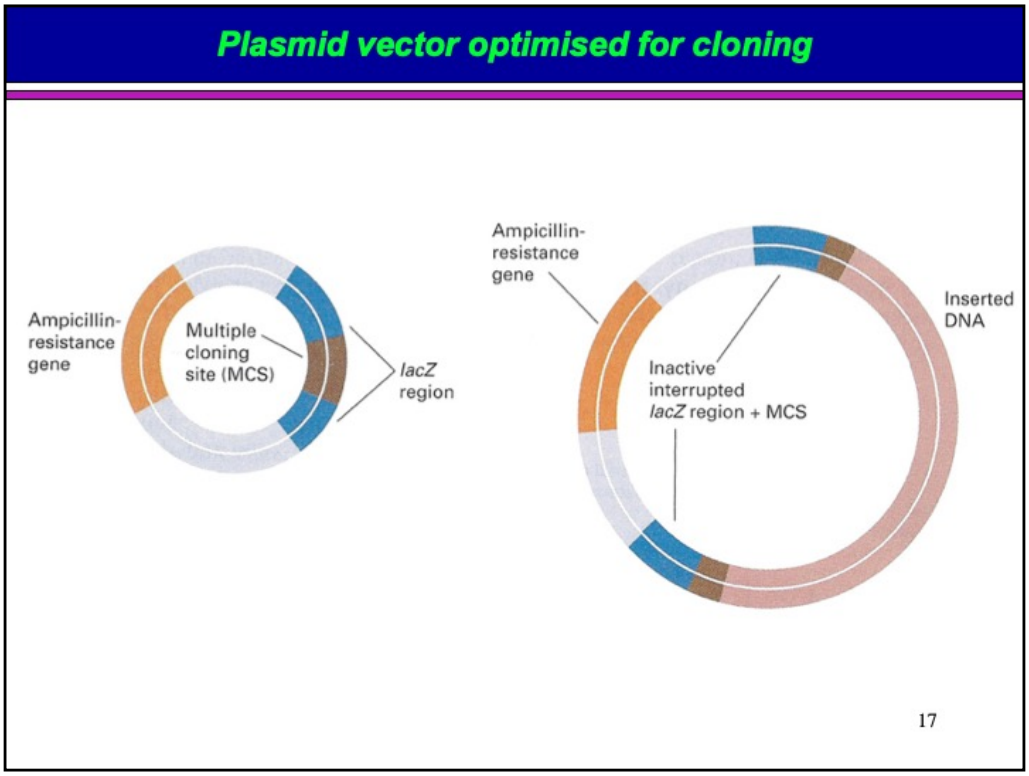
1.It has the pMB1 origin of replication, for relaxed replication control and high copy number.

2.It has only one antibiotic resistance gene (Amp) for selection and maintenance in a bacterial culture.

3.LacZ is the first portion of the lacZ gene, comprising the first 30 amino acid codons of the very large LacZ beta-galactosidase ORF (1021 amino acids). The lacZ amino acid codons (dark blue arrow) overlap the MCS (Multi-Cloning Site).

4.The LacR repressor binding site (orange) is just "upstream" (as we say) from the Lac promoter (red) and it also contains the operator site, which is part of the LacZ gene's promoter region/site. This expression system allows a cloned gene to be held 'silent' (not expressed at all), or to be switched on (expressed) at different rates (like water from a tap). The overall intention is to produce a protein product from the cloned gene, but under controlled conditions (more on this in later lectures). This control region will be explained in the next few slides.

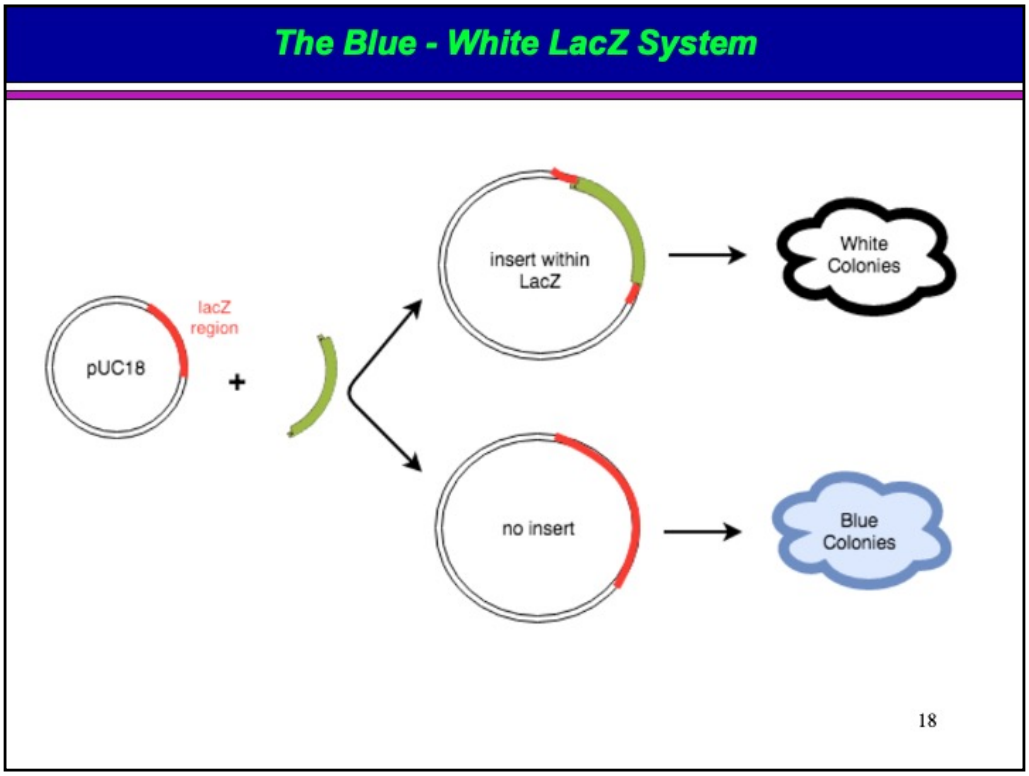
The MCS region contains an engineered short sequence of DNA with many restriction enzyme sites in tandem as indicated. The partner plasmid to pUC18, namely pUC19, is identical in all respects except for the opposite orientation of the MCS. This and most 'modern' plasmid cloning vectors do not use the antibiotic insertional inactivation system, and instead use a different type of insertional inactivation, hence the need for only a single resistance gene. A complementation system, based on a colony colour difference (blue Vs white) is used to discriminate between the original vector (blue colonies) and the recombinant clones (white), as illustrated in the next few slides.



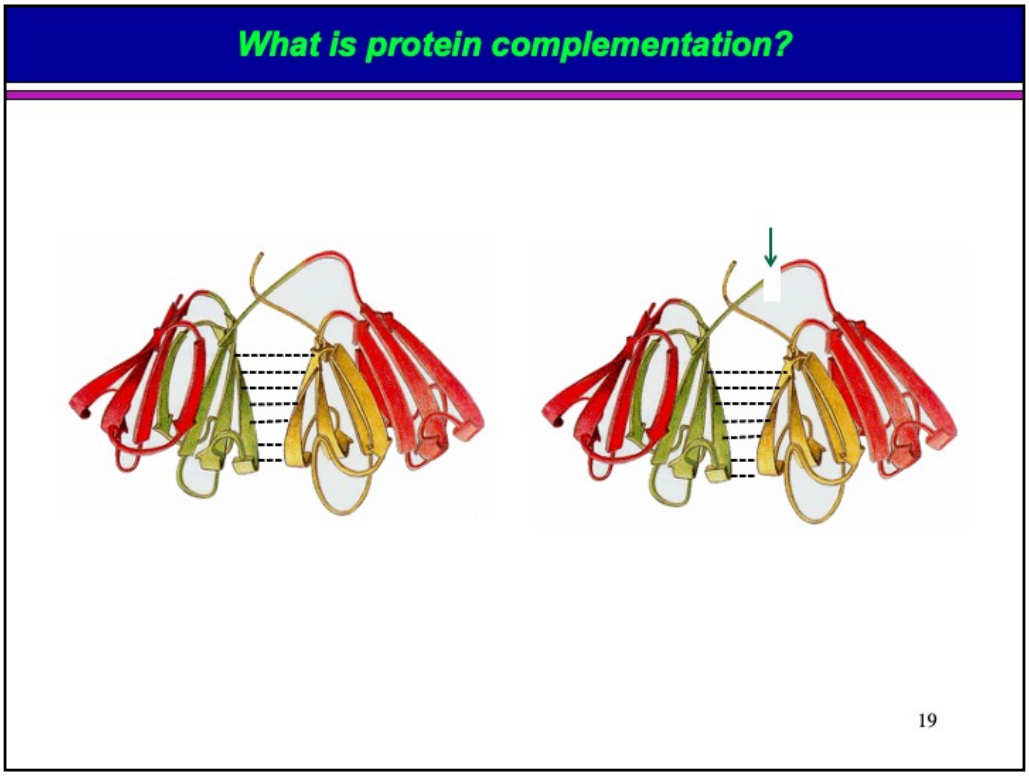
Following on from the last slide, this preview slide encapsulates the essence of the LacZ complementation/insertional inactivation system.

The vector at left depicts the composite LacZ-MCS cloning region.

Cloning into the MCS is the genesis of selection for recombinant clones in most modern vectors. The inactivated lacZ region in the recombinant plasmid at right (blue-brown region) is split/interrupted by the insertion of the foreign DNA fragment (pink). As you will see, this version of 'insertional inactivation' is a direct detection system, using blue-white colony differentiation to discriminate between religated vectors and recombinant plasmids. In addition to what is going on in the cloning vector, a defective LacZ gene in the *E. coli* host is also involved, as we will see in the next few slides.

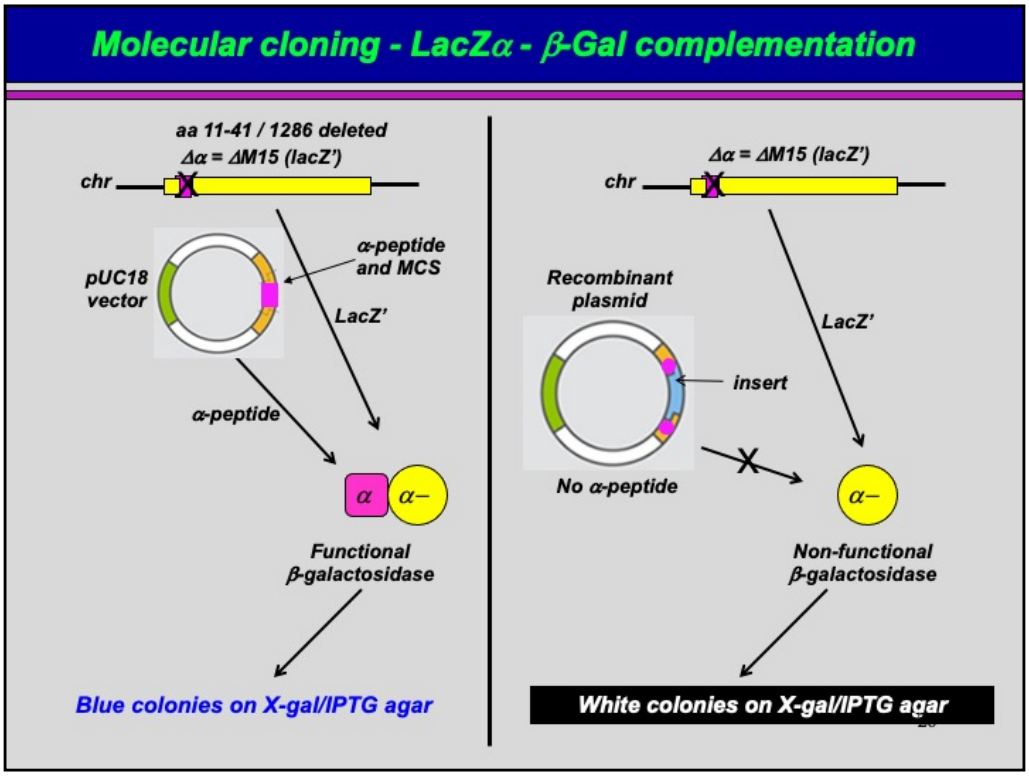


A simple scheme to indicate the recombinant/vector white/blue system. Note that the upper plasmid in the middle of this slide has an insert now ligated within the lacZ region. Transformants with recombinants such as these will appear white on agar plates containing the obligatory selective additives (Amp; IPTG inducer; X-Gal chromogenic substrate for LacZ = Beta-galactosidase).



Before we look at how LacZ insertional inactivation works, we need to revise or refresh our knowledge of **complementation** in proteins. Complementation in proteins occurs when two separate segments of the same original single polypeptide can associate into the same conformation and still carry out the same function as the single polypeptide version of the protein.

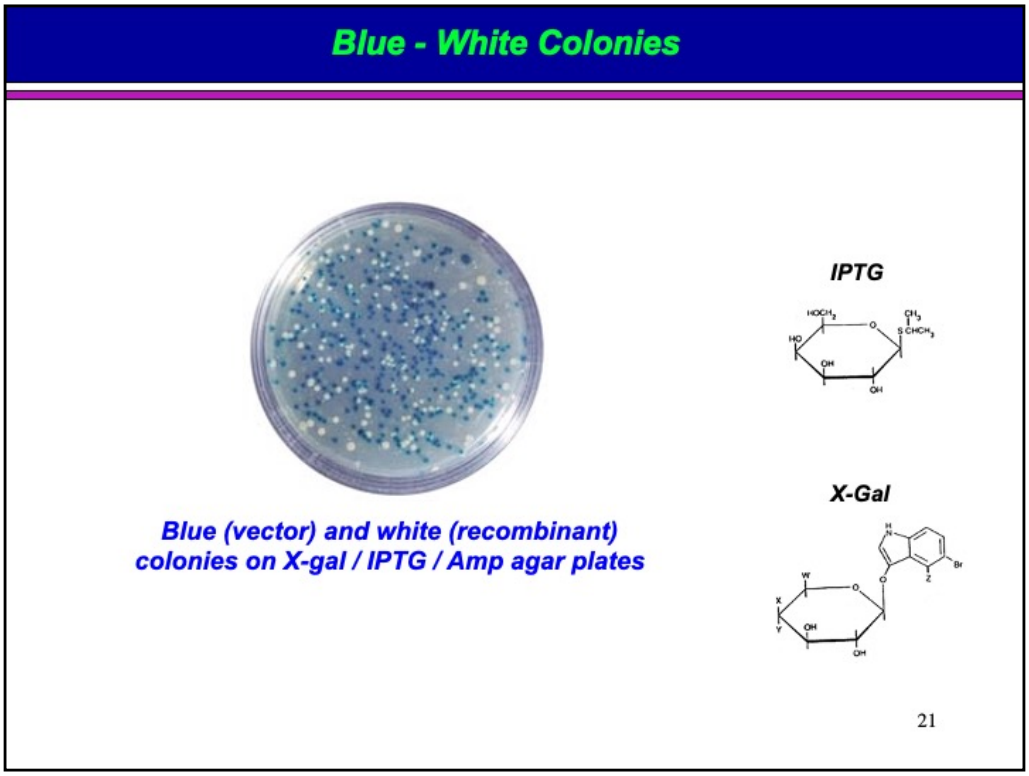
On the left is a contiguous single polypeptide folded into two near identical domains that are held together by a number of amino acid side chain ionic and H-bond interactions (dashed lines). On the right is the same protein cut in half in the linker chain (arrowed). However, the separate domains still associate non-covalently with all the side chain interactions, and the protein still works, in whichever way it does, to about the same activity level of the single polypeptide version. This is termed 'complementation'.



The LacZ blue-white agar plate selection method that works by insertional inactivation. Unlike the Tet-Amp system that depends on antibiotic resistance/susceptibility to discriminate between vector and recombinant clones, which is an indirect insertional inactivation method, this LacZ beta-galactosidase system uses colour (blue versus white) to discriminate between vectors and recombinants; and is therefore a direct selection method.

Left Panel: Elements of the cloning vector and the host *E. coli* chromosome combine to generate the active beta-galactosidase enzyme by **complementation**. Re-ligated vectors thus will supply an uninterrupted alpha-peptide (pink) that complements the defective enzyme (alpha-minus; yellow) being expressed from the chromosome, producing a functional galactosidase (left panel). This hybrid, complementing enzyme is able to hydrolyse the synthetic substrate X-gal, liberating indole that gives an intense blue colour to the colonies containing vector plasmids.

Right Panel: If an insert is ligated into the MCS of the vector, the alpha-peptide is insertionally inactivated (separated pink segments) and cannot complement the defective beta-galactosidase enzyme (yellow). Thus recombinants are identified by their white colony appearance on selection plates, as X-gal cannot be hydrolysed to produce indole. Like the previous antibiotic inactivation scheme, this one is determined in the agar plates on which the transformants are plated, with Amp, X-gal, and IPTG all included in the agar (next slide).

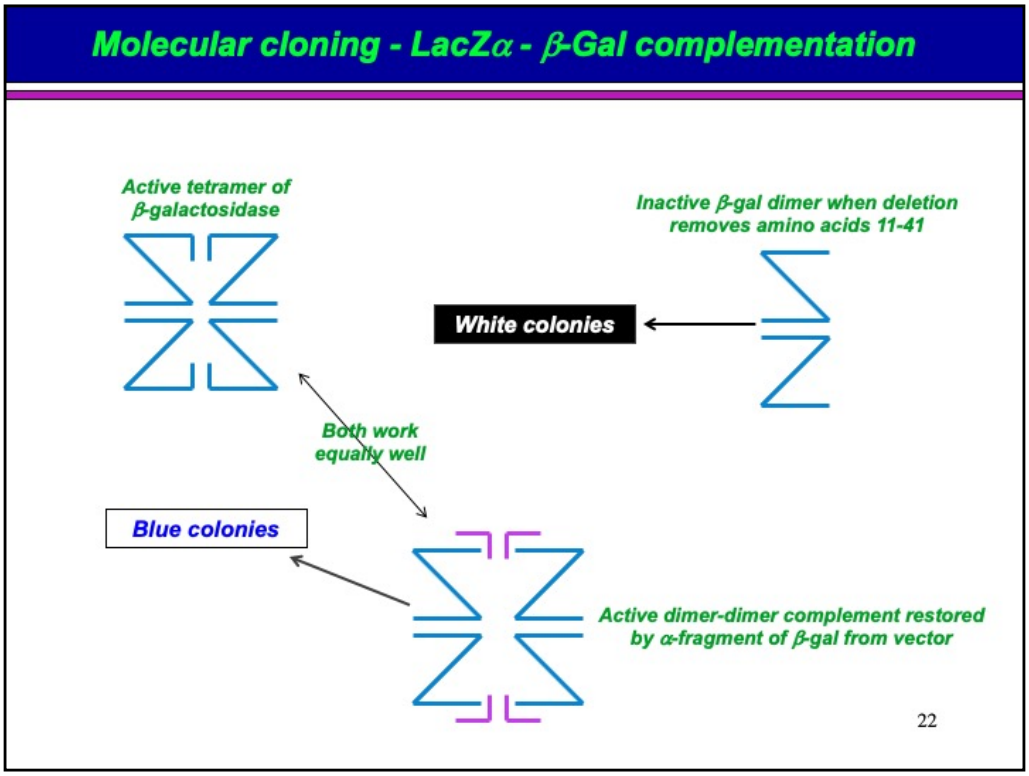


Blue vector and white recombinant colonies on ampicillin-X-gal-IPTG agar plates.

IPTG is the inducer molecule that removes the repressor from the operator-promoter on the vector pUC18 and enables read through uninterrupted transcription/translation of the LacZ alpha-peptide, which is then able to complement the defective beta-galactosidase transcribed and translated from the chromosome. The complemented enzyme (alpha plus beta components) is able to hydrolyse the synthetic substrate X-gal, releasing indole that gives the colonies a blue colour (vector background).

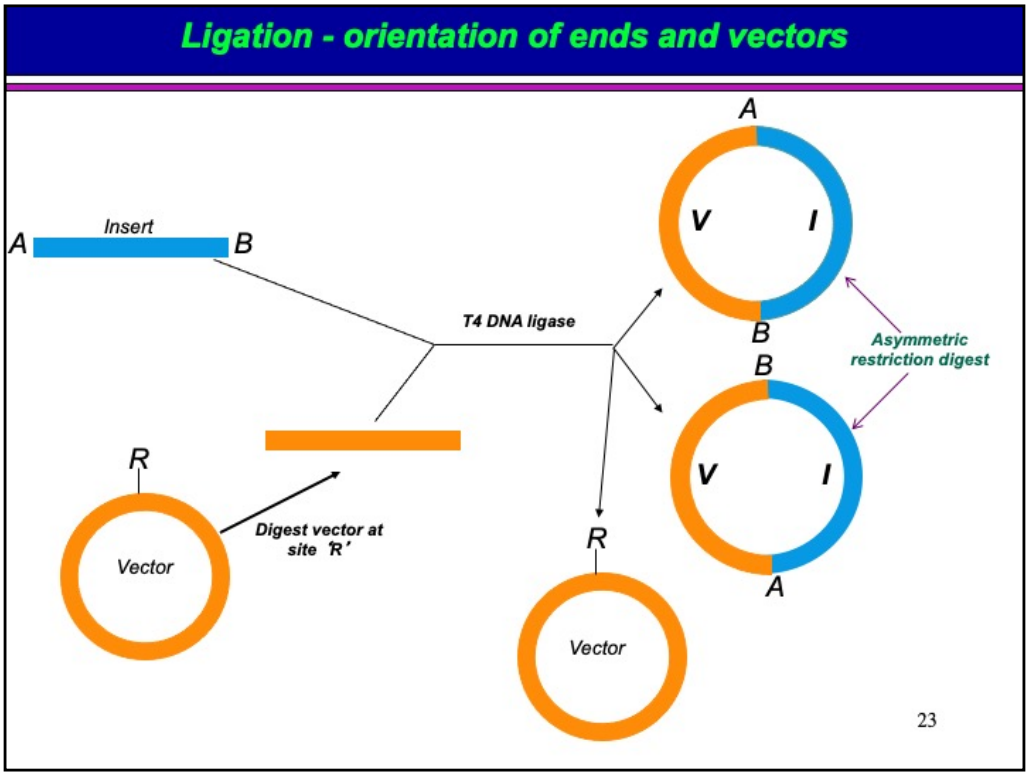
In the recombinants, the LacZ alpha-peptide within the MCS has been insertionally inactivated by the presence of the new cloned DNA fragment and cannot complement the defective beta-galactosidase, as the alpha segment is interrupted/split. There is no hydrolysis of X-gal and no indole or blue colour. The recombinant clones are white or cream coloured.

The structures of then synthetic inducer IPTG (Isopropyl β-D-1-thiogalactopyranoside) and chromogenic substrate X-gal (5-bromo-4-chloro-3-indolyl-β-D-galactopyranoside) are shown at right. The usual substrate for beta-galactosidase is the disaccharide lactose, whose glycosidic bond is hydrolysed to liberate galactose and glucose.



Complementation in beta-galactosidase. The complete enzyme only works as a tetramer or dimer of dimers (top left). If the alpha-peptide is deleted, the enzyme can only assemble as a dimer (top right) which is inactive. If the alpha-peptide is supplied by the vector (purple elbows), it can combine with the defective enzyme produced from the host chromosome and generate an active tetramer by non-covalent complementation of the two parts of the enzyme (bottom).

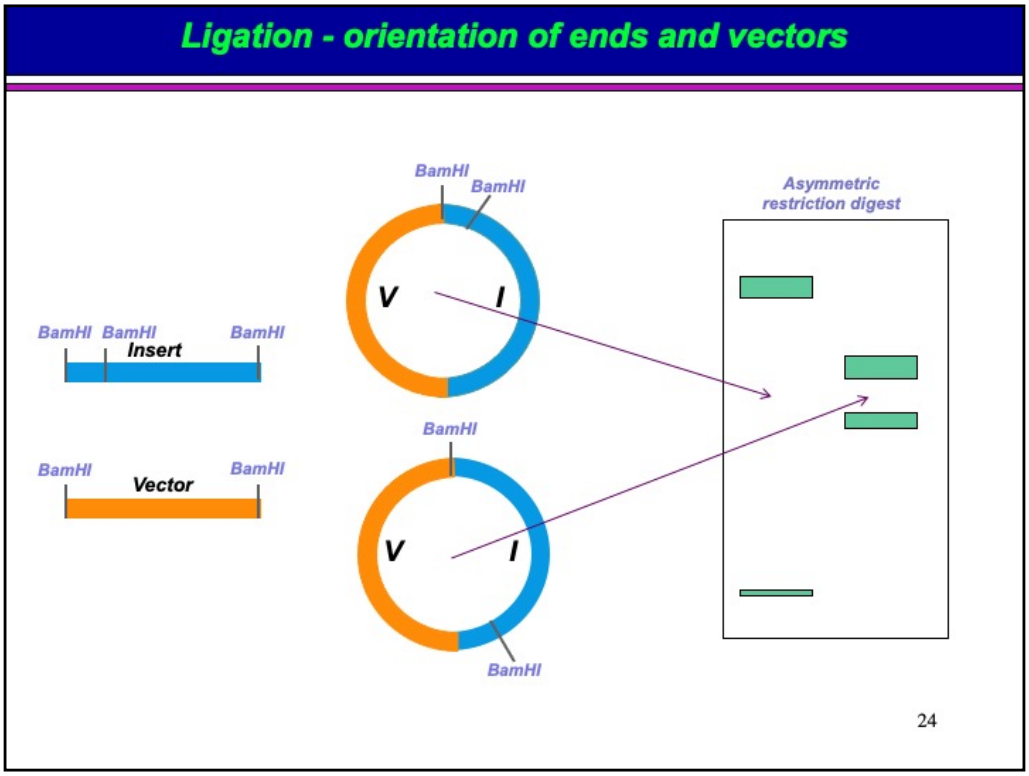
Why are we looking at this strategy in some detail (this and the next 4 slides)? Because most currently used vectors (plasmids and viruses) use this beta-galactosidase selection system for discriminating between religated vectors and recombinants.



Molecular cloning or genetic engineering, in which a new 'recombinant' DNA plasmid is formed, but with a 'twist', literally! The linear insert (A-B; blue) is to be ligated to the linearised plasmid vector (orange).

If the ends of the insert (A and B) have the same restriction site as that of the plasmid (i.e. all ends are compatible), then there are two possible products, shown at right, with the orientation of the insert opposite in each case to the fixed position of the vector. Is it possible to discriminate between the two populations of molecules? Yes, as we will see on the next slide.

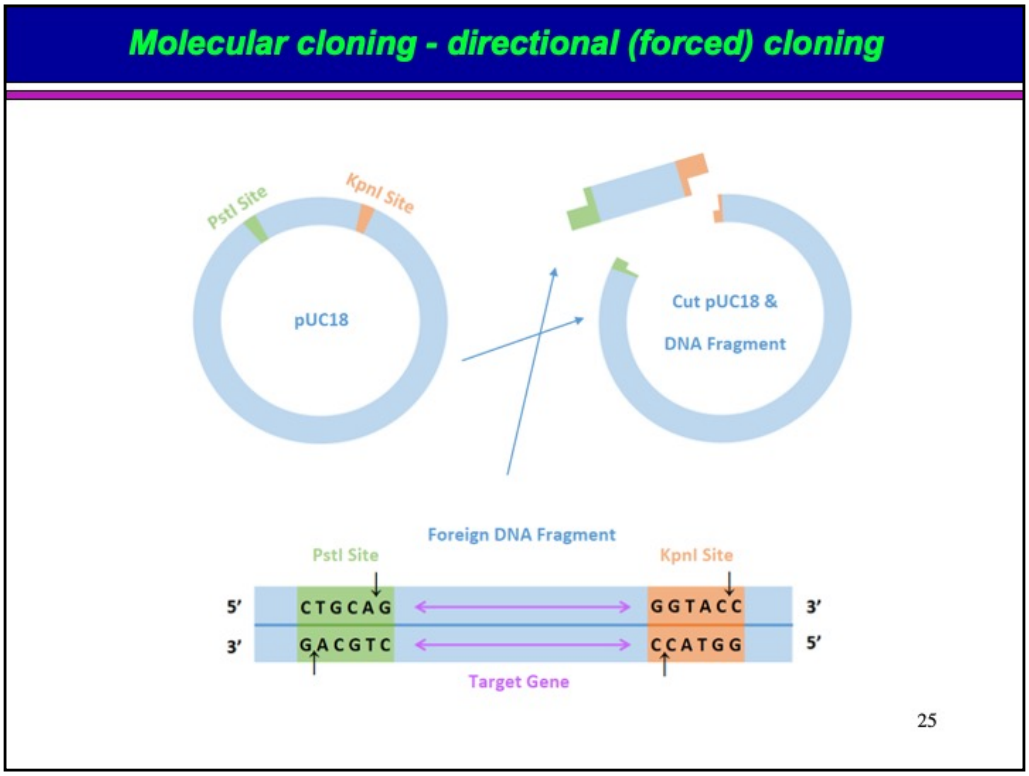
Despite the different orientations of the insert in the vector (50:50), a more pressing problem is that the original vector can comfortably re-ligate to itself (bottom right) and this will produce a very high population of vector clones that will dwarf the small number of recombinants, which are more difficult to generate as the ends are on different DNA fragments. We will look at a couple of easy ways to reduce the vector background to near zero in the few slides further on.



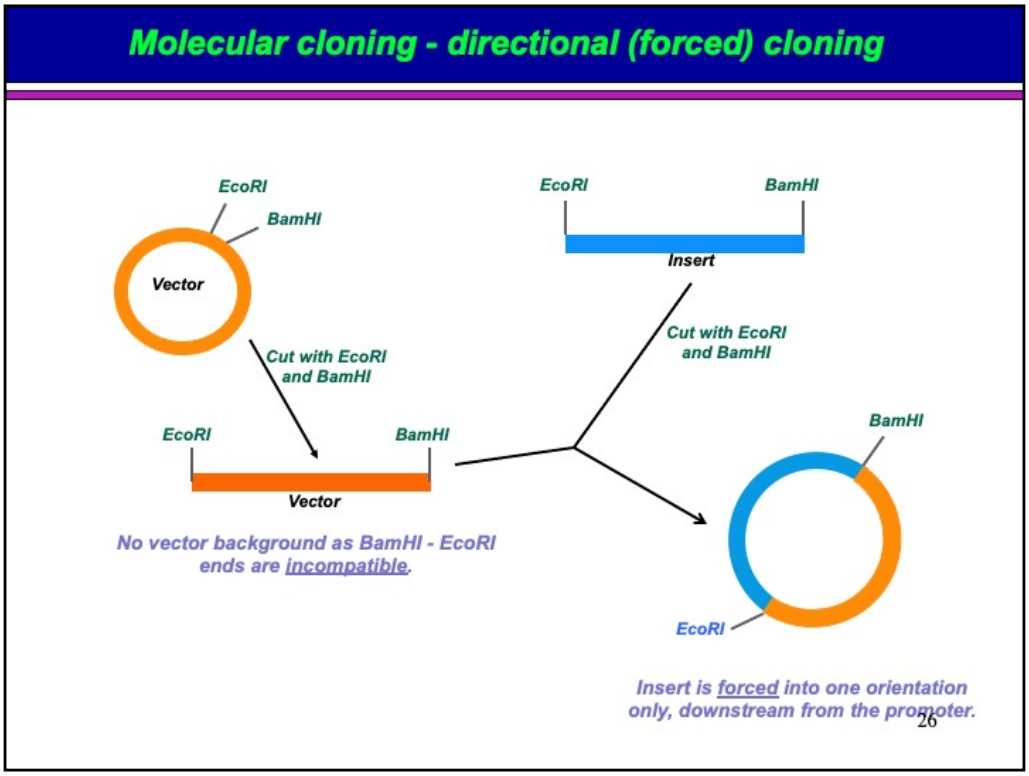
If the ends of the insert have the same restriction site as that of the plasmid (i.e. all ends are compatible), then there are two possible products, shown at centre, with the orientation of the insert opposite in each case to the fixed position of the vector.

We are able to work out the insert 'orientation' with respect to the vector, by carrying out an asymmetric restriction digestion of the two types of recombinant plasmids. We do this by making use of any asymmetric site within the insert. In this example, I have arbitrarily given the insert site the enzyme designation BamHI, for simplicity, but it can be a different enzyme, in which case a double enzyme digest would be required.

The agarose gel profile at right depicts the BamHI digests of the two types of plasmids (long indicator arrows). Note the simple and clear difference in restriction fragment sizes, as dictated by the positions of the asymmetric sites in the vector and insert.



A simplified scheme for directional or forced cloning using two different (incompatible) restriction enzymes. Note the site palindromes for PstI and KpnI, respectively, at the bottom. These sites are incompatible, meaning that the ends cannot be ligated to one another. The vector cannot ligate to itself anf this renders the background of vector clones as virtually zero.

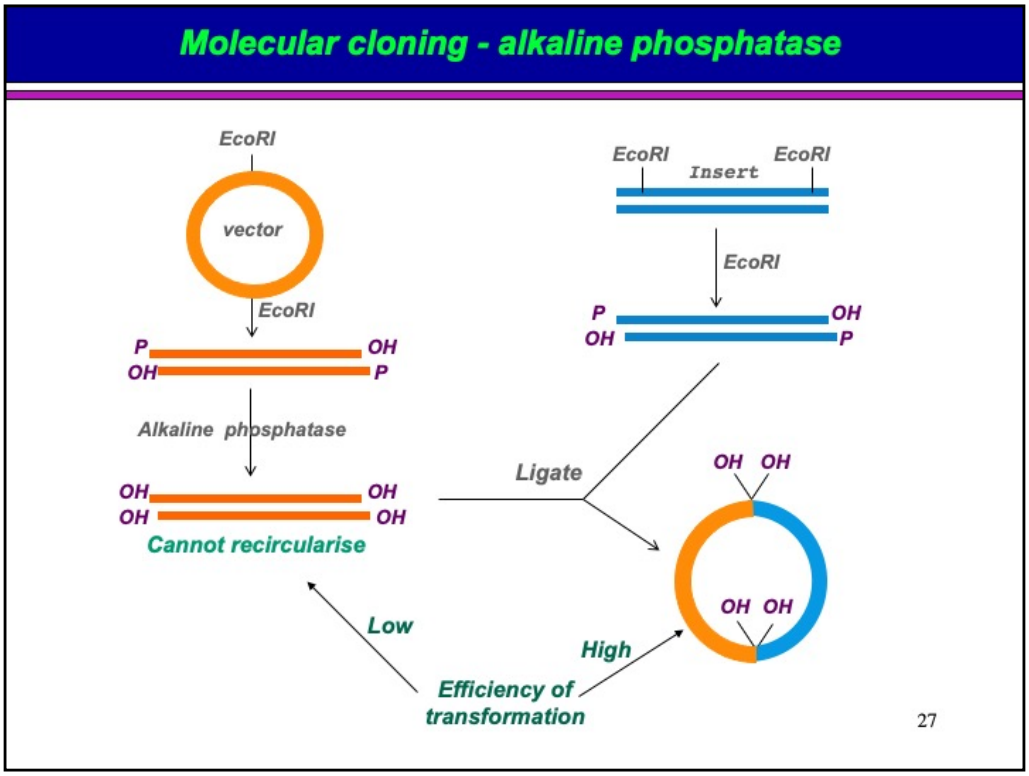


Directional or Forced Cloning has two important advantages in molecular cloning:

1.The first is the common theme we are now looking at – that is, reducing vector background to zero so that most or all transformants will contain recombinant plasmids. We can achieve this outcome in a very simple way by cutting the vector with two different (and incompatible) restriction enzymes as illustrated here for EcoRI and BamHI, though any incompatible pair within the MCS site of the vector may be used. The insert is prepared with the same two incompatible sites on either end. The ligation reaction will generate the recombinant plasmid shown at the bottom right of this slide. There is no background because the vector is incapable of religating – its ends contain incompatible restriction sites, and linearised vectors transform poorly, if at all, into *E. coli*. All transformants will therefore carry recombinant plasmids. We will be visiting this method in Experiment 6 (Molecular Cloning).

2.The second advantage of using this forced cloning scheme is that the insert is ligated into the vector in ONE (forced) direction only and this means that all recombinants (100%) are **exactly** the same. Directional cloning ensures that a cloned gene will have the correct orientation (5' to 3') with respect to an upstream promoter in the vector (not shown).

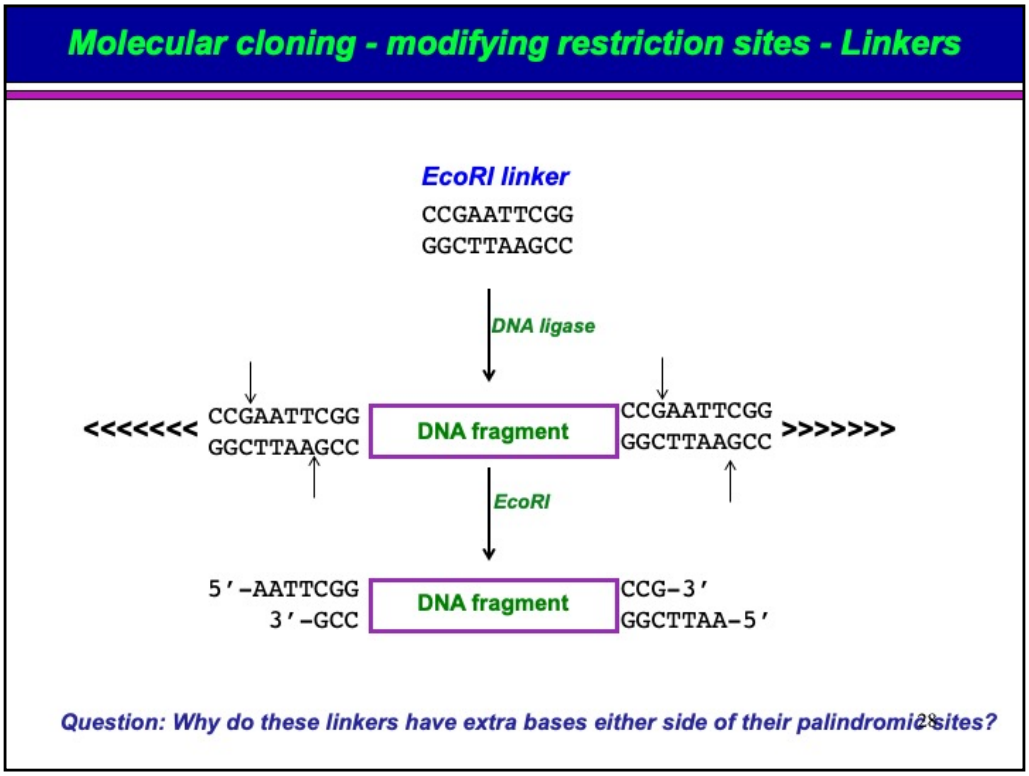
As effective as this method seems, in practice it has limited utility, as you are restricted (excuse the pun) to sets of enzymes that are found in the MCS of the vector, and that the insert can be generated with different restriction site ends. However, it works well enough when possible.



Another common method for reducing vector background, and one that we will explore in Experiment 6. Alkaline phosphatase is an enzyme that removes 5-prime phosphates from the ends of DNA molecules.

In the example above, the vector and insert are both prepared with EcoRI ends. The linearised vector is then treated with alkaline phosphatase, which removes the two diagonally opposite 5-prime phosphates (middle left in the slide), leaving the vector with four hydroxyl ends. The vector cannot be religated to itself and remains linear, because it is incapable of reforming the original phosphodiester bonds at the EcoRI cut site. The vector background is thus effectively reduced to zero.

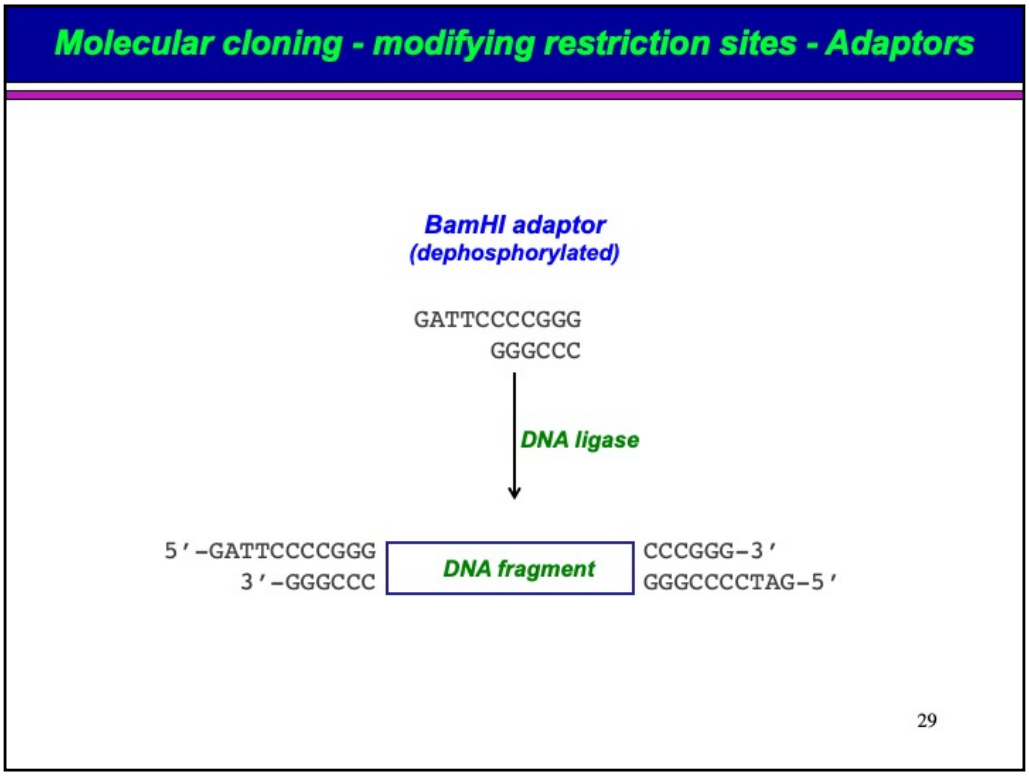
These phosphate-free linearised vectors can be ligated to the insert shown at right, because the insert still has its two 5-prime phosphates. Circular recombinants may be formed with diagonally opposite phosphodiester bonds (one on each strand) and diagonally opposite non-ligated hydroxyl ends (bottom right plasmid). These half-closed circular plasmids may be transformed into competent *E. coli* cells as efficiently as fully closed circular plasmids. The newly replicated recombinant plasmids in each cell, and all the resultant copies, will be fully closed circles, as usual.



The number of cloning options may be increased by using commercially available linkers (this slide) or adaptors (next slide).

Firstly, blunt-ended fragments are generated by using any restriction enzyme that will produce blunt ends. Then, commercially available blunt-ended 'linkers' are ligated to the ends of the fragment. There is a range of linkers available that have internal restriction sites for most of the common enzymes that generate staggered ends.

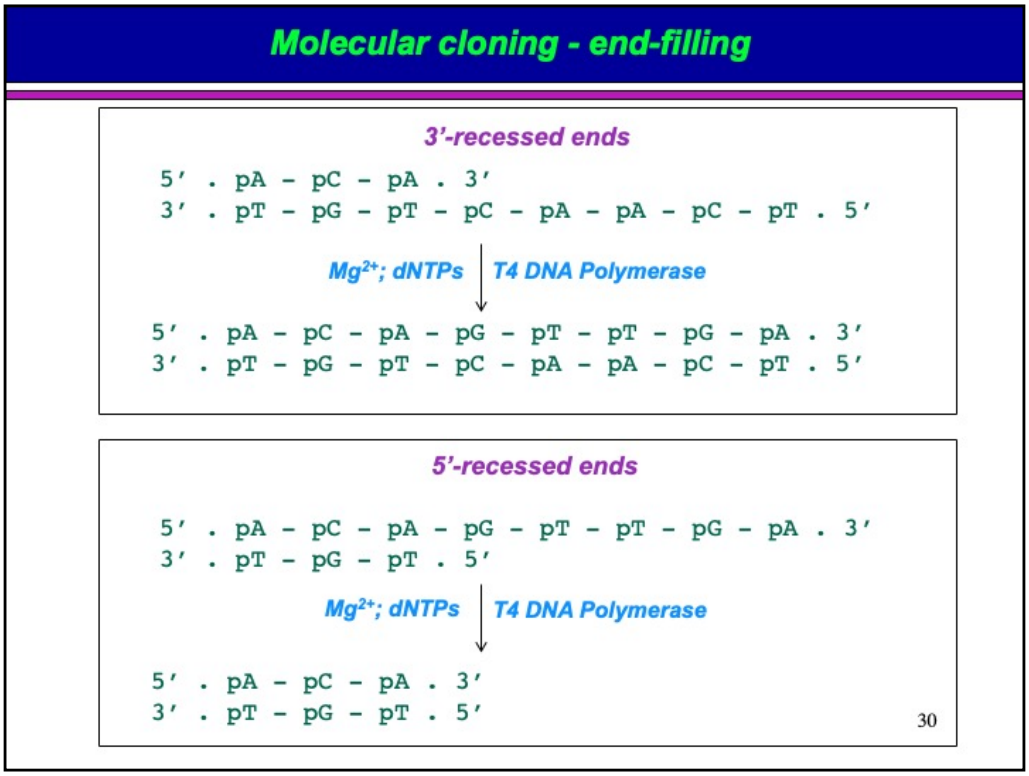
In the example in this slide, an excess of linkers is used so that all of the available ends in the fragment sample will be 'linked'. An excess of linkers drives the reaction and also results in multiple end-to-end (tandem) linkers being attached (left and right arrowheads above). However, when the ligated linkers are subsequently cut with EcoRI, all of the added linkers are digested away except for the last one at each end, which remain attached/ligated to the fragment, yielding 'designer' EocRI ends for cloning into a vector cut with EcoRI.



Adaptors provide a variation of the linker restriction site design illustrated on the previous slide. The only real difference is that adaptors have one blunt end and 'ready-made' restriction enzyme staggered sites at the other end that, once ligated to the blunt-ended DNA fragment, require no further treatment other than being used in a cloning scheme with a vector cut with the same enzyme as the adaptor.

Adaptors thus cut out the need for the restriction reaction step to generate the enzyme site in linkers; and are therefore easier to use, but more expensive - designer genes don't come cheap!

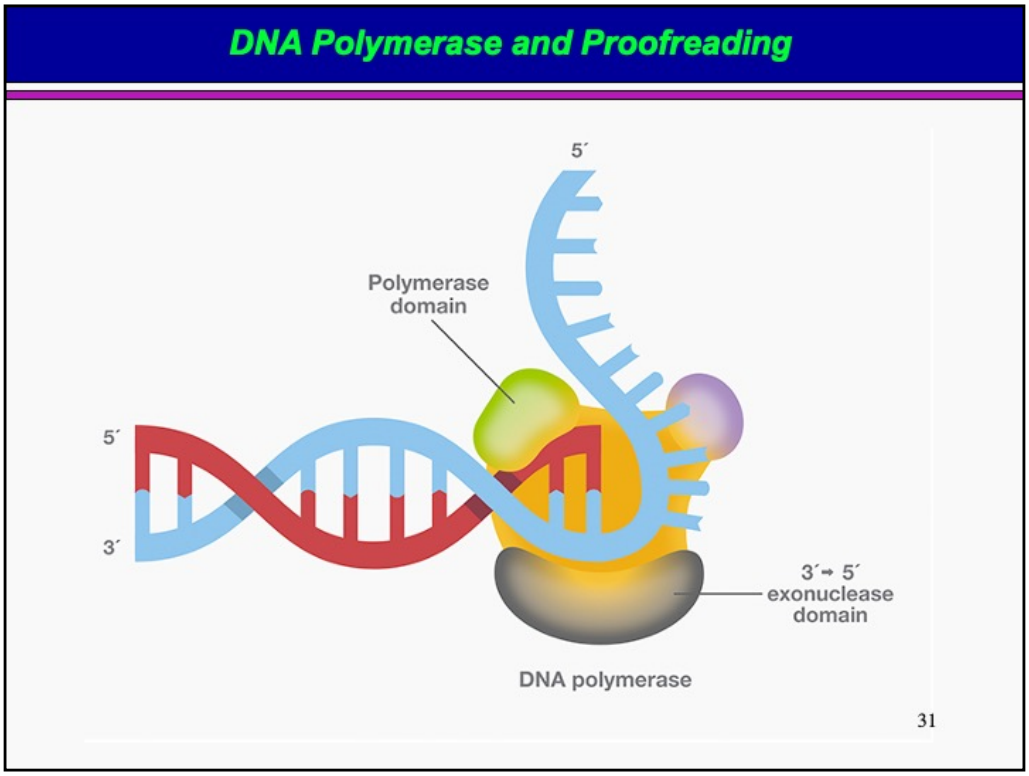
The other important point to note about using commercially available adaptors is that they are dephosphorylated – that is, the 5-prime phosphates have been removed by alkaline phosphatase digestion. This prior step is required to prevent the staggered ends of adaptors from self-ligating to one another, which would then require an additional restriction enzyme step.



Here, we are going the other way with designing DNA ends. This is a simple molecular scheme for converting staggered restriction enzyme ends to blunt ends.

Both types of staggered sites shown in the panels above can arise by using different restriction enzymes; for example, EcoRI will generate 3'-recessed ends and PstI will generate 5'-recessed ends. End-filling either 3-prime (top) or 5-prime (bottom) recessed staggered ends is achieved with the multifunctional enzyme T4 DNA polymerase, which is able to add nucleotides to a single-stranded recessed 3-prime end (top) or digest/excise nucleotides that overhang a recessed 5-prime site (bottom). This is the opposite of the linker-adaptor schemes illustrated in the previous two slides.

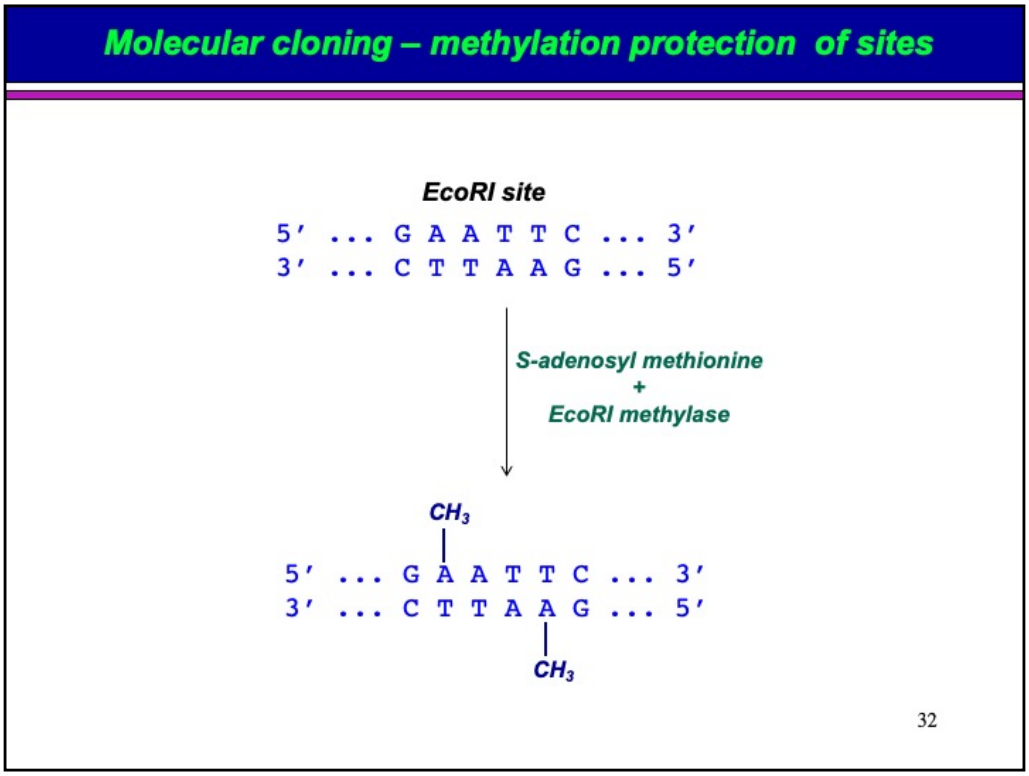
The next slide will illustrate the utility and functionality of enzymes such as DNA polymerase. It has two types of activities, both a polymerising function, which adds nucleotides, but only ever in one direction, namely 5-prime to 3-prime as in the top panel in this slide. Polymerases also have an exonuclease 'proof-reading' function, which can excise incorrectly added nucleotides in the opposite direction to the polymerising function, that is, the exonuclease excises nucleotides in the 3-prime to 5-prime direction, as in the bottom panel in this slide.



T4 DNA Polymerase catalyzes the synthesis of DNA in the 5´→ 3´ direction and requires the presence of template and primer. This enzyme also has a 3´→ 5´ exonuclease activity that is capable of paring back overhanging 3'-ends to the 5'-end at the double-stranded junction, as illustrated in the bottom panel of the previous slide.

Apart from that 'artificial' function that we make use of in molecular biology, its natural function is as a proofreader for the accompanying polymerase activity on the same multi-enzyme complex. If an incorrect base is incorporated, such as for example, T pairing with C rather than G, the polymerase domain pauses, the 3'-end of the DNA strand is partially 'melted' and redirected to the 3´→ 5´ exonuclease active site where the incorrect 'T' is removed. The strand is returned to the polymerase active site and the correct base ('G' in this case, opposite 'C' on the other strand) is inserted and the polymerase activity continues along the template strand.

Error rates for DNA polymerases are of the order of 1 in 10,000 to 1 in 1,000,000 nucleotides if the exonuclease activity is defective, but becomes much lower if the exonuclease is working, that is, an error rate of only about 1 in 10 billion nucleotides. Note in the figure above that DNA polymerases only add nucleotides in the 5-prime to 3-prime direction (red strand above) against the antiparallel template strand (blue strand above).



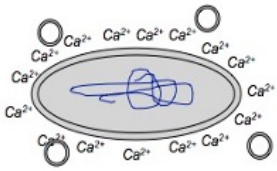
A final restriction modification scheme is one that mimics the methylase-restriction host bacterium system. Methylase adds methyl groups (CH3) to one of the nucleotide bases within the restriction site of it's partner enzyme, thereby preventing the host's restriction enzyme from digesting its own DNA.

In the example above, EcoRI methylase adds methyl groups to the opposite first adenine (A) bases across the EcoRI site. This site is no longer a substrate for EcoRI, as it is 'protected' or blocked from EcoRI binding and digestion. The bulky methyl groups simply prevent the EcoRI catalytic sites from access, that is, getting close enough to the substrate palindrome.

A molecular cloning scheme that would use this in vitro step would be one which requires a DNA molecule to be cut at a separate EcoRI site, but not at the now protected site, which could, for example, be within a gene sequence, whereas the non-protected EcoRI site would be external to the gene, in say the plasmid vector.

One unifying theme or take home message from the second section of this lecture is how we have co-opted DNA and enzymes to work in molecular biology in ways that mimic or are adapted or modified from their natural functions.

MCQ L6



This diagram depicts an E. *coli* cell surrounded with ice-cold ionised calcium chloride as free Ca^{2+} and several plasmid molecules (circles). What will happen, and why, if this mixture is given a short heat pulse at 42°C?

(A) The plasmid molecules will remain bound to calcium ions.
(B) One of the plasmid molecules will be transferred across the membrane into the cell.
(C) The plasmid molecules will dissociate from the calcium ions.
(D) All of the plasmid molecules will be transferred across the membrane into the cell.
(E) The calcium ions and the plasmid molecules will dissociate from the cell membrane.

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MCQ L6

5'.....A G C G T T C A G G.....3'
3'.....T C G C..... 5'

The DNA fragment above has an overhanging 3-prime end. It is reacted with T4 DNA polymerase, Mg^{2+} and a mixture of dNTPs. Which of the following statements is correct?

(A) It cannot be blunt-ended as DNA polymerases cannot add nucleotides to recessed 5-prime ends.
(B) An equal amount of 5-prime end-filled and 3-prime cut back molecules are obtained.
(C) The 3-prime overhang is cut back to the 5-prime end.
(D) The 5-prime end is end-filled up to the 3-prime end.
(E) It can be blunt-ended but only if the polarity of the strands is reversed.

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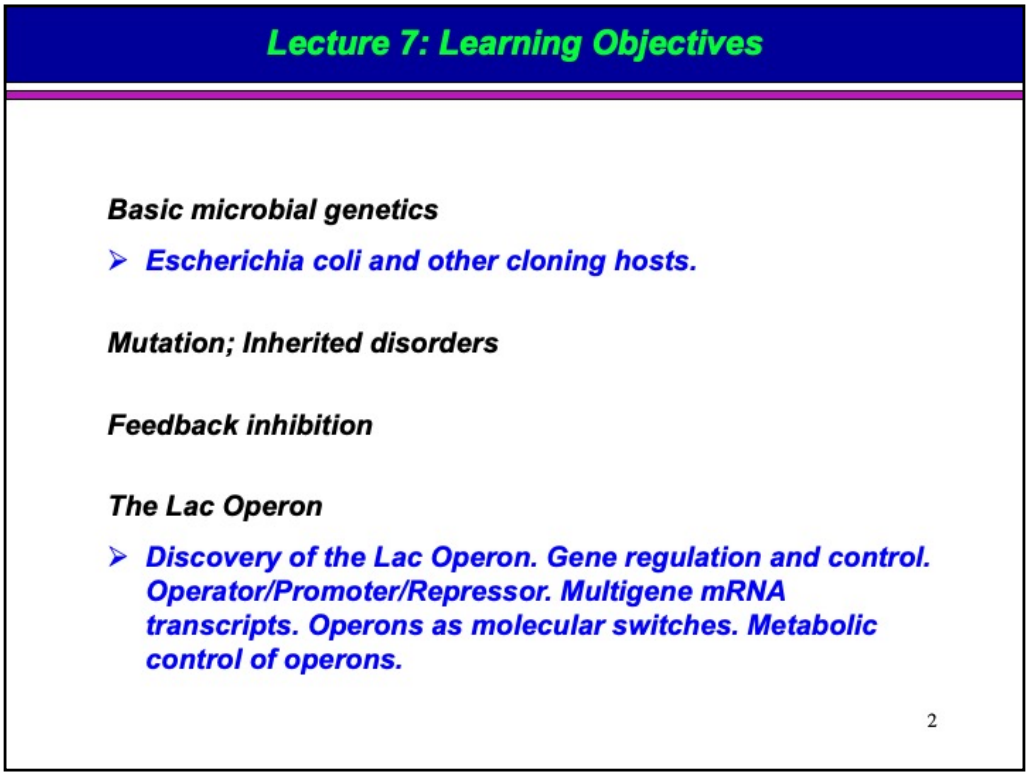
MOLECULAR BIOLOGY 1
Lecture 7

Introduction to microbial genetics

Regulation and control – the Lac Operon

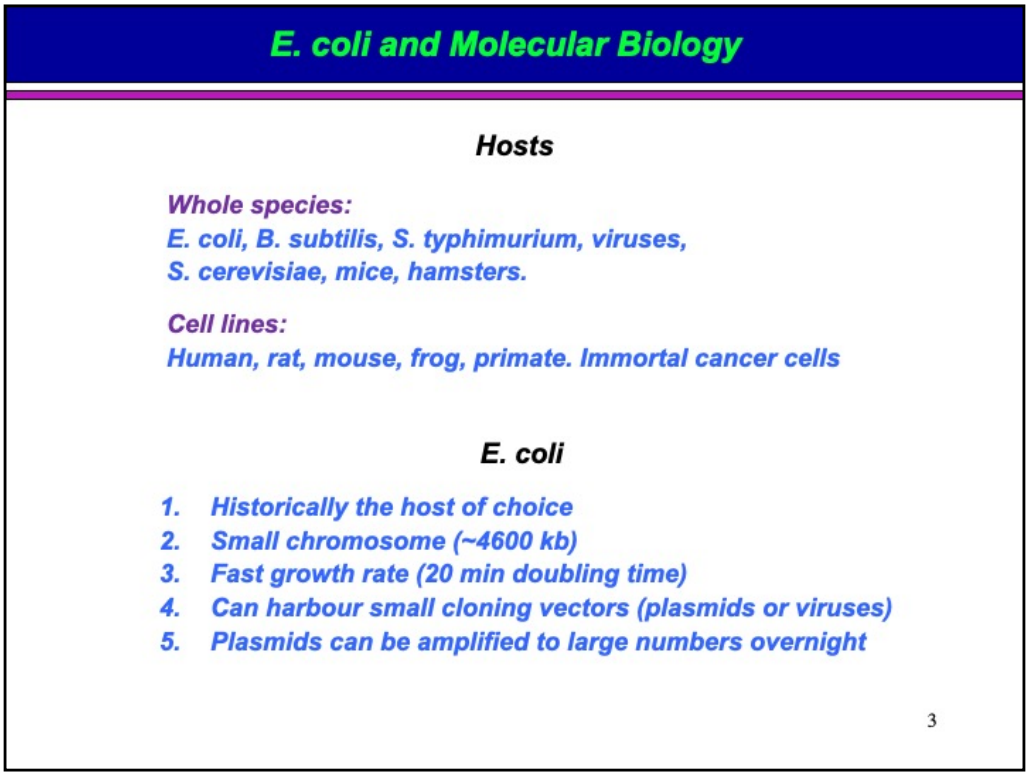
1



This is a very important lecture that alerts us to one of the major themes of molecular biology, namely, to understand how cells and organisms 'work' – in this case, how genes are regulated and how this is coordinated with responses to physiological molecular signals. Regulation and control of gene expression and associated processes such as cell signaling through receptors, transcriptional activators and repressors, hormones and small molecule regulators such as cyclic AMP, are at the heart of many disorders and diseases that afflict humans.

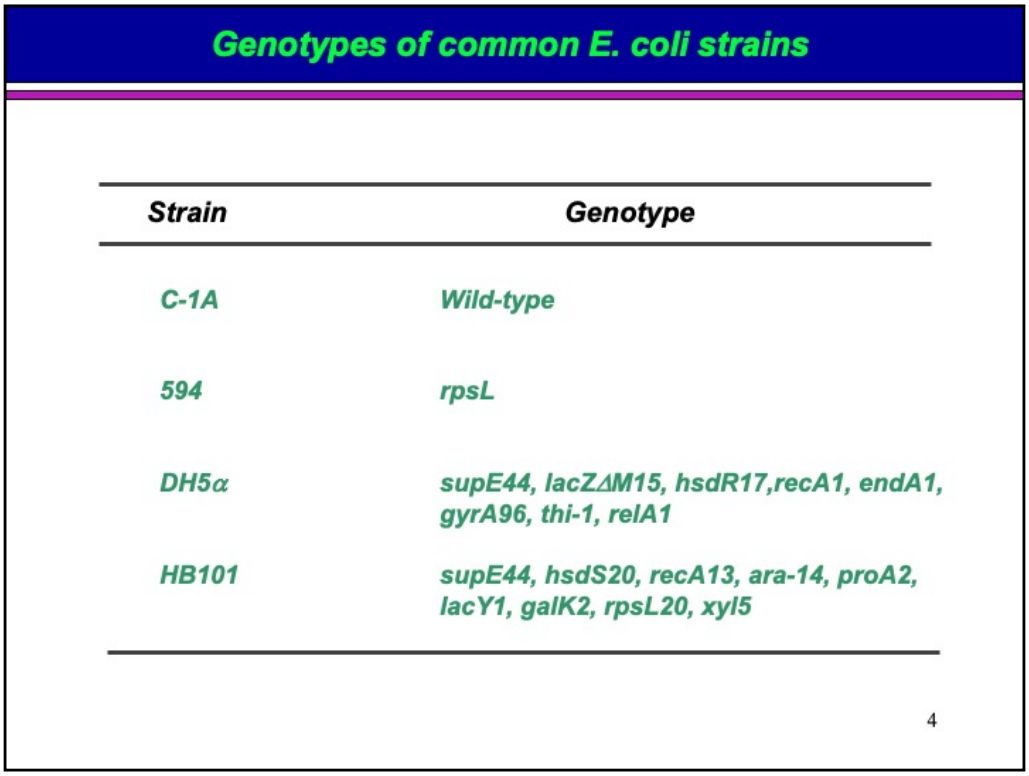
This lecture is therefore partly an introduction to 'How things work in cells and organisms'. It is also an overview of the coordinated regulation of metabolic processes, using the Lac Operon as the signature example.

From the perspective of molecular biology, the Lac Operon has been modified to work in plasmid/bacterial cloning systems, such as the LacZ-Alpha peptide complementation system we looked at in the previous lecture, which is the preferred recombinant identification method for most modern cloning vectors, and which will be used in Experiment 6 in the lab course for this subject.



Many hosts are available today for gene cloning and expression, and some of these are listed at the top of this slide, but the bacterium *E. coli* was historically the first and most prolific host used, chiefly because it was the most extensively investigated organism, especially during the 'genetic' age of research (~1900-1980). Often, E. coli is used as a prelude to work in other animals or cell lines.

E. coli is still used more than any other host and a list of its essential features is on this slide and the next. *E. coli* is therefore essential in any modern molecular or cell biology laboratory, but there are many genetic mutants or variants that have specific advantages for molecular cloning and expression, which we will look at briefly in this lecture; and in the last lecture in this subject.



Any organism can be defined by its genotype. The 'original' genotype is termed the 'wild-type' (WT) and any deviations from this represent different strains or mutants of the organism. The differences are termed 'mutations' and they can occur spontaneously (next slide) or be induced in a number of ways (see further on).

Here, several examples of *E. coli* strains are given, simply for the purpose of 'reading' genotypes (genes) and phenotypes (expressed characteristics or properties). Note that the full genotype will include all genes and sequences that can be expressed in any given organism. For the example of *E. coli* C-1A, there are about 3000 genes, yet its genome is referred to simply as Wild-type. This means that every gene in its chromosome is - by definition – normal Wild-type, and by extension, all of its expressed properties (phenotype) are typically 'normal'.

When an *E. coli* strain acquires one or more mutations, only the changed or mutant versions of the genes are listed as the 'genotype', as shown for the last three examples in the table above. The genotype of *E. coli* strain 594, for example, is wild-type for every gene except the one for rpsL (streptomycin resistance), which for simplicity is the one genotypic property given. Ditto for the other two strains and their several mutations that differ from Wild-type.

Spontaneous mutation test

Culture number (inoculum: 10^2 cells per ml of overnight culture)	Number of cells resistant to streptomycin* per 10^9 cells
1	0
2	8
3	0
4	766
5	2

* Streptomycin resistance in E.coli is caused by a point mutation in the rpsL gene for ribosomal protein S12.

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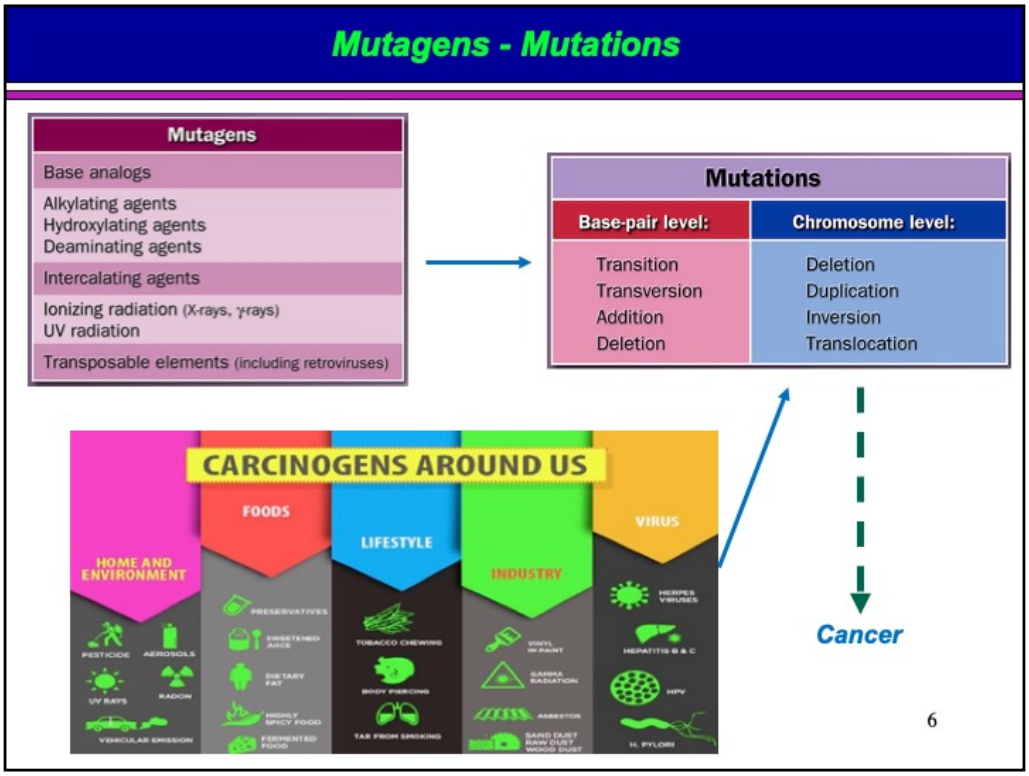
The table above is for 'identical' overnight *E. coli* cultures. Such mutations may occur spontaneously as indicated in the number of streptomycin resistant mutants above, which may occur at any time during the growth of the culture, with the mutation being inherited in all offspring.

Spontaneous mutations may occur anytime in any organism. Streptomycin works by binding to the 16S RNA, within the smaller of the two ribosome subunits, and distorts it in such a way that the mRNA cannot be transcribed.

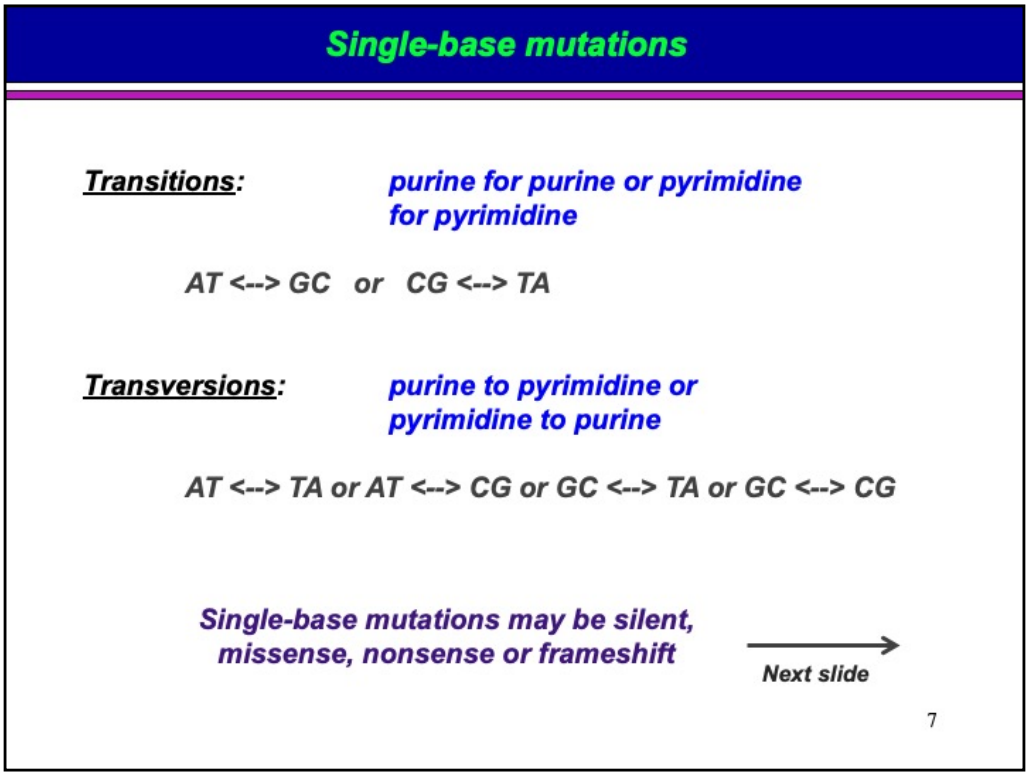
When spontaneous mutations occur in the S12 protein (*rpsL* gene) of the small subunit, the smaller subunit is changed in its configuration such that streptomycin is no longer able to bind to the 16S RNA and protein synthesis proceeds uninterrupted, with the *E. coli* strain now said to be resistant to the antibiotic.

Streptomycin, along with all other protein synthesis inhibitors, gives us an insight into the importance of conformation in binding/inhibition, in this case the conformational integrity of the smaller ribosome subunit, which is 'disturbed', that is altered, when S12 is mutated by a single point mutation (in one amino acid only) that alters its configuration and consequently the configuration of the smaller subunit 'scaffold', the result being that streptomycin can no longer bind at its 16S mRNA site.



Most mutations are not spontaneous, but are brought about by agents that induce mutations in DNA. Such agents include: chemicals, radiation, viruses, toxins and carcinogens (e.g. cigarettes; asbestos; dust and fumes from the air). These agents act mostly by changing the DNA, by simple single base changes (addition, subtraction, substitution), or by more robust changes that affect the chromosomes more tangibly, by deletion, insertion, inversion, or translocation of substantial segments of DNA (tens to thousands of nucleotide base pairs). Some mutagens are depicted in the top left panel, but to a large extent mutagens and carcinogens (lower panel) are not mutually exclusive, but overlap to a large extent.

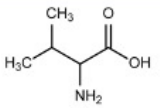
Carcinogens generally trigger a process termed apoptosis, 'programed cell death', with cell death itself termed necrosis. Apoptosis is a defence mechanism for cells to try to ward off the deleterious effects of carcinogens. However, prolonged exposure or abuse (egs. cigarettes, asbestos, sun tanning, chemicals, toxins etc) may eventually damage healthy cells, bypassing apoptosis, and leading to transformation to tumourigenesis, a process that induces irreversible changes in cells (malignancy). Transformed cells become 'immortal' because they replicate uncontrollably and grow and invade surrounding tissues as tumours, or circulate through the body's canal systems (lymph or blood).



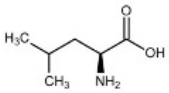
The simplest mutations to consider are single base changes that result in new base pairs when the DNA is replicated, if it is not repaired beforehand. If the base that is changed is one purine (A or G) for the other (G or A), this is termed a transition to a new base pair (e.g. AT to GC, see above). If a purine is changed to a pyrimidine (A or G to T or C) or vice versa, this is termed a transversion to a new base pair. Interestingly, such single base mutations may or may not be deleterious to the organism (see the next slide).

Single-base mutations

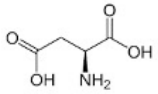
Codon	Change	Mutation
GTT (Val)	GTA (Val)	silent
GTT (Val)	CTT (Leu)	silent
GTT (Val)	GAT (Asp)	missense
TGG (Trp)	TGA (stop)	nonsense
(A)GTT (Val)	AGT (Ser) T--	frameshift



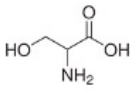
valine



leucine



Aspartic acid



serine

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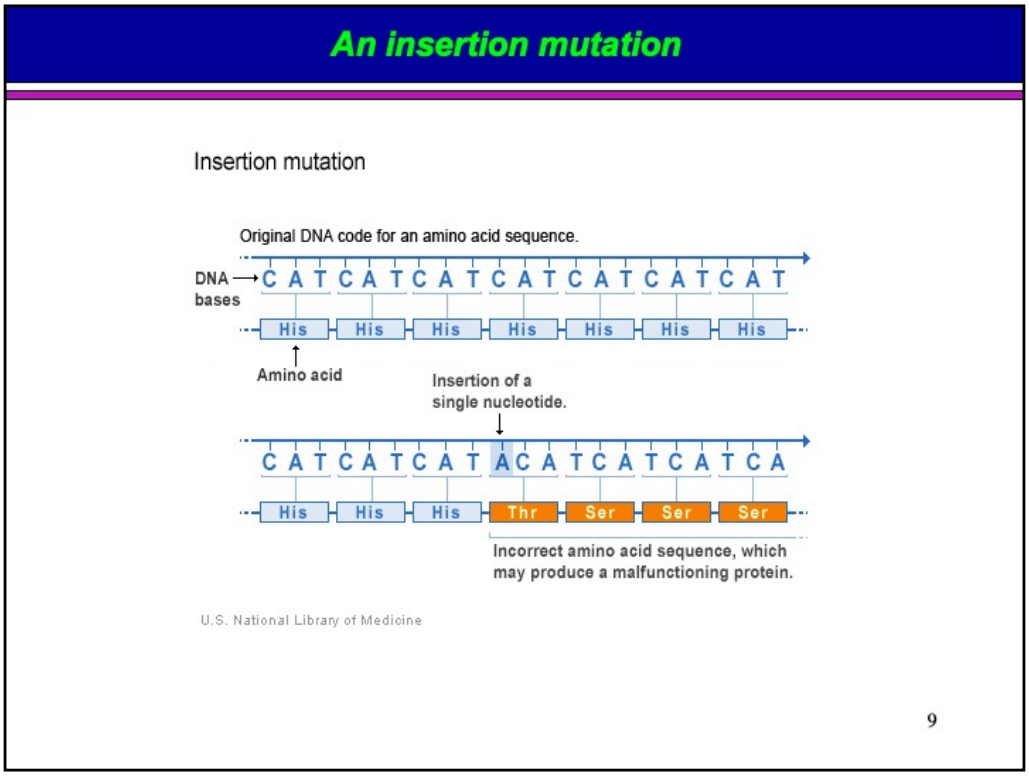
This table gives some examples of the four types of single base mutations, namely **silent**, **missense**, **nonsense** (all are substitutions), or **frameshifts** (insertions or deletions). Note that some of these can be harmless (silent), but many are deleterious, causing severe disruption to the nature of the gene-specified polypeptide. Missense, nonsense, and frameshift mutations all cause irreversible dysfunction in proteins. There are examples of these changes on the next few slides.

The silent mutations (first two rows) result from the redundancy of the Genetic Code – that is, several triplet codons specify the same amino acid or one that is similar in terms of the side chain size and polarity.

All of these mutations may occur in any part of a chromosome, and may or may not be benign (harmless), depending on the site. Silent substitutions are a benefit of the Genetic Code being redundant, even though in reality, the redundancy is mainly due to the need to use multiple codon amino acids, such as glycine, leucine, alanine etc more frequently than other amino acids (e.g. tryptophan; methionine) in polypeptides.

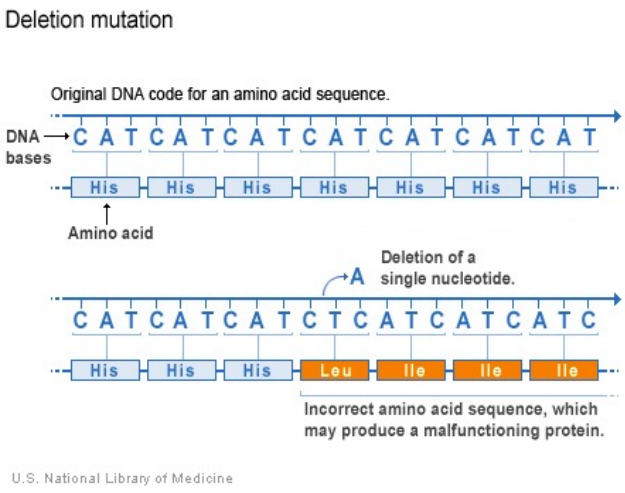
As illustrated here, codon changes by single base substitutions may or may not have severe consequences for the protein. Frameshifts within an ORF, on the other hand, are always deleterious.



The upper sequence code, which is a simple repetitive of histidine codons (CAT), is modified (mutated) by an insertion of an adenine nucleotide ('A') at the position indicated by the arrow in the lower sequence.

The consequence of this insertion is easy to see: The 4th 'CAT' codon is no longer 'in frame', as the inserted 'A' now becomes the first base of the 5th codon, which is now 'ACA' specifying threonine (thr) instead of histidine. The frame change is registered downstream with all the following codons changed to serines instead of histidines. The net result is a complete change and loss of integrity (identity) of the sequence downstream from the insertion.

A deletion mutation



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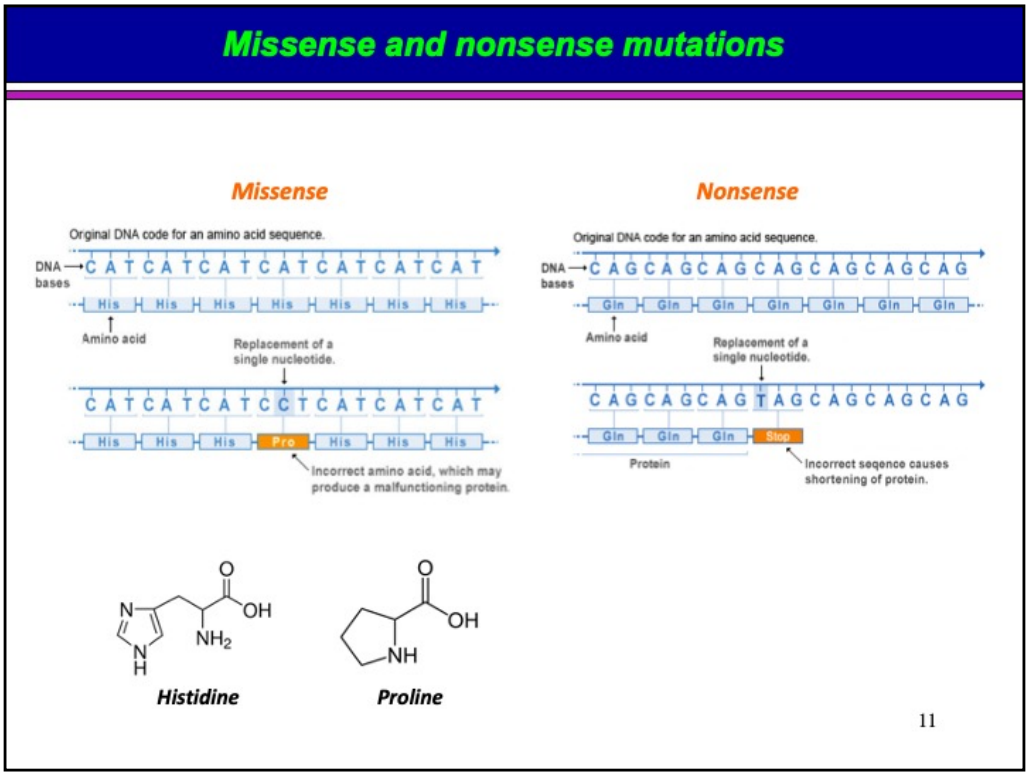
Single nucleotide deletions (lower sequence) are the reverse of insertions and the consequence for the polypeptide are catastrophic in terms of the production of a malfunctioning protein. As in the previous (insertion) example, all codons downstream of the single base deletion change to amino acids other than histidine.

Take home message for insertions and deletions is:

If you make a single mistake in a 'sentence', as we have seen in this and the previous slide, you jumble the text and make it essentially unreadable and unusable.

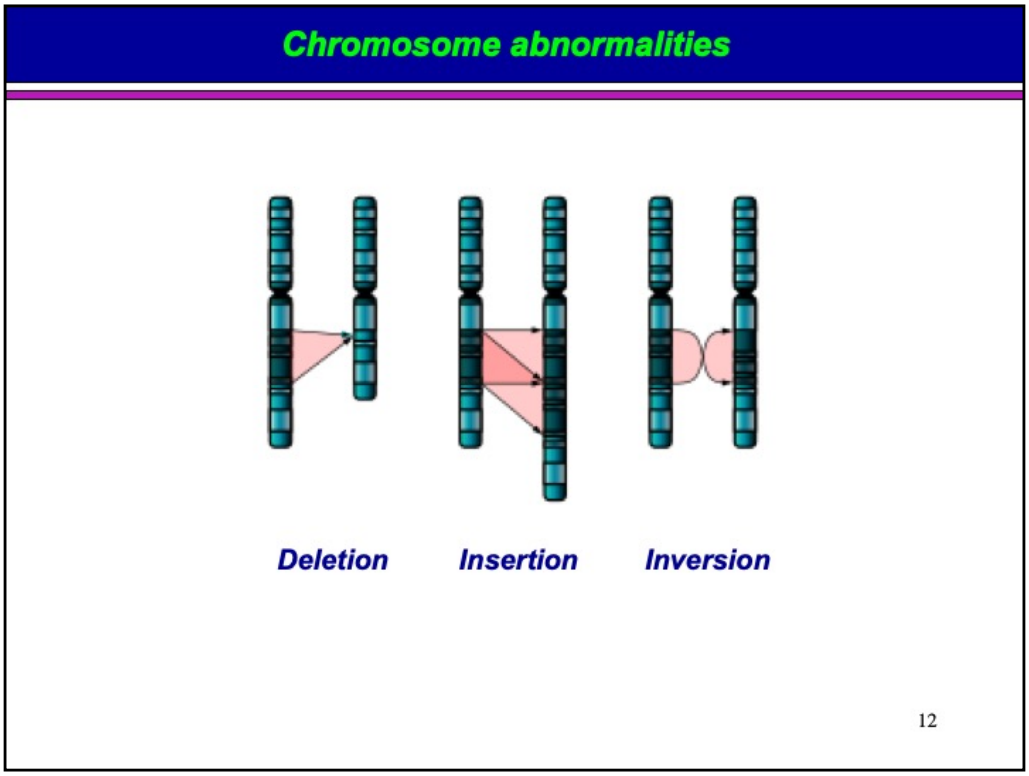


These two examples are of <u>substitutions</u>, or replacement of one nucleotide with another.

The substitution of an ‘A’ with a ‘C’ (on the left) produces a <u>missense mutation</u>, that is, histidine (CAT) is replaced/substituted by proline (CCT). The consequence for the protein may or may not be deleterious, but almost certainly is because, in this example, an amino acid with a partial positive charge side chain (Histidine) is replaced with Proline, which has a neutral side chain that is also a fixed ring structure that changes the direction of that section of the folded polypeptide.

The substitution of a ‘T’ for a ‘C’ (on the right) introduces the nonsense or stop codon TAG (refer back to the Genetic Code slide) in place of histidine, with catastrophic consequence of generating a shortened and non-functional protein.

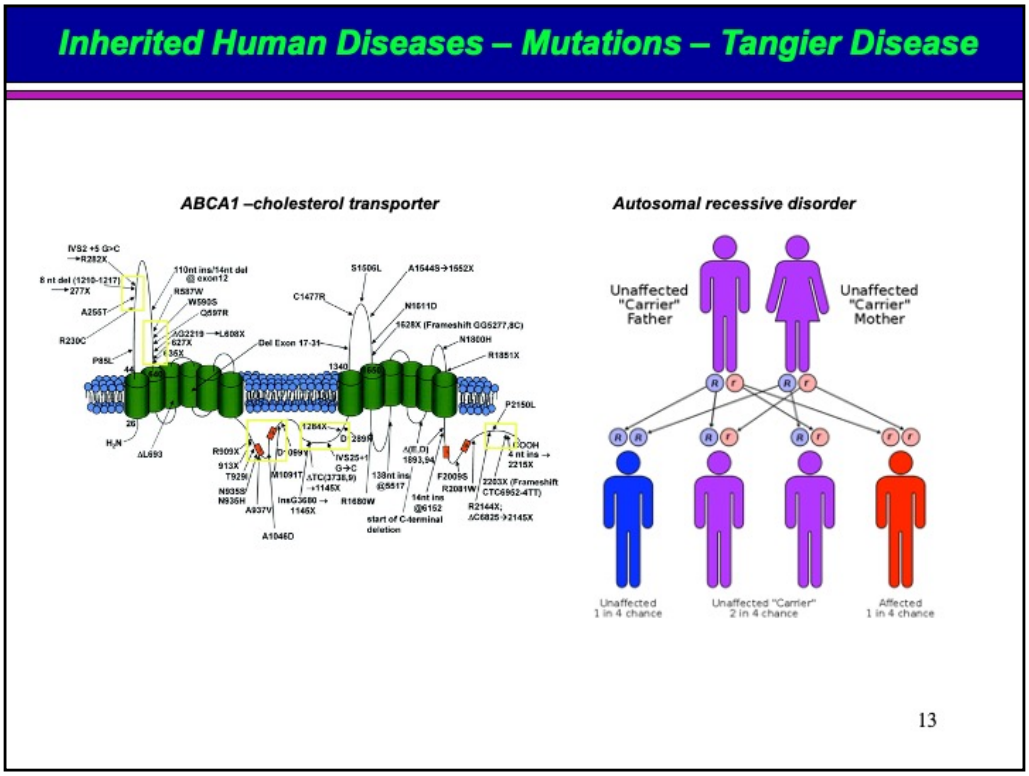
We have now looked at the four main types of single base mutations, namely insertion, deletion, and replacement or substitution. In these examples, almost all are deleterious in producing a non-functional protein. But, remember that unlike deletions and insertions, not all substitutions are deleterious (see the Table 3 slides back for examples of ‘silent’ mutations.



Chromosome abnormalities may take several forms and may occur at anytime, but most occur during cell division following meiosis or mitosis; or within somatic or germ line cells exposed to toxins or radiation. Chromosome abnormalities, disorders, irregularities, aberrations, or mutations, result in missing, extra, or irregular chromosomal DNA segments. Examples are the three changes depicted in the figure above.

Severe changes in chromosome, such as major transversions, deletions or duplications are beyond the scope of this subject, though we will look at some examples in a later lecture.

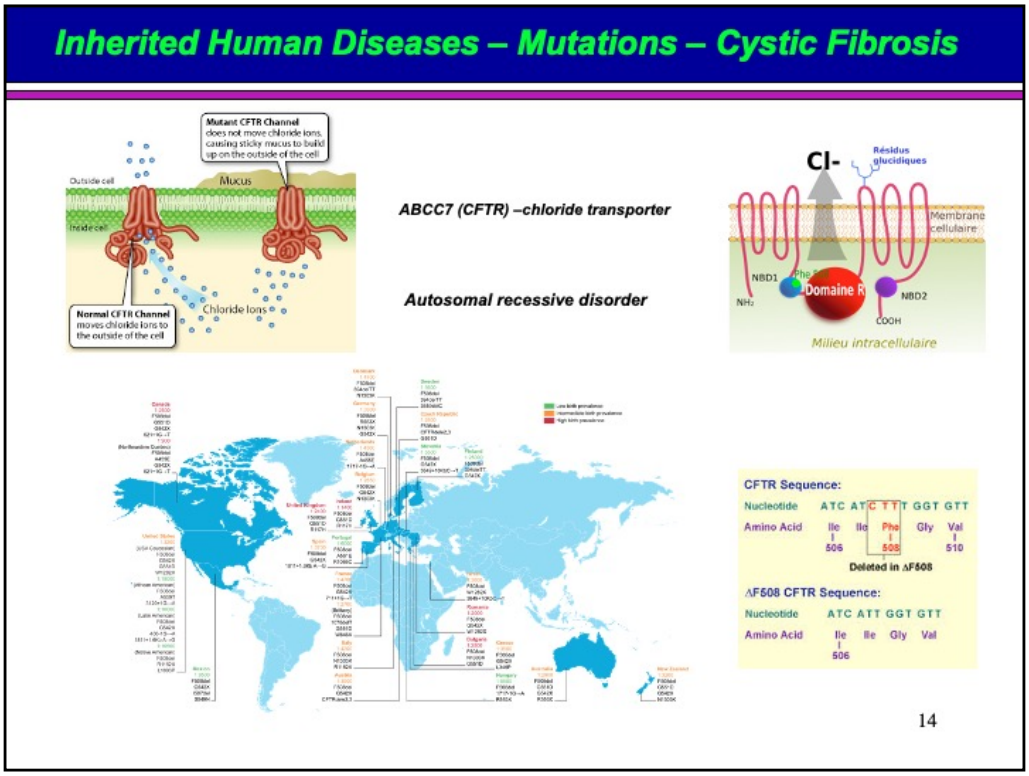
On the other hand, single base changes, such as single site mutations, as substitutions, deletions, or insertions, are not only the most commonplace types of mutations, but are important in the integrity and expression of genes, as we have seen in the last few slides.



Many genes may be affected by mutations and these are often inherited.

Left panel: The multiple sites of mutations (deletions, insertion, point mutations) give rise to a defective ABCA1 transporter for cholesterol export out of cells. They lead to a reduced level of the "good" HDL and gradual degenerative health such as atherosclerosis, enlarged liver, cloudy cornea. Can be mild (most cases not identified, or may lead to cardiovascular disease and Type II diabetes) to severe, resulting in the requirement for constant medication, and possibly early death, especially if associated with inappropriate diet and/or cigarette smoking.

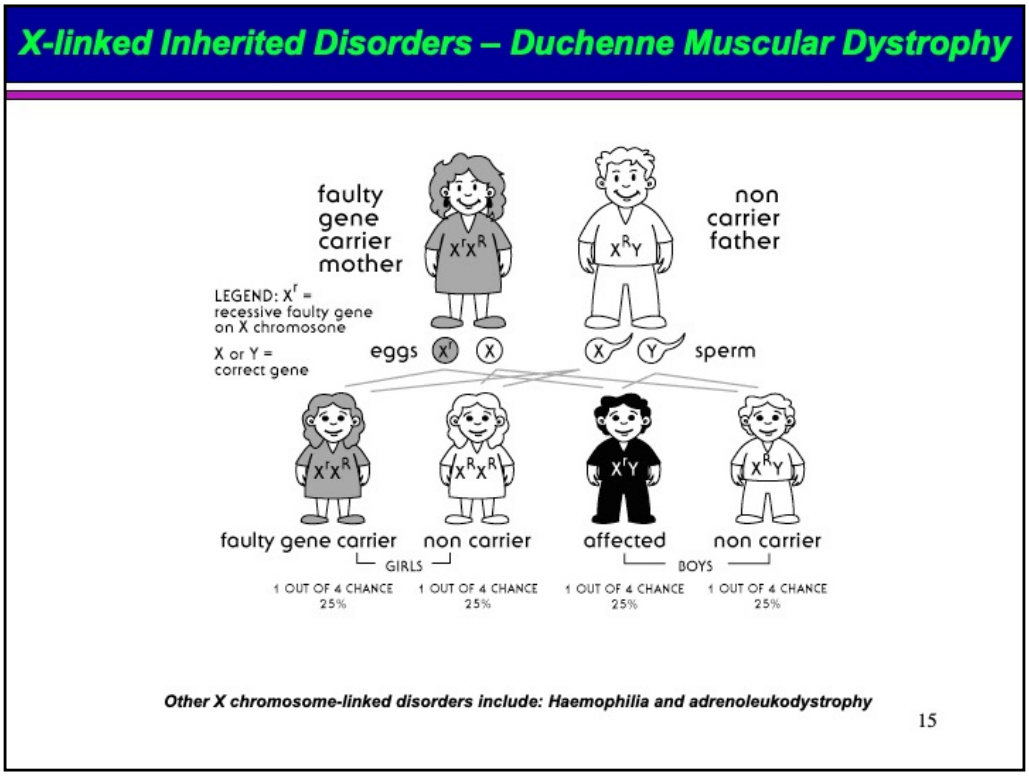
Right panel: It is an autosomal recessive disorder and is very rare with only about 100 cases worldwide at any time, but many more go undiagnosed due to the mildness of symptoms. An affected individual must inherit the defective gene from both parents, which adds to its rarity. First diagnosed in a boy on the island of Tangier off the Virginia coast in 1959.



Cystic Fibrosis is an inherited disease mostly found in Caucasians at a frequency of about 1:3000, with mutations/lesions giving rise to mild to severe symptoms such as constant coughing, shortness of breath, and recurrent infections, the last of which is the most serious.

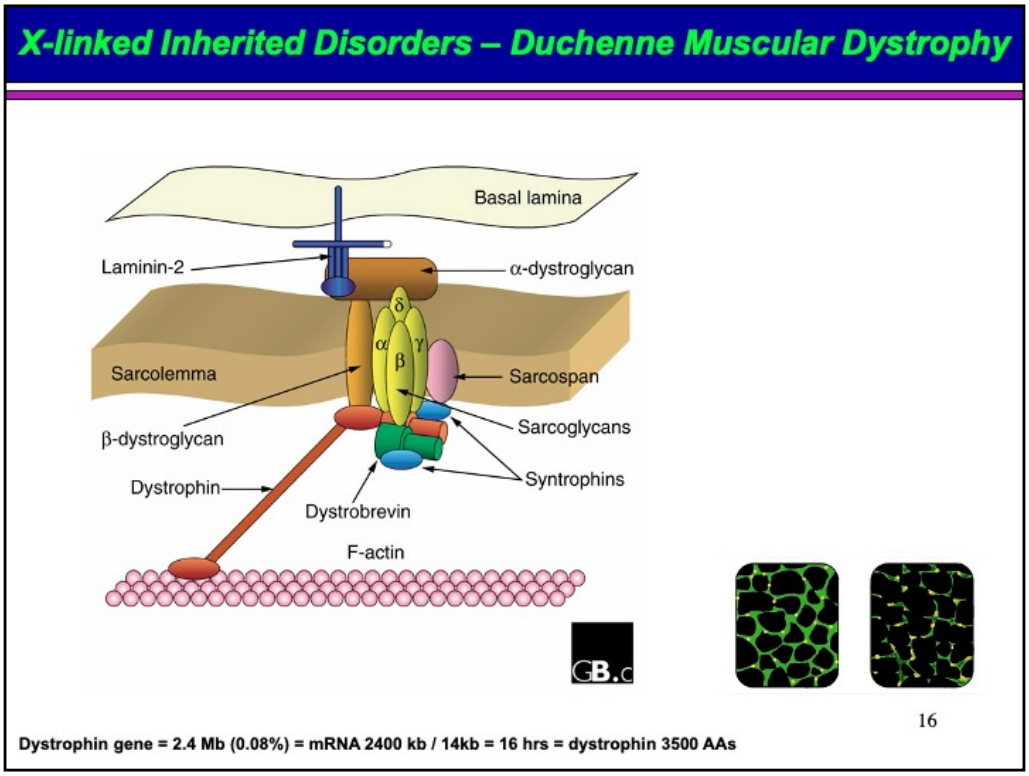
CFTR protein (Cystic Fibrosis Transmembrane Conductance Regulator) is a chloride channel essential for maintaining homeostatic water/salt balance in the epithelial fluid in the lungs and other organs (top left; top right). The defective protein cannot pump chloride and the free flow of ions and water in and out of cells is blocked, resulting in an imbalance of water, especially in the lungs, resulting in viscous mucus that is an ideal medium for bacterial biofilm growth. It is the bacterial infections, rather than the condition itself, that leads almost invariably to death, with a mean survival age of about 38 years.

Over 70% of the dozens of mutant alleles (see world map, bottom left) are due to a single deletion of the phenylalanine 508 residue (bottom right). This deletion results in CFTR misfolding and failure to insert into the membrane. One new drug (Vx770) can overcome another lesion that causes 4% of CFTR cases, and it is available at $US300,000 per annum. New CF correctors are in Phase III/IV trials with Vx-659 and Vx-455, which have the potential to help over 90% of CF patients.



X chromosome-linked inherited diseases are fairly commonplace. Duchene Muscular Dystrophy (DMD), in severe cases, may result in muscular degeneration, respiratory failure, and death. DMD affects males at a frequency of 1:3,500.

Haemophilia (blood clotting disorders), and adrenoleukodystrophy (adrenal gland failure, progressive brain damage, death) are other well known X-linked disorders.

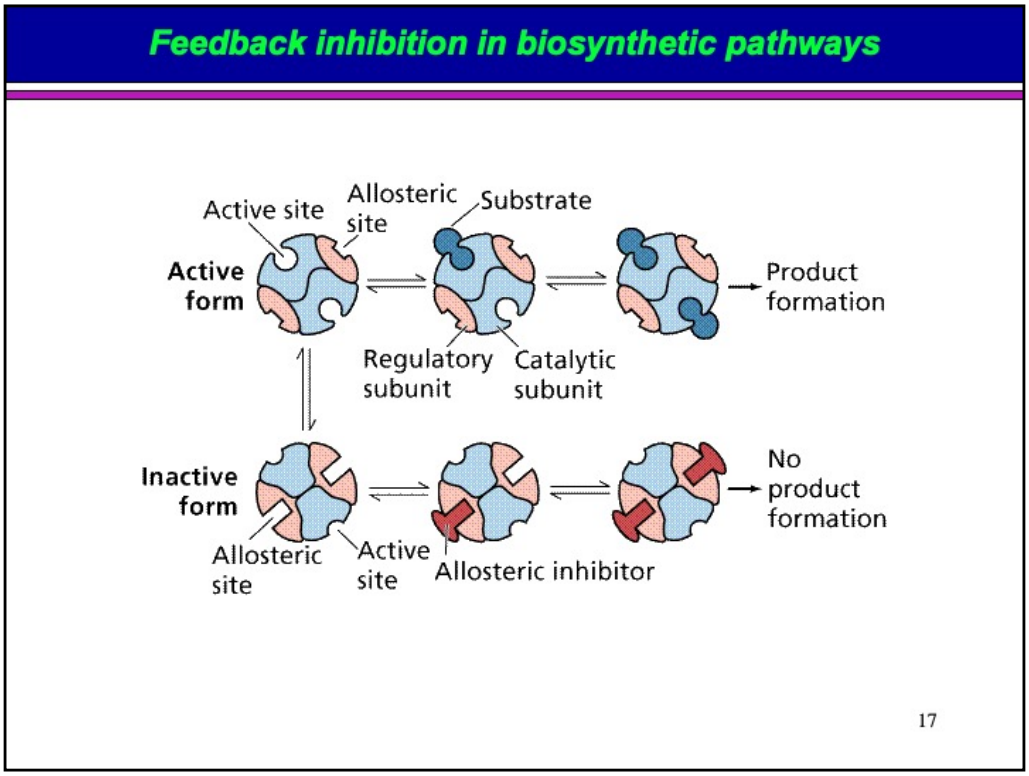


Dystrophin is a 'shock-absorber' protein that gives muscles their flexibility. Dystrophin connects the cytoskeleton (as F-actin microfilaments; see above) of each muscle fibre to the underlying basal lamina.

The dystrophin gene is the largest in the human genome at 2.4 Mb (0.08%). The primary mRNA transcript is 2,400 kb long and takes 16 hours to transcribe. The processed mRNA is 14 kb and encodes dystrophin, with 79 exons and over 3,500 amino acids.

Duchene Muscular Dystrophy (DMD) results from mutations in the dystrophin gene that are mild to severe, mostly single base changes and stop signals, and in a milder form, exon skipping. The small figures (bottom right) depict normal (left) and affected (right) muscle, with the dystrophin concentration decreasing and the tissue becoming disorganised, as seen in the right figure.

Becker Muscular Dystrophy (BMD) is a milder form of DMD in which one to many exons are deleted (in frame) leading to a shorter dystrophin rod that is still partly functional. In some instances over 2/3rds of the rod domain may be missing, but still functional due to the connecting end domains being present. Symptoms include, muscle weakness, difficulty in learning to walk, and in standing, unsteady gait, clumsiness, and walking on toes.



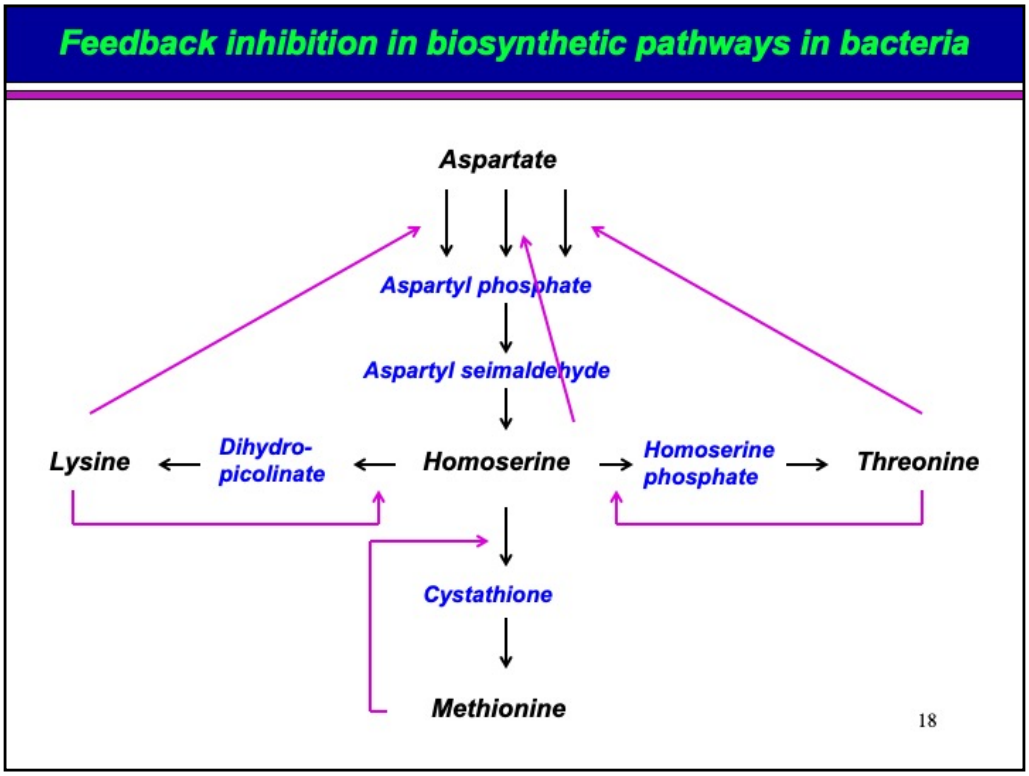
Feedback inhibition is a fundamental method of regulation and control of biosynthetic pathways.

The net effect of feedback inhibition is to cease conversion of substrate to product when an excess of product has accumulated.

In the example above, the enzyme 'active sites' are binding sites for substrate, and the regulatory subunit sites are the 'allosteric sites' for feedback inhibition by an inhibitor, which is often the end product of the reaction process.

Feedback inhibition only occurs when an excess of product accumulates. Note that when the product (allosteric inhibitor as indicated above) binds to the regulatory site, it changes the conformation (shape) of the enzyme in such a way that the 3-D geometry of the active site is altered and the substrate can no longer 'fit' into its site. The enzymatic activity ceases.

In this example, the feedback inhibition occurs on the same enzyme, but there are pathways that use many enzymes and often the feedback inhibition may occur at later steps or branches in the pathway – see the next slide.

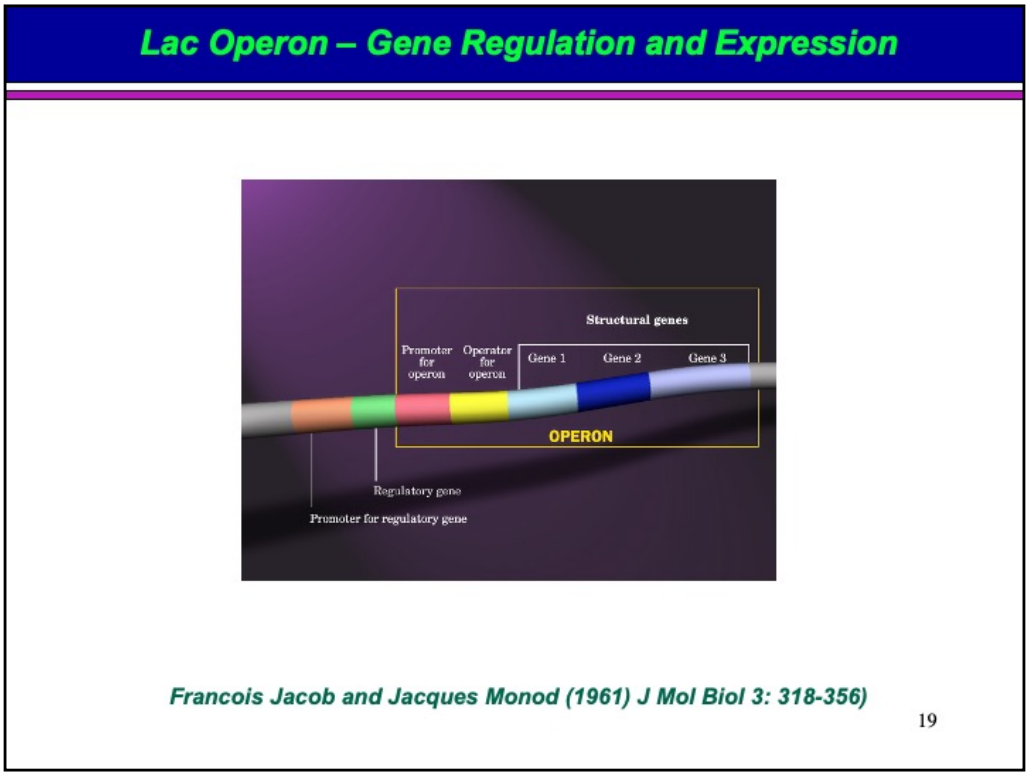


To begin our introduction to the regulation and control of gene expression in bacteria, we take firstly a simple example of feedback inhibition, which occurs at the substrate-protein level initially, then, as intermediate and end-point compound(s) in the pathway accumulate, the signal (excess of a product compound) is registered back to a regulatory site on one of the enzymes of the pathway and expression of some (sub-branches) or all genes of that pathway ceases.

In the example above, the synthesis pathways are delineated by black arrows, and feedback inhibition by pink arrows.

The 'signal' that switches off the genes is usually the product that, in excess, binds to a main enzyme of the pathway. This feedback inhibition is an example of allostery or allosteric regulation. How does it work? The major product of the pathway has a secondary binding site (allosteric or regulatory site) on one of the earlier enzymes of the pathway, that is, at a different site to the substrate site (see the previous slide). Binding at this 'allosteric site' causes a conformational change in the enzyme in such a way as to make the substrate site inaccessible, and downstream biosynthesis ceases.

Try this useful video link:
http://highered.mheducation.com/olcweb/cgi/pluginpop.cgi?it=swf::535::535::/sites/dl/free/0072437316/120070/bio10.swf::Feedback+Inhibition+of+Biochemical+Pathways



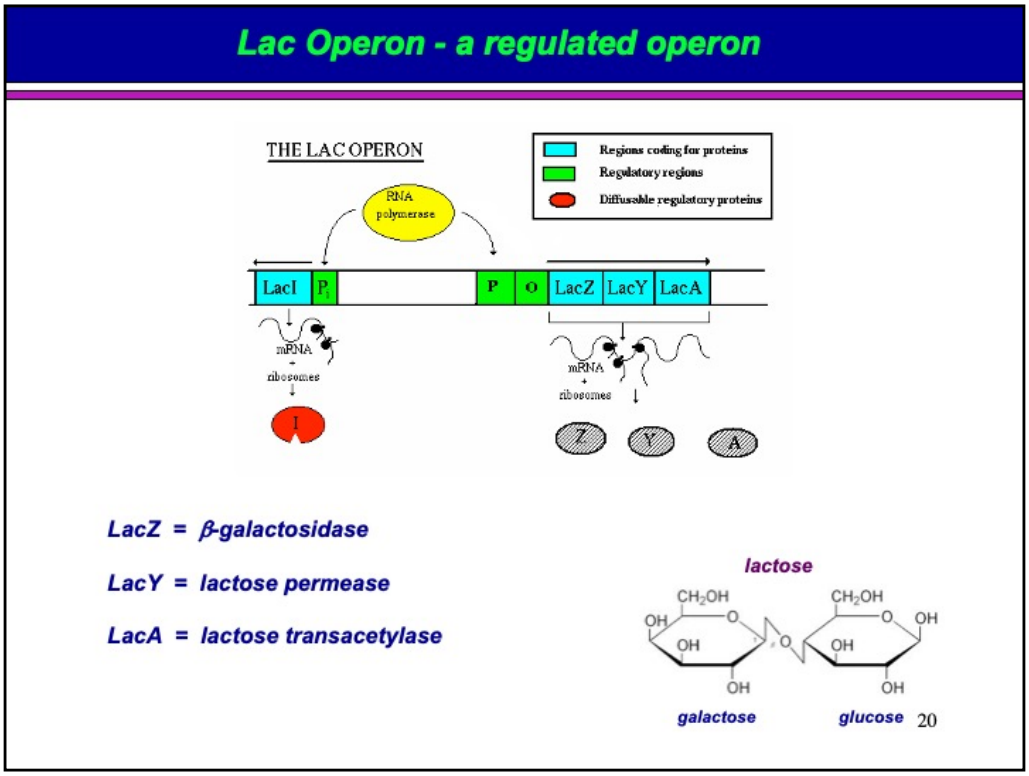
Now we come to the main topic for this lecture, which is regulation and control at the gene level. What we saw in the last couple of slides was regulation at the protein level.

In 1961, Francois Jacob and Jacques Monod proposed the (lac) operon model of gene regulation in bacteria – and won the Nobel Prize for Physiology & Medicine in 1965. The first discovered and best studied operon is the *lac* Operon in *E. coli*. This simple drawing shows the main elements, namely a regulatory gene (green) with its own promoter (orange), the operon's promoter (pink), the operator (yellow), and the downstream structural genes (blues) that are regulated in this operon (these features are not drawn to scale).

An **Operon** is a regulated gene system for a biosynthetic or processing pathway in which all genes/proteins are essential for the system to work.

There are two promoters, one (in orange) for the constitutively expressed operon repressor (regulatory) gene (green). Constitutively expressed means always switched on, which means that **in operons, the repressor protein is always present in excess**. The second and major promoter (pink) is for the series of linked structural genes that will be transcribed and translated together. The operator site (yellow) is just downstream from the operon promoter.

The major point to note about an operon is that it is regulated to be either on or off and the metabolic process or pathway that is controlled requires ALL of the structural genes – that are present as a series of tandem ORFs. It therefore makes perfect sense that the structural genes of the operon are regulated by a single promoter. Operon sequences are unique to operons and occur just downstream to the operon promoter.

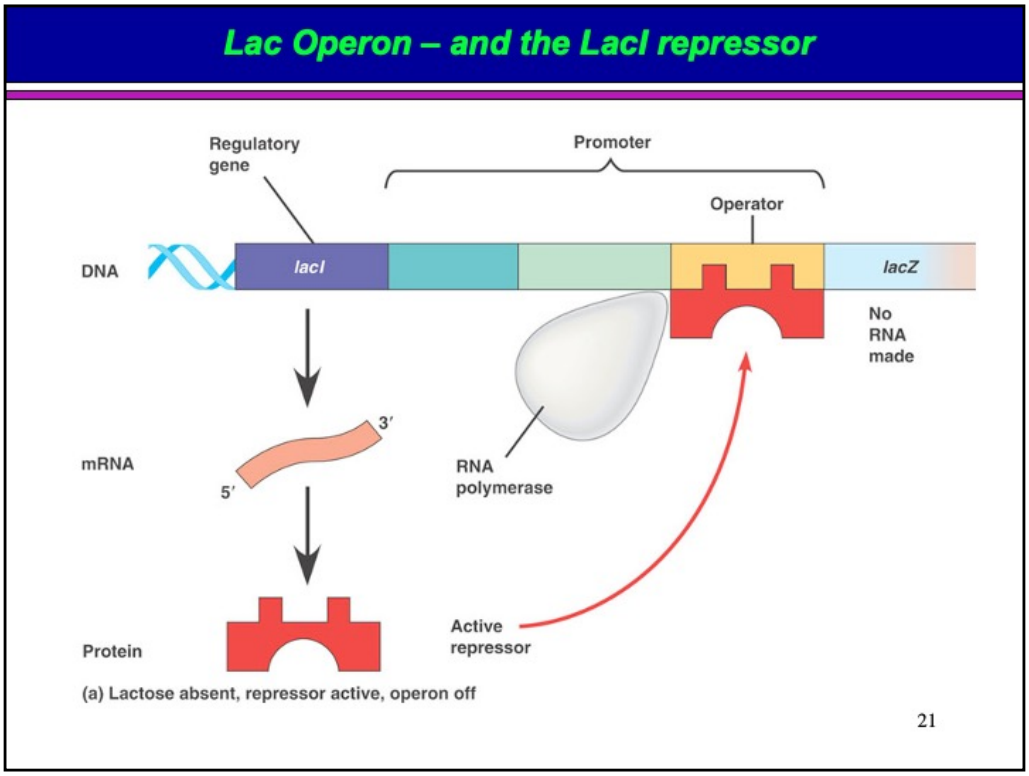


All of the elements in the Lac Operon are shown in this schematic. Bear in mind that the sole purpose of this bacterial operon is for the utilization of lactose (bottom right), as an alternative carbon source to glucose. This means that operons are usually in the 'off' position, when glucose is present, but the operon is switched 'on' only when lactose becomes available. An analogy is a desalination plant, which is not needed when water is freely available, but is switched on during long dry spells or droughts.

The enzyme that hydrolyses lactose is our 'friend' from the last lecture, namely beta-galactosidase (LacZ; 'Z') in the scheme above. It is joined here by its partners in the Lac Operon, lactose permease (LacY; 'Y') and lactose transacetylase (Lac A; 'A').

Other points to note in this slide are the promoters (green) for the LacI repressor gene (left) and for the operon (right). RNA polymerase (yellow) will transcribe from both promoters, at LacI constitutively (always), and at the operon's structural promoter (right) occasionally, that is, when the operon is switched on.

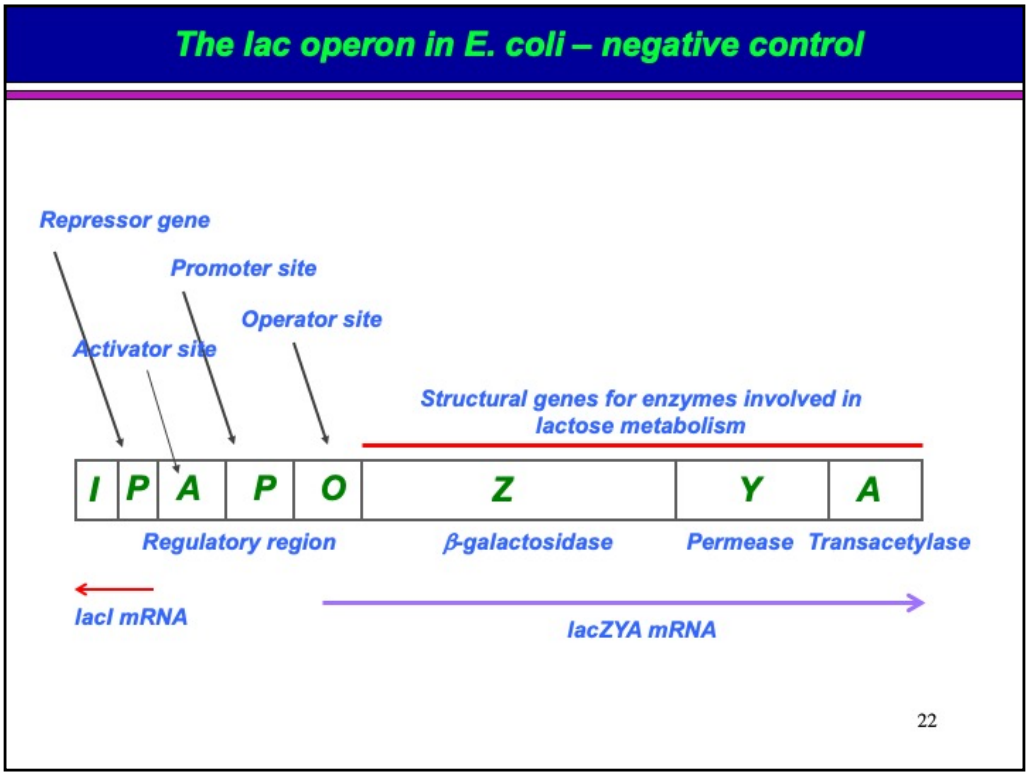
The Operon is **regulated by the Lac repressor protein** (in red) being bound at the operator site in the absence of lactose, which is the substrate for the operon. No substrate, no need for the operon to be working, or for the three proteins to be produced.



This diagram is another view of the basics of the regulated lac operon. The structural genes, beginning at LacZ (the other two are to the right, but not shown in this diagram), are either off all the time (lacI, constitutive) or on. In the scheme above, only the 'off' scenario is depicted.

The lacI regulatory gene is constitutive (orange mRNA made constantly) and encodes the repressor protein (red) that binds to the operator region at the downstream or 3-prime (3') end of the main promoter. The repressor literally and physically blocks the binding and passage of RNA polymerase at the promoter site, and consequently there is no transcription of the downstream structural genes for lactose utilization.

More details in the next slides, including how and why the operon is 'switched on'.



This drawing of the *E. coli lac* operon is not to scale with the regulatory region exaggerated so that it is easier to see all of the elements (this has been so for previous slides). One new feature of this representation (versus those earlier) is the 'activator site', whose importance will be seen later, so let's not confuse ourselves with too much new stuff just yet.

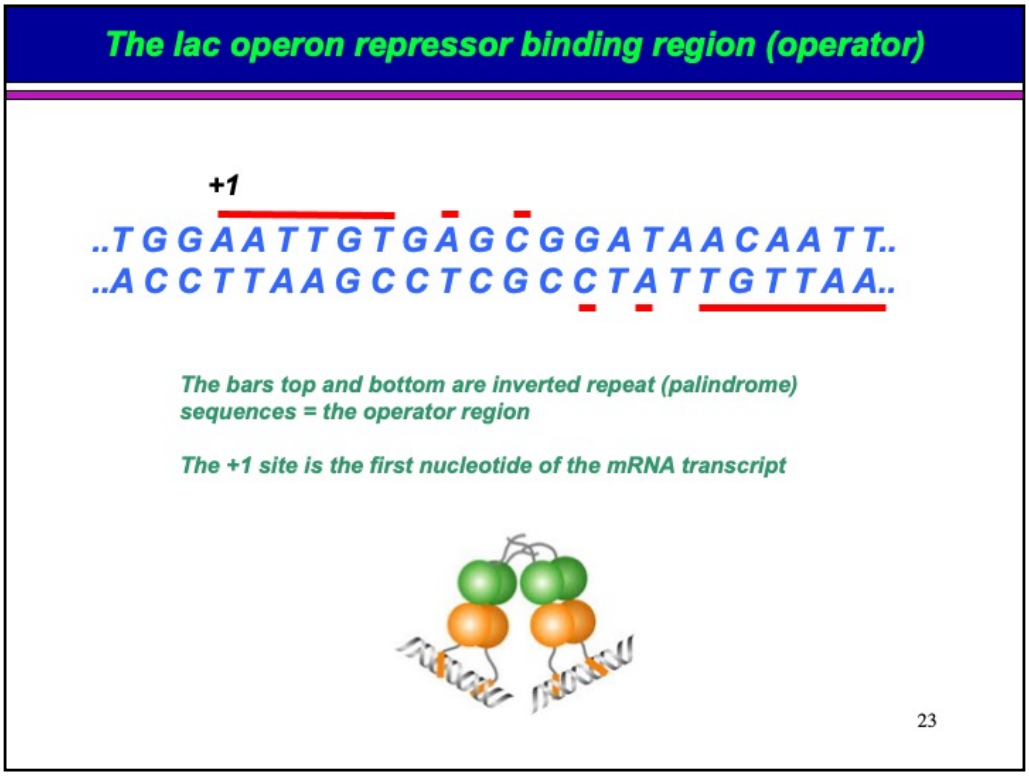
The *lac* operon is designed for one purpose only – the utilization of the disaccharide lactose as a carbon source, especially in the absence of the preferred carbon sugar, glucose, in *E. coli* and other bacteria.

The Lac Operon 'choreographs' the uptake, modification and hydrolysis of lactose into the monomer sugars glucose and galactose. The three structural genes work in concert: Lac permease (LacY) is a membrane protein that enables the active uptake of lactose into the cell. Transacetylase (LacA) is an enzyme that acetylates lactose, priming it for hydrolysis by the enzyme beta-galactosidase (LacZ).

Note that the regulatory region also contains the *lacI* repressor gene that has its own promoter and mRNA. The *lacI* gene is expressed constitutively, which means that it is always ON. The lac operon is therefore in the 'off' mode most of the time, as it is only required to be on when lactose is available.

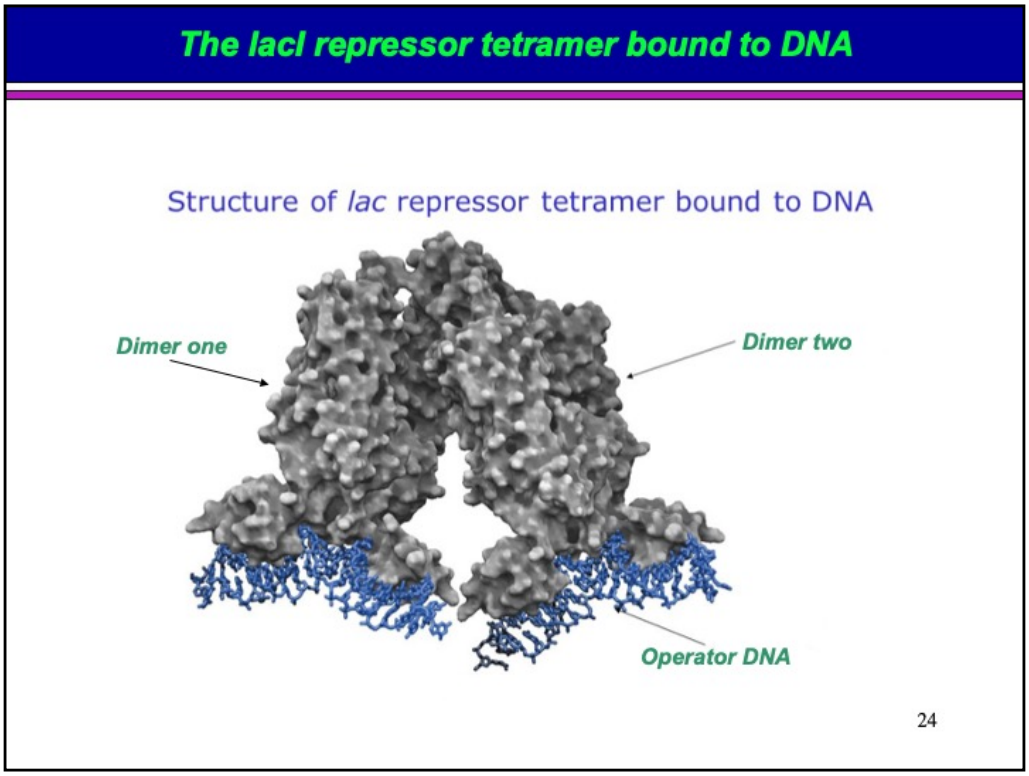
What we see here for the first time is the lacZYA mRNA transcript (long purple arrow) that comprises the three operon structural gene ORFs in tandem, from a single promoter.

Take Home Message: Operons generate a single long mRNA transcript for however many structural genes (and ORFs) lie in tandem downstream.



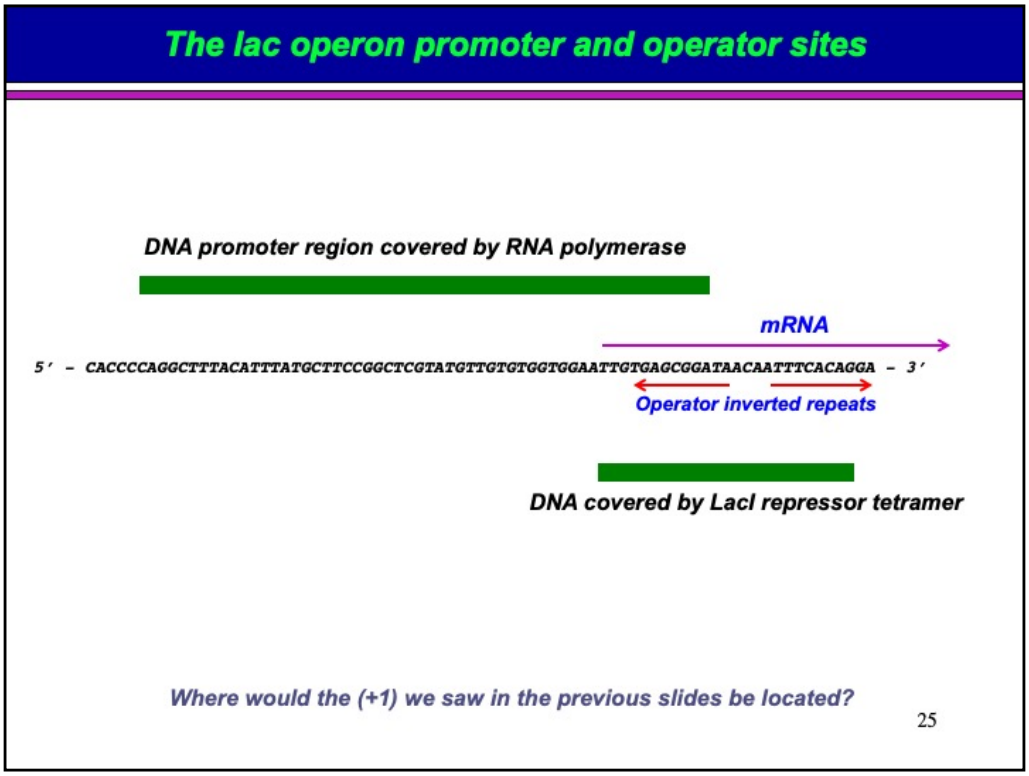
The lac Operator sequence is designed to bind a dimer of dimers (tetramer) of the Lac repressor protein (cartoon at the bottom of the slide). The Lac repressor protein is expressed constitutively by the *lacI* gene, and the repressor protein **only works** when it is assembled into a **tetramer**. The two dimer pairs of the tetramer bind to the inverted repeat sequences (red bars; shown as orange 'clamps' on the DNA in the cartoon).

The promoter for the operon is upstream from the operator site, that is, to the left of +1 and not shown here. The +1 position for the mRNA transcript is indicated on the top strand of the operator sequence.



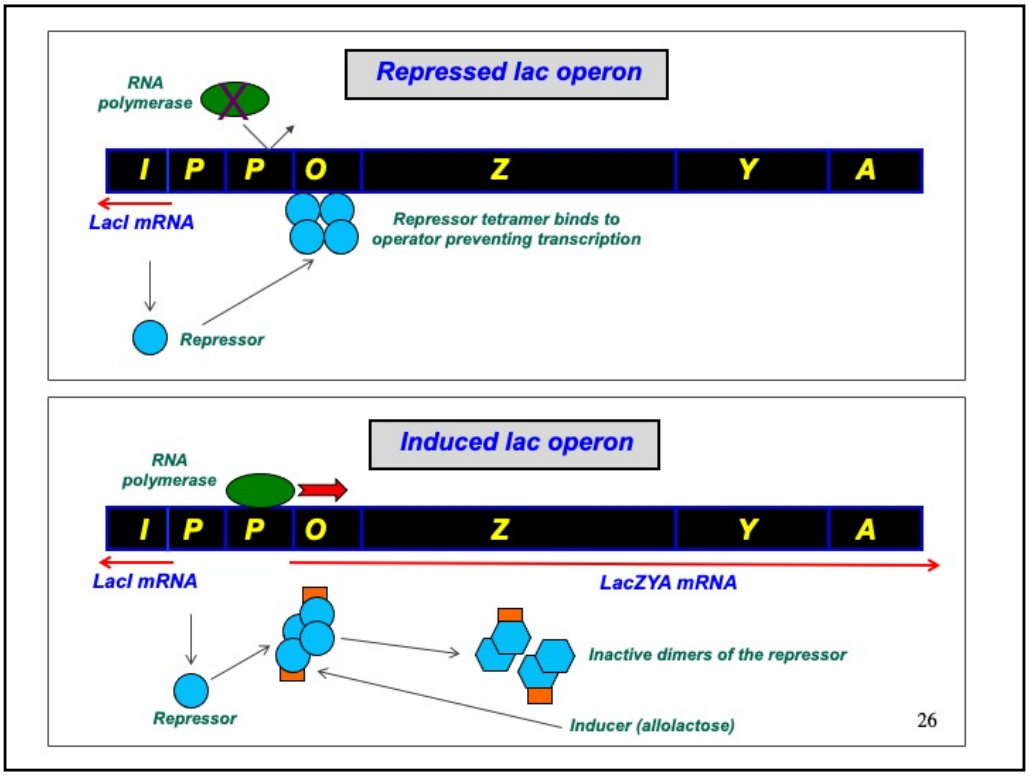
The binding of the Lac repressor dimer pair or tetramer to DNA. Each dimer is depicted in grey and the DNA in blue. This type of binding is typical of ALL dimeric proteins binding to double-stranded DNA. Another good example will be seen in a later slide, but this same process happens for many other proteins and enzymes, for example restriction endonucleases.

Notice how each dimer binds in an equivalent groove in which the operator repeats lie, and importantly how each attached dimer bends the DNA, making the nearby (upstream) promoter temporarily inaccessible to RNA polymerase binding – therefore no mRNA transcript! A physical conformational impasse is set up.



Note, in the diagram, only one situation is possible (only one green bar can be present). Either the repressor stops RNA polymerase binding (bottom green bar), or the repressor is absent (when lactose is present) and RNA polymerase can bind to the promoter site (top green bar).

This slide is intended to illustrate the proximity and overlapping of the promoter and operator sites in the regulatory region, which is why it is drawn with both green bars present.



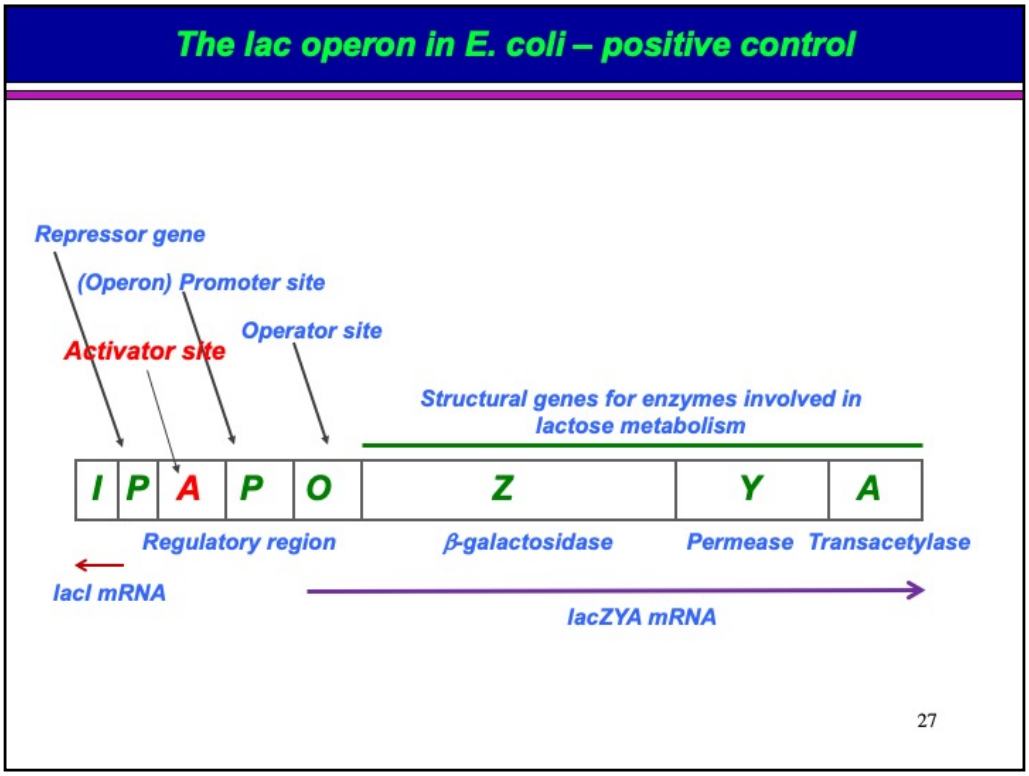
This slide summarizes the two states of the lac operon:

Top panel: OFF = repressed (absence of lactose). Repressed operon – no lactose, Lac repressor binds to operator, RNA polymerase cannot bind to promoter, no transcription of the structural genes.

Bottom panel: ON = induced (presence of lactose). Induced operon – lactose provided, constitutively transcribed and translated Lac repressor is 'titrated' by allolactose binding (orange squares; acetylated lactose) which changes the conformation of the repressor (from circles to hexagons) so that it can no longer bind to the operator inverted repeats. RNA polymerase is free to assemble on the promoter site and transcription of the operon ensues.

Note, for simplicity at this point, the activator site, shown a few slides earlier, is still being ignored.

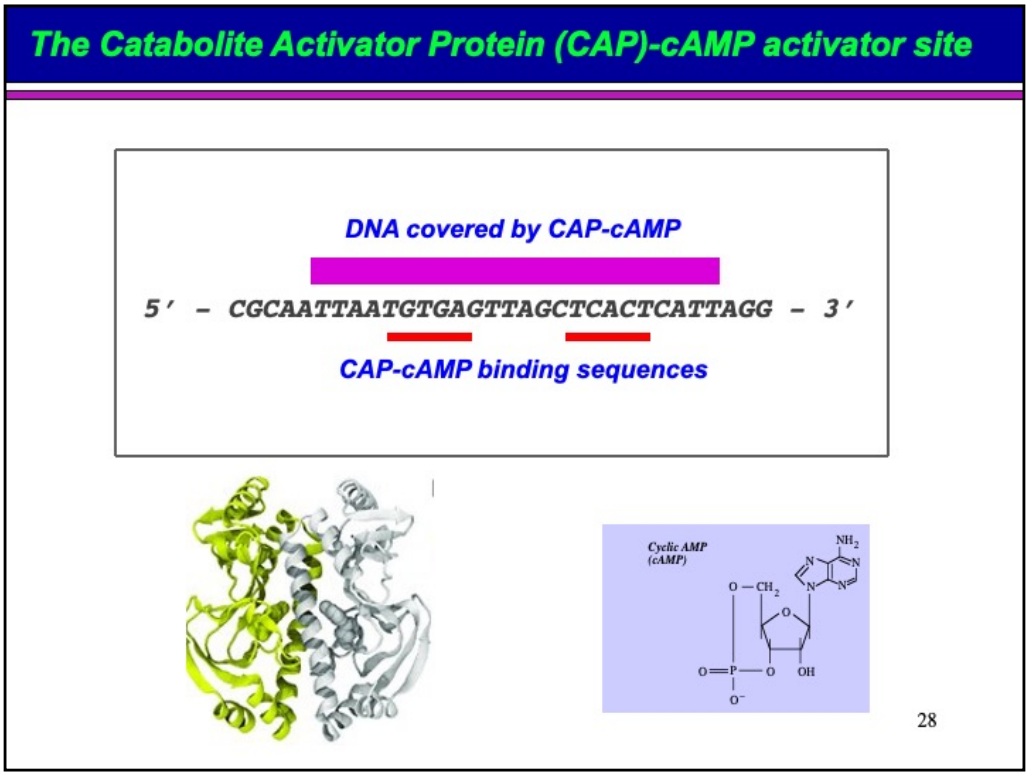
Note also that the mRNA for the LacI repressor is present in both panels, but the mRNA for the lac operon structural genes is only present in the bottom panel.



One new feature of this representation (versus those in earlier slides) is the 'activator site' (red) whose importance will be seen shortly. The *lac* operon is designed for one purpose only – the utilization of the disaccharide lactose as a carbon source, especially in the absence of the preferred carbon sugar, glucose.

To refresh what we know so far, the entire operon 'choreographs' the uptake, modification and hydrolysis of lactose into the monomer sugars glucose and galactose. The three structural genes work in concert: Lac permease (LacY) is a membrane protein that enables the active uptake of lactose into the cell. Transacetylase (LacA) is an enzyme that acetylates lactose, priming it for hydrolysis by the enzyme beta-galactosidase (LacZ). Note that the regulatory region contains the *lacI* repressor gene that has its own promoter and mRNA. The *lacI* gene is expressed constitutively, which means that it is always ON. The lac operon is therefore in the 'off' mode most of the time as it is only required to be on when lactose is available.

The Lac Operon seems to be nicely set up and poised to utilise lactose when it becomes available, so what is the function and importance of the activator site?



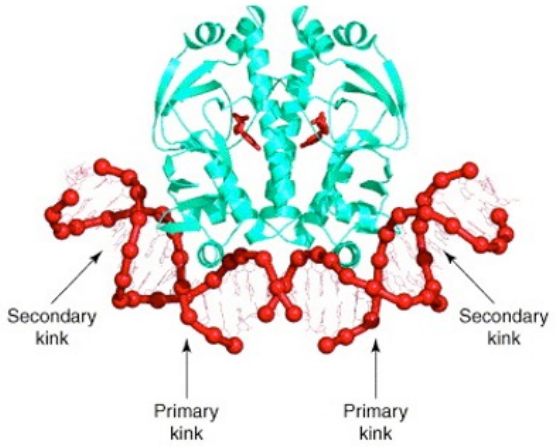
The activator site and its binding complimentary sequences (red bars).

The CAP-cAMP protein (bottom left) ONLY binds to the activator site when cAMP (bottom right) is bound to the CAP protein (Catabolite Activator Protein).

This is a **positive** metabolic signal or response to **low levels of glucose only** (note glucose, not lactose!). The cell has sensed (via raised cAMP levels) that it is being starved of an essential carbon source and one response to this is to 'look' for and use alternatives such as lactose – hence the activation of the *lac* operon.

When the cell senses that glucose is low, cAMP levels **rise** and this activates transcription and translation of CAP to which cAMP binds, causing a small conformation change that then enables it to bind to the *lac* operon activator site (illustrated on the next slide). In doing so, it binds to RNA polymerase (for want of better words – 'captures RNA Polymerase and holds it down on the promoter'), enhancing the binding of RNA polymerase (~50-fold) to the *lac* promoter site, and rapid transcription ensues.

The CAP-cAMP complex bound to DNA



These kinks have really got the DNA, which is sometimes referred to as the Lola bind

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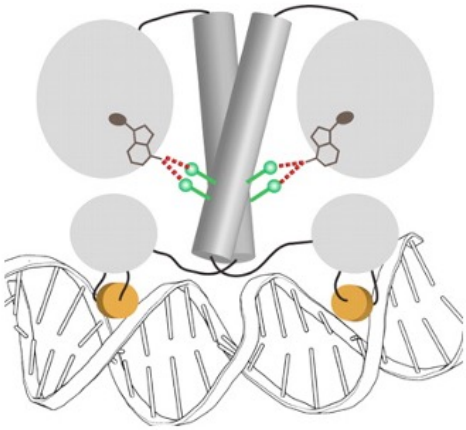
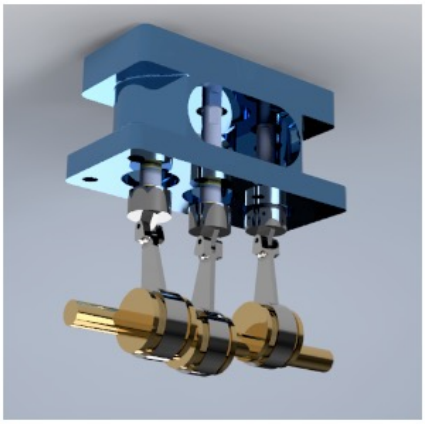
A view of the CAP/cAMP dimer (coloured aqua) with cAMP bound at each catalytic site (small red molecules within each half of the CAP dimer). The CAP-cAMP dimer now binds to DNA (red strands at bottom), which will only occur when cAMP occupies the sites either side of the central diagonal green helices.

Note – as was the case with the Lac Repressor – that the dimer binds to the grooved sequence repeats and bends the DNA. In this instance, the DNA bending and the CAP protein makes the promoter much more accessible to RNA polymerase binding.

To reiterate from the previous slide,

The CAP-cAMP complex bound to DNA





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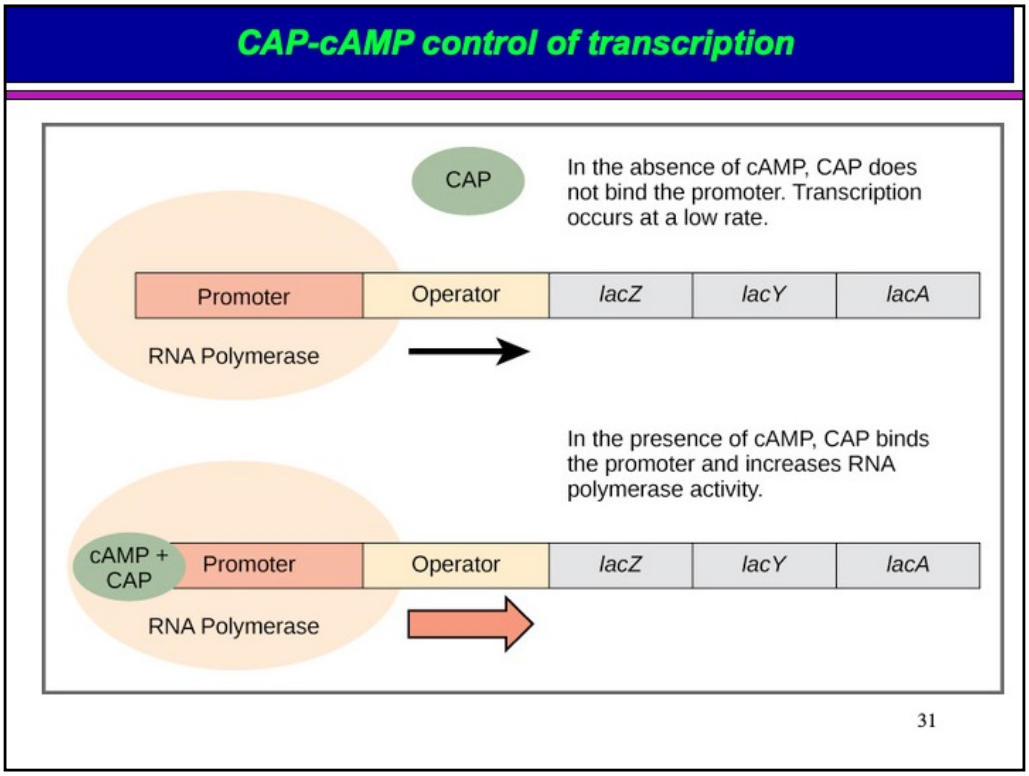
The left panel depicts a cartoon of the CAP-cAMP dimer complex, showing how the cAMP molecules allow the geometry of binding to DNA. cAMPs attach to the two central helices and align them in such a way that the two short helices (in gold) at the bottom of each monomer will now fit within the two large grooves of a DNA molecule at the precise activator repeat sequence sites, thereby enabling the twisting or bending of the DNA.

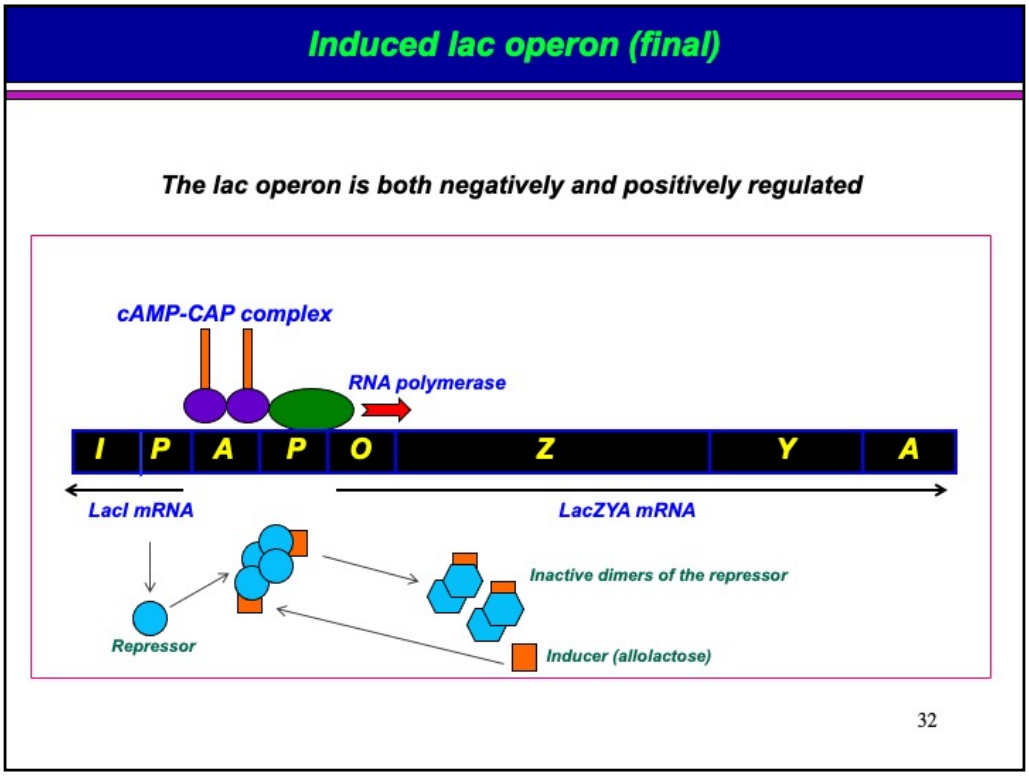
This is another example of allosteric control in that the binding of one cAMP molecule in one of the dimer monomers enhances the binding of the second cAMP, by a small change in the access portal for the cAMP molecule. The bound cAMPs now effect a small rotation of the two central helices, which in turn align the two (gold) DNA binding helices with the large groove of the DNA at the activator site.



The two scenarios in this slide are self-explanatory, with the exception that in each panel, the absence or presence of cAMP is also a signature for the metabolic glucose concentration, which is **opposite** to that of the level of cAMP. But there is a 'trick' in the system that is explained below.

So, with cAMP low, glucose is high and the CAP complex activation is not required; but with cAMP high, and glucose low, the CAP-cAMP complex binds to the activator site and is primed to stimulate transcription. The level of transcription in each scenario is depicted by the widths of the arrows under the Operators.

Note the use of the word 'primed' in the previous sentence. In the absence of glucose, the second scenario is set to go, but this can only happen if a source of lactose is made available, because the LacI repressor still needs to be dislodged from the Operator site.

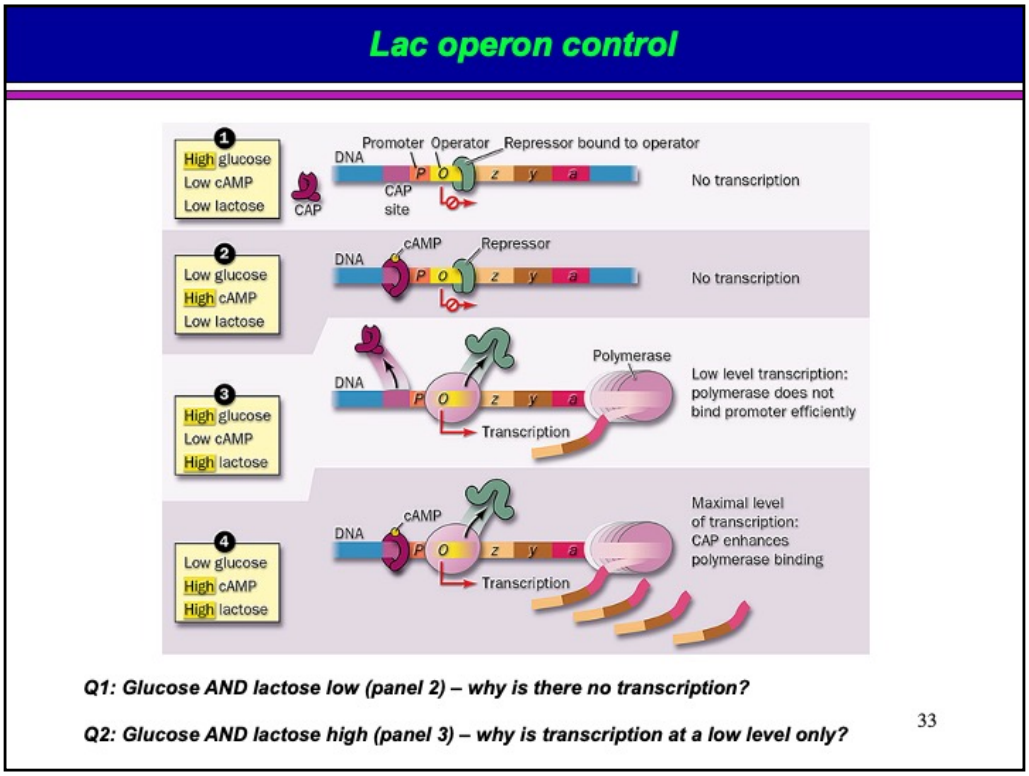


The complete induced state of the *lac* operon, with all interacting elements depicted.

The default or holding position is negative regulation (binding of Lac repressor to the operator site), which is not shown in this slide.

When glucose is low (and lactose high), the full positive regulatory state occurs, with maximal transcription to capture and utilize the new lactose carbon source, which is the state depicted in this slide.

The operon is said to be both negatively and positively regulated according to the dictates of carbon source and the catabolite cAMP level. Only one of four states is possible, as the next slide will show.



A schematic summary of the four possible metabolic states of the ON or OFF *lac* operon.

A low level of lactose, even in the presence of low or high glucose, will keep the Lac Operon in the off state. Negative Control, as per Scenarios 1 and 2.

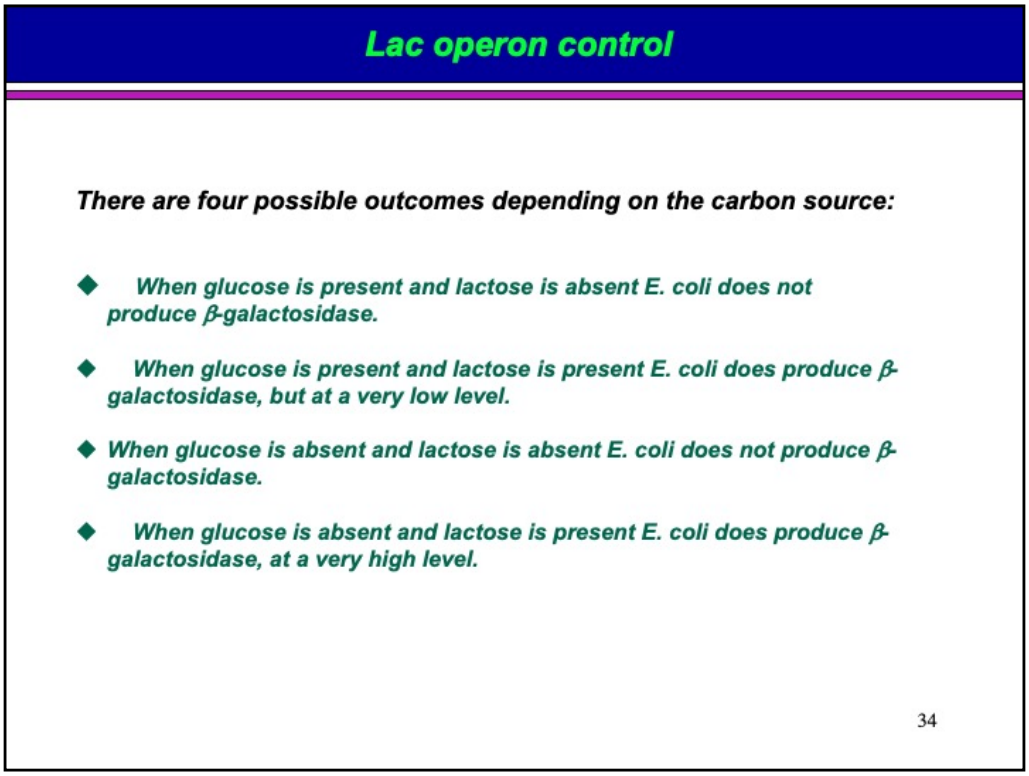
The levels of glucose and cAMP are opposite to one another and they have an influence on how this operon will work. Positive Control, as per Scenarios 3 and 4.

Two abiding questions come to mind when looking at these four metabolic scenarios:

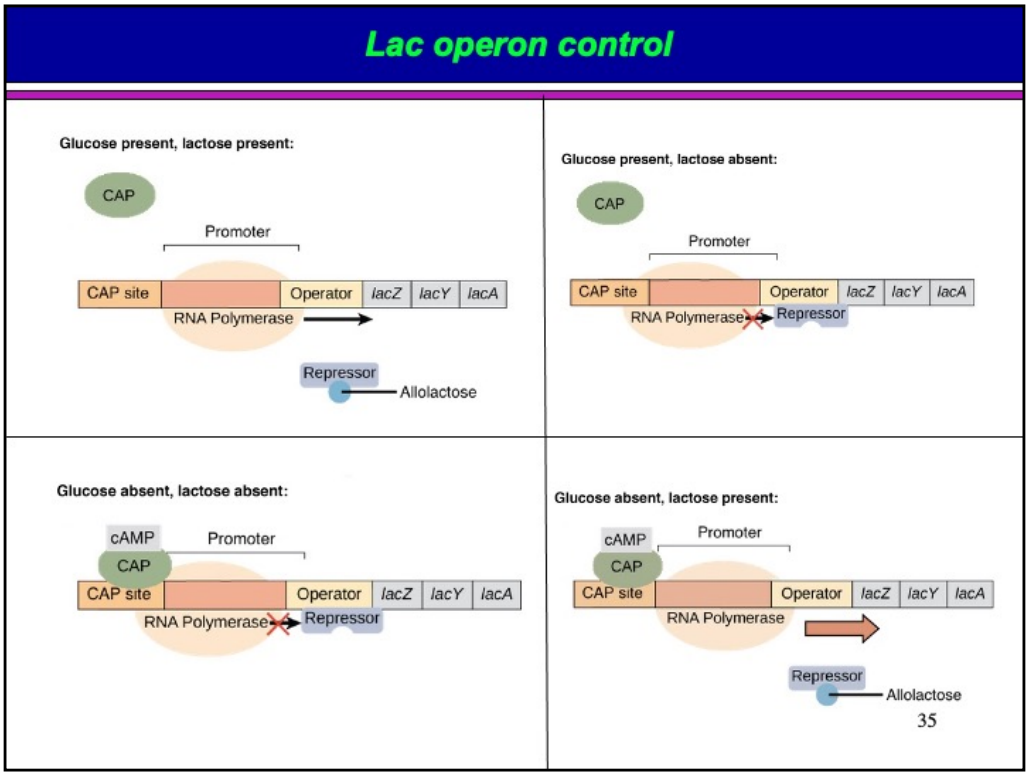
Why is there no transcription of the operon when both glucose and lactose are low?

Why is transcription low when there are high levels of glucose AND lactose?

If you can answer these two questions, then you know the Lac Operon very well!

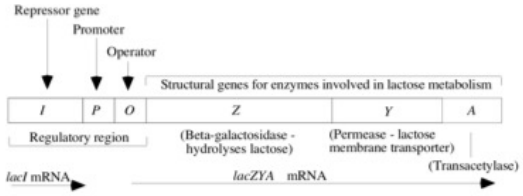


This slide summarises the four scenarios on the previous slide.



This slide is another way of summarising the four scenarios on the previous two slides. Note the sizes of the transcription arrows below the operon in each panel and whether the arrows are on or off, and the width of the last large orange arrow, which indicated the maximum transcription rate of the Lac operon.

MCQ L7



The scheme shows elements of the Lac Operon. The regulatory parts of the operon are the repressor gene, the promoter and the operator, all of which are required because…

(A) Glucose and lactose may be used simultaneously.

(B) The repressor will switch the operator on and off in response to glucose levels.

(C) If lactose becomes available, the operon will be switched on.

(D) It will only be switched on if there is no available lactose.

(E) The repressor will switch the operator on and off in response to any sugar source.

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MCQ L7

The dystrophin gene is the largest in the human genome, with the Open Reading Frame comprised of 79 exons. Can you speculate on the number of introns?

A. Yes, it is 80 introns.

B. Not really, because we don't know the size of the gene.

C. No.

D. Not really, because we don't know the size of the mRNA.

E. Yes, it is 78 introns.

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MOLECULAR BIOLOGY 1
Lecture 8

Polymerase Chain Reaction (PCR)

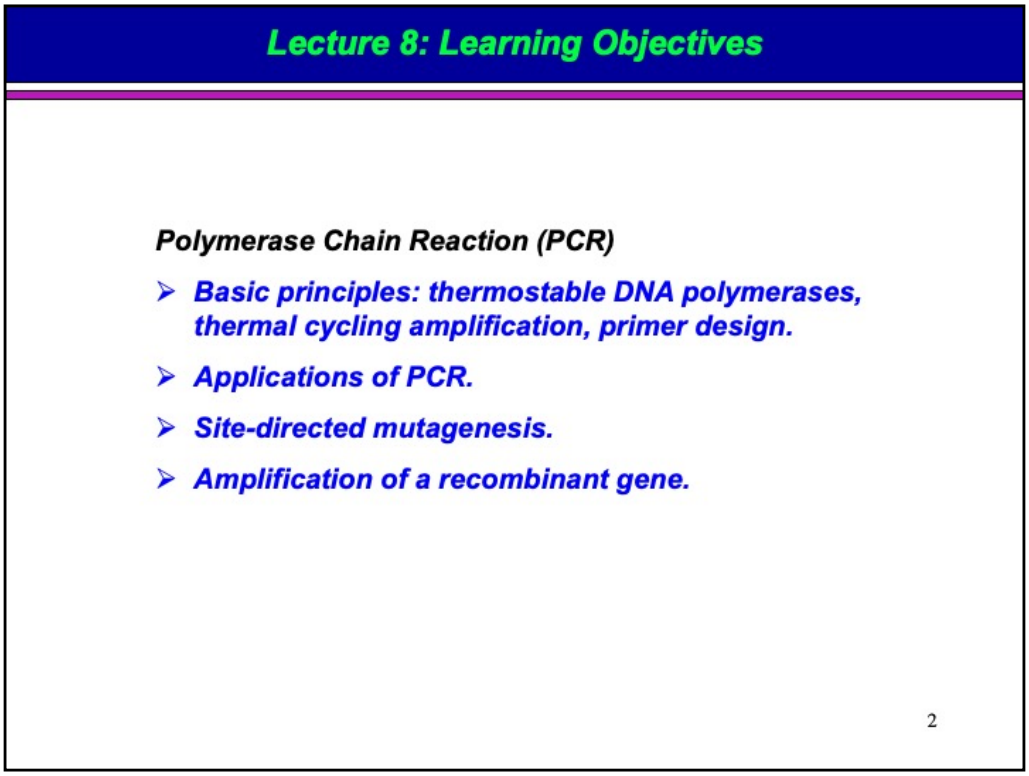
Applications

Site-directed mutagenesis

Amplifying a gene within a recombinant plasmid

1

This lecture will introduce the powerful method of the Polymerase Chain Reaction (PCR) and a few research applications.

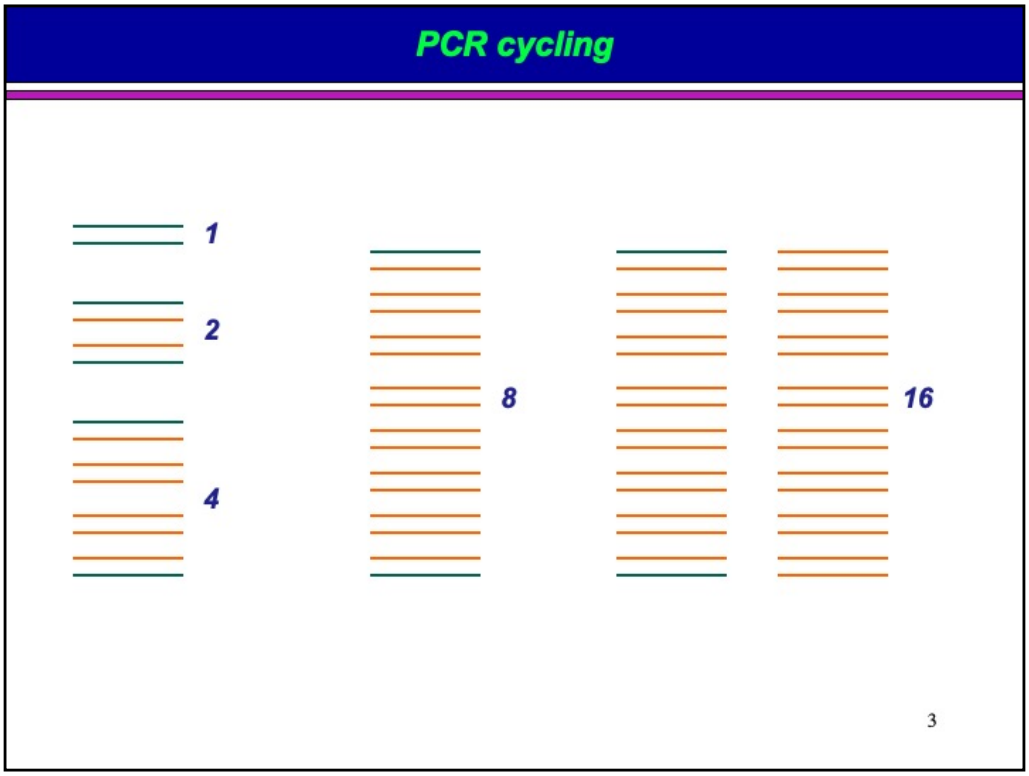


This lecture introduces one of the most powerful methods in molecular biology, namely the Polymerase Chain Reaction (PCR), which is a semi-automated technique for generating large amounts of DNA from any source (cloned genes; whole chromosomes; 'dirty' DNA from blood, cells, hair, fingerprints, etc, in forensics, archeology, paleontology, and medical diagnosing; or just plain research).

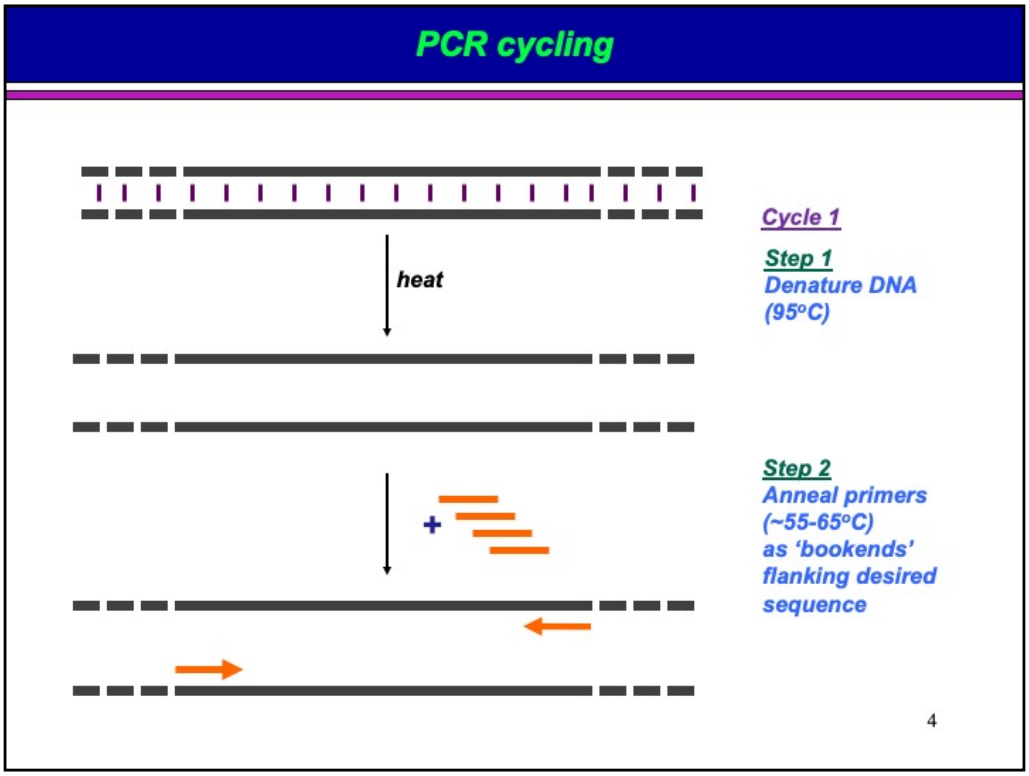
The basics of thermal amplification of DNA will be illustrated, including the genesis of this technology following the discovery of thermostable DNA polymerases and microprocessor controlled thermal cyclers during the 1990s.

We will look at several applications of the PCR and in particular at the very powerful method of site-directed mutagenesis, in which any codon for any amino acid in any cloned gene may be altered (substituted, deleted, inserted), enabling insights into how proteins/enzymes work.

Finally, we will look at a simple scheme to amplify a gene in a recombinant plasmid, which is the basis of the last experiment in the practical course (Expt. 7).



The principle of the PCR is to amplify template DNA in an exponential manner so that multiple copies are accumulated in a very short time, by a theoretical doubling of the number of molecules in each cycle. It is essentially simply a laboratory method of DNA replication that is accurate and rapid (as is DNA replication).

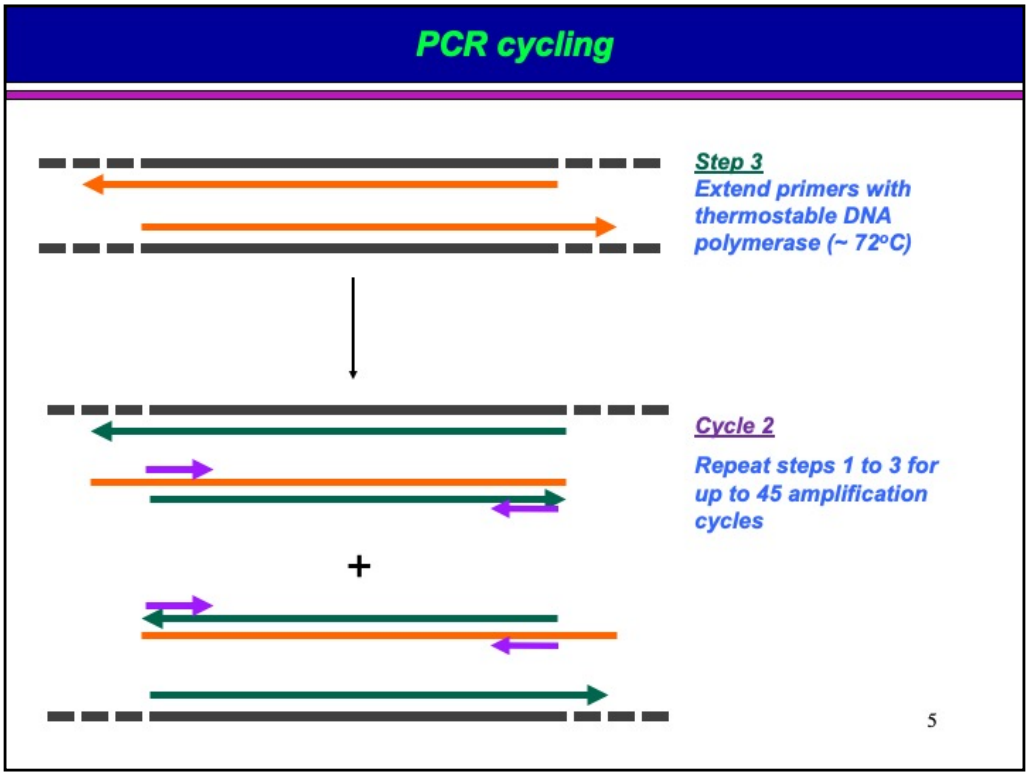


PCR cycling is performed in a special thermal cycling machine (see later) that is able to carry out the steps of each cycle very rapidly. All of the reaction contents are present in a thin-walled plastic tube before the cycling is commenced.

Cycle One begins at Step 1 (Denaturation) in which rapid heating to about 95°C is applied to the template to "melt" the double-stranded DNA by breaking the H-bonding between the complementary base pairs – Step 1 takes about 30-60 sec.

Step 2 is Annealing in which the temperature is taken back to about 55-65°C that allows an excess of primers (orange in the figure above) to bind to their respective short complementary sequences – usually but not always as ''bookends'. Step 2 takes 10-30 sec.

The intention in this example, and most PCR experiments, is to amplify only the part of the DNA template between the bookended primers.

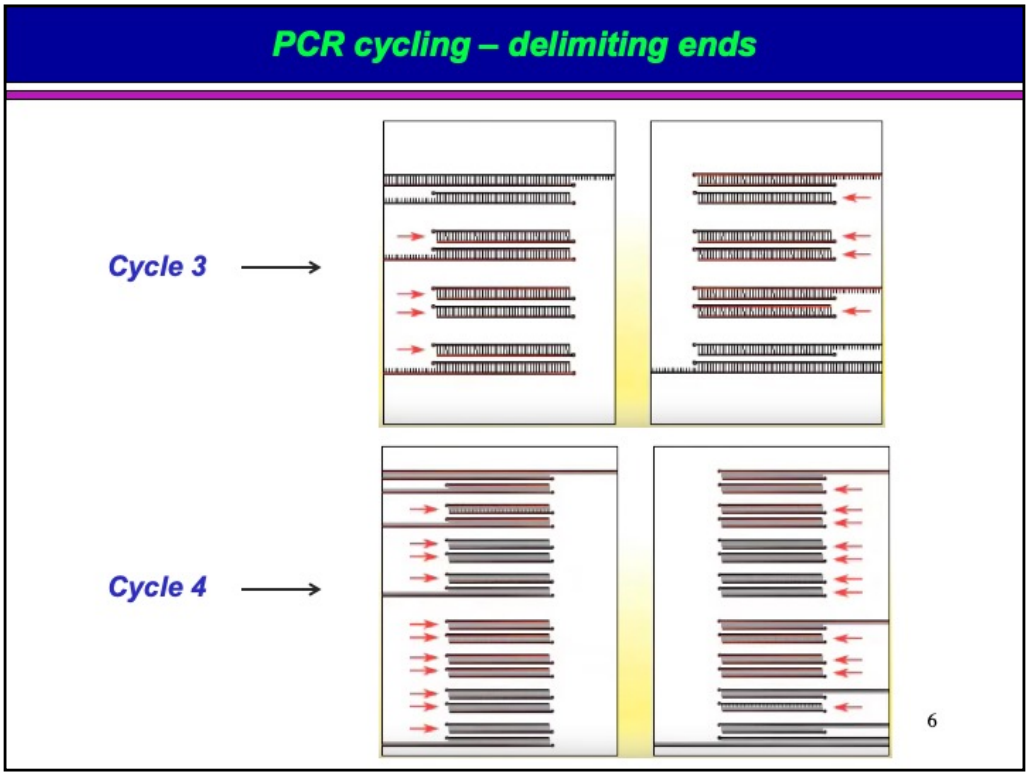


Step 3 is the Extension reaction in which a thermostable DNA polymerase is used to synthesize new single strands (in orange) by extension of the 3-prime ends of the primers along the template strands at the maximum allowable reaction temperature of the thermostable polymerase (~68-74°C depending on the type of DNA Pol used). The Step 3 extension time depends on the length of the template that is to be amplified, but is usually conducted for about 1 min per kb of DNA, up to a maximum of 30 kb, which is about the maximum length of template limit currently possible.

Cycle 2 starts the whole three-step process again using the original two template strands (dark grey) and the newly amplified strands (orange) to make new copies (green). Note that the orange strands will have been extended a little beyond the primers (purple) on the opposite strands, but in the second and subsequent cycles, the same primers bind at the same positions and the strands can only be extended to the ends of the new templates and so on for each successive cycle.

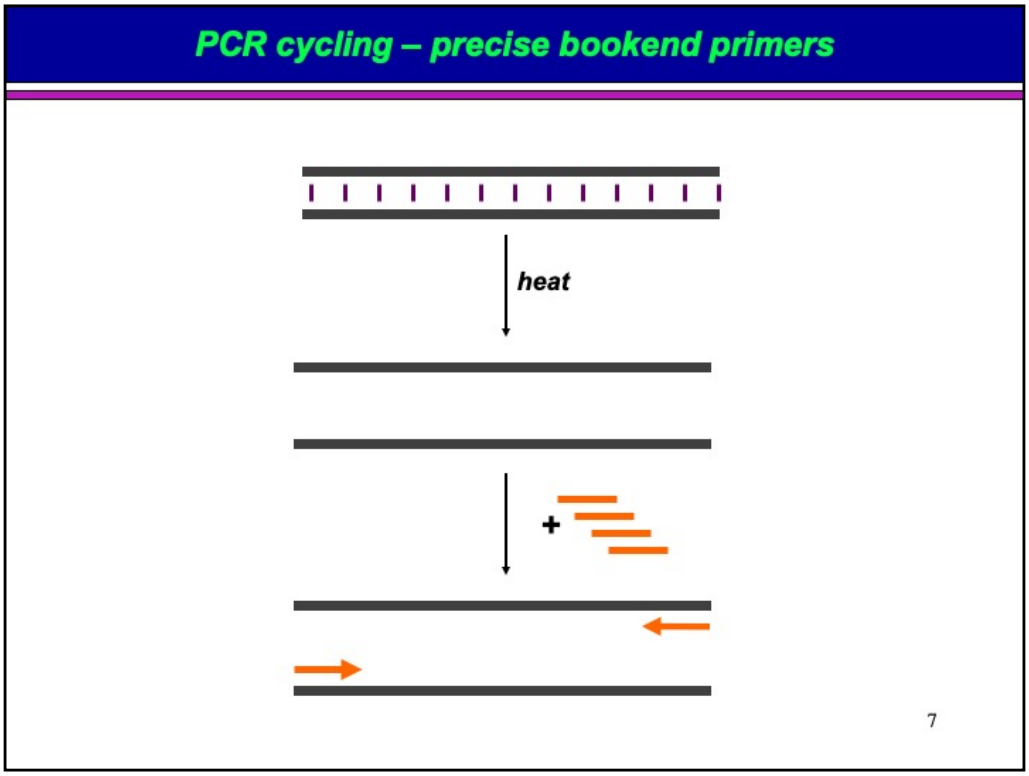
The result of this limited extending is that the newly amplified strands will progressively increase to much higher numbers of the same (bookended) length so that after however many cycles (15 to 45), essentially a single sized product band will be obtained and its size will be that delimited by the primers, as noted by the position of the primers (purple) on the new template strands above.

You can see this on the next slide; and in the second of three PCR You Tube videos under the 'Helpful Material' menu item on the left panel in UTS Online. The longer products only increase by two copies in each cycle, but the correct length products increase exponentially. For example, in Cycle 2 there are 4 longer products and 0 correct length products; cycle 3 = 6 longer, 2 correct; cycle 4 = 8 longer, 8 correct,……cycle 7 = 14 longer, 114 correct etc.



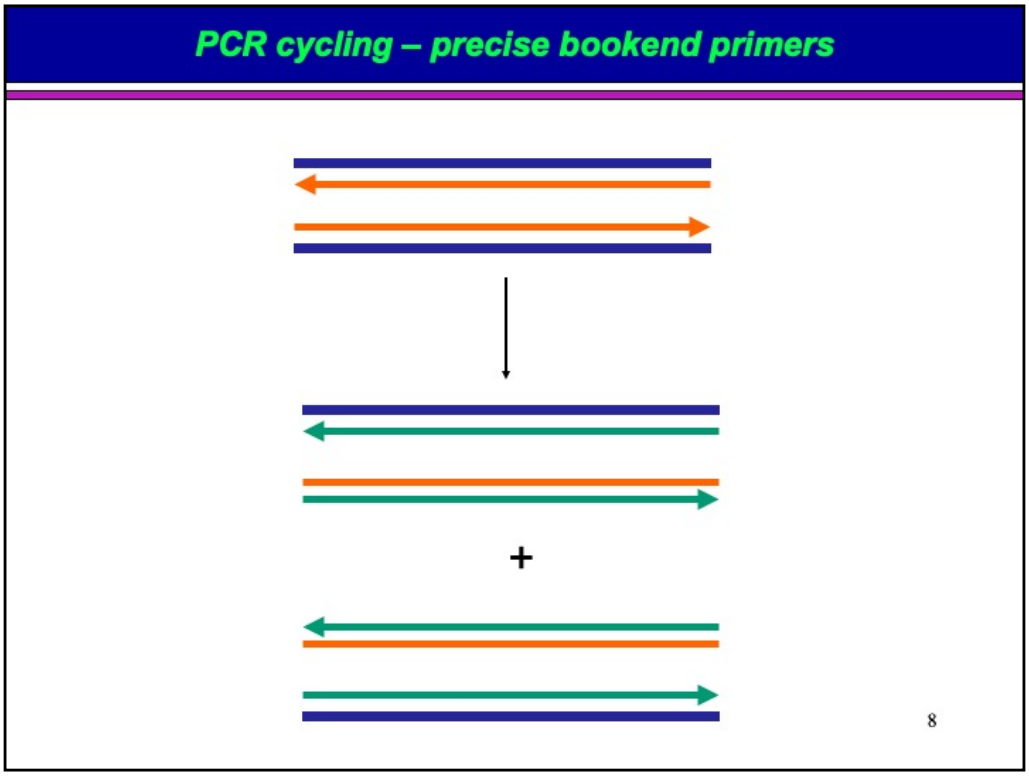
This slide shows the exponential accumulation of the delimited (bookended) products, versus fewer longer products from the original longer template strands. The longer products only increase by two copies in each cycle, but the correct length products increase exponentially.

In Cycle 3 above, there are 6 longer products and only 2 correct (precise length) products. Cycle 4 = 8 longer, 8 correct length……Cycle 7 = 14 longer, 114 correct etc.



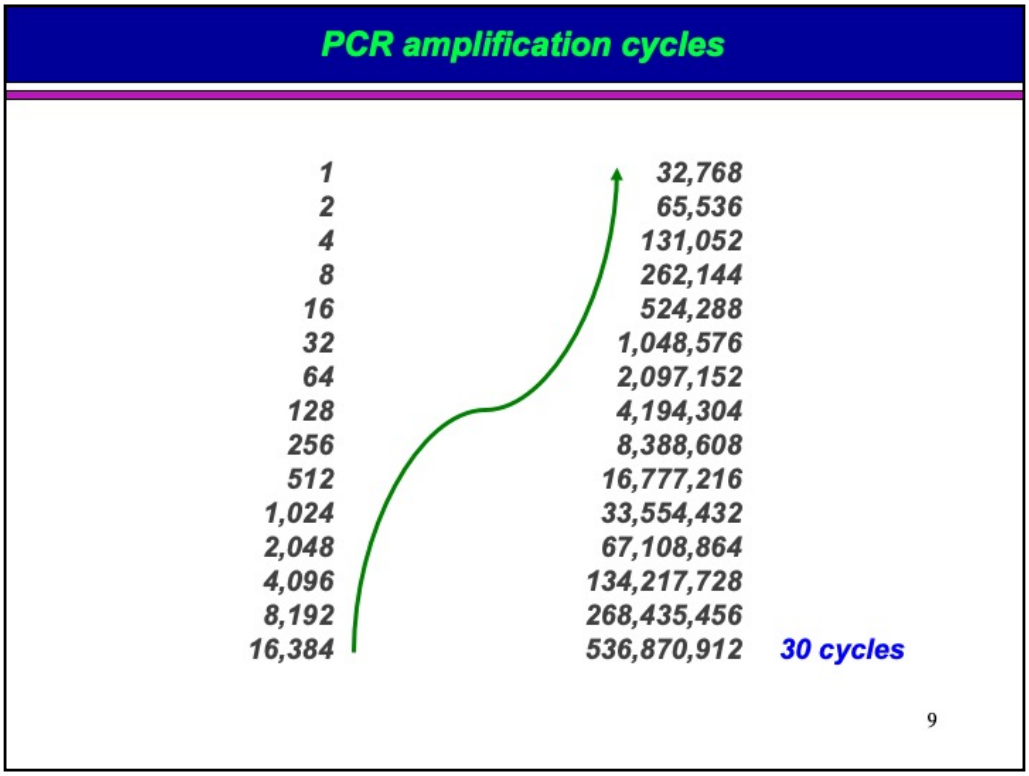
This slide depicts precise amplification of template by bookended primers that anneal at the ends of the fixed length template strands. This can only be accomplished if the primers can be designed to fit the ends of the target template without secondary structure issues (see further on for primer design).

In this scenario, there are no longer product molecules. Although not shown here, but easy enough to envisage, is the amplification of DNA in a plasmid with primers that bookend an inserted gene or fragment of DNA in that recombinant plasmid. Only the length of DNA between the primers is amplified, producing a fixed length linear product. You will see this on a slide near the end of this lecture; and in Expt. 7.



Completion of the first cycle indicating the precise limitation of the template extension (top; orange arrowed lines). PCR requires thermostable DNA polymerases to do the extension reactions for multiple cycles (~10 – 40) on double-stranded DNA (linear; or circular in plasmids).

The multiple repeat cycles of heating and cooling requires the enzyme to be robustly resistant to denaturation – more on this a few slides on.



Beginning with a single copy of a double-stranded template, 30 cycles will generate a theoretical yield of half a billion copies – usually taking just a few hours. In practice, this level of efficiency is never reached, but will be closer to 20-30% of the theoretical maximum yield depending on the reaction conditions and the number of cycles.

Again, in practice, the starting template concentration is not a single copy, but is usually at least a few nanograms (and up to 200 ng) of DNA. As an example, 1 ng of a 1 kb piece of DNA contains about 1 x 10^8 copies and if this is cycled for several hours, the amount of DNA produced will be a theoretical many hundreds of mg, but in the actual reaction, the gradual exhaustion of dNTPs and primers will slow the reaction and yield perhaps up to a few hundred µg at best. This is still an enormous amplification in just a few hours.

PCR reaction mix

Reaction mixture

- DNA template
- dNTP mix
- buffer + Mg^{2+}
- primer pair
- water
- DNA polymerase (last)

Cycling protocol

Step 1:	Denaturation of template at 95°C 30 - 60 sec
Step 2:	Annealing of primers at 56°C 30 sec
Step 3:	Extension of primers at 72°C 1 to 20 min

These three steps make up one cycle which is repeated up to 45 times taking from 2 to 6 hours

PCRs are performed in thin walled tubes in thermal cyclers with microprocessors for ramping the temperature very rapidly

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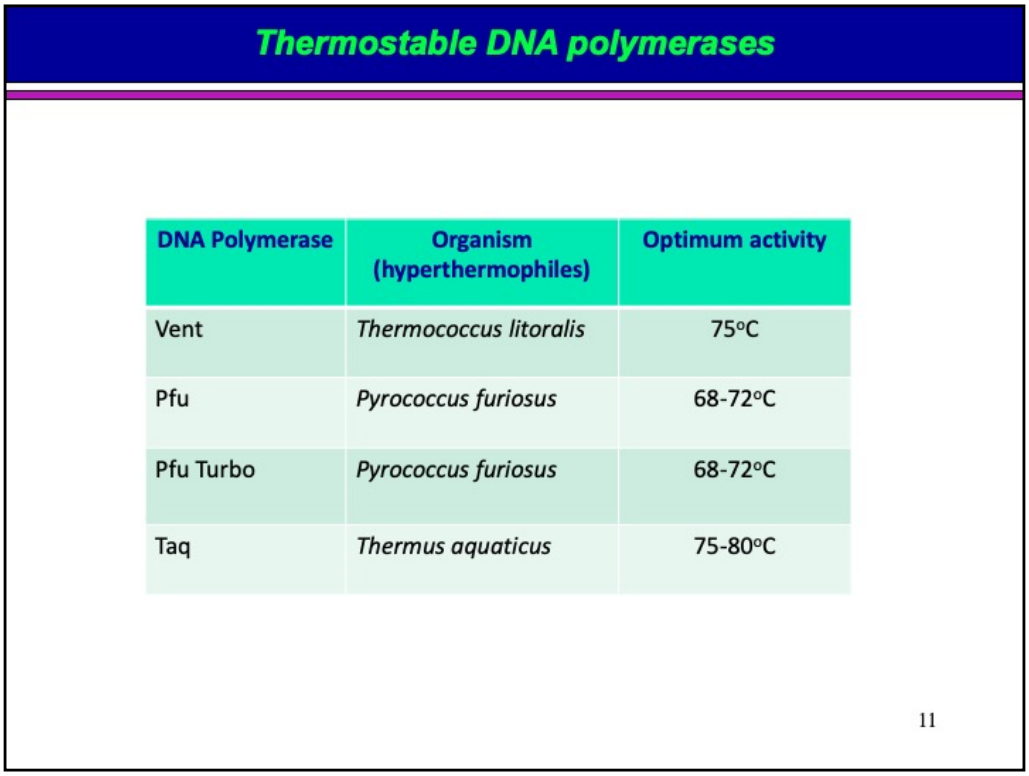
A typical reaction cocktail is shown on the left and cycling protocol parameters on the right. The total reaction mixture is adjusted with water to 25 to 50 µl.

All reaction contents are added at the start to a thin-walled narrow tube and not disturbed during the cycling. The individual steps of each cycle are short (30-60 sec, but up to a few minutes for long extensions). The thermal cycler microprocessor can ramp the temperature changes very rapidly so the the interval times between steps is very short (about 10 secs).

A normal 1.5 ml eppendorf tube is shown alongside a thin-walled 0.5 ml PCR tube. Although not obvious, the PCR tube is only about one third the thickness of the 1.5 ml tube. The thin walls are needed for the rapid heating and cooling. Although not shown, some PCR thermal cyclers accommodate 0.2 ml thin-wall tubes.



Some commercially available thermostable DNA polymerases used routinely in DNA cycle sequencing and the Polymerase Chain Reaction. They can add at least 100 nucleotides per second to a growing chain at the optimum temperature, but work very poorly, if at all, at 50°C and lower.

Taq polymerase is a “workhorse” enzyme as it’s proofreading exonuclease activity is far less accurate than the others.

Each polymerase includes a 5-prime to 3-prime exonuclease ‘proofreading’ activity that is required to replace occasional incorrectly inserted nucleotides. Incorrect bases are inserted at rates of 1:1000 to 1:10,000, which would be disastrous for the integrity of DNA copying. The exonuclease function of each DNA polymerase ensures that the error rate is far less frequent at 1:1,000,000 to 1:100,000,000. The lower the error rate, the higher the ‘fidelity’ of the enzyme.

Thermostable DNA polymerases

YEARS	PRODUCT(S)
1990-91	$Vent_R$® DNA Polymerase
1992	$Vent_R$® (exo-) DNA Polymerase Deep $Vent_R$™ DNA Polymerase
1993-1994	Deep $Vent_R$™ (exo-) DNA Polymerase
1995	9°N_m™ DNA Polymerase
2007-2008	Phusion® High-Fidelity DNA Polymerase
2011	Phusion® Hot Start Flex DNA Polymerase
2012	Q5® High-Fidelity DNA Polymerase Q5® Hot Start High-Fidelity DNA Polymerase

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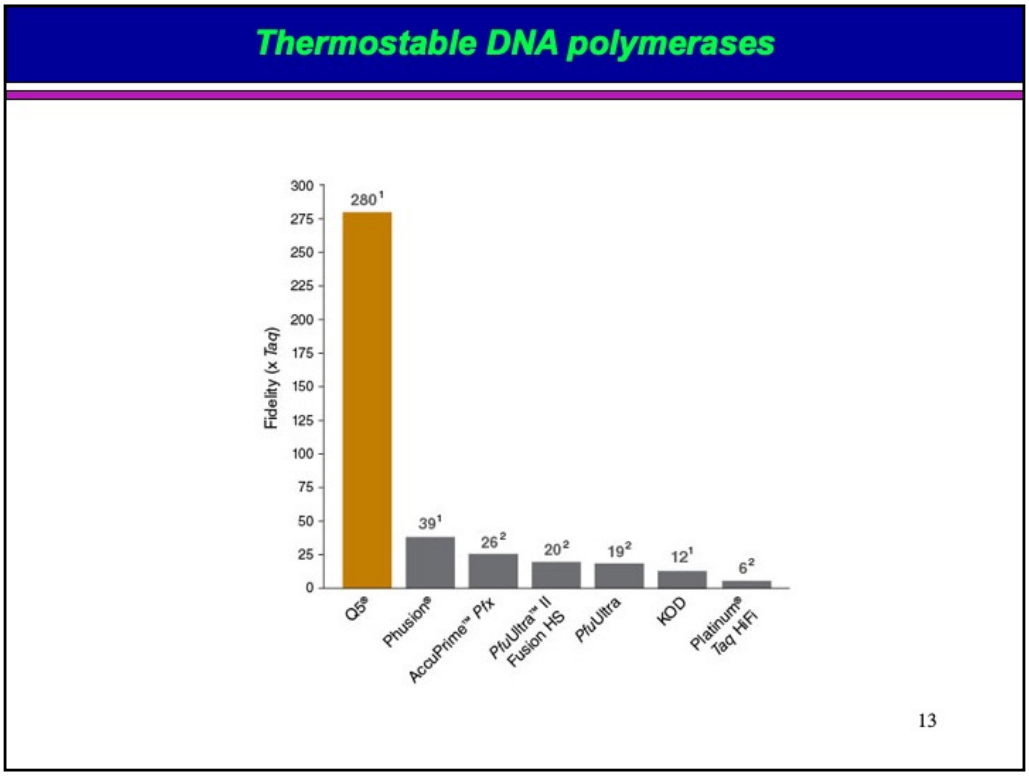
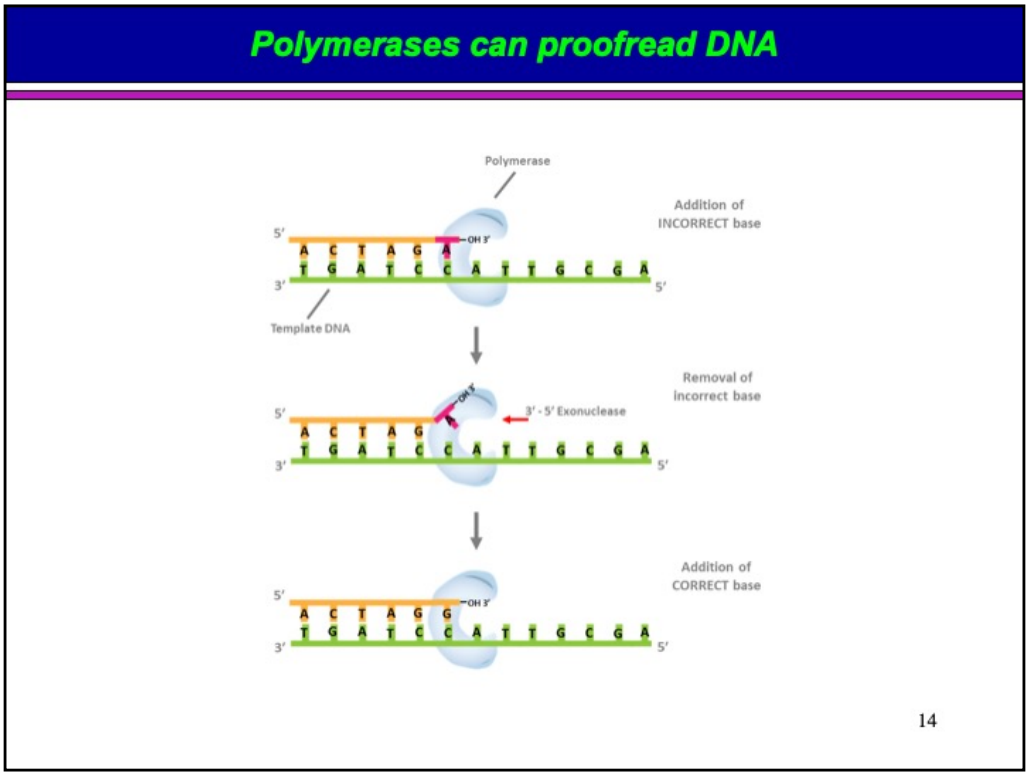
Some commercially available thermostable DNA polymerases, from New England BioLabs, used routinely in DNA cycle sequencing and the Polymerase Chain Reaction. They can add at least 100 nucleotides per second to a growing chain at the optimum temperature, but work very poorly, if at all, at 50°C and lower. The latest of these (Q5) is particularly accurate (see the next slide).



A comparison of the fidelity (accuracy) of several thermostable DNA polymerases. The number at the top of each column is the accuracy relative to Taq polymerase (set at '1', but not shown on this graph. Platinum Taq HiFi is an improved recombinant version (6x) of the original Taq Pol.).

The first discovered – and least accurate – is Taq polymerase. The last column in the figure has a slightly more accurate version, namely Platinum Taq HiFi polymerase. Taq Pols are mostly used for amplifications for which total accuracy is not required, for example, general cloning and sequencing. A recently introduced recombinant Taq polymerase is more accurate and robust than the original version.

Taq DNA polymerase has terminal transferase activity that results in the addition of a single nucleotide (adenosine, A) at 3'-ends of the extended/amplified DNA template strands. Only Taq has this facility and it is used in T/A cloning of linear products into a specially prepared vector plasmid (see further on).



Most DNA polymerases are capable of inserting thousands of nucleotides per minute against a single stranded DNA template. This extension polymerising reaction requires great accuracy or fidelity, of the order of a single error in 100,000 nucleotides or more. These enzymes are able to achieve the required high level of accuracy/fidelity because they are multifunctional with an oligomeric structure that has two different enzymes in a single oligomeric arrangement. The two activities are: polymerase in the 3’ - 5’ direction, and exonuclease in the 3’ - 5’ direction. This exonuclease is the proofreading enzyme and is illustrated in the figure above (middle panel) in which an incorrectly inserted nucleotide is removed and replaced with the correct base/nucleotide. This is not unlike the backspace correcting key on computer keyboards, which originated incidentally in now redundant typewriters.

In the previous slide, a number of thermostable polymerases are listed with comparisons of accuracy relative to Taq Polymerase (Taq Pol), the first thermostable polymerase isolated, from *Thermus aquaticus* in 1976. Taq Pol is not very accurate with an error rate of about 1 in 9,000 bases. Most other polymerases are from 6 to 300 times more accurate then Taq Pol. The simple reason for this difference is that Taq Pol doesn’t have the proofreading exonuclease function, whereas most other DNA polymerases have the exonuclease.

Typical thermal cyclers. Most modern instruments have blocks with up to 48 tubes capacity.

The PCR steps and cycling protocol can be programmed a single time and stored in the instruments for future use – and dozens of protocols may be stored in this way for different reactions and for multiple users.

Note the thermostated lids for 'hot starts'. A hot start means that as soon as the lid is closed and the cycling commenced, the lid is heated first to about 100°C, then the reaction contents at the bottom of the tube in the heating block are heated rapidly so that lower temperature 'mispriming' is eliminated. Mispriming is a situation in which a primer may bind to a different sequence on the template that is not completely complementary and this will generate by-products.

The second advantage of using the heated lid is to force any evaporated water from the reaction back down into the reaction mix at the bottom of the tube so that the concentration of the reagents is kept steady. The first PCR thermal cyclers did not have these heated lids and the evaporation problem was solved by overlaying the reaction liquid with an equal amount of mineral oil that prevented water vapour from forming above the reaction mixture.

Cycles required

Number of target molecules	Number of cycles required
3×10^5	~20
1×10^4	~25
1×10^3	~30
50	~40

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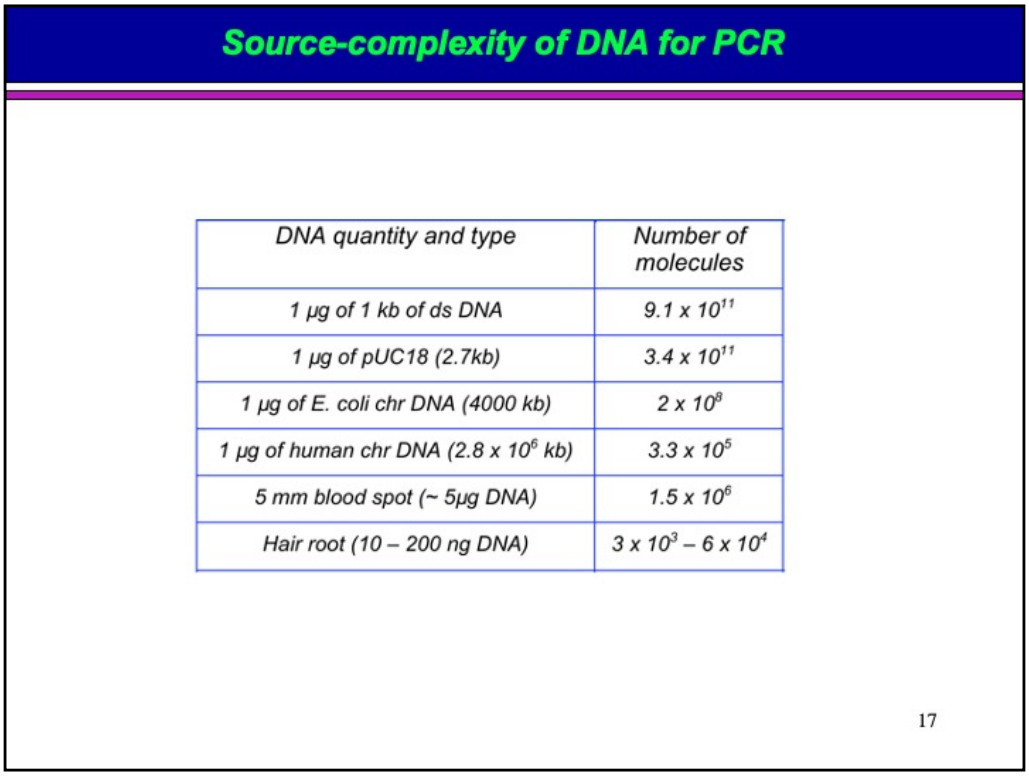
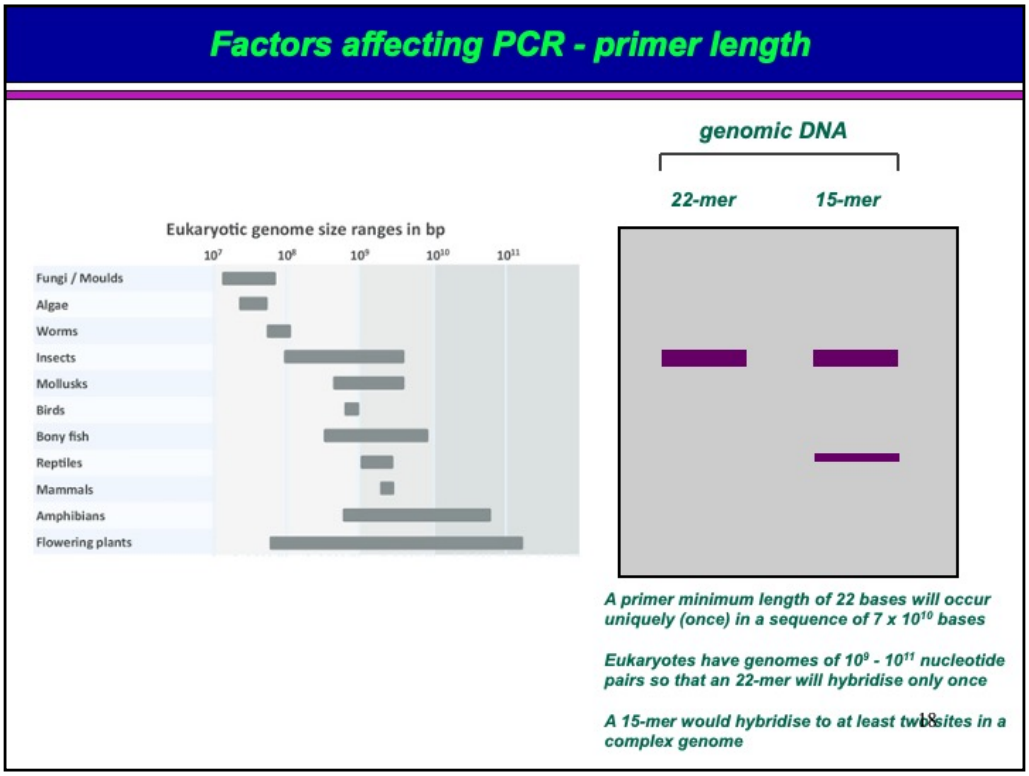
A simple ready reckoner table of cycle number versus initial target molecule number, in order to maximize yield.



A table of calculated molecule numbers versus source template DNA.

Most of these samples have the target DNA in large excess, so that in practice, much less DNA is used. The last two slides give precise numbers, but in practice, PCR amplifications are largely arbitrary, using nanogram (ng) amounts of DNA template and a large (molar) excess of primers.

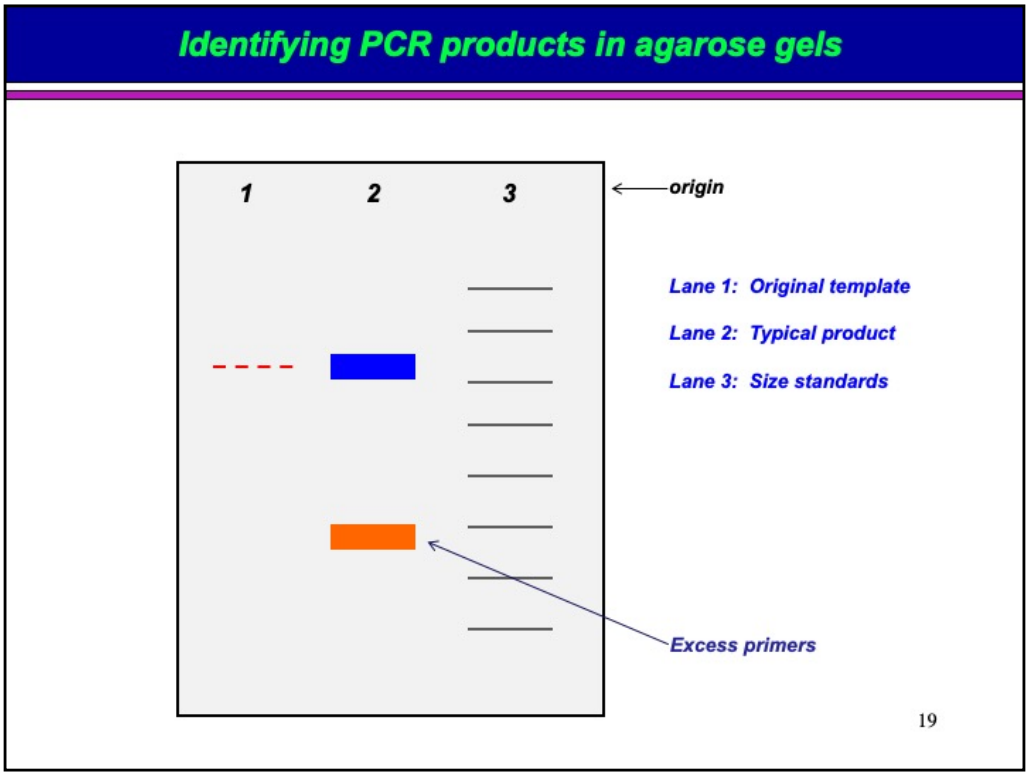
The overarching idea, in most cases, is to generate an amplified product that will be visible in an agarose gel, for downstream analysis, identification, or cloning. To give you an idea of the practical outcome, if you start with 10-20 ng of DNA template, which would not be visible in a stained agarose gel, then amplify the template for 20 to 30 cycles, there will be so much product (of the order of several to many µg) that as little as 5% of the product will be easily visible in a stained gel.



This slide demonstrates the need to calculate primer length carefully when designing a PCR strategy. The important determinant is the complexity of the target DNA template (left) – which is often an entire genome of a species. The PCR amplification process is so uniquely accurate that the whole DNA genome content is used for the reaction, rather than firstly fractionating or separating smaller templates – which is impossible anyway in many applications (forensics, paleontology, etc).

The gel profile (right) is easily explained by the template complexity and is very similar to the restriction enzyme substrate calculation – how many times will a certain sequence length occur in DNA. A 22-mer (where 'mer' is for 'oligomer') is long enough such that it will be represented only once in a complex genome, whose representative sizes are given in the left panel.

How do we work out these numbers? The same way as we did for restriction endonuclease cutting frequencies, that is, the chance of finding any given short sequence of DNA in a large DNA molecule (plasmid or chromosome) is a factor of the length of the given sequence. For instance, a 6-base sequence will occur at 1 in 4^6 (1 in 4092 nucleotides) and so on for longer sequences such as 1 in 4^{22}, which will occur only once in a complex genome (lane 1 in the gel figure). As we pare back the length of the oligomer primer (from 22 to 15), it turns out that the new shorter primer will be represented twice in the same genome (lane 2 in the gel). Here again, the calculation is the determining factor, with 1 in 4^{15} being the new term.



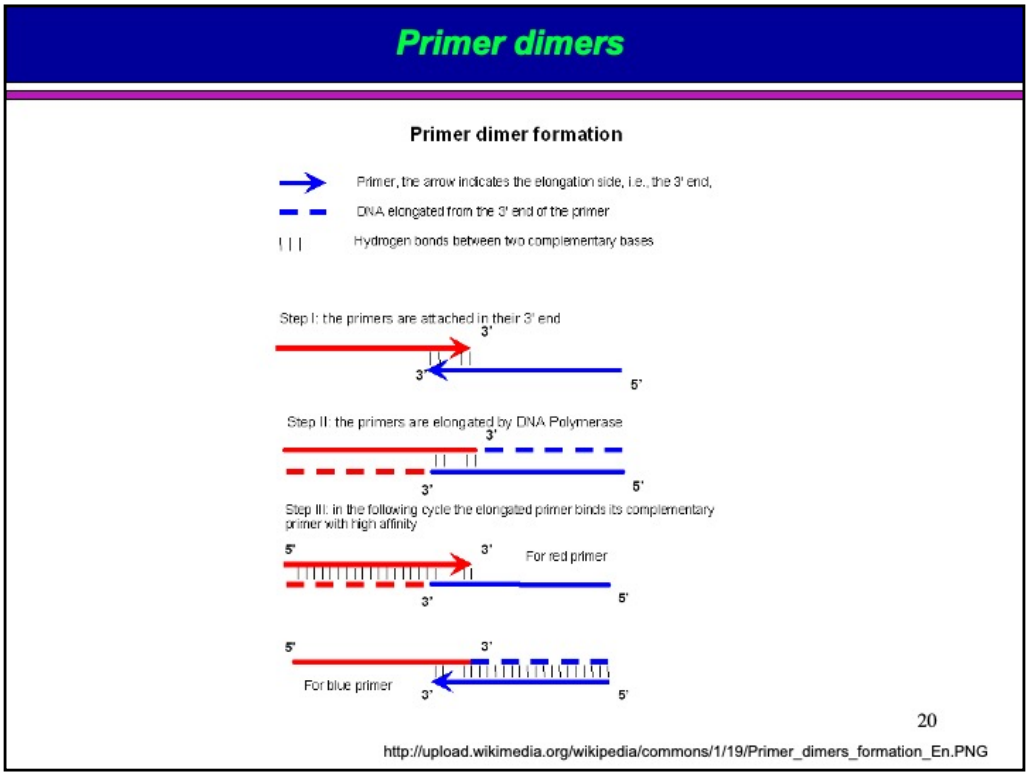
A gel profile of a PCR product for a typical amplification of a linear sequence.

Lane 1 contains a sample of the original template, which may be at the same position as the product band, but only if the template (dashed red line) and product are exactly the same length, which is rare, but is shown here simply to compare product vs template bands. Often, complex genomic DNA (millions of bases and multiple chromosomes) is used to amplify a 'signature' short sequence of only a few hundred bases, so the only 'control' template for comparison would be a faint band of genomic DNA higher up in the gel; and a bright, broad, but shorter amplified band lower down in the gel lane.

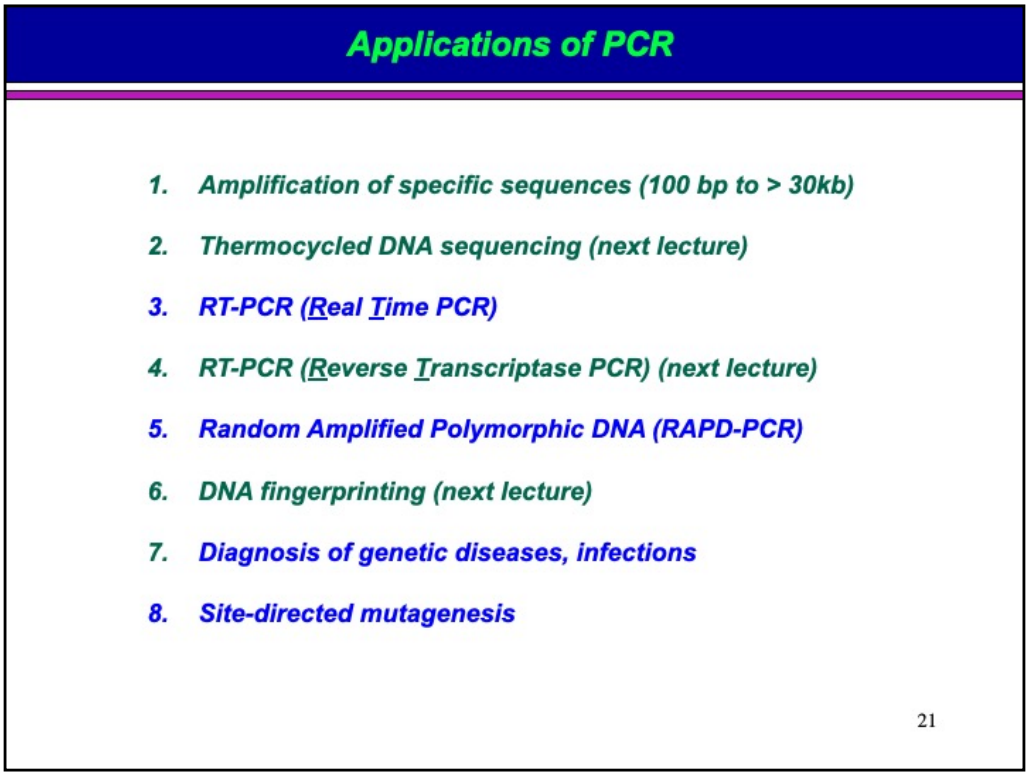
Whether the size of the PCR product is known (by primer design and the annealing positions or size of the predicted product) or unknown, a new very bright product band should appear in the gel (broad blue band), at a position that allows its mass to be confirmed by using size markers in the gel (lane 3). Despite this, there is a PCR 'caveat' that decrees that the product must be confirmed to be precisely the right one and this is usually done by DNA sequencing or restriction mapping. This proof is required, as mispriming (though supposedly rare) can generate a false product.

Lane 2 shows only about 5-10% of the amount of product (blue band), as this should be sufficient for identification. The second 'product' band (orange) is the excess of primers running lower in the gel.

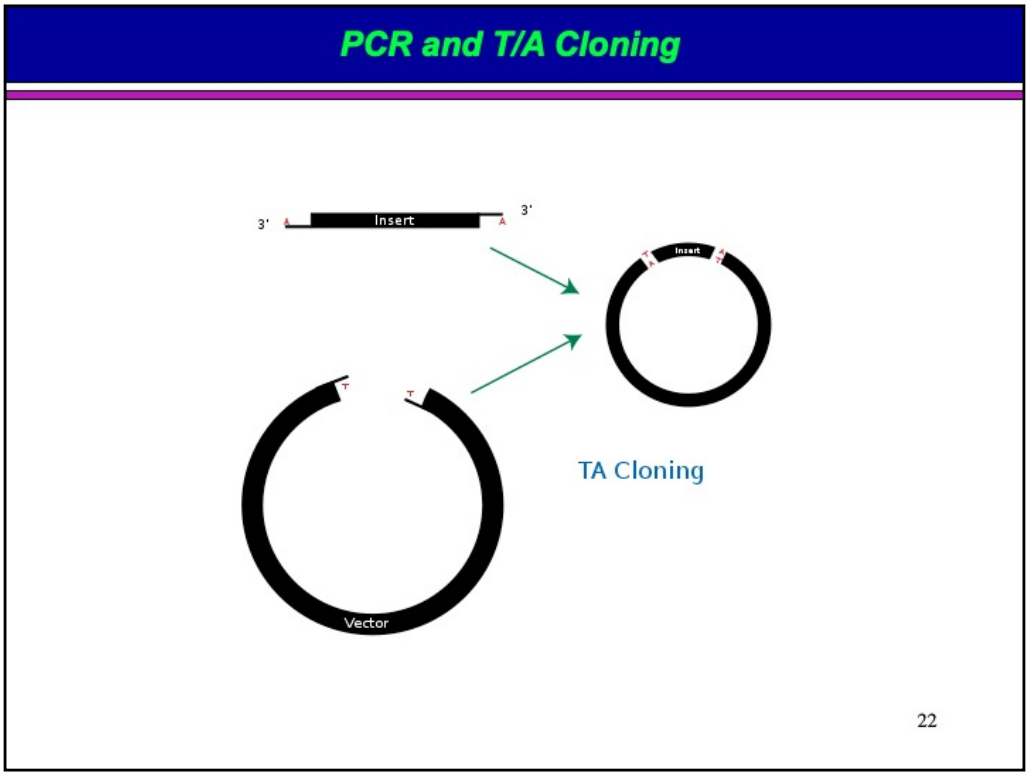
Linear PCR products may be cloned into vectors if they are blunt-ended, or in a T/A cloning vector in Taq polymerase PCRs. Alternatively, restriction enzyme site sequences can be incorporated into the primers so that the product can be easily cloned and sub-cloned, or if desired, linkers or adaptors may be attached to the blunt ended product after the PCR reaction. There is thus a multiplicity of continuations following a PCR experiment.



The partial overlapped complementarity of the primer pair shown here allows self-amplification and depletion of these reactants, leading to a primer-dimer product yield and lesser of the desired PCR product. However, this by-reaction is not always a problem, as enough of the actual expected product is produced in most applications.



There are many applications and strategies for PCR and a few of the more useful ones are listed here. Items in blue will be discussed in this lecture; items in green in the next lecture.



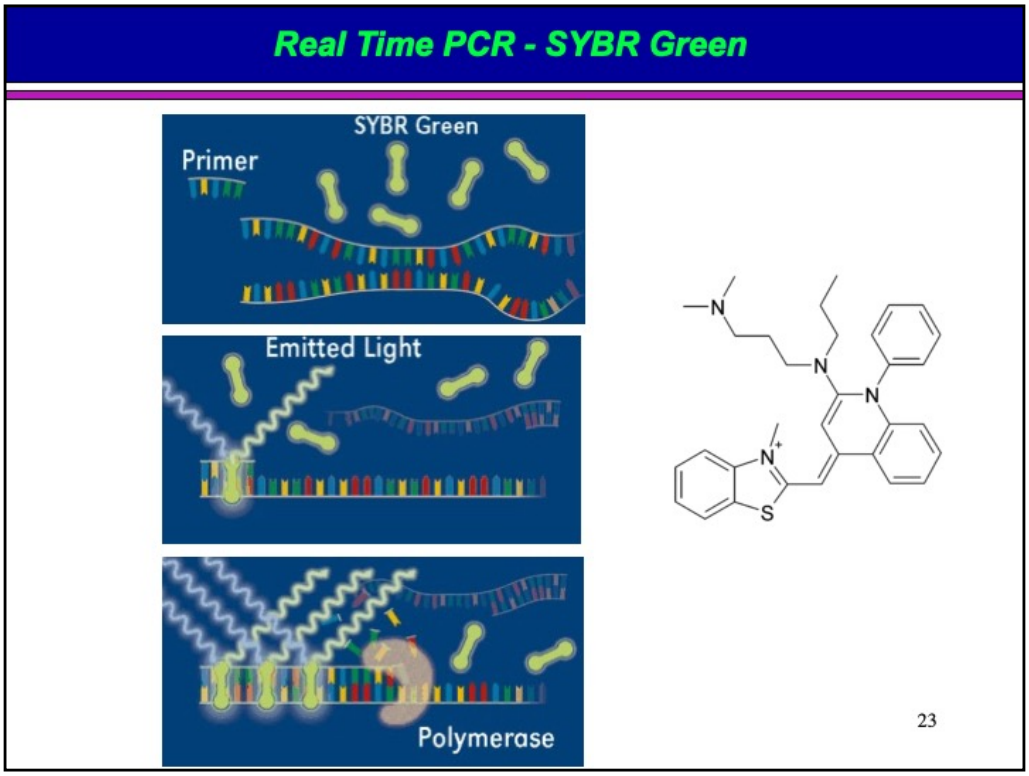
T/A cloning is a special feature of PCR amplifications, and it is governed by two factors.

Firstly, it makes use of the unique activity of Taq polymerase (and no other thermostable DNA polymerase) to add a single deoxyadenine base ('A') to the 3'-prime ends of the amplified template, as for example depicted by the insert in the drawing above.

The target vector is linearized and cut with a blunt-end restriction enzyme. The blunt-ended and linearised vector is then tailed with dideoxythymidine triphosphate (ddTTP) using the enzyme terminal transferase. This 'tailing' reaction leaves the vector with a single 3'-overhanging thymine residue on each end (large open vector at left). Commercial suppliers sell TA Cloning kits to adapt any given vector. Some companies 'spoil' molecular biologists (at a price of course!) by providing some pre-prepared T/A vectors.

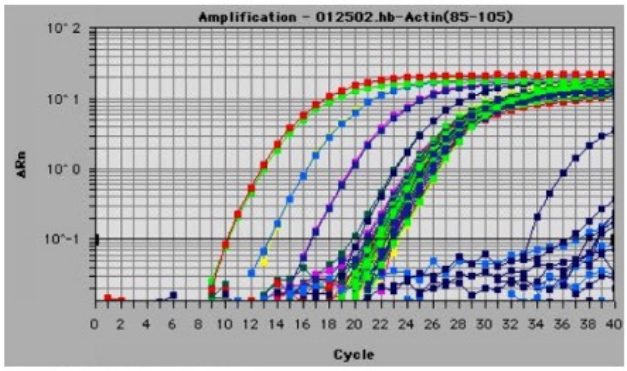
The insert and vector will anneal (as shown above at right) and may be transformed in this open circular format, or may be ligated prior to being transformed.

The second feature of T/A cloning is that it avoids the need for restriction enzymes to prepare and ligate vector and insert.



SYBR Green (shown at right) is a DNA staining dye and a typical fluor used in (Real Time) RT-PCR. It only emits visible green light, when irradiated by UV, when bound to double-stranded DNA (it does not bind to single-stranded DNA). The small amount of SYBR Green that binds to double-stranded primer-templates is measured as background fluorescence only and does not affect the large fluorescence signal that builds with an increasing amount of amplified double-stranded DNA product, to which the SYBE Green binds and fluoresces. This method thus shows the progress and yield of a PCR amplification over the time course of the reaction of up to several hours.

Applications of PCR - Real Time PCR



- *Real Time–PCR (RT-PCR) is a PCR that can be monitored in real time to gauge progress of the reaction evidenced as amplified product.*
- *Requires a dedicated PCR thermal cycler with a UV source and λ detector. Products are identified in real-time by SYBR green fluorescence.*

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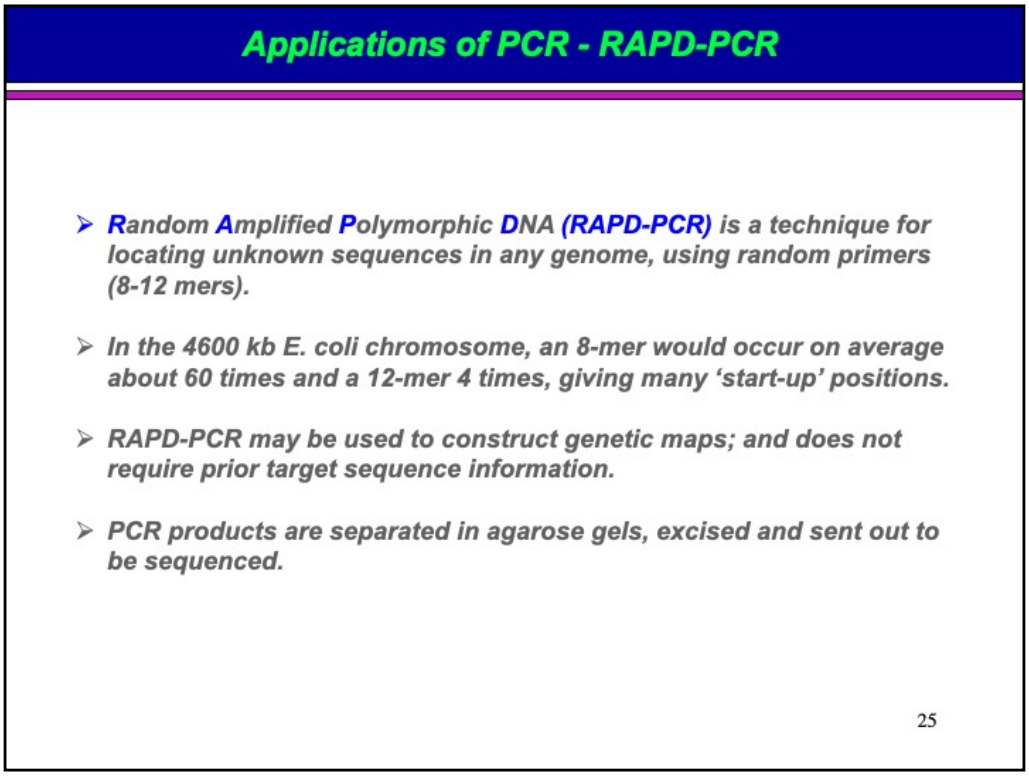
Real Time PCR (RT-PCR) is an adaptation of PCR that is able to monitor the progress of PCR reactions in a special thermal cycler that measures synthesized products by the SYBR green fluorescence derived from extended primers and new double-stranded DNA.

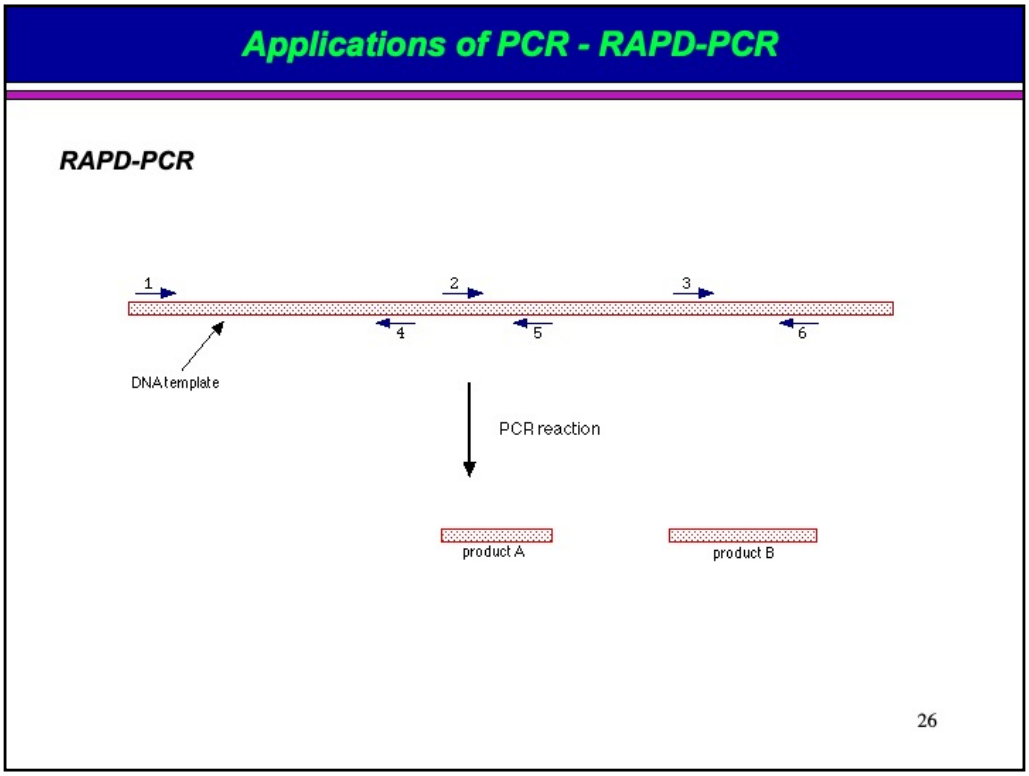
The principle is based on the presence of a "fluor" (SYBR green fluorescent dye) in the reaction mixture. The fluor is only able to emit visible (green) light when attached to double-stranded DNA that is irradiated by UV during the course of the amplification reaction in real-time, hence RT-PCR.

A screen monitor of all of the RT-PCR reactions (in different tubes) displays the real-time measurement of amplified products, by SYBR fluorescence, as in the figure above. The plot height, in log scale, indicates the success or otherwise of the amplification reaction. The yield of product can be quantitated for each sample in every cycle if the starting concentration of template is known.

Do not confuse this method with another type of 'RT-PCR' = Reverse Transcriptase-PCR, which is used to quantify gene expression by copying mRNAs into double-stranded DNA and amplifying these molecules to give differential yields of mRNAs that reflect gene expression levels. This other type of 'RT-PCR' will be introduced in the next lecture.



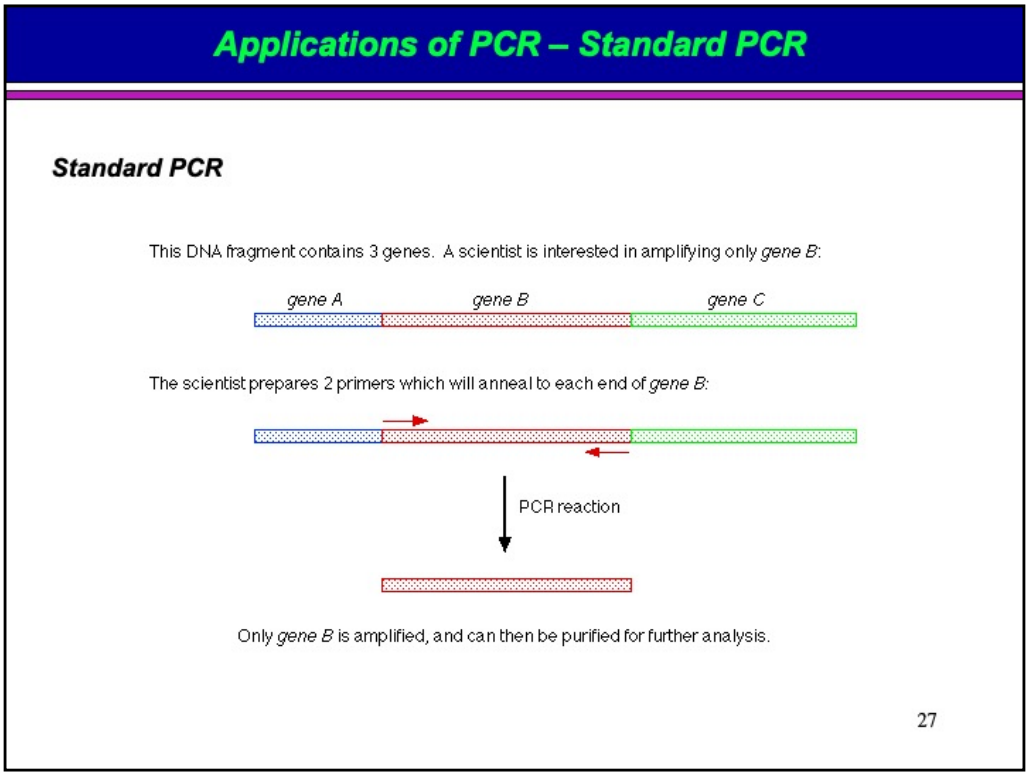
RAPD-PCR makes use of short, random, single-stranded, primers of 8-12 nucleotides (8-12 mers) that will anneal at multiple different positions in chromosomes that have not been sequenced or characterized previously. The PCR extended primers are able to provide amplified products that may be sequenced. See the next two slides to understand the difference between standard and RAPD-PCRs.



In RAPD-PCR, the attachment (annealing) of primers to complementary sequences along a chromosome results in many starting points for extension PCR amplification reactions, as depicted by short numbered arrows.

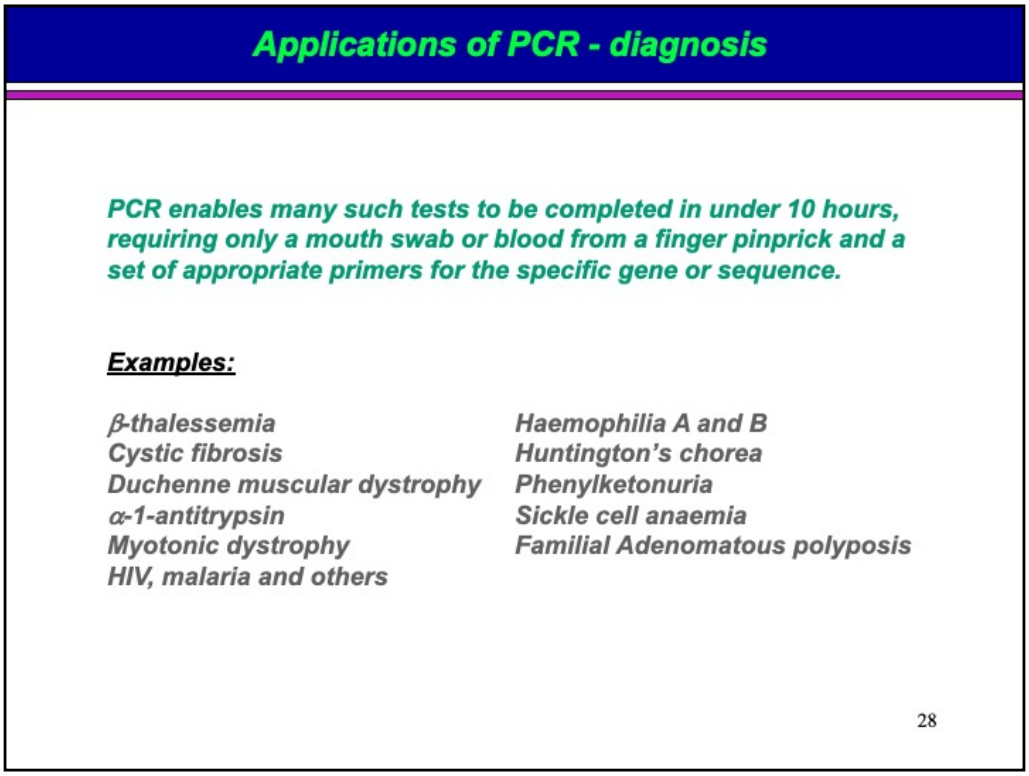
At the end of the reaction, the product mixture is digested with a generic four-base cutting restriction enzyme that will yield double-stranded fragments of manageable size (200 – 1000 base pairs). These fragments are run out in an agarose gel and the bands are excised. The agarose is removed and the purified DNA is sent away for DNA sequencing to provide new DNA sequences from a previously unknown genome (species). There will be gaps in the aligned but discontinuous sequences (know as contigs when they overlap) and these can easily be filled in later by gap-bridging DNA cloning/sequencing (next lecture).

Alternatively, the actual products, of variable length may be sent out for DNA sequencing, after blunt-end cloning into a typical small vector such as pUC18.



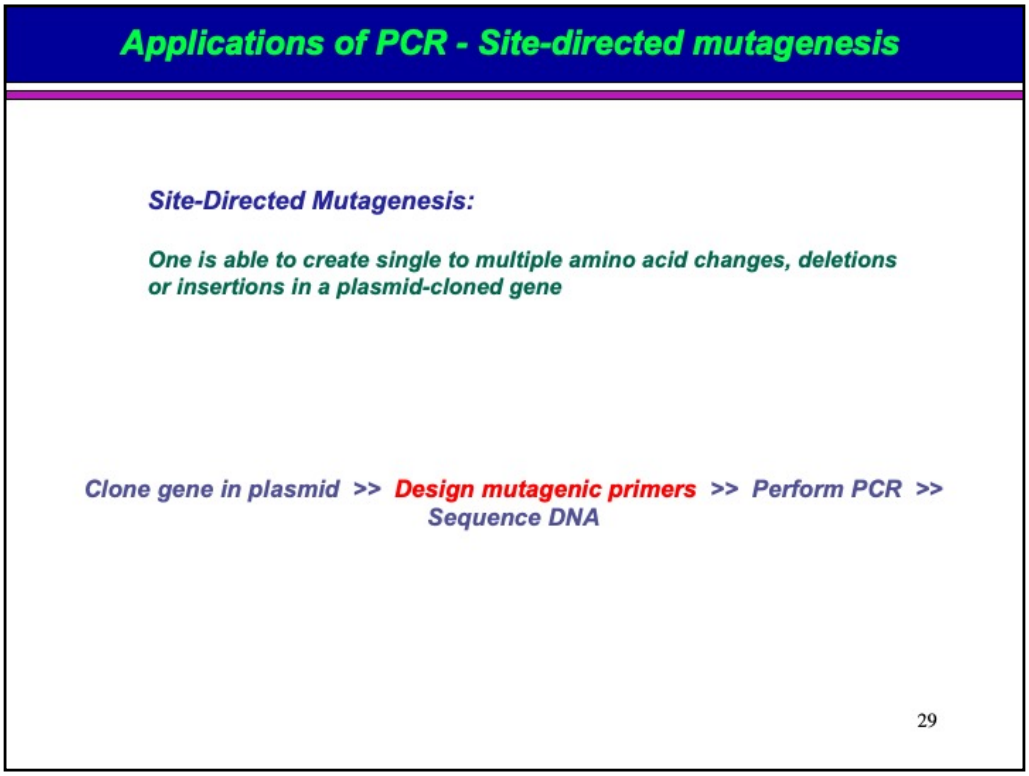
In a standard PCR reaction, specific "bookended" primers will only amplify the template that lies between the inward facing primers (red arrows). In the diagram above, only the central gene (B) is amplified.

Compare this specific PCR amplification to the nonspecific multiple PCR amplifications on the previous slide.



PCR is becoming increasingly prevalent in the diagnosis of genetic or inherited diseases, and infections. The process is simple and rapid and depends only on the source of template and available primers for the PCR reaction. Another 'branch' of this type of 'diagnosis' in to be found in a wealth of applications in forensics, paternity/maternity/family tree testing, ethnicity, paleontology, anthropology and population studies, fossils and archeology including ancient or extinct species and civilisations. For ethnicity, consider the 'Ancestry.com', as advertised.

The sole requirement for any of the medical applications listed here is a specific primer pair for the locus of interest.

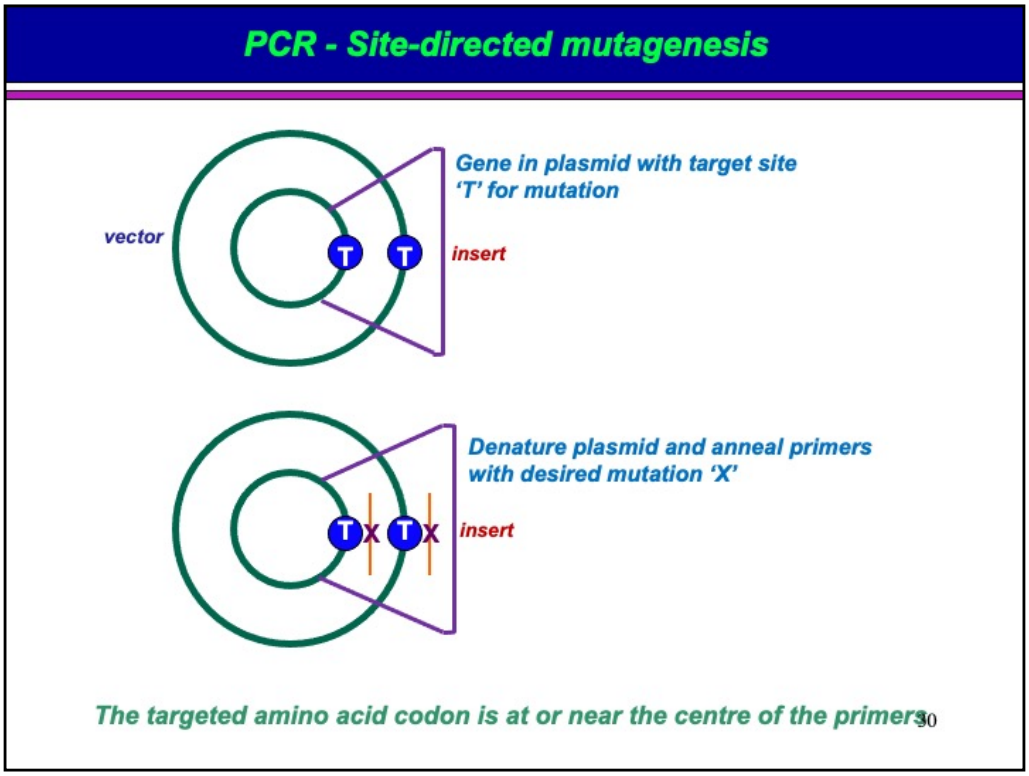


Site-directed mutagenesis is a special application of PCR that is a powerful and rapid method for changing one to several bases in any DNA sequence, but is especially applicable to investigating the importance or otherwise of certain amino acids in any given protein.

The conditions that must be met for the method to work are that the gene must be cloned into a plasmid, special 'overlapped' **mutagenic primers** need to be designed, the PCR reaction is carried out to amplify (replicate) the entire plasmid, and the substituted sequence mutation is identified by DNA sequencing (or in some cases, restriction digestion).

Note the seemingly 'silly' principle of this method that amplifies an entire plasmid of up to as many as 10,000 bases, just to incorporate and mutate (substitute) a single amino acid codon, which requires only one to three base changes, out of up to 10,000. Clearly, the method needs to be precise and efficient. Only very high fidelity thermostable DNA polymerases are used to ensure that the error rate of misincorporated bases is minimal.

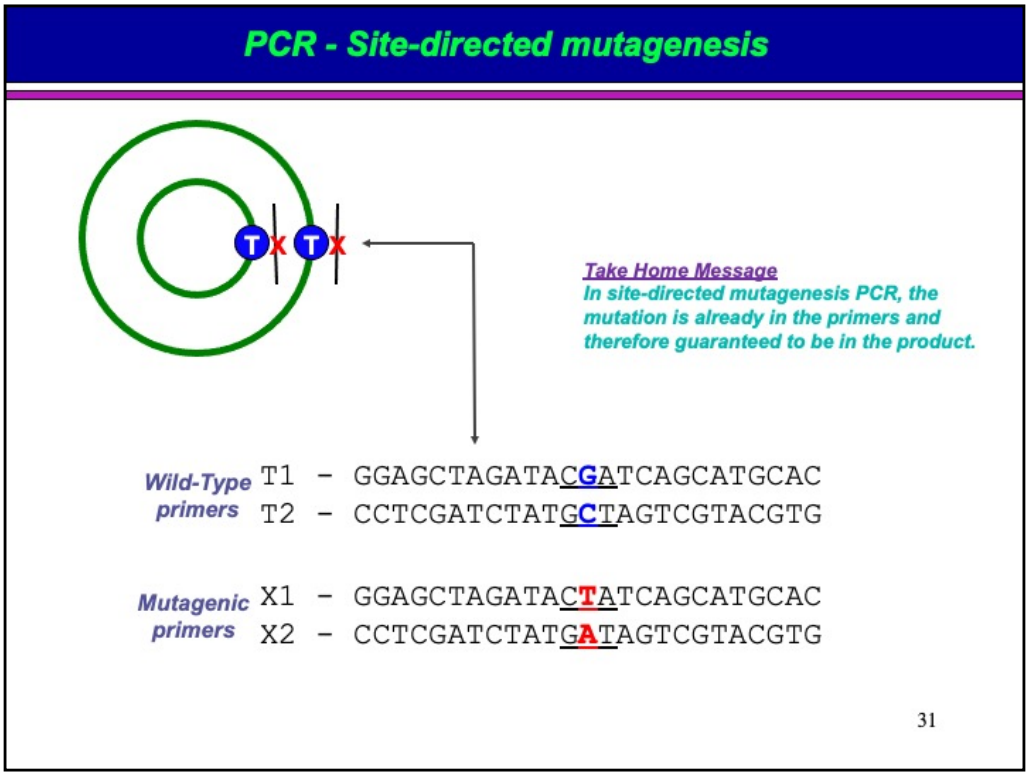
The other and main point to be made about this method of site-directed mutagenesis is that the actual mutation is introduced via the mutagenic primers and the PCR of the entire plasmid template is purely an amplification reaction with the mutation already in place in the primers that are about to be extended around the entire and otherwise unaltered plasmid template.



The primers ('X' in the bottom plasmid figure) are selected to be positioned within the ORF of the cloned gene insert in such a way that the amino acid codon to be substituted (site-mutated) is at or near the middle of the primers ('T'). Why? [Answer will turn up in a later slide and in Expt. 7]

All PCR reaction contents (DNA Polymerase, dNTPs, buffer, plasmid template, primers) are combined in a thin-wall tube and placed into the thermal cycler and the program selected and started. The plasmid is denatured at 95°C (PCR Step 1 of Cycle 1) but the interlocked plasmid strands do not completely separate. The primers are annealed at higher than normal temperatures (~65-68°C) in Step 2. Why? [Answer will turn up in a later slide and in Expt. 7]

If the object is to change (substitute/mutate) a single amino acid, then the primers are designed in such a way that the codon for the target amino acid lies near the middle of longer than normal primers (~25-45 bases).



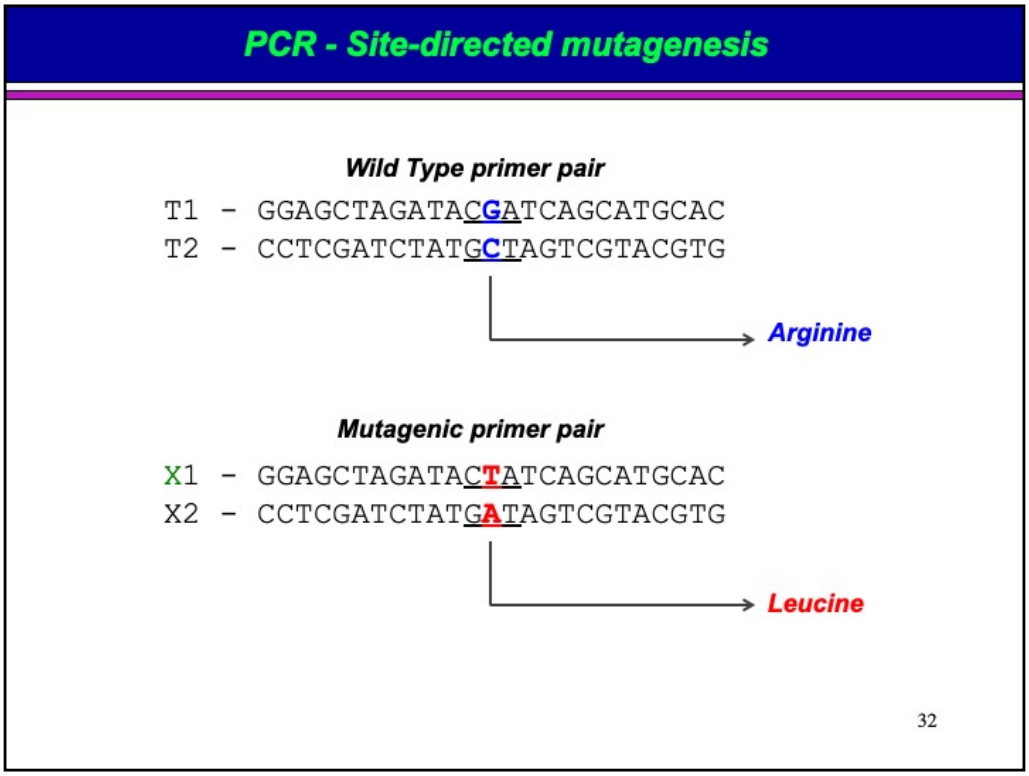
X marks the spot! The X1 and X2 primer pair, are shown here as short, vertical black lines against the plasmid template strands.

The mutagenic primer pair near the bottom of the slide contains the single base pair mutation (red in X1 and X2) to the wild-type base pair (blue in T1 and T2) that indicates the position of the planned **GC** - **TA** substitution in the ORF codon of the cloned gene.

The mutation, depicted here as a single base pair, can be two or all three base pairs of the codon if required; and is around the middle of the primer sequences. The primer design has special constraints (see further on).

In the example above, the site-directed mutagenic codon is underlined. The primers are aligned precisely and face away from one another in opposite directions in the plasmid – that is, they are antiparallel and complementary to one another and to the respective strands to which they anneal by complementarity.

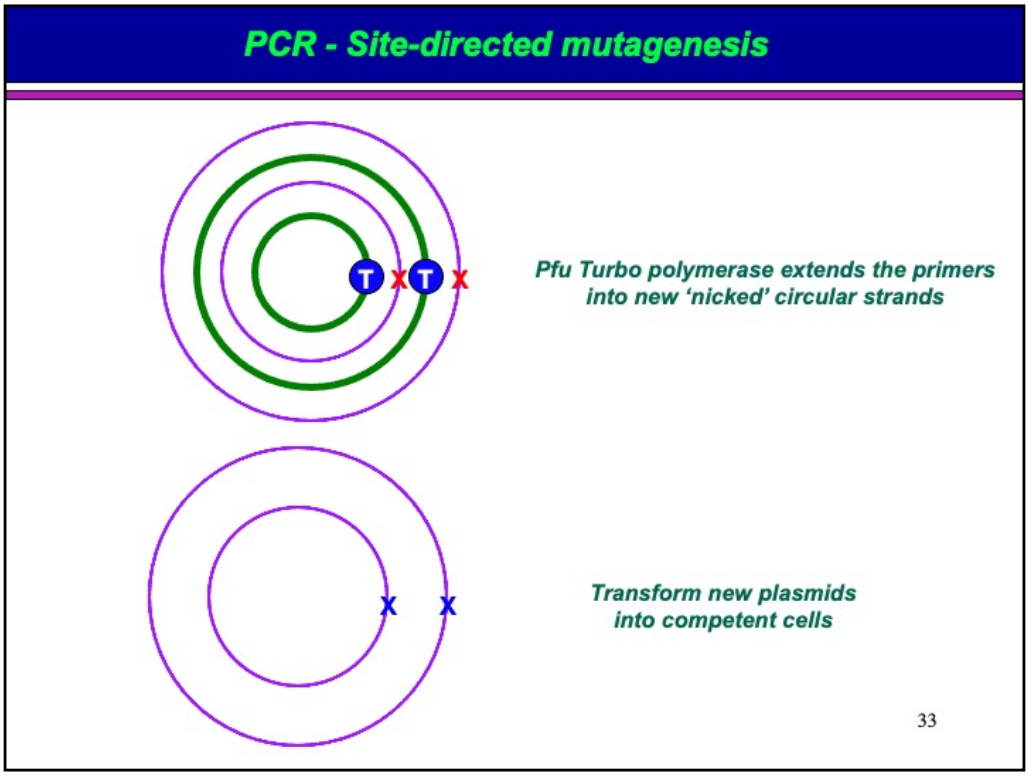
The unusual application of this method of changing codons is such that the entire plasmid sequence will be copied precisely, but with the required change (substitution) already present in the mutagenic primers! An enzyme such as Pfu Turbo DNA polymerase will extend these primers and incorporate them into the newly synthesized plasmid strands.



This slide illustrates the base pair substitution in a section of the ORF of a gene (blue to red) and the resulting amino acid codon change (underlined) in the protein. The underlined target codon is depicted in the wild type sequence and in the corresponding aligned mutagenic primers. A codon for arginine is substituted with a codon for leucine and the single base change that brings about this substitution is highlighted in blue and red text – a GC pair is changed to an TA pair.

The codon change should be near the centre of the mutagenic primers, so that the overlapping 'shoulders' will anneal to the template strands by precise complementarity. Although a simple point, one, two, or all three bases of a targeted codon may be changed to bring about the amino acid substitution in the ORF.

The desired substitution or change might require just a single base (as above), or two, or three bases, depending on the amino acid substitution and the closeness of the codon triple base sequences for the two amino acids. Examples of all three, that is, 1, 2 or 3 base changes in a target codon, with the changes underlined, are: Leucine (CTT) to Proline (CCT); Valine (GTC) to Aspartic Acid GAT); Threonine (ACT) to Arginine (CGC).



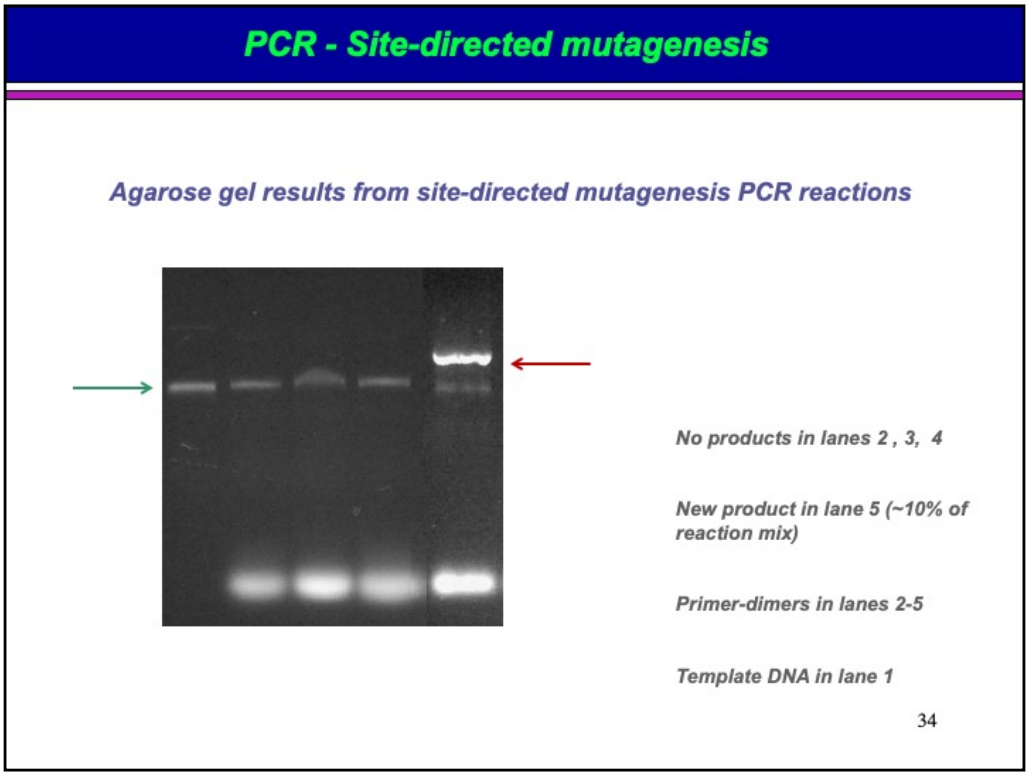
As in all PCR reactions, the extension step is carried out from the primers in the 5'-3' direction for 1 minute per kb of template.

In this particular case, in which the template is an entire plasmid, the last phosphodiester bond will not be joined, so that the product strands (in purple) are never covalently sealed. This is typical of **all** DNA polymerases in all cell types and processes, whether it is DNA repair, or replication, or molecular biology cloning, or PCR.

The 'nicked' (unsealed) double-stranded plasmids will still transform competent cells efficiently in order to propagate the new mutation in fully closed newly replicated plasmids, thanks to DNA ligase, which seals the final phosphodiester bond, and allows the cells to express the mutant protein.

Some questions arise from this special application of PCR:

- What happens to the wild-type template plasmids?
- If the primers are perfectly complementary to one another, rather than bookended at the ends of a DNA template, how does the reaction work when the primers will anneal to one another?
- Why do the substituted base(s) in mutagenic primers need to be near the middle of the primers
- Why do the primers need to be longer than usual - instead of the customary 12-18 bases, they are 25-45 bases long?



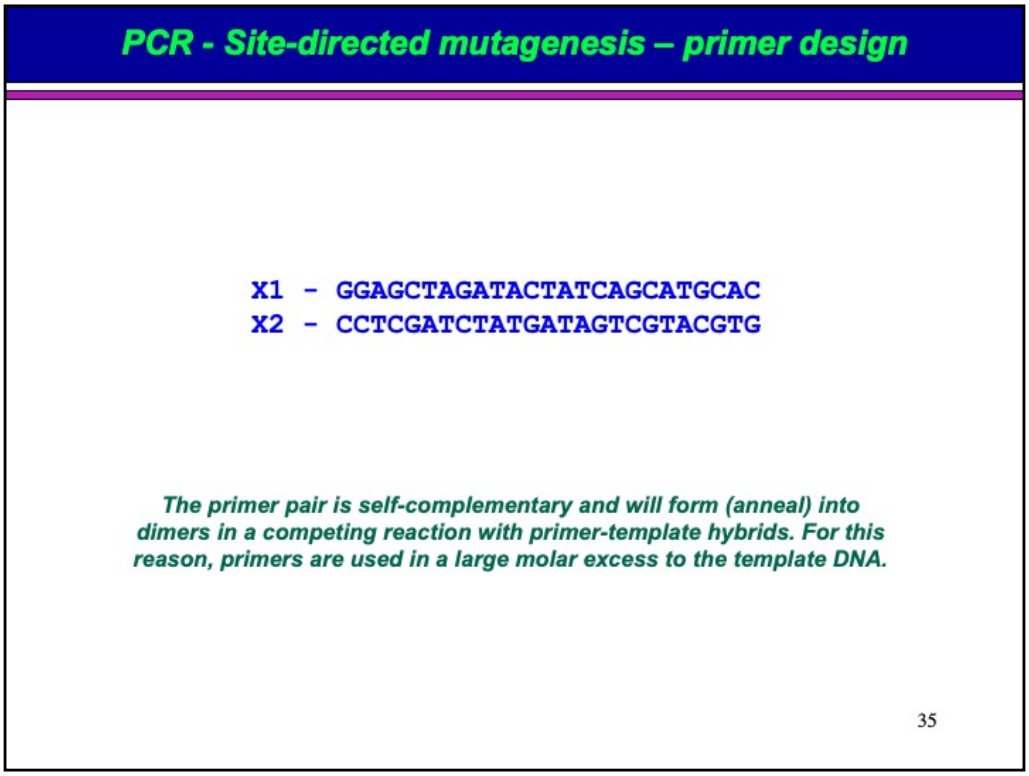
Typical results for a site-directed mutagenesis PCR experiment using a plasmid as template.

In this case, the template plasmid is present at a higher than normal concentration so that its position in the gel is visible (green arrow); and to differentiate its position from that of the more intense product band in the last lane (red arrow).

Note the excess dimers from each of the PCR reactions running near the end of the gel.

Note the position of the product band. Why is it higher up the gel than the template band, given that it is exactly the same size/mass, except for the amino acid substitution?

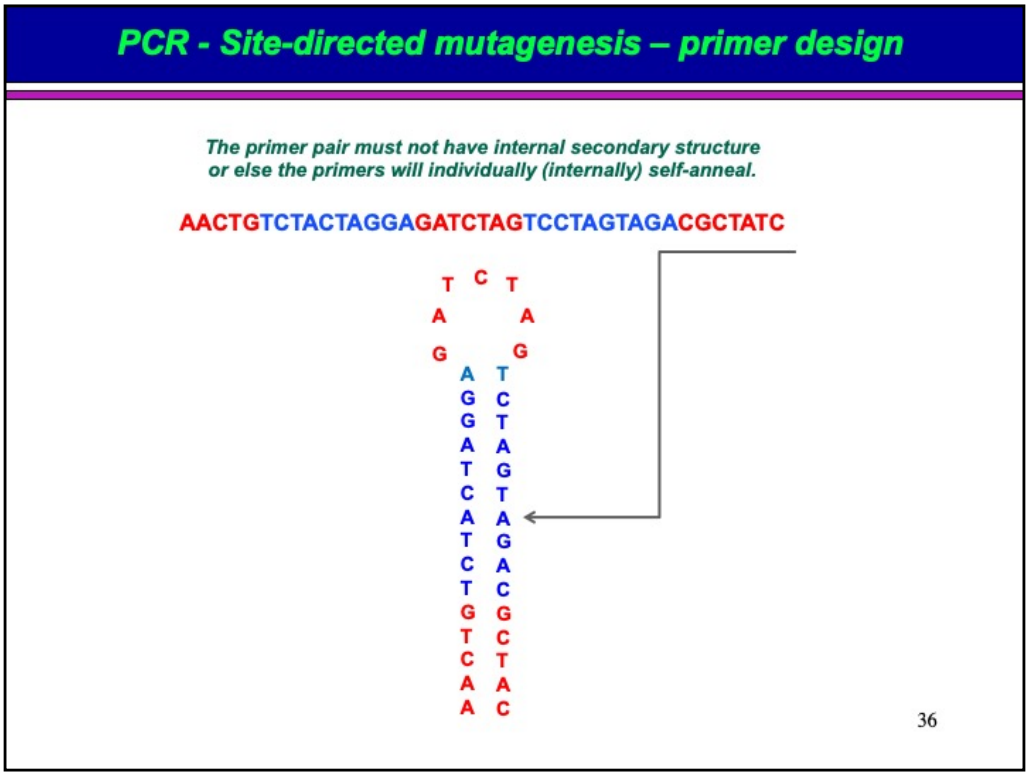
Another thing to be aware of is that only about 5-10% of the product reaction mix is run in the gel. If the amount of total template in the PCR is about 1/10 of what is shown in this gel, and the product band is only 5-10%%, then the difference between the original template and the product is at least 100 times what is shown between the bands in this gel, which makes the amplification PCR very effective.



The primer pair (X1, X2) is always self-complementary in site-directed mutagenesis.

The primers form into dimers spontaneously at the annealing temperature. Note this is not like the occasional overlapped primer-dimers formed in PCR (see earlier slide), as these do not give rise to newly synthesized primer-dimers, but simply self-anneal, thereby reducing their effective concentration in the PCR reaction.

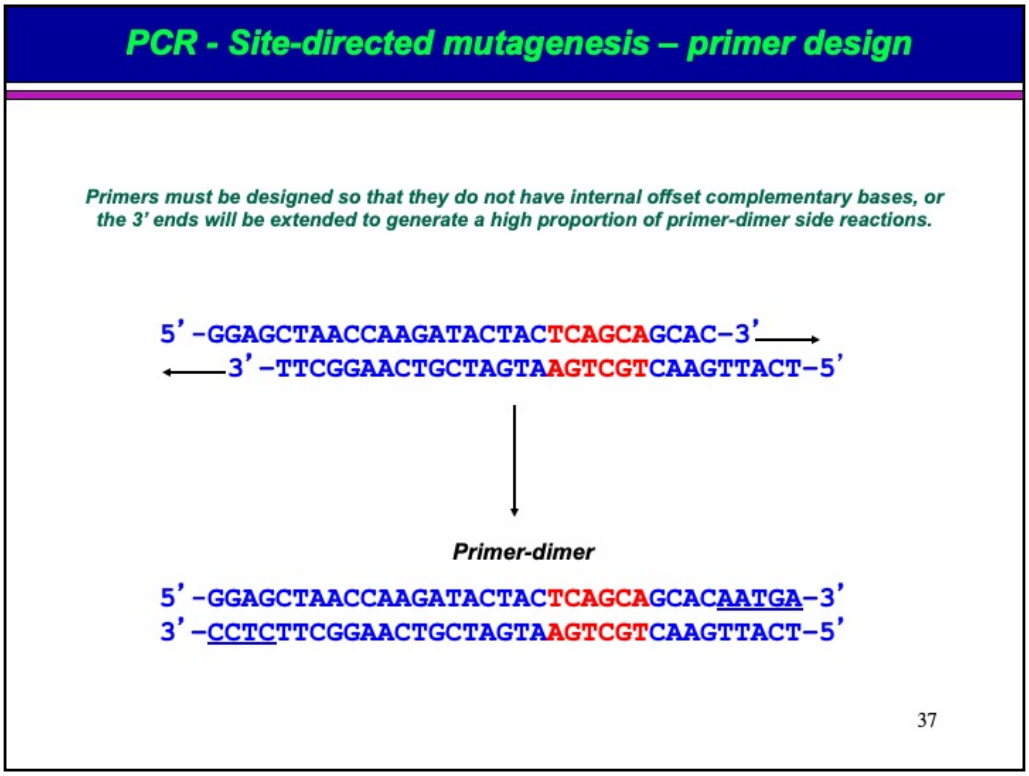
This competing reaction is not a difficult problem and is surmounted by using the primers in large excess. Nevertheless as the amount of new product accumulates, a significant portion of the primers will be 'lost' as they are incorporated into the product, or continue to self-anneal, and the reaction will be less efficient over time. This is one reason, along with the depletion of dNTPs, why the theoretical exponential doubling of product mass per cycle is never achieved in practice.



Internal secondary structure is a major concern in primer design. In the example above, this primer has short, internal complementary sequences (in blue) and will form (anneal) into the stem-loop structure shown, in addition to annealing to the targeted template strand. The same will happen to the complementary primer (not shown here).

This problem may be mild, moderate, or severe and if the latter two, requires that the primer sequences be redesigned – usually by sliding the sequences slightly along the template, or by making the primers longer (they can be a minimum of 20 bases but can be up to 50 bases in length), or by using a combination of these modifications.

Mild to moderate secondary structure might be overcome during the PCR cycling by higher annealing temperatures (~65-68^{o}C Vs ~55^{o}C for standard PCRs), but the product yield may be be low to very low (any amount of product is acceptable in a simple cloning experiment, as you will not need a very high number of amplified copies). Nevertheless, you cannot simply transform the mixture without earlier confirmation of a visible product band (however faint) in the gel, or else much follow up work will be a waste of time if the clone is not there, because the primer pair was mostly unavailable due to the competing internal secondary annealing of each primer; along with the usual complementary dimer formation.



If there is a short, internal complementary sequence in each primer (red bases), then you can expect a high proportion of primer-dimer products, depicted as an intense band low in the agarose gel. The filled in bases of the primer-dimers are underlined in the primer dimer (lower set of sequences). Such a situation will 'dilute' or sequester the primers from the expected annealing to template and the product yield will be very low.

Here again, further 'dilution' or reduction of product will ensue and will also be due to the additive effect of the primer-dimer reaction and the usual self-annealing of the fully complementary primer pair.

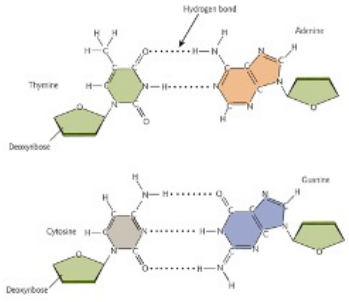
PCR - Site-directed mutagenesis – primer design

A third and important consideration in primer design is the base composition at the 3-prime end of the primer.

5'-GGAGCTAACCAAGATAGCACACCGG-3'

5'-CCTCTTCGGAACTGTCGTCATTTGT-3'

5'-GGAGCTAACCAAGATACTACTATTA-3'



Why aren't the 5'-ends of the primers as important as the 3'-ends?

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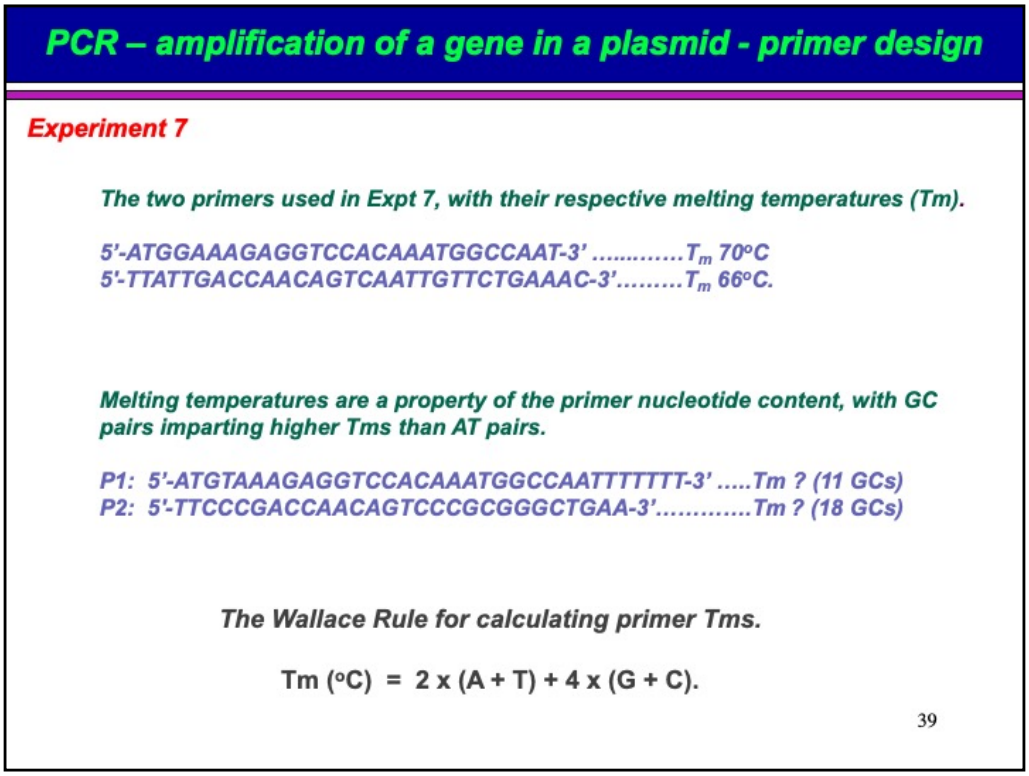
The order of priming efficiency decreases from top to bottom, with only the first (in blue) really suitable for an efficient PCR experiment.

Priming is most efficient – and the yield greatest - when the last one to several bases at the 3-prime end are Gs or Cs, as these, with 3 H-bonds versus only 2 for A or T, depicted in the pairings at the bottom, will anneal more tightly to the template strand and provide a stronger binding efficiency for the DNA polymerase extension reaction.



Examples of melting temperature considerations in primer design.

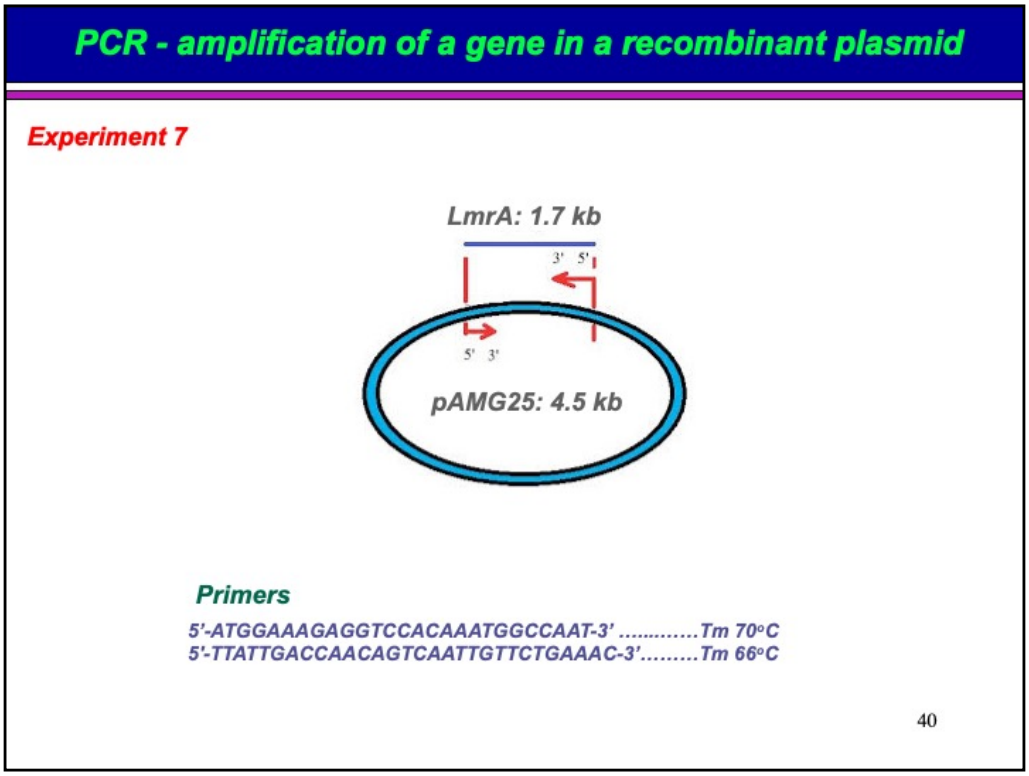
Whilst not always possible, the primer Tms should be as close as possible to one another for efficient annealing to the two template strands. The two factors that most affect Tms are the lengths of the primers and their GC/AT content.

In some cases, a compromise will work, as for example in Expt 7, for which the primer pair is depicted at the top of this slide. Note that the first (top) primer (Tm 70°C) is 3 bases shorter than the second primer (Tm 66°C), yet has a higher melting temperature (Tm). This is because there are 12 GC pairs in the first primer, but only 10 GCs in the second primer. This "rule" will not hold if the two primers are of significantly different lengths, say 10 or more base pairs length difference, because the Tm does go up as the length of the primer increases, but since most primer pairs are of nearly the same lengths, the GC content is the deciding factor in most Tm calculations.

In Expt 7, using the top primer set, a product band is obtained at 65°C, but not at 70°C. Why?

The second primer pair, shown in the middle of this slide, illustrates the GC content 'rule' more clearly, with the number of GC pairs very different (11 for P1 versus 18 for P2). This primer pair is unsuitable for PCR. Why?

A simple formula for calculating primer Tms is the Wallace Rule, shown at the bottom of this slide.

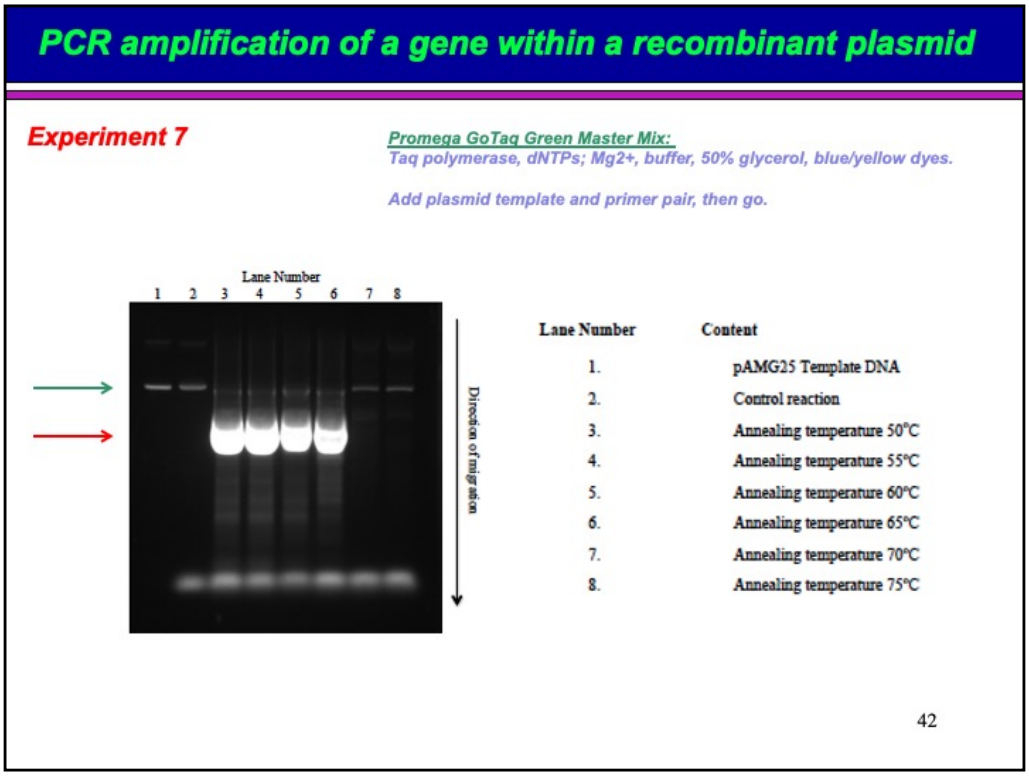


This illustrates the process by which a segment within a plasmid is amplified, using primers that are 'bookended' (facing each other on opposite strands) either side of the segment of interest, in this case the cloned LmrA gene (~1.7 kb), which is copied over and over through the PCR cycling protocol. Note that only the bookended gene and not the plasmid, is copied/amplified. [In site-directed mutagenesis, everything was copied.]

The primers and their melting temperatures (Tms) are shown at the bottom. This plasmid, comprising the original vector pUC18 and the cloned LmrA gene and now known as pAMG25, will be the template for Expt 7, in which the LmrA gene will be amplified.

This slide shows Promega vials of the Go Taq Green Master Mix (GTGMM); and a sample being loaded into a gel lane ('Before', green) and during the gel run ('After', blue and yellow).

Note the approximate size partitioned positions of the blue and yellow dyes, which clearly must be of very different molecular masses.



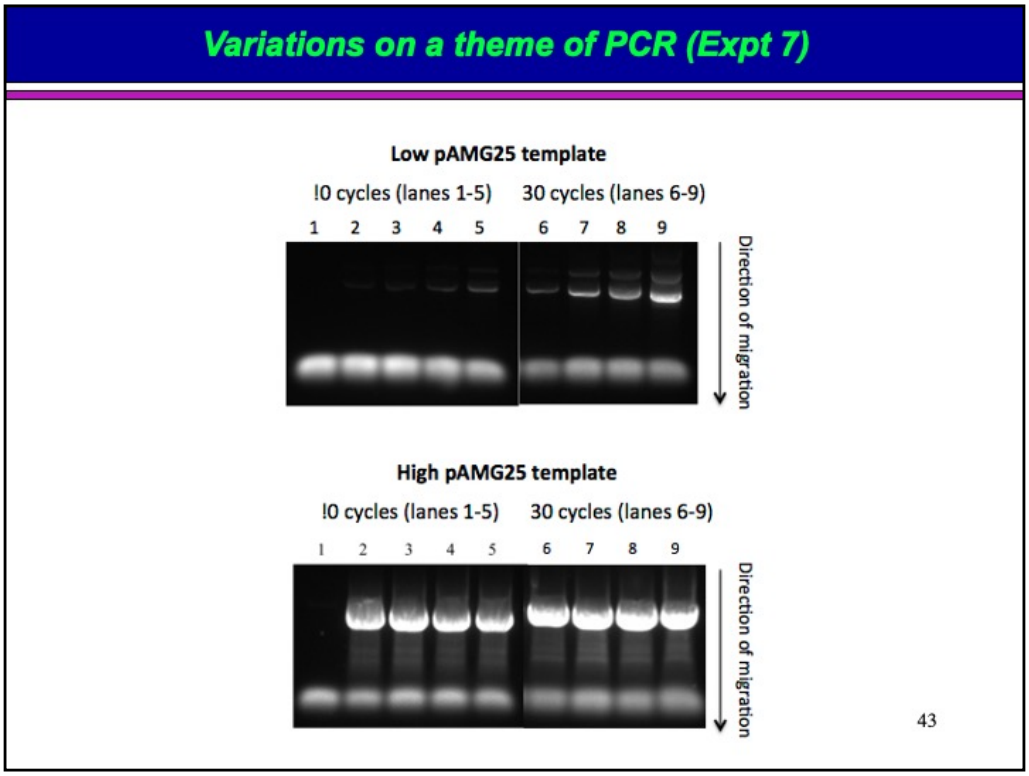
This slide comprises the result sheet for Experiment 7, in which a gene (lmrA ~1.7 kb) within the cloning vector pUC18 (~2.7 kb) is amplified by PCR using the Promega GoTaq Green Master Mix (GTGMM) protocol.

Complete details are in the lab manual, but it is worth mentioning here that the GTGMM (master mix) is ideally suited for PCR. The mix contains Taq Polymerase, reaction buffer, Mg^{2+}, dNTPs, 50% glycerol and two dyes, blue and yellow, that combine to give a green colour to the GTGMM.

The reaction is started once a primer pair and template DNA are added to the master mix and the PCR amplification begins in the thermal cycler. Once the reaction is completed after ~30 cycles, an aliquot (~10%) is added directly to an agarose gel, with no need for sample dye mix, as the GTGMM master mix already contains glycerol to provide the required density for the sample to settle in the wells, and the blue and yellow dyes for monitoring the progress of the sample within the agarose gel.

The contents of each lane in the gel above are given in the legend on the right. Note that the template plasmid band (green arrow) is running nearer the origin than the PCR products (red arrow; bright bands in the central four lanes). The second lane is a control without the master mix, so only the template and primers are visible in this lane. The other lanes are as indicated in the legend on the right.

Note that **no** product is obtained at the two highest annealing temperatures. Why not? (Hint, go back two slides).

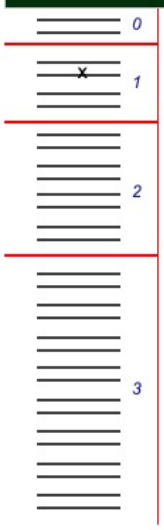


In Experiment 7, the annealing temperature for the primers was the variable parameter. This slide depicts the same experiment, but with the annealing temperature at an ideal and constant 60^{o}C, and the variables being the number of cycles (10 or 30). The mass of template plasmid DNA was also varied.

Lane 1 in the two panels above are controls without template DNA. The low template gels (top panel) have 1, 2.5, 5, and 10 ng of template DNA in lanes 2-5 (10 cycles) and lanes 6-9 (30 cycles). In the lower panel for high template DNA, the masses of template across lanes 2-5 and 6-9 are 10, 25, 50, and 100 ng, respectively.

In a nutshell, what these results show is that 10 cycles is too few for template masses below 10 ng, but fine for masses of 10 ng and higher even for only 10 cycles, and certainly for anything higher up to 30 cycles. However, even the low template products (lanes 6-9 upper panel) yield visible amplified template DNA, which can be used for cloning or sequencing purposes etc.

MCQ L8



The scheme on the left shows the first three doubling cycles in a PCR experiment. Say a base mutation error was introduced in the first doubling cycle, as indicated by the 'x' on one of the new strands in Cycle 1. What percentage of the double-stranded product molecules will contain the mutation in Cycle 3?

(A) 100%.

(B) 50%.

(C) 25%.

(D) 10%.

(E) 0%.

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MCQ L8

The two primers used in Expt 7 are as indicated with their respective melting temperatures.

5'-ATGGAAAGAGGTCCACAAATGGCCAAT-3' .....Tm 70°C
5'-TTATTGACCAACAGTCAATTGTTCTGAAAC-3'.....Tm 66°C

Why is the Tm of the first primer higher (at 70°C) than that of the second one (at 66°C), given that the second is three nucleotides longer than the first?

(A) Tms do not depend upon the number of nucleotides.

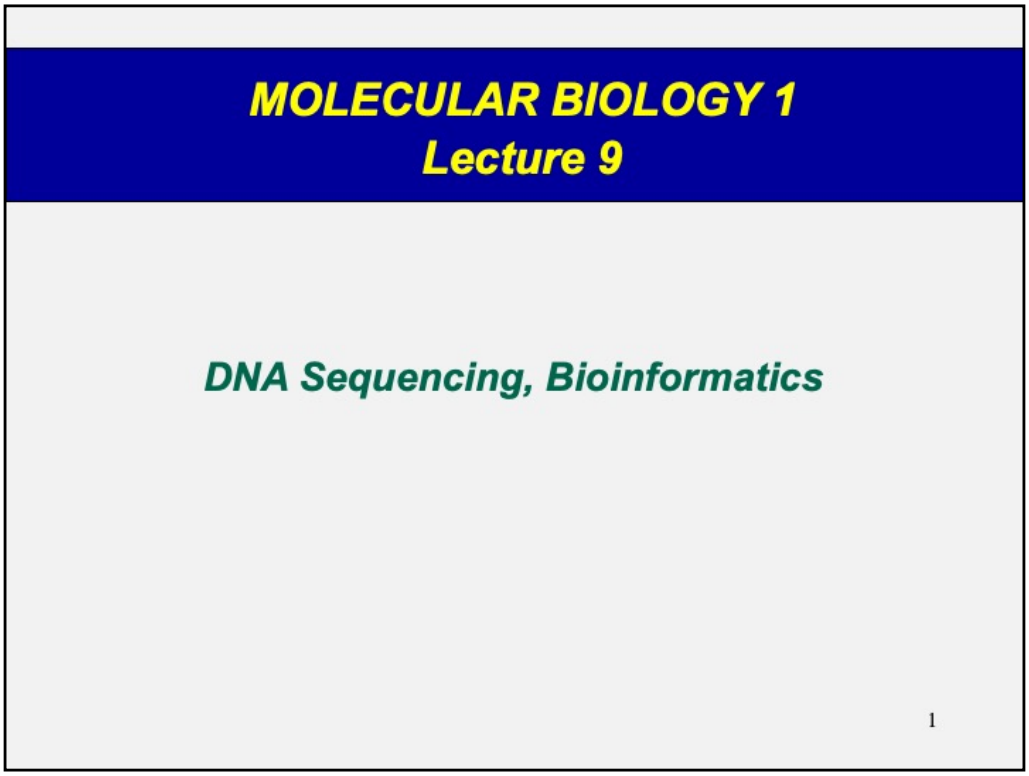
(B) Tms are measured by a formula that does not account for nucleotide numbers.

(C) The first primer has a higher AT content than the second primer.

(D) The second primer has a higher GC content than the first primer.

(E) The first primer has a higher GC content than the second primer.

45



A very important method/tool in Molecular Biology is DNA sequencing, which will be explored briefly in this lecture, going through the basics of the science and the methodologies.

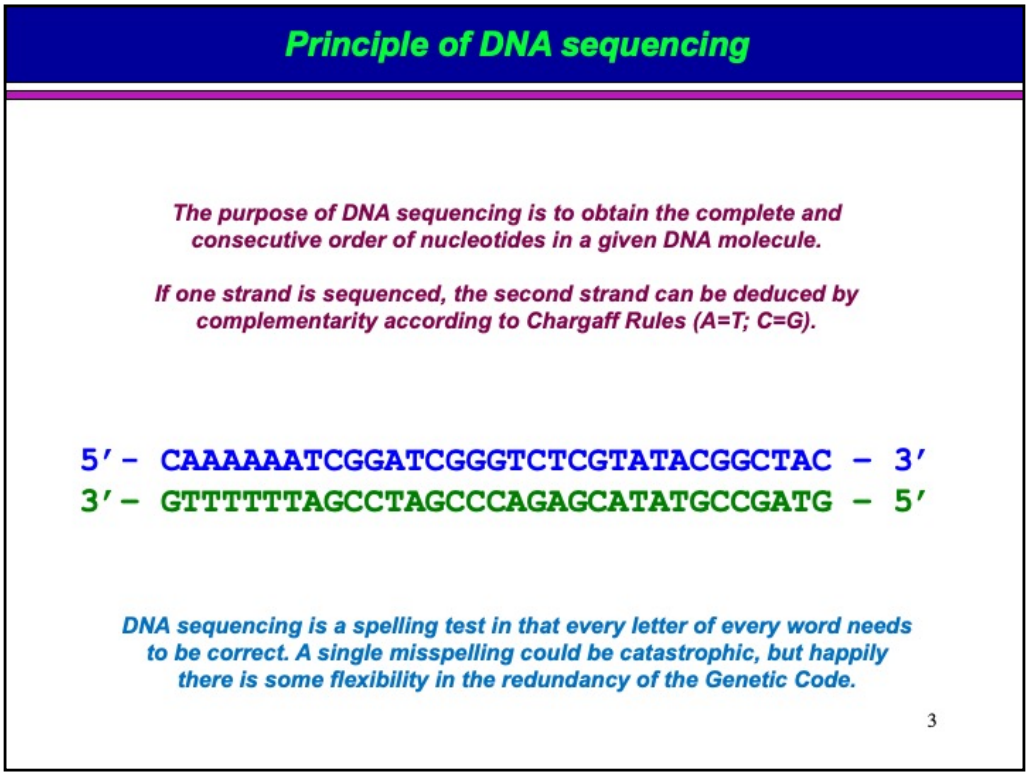
Lecture 9: Learning Objectives

DNA sequencing and Bioinformatics

- ***Principle of DNA sequencing.***
- ***Sanger dideoxy chain terminating sequencing.***
- ***Cycle sequencing. Thermostable DNA polymerases.***
- ***Shotgun sequencing.***
- ***Bioinformatics and associated websites and software.***
- ***DNA fingerprinting.***

2

The learning objectives for this lecture are listed above and will be elaborated in enough detail to encompass the basics of different aspects of DNA sequencing and analyses of the sequences.



In one sense, DNA sequencing can be thought of as reading the "text" of DNA as individual letters/nucleotides in the correct order from the 5-prime end of each template to the 3-prime end.

The principle is based – like DNA replication – on the complementarity of the two antiparallel strands of DNA, essentially as defined by Chargaff's Rules of A opposite T and C opposite G. If the DNA nucleotide sequence of one strand is known (e.g. the top strand in the figure above), then the (antiparallel) nucleotide sequence of the bottom strand may be deduced by complementarity, as depicted in the green sequence.

The natural biological processes of DNA replication, transcription, and repair achieve perfect complementarity in order to preserve the integrity of DNA in the genomes of species; and for that matter, in viruses and plasmids.

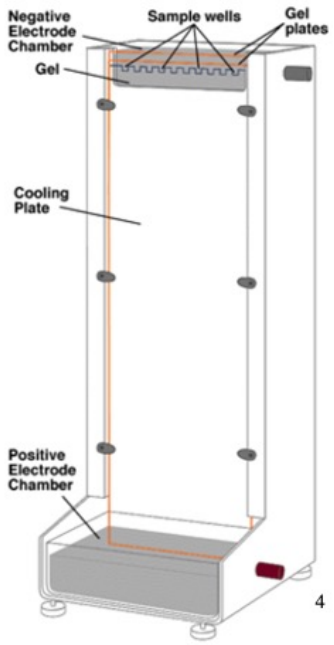
In molecular biology, DNA sequencing and the polymerase Chain Reaction (PCR) achieve the same outcome, that is, the matching of complementary nucleotides to those on the template strands.

You can think of a DNA sequence as a type of spelling test. However long the sequence (word), the two strands must be perfectly complementary ('spelled' correctly). A single base error is a misspelling of the sequence.

Principle of classical DNA sequencing

DNA sequencing reaction mixtures were run in long, thin urea-polyacrylamide gels, which were exposed to X-ray film to identify β-emissions from labelled fragments.

Labelling of fragments is with: α^{32}P-ATP, α^{33}P-ATP or α^{35}S-ATP

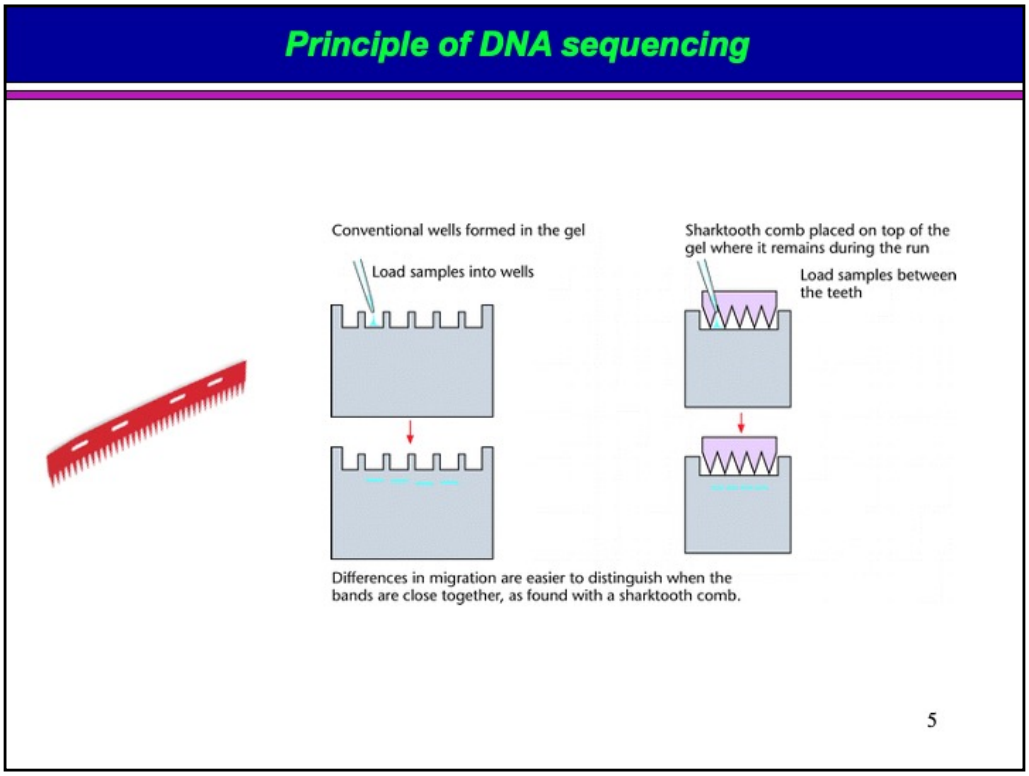


We will look briefly at classical DNA sequencing, namely Sanger Chain Terminating Sequencing, which is still used. Classical sequencing uses radioisotope labels on one only of the four nucleotides, namely ATP, and it can be any of three isotope types that vary in beta emission strength.

Notice the drawing of the rather long (up to 60 cm) vertical apparatus, with electrodes and buffer chambers at either end, sample wells in the top chamber, and a cooling plate at the back of the sandwiched glass plates. The cooling plates disperses most of the heat generated by running these gels at very high voltage (~2000v). The gels are very thin (0.2 to 0.4 mm) and the long gels require high voltages to move the fragments during a reasonable time frame of 2-4 hrs. The heat needs to be dissipated so that the gel bands are not distorted.

The gel itself is made of 4-6% polyacrylamide with 8M urea; and is cast (poured and set) between two glass plates.

The polyacrylamide forms much smaller pores than agarose as the individual fragment bands are quite short and will not resolve in agarose gels. The urea is needed to keep the single-stranded fragments denatured and separated as they run through the gel.

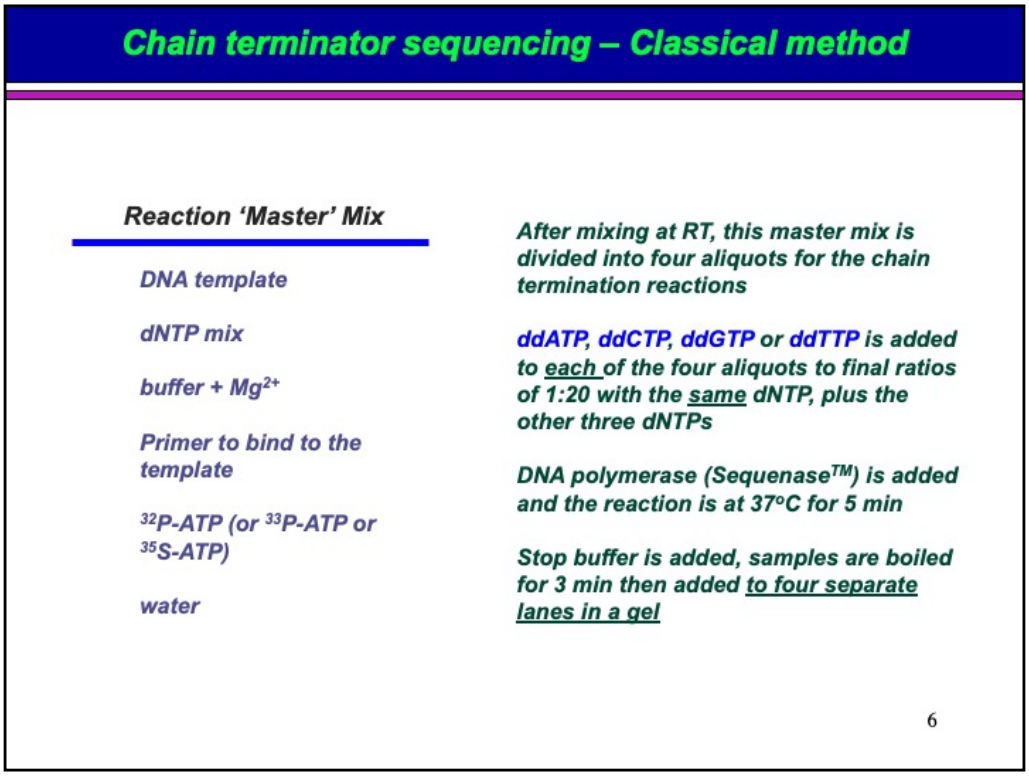


Samples are loaded in 'sharks tooth' comb wells, which may provide as many as 32 to 48 lanes. A typical sharks tooth comb is shown at left.

Note the differences between conventional and DNA sequencing gels.

Why sharks teeth combs? In conventional gels (middle left), the samples are loaded into lanes/wells pre-formed in the gel by the comb, which is removed for sample loadings. In DNA sequencing gels, the 'lanes' are not formed in the gel, but are the spaces between the teeth of the comb, which is inserted just into the top of the gel, where it remains during the loading and run. The samples are therefore in the buffer just on top of the gel, and so the comb must remain in place to allow the samples to enter the gel.

The sharks tooth comb 'effect' is to enable the bands to be right alongside one another with virtually no gaps between the ends, so that their different positions are easier to resolve, as depicted by comparing the light blue bands in conventional and sequencing gels in the bottom two gel drawings.

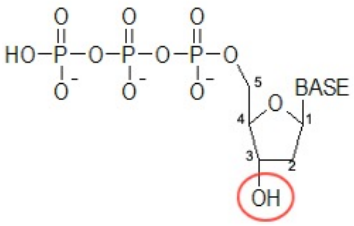


A 'master' reaction mix is shown on the left. It is subdivided into four ddNTP reactions to which a special DNA Polymerase (' DNA Sequenase') is added and reaction carried out for 5 mins at 37°C. In this short time, the enzyme is capable of adding several hundred bases per minute, more than enough to read sequences of up to 1000 bases from the end of the primer.

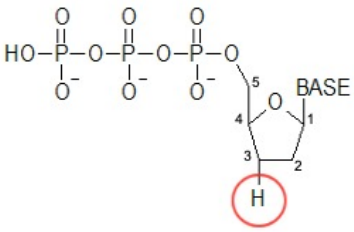
The stop buffer contains the two running dyes, EDTA to chelate Mg^{2+} (stopping the enzyme reaction), the denaturant urea, and sucrose for viscosity for loading the gels.

At the completion of the sequencing reaction, the four mixes are heated for 3 mins at 95°C just before they are added to the gel, in order that the different lengths of extended sequences are detached from the single length template strand. The gels are run at very high voltage (2000v) for 90-120 mins. The high voltage generates heat (the plates and gels become heated to about 60°C) that, together with the 8M urea in the polyacrylamide gel, keeps the extended sequences single-stranded.

Sanger dideoxy chain terminator sequencing



deoxynucleotide triphosphate



dideoxynucleotide triphosphate

The principle of Sanger chain terminating DNA Reaction sequencing is based on the insertion of chain terminating nucleotide bases at every position for that specific nucleotide along template strands.

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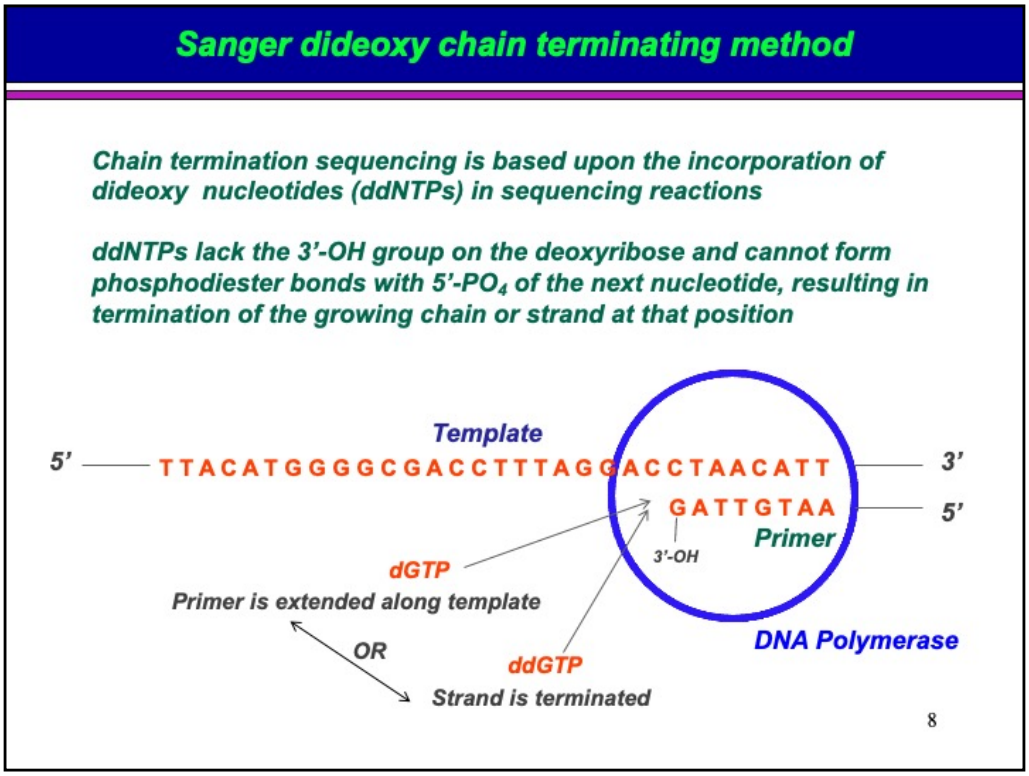
The nucleotide on the left is a deoxynucleotide triphosphate (dNTP). It contains the sugar deoxyribose, without the 2'-carbon hydroxyl group, typical of DNA.

The nucleotide on the right is a **dideoxynucleotide** (ddNTP), as it has lost a second hydroxyl group, namely the one on the 3'-carbon of 2'-deoxyribose (now 2',3'-dideoxyribose). Dideoxynucleotides cannot be added to growing DNA sequences as they cannot form the linking 3'-5' phosphodiester bond between adjacent nucleotides. Thus, chain terminating DNA sequencing – simple and clever! All DNA polymerases require 5'-phosphaes and 3'-hydroxyls as substrates for the formation of the phosphodiester bond.

By including dideoxynucleotides in the mix of dNTPs in DNA sequencing reactions, there will be occasional and regular insertions of dideoxynucleotides into the growing sequence, by virtue of the 5'-phosphates linking to the deoxynucleotide ahead. However, the chain/sequence must terminate at this point because there is no 3'-hydroxyl group on the incorporated/inserted dideoxynucleotide for DNA polymerase to continue the sequence.

Only the nucleotide on the left can be incorporated into a growing DNA chain or sequence, because it contains the vital 3'-hydroxyl group that links to the 5'-phosphate of the next deoxynucleotide in the chain.

Take home message: The principle of Sanger chain terminating DNA sequencing is based on the insertion of chain terminating deoxynucleotide bases at every position for its partner nucleotide along a template sequence. The Chargaff rules still apply – A=T and C=G, even if one of any of these bases is a deoxynucleotide.



The principle of the Sanger dideoxy chain terminating method of DNA sequencing, shown here for the 'G' reaction. It works in two parts:

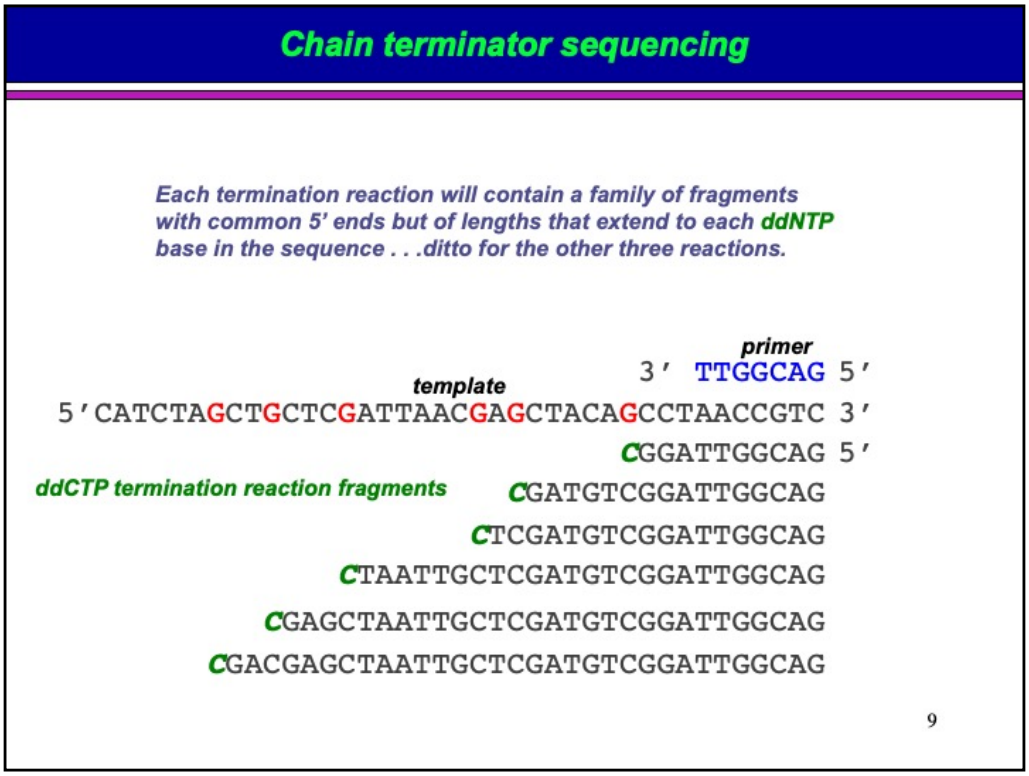
1.The method relies on the regular incorporation of chain-terminating dideoxynucleotides (ddNTPs). The rate of incorporation of ddNTPs is determined by the need to terminate the growing chain at every C on the template, i.e. at every 'G' in the growing sequence out from the primer (see figure above), where a ddGTP is incorporated and which will terminate the chain at that point.

2.The method simultaneously requires that the primer be extended to every subsequent 'C' in the template, with a dGTP being inserted at each 'C'; in adition to the ddGTP chain terminating insertions, described in point 1 above.

The simultaneous effect of termination and extension is achieved by using a 20:1 ratio of dGTP:ddGTP, which enables the chain to be both extended and teriminated along separate (but identical) template strands, out from the primer. A similar situation applies for the other three base reactions.

Note that all four dNTPs are required in each of the four reactions, but only one of the ddNTPs. So, there are 4 different reactions/mixes, comprising the 4 dNTPs and a different ddNTP for each. It is therefore the ddNTP that "names" the base position in the sequence and lane designation in the gel.

Note in this example, the starter primer will be extended along the template strand in the 5-prime to 3-prime direction by DNA polymerase (blue circle). The enzyme only binds to double-stranded DNA which is one reason the primer is required. A second reason for the primer is that it flags the starting point of the DNA to be sequenced.

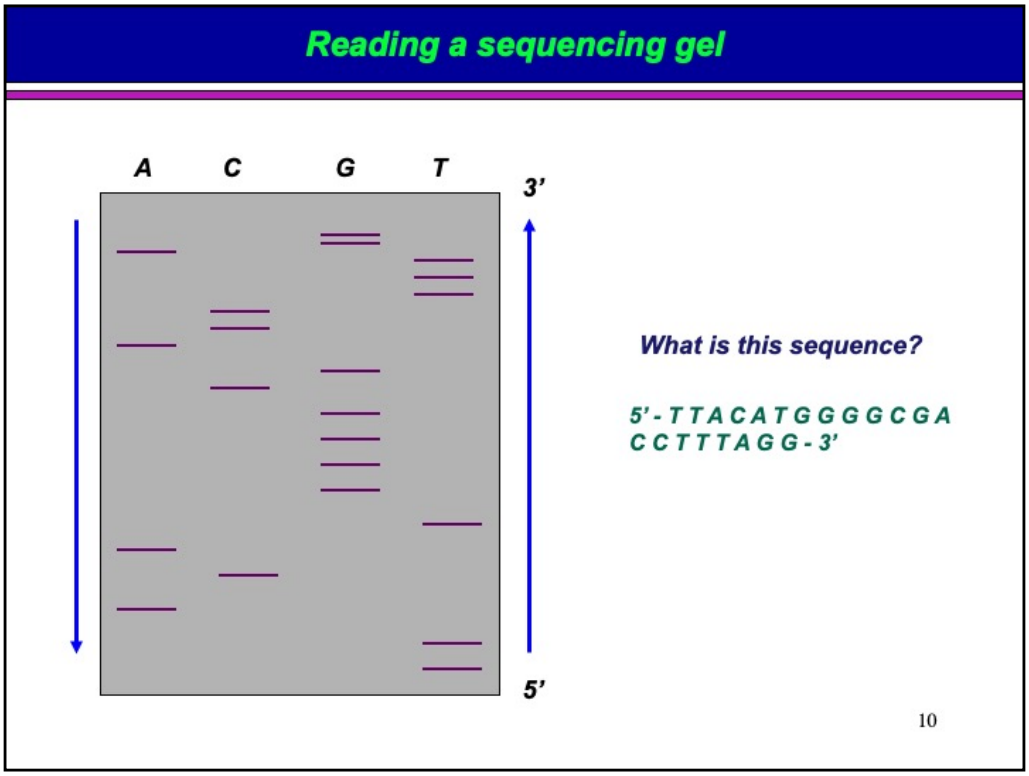


This slide depicts the example of a growing set of ddCTP terminated fragments that indicate the positions of every 'C' in the sequence (opposite a 'G' in the template).

Every extended fragment begins from the common 5-prime end of the primer (shown above the template in blue). An excess of these primers binds in exactly the same position on different template strands, but for simplicity, shown here as a single template strand, under the blue primer.

In each reaction, there are many tens of thousands of identical template strands and therefore a vast number of extended and terminated fragments. The many thousands of terminated fragments of the same size form a sharp band that runs to the same position in the sequencing gel; as do bands for every other terminated sequence at other positions.

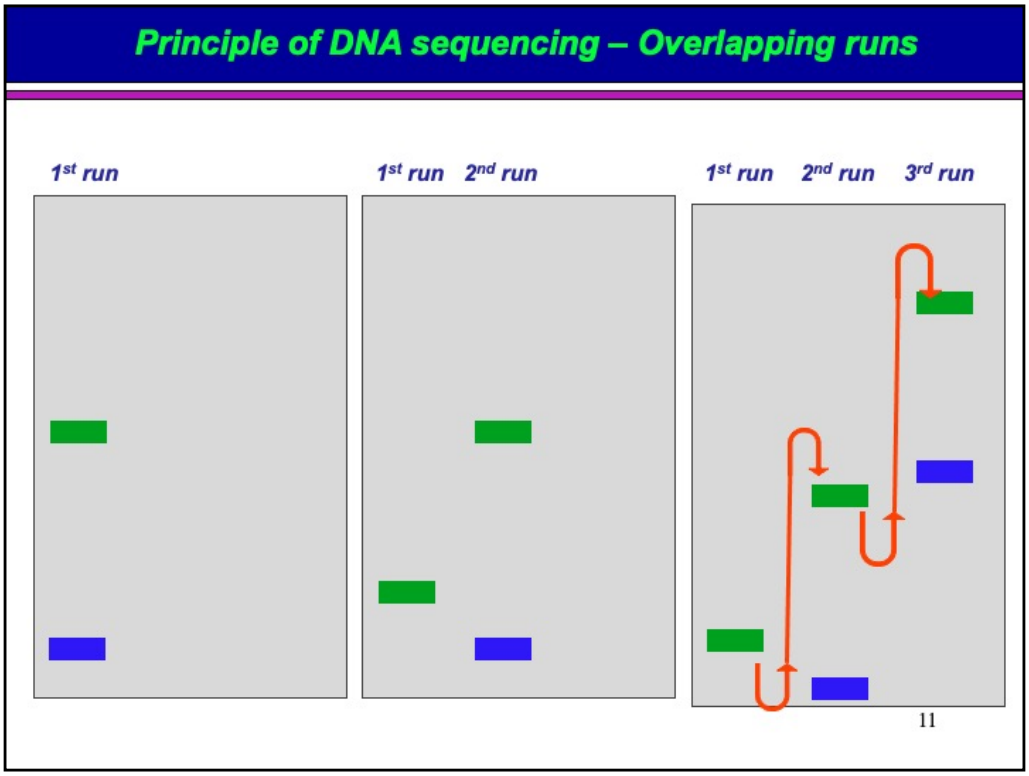
Similar new and extended strands are generated in the other three dideoxy NTP reactions (for A, G, and T). This means that the 4 reactions collectively will contain terminating dideoxy NTPs for every nucleotide along the DNA template. In this way, every nucleotide along the template is 'read' in the correct contiguous sequence, and the sequence order will appear in the gel, an example of which is on the next slide.



How to read a DNA sequence from a set of the four reactions for a single template.

Note the lane designations for the four DNA (dideoxy) bases (from the four ddNTP reactions) and the polarity and order of the sequence fragments in the gel.

Take Home Message: DNA sequencing in gels is so precise that the different bands of extended sequence differ from one another by a single nucleotide, which is why the sequence is able to be read for every consecutive nucleotide (letters in a sentence, as it were).



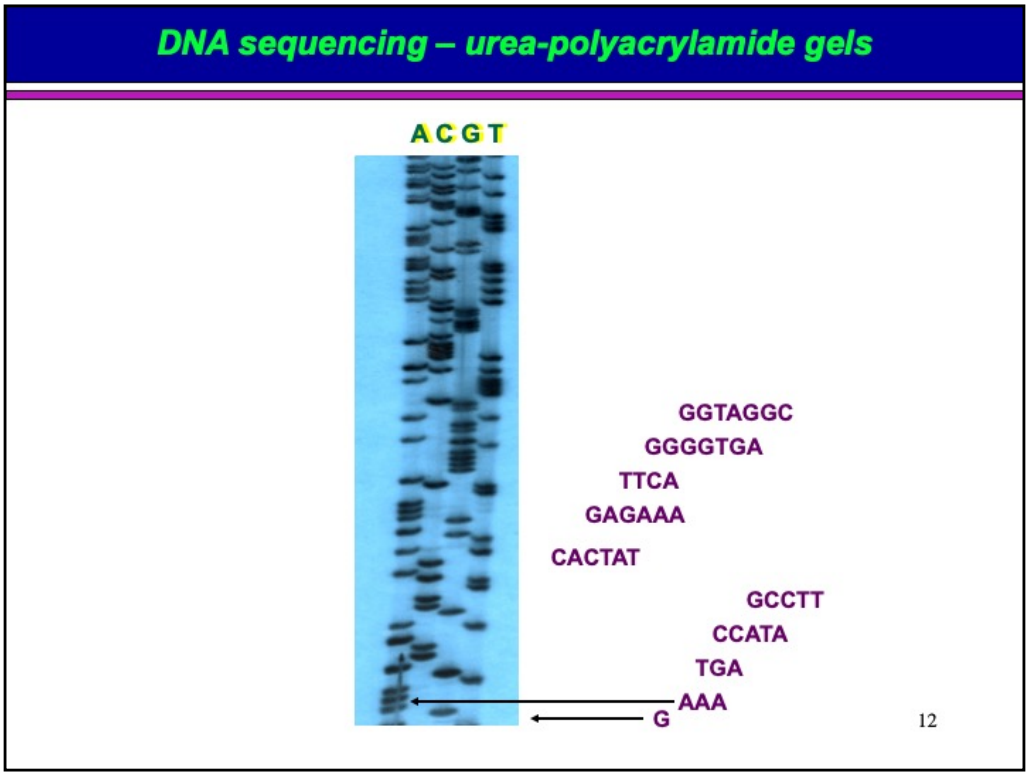
DNA sequencing requires long denaturing gels for resolving small fragments that differ in length by a **single base**; and for reading a significant portion of the sequence. For this reason, agarose cannot be used; instead, 4-6% polyacrylamide gels (plus 8M urea as denaturant) are used, as they have smaller pore sizes that are able to resolve very short DNA fragments.

Though gels are long, base sequences can only be read in about the bottom two-thirds of a gel, chiefly because the longer fragments are more compressed nearer the origin and are not clearly separated enough to resolve.

The lack of resolution of fragment bands in the upper half of the gel is offset by using overlapping runs. This enables read through of much more sequence (as terminated fragments), thus taking advantage of the amount of sequence generated in the reaction mixtures, The same reactions (and therefore sequences) can be loaded twice and even three times across a single gel, in which only four lanes are required for each reactions for A, C, G, and T; as depicted above. It is possible to have as many as 36 lanes in a typical gel, so that up to 9 separate, or 2 or 3 double or triple reaction mixes may be run in one gel.

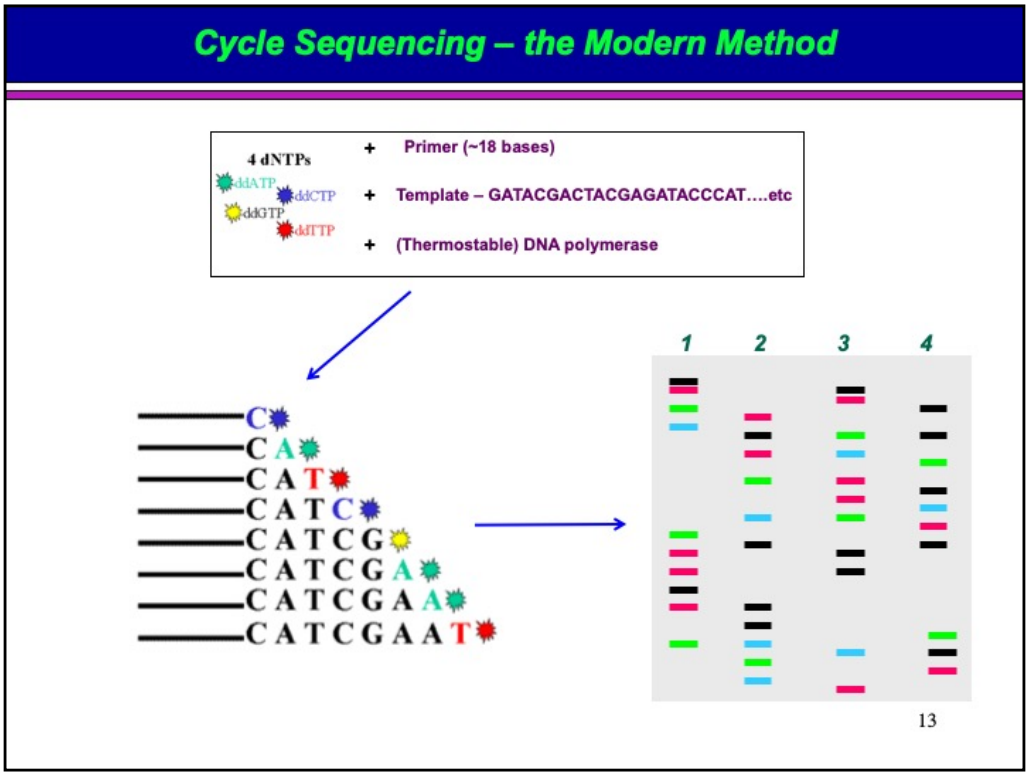
This slide depicts how the overlapping runs for the same reaction are spaced out in the gel, by loading the same samples about 30 mins apart. Notice that the xylene cyanol and bromophenol blue marker dyes indicate the relative positions and spread of the resolved fragments, from smaller near the bottom to larger nearer the top. A triple run of the same reaction set – and therefore single DNA template – with overlaps is indicated by the positions of the running dyes and connected orange arrows in the third panel.

This staggered set of runs of the same sequences increases the span of readable sequence from the gel, which can be up to 600 bases from a single gel triple run such as illustrated here.



An example of a developed 'X-ray' film of a sequencing gel. The reaction mixtures contain one radiolabelled isotope nucleotide (^{32}P-ATP; ^{33}P-ATP; or ^{35}S-ATP). The isotope is incorporated into every extended and terminated fragment wherever there is a 'T' on the template strand. Thus all the newly extended sequences are "hot"; and will show as darkened bands on the developed X-ray film.

After the gel is run, the facing glass plate is removed with the gel still adhered to the backing plate. The exposed gel/plate is then fixed in 10% acetic acid, wrapped in gladwrap, and placed in an X-ray film cassette box with a sheet of X-ray film against the gel and left for several hours to 2 days depending on the strength (higher beta-emissions) of the isotope used. The film is developed so that the latent "hot" bands deposit silver on the exposed X-ray film from the developer solvent in an otherwise clear background.



Cycle sequencing is a modification of the traditional Sanger dideoxy terminating sequencing method. The principles are the same as in Sanger sequencing. Dideoxynucleotides (ddNTPs) are used in a polymerization reaction to create a set of DNA fragments with dideoxynucleotides at the 3' end of each terminated fragment (top box and left bottom).

A key difference is that cycle sequencing employs a thermostable DNA polymerase and thermal cycler, similar to those used in the PCR. The advantage of using such a polymerase is that the sequencing reaction can be repeated over and over again in the same tube by heating the mixture to denature the DNA and then allowing it to cool to anneal the primers and polymerize new strands. Far less template DNA is needed than for conventional sequencing reactions, because the terminated fragments are amplified simultaneously with the extension/termination reactions.

The second key difference in Cycle Sequencing is that isotopes are no longer used – instead each of the four ddNTPs is labelled with a difference fluorescent tag which emits light at different wavelengths. In this way, all four reactions take place in a single tube (top box) and are subsequently loaded into a single gel lane (bottom right), because each different terminated band is able to be read (discriminated) as it passes a UV beam near the end of the gel (see further on). Each of the four fluorescent tagged ddNTPs have different fluorescent maxima and can be detected independently by the laser/detector discriminator (see further on).

An aliquot of the reaction products, as four combined reactions in a single sample, is added to a single lane of a urea-polyacrylamide sequencing gel (bottom right gel figure). Other combined reactions may be loaded into other lanes of the same gel (lanes 1 to 4; bottom right gel figure). [In comparing the tags (bottom left) to the gel band colours (bottom right), dark blue and yellow (left) are depicted as light blue and black (right)].

PCR thermal cyclers and Sequencing Gels

Thermal cyclers have thermostated lids for "hot starts" to avoid template mispriming.

Modern DNA sequencing gels are much shorter.

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Thermal cyclers (left) that we saw in the previous PCR lecture. These are used for cycle sequencing as well.

A typical DNA gel sequencing apparatus is shown at right. Notice that they are much shorter (~15-20 cm) than 'traditional' sequencing gels (~60 cm). The reason for this is that longer gels needed to be read along the length of the gel, whereas these so-called 'modern' gels rely on the DNA bands resolving enough over the length of the gel to be 'read' individually as they pass though a laser beam near the bottom of the gel.

Thermostable DNA polymerases

DNA Polymerase	Organism (hyperthermophiles)	Optimum activity
Vent	*Thermococcus litoralis*	75°C
Pfu	*Pyrococcus furiosus*	68-72°C
Pfu Turbo	*Pyrococcus furiosus*	68-72°C
Taq	*Thermus aquaticus*	75-80°C

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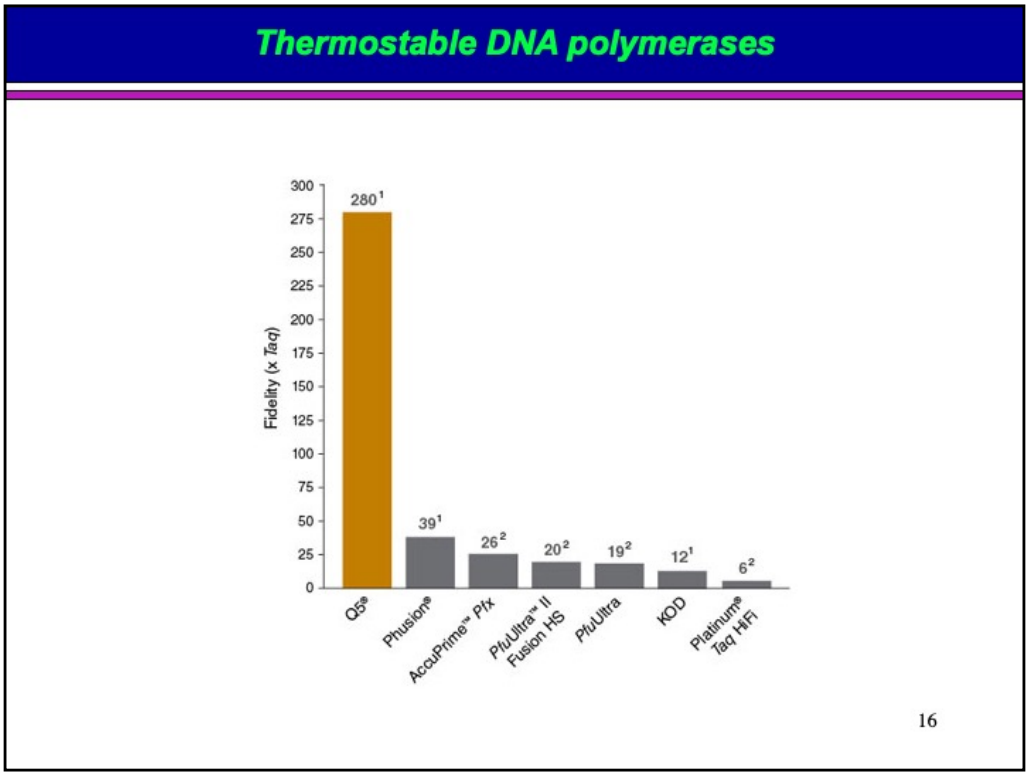
Some commercially available thermostable DNA polymerases used routinely in DNA cycle sequencing and the Polymerase Chain Reaction. They can add at least 100 nucleotides per second to a growing chain at the optimum temperature, but work very poorly, if at all, at 50°C and lower.

Each polymerase also includes a 5-prime to 3-prime exonuclease 'proofreading' activity that is required to replace occasional incorrectly inserted nucleotides. Incorrect bases are inserted at rates of 1:1000 to 1:10,000, which would be disastrous for the integrity of DNA copying. The exonuclease function of each thermostable DNA polymerase ensures that the error rate is far less frequent at 1:1,000,000 to 1:100,000,000. The lower the error rate, the higher the 'fidelity' of the enzyme.

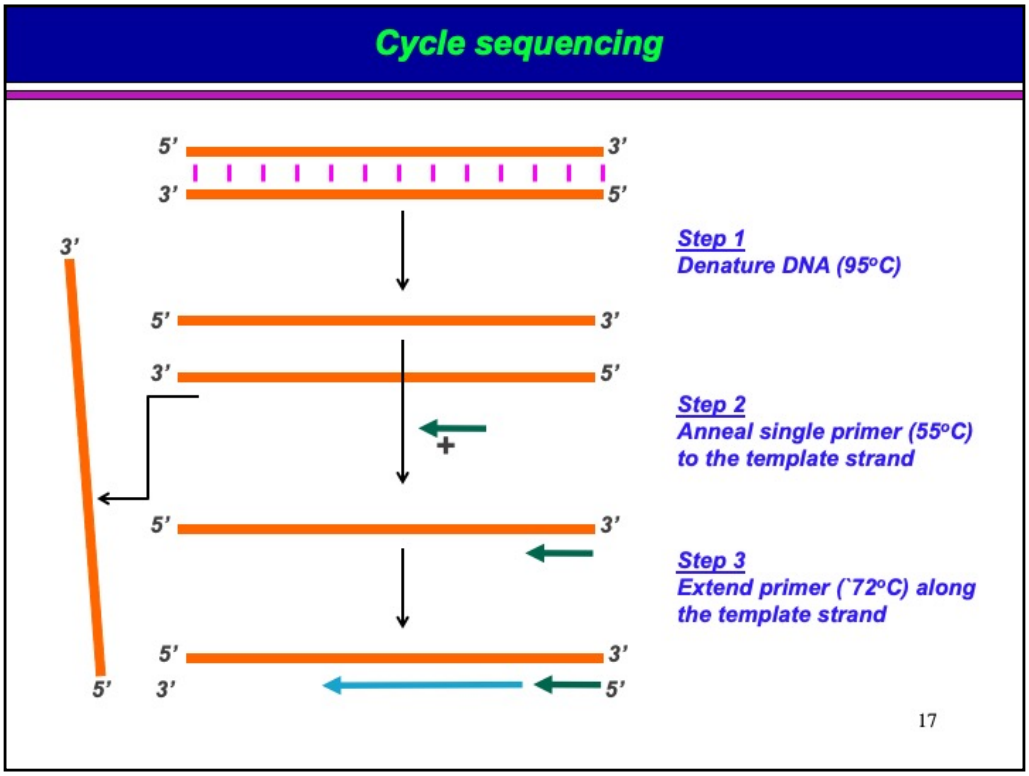
The original Taq polymerase is a "workhorse" enzyme as it's proofreading exonuclease activity is far less accurate than the others. Newer recombinant versions are more accurate.



A comparison of the fidelity (accuracy) of several thermostable DNA polymerases. The numbers at the top of each column is the accuracy relative to Taq polymerase (ordinate axis).

The first discovered – and least accurate – is Taq polymerase (last column), now mostly used for amplifications for which total accuracy is not required, for example, general cloning and sequencing. A recently introduced recombinant Taq polymerase is more accurate and robust than the original version.

Taq DNA polymerase has terminal transferase activity that results in the addition of a single nucleotide (adenosine, A) at 3'-ends of the extended/amplified DNA template strands. Only Taq has this facility and it is used in T/A cloning of linear products into a specially prepared vector plasmid (see PCR lecture).

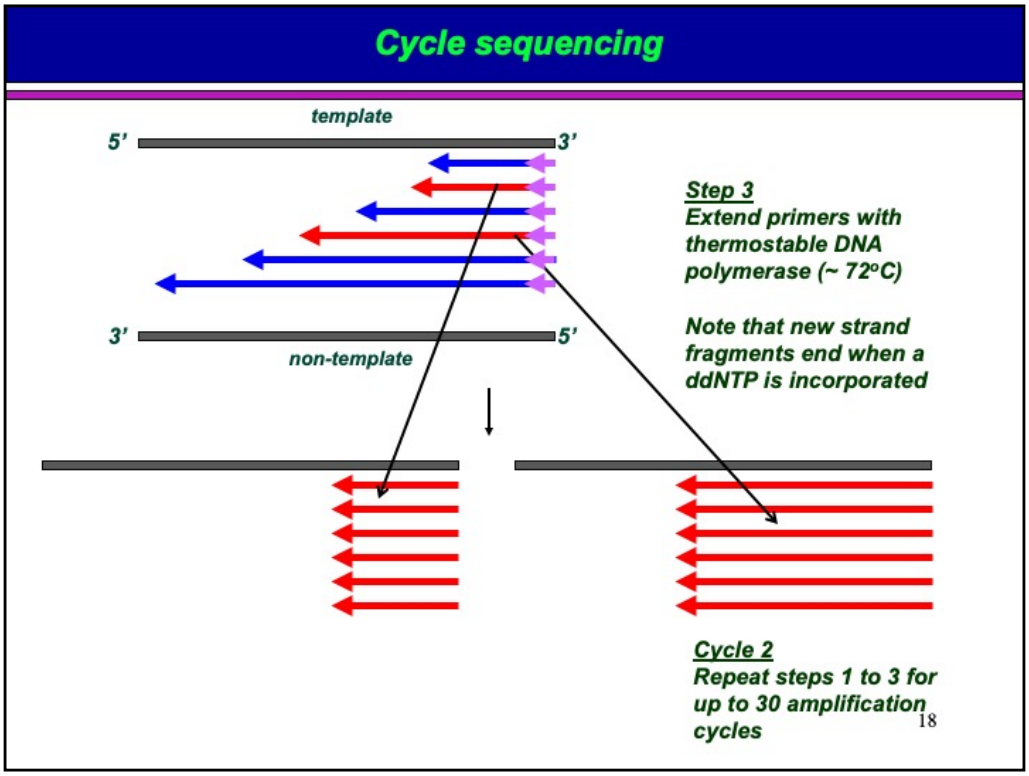


Cycle sequencing on a double-stranded template using strand-specific primers (short green arrows), a thermostable polymerase and temperature cycling, with steps 1 (denature at 95°C) and 2 (primer annealing at 55°C) shown in this slide, and the beginning of step 3.

Step 3 is the extension reaction (long light blue arrow). The reactions are performed in very thin-wall plastic tubes of 0.5 ml capacity with the reaction volume being 50 µl. The tubes are placed in a Thermal Cycler that has a microprocessor capable of ramping temperatures between 95°, 55° and 70°C very rapidly (5-10s per temperature step).

Note the 'discarded' (upright) strand on the left, which represents the non-used (non-template) strand, which is the lower strand of the original pair shown at the top of the slide. The 'discarded' strand is often actually sequenced in a separate reaction, using a second primer that will anneal near the 3'-end.

Reminder Hint! Although the strand at left is the discarded non-template strand, it is in fact the sense or coding strand that will be identical to the newly sequenced extensions.



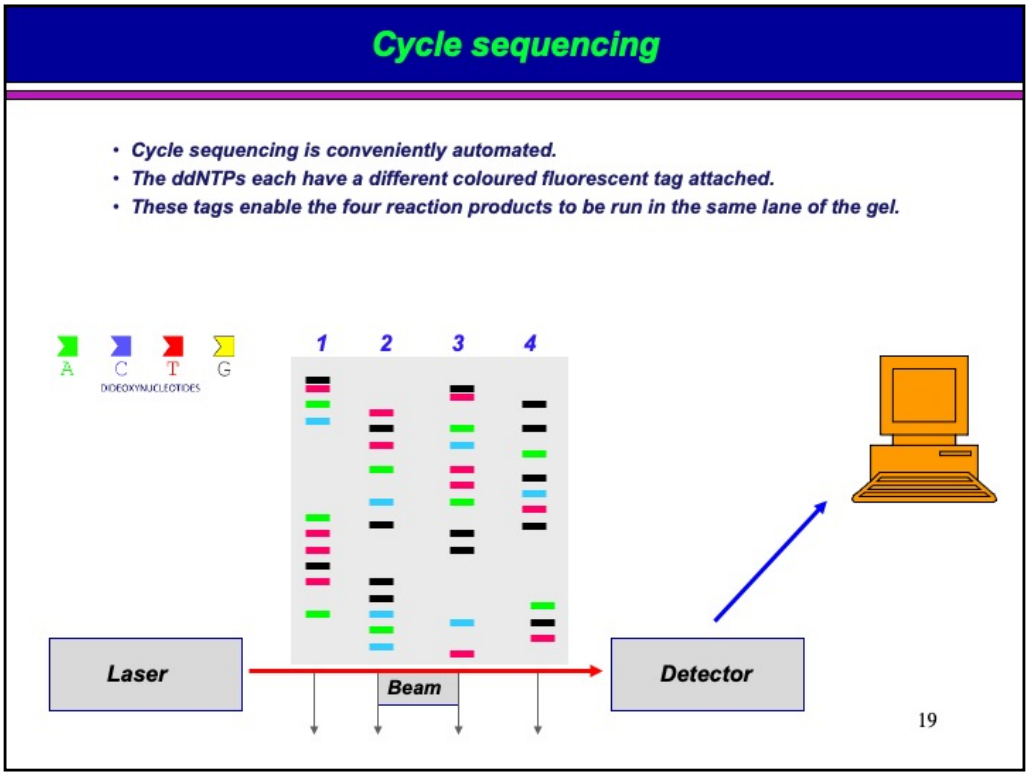
The chain terminated extension reaction (Step 3), performed at ~72°C. The duration of each step is 30 sec for steps 1 and 2 (previous slide) and 2-4 min for step 3 (above).

The three steps of each cycle are repeated up to 30 times in order to generate – amplify – the product fragments many thousand-fold so that each fragment becomes a prominent 'band' that can be detected by the instrument laser.

The diagram at the bottom (left and right) shows two of the six extended fragments from the top (red and blue) as a multiplicity of fragments terminated at two different positions on the same template (red arrows only). In reality this will happen on thousands of template copies so that the ultimate 'population' of terminated fragments is amplified exponentially (2 becomes 4 becomes 8 etc).

In summary, the chief difference between standard DNA sequencing and Cycle sequencing is:

All four dideoxy NTPs are labelled with a different fluorescent tag so that the 4 reactions can be performed in a single tube and run in a single gel lane. The thermal cycle amplification ensures that enough of each fragment is synthesized and therefore able to be easily detected.

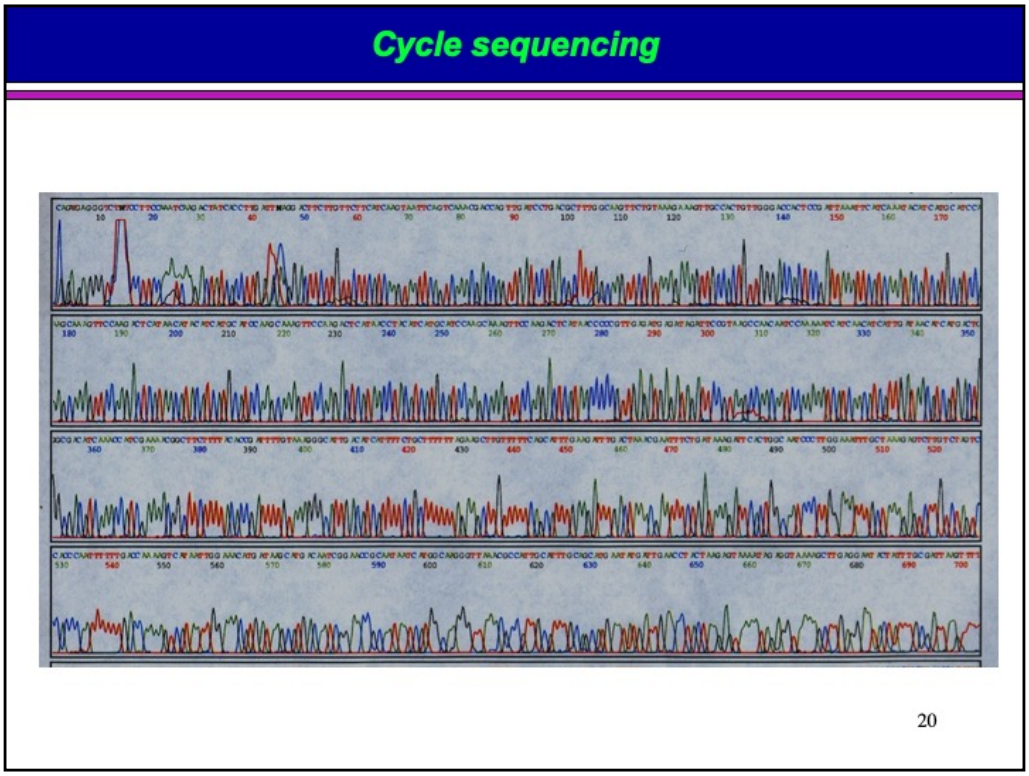


Simple set-up for cycle sequencing.

With isotope labelling in classical DNA sequencing, only the one same base is labelled (e.g. ^{32}P-ATP), requiring four separate gel lanes for the ddNTP reactions. In the case of fluorescent tagging, as illustrated above, each ddNTP has its own fluorescent colour tag (green, blue, red, yellow; shown at left). Note that black is used in the diagram above and in sequence readouts for clarity, as yellow is too feint to see on a white background.

The four separate reactions are combined and an aliquot added to a single gel lane as indicated above, where each numbered lane is a separate template and sequence for all four bases, A, C, G, and T. The bands in each lane are not read until they pass through - and interrupt - a laser beam at the bottom of the gel. In this way, the gel can be shorter (15-20 cm) and the apparatus more compact.

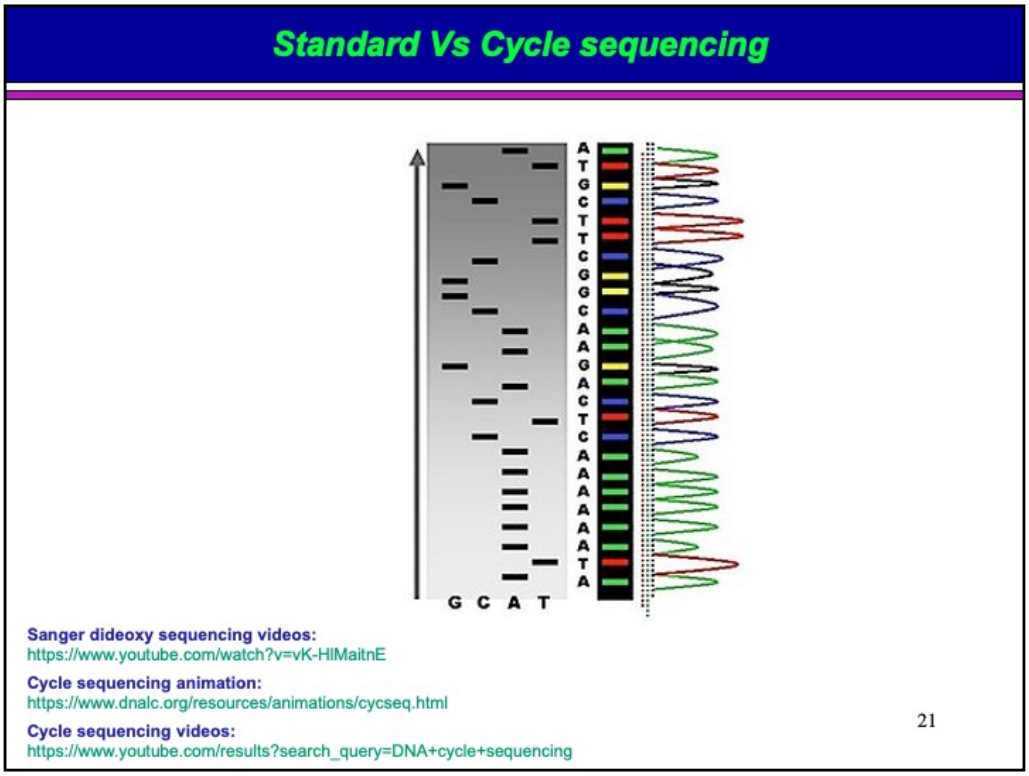
The Detector has a discriminator, essentially a spectrofluorimeter capable of detecting the four different tags, because it has the four different wavelengths set for the four fluorescent maxima. Once detected, the fluorescence is converted to an electrical signal for each of the four bases, and the consecutive order of the bases is deposited in word files and graphical readouts in the computer (see next slide for a typical readout).



A single-lane cycle sequence readout from a recombinant plasmid, sequenced from a universal primer in one direction (i.e. along one strand).

Colored peaks indicate the tagged and numbered bases shown above the peaks. An 'N' indicates that the base at that position could not be read – this usually happens very near the primer start point, but also near the end of readable sequence.

In this example, clear consecutive sequence can be read from about bases 50 to 700+; a span of at least 650 bases.

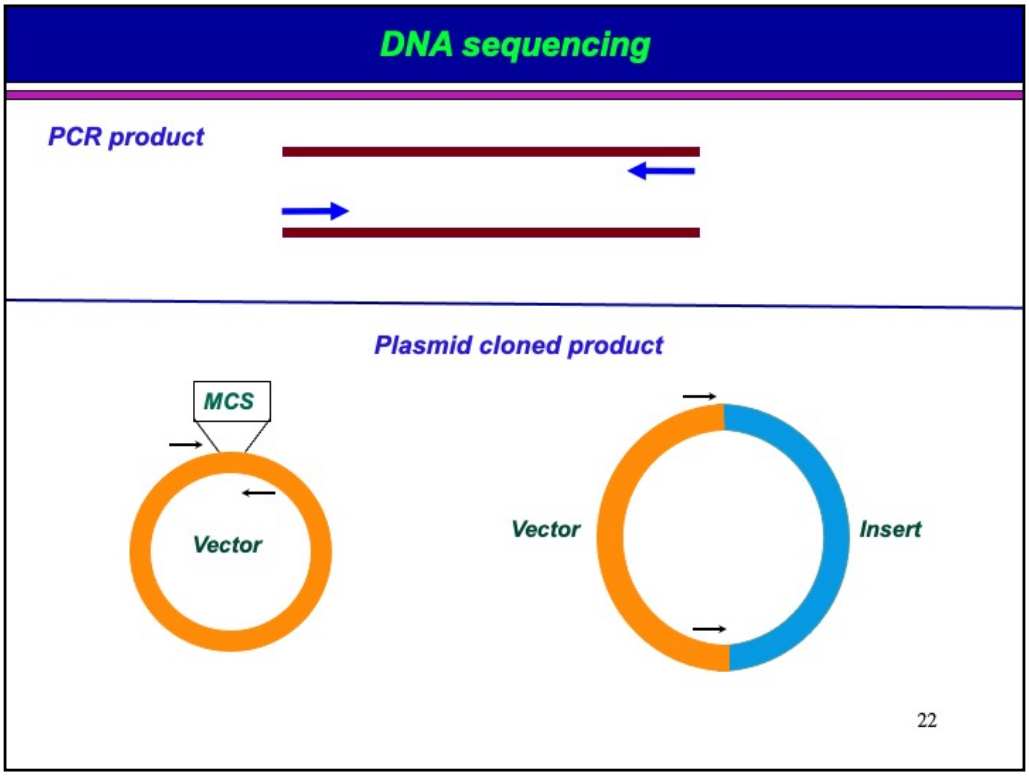


Comparison of standard sequencing (left) and cycle sequencing (right). The four-lane to one-lane transition is shown for the two different sequencing methods.

In standard DNA sequencing, only one nucleotide is labelled (dA$T^{32}P$), so each terminating base reaction must be run in four separate lanes of the sequencing gel (left).

In cycle sequencing, each ddNTP is labelled with a different fluor tag and all four reactions may be run in a single lane of the gel (right).

The links shown at the bottom of this slide are to very useful animations and videos on DNA sequencing.

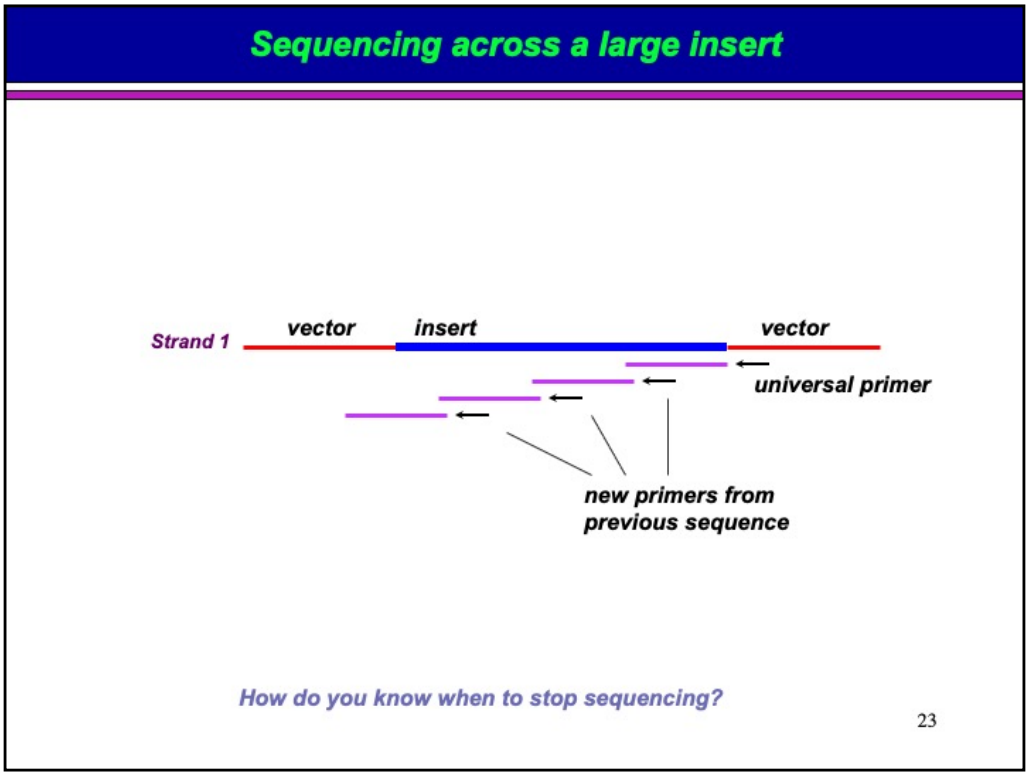


Technical note about sequencing: The best and most error-free enzyme for 'normal' sequencing is "Sequenase", which is used in classical DNA sequencing, but several accurate thermostable DNA polymerases are now available for Cycle Sequencing, as seen in earlier slides.

Top panel: Linear PCR products can be sequenced directly without cloning into plasmids, by using the same PCR primers to extend against each template strand. If DNA is copied or amplified using the Polymerase Chain Reaction (PCR), then the sequencing reactions can be performed directly on the product using the same PCR amplification primers as 'bookends' (inward-facing blue arrows), so that all of the newly amplified DNA sequence between the primers can be read.

Bottom panel: In a similar sense, anything cloned into a plasmid vector at the MCS (multicloning site) is also 'bookended' between the two commercially available universal vector primers (short black arrows) that may be used to sequence into the cloned insert along each strand in opposite directions. The primers are complementary to short vector sequences on each strand, either side of the MCS. Note the positions of the primers in the vector and in the plasmid that now contains an insert, with the primers facing the insert on either strand.

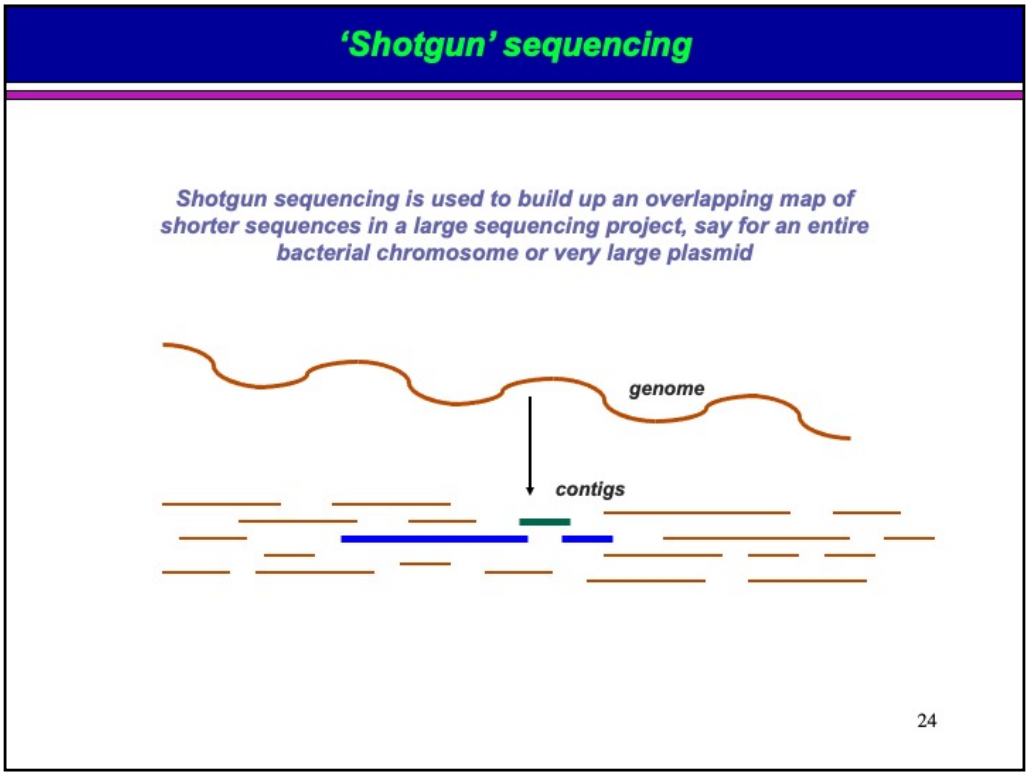
Cycle sequencing enables sequences to be obtained directly from recombinant plasmids, because the thermal cycling steps denature the two plasmid strands during each cycle so that one plasmid strand acts as the template for its primer in the presence of the other strand, which may be sequenced in a separate reaction with its own specific primer.



This slide illustrates the method of 'primer walking' to sequence sections (contigs) of a large insert (blue). For simplicity, this is a linearised representation of a circular recombinant plasmid. The ends of the vector (orange) are joined up to form to form a circular molecule.

A long insert (blue line) of say 2-3 kb or more can be sequenced from the adjacent vector by doing overlapping sequencing runs for which each additional primer sequence (right to left short black arrows) is determined from the end of the previous sequencing run (pink newly sequenced contig segments). This is known as 'primer walking'.

The sequenced contigs are added in overlapping segments until the end of the insert is reached. The process is repeated for the second strand, in this case sequencing across the insert in the other direction, so that the two opposite sequences match perfectly.

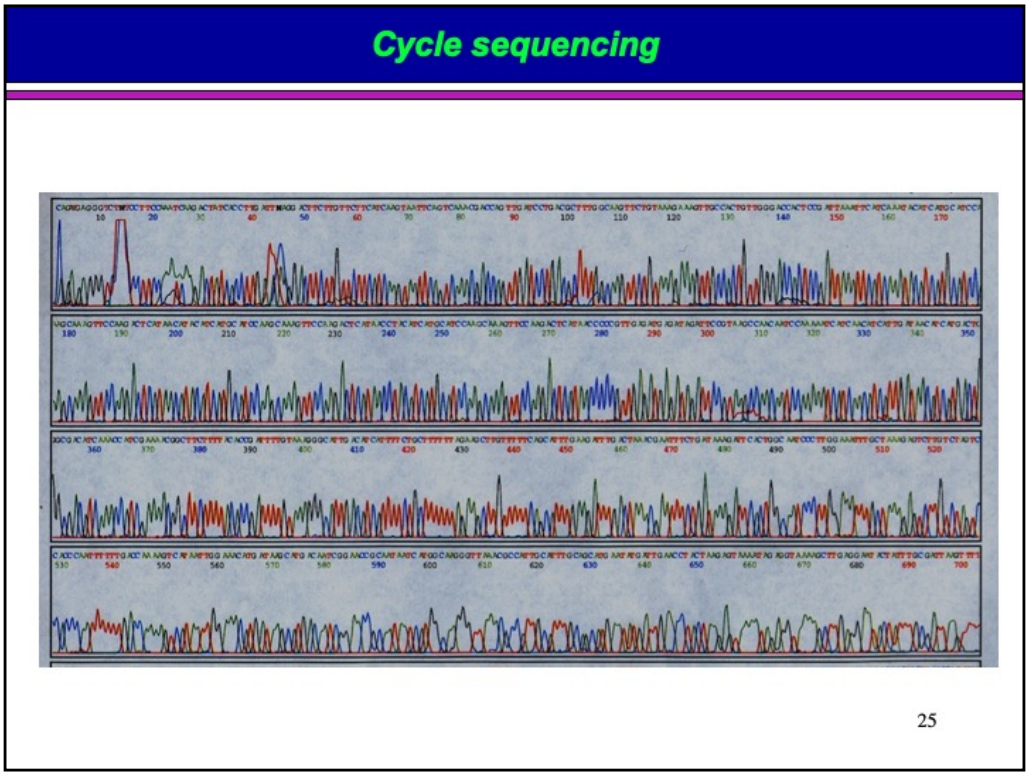


A genome sequencing project will take a very long time to complete using the processive overlapping approach indicated for long inserts on the previous slide.

Instead, a "shotgun" approach is used in which the chromosome is fragmented by restriction digestion, or mechanical shearing, then either cloned into a plasmid vector for individual sequencing runs, or sequenced as linear fragments using short (~6-8 bases) random primers to attach at multiple sites on most of the fragments. This method is essentially akin to the RAPD-PCR method we looked at briefly in the previous lecture.

During the sequencing project, gaps are identified by a contig overlapping assembly program and these gaps can be filled in by sub-cloning 'missing' (unsequenced) fragments into a plasmid, or by designing primers to each end of the "missing" fragment from adjacent fragments.

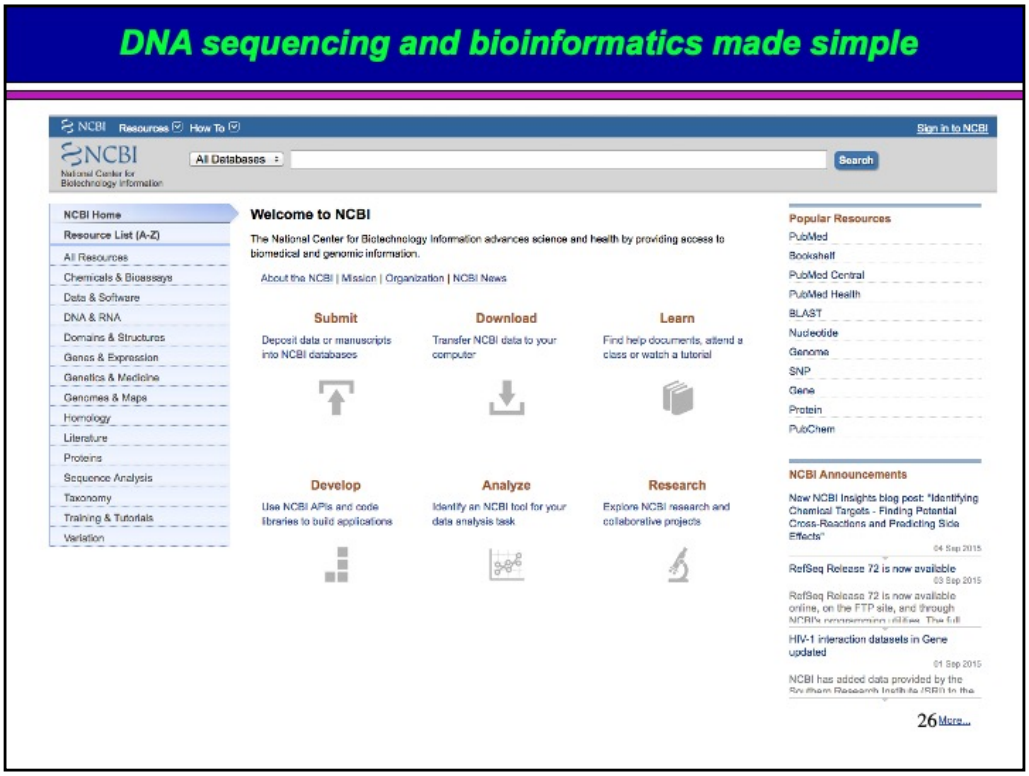
As an example, the green fragment sequence in the above slide is obtained by sequencing in from the adjacent blue contigs; or the green fragment is cloned into pUC18 and sequenced using the pUC18 universal primers, as illustrated two slides back.



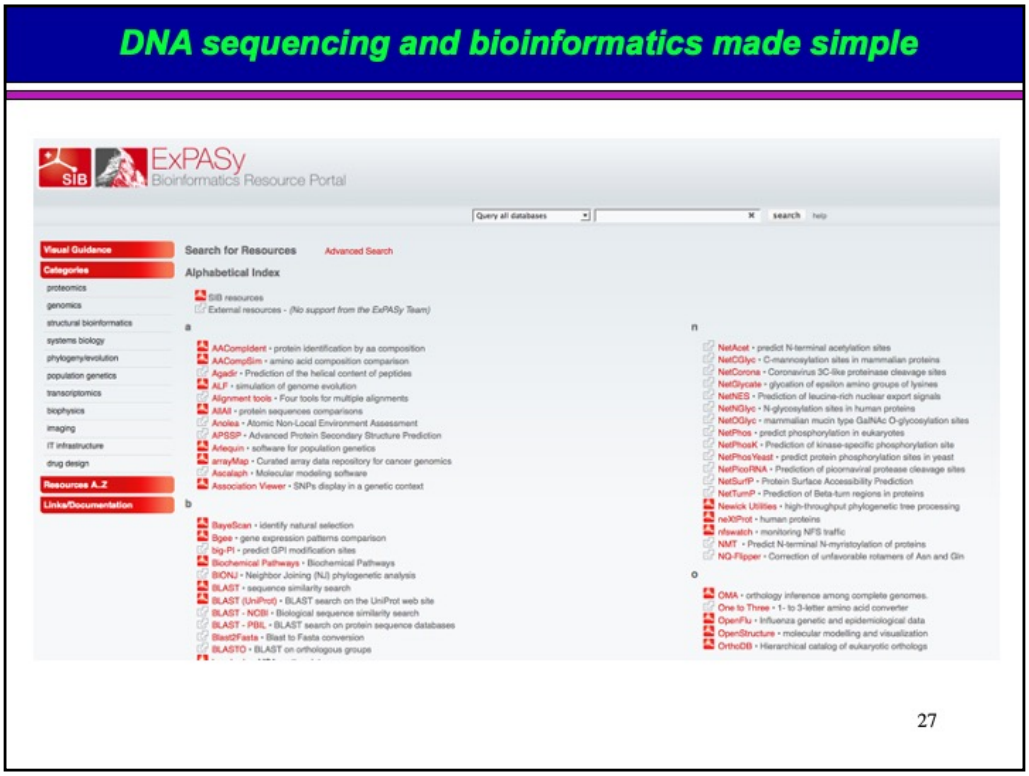
A single-lane cycle sequence readout from a recombinant plasmid, sequenced from a universal primer in one direction (i.e. along one strand).

Colored peaks indicate the tagged and numbered bases shown above the peaks. An 'N' indicates that the base at that position could not be read – this usually happens very near the primer start point, but also near the end of readable sequence.

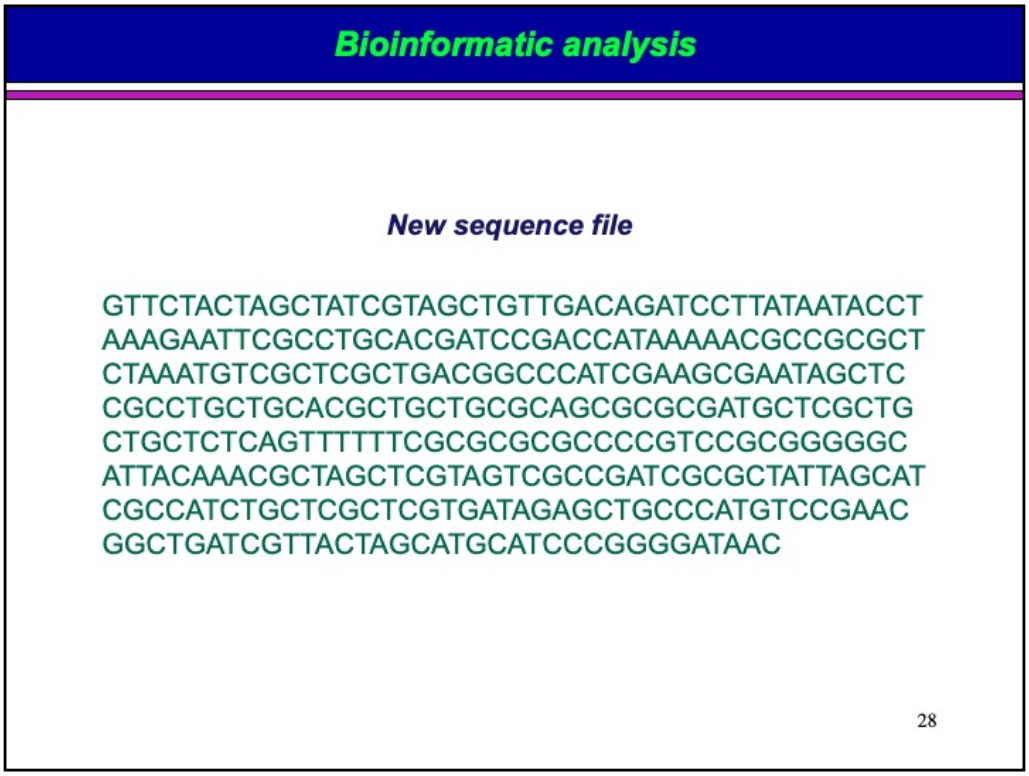
In this example, clear consecutive sequence can be read from about bases 50 to 700+; a span of at least 650 bases.



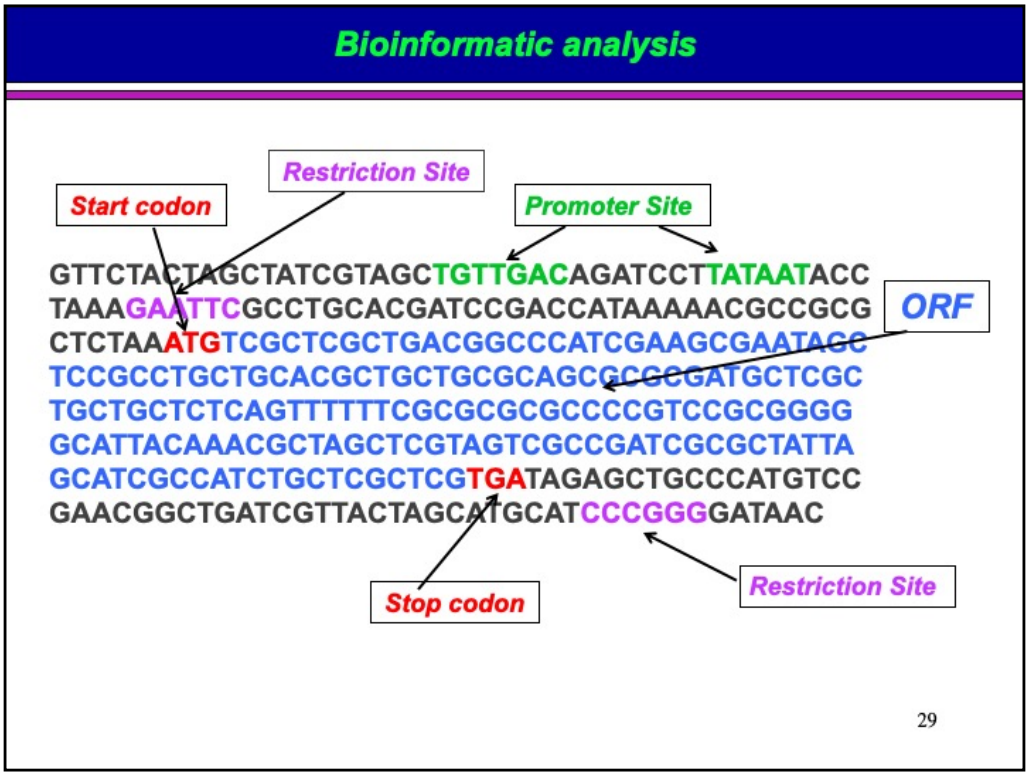
NCBI is a major Bioinformatics site with a wide range of database and software links, including Genbank, the main DNA database, and Uniprot, the main protein database.



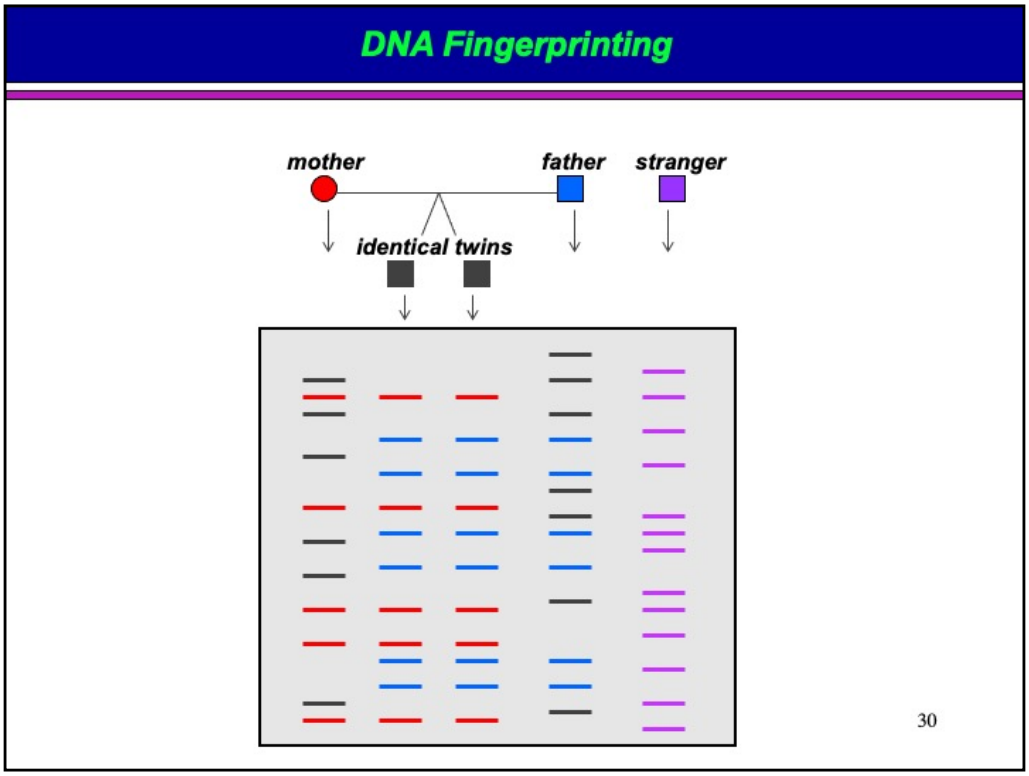
ExPASy is, like the NCBI on the last slide, a major Bioinformatics portal, with links and access to all databases and software options. This is, like the previous slide, a snapshot of the Homepage.



A very brief introduction to bioinformatic analysis. Quite simply, there are software programs and helpful sites (e.g. NCBI and ExPasy, shown in the previous two slides) that will do the analyses for you. The arbitrary and unlabelled sequence above can be identified and "filled in" using free (online) software programs. This sequence may then be labelled as indicated on the next slide.

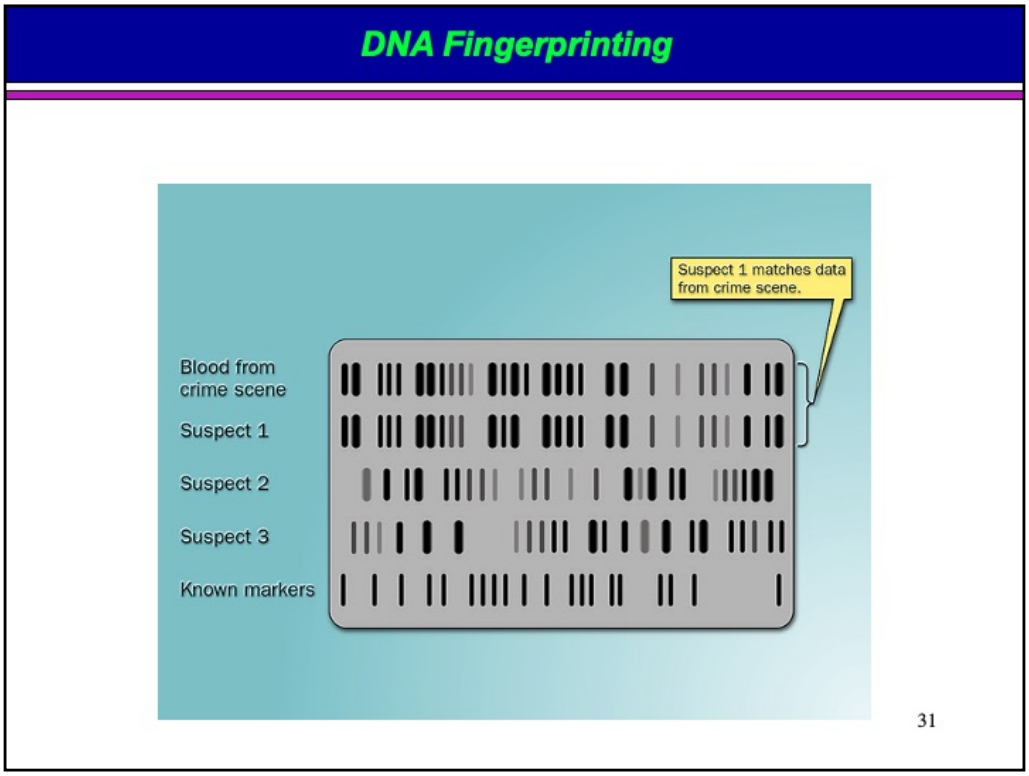


Self-explanatory IDs on this DNA sequence. Sequences can be searched by a number of software devices - bioinformatic analysis – 'Blast' searches for related sequences, for open reading frames, start and stop codons, promoters, termination signals, restriction sites etc.

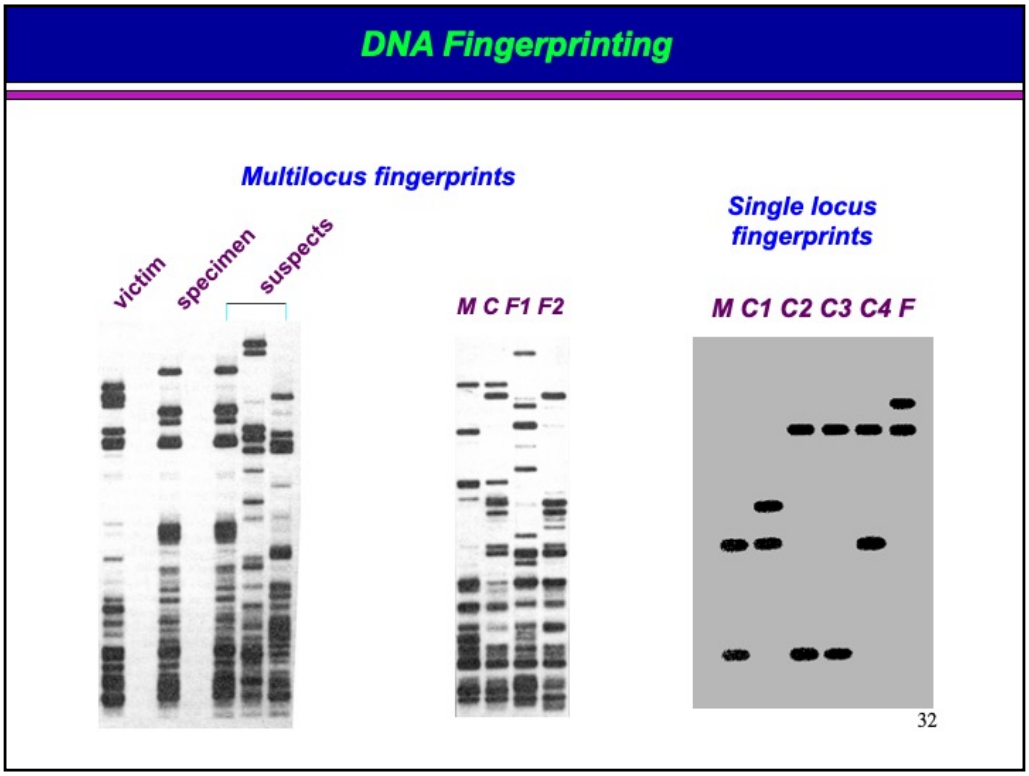


DNA fingerprinting is a technique that has been made possible by the accumulation of DNA sequences in databases such as GenBank; and it has allowed matching of sequences from any source!

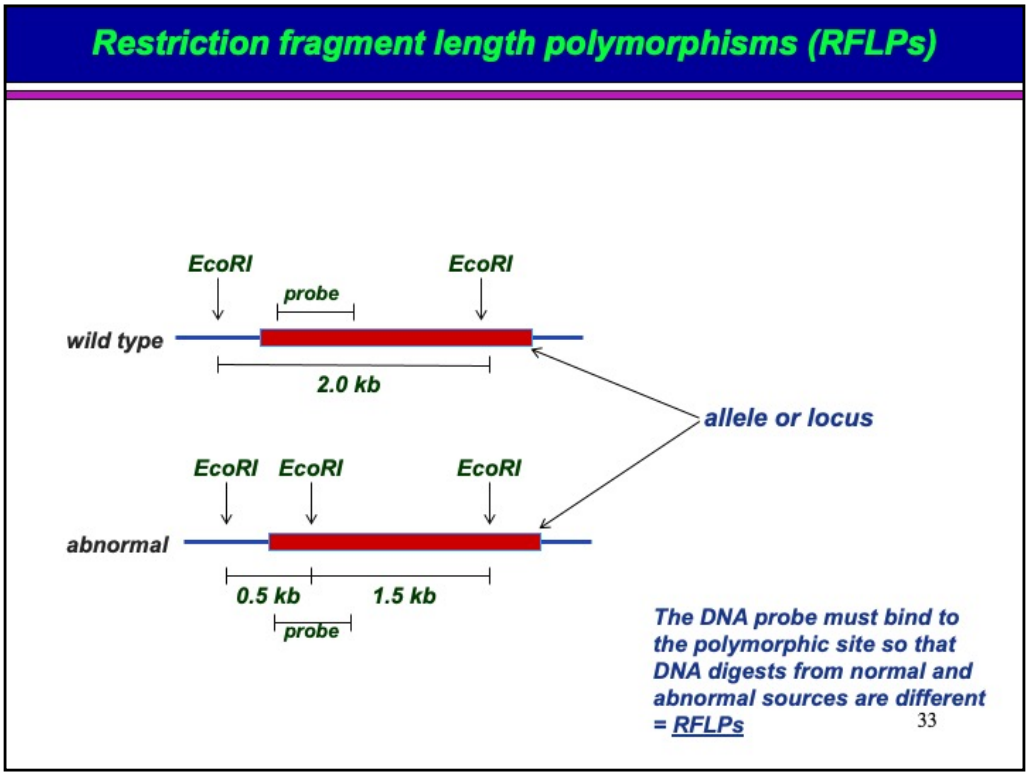
This facsimile diagram of a gel set of DNA bands is a typical familial DNA fingerprinting example ('bar code'). All bands for the twins are the same; and can be apportioned to either parent.



Self-explanatory forensic DNA fingerprint.

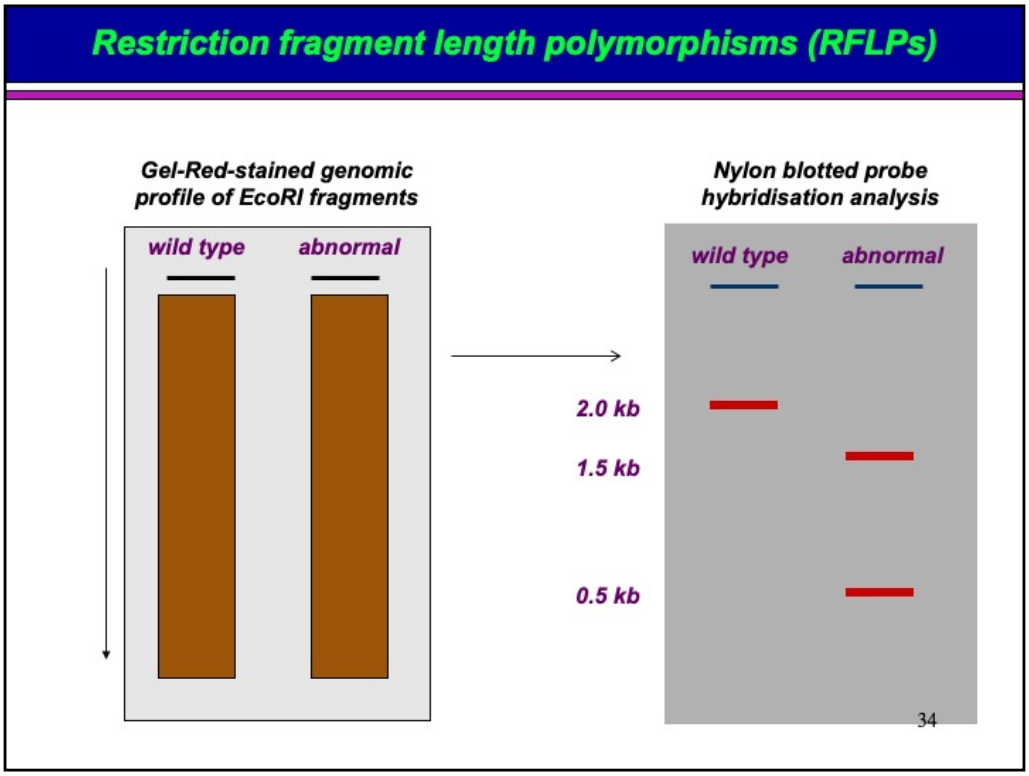


Examples of multilocus and single locus DNA fingerprinting (profiling) analyses.



RFLPs are used for detecting genetic abnormalities or for fine mapping of alleles. The difference in alleles is usually down to a single base change that may be detected as the removal of, or extra, restriction site in or near the locus.

These WT and mutant alleles are easily distinguished by the difference in EcoRI fragment sizes (RFLPs). A probe that binds to both loci across the junction of the new EcoRI site will identify either a 2 kb EcoRI band (WT) or 0.5 and 1.5 kb bands (abnormal). See the next slide for the RFLP gel analysis.



Discriminating WT and mutant alleles using a locus-specific probe that identifies different RFLPs.

In order to interpret how this RFLP analysis works, you should also look back to the previous slide to identify the positions of the EcoRI sites in the two loci.

The basic determining factor for this RFLP is the occurrence of an extra EcoRI site in the abnormal locus. This gives rise to two fragments of 1.5 and 0.5 kb, respectively in the abnormal locus (lane 2, gel profile on right). On the other hand, the normal allele has a single EcoRI site that generates only a single 2 kb fragment in the gel profile (lane 1).

Note that the fragments depend on an external EcoRI site as well (see previous slide), thereby generating the bands/fragments seen in the gel above. The mutant allele has an extra EcoRI site, but requires the external site to generate the two fragments seen in lane 2 of the gel. Ditto for the normal allele, with one internal and one external EcoRI site and a single band.

MCQ L9

A student has identified a new peptide hormone and clones the DNA containing the hormone's gene sequence. The clone is sequenced and is as indicated below. The student doesn't bother to analyse the sequence using bioinformatics software, but instead identifies the Open Reading Frame for the peptide gene by eye, marking the start and stop codons in bold type. From this analysis, we can conclude that the…

AAAGAATTCGCCTGCACCATAAAAACGCCGGCTCTAA**ATG**TCGCTCGCTGACGGCCCATCG
AAGCGAATAGCTCCGCCTGCTGCACGCTGCTGCGCAGCGCGCGATGCTCGCTGCTGCTCTC
AGTTTTTTCGCGCGCGCCCCGTCCGCGGGGGCATTACAAACGC**TAG**CTCGCTAGTCGCCG
ATCGCGCTATTAGCATCGCCATCTGCTCGCTCGTGCGATAGAGCTGCC.

(A) Open reading frame has been correctly identified.
(B) Student has incorrectly identified the stop codon and it should be the one shown in the box.
(C) There is no open reading frame in this sequence and the wrong insert has been cloned.
(D) Student has incorrectly identified the stop codon and it should be the one underlined.
(E) Sequence is incorrect because only one strand was sequenced.

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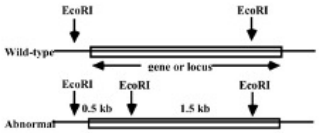
MCQ L9



The diagram above depicts the pattern of EcoRI sites in wild-type and abnormal alleles of a particular gene. This allows the detection of the abnormality by restriction fragment length polymorphisms.

This polymorphism can be detected in a southern blot of an agarose gel of genomic DNA cut with EcoRI because...

(A) A 2.0 kb EcoRI probe will detect the abnormality.
(B) A 0.5 kb EcoRI probe will detect the abnormality.
(C) A probe that straddles the central EcoRI site will detect the abnormality.
(D) All of the above.
(E) None of the above.

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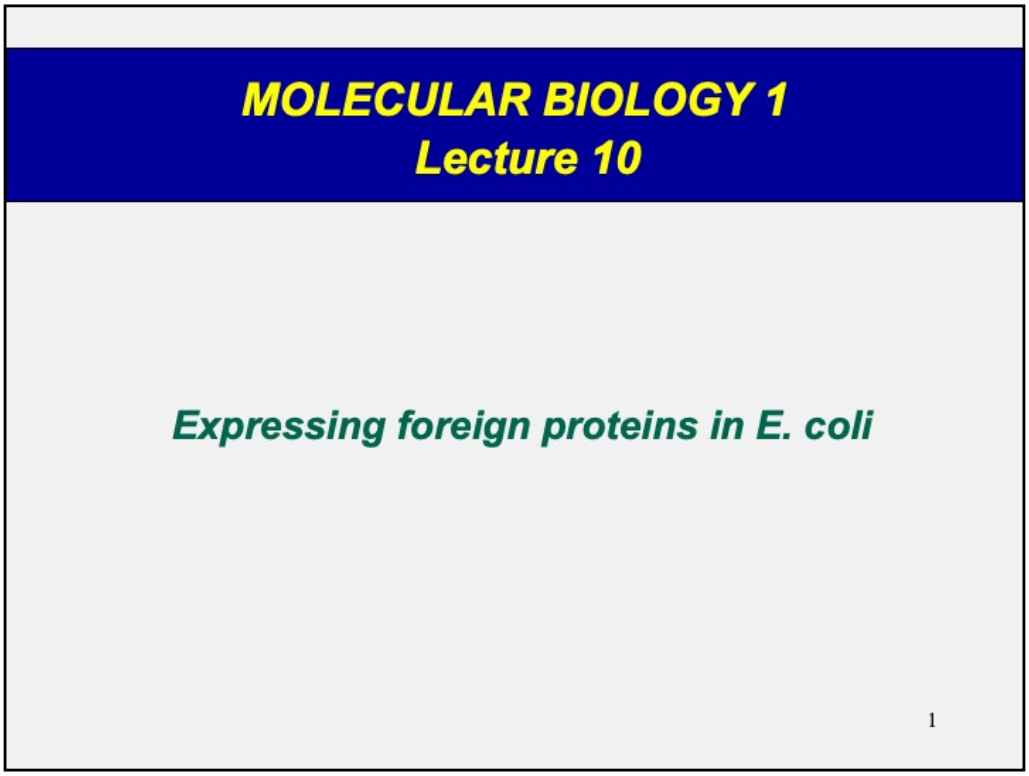
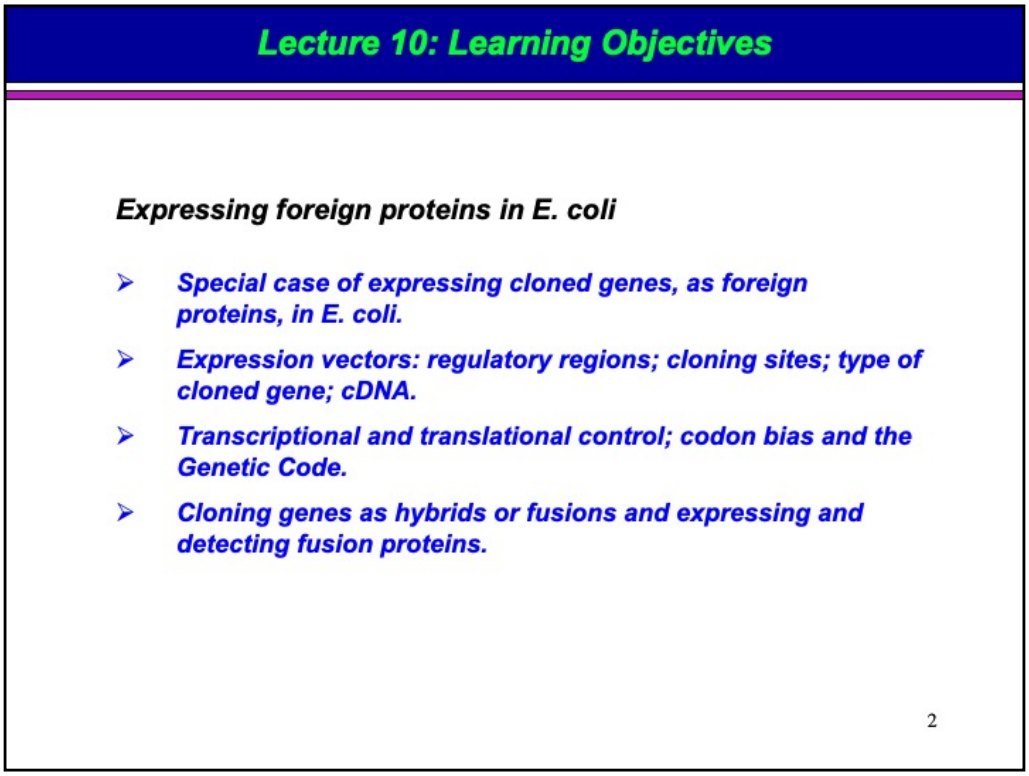


The theme of this lecture is to take molecular cloning of recombinant plasmids to the obvious point of expressing proteins from cloned genes, using the advantage of regulated control of expression within the host plasmid for the controlled production of protein.

This seems like a simple task, with the cloning scheme to obtain the recombinant plasmid seemingly the major hurdle, but in practice, the expression of foreign (heterologous) proteins in a host such as *E. coli* is peppered with pitfalls and obstacles that will be explored in this lecture.

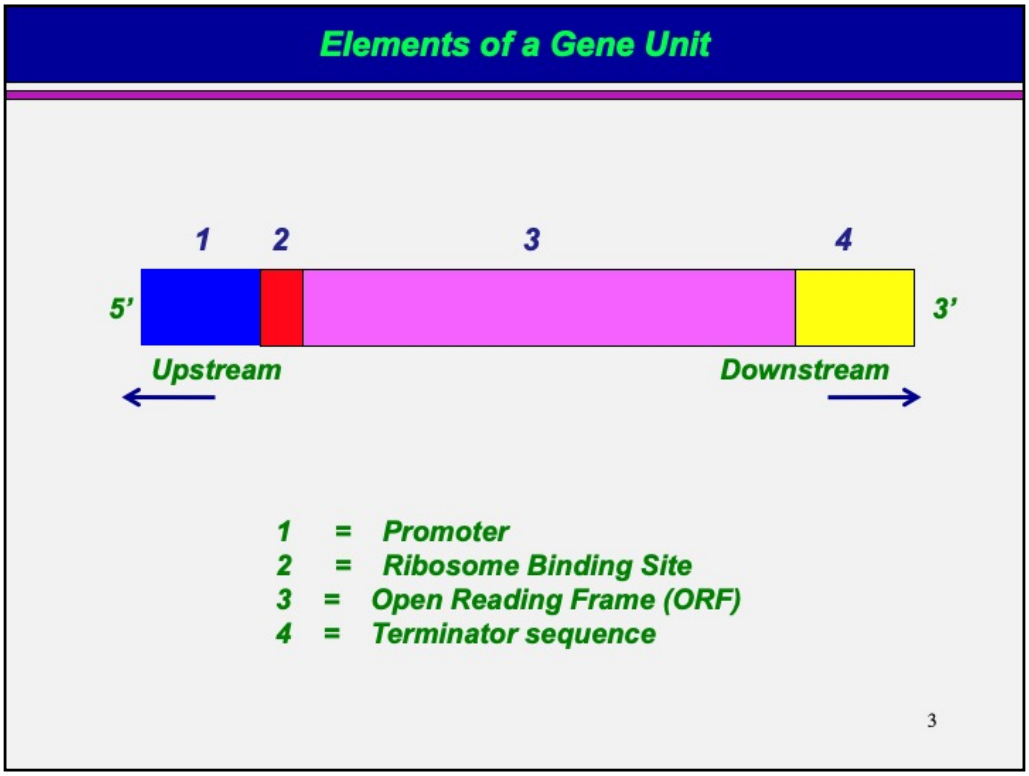


In Experiment 6, you will clone your own 'gene' in a plasmid and introduce it into *E. coli*. As lab work goes, this will be a relatively simple exercise with a positive outcome, hopefully.

However, when you spread the molecular biology of gene cloning far enough to encompass the cloning of any gene from any species, then the technical side of things becomes a little more involved and difficult. There is a duality of issues to consider, but the chief of these is the fact that though *E. coli* is an ideal cloning 'vessel', it will only express genes that have a promoter that works in *E. coli*.

As most cloning exercises involve genes from other species, that is so-called foreign genes, we can describe the foreign proteins expressed as 'heterologous' proteins, because they do not belong to the *E. coli* world, as it were. The requirements for expressing such foreign genes are much more demanding and require special expression vectors with regulatory regions (promoter-operator-inducer) that will work in *E. coli*.

This lecture will explore this theme in some detail. A number of topics/methods are involved and these are listed as the learning objectives on this slide.



We think of a gene in terms of the translated protein or polypeptide product, but all genes have defined elements that control the expression of the gene, such as:

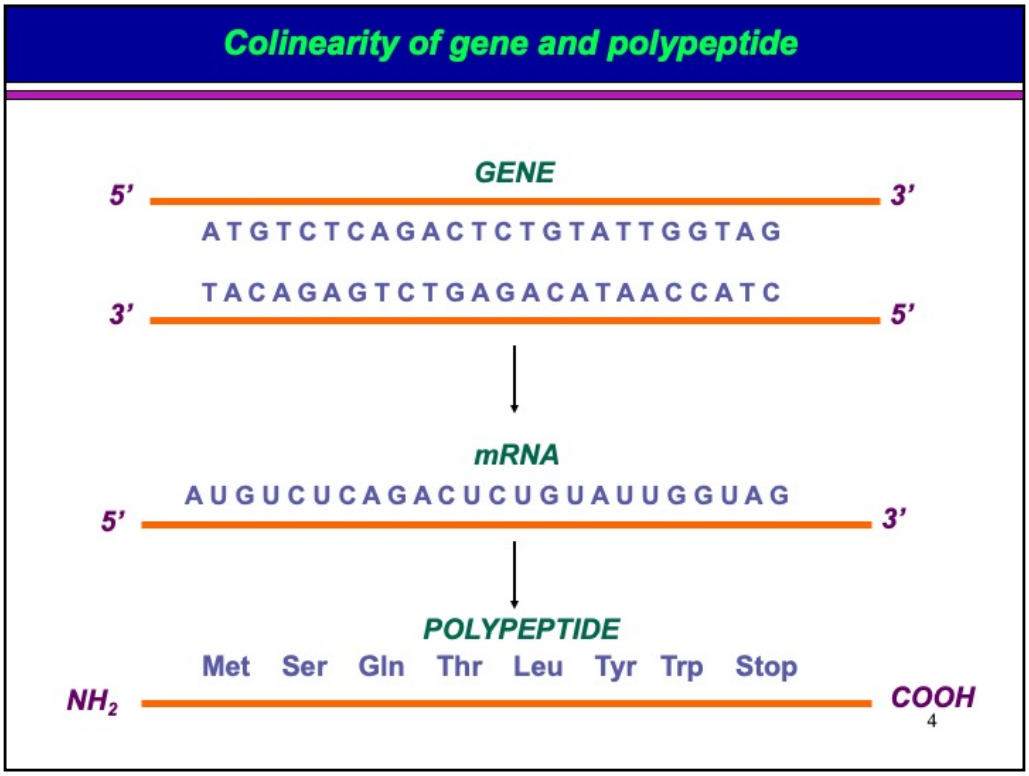
The initiation of transcription, as the binding of RNA polymerase to the Promoter site (blue).

The ribosome binding sequence that directs the RNA transcript to the protein translation site on the ribosome (red).

The termination signal that enables RNA polymerase to be disengaged from the mRNA transcript (yellow).

The fourth element – the Open Reading Frame (pink) – is the set of triplet nucleotide codons that specify the sequence of amino acids for the translated polypeptide; and this is what we usually mean when we refer to a 'gene'.

In reality, a gene comprises all of the elements depicted above, and often others, such as signal sequences for exporting proteins outside of the cytoplasm and/or out of the cell, and activator or repressor regions that we find in more complex eukaryotes and which might be located remotely from the gene. Though technically not part of the gene and its framing sequences, these regions are still considered germane to the control of expression of the gene(s).

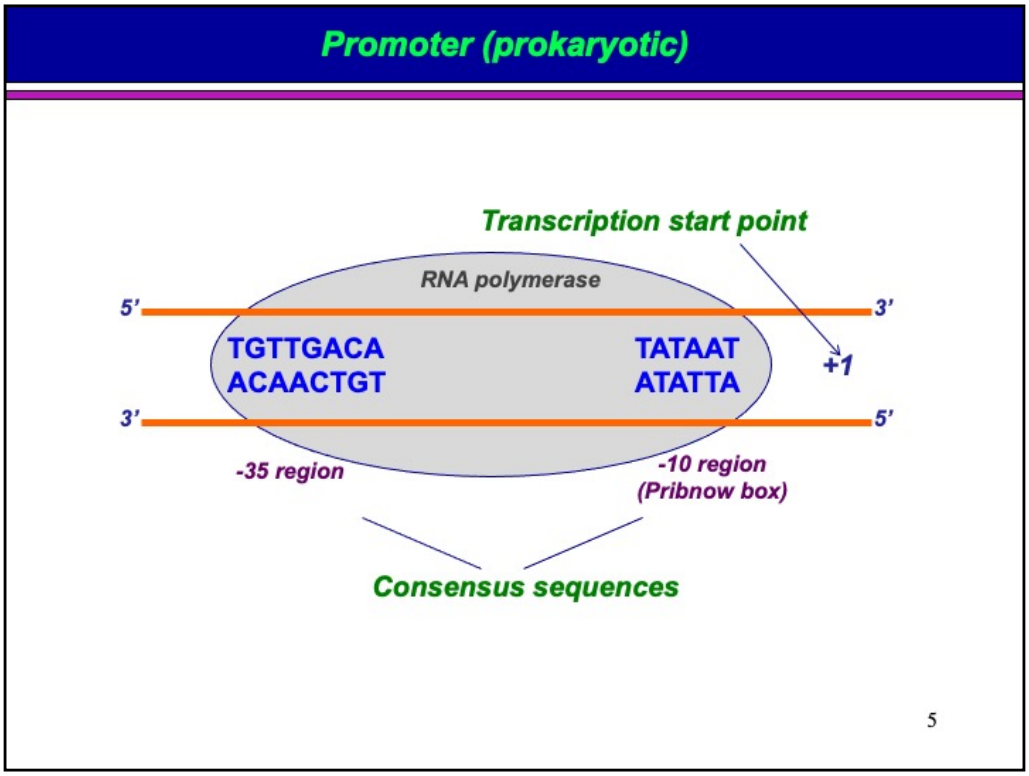


Within the Open Reading Frame (ORF) of a gene, each triplet codon specifies an amino acid in the translated polypeptide and the two – triplet codons and amino acids – are collinear, that is, they are aligned precisely.

The messenger RNA (mRNA) transcript triplets are identical to the triplets of the gene's ORF strand (upper sequence, top), but complementary to the triplets of the template strand (lower sequence).

The template strand is the 'go between' strand that enables the gene sequence to be copied exactly, by the principle of complementarity. This means that the mRNA is actually copied from the strand opposite to the gene sequence strand, in order that the mRNA is identical to the gene sequence.

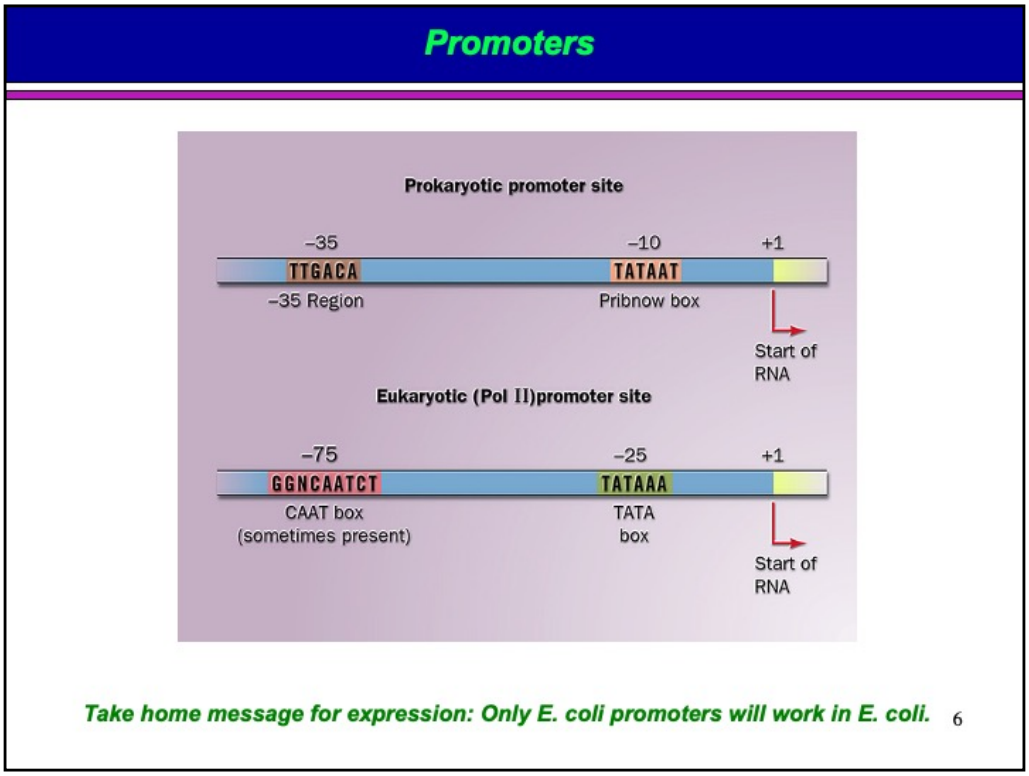
Note that the gene sequence and mRNA have the same strand polarity (5' - 3'), which is opposite to the template strand polarity (3' – 5'). Lastly, notice that in RNA, the base uracil (U) substitutes for thymine (T).



The two consensus sequences(blue text) are spaced at the correct distance for the interactive binding of RNA Polymerase (grey oval) that will begin transcription of the gene sequence. These consensus nucleotide bases align non-covalent interactions between the consensus bases and RNA Polymerase.

'Consensus' means that many genes have variations of these sequences and the differences decrease the number of binding contacts with RNA Polymerase, making the promoter less effective or weaker, in that transcription is not as vigorous as for higher consensus sequence promoters.

This is a form of regulation in that cell functions that require many copies of certain proteins (e.g. repair proteins; haemoglobin etc) will have strong promoters (high consensus), while weaker promoters are used for genes that only need to produce small numbers of a specific protein for a shorter time (e.g. DNA polymerase III, the chromosome replicating enzyme).

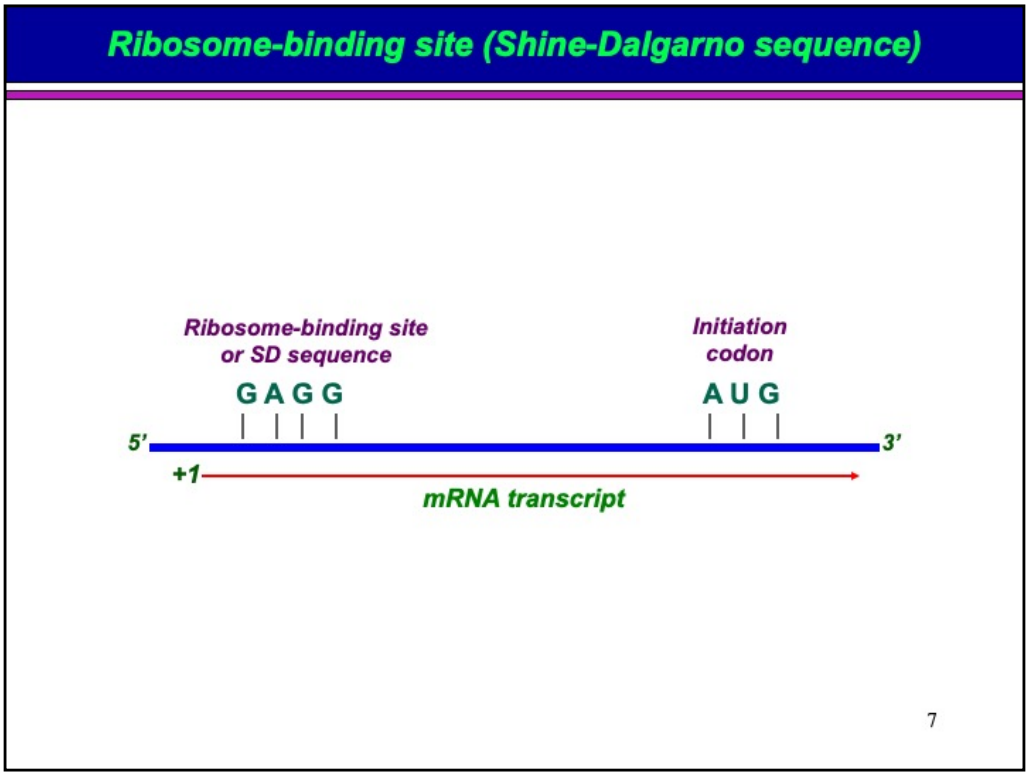


A comparison of consensus promoter sequences in prokaryotes and eukaryotes (not to scale).

Note the different minus region sequences (especially at -35 and -75), which are reflective of the different RNA polymerases in each kingdom.

The two polymerases have different 'footprints'. The footprints are designated as 'minus' regions because promoter DNA is only a binding site and is not copied or transcribed into messenger RNA.

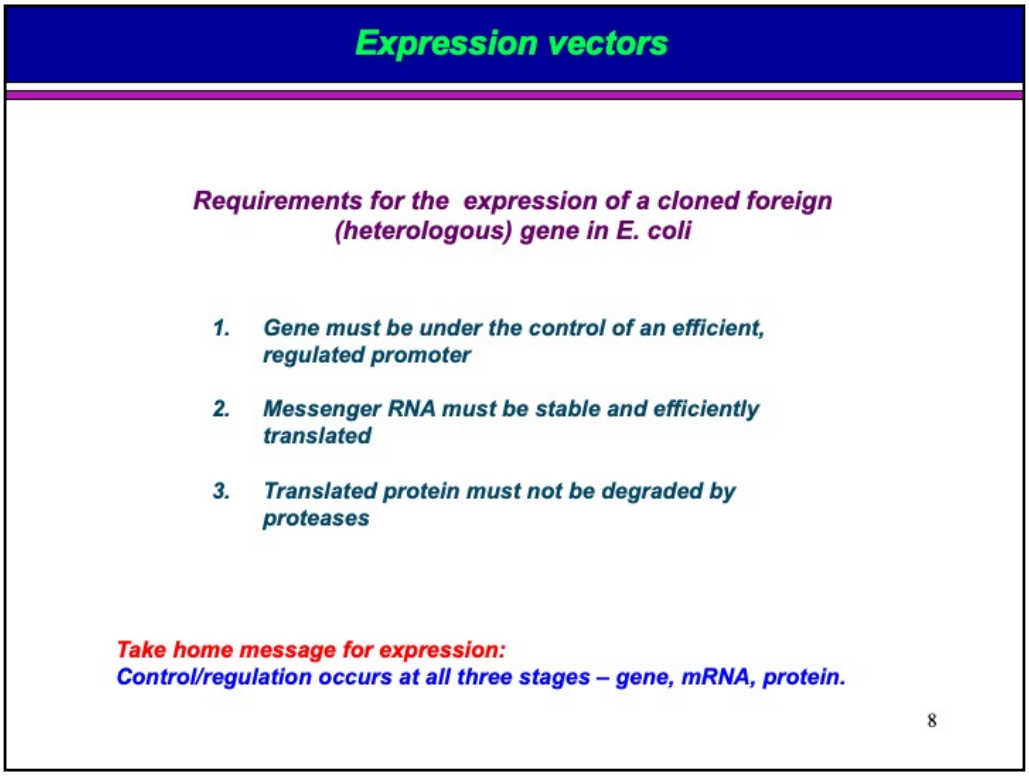
The actual start points of the mRNAs are shown to the right of the figure at the +1 position, which will be nucleotide #1 in the mRNA transcript. The crucial point that brings this comparison to the molecular biology world of protein expression is that only *E. coli* promoters (and some closely related Gram negative bacteria) will work in *E. coli*.



The four-base SD consensus sequence (GAGG) is complementary to the end of the 16S ribosomal RNA (CTCC), which is the 'feed in' point for the mRNA transcript to be 'read' (translated) into a protein.

Just like with promoter sequences, the degree of consensus determines whether the transcript will be held more tightly and therefore be more likely to be translated and make more of the protein.

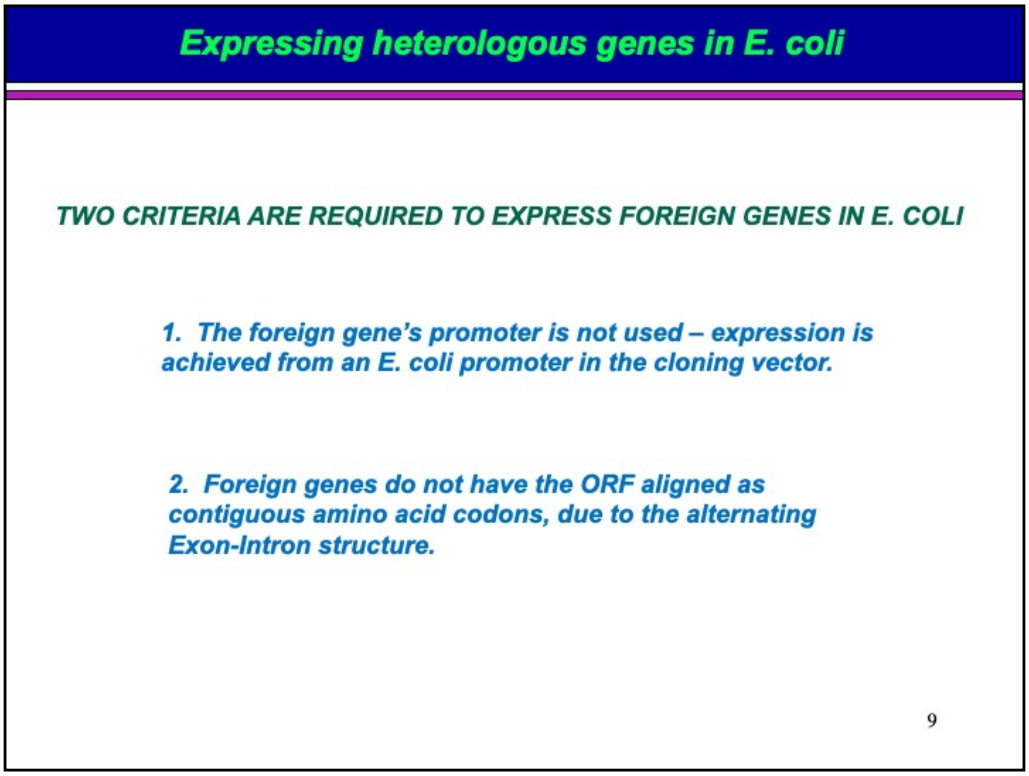
Note the position of the mRNA +1 nucleotide a little upstream from the initiation codon (AUG = methionine).



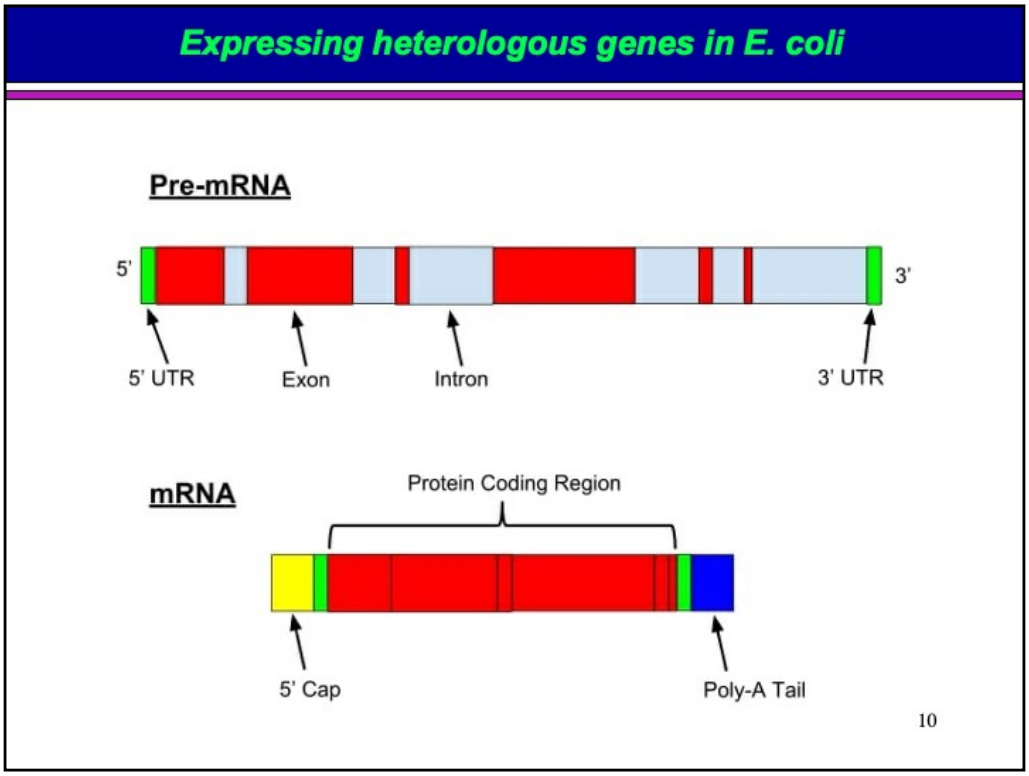
A cloned gene in a recombinant vector needs to be transcribed and translated successfully and efficiently in the *E. coli* host. This can only be achieved when the three criteria listed in this slide are met.

Each of these conditions will be dealt with in turn in the following slides. As you will see, the logistics of the cloning, regulation and expression all need to be attuned to the physiological requirements of the host bacterium in managing the production of large amounts of a foreign protein.

There is a measure of control/regulation of expression at all three stages, namely the gene, mRNA, and protein levels.

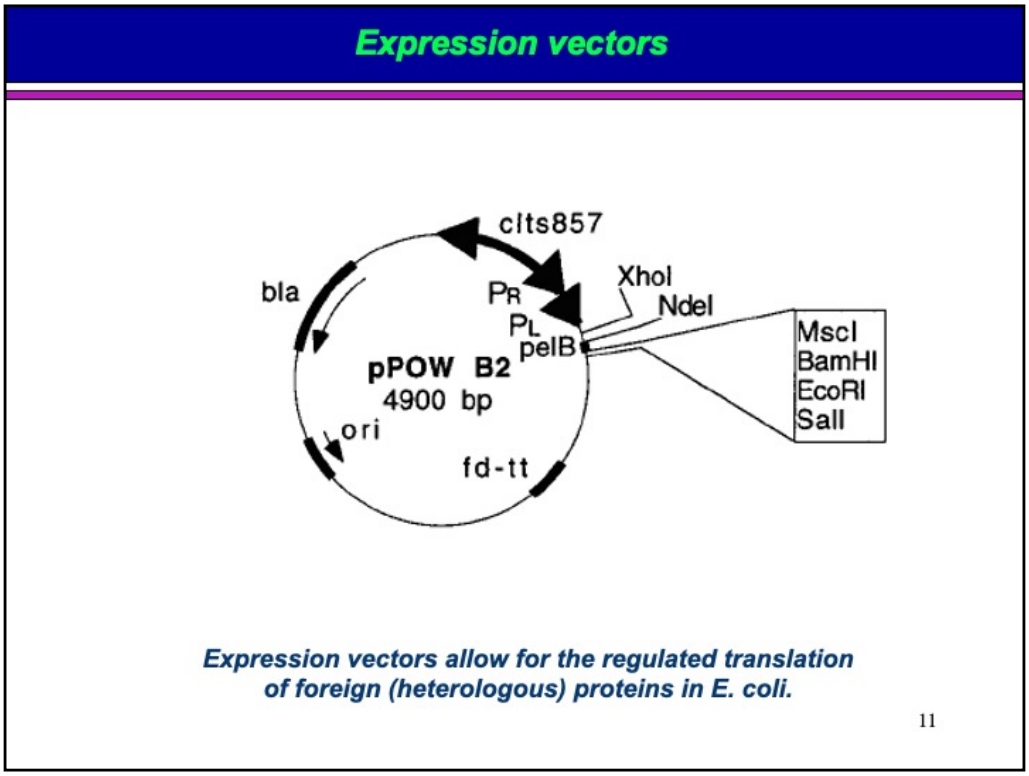


These are the two considerations required for expressing foreign (heterologous) genes in *E. coli*.



This slide depicts the primary or pre-mRNA (top) found in the nucleus after transcription, and the mature mRNA (bottom) which more closely resembles typical mRNAs from prokaryotes, with the exon cassettes joined to provide a contiguous ORF of amino acids.

The UTRs are typically found in eukaryotic genes and pre-mRNAs and they are not translated (UTR = untranslated region). The 5'-Cap and Poly-A tail are also typical for eukaryotic mRNAs.



Some typical features of an expression vector (pPOWB2), namely: The origin of replication (ori), an antibiotic resistance selection marker (amp), and a multicloning site for positioning the inserted gene.

The features that determine that this is an expression vector are: strong (tandem right/left lambda phage; P_R, P_L) promoters, the lambda repressor gene (cI ts857), and the transcription terminator (from the 'fd' bacteriophage; fd-tt). The tandem promoters are very strong and need tight repressor control to regulate expression, lest the accumulating protein compromise the physiological well-being of the bacterial cell, by slowing growth, or even triggering a 'death' response.

This repressor gene has a temperature-sensitive single site mutation at position 857 that substitutes one amino acid for another, making the repressor protein heat labile – it is fully functional at 32°C, partially at 37°C, as it gradually loses its conformational stability, and becomes fully dysfunctional at 42°C. Expression of a cloned gene is zero at 32°C, moderate at 37°C and at a maximum at 42°C.

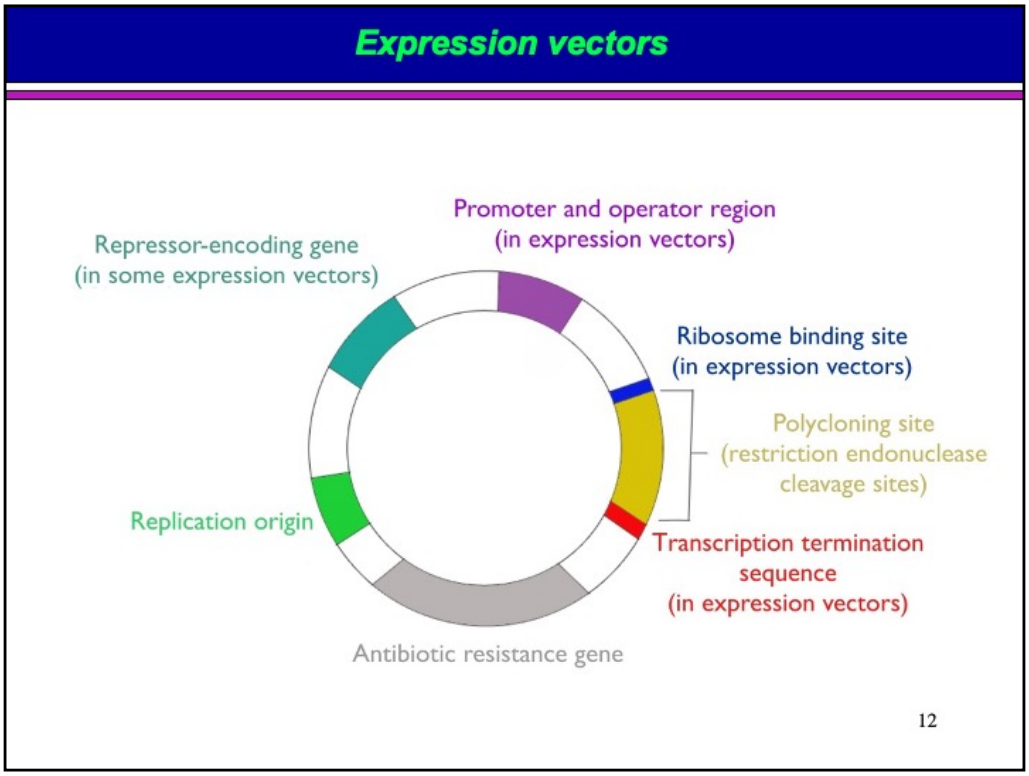
Why do expression vectors need regulated promoters? Three reasons:

(i) Strong regulated control of expression is required for efficient and productive translation (some bacterial promoters are weak);

(ii) Foreign genes cloned into *E. coli* via an expression vector will have promoters that will not work (from any species other than gram-negative bacteria); and

(iii) Many genes generated from PCR reactions or from mRNA – so called "complimentary DNA or cDNA – consist only of the open reading frame (ORF) of the gene and lack all other elements (promoter, ribosome binding site, transcriptional termination sequence).

Take home message: Expressing cloned genes in *E. coli* is a trade off between moderate and maximum protein expression. A simple analogy is a tap from which the flow of water may be nil, a slow drip, moderate or gushing flow.



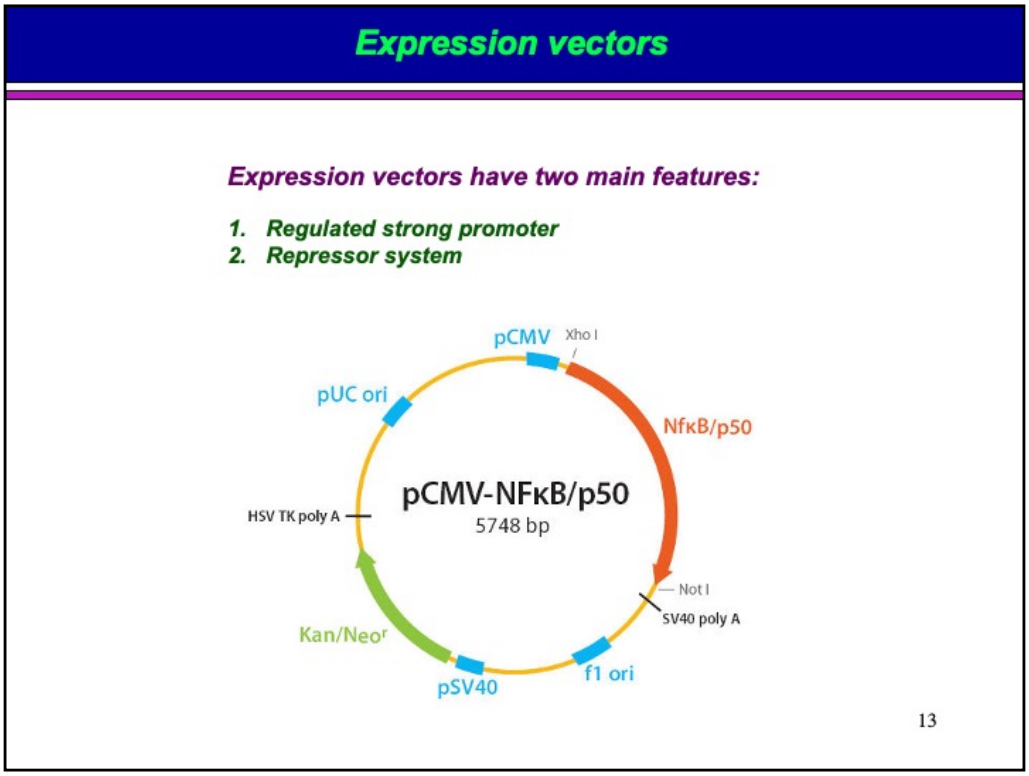
Typical features of an expression cloning plasmid vector with the main elements labelled, though not drawn to size.

Starting clockwise from the top, the promoter/operator is one that will work in *E. coli* (and is most commonly, but not always, the LacI/LacZ system that we studied in earlier lectures). The cloned gene will not have its own promoter, because most promoters (from other species) will not work in *E. coli*, and even those that do are best left out (deleted in the cloning phase) and the vector promoter used because it is regulated.

Expression is switched on by adding an inducer that titrates the repressor protein off the promoter/operator region, by altering the shape of the repressor so that it can no longer bind to the operator site. The repressor protein is unregulated, that is, it is produced constitutively (all the time) from the repressor gene (at 11 o'clock on the plasmid face above).

The RBS is needed at the 5' end of the cloned gene's ORF so that the mRNA transcript can bind to the ribosome for translation of the ORF into a polypeptide. The polycloning site (with a battery of restriction enzyme site choices) is in the yellow section of the plasmid, which is also where the cloned gene will be inserted.

The transcriptional terminator is required to both terminate the mRNA transcript, and prevent transcription read-through into the replication origin, which would destablilse and destroy the plasmid. Finally there are the antibiotic selection gene (in grey) and the replication origin to enable relaxed multiple copying of the expression vector.

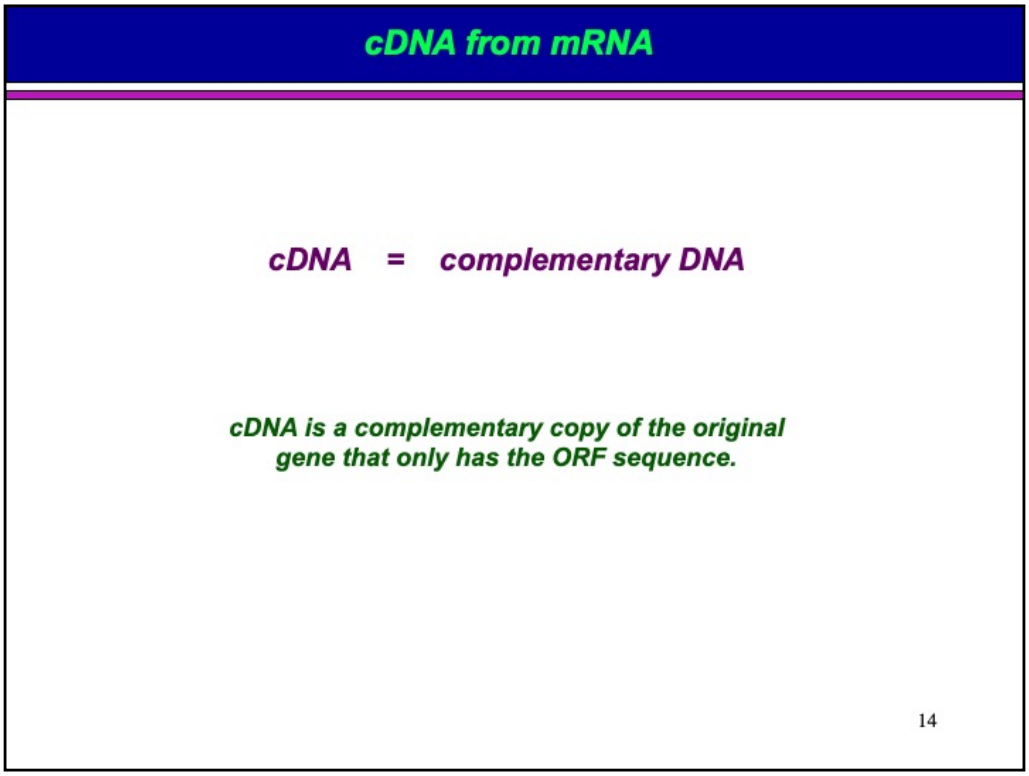


Expression vectors are specially designed cloning vectors for the cloning and expression of genes from any source – bacterial, vegetable, animal. Since most cloning vectors are used or passaged through traditional hosts such as *E. coli*, the expression vector requires a regulated promoter-repressor system.

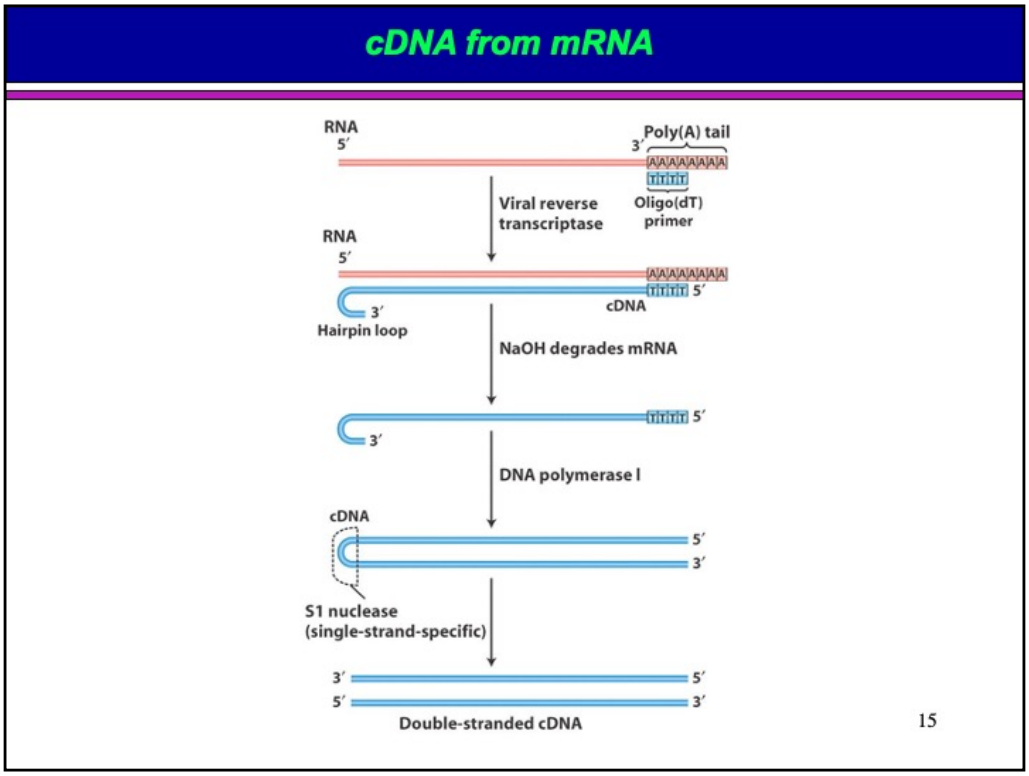
Many expression vectors are dual expression vectors – that is, they have promoters for both prokaryotic (pUC ori) and eukaryotic (pCMV) hosts so that foreign genes may also be expressed in eukaryotic host systems.

Either way, the regulated promoter system is required to switch the transcription of the cloned gene on or off and at different rates (like a tap). If the gene is overexpressed, it often will compromise the metabolic stasis of the host cells and cause the cells to slow their growth or even die, thereby making the cloning exercise a failure.

In addition, most gene promoters will not work across species (within and between prokaryotic and eukaryotic hosts) and therefore, expression vectors carry their own 'generic' regulated promoters that will work in bacterial and higher expression host systems, such as the example above.



cDNA is the copied gene ORF that will be ligated just downstream from the vector's promoter region. The cloned gene will not have its own promoter. Why cDNA? Look at the second item in this slide.



Traditional method of preparing cDNA from mRNA – note that eukaryotic genes cannot be used in cloning and expression, as they have an intron-exon arrangement of 'cassettes', with introns disrupting the collinear open reading frame (ORF) sequence of amino acid codons.

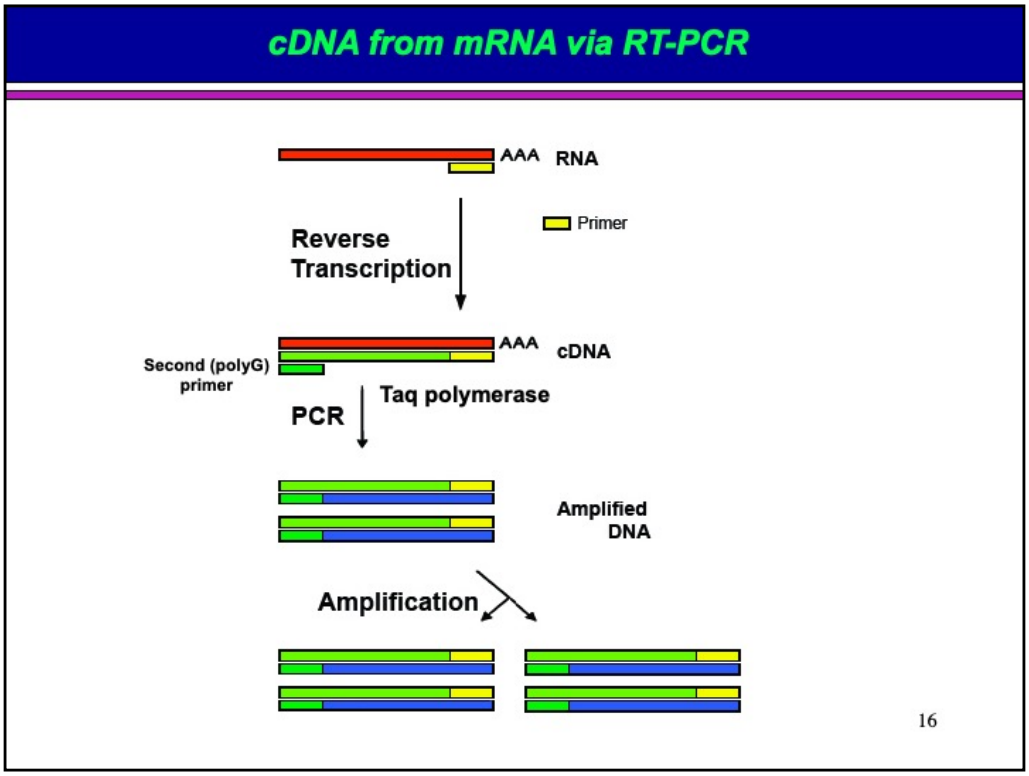
The mature mRNA is spliced from the much longer pre-mRNA and has only the ORF sequence and is also promoter-less.

In the figure above, the mRNA is copied using an RNA-specific 'reverse transcriptase' (an RNA-dependent DNA polymerase) and a 'universal' oligo dT primer, which is paired to the polyadenyate (AAAAAAA….) tail of mRNAs in eukaryotic genes.

The oligo(dT) primer is extended into a DNA complementary strand against the mRNA template. The mRNA is then degraded using mild alkali or digestion with ribonuclease H.

The first DNA strand, with the hairpin 3'-end serving as an internal primer, is copied (by complementary base pairing) into the second DNA strand, this time by a conventional DNA polymerase. The hairpin is then cut by S1 nuclease, leaving a complete cDNA copy of the mRNA and therefore of the original gene (intron-free) with the exons contiguous as the ORF.

This method may be used for prokaryote or eukaryote mRNA sources.



This slide combines the normal method of cDNA synthesis (from mRNA) with PCR amplification in order to generate a large number of cDNA copies from many different mRNAs so that a library of cDNAs may be cloned.

The first cDNA strand is synthesised from mRNA by reverse transcriptase, using polyT primers. A special MMLV reverse transcriptase is used, as it adds a few 'Cs' to the 3'-end of the newly synthesised cDNA strand, which though not visible above, is positioned in the new (green) strand on the left, where a polyG primer binds ahead of the second strand synthesis.

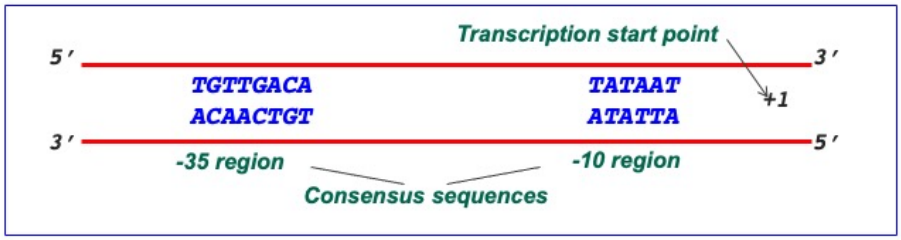
A thermostable DNA polymerase copies the new template DNA strand into a complementary DNA strand and continues during the PCR amplification of template strands, by using alternate polyG and polyA primers for each of the two cDNA strands, as shown at the bottom of the slide.

Note that this method of preparing cDNA, though different to the reverse transcriptase method shown on the previous slide, is otherwise similar in generating cDNA 'libraries' (of ORFs) from the population of mRNA transcripts. Either way – from these two slides - of generating cDNA, is followed by ligating the cDNAs into expression vectors.

(1) Factors affecting expression of cloned genes - promoters

Expression vectors contain strong regulated promoters:

1. ***lacI-lacZ (induced by IPTG)***
2. ***trp-lac (tac) (induced by IPTG)***
3. ***c_I-$P_R$$P_L$ (induced at 42°C)***



Strength of a promoter depends on degree of consensus and distance between regions.

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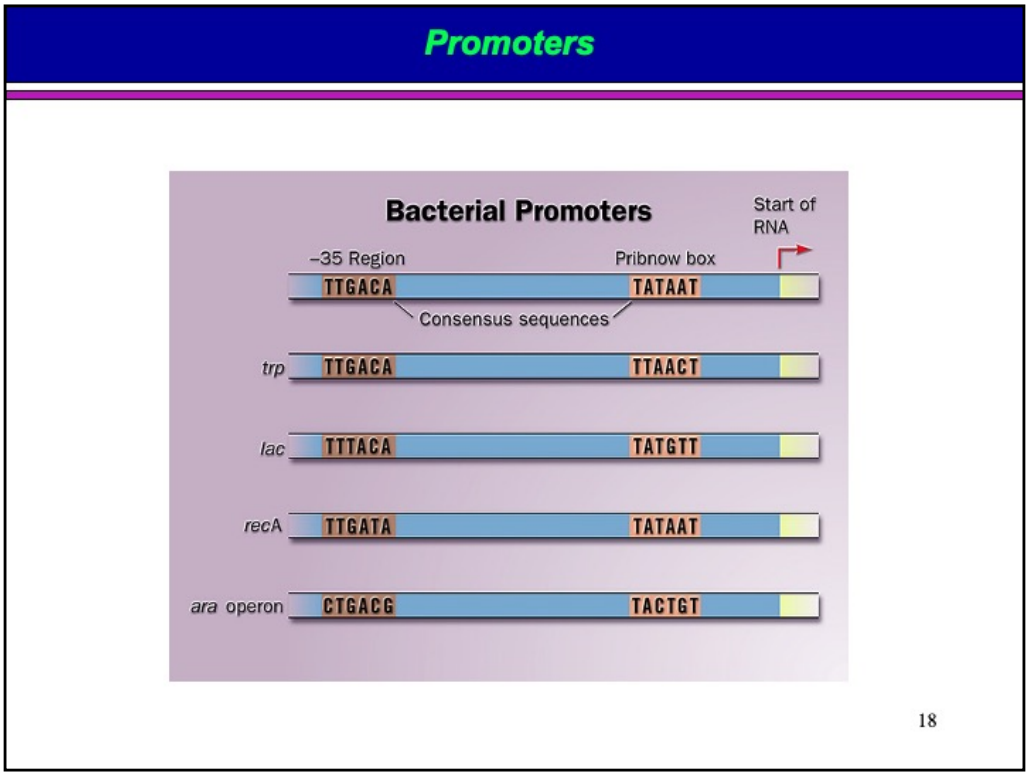
Three typical promoters used in cloning (expression) vectors are shown in this slide.

The first (lacI-LacZ) is found in most vectors (including pUC18/19) but is a little 'leaky', meaning that a basal level of transcription is always occurring despite the lacI repressor inhibition at the promoter-operator site. Ditto for the hybrid trp-lac (tac) promoter that is several-fold stronger than the purely lac promoter. Lastly the lambda system found in only a few expression vectors is very tightly controlled, with no expression at lower temperatures (30 to 35°C; tap is off!) but increasing expression from 37° to 42°C (tap is on, from moderately fast flow to gushing).

The bottom half of the slide summarises a revision point about the strength of promoters being aligned with the degree of consensus of the RNA polymerase consensus binding motifs at minus 10 and minus 35.



The two consensus sequences are spaced at the correct distance for the interactive binding of RNA Polymerase (RNA Pol) that will begin the transcription of the gene sequence. These consensus nucleotide bases either make or align non-covalent interactions between the bases and RNA Polymerase.

'Consensus' means that many genes have variations of these sequences and the differences decrease the number of binding contacts with RNA Polymerase, making the Promoter less effective or weaker in that transcription is not as vigorous as for the consensus sequences.

The net result is fewer transcripts and fewer copies of the polypeptide. This is a form of regulation in that cell functions that require many copies of certain proteins (e.g. repair proteins; haemoglobin etc) will have strong promoters, while weaker promoters are used for genes that only need to produce small numbers of a specific protein (e.g. DNA polymerases, chromosome replicating and repair enzymes).

Of course, for purely expression purposes, a regulated strong vector promoter-repressor system is used with the cloned ORF amino acid codons (either as a cDNA, or prokaryotic gene) immediately downstream.

(2) Factors affecting expression - Ribosome binding site

The effect of the RBS on the efficiency of translation may be affected by:

1. *The degree of complementarity between RBS and 16S rRNA*
2. *Spacing between RBS and AUG codon*

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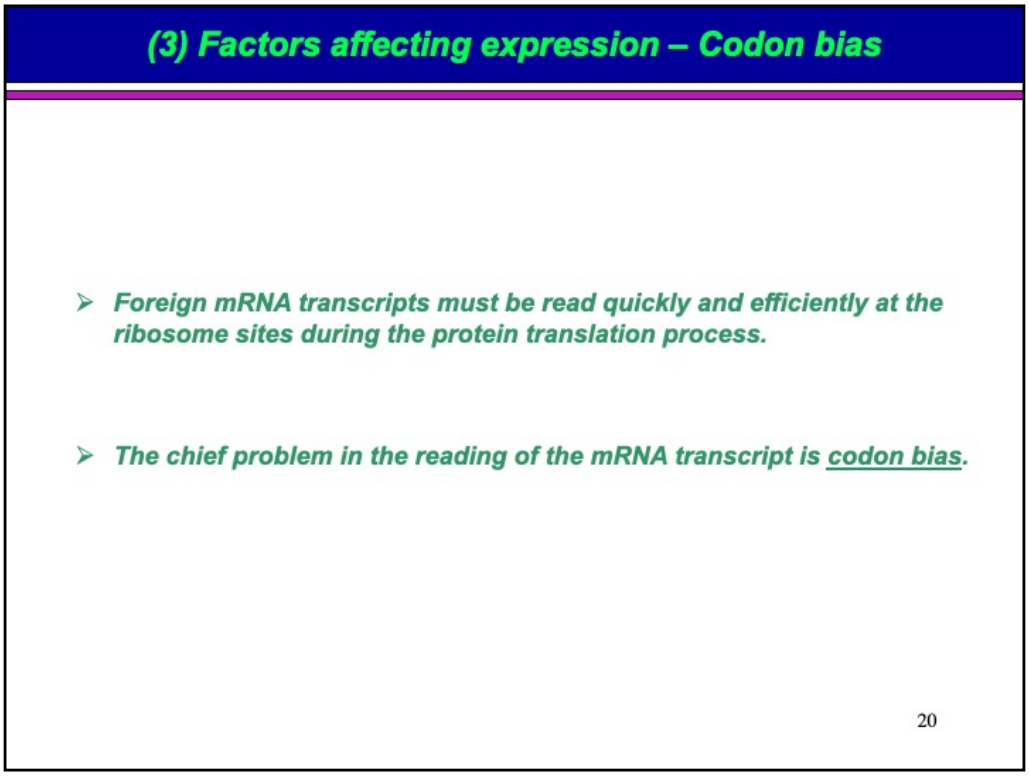
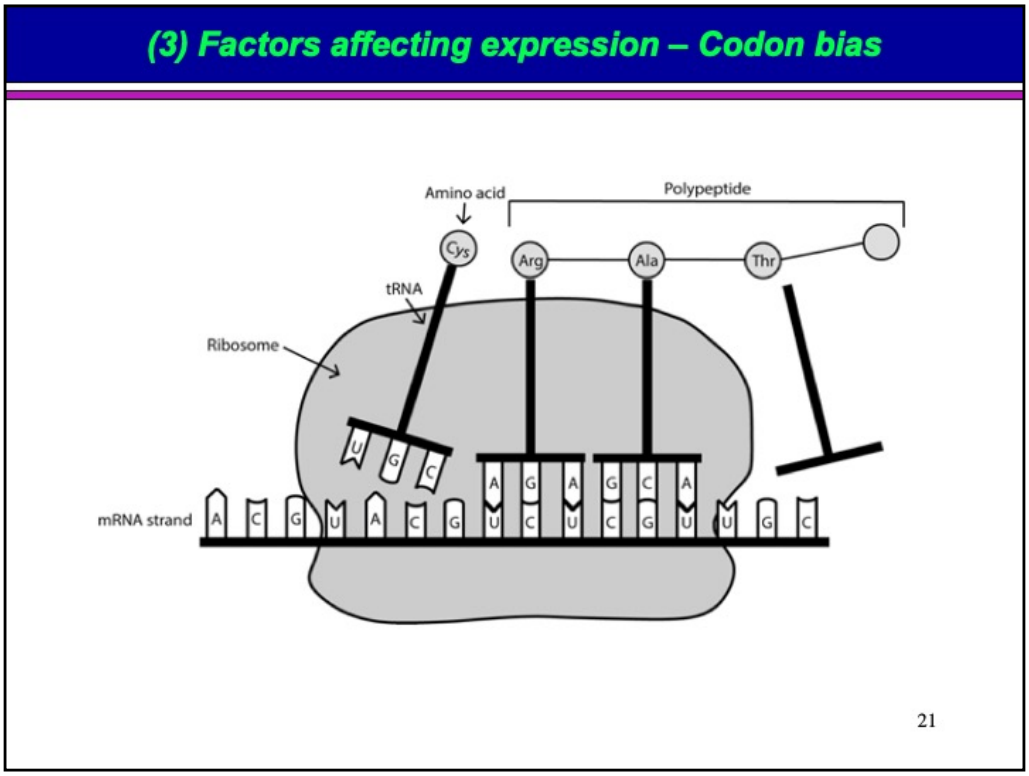
The position of the ribosome binding site (RBS) and initiation codons are shown in relation to the +1 nucleotide of the mRNA transcript and the upstream promoter (not shown, to the left).

Translational expression of the cloned gene is also subject to efficiency 'rules' indicated in the two points (bottom of slide) that govern the RBS and initiation codon sequences. Higher protein expression will result from the closest adherence to these rules.



Transcripts of mRNA must be read/translated quickly. Any pause or stoppage in the rate of translation will result in ribonuclease attacking the mRNA and proteases attacking the nascent polypeptide in order to remove them from what the cell perceives as a 'dysfunctional' ribosome/translational assembly. In other words, clear the blockage and carry on elsewhere.

The chief cause of pauses or stoppages in reading mRNA transcripts is **codon bias**, which is a result of evolutionary divergence between prokaryotes and eukaryotes. To understand what this is, we need to revisit the Genetic Code, amino acid codons, and tRNAs, on the next few slides.

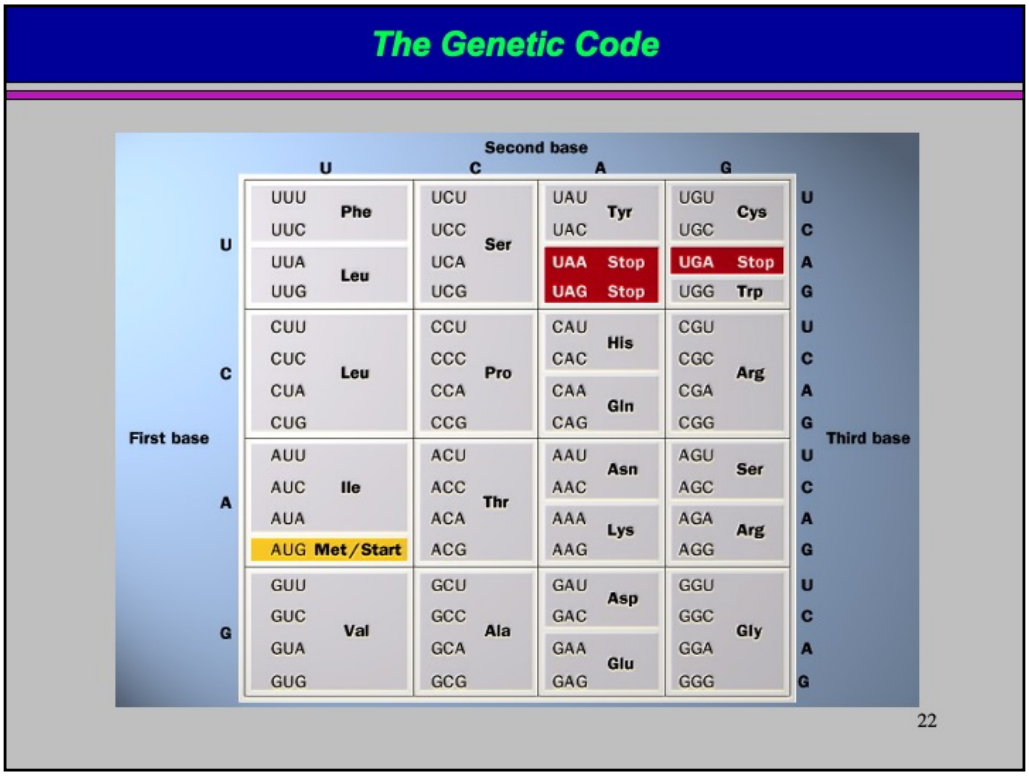


This is a simple diagram of the translational process of polypeptide synthesis.

As the mRNA is threaded between the ribosome subunits, amino acid bearing tRNAs are captured and positioned with their anticodons binding to the complementary mRNA codons. The amino acids at the other end of the tRNAs are then joined by peptide bond synthesis to the growing amino acid chain.

Peptidyl transferase is an enzyme that catalyzes the addition of amino acid residues to synthesise the polypeptide chain in protein synthesis. It is located in the large ribosomal subunit, where it catalyzes the peptide bond formation between two bound tRNAs (Arg-Ala in the figure above).

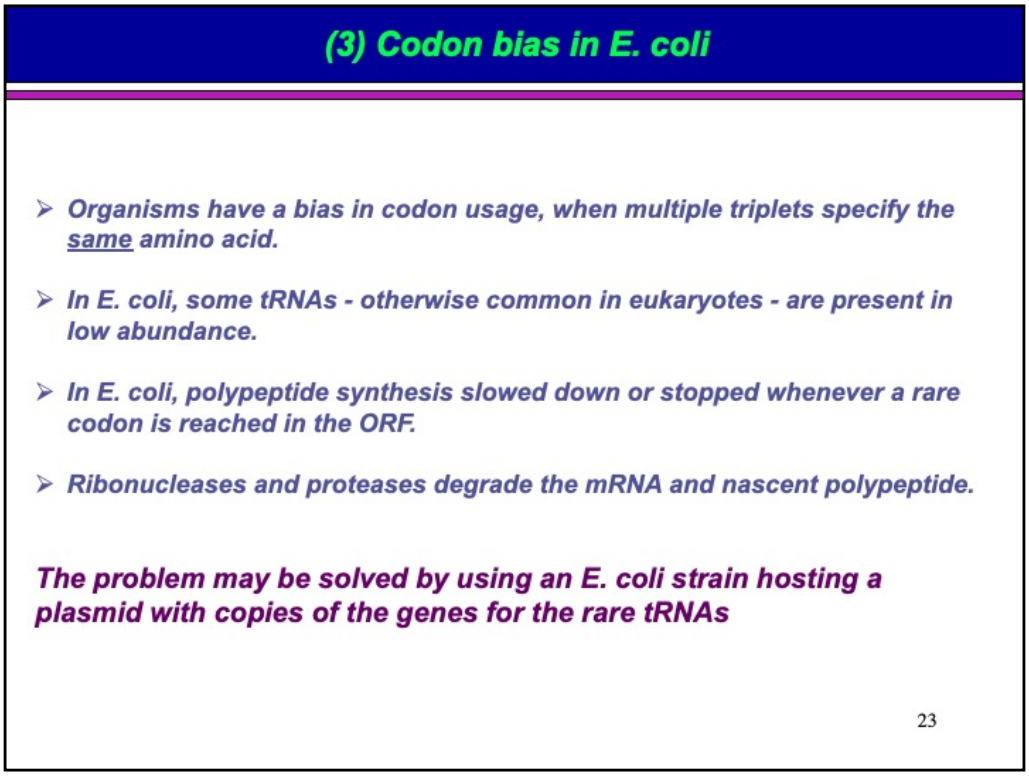
Despite the 'enzyme' name, peptidyl transferase is actually a catalytic RNA molecule, a so-called ribozyme, and not a proteinaceous enzyme.



The 'Genetic Code' of triplet codons of which there are a total of 64 (4x4x4).

As you can see, there are three stop or nonsense codons (UAA; UAG; UGA) and a single start codon (Met). Tryptophan also has a single codon (UGG).

However, the remaining 59 available codons are distributed among the other 18 amino acids, resulting in a range of codon numbers for each amino acid (from two to six), making the Code 'degenerate', meaning, in this case, more two or more codons for each amino acid among the extra codon choices. How this Code is used and 'adjusted' in molecular cloning will be made apparent on some of the following slides.



There is an amino acid 'codon bias' in the Genetic Code distribution of amino acid tRNAs in different species – a consequence of evolution and the relatedness and/or separation of species in time.

This means that most species have a propensity to use only some of the multiple (redundant) codons available and therefore some tRNAs are less or more abundant than others. The consequence of this is that when a foreign gene is to be artificially transcribed and translated in *E. coli*, some codons from the foreign gene are not abundantly represented, if at all, in the new (*E. coli*) host. This results in stalling at the ribosome during translation of the mRNA into a polypeptide, as the rare or absent tRNA is not there to add the next amino acid to the growing protein chain quickly enough.

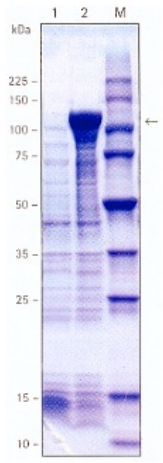
The cellular physiological stasis is set up to recognise this translational stalling as a problem or deficiency that must be corrected so that normal metabolic processes may be resumed. As a result, ribonuclease and protease attack the stalled mRNA and polypeptide, respectively, and they are progressively digested so that the ribosomes may be disassembled and reused.

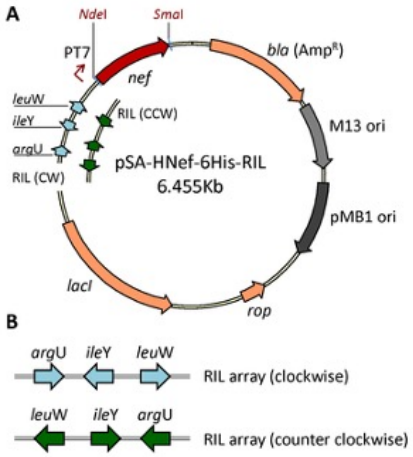
Take home message: Any cloned gene subject to this stalling process will therefore either not be expressed as a protein or will be expressed in very poor yield. The solution is a simple one – special *E. coli* strains for molecular cloning are provided with a resident plasmid containing copies of the genes for rare *E. coli* tRNAs.

(3) Codon bias in E. coli

In E. coli, eight tRNAs are rare.

Amino acid	Codon	Fraction in all genes
Arg	AGG	0.022
Arg	AGA	0.039
Arg	CGG	0.098
Arg	CGA	0.065
Arg	CGU	0.378
Arg	CGC	0.398
Gly	GGG	0.151
Gly	GGA	0.109
Gly	GGU	0.337
Gly	GGC	0.403
Ile	AUA	0.073
Ile	AUU	0.507
Ile	AUC	0.420
Leu	UUG	0.129
Leu	UUA	0.131
Leu	CUG	0.496
Leu	CUA	0.037
Leu	CUU	0.104
Leu	CUC	0.104
Pro	CCG	0.525
Pro	CCA	0.191
Pro	CCU	0.159
Pro	CCC	0.124





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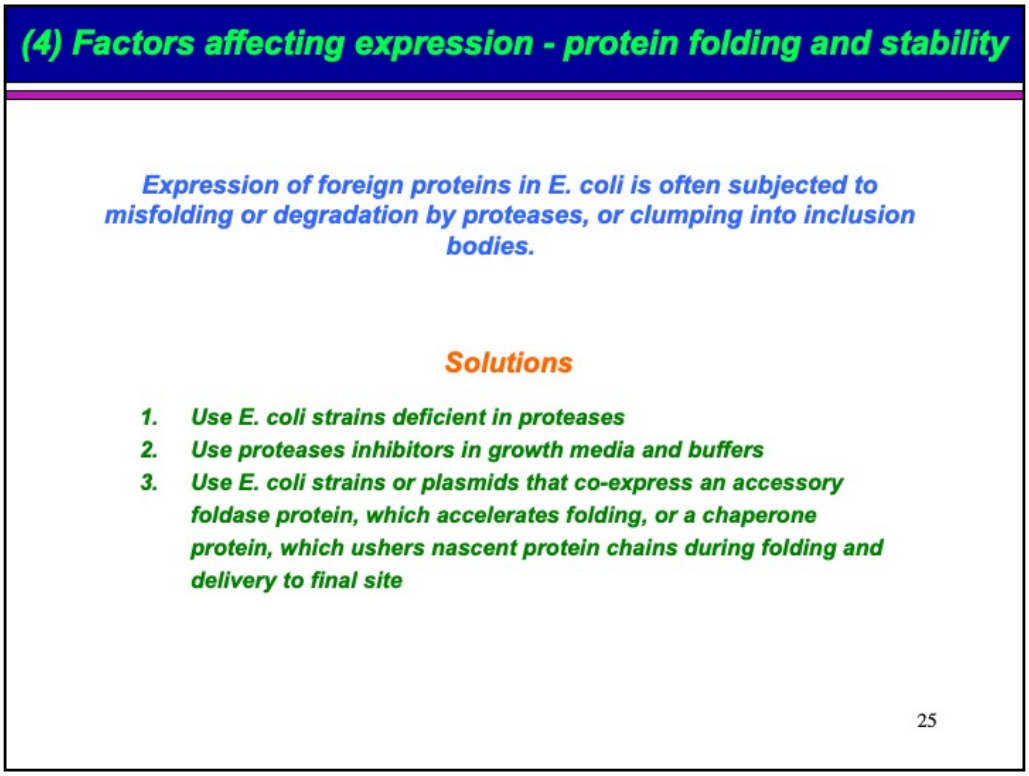
Codon bias in *E. coli* is illustrated in this table (left), in which eight codons are of low to medium abundance and therefore are represented by low abundance tRNAs (shown in red). This low abundance effect causes stalling of translation and degradation of mRNA and the nascent (newly synthesised) polypeptide.

The gel in the middle of this slide shows degraded protein in lane 1, where the codon bias effect has stalled translation. Lane 3 contains molecular size marker proteins. In lane 2, a significant band for the complete protein is visible at about 120 kDa; and this excellent result was achieved by using the available cloning tools, namely helper plasmid tRNA genes with the rare tRNAs expressed, with one such helper plasmid vector shown on the right.



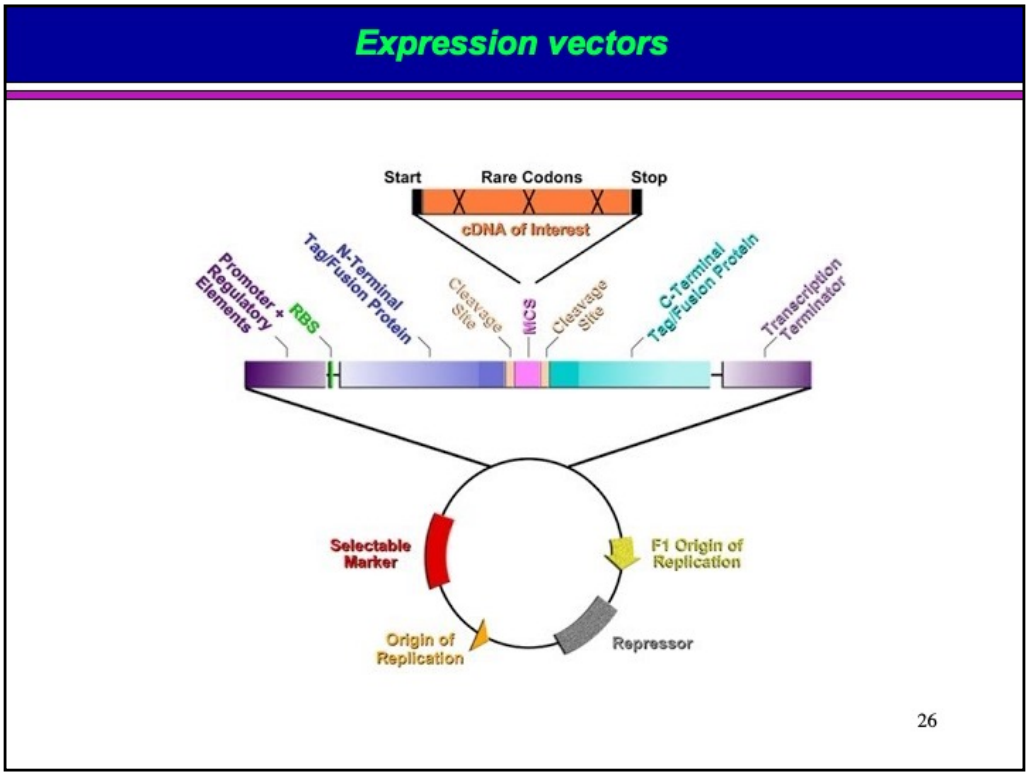
We have looked briefly at ways of maximising expression with promoters and at the stability of mRNA transcripts and codon bias/translation efficiency. We turn now to the last stage of efficient heterologous protein expression in *E. coli*, namely avoiding protease degradation of the translated protein.

Pre-1995, efforts to clone and express heterologous (foreign) genes in *E. coli* were beset by a number of problems. The two major obstacles were the degradation of foreign proteins by host cell proteases and/or the rapid accumulation of excess foreign proteins into small, aggregated, insoluble particles called inclusion bodies. If the amount of translated (foreign) protein begins to accumulate, degradation is still possible due to the protein, being foreign, accumulating too quickly, or not folding quickly enough.

The solutions to these problems were relatively easy. Proteases inhibitors added to the growth medium would enable foreign proteins to survive protease digestion. Clumping into inclusion bodies was more difficult to avoid, but these inclusion bodies could be recovered and dissolved in detergents to obtain the intact protein (see below).

More recent commercial innovations have made new *E. coli* strains available specifically for expressing foreign proteins. These strains are deficient in proteases and carry a plasmid that expresses a foldase protein that accelerates folding and therefore reduces the occurrence of inclusion bodies; and of protease digestion, if the *E. coli* strain is not protease deficient.

Alternatively, one can recover inclusion bodies and use a commercial foldase kit to recover the protein. These kits have strong denaturants to solublise the inclusion bodies, oxidising and reducing agents, and a foldase enzyme. While this is a 'hit-and-miss' proposition, the process is often successful.

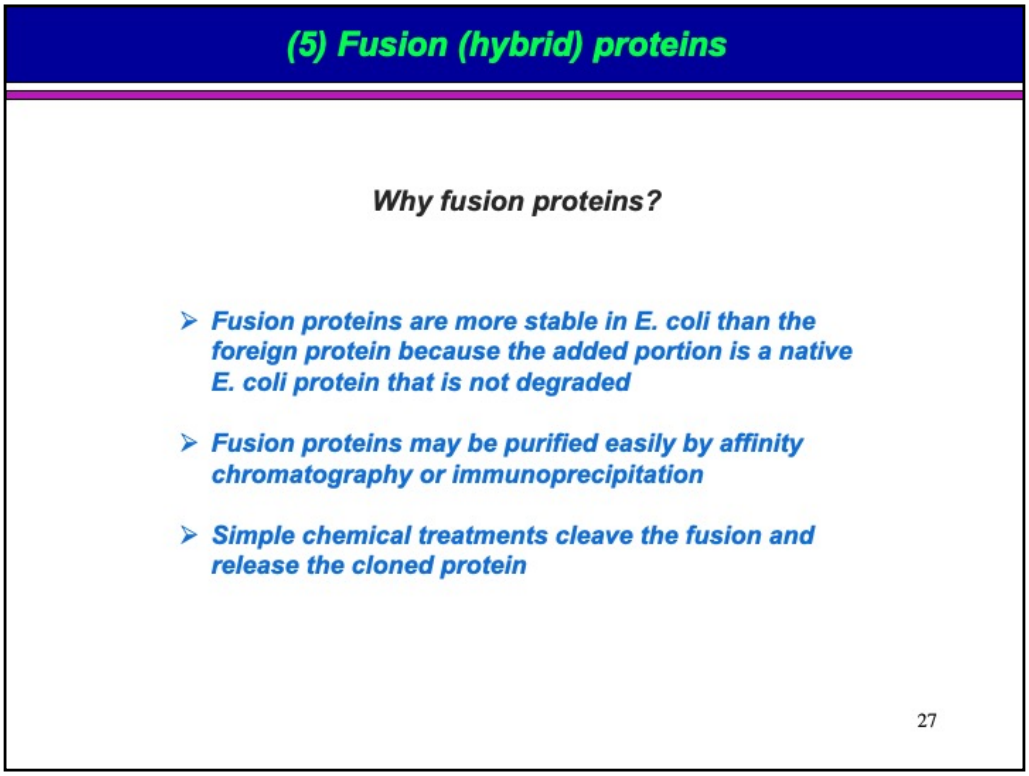


Typical features of a 'sophisticated' expression cloning plasmid vector, showing all possible elements of the plasmid, though not all will be present in all expression vectors.

The inserted cDNA is shown at the top (orange) with rare codons marked. The codon deficiency is solved by employing a tRNA helper plasmid in the *E. coli* host, as described on a previous slide.

Cleavage restriction sites are shown either side of the inserted cDNA for easy removal of the insert and transfer to another vector.

The N- and C-terminal fusion protein junctions either side of the insert (purple, green) and MCS (pink) are fusion hybrid options for the cloned cDNA ORF polypeptide. This hybrid method will be explained in the next few slides.



Fusion (hybrid) proteins are a common option in molecular cloning and expression. They are useful for two reasons:

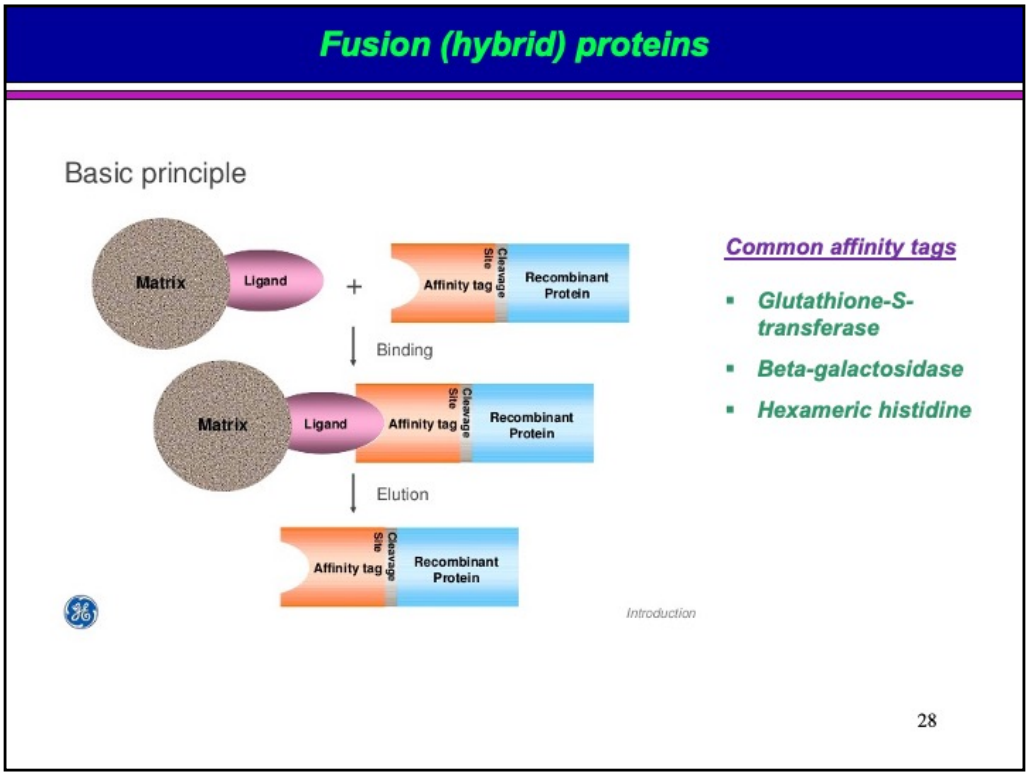
Firstly, the *E. coli* native protein that is fused N-terminally to the protein of interest is recognized by *E. coli* as 'self' despite the foreign protein adduct; and this means that protease digestion is avoided and proper folding is more likely, as the native protein will fold correctly, allowing time for the foreign half of the hybrid polypeptide to do the same.

Second advantage of fusions is the ease of purification that uses a monoclonal antibody or a ligand to purify the hybrid recombinant protein by immunoprecipitation or affinity chromatography, after which the hybrid may be cut precisely at the junction of the two proteins, thereby releasing the free and purified foreign protein.

This form of cloning and expression is especially useful and popular in instances where the foreign protein is either unknown or does not have a ligand or antibody for purification proposes.

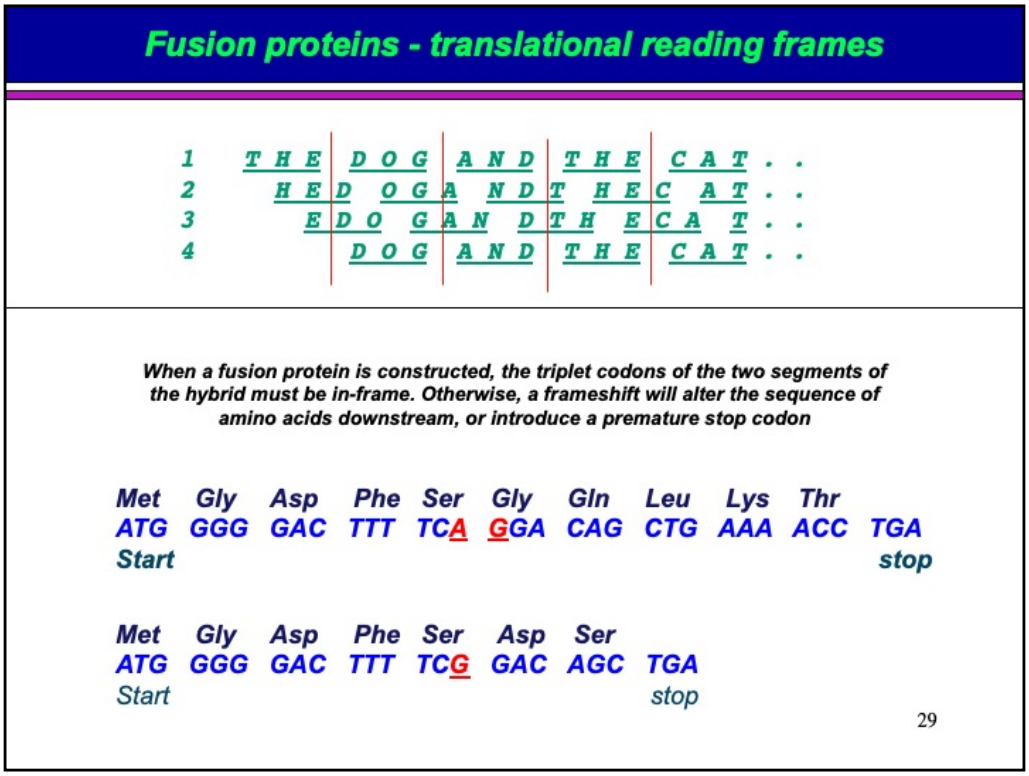
Another feature of hybrids is that some can be directly assayed for expression levels by a reporter enzyme reaction afforded by the *E. coli* fusion protein (e.g. beta-galactosidase may be assayed by blue indole production from X-gal).

Some fusions are designed only for purification purposes and these are not true protein hybrids, but rather are short peptide adducts (N- or C-terminal) that bind to affinity chromatography columns (e.g. a polypeptide with a 6x His adduct will bind to Nickel beads in an affinity column, then be eluted with imidazole).



A simple scheme to indicate the affinity purification of a recombinant, fused (hybrid) protein, using beads with an attached ligand that will bind the tagged target protein.

The affinity tag may be one of several candidates depending on the identification and purification strategy, but the major tags are reporter proteins which may be assayed to indicate quantification of expression, such as Glutathione-S-Transferase, or beta-galactosidase, or a tag just for the purification process, such as a hexameric histidine tag (6x His) (see slides further on).



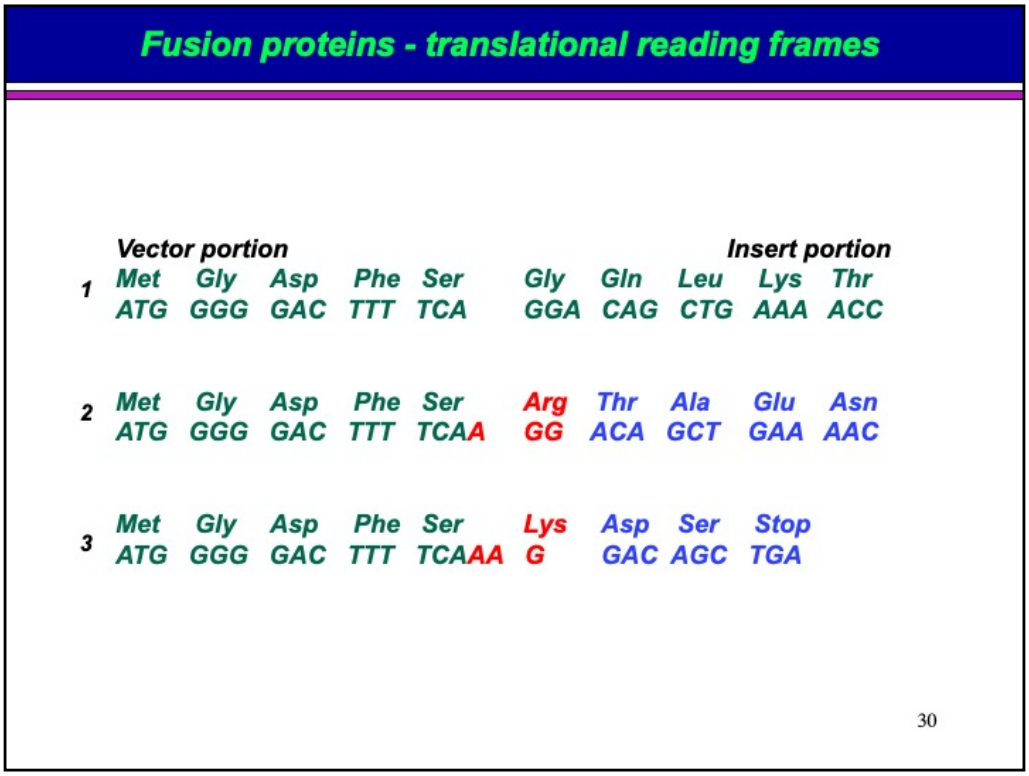
How are fusions made? There are two simple aspects to a fusion hybrid scheme:

Firstly, the genes for the native and foreign proteins need to be placed in tandem and contiguously in the recombinant plasmid. The native protein reporter gene or short adduct is already in the specially constructed cloning expression vector. The foreign protein (usually as an ORF cDNA) will be introduced at a MCS that is located at the 3-prime end of the native gene/adduct (C-terminal to the vector ligand/adduct).

Secondly, there is an absolute requirement that the foreign and native genes/adducts must be joined in-frame to one another; that is, with the triplet amino acid codons of the native gene or adduct continuing in codon (triplet) frame into the foreign gene. The stop codon for the native gene is missing so that the new hybrid will be contiguous – as an uninterrupted 'double' polypeptide, with the stop codon being supplied at the end of the foreign fused protein.

This slide refreshes our knowledge of what it means for protein primary sequences to be in-frame. You can see how the frame (sense) of a sentence changes in the top half of this slide.

This is illustrated for an actual amino acid sequence in the bottom half of the slide. If the first codon sequence loses the 'A' in the third position of the Serine codon (underlined and in red), then the codon frame shifts one base to the right (with the first 'G' from Gly replacing the deleted 'A'), producing the new and shorter sequence shown at the bottom. Here, the first G (underlined and in red) now becomes the third base in the new TCG codon that also happens to be a Serine codon, as occurred in the top sequence, but the sequence frame changes drastically downstream, with different amino acids specified by the frame shift and, in this example, an earlier stop codon is introduced.



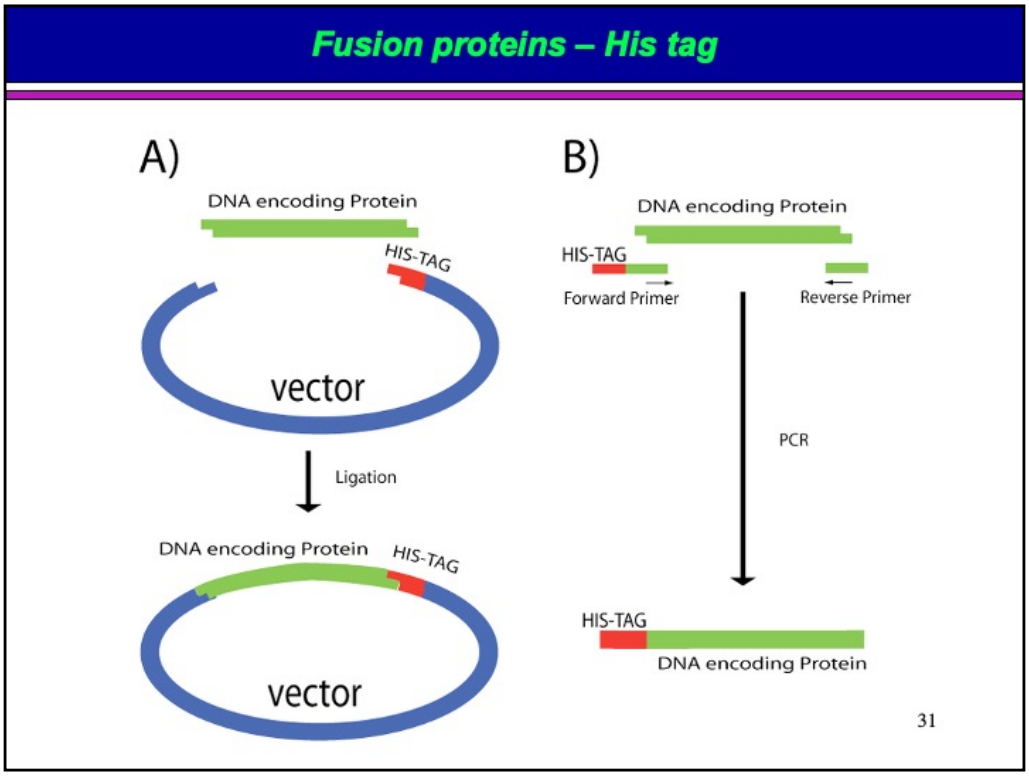
Commercial cloning vectors for generating hybrids or fusions are provided in sets of three, with each cloning junction in one of the three possible reading frames.

Since the junction of the native and foreign portions cannot be guaranteed to be precisely contiguous, we can ensure that they are joined in-frame by performing the cloning steps in all three vectors, only one of which will be in the correct in-phase triplet frame. How do we know which one it is?

When the three recombinants are constructed and expressed, SDS-polyacrylamide gel electrophoresis will identify the correct hybrid by its molecular mass position in the gel, or by assaying the activity of the reporter protein. The two out-of-frame fusions will not produce a hybrid protein. If all of the DNA sequence around the fusion site is known, you can easily work out which of the three versions will be in frame and use only that vector.

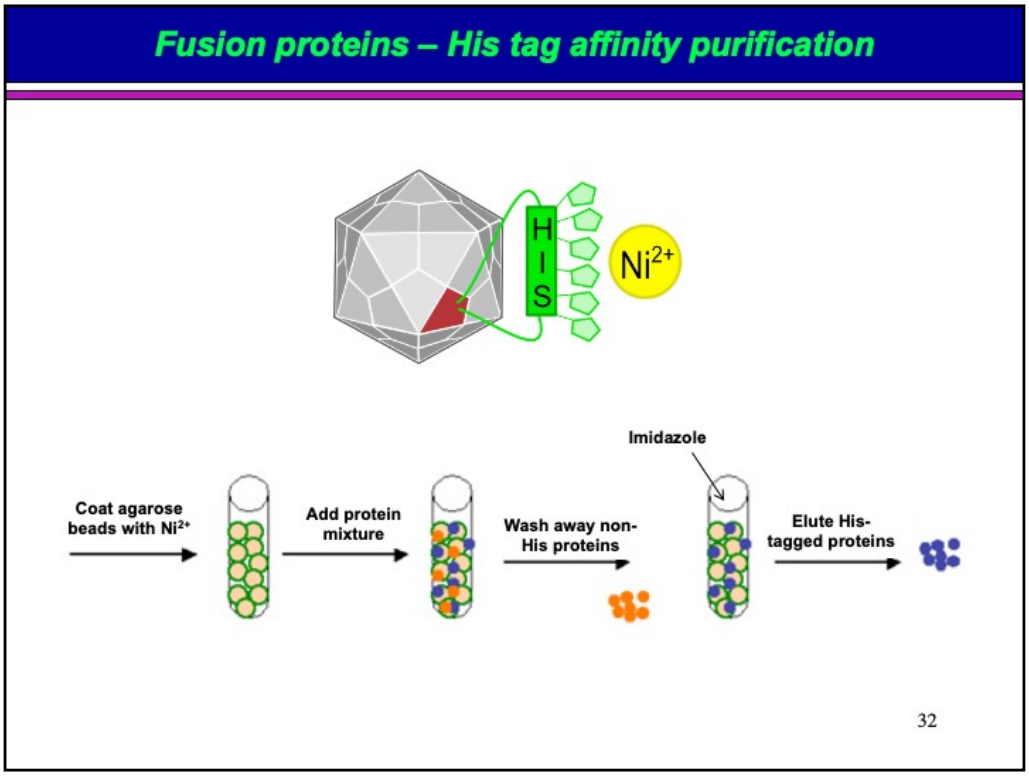
In this slide, we can see how a frameshift will affect the fusion junction of the hybrid proteins, for which the vector (left portion) provides the native codons (only the last few C-terminal codons are shown) and the insert portion provides the foreign codons (only the first few N-terminal codons are shown).

The Serine-Glycine junction in the first line is the contiguous in-frame read through that generates the correct hybrid of native and foreign fused proteins. The second and third lines show the difference in the codon read through if the junction is one or two bases out of frame. These make the foreign protein sequence unrecognizably incorrect.



Two ways of cloning a His-tagged gene. In (A), the gene is ligated directly into a suitable expression vector with a 6x His tag in the vector (note, all three frames need to be checked for continuity of the ORF with the His sequence, as we saw in previous slides).

In (B), a His tag may be added to the N- or C-terminus of a gene using a PCR reaction, where one of the primer pairs used contains the 6x His sequence on the end of one of the primers. The amplified and His-tagged sequence may then be cloned into an expression vector.

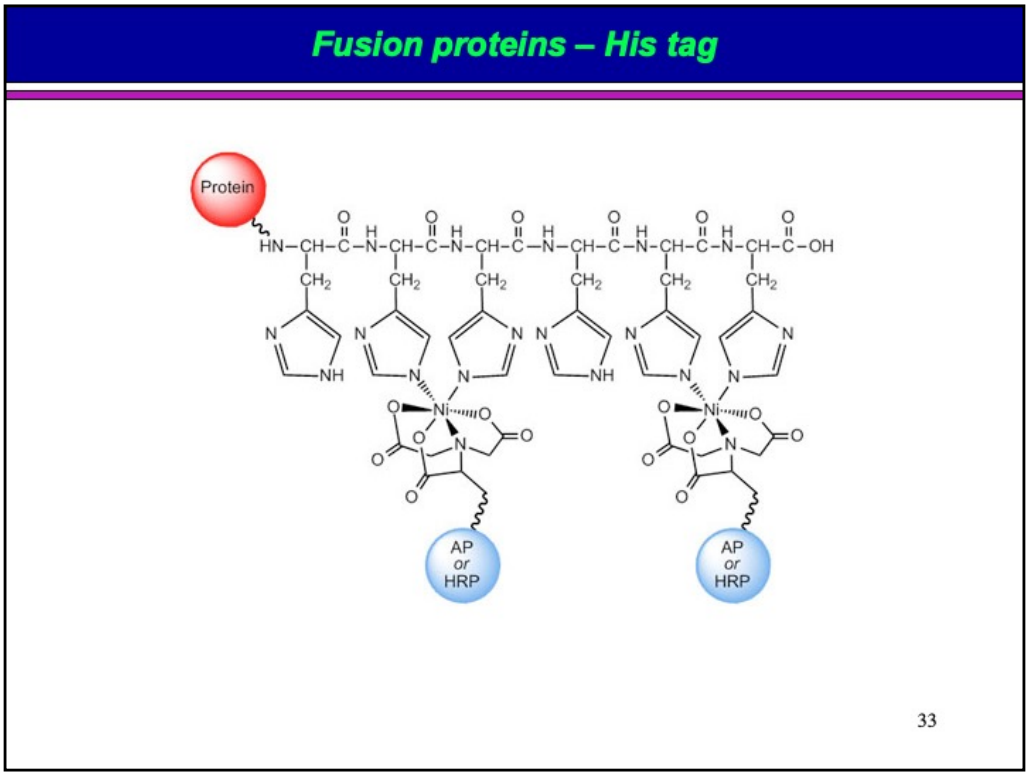


A His-tagged protein (top) may be easily purified from cell lysates by nickel affinity chromatography (bottom).

Although histidine has a side chain that bears a weak positive charge, and nickel is also positive (Ni^{2+}), it is the (non-charged) imidazole ring of the histidine side chain that binds to the Ni^{2+} beads in a column.

Non-His adsorbed proteins (orange) are washed through with an excess of column buffer.

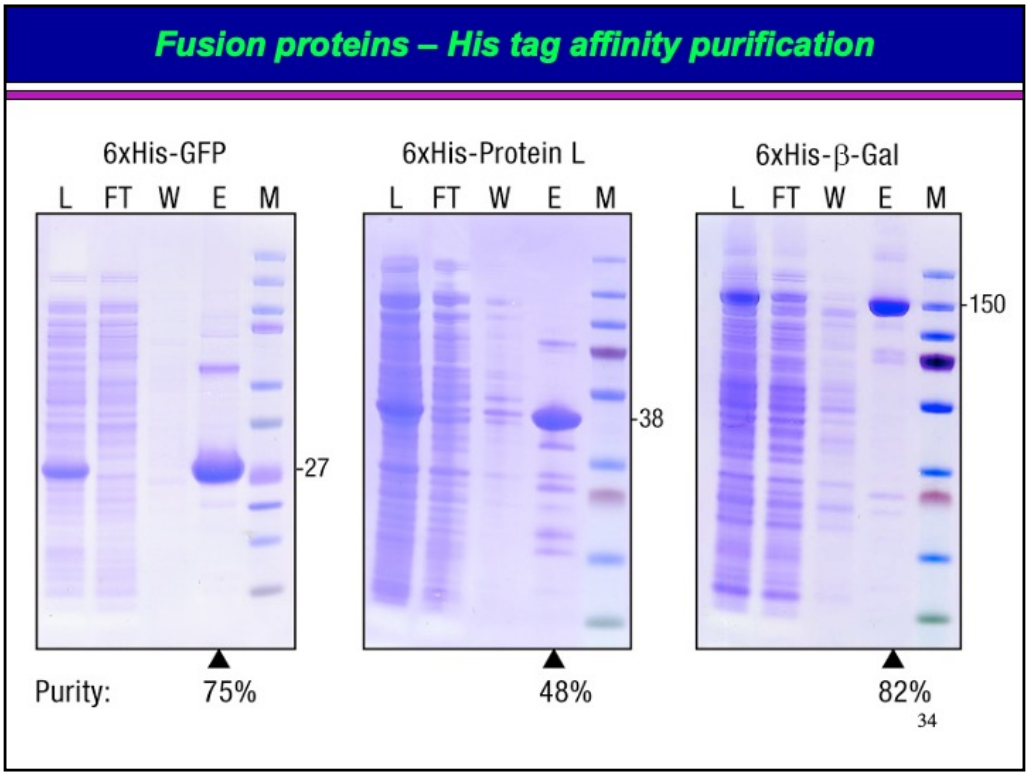
The bound purified protein is then eluted from the beads by using buffer containing an excess of imidazole (the histidine side chain as a separate compound) that saturates the Ni^{2+} beads by counter binding in place of the now released -6xHis hybrid protein (blue).



The most common method of producing fusion proteins for affinity purification is to fuse a hexameric (6x) His tag to the N- or C-terminal portion of the cloned protein of interest, as depicted in this slide.

There are simple ways to detect the tag such as indicated here with signature blue colour or UV light produced from substrates that react with bound (tagged) alkaline phosphatase (AP) or horse radish peroxidase (HRP).

The tagged protein may be monitored and purified from a cellular lysate of many proteins using a column purification process, and by running samples in polyacrylamide gels and detecting the His-Tag by immunoblotting with anti-His monoclonal antibody (next slide).

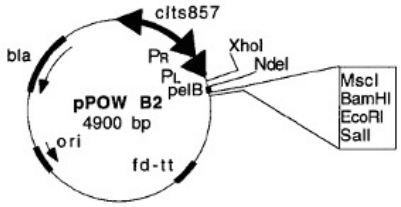


Three examples of purified His-tagged proteins from cell lysates on an affinity column.

L is lysate; FT is flow through; W is wash; E is imidazole eluted protein; M is MW standards.

Note the enhanced concentration/purity of the tagged proteins in the 'E' lane of each example.

MCQ L10



The two features of this plasmid that identify it as an expression vector are the….

(A) Repressor gene and multicloning site.
(B) Ampicillin resistance gene and promoter-operator.
(C) Repressor gene and promoter-operator.
(D) Multicloning site and promoter-operator.
(E) Repressor gene and transcriptional terminator.

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MCQ L10

Fusion or hybrid recombinant proteins are used in molecular biology chiefly as a means of stabilizing the foreign protein in the bacterial host, or for easy purifications. C- or N-terminal portions of a reporter protein or His-tag sequence are provided in the vector. The vector is actually supplied in three versions and this is because….

(A) It provides more options for the cloning strategy.

(B) Three vectors provide a wider range of multicloning sites.

(C) Each of the three vectors contains a different reporter protein or His-tag for the hybrid construction.

(D) The three vectors provide all possible frames for the hybrid construction.

(E) Each of the three vectors contains a different regulatory system to allow for differential expression of the recombinant protein.

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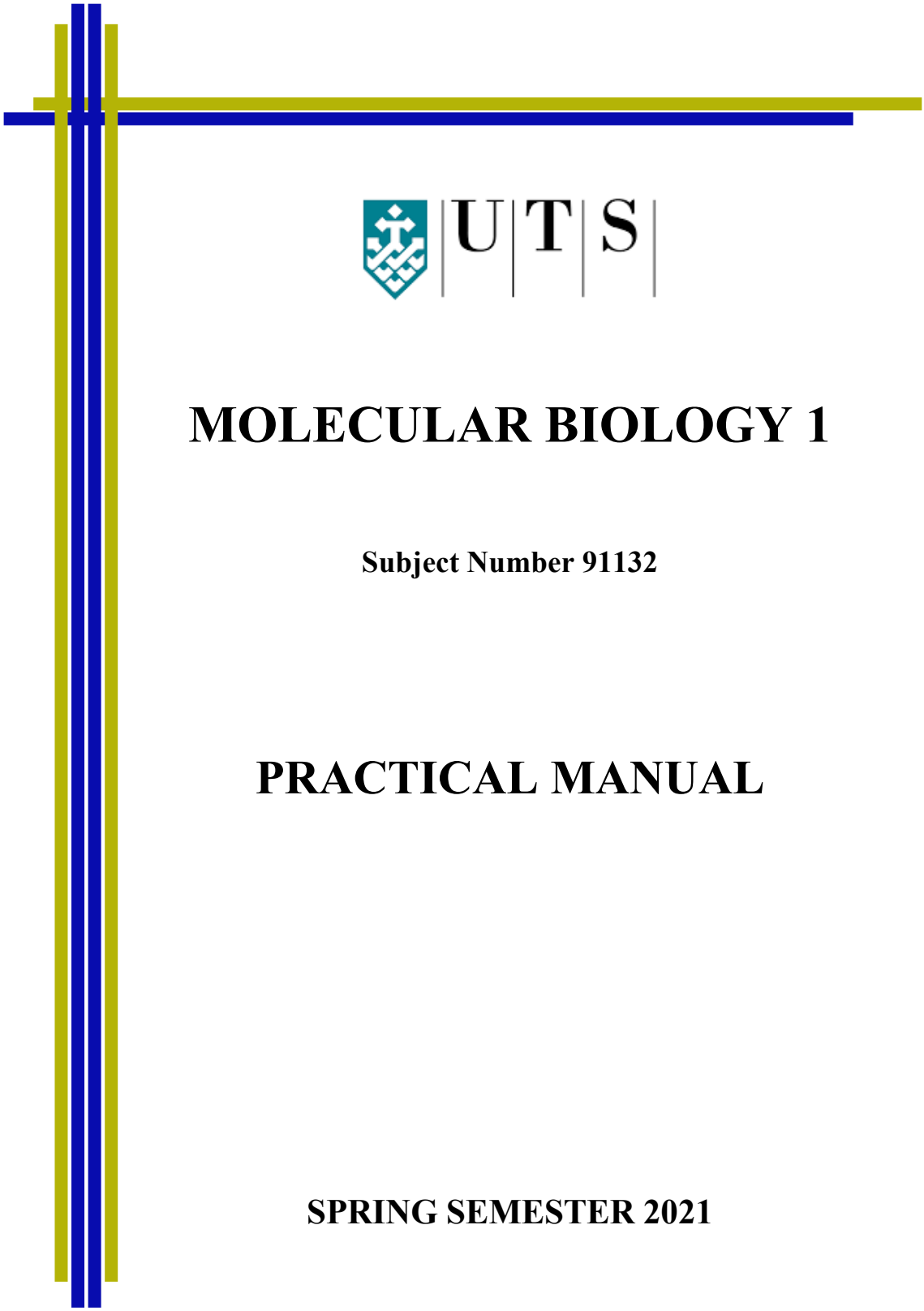

MOLECULAR BIOLOGY 1

Subject Number 91132

PRACTICAL MANUAL

SPRING SEMESTER 2021

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GENERAL INFORMATION

The details in the next few pages are an extension or complement to the Subject Outline and other information posted in UTS Canvas for individual subject tasks. **Frequently Asked Questions** (FAQs), which you will find in this lab manual (page 17), is the best first port of call for questions and queries about the subject that aren't answered elsewhere.

SUBJECT DETAILS

UTS: Faculty of Science; Spring 2021; City Campus
Credit points: 6cp; Result type: Marks / Grade
Pre-requisites: 91161 Cell Biology and Genetics

SUBJECT COORDINATOR

Name: Tony George
Phone: 9514 4158
Room: CB04.07.334
Email: tony.george@uts.edu.au

SUBJECT DESCRIPTION

This subject introduces the basics of molecular biology and an understanding of the key concepts underlying the experimental techniques of DNA manipulations in a molecular biology laboratory. Topics covered include: DNA and RNA isolation, restriction enzymes, DNA ligation, cloning strategies; southern, northern and western blotting; and an introduction to DNA sequencing and PCR. This subject encourages students to become adept at the techniques required for molecular analysis in a modern scientific laboratory and provides a foundation for more advanced molecular biology study.

TEACHING AND LEARNING STRATEGIES

The integrated, interactive teaching modes in this course are designed to give you ownership of learning in this course. The integrated lecture-MCQ tutorials are focused on 'demystifying' MCQs thereby juxtaposing the lecture/tutorial MCQs with the final examination MCQ format.

A list of useful You Tube teaching videos and links as background to the lectures and experiments is on pp 15-16 of this Lab Manual – 'Teaching Aids' -and is also in the Week 1 menu as a pdf. To explore such sites and videos further, you may access other videos that are displayed on the right screen of a video site; or type the topic into Google and explore 'animation', 'images', or 'tutorial'. You will find a wealth of knowledge explained in these, plus links to other useful sites. Such self-exploring will give you a broader and deeper view into each topic. These are especially useful in the 4 weeks of Online teaching.

One of the chief issues that concern some students in this subject is the amount of 'new' molecular biology terms and 'jargon'. This is unavoidable, but the Subject Coordinator has designed MB1 in such a way that the new terms and jargon are easily incorporated into your

lexicon, surprisingly without needing to rote learn everything. Information is always provided for and within tasks. In addition, you will find a very helpful 'Glossary of Molecular Biology' terms in this manual (from page 8).

MINIMUM REQUIREMENTS

Students are required to obtain an overall grade of 50%.

ASSESSMENT TASKS

Practical Reports	60%
Final MCQ Examination	40%

Special Conditions for Online Work (August, Weeks 2-5):
You should note the revised laboratory schedule on p6, which is also in the Subject Outline Addendum. The first 4 weeks of semester constitute Online work only for all students. In addition to information in the revised schedule, there will be announcements, emails and instructions to help you in the Online period.

As a quick help-as-you-go instruction here, it is better to take yourself through 2 lectures per week for the first 4 weeks. You may refer to and work through any of the lectures at your own pace and at any time during semester.

Experiments 1 and 6 will be Virtual Experiments, even though they appear in this manual as lab expts. A certain degree of latitude will be given to you in completing and submitting your seven (7) lab reports. However, all students should have completed and submitted the Expt 1 report before the end of August. Online students should complete and submit reports in the order in which they appear in this manual, except for Expt 6 which many be submitted out of sequence, by online (and lab attending students) at your earliest opportunity to do so. Expt 6 will require some information from several lectures, as well as information within the experiment itself in this manual. but so as not to take time away from the other 5 reports.

Lab Manual and Practical Reports:
Please refer to the FAQs, Subject Outline, and notes below. The HIVE is paperless so do not printed matter to the labs. You will use the tablets provided to view the Lab Manual.

Generic Results Sheets and Report Questions are to be found at the end of each experiment in this Lab Manual. The reports are short-answer questions only, but they cover all aspects of the experiments and range beyond to related topics. Marks and clues are at the end of each question.

Reports need to be completed as soon as possible after the conclusion of the experiment; and uploaded via the 'Assignments' menu item in Canvas. Reports must be in Word format only (.doc or .docx), with your name and student ID clearly indicated at the top. They will be marked and available to view within a week. Your marks will appear in Gradebook, accessed via the 'Marks' menu item in Canvas. I supply 'model answers' for the demonstrators. Should you think that you have not been marked fairly on any question(s), please contact the Subject Coordinator, not your demonstrator. The Subject Coordinator will ensure fairness in disputed marking. Marks and Clues are supplied for each report question. The number of questions and total marks reflect the time spent and difficulty of each experiment.

*Copied answers, from another person or from previous years model answers will attract a mark of **zero**. Copying/plagiarism is not permitted at UTS. You are encouraged to work together on reports, but write your answers in your own words, not an agreed word for word answer.*

It should not escape your attention that you will benefit immensely during the semester by beginning your reports for all experiments before their completion. This is possible because though all reports bear some essential reliance on the experimental outcome, and Result Sheets, many of the questions range around and beyond the results in a theoretical and problem-solving fashion; and these answers can be worked on independently of the results.

Lectures:
There are notionally two hours of lectures each week. However, there will not be any lecture recordings, nor will lectures be given live in Zoom. Instead, the lecture notes/slides will be available in UTS Canvas for Weeks 2-11, meaning that you have the option of downloading and viewing PDFs of the slides/notes for each lecture on a weekly basis, or at any time and for as many times as desired during the semester. At the end of each lecture there are a couple of practice MCQs, typically taken from past exams and included as a guide to what the Final Exam MCQs might look like for each topic. The lectures are designed around the MCQs, based on the lecture material, but let me stress that these MCQs will not be used in this year's Final Exam. Therefore, in place of formal lecture sessions will be an opportunity for you to view the 10 lectures in sequence and/or as often as you like. Feel free to take my invitation to ask questions about lecture topics at any time by email, face-to-face, or in Zoom sessions. In lieu of 'live' lectures, I will post Zoom invitations for one hour slots during the notional two hour lecture sessions on Mondays and Tuesdays in Weeks 2-11. The Zoom sessions will be at 16.00 pm – 17-00 pm. If you miss out on a Zoom session, you may 'catch up' with any questions you might have about any lecture – and at any time during the semester – by emailing me or arranging a meeting alone or in small groups of students with similar queries.

Final Examination:
A two-hour exam on all sections of the subject (lectures; practicals) will contain 40 MCQs (no negative marking). The final examination is meant to distil and test your knowledge of the course content. MCQs are compiled in a way that best serves this purpose, relying on analytical thinking and in only a small way, if at all, on rote memory. A couple of Practice MCQs, based on each lecture, and taken from previous exams, are to be found at the end of each set of lecture slides/notes. These will be a barometer of familiarity with the subject content and a means of revision for final exam, which will comprise similar MCQs.

The examination will be conducted under 'Restricted Open Book' conditions. You may bring one A4 page into the exam. You can hand write or type notes on both sides in any font size, but the text must be in English, no MCQs allowed, and your A4 page must be surrendered with the exam paper when you leave. It will be checked by me and if any indiscretions are found on your A4 sheet, a report will be prepared for the Associate Dean (Teaching and Learning), who will contact you in due course.

FEEDBACK

Should you wish to query marks, do not ask your demonstrator. Contact the Subject Coordinator who will adjudicate all marking disputes, to ensure accuracy and fairness for all students.

Please ask the Subject Coordinator for more comprehensive feedback or help at any time.

LABORATORY SCHEDULE

This program is also given in Canvas. Only the 4 Online teaching weeks section of this schedule (Weeks 2-5) is given in the Subject Outline Addendum. If the lockdown is eased and campus access is possible, then Weeks 6-12 below will be applied.

WEEK	DATES	EXPERIMENTAL WORK
1	Aug 2-Aug 6	No classes: Read all of the information in Canvas Week 1. Prepare for Expt 1 by reading the introductory notes. Ditto for following experiments in the week before they commence.
2	Aug 9-13	Online only: Lecture Zooms; Lab manual instructions; watch teaching videos in sequence; use teaching links to broaden topic coverage; prepare to do Virtual Expt 1 (Agarose Gel Electrophoresis).
3	Aug 16-20	Online only: Lecture Zooms; Lab manual instructions; watch teaching videos in sequence; use teaching links to broaden topic coverage; prepare to do Virtual Expt 1 (Agarose Gel Electrophoresis).
4	Aug 23-27	Online only: Lecture Zooms; Lab manual instructions; watch teaching videos in sequence; use teaching links to broaden topic coverage; prepare to complete and submit Virtual Expt 1 (Agarose Gel Electrophoresis). Begin Virtual Expt 6 (Molecular Cloning).
5	Aug 30-Sept 4	Online only: Lecture Zooms; Lab manual instructions; watch teaching videos in sequence; use teaching links to broaden topic coverage; complete and submit Virtual Expt 1 (Agarose Gel Electrophoresis). Continue Virtual Expt 6 (Molecular Cloning). Complete and submit in this or any of the following weeks.
6	Sept 6-10	Lab Classes begin: Expt 2: Isolation of total RNA (Day 1). [Gel]
7	Sept 13-17	Expt 3: Isolation of genomic DNA (Day 1).
8		STUVAC – NO CLASSES
9	Sept 27-Oct 1	Expt 3: Isolation of genomic DNA (Day 2). [Gel] Expt 4: Transformation of plasmid DNA (Day 2).
10	Oct 4-8	Expt 4: Transformation of plasmid DNA (Day 3). [Gel] Expt 5: Restriction enzyme digestion of DNA (Day 1).
11	Oct 11-15	Expt 5: Restriction enzyme digestion of DNA (Day 2). [Gel] Expt 7: PCR amplification of a plasmid gene (Day 1).
12	Oct 18-22	Expt 7: PCR amplification of a plasmid gene (Day 2). [Gel]
13	Oct 25-29	No Lab Work. Expt 7 report due.

LAB REPORT ASSESSMENT MARKS

Note in the table below that the seven experiments vary in the allocation of marks. This is deliberate. The intention is to "back-end" the bulk of the marks for the later, longer, and more difficult experiments. This also has the added advantage of enabling you to work your way into the correct way to approach and complete your reports.

Should you not do well in Expt 1 and possibly Expt 2 say, then you will not have "sacrificed" many marks. You should then do better in the following experiments, based on taking the time to do your report ahead of the due date, with deference to the clues, help among your friends (but writing the answers in your own words), and feedback if needed.

Your total out of 100 marks will convert to a direct assessment percentage out of 60, which will appear in Gradebook, accessed from the 'Marks' menu item in Canvas.

Experiment	**Marks**
1. Agarose gel electrophoresis: DNA conformation and size.	5
2. Isolation of RNA.	10
3. Isolation of genomic DNA.	10
4. A: Transformation of *E. coli* with plasmid DNA. B: Isolation and purification of plasmid DNA.	15
5. Restriction endonuclease digestion of DNA.	20
6. Molecular cloning of recombinant DNA.	20
7. PCR amplification of a plasmid gene.	20
Total	**100**

LABORATORY SAFETY

In addition to the Subject Risk Assessment document, please note the following.

Lab Coats; Eye Protection; Gloves; Footwear; Bags.
Lab coats must be worn at all times. Safety glasses and gloves must be worn when handling dangerous solvents such as phenol and chloroform. Normal spectacles are a suitable substitute. No thongs, sandals, or open-toe footwear are permitted. Designated areas are available for bags. Loose hair should be tied with a ribbon or clip.

Face Masks.
These are optional but you must supply your own if you chose to wear one.

First Aid and Accidents.
First Aid Boxes and First Aid Officers are available for minor injuries. All accidents, however minor, must be reported to a demonstrator or technical staff member. An accident report form is to be completed. Fire extinguishers, blankets and sprinklers are located in each laboratory.

Emergency Evacuation.
If the fire alarm rings, all classes are to cease immediately, and the evacuation drill will be carried out. Floor and stair wardens will assist you to leave the building via stairwells and exit doors. Do not use the lifts! Proceed to the designated waiting area away from the building.

Please also refer to the Risk Assessment Document in Week 1

TEACHING AIDS
[YouTube Videos; Teaching Sites/Links]

1. Revision of nucleic acid structure and properties. Gene structure.
https://youtu.be/MOtRqBs0jxE

2. Agarose gel electrophoresis. Methods for the extraction and purification of nucleic acids: Genomic DNA; plasmid DNA. Isolation of RNA.
https://youtu.be/vq759wKCCUQ
https://youtu.be/QEG8dz7cbnY
https://youtu.be/6QYgN-toA1A

3. Methods for the extraction and purification of nucleic acids: Genomic DNA, Plasmid DNA, Isolation of RNA.
https://youtu.be/O4oLyd2mZv8
https://youtu.be/nsv8py5ujA0

3. Essential microbial genetics. Regulation and Control - the lac Operon.
https://youtu.be/h_1QLdtF8d0
https://youtu.be/oBwtxdI1zvk

4. DNA sequencing. Bioinformatics.
https://youtu.be/vK-HlMaitnE

5. Restriction Endonucleases. DNA Ligation. Transformation and electroporation.
https://youtu.be/kRUBRrd6Lyc
https://youtu.be/fJzGGHBRwFc
https://youtu.be/xMBC3q_CbTA

https://www.khanacademy.org/science/biology/biotech-dna-technology/dna-cloning-tutorial/a/restriction-enzymes-dna-ligase

https://youtu.be/7Ul9RVYG5CM

6. Molecular Cloning Strategies.
http://bootstrike.com/Genetics/Cloning/cloning_strategies.php
https://youtu.be/sjwNtQYLKeU
https://youtu.be/b9Lm2TUDN2M
https://youtu.be/W6x90HWkL7M

7. Vectors for molecular cloning. Expression Vectors. Factors affecting expression of heterologous proteins in E. coli.
https://youtu.be/g0eMysE3TTs
https://www.hindawi.com/journals/isrn/2013/590587/

8. Polymerase Chain Reaction (PCR). Primers.
https://youtu.be/matsiHSuoOw
https://youtu.be/FWRxXcoM0hk
https://www.promega.com.au/resources/tools/biomath-calculators

9. Site-directed mutagenesis.
https://youtu.be/WYAvlvx786w

10. DNA Fingerprinting. RFLPs.
https://youtu.be/AkBUriMK9u8
https://www.ncbi.nlm.nih.gov/genome/probe/doc/TechRFLP.shtml
http://forensicoutreach.com/library/5-real-life-cases-where-dna-profiling-changed-everything/

11. Virtual Labs.
A range of virtual labs across many science disciplines. To get to the Molecular Biology virtual lab click on "Biotechnology and Biomedical Engineering". Then if you scroll down you should be able to see Molecular Biology Virtual Lab I and. Molecular Biology Virtual Lab II. These labs cover a number of relevant topics including gel electrophoresis and PCR. Each topic includes pages on theory, procedure, self-evaluation, simulator, assignment, reference, video and feedback. To access all areas you will need to sign up, although this is free and very quick!

http://learn.genetics.utah.edu/content/labs/gel/
(A click through video showing a virtual experiment)

12. A Learning Centre library of science animations that may help you understand a number of concepts.
https://www.dnalc.org/resources/animations/

13. A number of online modules about DNA.
http://www.dnai.org/index.htm

14. A cool animation showing EcoRI cutting DNA.
https://youtu.be/piGOuud3f40

15. Genetic engineering - Molecular biology in the real world. Includes DNA ligation method.
https://youtu.be/3IsQ92KiBwM

16. CRISPR: It's all over the science news. But what is it? and how can we utilise this technology?
https://youtu.be/jAhjPd4uNFY

FREQUENTLY ASKED QUESTIONS (FAQs)

This is an accessible summary and easy guide to the important issues you will encounter in Molecular Biology 1. Although everything required will be wholly or partly in **Canvas**, the **Subject Outline**, and the **Lab Manual**, these FAQs should be your first port of call when unsure of what to do, how to interpret information, or gather answers to your questions, under each of the headings below.

EMAIL

Why hasn't my email to the Subject Coordinator been answered?
I endeavor to answer all emails in a reasonable turnaround time. If you do not receive a response within 24 hours, then the reason is almost certainly because the answer to your query is already at hand in one or more of: Subject Outline; Lab Manual introductory pages; or emails or announcements to students.

Always have this FAQs document at the ready and if the answer(s) are not here, or in one of the places listed above, then and only then, contact the Subject Coordinator, Tony George: tony.george@uts.edu.au

WHERE ARE ALL THE FILES AND LEARNING TOOLS THAT I WILL NEED?

Begin at **Modules** in **Canvas**

Week 1: The following files will be placed in the Week 1 Schedule: MB1 Intro Slides PDF; Lab Manual PDF; Frequently Asked Questions (FAQs); MB1-YouTube Links.pdf (YouTube Videos, Virtual Lab and Learning Links). These YouTube links are a rich source of relevant knowledge and information that reinforces and advances the formal subject content.

Weeks 2-11: Each of these weeks (except for StuVac) will have listings for Lecture Slides/Notes PDF files and tentative Lab Practical Online and Lab Schedules.

LECTURES - MCQs

What will I get out of the lectures?
You should go through the lecture slides and explanatory notes below each slide at your own pace. The slides/notes are more useful and accurate than recorded lectures which would really only be a rough a guide to walk you through the lecture slides.

There will not be any lecture recordings – Reason: These are only helpful when there are slides with no accompanying notes, or as a follow up refresher to face-to-face lectures, which for reasons beyond my comprehension have been discontinued at UTS. I have spent quite some time writing clear explanatory notes under the slides – and these notes are more precise and accurate than my trying to talk to the slides with the usual 'ums' and 'ahs', pauses, and forgetting to mention every important point.

'Live' lecture sessions during the scheduled lecture sessions will not be used for the simple reason that they are very poorly 'attended' from remote locations. In place of recordings or 'live' lecture sessions, there will be a Zoom session each week for you to ask questions about that week's lecture, or from past lectures as we progress through the semester. The Zoom sessions will be set to cover a part of the formal lecture session time. These times will be on Mondays and Tuesdays at 16.30 pm– 17-30 pm (excluding StuVac week -Sept 20,21). Links to Monday and Tuesday Zoom Lecture Q/As will be announced and posted in Canvas. You may of course dial in to either or both weekly Zooms. Please note: If you miss out on a lecture Zoom session, you can 'catch up' with any questions you might have about any lecture – and at any time during the semester – by email or arranging a meeting alone or in small student groups with similar queries. In addition to lecture slides, there are many 'YouTube' video and 'Learning Centre' links in the PDF MB1-YouTube Links in the Week 1 Schedule. This file will also be found in the Lab Manual.

Every lecture PDF slide file has two MCQ slides at the end. These MCQs are practice exam MCQs, typically from past exams. Note: These are only indicative MCQs and will not be in the final exam paper. Answers will not be supplied! You should also note that much of the lecture material parallels the practical course.

EXPERIMENTAL REPORTS

Which Lab Session am I in?
Lab session places are allocated on a first come first served basis until the session is full. You will be able to switch labs only if a space becomes available and NOT AFTER WEEK 2 unless you liaise with the Subject Coordinator.

What does the Lab Course entail?
There are 7 experiments over 10 weeks. Due to the Covid-19 restrictions, two experiments will be wholly Online (Expts 2 and 6). The other 5 experiments will be dovetailed into the last 6 weeks of semester for both labs and online students.

Where do I get help for the report questions?
The reports rely on both the generic Results Sheets and extended content that comes mostly from the pages of the experiment, as introduction notes and protocol steps, and from the relevant lectures. Additional help is provided by clues at the end of each question. These clues are especially useful and are intended to save time by directing you to the source of the answer and/or a way of solving the question.

How useful are Online sites such as StuDocu or MB1 report searches?
In the past several years, quite (unlawfully) useful, but increasing unhelpful, as the databank of new report questions is being expanded. This year, most of the report questions are significantly different and/or new.

Answers to the report questions must be written in your own words and must not be copied from another student or from any online site. By all means, combine with others to work out the report answers, but write them in your own words to avoid the discovery of similar or identical reports, which carry heavy penalties.

To whom do I give my report?
Reports in Word format only (.doc, .docx) have to be uploaded alongside your name via the 'Assignments' tab in Canvas. Demonstrators will access your report via this portal only.

Your name must be written clearly on your report or it will not be marked. Your report marks will appear both on the marked returned report and in Gradebook in Canvas (via the 'Marks' menu link).

If you email your report to the Subject Coordinator to pass on to your demonstrator, your email (and attached report) will go into the computer bin.

Why haven't I any substantial feedback for my lab reports?
Quite simply, and as mentioned in the Subject Outline and Lab Manual, demonstrators are given my generic answers to the report questions and they therefore will award ticks/crosses and marks according to the degree of correctness of your answers. They will also be instructed to write a brief summary comment on your total mark for the experiment (e.g. excellent, fair, could do better, seek assistance etc). Note that since the reports comprise questions only, the answers are usually precise and brief and there is not much leeway for flexibility or substantive text beyond the scope of the actual answers. This is also the reason why there is no Rubric for reports. Additional feedback will be given upon request (email or face-to-face) by the Subject Coordinator only, as this ensures a fair outcome for all students.

FEEDBACK

When and what type of feedback is given?
Feedback is an essential tracking device for students and should align with assessment outcomes. Feedback for lab reports will be generic, that is, by way of very brief correct answers model answers posted in the Practical Contents Folder. Feedback for the assignments will be by way of specific comments after they are marked, with the grade and comments published in Review.

Feedback on any aspect of Molecular Biology 1 is always available on request (email or meeting) from the Subject Coordinator at any time during the semester.

FINAL EXAMINATION

What is the format for the final paper?
Forty (40) MCQs in 2 hours with 10 minutes reading time.

Is the Final Exam Open Book or Restricted Open Book? What materials are permitted?
The final paper is 'Restricted Open Book'. You are allowed to bring one A4 page and may use both sides hand written or printed in English. No MCQs or variations thereof are permitted on your A4 sheet.

You will surrender your A4 notes page along with the examination paper when you leave. If there are embedded or disguised MCQs, or other plagiarized material, your entire paper may be cancelled, as per UTS examinations rules.

Calculator (simple type only).

OVERALL ASSESSMENT

What are the assessment rules for MB1?
Final Paper (40%). Lab Reports (60%).
To pass MB1, you will need to obtain an overall total assessment mark of 50%.

PLAGIARISM

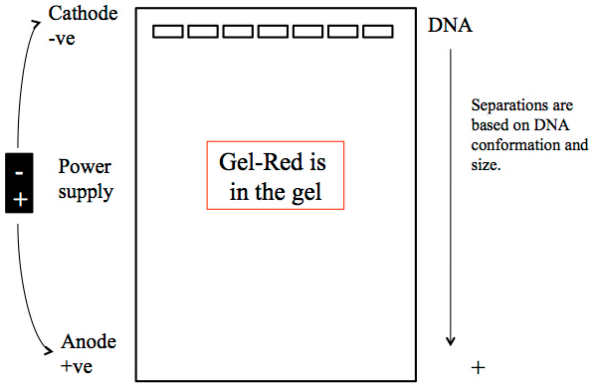
What is plagiarism?
Plagiarism is cheating and is not permitted at UTS in any guise. Please note, the Subject Coordinator and the Faculty Learning Team are constantly scanning the Internet for subject document solutions/answer posted by present or past students. StuDocu and other links will not help you. MB1 has long abandoned questions/answers found at sites such as these.

VIRTUAL EXPERIMENT 1

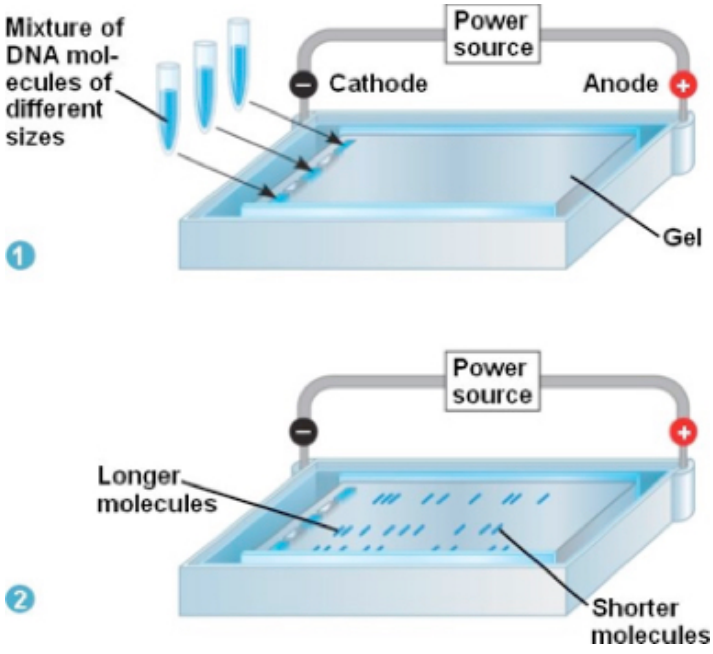
AGAROSE GEL ELECTROPHORESIS:
DNA CONFORMATION AND SIZE

Background

This experiment introduces students to one of the fundamental methods in molecular biology for analysing DNA molecules of different sizes or conformations. Agarose gel electrophoresis is chiefly used to separate, identify, or purify DNA molecules.



Electrophoresis is carried out in a horizontal gel apparatus containing a pre-cast agarose gel that is submerged in an appropriate buffer in the gel tank. DNA samples are added to wells at the cathode end of the gel, because, being negatively charged they migrate to the anode. When current is applied, the DNA migrates through the gel as a function of conformation and size. The method is simple, rapid, and capable of resolving complex mixtures of DNA or RNA. The positions of discrete DNA bands can be located *in situ* by staining with the fluorescent intercalating dye ethidium bromide (EtBr) or the fluorescent dye (Gel-Red™). The DNA-dye complexes are visualised by irradiation of the gel with ultraviolet light at 254 or 302 nm. The absorbed UV light causes the dye to fluoresce at 590 nm in the red-orange region of the visible spectrum. A black and white photo exposure may be taken as a record of the DNA bands in the gel.



The migration rate of DNA molecules through a gel is dependent on two main parameters:-

i The agarose concentration. The approximate separation range of agarose gels:-

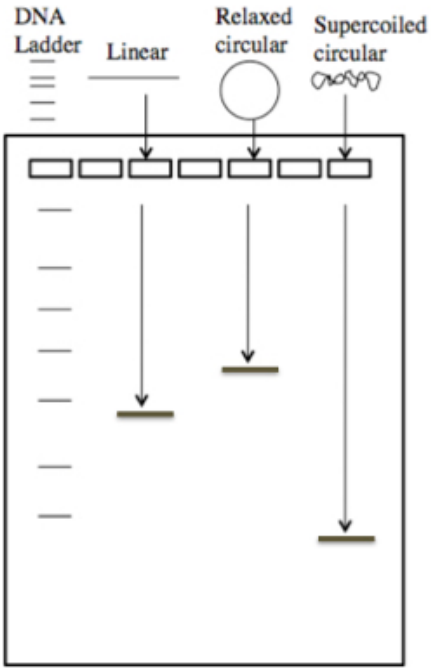
amount of agarose in gel (%)	efficient range of separation (kilobase pairs)
0.3	5.0 ---› 60
0.7	0.8 ---› 10
1.2	0.4 ---› 6
2.0	0.1 ---› 3

ii The conformation of the DNA. Closed circular (supercoiled), open circular (relaxed) and linear DNAs of the same molecular size migrate through agarose at different rates, that are defined by the differences in conformation (spatial displacement or density).

In this experiment, the chief aim is to learn the method of agarose gel electrophoresis; in particular, casting of the gel, loading of samples, running, staining and visualising the resolved DNA bands.

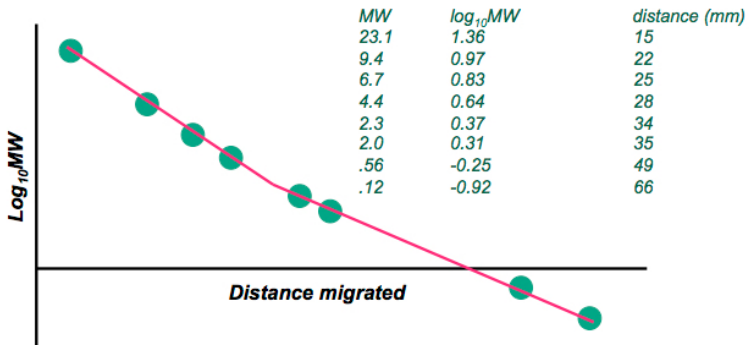
A second aim will be to test and prove premises (i) and (ii) above. DNA samples will include the three conformations, namely linear, relaxed (open) circular, and supercoiled, of the same small plasmid, pAMG11, and linear duplex DNA standards.

This stylized gel diagram shows the typical results for this experiment, except that the plasmids are in different lanes. Note that the migration rates of the three pAMG11 plasmids, of precisely equivalent mass, depends solely on the different conformational geometry of the plasmids. The DNA ladder in lane 1 represents double-stranded linear fragments of DNA; and these are used to calculate the size of the plasmid, by setting up a standard curve and using the linear plasmid band for the size estimation. Though difficult to envision by looking at simple drawings of the three pAMG11 conformations, they migrate end-on-end and according to conformational differences, which are restricted by the agarose pore size. Supercoiled plasmids are very tightly and almost inflexibly constricted as rods. The linear version of the same plasmid, though also rod-shaped, is freely flexible and will bump into the walls of the pores, causing a



'tumbling effect that slows the migration rate. The floppier relaxed circles migrate slowest, as they bump into the walls much more frequently than the other two configurations.

A standard curve is drawn by plotting the distances migrated from the origin to the DNA standard bands in an agarose gel, versus the log10 of molecular masses, as illustrated in this example for pUC18. Alternatively, this plot could be drawn on semi-log paper as MW values directly on the log scale and distances on the linear scale. The size of an unknown DNA molecule is estimated from the standard curve by drawing a vertical line from its distance migrated up to the line and across to the ordinate axis from which its molecular mass is interpolated. An important point about standard curves is that a broad range of standards will migrate in two (or more) modes as shown above. The larger DNA bands run a little slower than the smaller bands – due to the frictional resistance caused by the restrictive pore size on the larger molecules – and this is reflected in the standard line having two slightly different gradients. The standard "curve" line is drawn through every point, and never as a line of best fit, as this will affect the accuracy of estimates in different sections of the standard line.



Experimental protocol

Pre-cast 0.7% agarose gels will be used with the DNA dye 'Gel-Red' incorporated in the gel. Gel-Red is a much safer and less toxic dye than ethidium bromide, but it shares a similar structure and staining process for the ultraviolet light detection of the DNA or RNA bands.

Sample loading

1. The DNA samples to be placed into the lanes are:-
   P [supercoiled + linear + relaxed conformations of the plasmid, pAMG11]
   S [DNA linear standards: 23.1, 9.4, 6.7, 4.4, 2.3, 2.0, 0.56, and 0.12 kb pairs]

2. Load 10 µl (~200 ng) of the sample labelled **P**. Standards (**S**) will not be added to the gel, but they will be depicted in the Expt. 1 Results Sheet agarose gel photo. Make sure that you write your name on the lane number sheet.

The samples also contain loading dye solution (40% sucrose, 0.025% bromophenol blue / xylene cyanol, 100 mM EDTA, pH 8). The sucrose gives viscosity so that the sample remains in the gel slot until the run is begun; the EDTA protects the DNA samples from nucleases; the dyes enable the sample to be seen while it is loaded, and also tracks the progress of the run through the gel.

Electrophoresis

1. Carefully affix the apparatus lid ensuring that the sample wells are at the cathode end of the tank and that the electrodes are plugged into the correct terminals of the power supply.

2. Electrophoresis is carried out at 90 to 120 volts for about 1 hour, or until the bromophenol blue (BPB) dye front is about 1-2 cm from the end of the gel. Caution! Turn off the power before removing the gel.

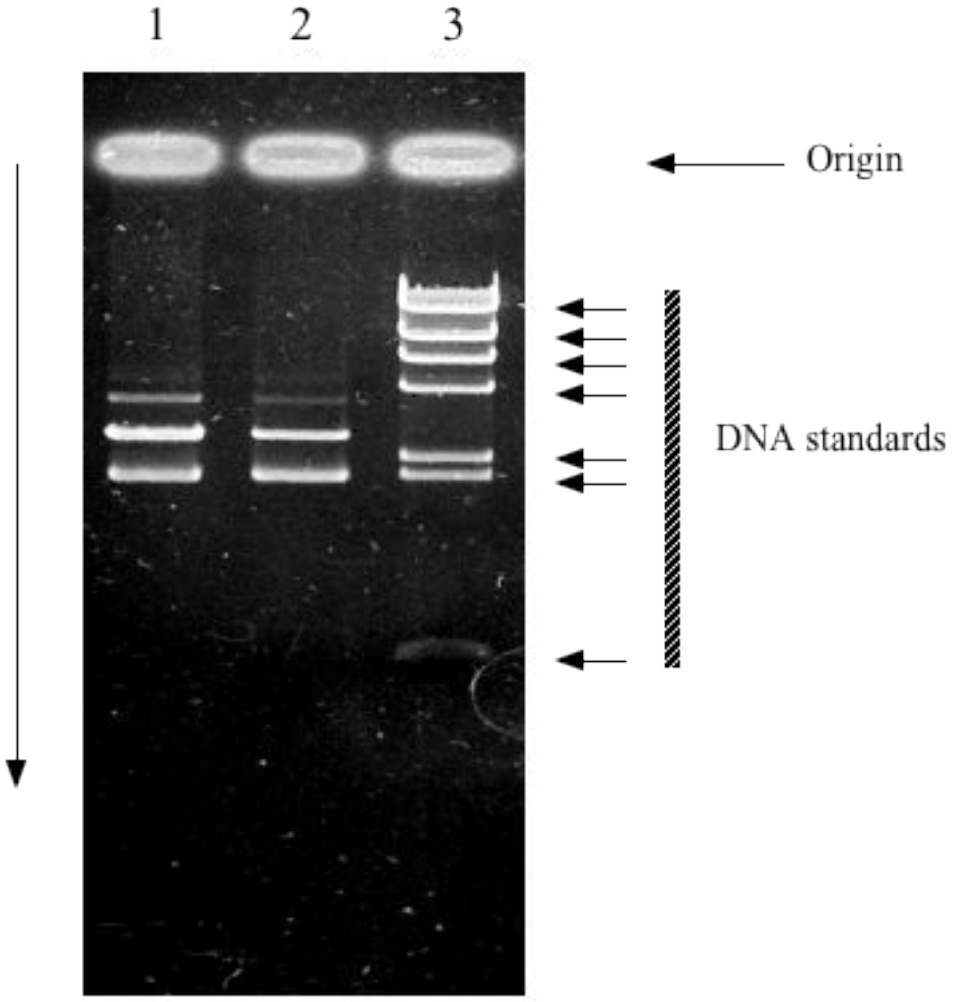
UV visualization

1. Demonstrators will transfer the gels to the UV transilluminator units so that the stained DNA bands may be visualised. The excitation UV light at 302 nm is absorbed by the Gel-Red and emitted (fluoresces) as bright orange light at 540 nm.

2. The DNA has been stained during the run because a DNA staining dye was added during the casting of the gel. EtBr is a powerful mutagen and is no longer used. Gel-Red stains the DNA in the same way and is far less toxic than EtBr and is non-mutagenic.

The Result Sheet is on the next page and is followed by the Experiment 1 Report Questions on p26. Your report is due in one week, uploaded into the Assignments folder in Canvas.

RESULT SHEET: EXPERIMENT 1



Explanatory Notes:

1. Lanes 1 & 2 are both of pAMG11 in all three conformations.

2. Lane 3 contains the linear standards whose sizes are given in the Expt Protocol; Step 1, 'S'.

3. The largest (first) standard band in lane 3 looks like two bands but is in fact only one as the arrow indicates. The band distortion is an artifact caused by the depth of the gel.

4. The first plasmid band – relaxed circles – is faint and sometimes not visible. The amount present is a reflection of: (i) the extent of DNA shearing during the plasmid isolation process; (ii) the size of the plasmid; and (iii) the amount of DNA added in the gel lanes.

5. Only supercoiled and some relaxed plasmid DNA occur in cell cultures. The linear band in this gel was obtained artificially by digesting circular DNA to make it linear, and then adding it back to the circular plasmid DNA preparation. In this way, we may observe all three plasmid conformations in the same gel lane.

Experiment 1 Report Questions

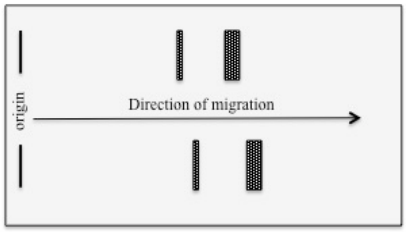
The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

Q1. Use the Result Sheet gel photo to draw a standard curve of $\log_{10}$[kb pairs] versus distance from the origin wells in millimetres (mm) to each of the DNA linear standards, on normal scale graph paper. Or, plot kb pairs versus distance migrated on semi-log graph paper. You must include your data table of distances versus standards. Use either plot to determine the molecular size of plasmid pAMG11 in kilobase or base pairs. [1.5 marks]
[Clue: This is right out of the lecture notes and preamble to this experiment.]

Q2. Say you were only given only two standards (instead of 8) for this experiment and you were asked to plot just these two standards and use the graph to make a reasonable approximation of the size of pAMG11. (a) Which two standards would you choose? (b) Explain your answer. [1 mark]
[Clue: Location, location, location!]

Q3. In the diagram below, a sample of the same plasmid is run in each lane. One sample has Gel-Red added to the plasmid sample, the other not. The leading (broader) band in each case is supercoiled plasmid, the trailing band is relaxed plasmid. Which lane, upper or lower, has the plasmid plus Gel-Red? Explain your choice. [1 mark]
[Clue: Lecture notes on Gel-Red-DNA intercalation.]



Q4. Gel-Red binds to DNA as the basis for UV visualization. The structure of Gel-Red bears a positive charge and planar aromatic rings. How do each of these assist in Gel-Red binding to DNA? [1 mark]
[Clue: Lecture notes on Gel-Red.]

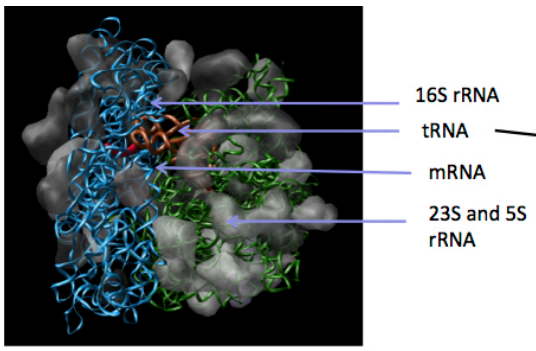
Q5. The Result Sheet for this experiment depicts only 7 standards in the gel photo, yet the list given in Step 1 of the Experimental Protocol above has 8 DNA linear standards. Explain this apparent anomaly. [0.5 mark]
[Clue: 8 into 7 will not go.]

EXPERIMENT 2

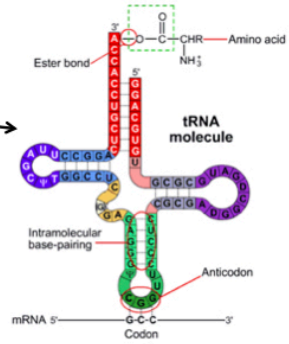
ISOLATION OF RNA FROM *E. coli*

Background

Ribonucleic Acid (RNA) is present in cells as several different types of molecules or structures. The major RNA species are: Ribosomal RNAs (rRNA) consisting of two very long RNA molecules that associate with many proteins to form the two composite ribosome subunits, and a third smaller rRNA. These are known as 28S, 18S and 5.8S rRNAs in eukaryotes, and 23S, 16S and 5S rRNAs in prokaryotes. The RNA molecules in the ribosomal subunits have extensive secondary structure, which helps to form latticed assemblies with the attached proteins/enzymes. rRNAs constitute about 70 to 80% of total cellular RNA. Cells also contain about 30 small (about 90 nucleotides each) transfer RNA molecules (tRNAs), which constitute about 5 to 10% of the total RNA within cells. These molecules are single-stranded but self-assemble into clover-leaf-like structures with extensive secondary structure (intra-double-strandedness). tRNAs are about half the size of the small rRNAs.

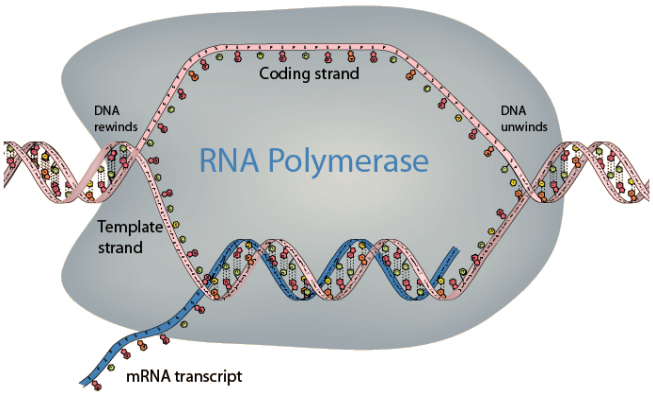


E. Coli 70S Ribosome
Role: Translate mRNA into polypeptides



tRNAs are half the size of 5S rRNA but they run almost together in agarose gels

The third major type of cellular RNA molecule is messenger RNA (mRNA), making up about 1 to 5% of the RNA of cells. Unlike rRNAs and tRNAs which have a small number of defined lengths and conformations, mRNAs form a large heterologous class of single-stranded molecules. Each mRNA is transcribed from a specific gene sequence and thus there are at least as many mRNAs as there are genes. mRNAs are transiently present in cells, with variable short half lives, dictated by the rate and quantity of



transcription/translation required for polypeptide production. To this end, mRNAs are also variously abundant with anything from a single copy to thousands of copies, depending on the needs of the particular cell type. For example, globin mRNAs are the most abundant in erythropoeitic haemoglobin-producing cells. Many mRNAs are expressed only at certain times in the cell cycle, again according to need. Transcription of DNA into mRNA is therefore a dynamic and variable process.

In Molecular Biology, mRNAs are the most important of the RNA types since they can be isolated from cells and used as templates to synthesise complimentary DNA (cDNA) libraries from any type of cell. This is a particular advantage with eukaryotic cells since the processed mRNA is copied into a cDNA with an uninterrupted translational protein reading frame. That is, the usually very long intervening intron sequences that are present within the genes in eukaryotic DNA are absent from the processed mRNA molecules.

Experimentally, the isolation of RNA is relatively simple but the single biggest problem is the preservation of the integrity of the mRNAs, some of which are present in very low copy numbers. Unlike double-stranded DNA, single-stranded RNA is susceptible to degradation by ribonucleases (RNAses) that are ubiquitous in cells, in buffers and on apparatus, and on the hands of laboratory workers. Thus the most difficult task in preparing a largely undegraded RNA preparation is inhibiting or eliminating the sources of RNAse contamination.

The best isolation methods are those that use a minimum of steps and begin by rapidly inactivating RNAses, which is achieved in this protocol by phenol in the cell lysing buffer. The cell lysate is added to a 'PLG Heavy' tube, containing a colloidal 'phase-lock' gel, which is of intermediate density between aqueous and organic solvent phases. When the cell lysate is centrifuged at high-speed in the PLG tube, the aqueous layer (containing all of the RNA) remains on top of the gel, while the major contaminants (chromosomal

1. The PLG tube with the colloidal gel at the bottom. The gel is intermediate in density between water and organic liquids such as phenol and chloroform.

2. Mix by shaking back and forth with thumb and forefinger firmly on the cap and bottom of tube. This slide shows the three layers before emulsifying (mixing) and centrifuging. Cell suspension (upper); phenol-chloroform (middle); PLG (bottom).

3. The mixed and centrifuged PLG tube, with the cleared aqueous (RNA containing) layer (top); PLG (centre); and phenol/chloroform layer (bottom). Note, the chromosomal DNA, membrane and protein debris are compacted into a pellet in the PLG and phenol/ chloroform layers.

DNA, cell membrane debris and proteins) are collected at the gel interface. The organic mixture (phenol/chloroform) is heavier than the gel and forms the bottom layer in the PLG tube.

In this experiment, a bacterial culture will be used to isolate total RNA, which will then be run in a denaturing agarose gel to assess the quality of the preparation. Although the integrity of individual mRNAs cannot be directly checked, the identification of the three major cellular RNA species as discrete bands in a denaturing agarose gel will give a good indication of the degree of success of the isolation procedure.

A denaturing agarose gel is needed for RNA samples. The RNA is firstly denatured (rendered single-stranded; breakage of H-bonds) by heating at 90^{o}C for 2-3 min in the special FFMB gel loading buffer, which contains the denaturants formamide and formaldehyde to assist and maintain the denatured status of the RNAs. The now single-stranded RNA (rRNA, tRNA, and mRNA) is added to a gel slot and subjected to electrophoresis. The agarose gel contains guanidinium isothiocyanate (at 5 mM) and the presence of this denaturant prevents H-bonds reforming in the RNA and therefore maintains and allows the RNAs to migrate in the gel as single-stranded molecules.

BEFORE YOU BEGIN: A NOTE ABOUT CENTRIFUGES

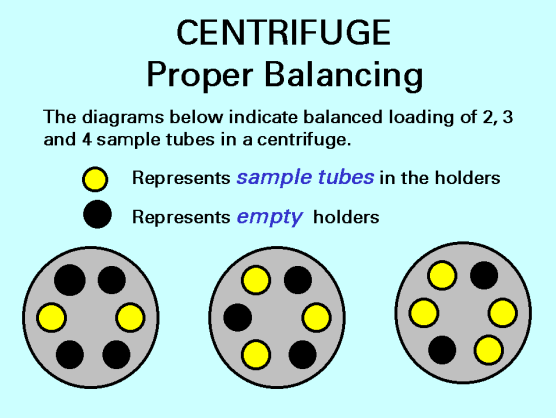
Mini centrifuge
Mini-fuge

High speed centrifuge

These two types of centrifuges will be used in experiments from here on. The one on the left is a simple low-speed (600 rpm) mini-fuge, used exclusively to bring contents in a tube to the bottom. It is not for collecting cells or nucleic acids as pellets; and will be used sparingly.

The two centrifuges on the right are examples of high-speed (~12,000 rpm) centrifuges and these will be used each week for pelleting cells, or nucleic acids, or removing cellular debris. Due to the high speeds generated, it is imperative that the inserted sample tubes be balanced (in volume) and placed in the rotor correctly, as indicated in the two diagrams below.

How to balance centrifuge tubes in different rotor configurations.



Experimental protocol

Isolating total RNA (Work individually)

1. Prepared in advance. Grow up *E. coli* cultures at 37°C in LB to an $A_{530\ nm} = 0.8$.

2. Place 1.0 ml of the cell suspension into a polypropylene tube and centrifuge in a high-speed centrifuge for 30 sec to 1 min. Discard the supernatant completely (use a pipette to remove the last bit of supernatant).

3. Resuspend the pellet completely in 50 µl of 1 mM Tris (pH 7.5) – 0.1 mM EDTA (TE) buffer. This is best done by pipetting up and down a few times, or flicking the tube.

4. Go to the fume hood where a demonstrator will dispense 200 µl of phenol/chloroform (buffered with 0.5 M Tris pH 7.5 – phenol/chloroform is in the bottom layer) into your 1.5 ml tube. Close the tube fully and return to your bench.

5. Mix the tube for 10-15 seconds, by holding the tube between thumb and index finger and mixing back and forth thoroughly, away from your face and other students. DANGER: Phenol is a strong corrosive chemical; ensure tubes are properly sealed.

6. Pipette the cell lysate contents of your tube into a labelled yellow PLG1-Heavy tube. Spin for 2 mins.

7. You or a demonstrator will then add 200 µl of chloroform to the same PLG1 tube, close fully and mix as in Step 5 then spin for 2 mins. NOTE: Chloroform is not nearly as toxic as phenol (now safely under the PLG gel) but still exercise caution!

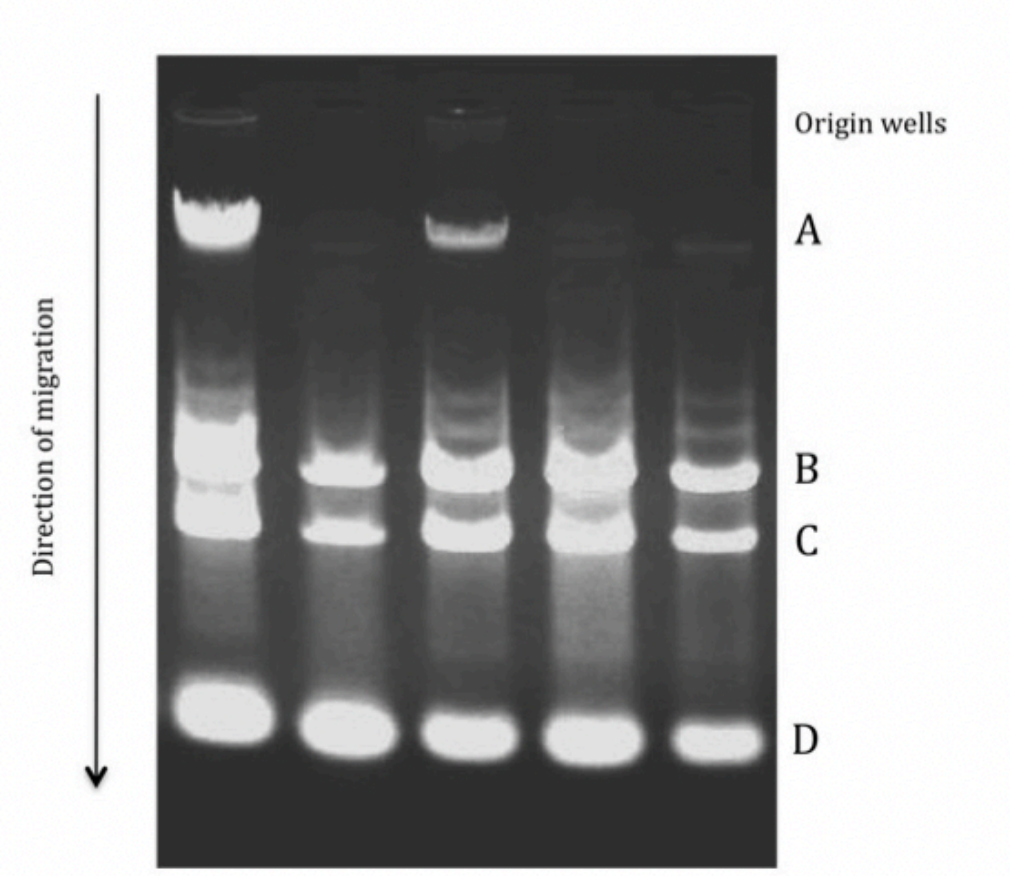
8. Add 20 µl of the FFMB dye mix [formamide + formaldehyde + 50x TAE at 10:3:1; sterile glycerol (30%); bromophenol blue dye (0.1%)] to the aqueous layer above the gel plug in the PLG tube. Note that the organic solvents are now below the gel plug.

9. Remove about 30 - 40 µl of the sample and place into a new 1.5 ml polypropylene tube. Heat this sample at 90^{o}C for 2 mins.

10. Add 15-20 µl of the sample to one of the gel slots.

11. Run the 1% agarose denaturing gel at 140 volts until the bromophenol blue dye front reaches about halfway. Remove (demonstrators do this) and visualize under UV light on the GelDoc UV set-up.

RESULT SHEET: EXPERIMENT 2



Experiment 2 Report Questions

The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

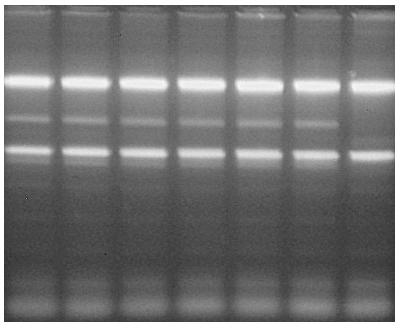
Q1. From the 'Results Sheet' for this experiment, identify the components of the 4 major bands, A to D. [1 mark]
[Clue: Lecture 2.]

Q2. Why don't we see separate sharp bands for the mRNAs like we do for the major ribosomal RNA (rRNA) bands on the Result Sheet? [1.5 marks]
[Clue: Lecture 2.]

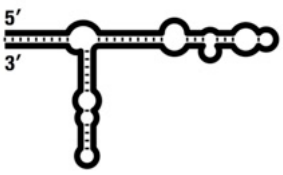
Q3. The PLG tube is a useful device in this experiment. What is the main reason for using the PLG tube? [1.5 marks]
[Clue: Lecture 2 and Lab Manual Expt 2.]

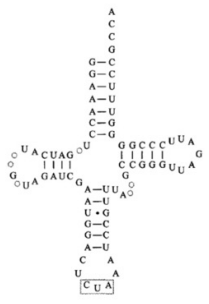
Q4. Before loading your RNA sample into the agarose gel, it is heated to 90°C in the FFMB mix for a few minutes. We know that heating nucleic acids to 90°C will denature them by breaking hydrogen bonds. This being the case, why are formamide and formaldehyde present in the FFMB mix? [1 mark]
[Clue: It takes time to go from the heating step to the loading of the gel.]

Q5. The denaturing agarose gel photo below is a profile of total RNA from several eukaryotic sources. Note that the three major bands (28S, 18S, 8S), equivalent to prokaryotic bands (23S, 16S, 5S), are present as: The two brightest bands near the top and middle of the gel, and the fuzzy band at the very bottom. Note that there are several additional bands other than the main three rRNA bands – compare with the Result Sheet bands. What exactly might these extra bands represent in this eukaryotic gel? [2 marks]
[Clue: Eukaryotic mRNA is more complex than prokaryotic RNA.]



Q6. Below are cartoon drawings of a typical 5S RNA and tRNA. These two types of RNAs run together as a single band in a denaturing agarose gel. Can you explain why these seemingly different molecules run together? [**1 mark**]
[Clue: What would both look like as single stranded molecules?]





Q7. Guanidinium isothiocyanate in the agarose gel prevents the ribosomal and transfer RNA molecules from renaturing, and it inhibits ribonucleases. Guanidinium isothiocyanate is a common component of some cell lysing buffers for isolating total RNA. Why wasn't it used in the lysing solution in this experiment? [1 mark]
[Clue: What does the phenol do?]

Q8. The polar and non-polar phases from the phenol-chloroform cell lysate are separated in the PLG tubes. What property of the PLG (Phase-Lock Gel) is required to bring about this separation? [1 mark]
[Clue: Density of solvents.]

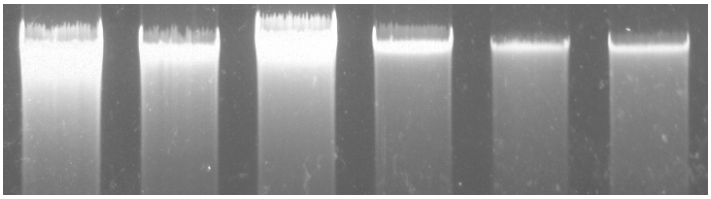
EXPERIMENT 3

ISOLATION AND PURIFICATION OF GENOMIC DNA

Background

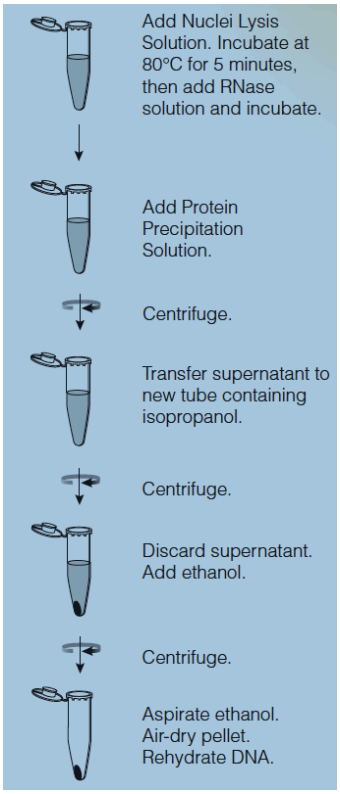
Chromosomal or genomic DNA from mammalian tissue or cell lines, from plant tissue, or from yeast, fungal, and bacterial cultures, is isolated and purified in essentially the same manner. The major differences are in the components of the cell lysing solution, and in limiting the number of handling operations for very high molecular weight DNA. The physical properties of genomic DNA - rigidity, viscosity, and long length to diameter - render the DNA highly susceptible to mechanical shearing from pipetting and mixing/vortexing. Severe isolation methods will yield DNA with an average size of 20-50 kb. But for good quality preparations, much larger fragments are needed (from 50-400 kb). In this case, much milder lysis and handling methods are employed, such as less vigorous mixing and pipetting, or milder isolation methods. Genomic DNA from a cell lysate can be spooled directly onto a glass rod at an ethanol/aqueous interface, and then dialyzed to remove salts (precipitation and resuspension would enhance shearing). Alternatively, the DNA can be precipitated from a crude lysate, resuspended and dialyzed, or purified further by phenol/chloroform extraction or caesium chloride-ethidium bromide (CsCl-EtBr) density gradient centrifugation. The last of these provides very pure DNA that can be stored indefinitely at 4^{o}C as the caesium ions protects the DNA from damage by nicking.

Alcohol (ethanol or isopropanol) and high salt (0.1 to 0.5 M) assisted precipitation of high molecular weight nucleic acids is very rapid at room temperature with a stringy white precipitate appearing almost instantly. Residual salt can be washed from with 70% ethanol, which dissolves salt without affecting the DNA pellet. Though isopropanol is more effective than ethanol in precipitation DNA, it is less volatile and any residue clinging to the DNA pellet is easily dissolved in the ethanol wash. Smaller DNA molecules or more dilute DNA solutions require longer precipitation times, low temperatures, and long centrifugation times for good recovery. In the diagram, chromosomal DNA samples are run in an agarose gel. The first three lanes have larger amounts of DNA than the last three lanes, but the sizes of the fragments and distributions are about the same, indicating that the samples are otherwise similar except for yield. One point to note is the streaky or striated appearance of the



tailing edges of the bands, which is a signature feature of chromosomal DNA in agarose gels. The unevenness of the tailing edges reflects differences in the lengths of the DNA molecules, which run end on end in the gel. Note that the leading edges are lined up and sharp because most of the DNA fragments are approximately of similar size about a typical bell curve and the current and buffer ions enable the leading edges to migrate at the same rate. You might also notice a haze in front of the main bands, and especially in lanes 1 to 3. These are DNA fragments of minor abundance that migrate faster than the main band because they are smaller and are spread along a large length of the gel lane to indicate their different sizes.

The Wizard Genomic DNA Purification Kit is based on a four-step process. The first step in the purification procedure lyses the bacterial cells. A ribonuclease (RNase) digestion step is included to remove RNA. The cellular proteins are removed by a salt precipitation step, which precipitates the proteins but leaves the high molecular weight genomic DNA in solution. Finally, the genomic DNA is concentrated and desalted by isopropanol precipitation. In this experiment, a general and reliable method (*Wizard Genomic DNA Purification Kit*) is used to produce bacterial genomic DNA free from contaminants. A brief schematic of the isolation and purification process is given in the diagram at right. Note, in the DNA protocol, the major contaminants are: proteins (cytosolic and membrane), lipids and polysaccharides, all of which are precipitated in the lysis solution and removed/pelleted in a high-speed centrifugation step; and RNA, which is digested by ribonuclease into individual ribonucleotides, which remain in the supernatant when the DNA is precipitated with isopropanol (plus salt, which is present in the DNA solution from the earlier reagents). An ethanol (70% w/w) wash step is followed by final centrifugation and drying steps. The total yield of *E. coli* genomic DNA should be up to 20 µg from 1.0 ml of an overnight culture at a density of ~2 x 10^9 cell/ml.



Experimental protocol

<u>Isolation of crude genomic bacterial DNA</u> (Day 1) (work individually)

1. Bacterial cultures have been grown to saturation in LB broth overnight. Make sure you mix the culture suspension before withdrawing your sample for Step 2.

2. Add 1.0 ml of the culture into a labelled 1.5 ml tube and centrifuge it for ~30 seconds in a benchtop high-speed centrifuge.

3. Drain the pellet completely. If necessary, use a pipette or Kimwipe tissue to remove the last amount of supernatant on the downturned lip of the tube.

4. Add 600 µl of Lysis Solution (LS) and pipette gently to resuspend the cell pellet completely.

5. Incubate at 80ºC for 5 mins then cool to room temperature (RT).

6. Add 3 µl ribonuclease (RNase). Mix gently then incubate at 37ºC for 15 mins. Cool to RT.

7. Add 200 µl of Protein Precipitation Solution (PPS). Vortex briefly (~5 sec), making sure to hold the capped tube at the top so it does not open.

8. Incubate on ice for 5 mins.

9. Centrifuge at high-speed for 3 mins. While this spin is on pipette 600 µl of isopropanol into a new tube and label the tube with your mark/initials.

10. Using a P1000, transfer most of the supernatant (avoid the pellet) to the tube containing the isopropanol. Mix the tube back and forth a couple of times. [You may see the appearance of a small stringy aggregated DNA “particle”].

11. Centrifuge at high-speed for 2 mins at RT. Decant (drain off) the supernatant. Use a Kimwipe to remove droplets from the upturned tube.

12. Add 600 µl of 70% ethanol. Cap the tube and mix back and forth gently once or twice.

13. Centrifuge at high-speed for 2 mins. Drain off the supernatant. Use a Kimwipe to remove droplets from the upturned tube.

14. Evaporate any residual ethanol by placing the open tube in a 65ºC heating block for 2 mins.

15. Rehydrate the DNA pellet by adding 50 µl of TE buffer (100 mM Tris pH 7.5; 1 mM EDTA), cap the tube and flick it a couple of times to dislodge the pellet.

16. Place your capped and labelled tube back into the 65ºC heating block. It will be collected later and held at 4ºC until Day 2 next week.

Day 2

Agarose gel electrophoresis of crude chromosomal DNAs

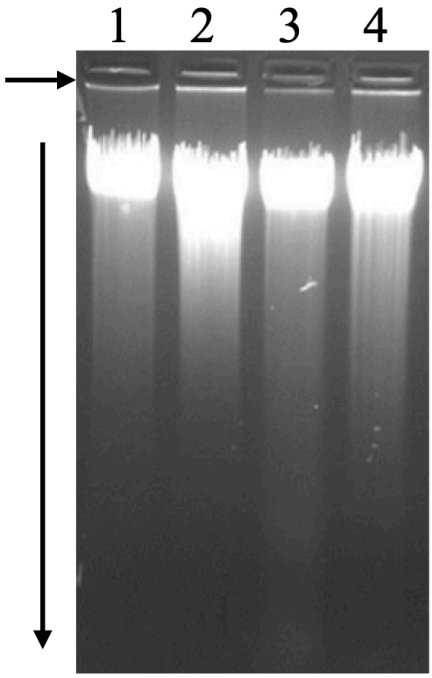
Agarose gels of 0.6% containing Gel-Red have been pre-cast.

1. Add 10 µl of running dye solution to your sample, mix by pipetting up and down. Apply ~15-20 µl to a well in a submarine gel in an electrophoresis tank.

2. Attach the tank lid (the negative cathode terminus should be at the well end of the gel) and apply 120 volts for about 40-60 mins.

3. Turn off the power supply, remove the gel to a UV transilluminator box, and observe the distribution and position of the Gel-Red stained DNA.

RESULT SHEET: EXPERIMENT 3



This diagram is a stylized of an agarose gel profile of four lanes of chromosomal (genomic) DNA purified as performed in this experiment.

The short horizontal arrow indicates the origin wells and the long vertical arrow the direction of migration of the DNA in the applied electric current.

Experiment 3 Report Questions

The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

Q1. Calculate the theoretical yield of chromosomal DNA from a 200 ml saturated culture using the following data. (Must show working). [1 mark]

Volume of culture:	200 ml
Cell density:	2×10^9 cells / ml
Size of *E. coli* chromosome:	4600 kb (4600×10^3 bp)
MW of ds DNA molecule:	(no. of base pairs in the chromosome) x (650 daltons / bp)
Atomic mass unit (dalton):	1.66×10^{-24} g

[Clue: A straightforward multiplication calculation.]

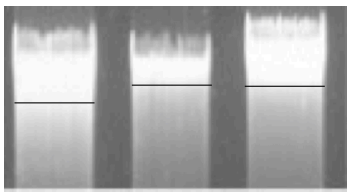
Q2. The two agarose gels below have genomic DNA preparations from two separate isolations. Can you say what, if anything, is wrong, and why, with the A and B gel profiles? [2 marks]
[Clue: Check Lecture 3 notes and/or introductory notes above.]

A

B

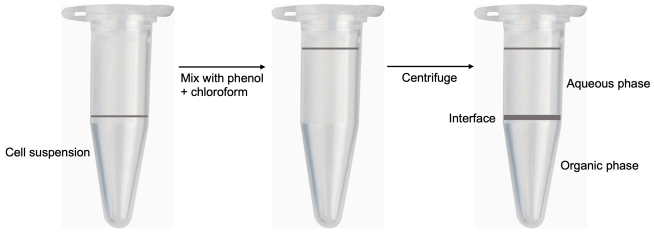
Q3. On the result sheet for this experiment, note that in each of the four lanes, the major DNA bands near the origin are accompanied by less intense fuzzier regions down most of the rest of each lane. Can you say what causes these fuzzier regions, which are seen typically in chromosomal DNA profiles in agarose gels? [1 mark]
[Clue: Check Lecture 3 notes and/or introductory notes above.]

Q4. The diagram shows high molecular weight chromosomal DNA migrating in an agarose gel. Note the streaky or striated appearance of the tailing edges of the bands, which is a signature feature of chromosomal DNA in agarose gels. The leading edges are marked with horizontal lines. Explain briefly, (a) Why the tailing edges are streaky/striated, and (b) Why the leading edges are straightened or lined up at the indicated horizontal lines? [2 marks]
[Clue: Check Lecture 3 'spaghetti' notes.]



Q5. Referring back to the DNA purification scheme in the introductory notes to this experiment, what exactly is happening at the two alcohol stages? [2 marks]
[Clue: Check Lecture 3 notes and/or introductory notes above.]

Q6. The scheme shown here is an alternative method for purifying genomic DNA. Pelleted cells are resuspended cells in a buffer (step 1, left), then mixed with phenol/chloroform (middle) and finally centrifuged at high speed (right). Can you describe exactly what happens in second (middle) and third (right) steps? [2 marks]
[Clue: Check Lecture 3 notes.]



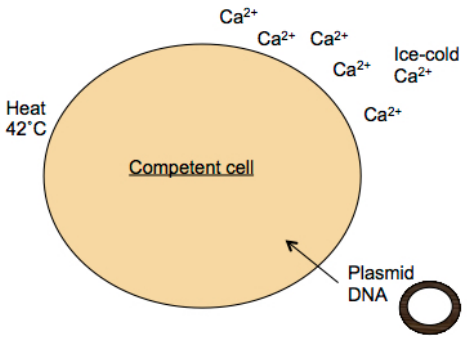
EXPERIMENT 4

A: TRANSFORMATION OF *Escherichia coli* WITH PLASMID DNA
B: ISOLATION AND PURIFICATION OF PLASMID DNA

Background

A: Transformation

In 1970, Mandel and Higa showed that bacteria treated with ice-cold calcium chloride solution were capable of taking up bacteriophage lambda DNA. The same method was used by Cohen *et al* in 1972 to transform competent bacteria with plasmid DNA. The passage of the DNA across the cell membrane - transformation - is a process about which little is known. The exposure of cells to divalent cations at low temperature, and a subsequent short heat shock, makes the cells transiently permeable to DNA, which has been loosely adsorbed to the cells in the presence of Ca^{2+}. Whilst the exact mechanism remains largely unknown, some features of the process have been identified. These are:



Plasmids transform logs more efficiently than linear

Smaller plasmids transform more efficiently than large plasmids

i It is facilitated by divalent cations, chiefly Ca^{2+}. Cells pre-incubated in calcium chloride solution at 0°C are said to become competent to transformation by DNA. Even so, only 3 to 10% of the cells become truly competent to accept DNA. Competence is also transient, reaching a maximum at about 24 hrs, then falling away rapidly beyond two days for cells refrigerated in $CaCl_2$ solution. Preservation of competence by freezing at minus 70°C is an expedient alternative to repeatedly preparing fresh cells.

ii A short heat shock treatment [2 mins at 42°C] greatly enhances the uptake of DNA.

iii The frequency of transformation is higher by several logs for circular than for linear DNA.

iv The frequency of transformation is almost exponentially dependent upon the size of the transforming DNA plasmid. For example, a two-fold increase in plasmid size is accompanied by about a four-fold decrease in transformation, but there is an upper limit.

Beyond a plasmid size of about 10 kb, the transformation begins to level off and virtually ends at about 30 kb. However, strain improvements and the technique of electroporation have increased the size limit to near 100 kb.

v Relaxed plasmids are transformed at the same frequency as supercoiled plasmids.

vi Optimum transformation conditions occur at low cell density, that is, in early log phase.

vii The DNA concentration is critical. 10 to 50 ng of plasmid DNA per 100 µl of competent cells gives ideal results. Each two-fold increase in DNA beyond this amount reduces the transformation frequency by at least the same factor (membrane pore "clogging").

B: Isolation of plasmid DNA

Small plasmids occur in multiple copies per cell, so that a small volume of the cell culture will yield enough plasmid DNA for most purposes such as gel electrophoretic detection, restriction digestion, ligations, sub-cloning, PCR, and DNA sequencing. However, it is possible to increase the plasmid yield up to about 100-fold by the process of amplification. The cell culture is grown to mid-log phase at which point a protein synthesis inhibitor (chloramphenicol or spectinomycin) is added. Incubation is continued overnight during which the cells cannot grow or divide, but the replication of plasmids continues unhindered using the pool of enzymes, metabolites and nucleotides present in the cell cytoplasm. When the culture is harvested, the lysis and purification steps are made simpler by the small cell mass, and a much larger and cleaner plasmid yield obtains.

The small size and generally supercoiled conformation of plasmids render them resistant to mechanical shearing forces, and fairly vigorous manipulations may be used. Nonetheless, all procedures for the isolation and purification of plasmids require that the final plasmid solution be essentially free of cellular contaminants such as cell envelope fragments, proteins, RNA, and chromosomal DNA. You will recall that Experiment 3 describes the isolation of high MW chromosomal DNA, yet the major difference in isolating plasmids is the removal of the chromosomal DNA itself.

Plasmid isolation procedures exploit the two major differences between the DNA of small plasmids and chromosomes, which are:

i The bacterial chromosome is much larger than plasmids used in molecular biology (the *E. coli* chromosome is approximately 4600 kb; plasmids are in the range 2 to 10 kb).

ii Chromosomal DNA is released from lysed cells as broken, linear fragments, whereas plasmids are mostly in closed circular supercoiled or relaxed circular forms.

All plasmid protocols include a step in which the chromosomal DNA is removed or separated from the lysed bacterial culture. This is achieved by heat or alkali treatment. Linearized chromosomal DNA is easily denatured (hydrogen bonds broken) and fragmented in alkali and precipitated as a complex with membrane protein. In contrast, because the plasmid strands are interlocked closed circles, they are only transiently denatured and regain their native conformation when returned to neutral pH; provide that the plasmids are small enough to survive the denaturation step. In the isolation of small plasmids from most *E. coli* strains, multimers are often produced, some of which may be visible in the agarose gel profile – this is a feature of the replication of plasmids in the cell and not of the isolation process.

The main protocols for the extraction of plasmid DNA are:

i Lysis by boiling. A method which produces crude plasmid DNA but retains most of the chr DNA in the centrifuged pellet of cell debris. Organic solvent extraction of proteins/lipids and ethanol precipitation are used to improve the purity of the plasmid DNA.

ii Lysis by mild detergent. A detergent such as SDS or Triton X-100 is used in combination with proteinase K to disrupt the cell membrane and release the cytoplasmic contents. The very crude lysate can be further purified by CTAB/NaCl precipitation of protein/polysaccharide/chr DNA complexes, and by extraction and precipitation of the remaining plasmid solution.

iii Lysis by alkali. This is the method of choice for the isolation of small plasmids as it is rapid, can be used on very small culture volumes, and removes all of the chromosomal DNA. The cell suspension is rapidly lysed with SDS/NaOH. The high pH (>12) of this denaturing step is followed by a neutralization step with potassium acetate (ph 4.5), returning the pH to about 7-8. This step produces a precipitate (“snow”) of most of the membrane fragments, proteins, lipids, and denatured chr DNA that is enhanced by the presence of SDS and potassium. Plasmid purification is carried using a Promega Wizard kit (see below).

There are now a range of commercial products designed to improve even further on the methods above, mostly by simplifying the manipulations following the lysis step in method (iii) above, and resulting in very good quality plasmid DNA that is free of protein, chr DNA, and RNA. These commercial products use small discs of hydroxylated silica resin embedded in small columns. DNA binds to the silica, but proteins are washed through the resin with a salt-ethanol solution. The bound plasmid DNA is then eluted (released) in a small volume of water or TE buffer. Note, that even in this commercial method, the chromosomal DNA and RNA are removed in the alkaline lysis step, before the cleared (centrifuged) lysate is added to the column. The Promega Wizard™ plasmid purification kit is ideal for small culture plasmid preparations (<10 ml), and we will be using this kit in Experiments 4 and 6.

If further purification of plasmid DNA is desired, or for large-scale preparations (>100 ml cultures), this can be done by ultracentrifugation, even if the lysate is quite crude in containing large amounts of protein, polysaccharide, RNA, and chromosomal DNA. During a CsCl density gradient run, proteins complex with EtBr and remain at the top of the tube. RNA is removed as a pellet at the bottom, and chromosomal DNA, if present, bands at a higher position - but lower buoyant density - than the denser supercoiled plasmid DNA (but relaxed and linear plasmids band at the same position as chromosomal DNA). For most purposes, this degree of purification is unnecessary except for the most delicate requirements (eg. very long-term storage at 4°C).

In this experiment, the *E. coli* strain DH5α has be used to prepare competent cells by the cold calcium chloride method. Aliquots of competent cells will be transformed with individual plasmids of different sizes – chosen so that frequency of transformation of multi-sized plasmids may be compared. In the second part of the experiment, you will isolate and purify plasmids from representative transformants, in order to score transformation frequency against size.

The transformation section (Part A) uses three multicopy plasmids of different sizes, designed to test the principle of plasmid size versus transformation efficiency. The plasmids are:

i. pUC18 (2686 bp; ~2.7 kb).

ii. pRSV-Rev (4174 bp; ~4.2 kb).

iii. psPAX2 (10703 bp; ~10.7 kb).

Plasmids will be supplied at the following concentrations and volumes: pUC18 (10ng/2µl); pRSV-Rev (16ng/2µl); psPAX2 (40ng/2µl).

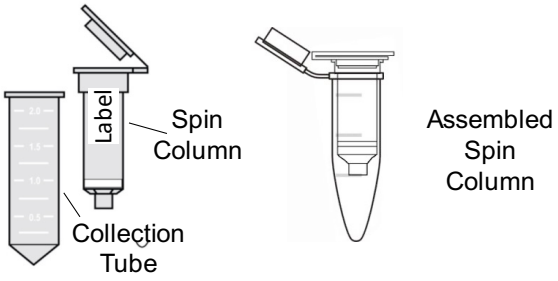
Maps of these three plasmids are depicted overleaf.

These are the three plasmids that will be transformed. The smallest (pUC18; 2686 bp), middle sized (pRSV-Rev; 4174 bp), and largest (psPAX2; 10703). They approximately double in size from one to the next and these size differences will affect the transformation frequency.

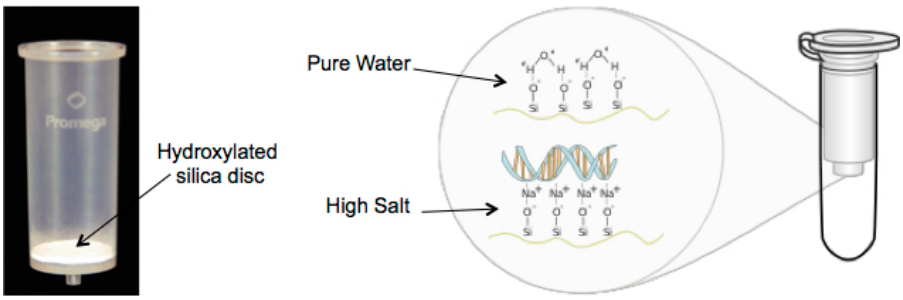
In Part B, you will be using a commercial kit for the plasmid isolation procedure. The Wizard SV Plus Minipreps DNA Purification System is designed to prepare pure plasmid DNA from small overnight cultures of 1 to 10 ml.

The kit uses the basic alkaline lysis procedure to produce a cleared lysate that is free of proteins, chromosomal DNA and RNA. The resuspension buffer (Solution 1) contains RNase; Solution 2 is NaOH/SDS lysis buffer; and Solution 3 is the neutralizing potassium acetate buffer.

The cleared lysate is centrifuged through a mini-column that contains a hydroxylated silica disc to bind the plasmid DNA, but allows flow through of the rest of the lysate. The bound DNA is washed with a salt-ethanol buffer to remove residual contaminants; then eluted from the disc with a small amount of sterile water. The purified plasmid DNA may now be used in any number of ways, such as checking in an agarose gel, for DNA sequencing, PCR, or molecular cloning. The mini-column components are depicted thus:



The diagram below depicts the basic principle of the silica column plasmid binding/purification procedure. The cleared lysate is centrifuged through the column into a collection tube. The



plasmid DNA binds to the hydroxylated silica disc in a high salt environment, and all other unwanted contents of the lysate (extraneous ions, salts, and small molecules) flow into the collection tube. The wash solution (70% ethanol plus salt) step removes and adsorbed material other than the plasmid DNA. Finally, the DNA is eluted in a small volume of water. The binding of DNA in high salt and its release in pure water is indicated in the diagram. A full accounting of the process is covered in the relevant lecture.

TIME WILL NOT PERMIT US TO PERFORM PART A. BEGIN AT PART B (DAY 2)

Part A: Experimental protocol - transformation of plasmid DNA

Preparation of competent cells. The competent cells will be prepared by technical staff in advance as a matter of expediency, but the method of preparation is given below (FYI).

1. A small overnight culture of DH5α is used to inoculate 1 litre of fresh medium [LB+10 mM $MgSO_4$] at 1:100. This is shaken vigorously (aerated) at 37°C until it reaches early to mid-log phase [An $A_{600\ nm}$ of approx. 0.4].

2. Chill the cells on ice for 10 mins.

3. Transfer the culture to centrifuge bottles and spin at 5000 rpm for 5 mins at 2°C. Decant the supernatant and resuspend the pellet in 100 ml of ice-cold 50 mM $CaCl_2$. Leave on ice for 15 mins. Repeat with another 100 ml of ice-cold 50 mM $CaCl_2$.

4. Centrifuge. Resuspend the pellet in 10 ml of ice-cold 15% glycerol / 50 mM $CaCl_2$.

5. Dispense 1 ml lots of competent cells into pre-cooled 1.5 ml tubes and freeze immediately in a dry ice / ethanol bath. Hold these at minus 70°C until needed.

Transformation procedure (Day 1) (Work in pairs)

1. NOTE: Maintaining the competent cells and plasmids on ice is crucial in the first stage. Do not warm up the cells or plasmids; keep both on ice until the heat shock step. Mix and then pipette a 50 µl aliquot of the thawed competent cells suspension into a chilled 1.5 ml tube in an ice bucket containing 5 µl of the plasmid assigned to your bench.

2. Mix gently and leave on ice for 20 mins.

3. Place the tube in a 42°C heating block for 2 mins.

4. Add 1.4 ml of LB broth (containing 0.2% glucose) to the tube and transfer the contents to a 15 ml Falcon tube. Place the tube in a water bath shaker at 37°C for 15 mins. During this incubation, label the undersides of two LB agar plates containing ampicillin at 100 µg/ml.

5. Transfer the culture to a 1.5 ml tube and centrifuge briefly (30-60 sec) to pellet the cells. Remove all of the supernatant by pouring it into the discard bucket and while holding it upside down over the bucket, remove the last droplet of broth with a P100 or P1000 pipette.

6. Add 100 µl of fresh broth to the tube and resuspend the pellet, using a P100 to pipette up and down to help. Then add 50 µl to one of the plates and the rest of the suspension to the

second plate. Spread the aliquots around the plates with the plastic hockey sticks provided. Place the plates upside down in the plastic tub on the side bench.

7. Incubate all plates at 37°C for 24 hours, then hold at 4-6°C until Day 2.

Part B: Experimental protocol - Isolation of plasmid DNA by the alkaline lysis method

Day 2

Count the number of transformants on each plate and enter the numbers on the Class Result Sheet at the Lab Leader console. Do this during centrifuge runs during the plasmid isolation and not at the start of the session.

(For the subsequent plasmid isolation procedure below, work individually)

1. Randomly selected transformants will be picked in advance by technical staff and grown overnight as saturated cultures in LB broth + glucose (0.2%) + ampicillin (50 mg/ml).
2. Centrifuge 1.4 ml of one of the cultures for 1 min. Decant all of the supernatant then proceed to isolate plasmid DNA using the Wizard mini-plasmid isolation system as follows.
3. Resuspend the cell pellet in 250 µl of Solution 1, completely.
4. Add 250 µl of cell lysis Solution 2, cap the tube and mix, firmly, but not vigorously, back and forth a few times. Add 10 µl of alkaline protease and mix briefly. Leave at room temperature for 4 mins. The cell suspension should clear and become viscous. The alkaline protease inactivates endonucleases.
5. Add 350 µl of Solution 3, cap the tube and mix vigorously (back and forth several times). The "snow" precipitate appears and the solution viscosity diminishes. [The precipitate contains K, SDS, proteins, membrane fragments and lipids, and chromosomal DNA.]
6. Centrifuge for 5 mins at maximum speed. During this step, place a Spin Column in a 2 ml Collection Tube. Label the side of the Spin Column, not the 2 ml tube.
7. Decant the cleared supernatant into the Spin Column. Centrifuge for 1 min.
8. Remove the column and discard the flow through. Re-insert the column and add 750 µl of Wash Solution. Centrifuge for 1 min.

9. Remove the column and discard the flow through. Re-insert the column and add 250 µl of Wash Solution. Centrifuge for 2 mins. During this spin, obtain a 1.5 ml tube and label it clearly.

10. Remove the column and discard the 2 ml tube. Place the Spin Column into the labelled 1.5 ml tube. Add 75 µl of sterile water, then centrifuge for 1 min. Place the tube cap to the side, as otherwise the centrifuge lids will shear off the open tops.

11. This time discard the Spin Column and cap your sample 1.5 ml tube, label it and place in the collection box to be stored at -20°C until next week.

Day 3

1. Pipette 20 µl of your plasmid sample into a new tube along with 6 µl of dye mix. Load this into a lane of a 0.7% agarose gel and run during the session. View the gel on a UV transilluminator box to see and identify your plasmid.

RESULT SHEET: EXPERIMENT 4

SECTION A

Plasmids	Numbers of transformants
pUC18	1690
pRSV-Rev	590
psPAX2	140

Numbers in the Table have been adjusted for the aliquots of the transformation mixture that were plated. Therefore, the numbers represent transformants/ml of culture, which can be related directly to the mass (in ng) of each plasmid. Note, these are the results from last year's labs.

SECTION B

This section is not required for your report, but in included for interest. Only three plasmids from each type of transformation were purified and therefore these numbers have no statistical significance, even though they would be broadly in keeping with the expected results if enough of them were screened.

Experiment 4 Report Questions

The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

Q1. Different masses of the three plasmids were used in the transformations, namely: pUC18, 20 ng; pRSV-rev, 32 ng; and psPAX2, 80 ng. (a)What is the idea behind using these respective masses for the plasmid transformations? (b) What would the transformation results look like if the same mass (say 20 ng) of each plasmid was used? [2 marks]
[Clue: (a) Mass versus molecules; (b) Mass versus molecules.]

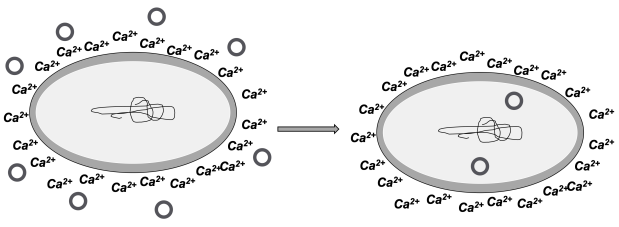
Q2. Transformation frequency is expressed as the number of clones per µg of plasmid DNA. Use the information in the Result Sheet to calculate the frequencies for each of the plasmid transformations. [1.5 marks]
[Clue: Direct calculations equating plasmid mass to transformation numbers and adjusting to a defined mass of 1 µg.]

Q3. In the bacterial transformation protocol, the incubation step in ice-cold Ca^{2+} is followed by a short heat pulse at 42°C. Explain exactly what is happening in these two steps. [2 marks]
[Clue: Check relevant lecture notes and Intro notes to this Expt.]

Q4. You are given two plasmids. They are, respectively, the low copy number pBR322 (4.4 kb) and the high copy number psPAX2 (10.7 kb). They are transformed into *E. coli* competent cells and one transformant of each plasmid is selected and grown overnight at 37°C as a 5 ml culture. These cultures are then used to purify the two plasmids, and a 10 µl aliquot of each is run in an agarose gel. What can you say about the comparative appearance of the Gel-Red stained plasmid bands in the gel? [1.5 marks]
[Clue: Amount of DNA.]

Q5. Twenty nanograms (20 ng) of pUC18 (~2.7 kb), pRSV-rev (~4.2 kb), and psPAX (~10.7 kb) are run in a single lane of an agarose gel, as indicated in the gel profile below, with the direction of migration left to right. What can you say about the number of Gel-Red molecules bound to each of these plasmid bands? [2 marks]
[Clue: Size versus staining.]

Q6. The drawing below depicts a before and after event (left to right) of a small plasmid (small circles) being transformed into Ca^{2+} competent *E. coli* cells.



Two students have different views of this transformation event. The first student says that as transformation is relatively arbitrary and that two plasmids from the pool of eight surrounding the cell on the left have been transformed into the cell on the right. The second student thinks that only one plasmid on the left has been transformed and the other one has appeared when the single plasmid replicated. Which student is correct and why? [2 marks]
[Clue: How many plasmids are able to transform a single *E. coli* cell? Try lecture notes.]

Q7. In the purification section, the plasmid is purified from other components of the cell lysate by being bound to hydroxylated silica in the column disc, in the presence of a buffer that contains salt. After the washing steps, the plasmid is released from the silica disc in pure water. Explain briefly how this binding/release process works. [2 marks]
[Clue: Intro notes to this Experiment and relevant lecture slides/notes.]

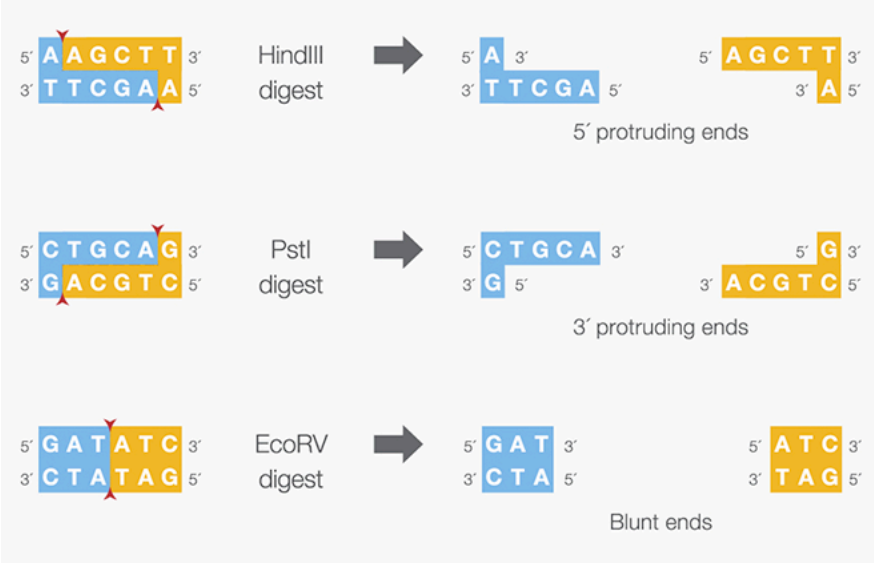
EXPERIMENT 5

RESTRICTION ENDONUCLEASE DIGESTION OF DNA

Background

Restriction endonucleases are enzymes that recognize specific sequences within double-stranded DNA, and cut or 'restrict' phosphodiester bonds in the two strands, generating new 5'-phosphate and 3'-hydroxy termini. The recognition sites are mostly tetranucleotide (4-base) or hexanucleotide (6-base) palindromes with axes of dyad (twofold) symmetry.

The palindromic sequence in each strand is the same but in opposite orientations (5'--->3'). Each enzyme generates cuts that are staggered or asymmetric between the first or last two bases of each palindrome, producing cohesive ('sticky') protruding 5' or 3' ends; or symmetrical cuts at the centre of each palindrome, producing blunt ('flush') ends. Examples of the two types of staggered cuts and a blunt cut are shown in the diagram. In a random sequence of DNA, a given 4-base palindrome will occur on average once every 4^4 bases (256), and a 6-base sequence once every 4^6 bases (4096). Of course, in practice these frequencies vary because the DNA molecule being cleaved is either not long enough, or not random enough, or the recognition sequence of the enzyme being used is either under- or over-represented in a particular DNA sequence or genome.

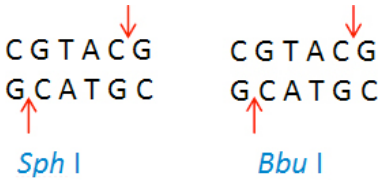


Each restriction enzyme has a set of optimal reaction conditions. The major variables are the temperature of incubation and the salt composition and pH of the buffer. Most enzymes are optimally active at 37°C but some work best at lower or higher temperatures (25°C, or 55 to 75°C). All restriction enzymes have salt (Na, K, Mg, or Mn) and ionic strength requirements (low, medium, or high), and most commercially supplied enzymes will work with one of three

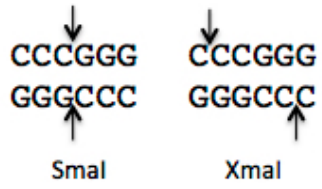
"universal" reaction buffers that vary mainly in their salt content. This avoids manufacturing specific buffers for each enzyme. By definition, one unit of enzyme is the amount required for the digestion of 1 µg of DNA to completion in 15 minutes in the recommended buffer and at the optimum temperature in a 50 µl reaction volume.

Restriction endonucleases are isolated from bacteria. Their purpose in the host organism is to maintain the fidelity of the genome by restricting foreign, invasive DNA. The host DNA is protected from self-digestion by the host-methylation system - methylases add bulky methyl groups to adenine or cytosine bases within the recognition palindrome, thus protecting these sites from the host restriction enzyme, whereas the foreign DNA is unmethylated in the same positions and is therefore restricted, then degraded by host exonucleases.

There are now some several hundred known restriction endonucleases and they are named from the originating microorganism. Although most sequences are unique, there are a significant number of enzymes from different organisms that recognize the same sequences, and these are called isoschizomers (e.g. SphI and BbuI).



Neoschizomers have the same palindrome recognition substrate site, but the cuts are in different positions, as shown here for the pair SmaI (producing blunt ends) and XmaI (staggered ends) . Some restriction enzymes have 4-base sites that are within another enzyme's 6-base site (e.g. Sau3A [GATC] and BamHI [GGATCC]).

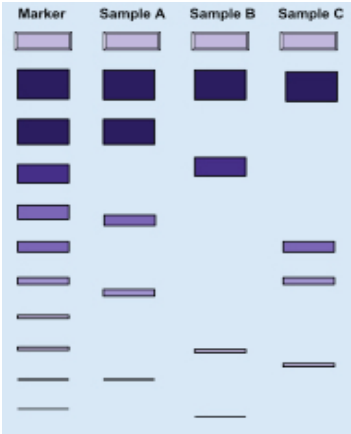


Restriction endonucleases are the focal point from which most molecular biology methods emanate. Molecular cloning, recombinant DNA technology, and genetic engineering (all 'roses by any other name') depend upon the cutting of DNAs from different sources and the joining of the fragments to produce hybrid, cloned molecules. In this way genes and small sequences of DNA can be isolated from much larger molecules and examined or analyzed in a number of ways. The method works because any given enzyme will generate compatible termini or defined ends on molecules that can be ligated or joined. Molecular cloning / recombinant DNA will be dealt with in detail in Experiment 6.

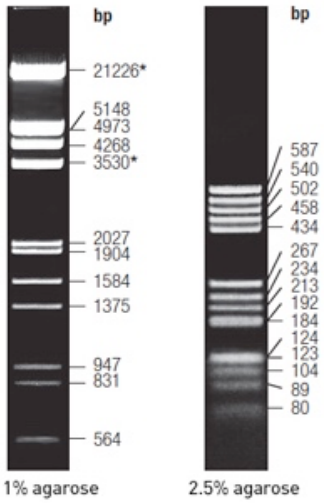
In this experiment we will examine the different fragment profiles produced upon digestion of DNA by representative 4- and 6-base cutters, namely HaeIII and EcoRI. Two DNA molecules of vastly different sizes will be digested. These are, *E. coli* chromosomal DNA (4600 kb per chr

molecule) and pUC18 (2.69 kb), a plasmid vector commonly used in molecular cloning. The families of fragments generated from the digests will then be analyzed in agarose gels.

A few points about restriction digest fragments should be noted here. For example, if a long linear double stranded DNA molecule, or a plasmid is digested with a restriction enzyme, a family of fragments of different sizes is yielded, as depicted in this diagram. The first lane contains size markers and the other lanes contain digested fragments. However, all 5 lanes show the same feature of the fragments, that is, they look broader nearer the origin and thinner lower in the gel. Though only a drawing, it is nevertheless a facsimile of native gels/DNA fragments. The width of each band and darker or lighter hues reflect the lengths of the fragments and the amounts of Gel-Red bound. Despite differences in fragment lengths and staining, the single source DNA means that each band contains <u>exactly the same number of DNA molecules</u>. This 'stylised' representation of DNA fragments in agarose gels helps explains the ideas above, but in reality, the traces can be quite different in appearance, depending on the sizes of fragments and density of the agarose gel.



For example, take an actual gel run of restriction fragments from different sources, such as in the diagram at left. Here, the set of fragments in the left lane resemble those in the first lane of the diagram above; and this is chiefly because the fragment sizes are large enough and the gel percentage low enough (1%) to resolve the fragments in typical profile with broader and brighter bands higher in the gel. The second lane of this gel has a profile of very small DNA fragments and this is reflected by the need to use a very high percentage agarose gel (2.5%). Though not obvious, the same 'rules' apply to these DNA bands, with the upper bands being larger with more Gel-Red adsorbed, and gradually decreasing down the lane to the shorter fragments. These are only faintly stained with Gel-Red, which reflects their smaller sizes, but notice how the bands are broader than the larger ones higher up.



This feature is due to the very small fragments diffusing in the agarose because the pores of even this high percentage gel are still big enough to allow the small DNA fragments to move too

freely through the gel. If this had been a more 'normal' gel, of say 1%, the lower bands would have diffused even more and would barely be visible. However, this is a rarer, unusual anomaly due purely to the use of very small/short DNA fragments in lane 2.

Experimental protocol

Day 1 (work in pairs)

Restriction endonuclease digestion

1. NOTE 1: There will be reagents for both Expts 5 and 6 in your ice buckets. These will be labelled, but also easily identified by different coloured Eppendorf tubes.

2. NOTE 2: The reagent tubes in the ice buckets contain very small volumes which two or more groups need to use. Before you take out your small volume, centrifuge the tube (against a balanced empty tube) in the mini-spin (dome) centrifuge at the end of your bench.

3. NOTE 3: These reactions require very small volumes (~1-2 µl), so be careful to set and use the P10 pipettor correctly – do not push down past the first (soft) stop point, or you will draw up more than the required volume. Push down past the soft stop position to expel the small aliquot into the reaction tube. Best to place these aliquots around the side of the tube so that you can see that they have actually been expelled from the pipettor.

4. Set up a restriction digestion in a 1.5 ml tube as indicated in the table below. The Table is a composite of four different types of reactions, so be very careful to add the correct reagents to your reaction, which will be one only of the four digests.

Contents	Tube (volumes in µl)			
	1	**2**	**3**	**4**
pUC18 (250 µg / ml)	-	-	1	4
chr DNA (2.5 mg / ml)	2	2	-	-
HaeIII 10 x buffer	-	1.5	-	1.5
EcoRI 10 x buffer	1.5	-	1.5	-
HaeIII enzyme	-	1	-	1
EcoRI enzyme	1	-	1	-
Water (to 15 µl)	10.5	10.5	11.5	8.5

5. Centrifuge briefly in a QuickSpin low-speed benchfuge (5 seconds) to bring the contents to the bottom, then mix by gently tapping each tube.

6. Incubate at 37^{o}C for 40 mins. Add running dye solution (6 µl) then inactivate the enzymes at 70^{o}C for 10 mins in a dry heat block. Remove the tube from the bath and place in the box for storage at -20^{o}C until Day 2.

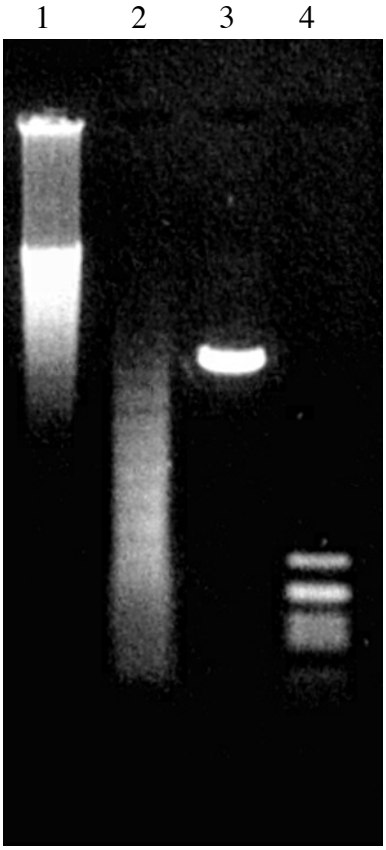
Day 2

Agarose gel electrophoresis of restriction digests

1. Collect and thaw the reaction mixtures from Day 1. Since each bench has the same digest, apply samples in different gels. The gel sheets will be labelled so that two samples from each bench are loaded into each gel.

2. Apply the samples to a pre-cast 0.8 % agarose gel (+ Gel-Red) and run at 110 volts for until the BPB dye front is about 3 cm from the end of the gel. Examine the gel on a UV transilluminator.

RESULT SHEET: EXPERIMENT 5



Lane 1: chr DNA / EcoRI

Lane 2: chr DNA / HaeIII

Lane 3: pUC18 / EcoRI

Lane 4: pUC18 / HaeIII

The degree of shading of the two chromosomal profiles indicates Gel-Red fluorescence.

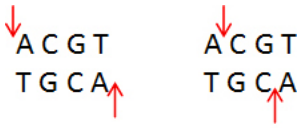
Experiment 5 Report Questions

The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

Q1. For the chromosomal lanes only on the Result Sheet, why are the HaeIII bands fainter in staining intensity than the EcoRI bands? [2 marks]
[Clue: Should be self-evident! However, Lecture notes as usual should help.]

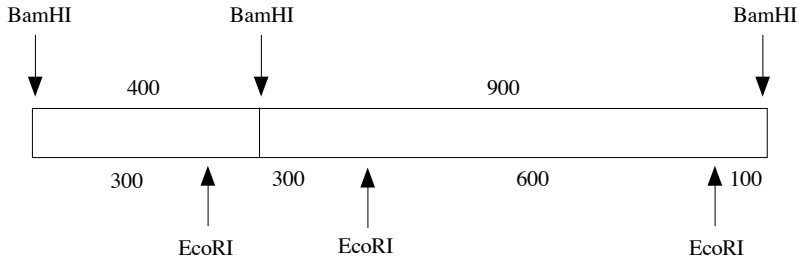
Q2. Shown below are the same two palindromic restriction sites. The arrows indicate different site specificities by two enzymes. To which subclass of restriction endonucleases do these two enzymes belong? [1 mark]
[Clue: Answer is on one of the lecture slides.]



Q3. You wish to clone a tetracycline resistance gene from one plasmid to another and you select the enzyme BamHI because it has a unique site in the second (cloning) plasmid. However, your efforts fail to generate tetracycline resistant clones. What is the likely problem? [1 mark]
[Clue: Think about Open Reading Frames.]

Q4. You will notice the very different stock concentrations of pUC18 versus chromosomal DNA used in the digestions, as per the table in the protocol section for this experiment. Why is there a large difference in the masses of plasmid and chromosomal DNAs used in these digests? [2 marks]
[Clue: Seeing is believing.]

Q5. The drawing below is of a fragment of DNA with BamHI and EcoRI sites as indicated. The distances between the arrowed sites are given in base pairs above and below the fragment for BamHI and EcoRI, respectively.



If this DNA fragment is digested with both enzymes and the products run in an agarose gel. (a) How many DNA bands would be seen in the stained gel? (b) How many actual fragments are generated? [2 marks]
[Clue: Scale drawing and therefore simple 'eyeball' measuring.]

Q6. A student performs a restriction digest on a plasmid, then runs a sample of the uncut plasmid (lane 1) and the digested plasmid (lane 2) in an agarose gel, as depicted below. The student cannot decide if the digest is complete or incomplete. Which is correct? Explain your choice of answer. [2 marks]
[Clue: The position of the digested fragments is crucial, especially the lower one.]

Q7. A new plasmid (pAMG22) has been isolated and it is 6 kb. The plasmid is treated with 4 different restriction enzymes singly and in all double digest combinations. The fragment sizes from the digests are listed below. Draw a circular restriction map of pAMG22 based on this information, placing the single BamHI site at 12 o'clock. Note, you must show the working calculations or reasoning that allowed you to arrive at the final map. [4 marks]

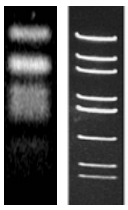
EcoRI = 6 kb; BamHI = 6 kb; PstI = 6 kb; SstI = 0.8 kb, 1.9 kb, 3.3 kb.
EcoRI + BamHI = 1.9 kb, 4.1 kb.
EcoRI + PstI = 2.6 kb, 3.4 kb.
BamHI + PstI = 1.5 kb, 4.5 kb.
SstI + EcoRI = 0.8 kb, 1.4 kb, 1.9 kb, 1.9 kb.
SstI + BamHI = 0.3 kb, 0.5 kb, 1.9 kb, 3.3 kb.
SstI + PstI = 0.7 kb, 0.8 kb, 1.2 kb, 3.3 kb.

[Clue: Looks complicated, but easy if you use clockfaces to position the sites, beginning with the three single cut enzymes (first line) then adding in the SstI sites. Put one of the single sites at 12 o'clock.]

Q8. A DNA fragment with SmaI ends, obtained from a partial SmaI digest of a longer fragment, is ligated into the unique SmaI site of the cloning vector, pUC18. A subsequent complete SmaI digest of new recombinant plasmid yielded a total of four SmaI fragments. How many SmaI sites are there in this recombinant plasmid? [2 marks]
[Clue: Draw the plasmid and put in the sites.]

Q9. Go to the Online Snapgene site and select the plasmid pcDNA3. The link is:
https://www.snapgene.com/resources/plasmid-files/?set=basic_cloning_vectors
Which restriction enzymes might you use to subclone the NeoR/KanR gene from pcDNA3? [2 marks]
[Clue: Choose the most appropriate sites either side of the gene. Here, appropriate means sites nearest the gene and in uncoded regions.]

Q10. Below are two profiles of the same pUC18 HaeIII restriction digest in separate gels. The first profile (left) is the one from the Result Sheet and careful inspection might identify 5 or 6 HaeIII fragments. The profile on the right shows 8 HaeIII fragments, which happens to be the number of HaeIII sites in pUC18. (a) Where are the "missing" two HaeIII fragments in the left profile? (b) Why do the two profiles look so different? [2 marks]
[Clue: Both answers straight out of lecture slide/notes.]



VIRTUAL EXPERIMENT 6

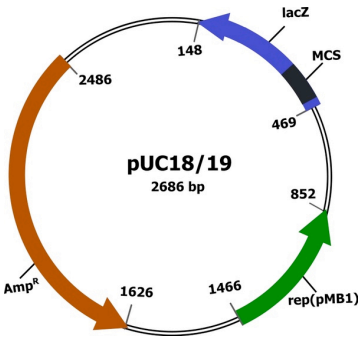
MOLECULAR CLONING OF RECOMBINANT DNA

Background

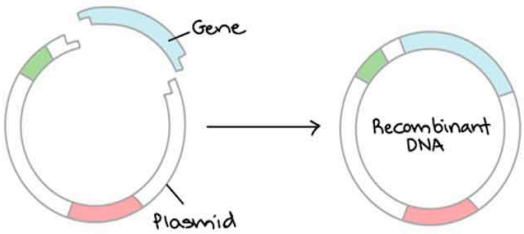
Plasmids are small extrachromosomal DNA replicons. The largest is about 400 kb in size. They occur in 1 to 2 copies per bacterial host cell. The smallest plasmids, which include all of those constructed as cloning vectors, are about 2 to 6 kb in size, occur at 5 to 200 copies per host cell.

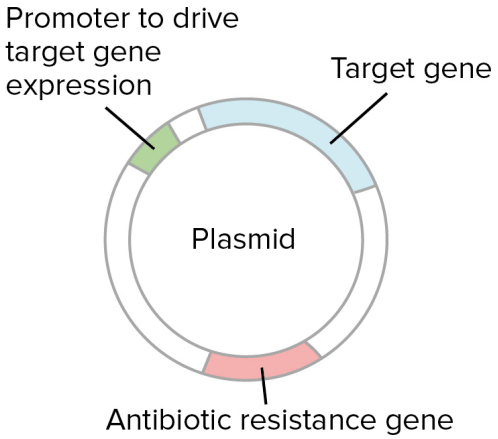
Plasmid vectors have been designed to contain some or all of the following properties:

i. Selectable phenotypic markers, eg. ampicillin resistance;
ii. Relaxed replication control for a high copy number;
iii. Small size enabling large fragments to be accommodated;
iv. Unique and plentiful restriction endonuclease sites (MCS);
v. Provision for the indirect or direct selection of cloned inserts.



pUC18 fits the bill for a typical small cloning vector (shown here for the pUC18/19 pair). A simple cloning scheme is indicated in the diagram below. pUC18 also works in the LacZ blue/white selection system to identify recombinant plasmids versus religated plasmid vectors.

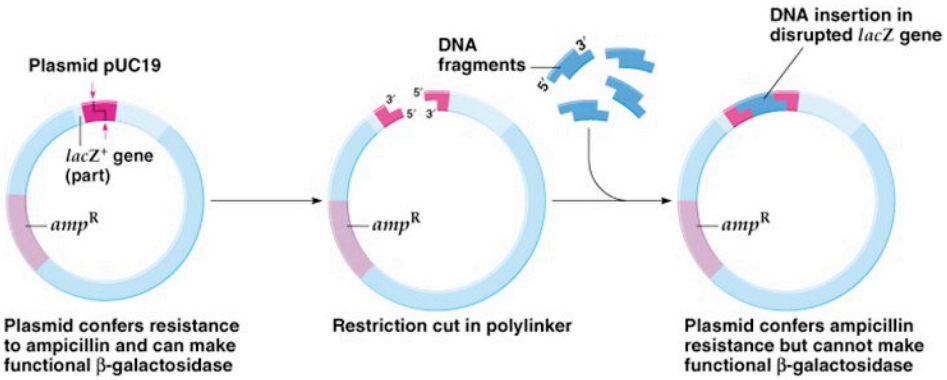




LacZ blue/white indirect selection system

This is insertion into, and inactivation of, the gene for β-galactosidase (*lacZ*) with attendant loss of the ability of recombinant clones to hydrolyze a chromogenic substrate X-Gal [5-bromo-4-chloro-3-indolyl-β-D-galactoside]. Inactivation is achieved because the multi-cloning site is within the N-terminal α-peptide part of the *lacZ* gene. In the presence of the inducer IPTG

[isopropyl-β-D-thiogalactopyranoside], hydrolysis of X-Gal releases indole, which gives vector-only colonies a blue colour. If an insertion has inactivated the β-galactosidase gene, the active enzyme cannot be produced and X-Gal is not hydrolyzed. Thus, colonies containing these recombinant plasmids are white. This type of selection is performed in a β-galactosidase α-peptide gene deletion strain (ΔM15 *lacZ*) containing the defective *lacZ* gene in the chromosome, which can complement the vector α-peptide to produce an active enzyme. A more complete description of the blue/white LacZ/X-gal selection system is given in the relevant lecture.



DNA ligation is a complex process to control, even though the simple objective is to join compatible cohesive or blunt restriction ends. In practice, cohesive ends will ligate at least ten-fold more often than blunt ends and thus it is always best to use restriction enzymes that generate cohesive termini. If blunt ends are unavoidable, there are ways to improve the yield of ligated products, as follows:

i Some ligation buffer cocktails and conditions assist the ligation of blunt ends, for example by including polyethylene glycol to increase molecular crowding.

ii A second method uses linkers. These are short, synthetic oligomers, which form blunt ended duplexes containing restriction sites for enzymes that will produce cohesive ends. The oligomers ligate efficiently to blunt ends when added at high concentration; and subsequent restriction provides cohesive ends for ligation.

The enzyme T4 DNA ligase catalyses the formation of phosphodiester bonds between adjacent 5'-P and 3'-OH termini in double-stranded DNA. There is a requirement for Mg^{2+} and ATP. Its optimum temperature is 37°C, but most reactions are conducted at 15°C (which is just above the mid-point melting temperature of cohesive ends).

In simple ligations, in which a single recombinant molecule is generated from insert plus vector molecules, or in which a linearized plasmid needs to be rejoined as a circle, the desired product will invariably occur in a complex mixture of linear and circular molecules, and concatenated molecules (long, linear molecules of many fragments joined end to end). The three most critical parameters in ligation reactions between vector and insert are:

i The concentration of ends.
ii The respective concentration of two different molecules, vector and insert, when applicable.
iii The concentration of added ligase.

Ligation is a compromise of two competing reactions:

i End-to-end joining of two different molecules (i.e. bimolecular concatemerization).
ii Joining of opposite ends of the same molecule (i.e. unimolecular cyclization). There is a formula for calculating the optimum concentrations of reacting ends, but commercial T4 DNA ligases come with conditions and protocols for different types of ligation reactions.

The plasmid pUC18 will be used as the vector in this experiment. Bacteriophage lambda DNA (cut with HindIII) will be the source of insert DNA, for reasons described below. This experiment consists of several stages in a typical molecular cloning strategy:

i restriction digestion of vector and insert.
ii ligation of restriction fragments.
iii transformation of ligated recombinant molecules into a competent bacterial host.
iv identification of recombinant clones by loss of β-galactosidase activity [IPTG / X-Gal white instead of blue clones].
v physical identification of recombinant plasmids by agarose gel electrophoresis.

A HindIII digest of lambda DNA will generate eight fragments of 23.1, 9.4, 6.6, 4.4, 2.3, 2.0, 0.56 and 0.13 kbp, respectively. Remembering that transformation frequency is inversely proportional to plasmid size, it is expected that most recombinants will contain only the smaller lambda inserts. Even so, the ligated vector alone will be represented in far greater numbers than even the smallest recombinant. This creates a problem in distinguishing the smallest recombinant plasmid (2.81 kb) from the religated vector (2.68 kb) in an agarose gel as they migrate almost together. However, this problem can be solved by noting the colour difference of the transformant colonies on the agar selection plates. If the clone is blue, it is the pUC18 vector in

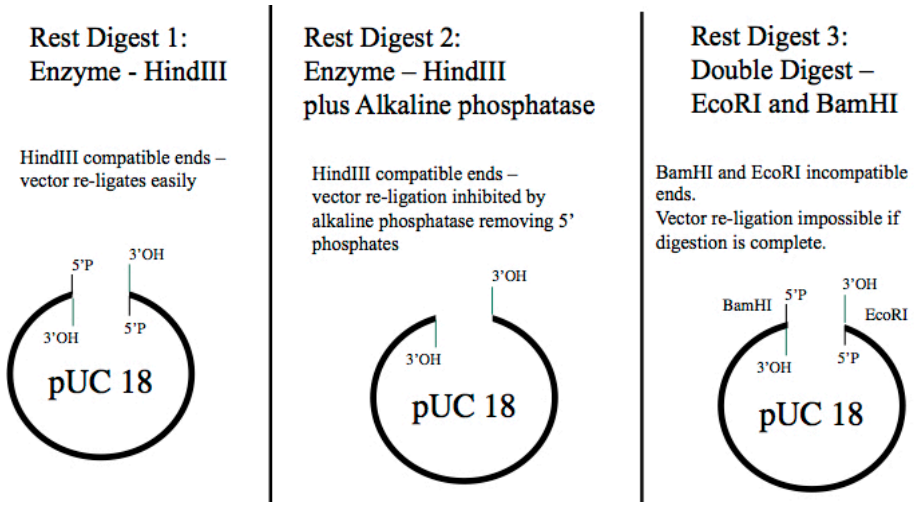
the colony, if white, and the gel shows a plasmid about the same size as pUC18, it is smallest recombinant, with the 0.13 kb lambda fragment ligated to pUC18.

Consider, for example, that the aim of this experiment was to isolate a recombinant the largest lambda insert, in which case a great number of transformants would need to be screened, so it would be useful to eliminate the high vector background. Two ways of achieving this are:

i Directional cloning is an efficient means of producing recombinants. In this method, insert and vector are digested with two restriction enzymes that produce incompatible termini. Linearized vector molecules cannot be relegated, but recombinant molecules can be circularly ligated with inserts directionally aligned.

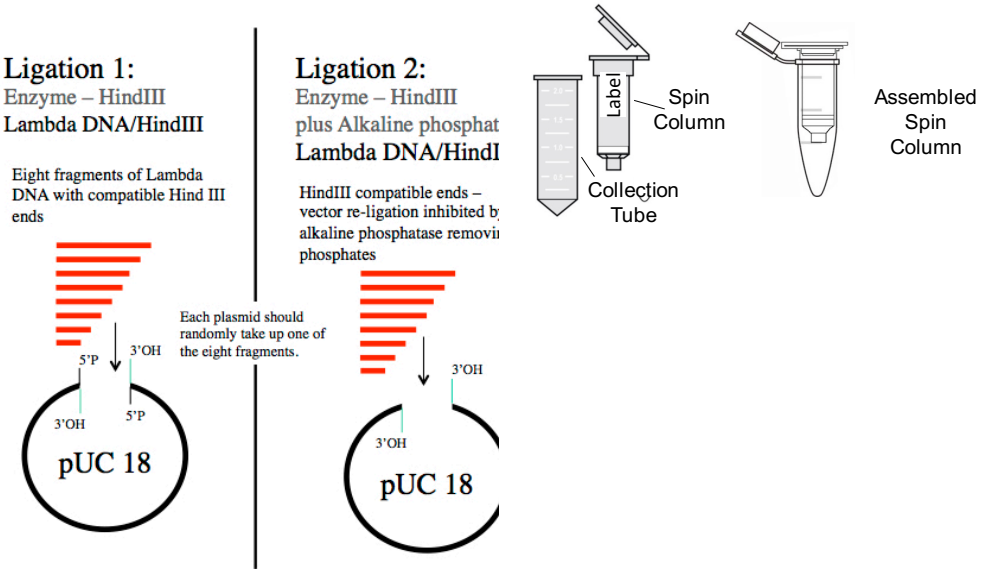
ii The background of blue vector transformants is reduced by alkaline phosphatase treatment of the restricted vector before it is ligated to an insert. This also should increase the white/blue ratio of transformants. Alkaline phosphatase removes 5'-phosphate groups from diagonally opposite ends of each strand, and as a result, the ends of the vector cannot be religated to one another, but can be ligated to the insert, and then transformed efficiently.

In the protocols below, you will be testing three different types of ligation reactions. Ligation 1 will be looking at the pUC18 vector cut with HindIII and later ligated to lambda HindIII fragments. Ligation 2 will be using vector cut with HindIII and treated with alkaline phosphatase, which removes the 5-prime phosphate groups from the restricted ends. This dephosphorylated vector will then be ligated to lambda HindIII fragments.



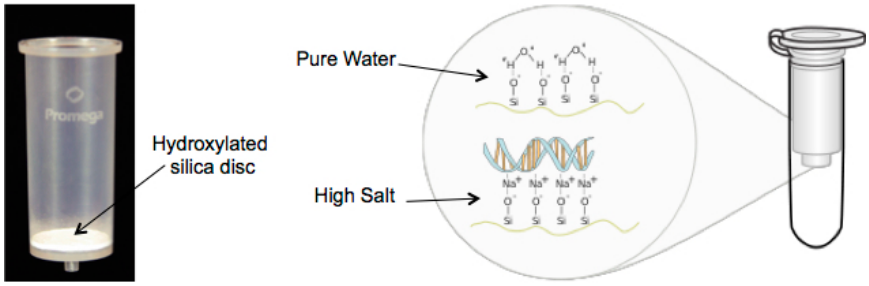
Ligation 3 will be looking at the concept of directional cloning using the vector doubly digested with BamHI and EcoRI. Here there will be no need to do an insert cloning step as the vector

background should be zero. Therefore, if an insert were used, all of the resulting clones would be white. Instead, the cut vector will be ligated anyway and the resulting transformation and number of clones will indicate the effectiveness of this method of vector background minimization.



In the last part of this experiment, we will be using a commercial kit for the plasmid isolation procedure. The Wizard SV Plus Minipreps DNA Purification System is designed to prepare pure plasmid DNA from small overnight cultures of 1 to 10 ml. It is the same as the one used in Experiment 4 to isolate/purify plasmid DNA from cleared cell lysates.

The kit uses the alkaline lysis procedure to produce a cleared lysate that is free of proteins, chromosomal DNA and RNA. The resuspension buffer (Solution 1) contains RNase; Solution 2 is NaOH/SDS lysis buffer; and Solution 3 is the neutralizing potassium acetate buffer. The cleared lysate is centrifuged through a mini-column that contains a hydroxylated silica disc to bind the plasmid DNA, but allows flow through of the rest of the lysate. The bound DNA is washed with a salt-ethanol buffer to remove residual contaminants; then eluted from the disc with a small amount of sterile water. The purified plasmid DNA may now be used in any number of ways, such as in an agarose gel, for DNA sequencing, or molecular cloning. The silica disc binding principle is repeated here (from Expt. 4 and the relevant lecture).



Experimental protocol

Day 1 (work in pairs)
Restriction digestion and alkaline phosphatase treatment

1. NOTE 1: There will be reagents for both Expts 5 and 6 in your ice buckets. These will be labelled, but also easily identified as different coloured Eppendorf tubes.

2. NOTE 2: The reagent tubes in the ice buckets contain very small volumes which two or more groups need to use. Before you take out your small volume, centrifuge the tube (against a balanced empty tube) in the mini-spin (dome) centrifuge at the end of your bench.

3. NOTE 3: These reactions require very small volumes (~1-2 µl), so be careful to set and use the P10 pipettor correctly – do not push down past the first (soft) stop point, or you will draw up more than the required volume. Push down past the soft stop position to expel the small aliquot into the reaction tube. Best to place these aliquots around the side of the tube so that you can see that they have actually been expelled from the pipettor.

4. Set up a restriction digestion in a 1.5 ml tube as indicated in the table below. The Table is a composite of three different types of reactions, so be careful to add the correct reagents to your reaction, which will be one only of the three digests.

Additions	Digest 1 (µl)	Digest 2 (µl)	Digest 3 (µl)
pUC18 (100 ng)	1	1	1
10 x HindIII buffer	1	1	-
10 x BamHI/EcoRI buffer	-	-	1
Alkaline phosphatase	-	1	-
HindIII	1	1	-
EcoRI	-	-	1
BamHI	-	-	1
Water (sterile)	7	6	6
Total volume	10	10	10

5. Spin briefly (low-speed minifuge). Incubate the reaction at 37°C for 40 mins.

6. Inactivate the enzymes by incubating in a 70°C heating block for 12 mins. Remove the tube from the bath and place in the box for storage at minus 20°C until Day 2.

Day 2 (work in pairs)

Ligation and transformation

1. Set up a ligation reaction in your sample tube as indicated in the table below according to your Ligation Number from Day 1. [The first 10 µl is your prepared vector from last week].

Additions	Ligation 1 (µl)	Ligation 2 (µl)	Ligation 3 (µl)
pUC18/HindIII	10	-	-
pUC18/HindIII dephosphorylated	-	10	-
pUC18/BamHI,EcoRI	-	-	10
Lambda DNA/HindIII (100 ng)	1	1	-
2 x ligation buffer	10	10	10
DNA ligase	1	1	1
Total volume	22	22	21

2. Spin briefly to mix and leave at room temperature for 25 mins. The ligation reaction is rapid because a special buffer is used to cause molecular crowding of the DNA molecules in the reaction and this reduces the half-life of the reaction from many hours to well under an hour. The ligation components are from the LigaFast™ kit (Promega).

3. Heat inactivate the DNA ligase for 3 mins at 70°C. Whilst unnecessary, this heat inactivation step has been shown to improve the transformation efficiency about twofold.

4. Remove 15 µl of the ligation mixture and add it to a tube of 75 µl of competent cells on ice. Hold on ice for 20 mins.

5. Heat shock for 2 mins at 42°C.

6. Add 1.5 ml of LB broth (+ 20 mM glucose) to the transformed cell suspension, transfer the contents to a 15 ml Falcon tube, and incubate for 12 mins at 37°C with vigorous aeration.

7. Label two LB agar plates with your name. [The plates also contain glucose (0.2%); ampicillin (50 µg/ml); X-gal (500 µg/plate - stock solution is 10% in DMF); and IPTG (200 µg/plate - stock solution is 100 mM).]

8. Spread 2 x 150 µl (for Ligation 1), or 2 x 500 µl (for Ligations 2 or 3) from the culture onto 2 labelled plates.

9. Put the plates right side up in the collection tub. The plates will be incubated at 37°C for 24 hours then at 4°C until Day 3.

Day 3

Scoring of transformants: Count blue and white transformants on the selective plates. Do this during a break in the protocol. Adjust counts to CFU per 1.0 ml before entering on the sheets.

The plasmid isolation scheme is as follows: (Work individually)

One student from each bench should use a blue colony cell suspension for the plasmid isolation, and the others different white colony suspensions. Write down the clone colour written on the culture tube – blue or white – before you begin the plasmid isolation procedure.

1. This step has been done in advance by the technical staff. Pick random white and blue colonies from the X-gal plates and inoculate into separate, labelled LB (+glucose) broths + ampicillin (25 µg/ml). Incubate these broths overnight with aeration at 37°C.
2. Centrifuge 1.4 ml of one of the cultures for 1 min. Decant all of the supernatant then proceed to isolate plasmid DNA using the Wizard system as follows.
3. Resuspend the cell pellet in 250 µl of Solution 1, completely.
4. Add 250 µl of cell lysis Solution 2, cap the tube and mix, firmly, but not vigorously, back and forth a few times. Add 10 µl of alkaline protease and mix briefly. Leave at room temperature for 4 mins. The cell suspension should clear and become viscous. The alkaline protease inactivates endonucleases.
5. Add 350 µl of Solution 3, cap the tube and mix vigorously (back and forth several times). The "snow" precipitate appears, and the solution viscosity diminishes. [The precipitate contains K, SDS, proteins, membrane fragments and lipids, and chromosomal DNA.]
6. Centrifuge for 5 mins. During this step, place a Spin Column in a 2 ml Collection Tube. Label the side of the Spin Column, not the 2 ml tube.
7. Decant the cleared supernatant into the Spin Column. Centrifuge for 1 min.
8. Remove the column and discard the flow through. Re-insert the column and add 750 µl of Wash Solution. Centrifuge for 1 min.
9. Remove the column and discard the flow through. Re-insert the column and add 250 µl of Wash Solution. Centrifuge for 2 mins. During this spin, obtain a 1.5 ml tube and label it clearly.

10. Remove the column and discard the 2 ml tube. Place the Spin Column into the labelled 1.5 ml tube. Cut the top off the tube. This is essential, as otherwise the centrifuge lids will shear off the open tops. Add 75 µl of sterile water, then centrifuge for 1 min.

11. This time, discard the Spin Column and transfer the sample to a new 1.5 ml tube with a lid, label and place in the collection box to be stored until next week.

12. Place your plasmid DNA sample in the collection box until Day 4. Hold these at minus 20°C.

Day 4

Agarose gel electrophoresis of plasmids

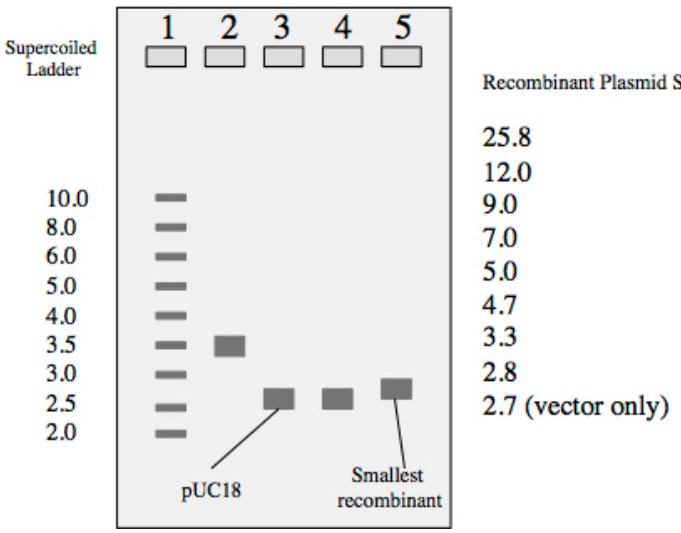
1. Add 15 µl of running dye to your plasmid sample from Day 3. Load 20 µl into the well of a 0.7% agarose gel. Supercoiled plasmid standards will be added to lane one of each gel.

2. Subject the samples to electrophoresis at 100 volts for 60-90 mins.

3. Examine the gel on a UV transilluminator. Identify the size of your plasmid by comparison with the supercoiled plasmid standards and enter your result on the Board.

The supercoiled plasmids standard ladder in kb pairs is: 16.2, 14.2, 12.1, 10.1, 8.1, 7.1, 6.0, 5.0, 4.0, 3.0, and 2.1. The Lambda HindIII fragment sizes in kb pairs are: 23.1, 9.4, 6.6, 4.4, 2.3, 2.0, 0.6, 0.13. The recombinant plasmids sizes are obtained by simply adding the vector (pUC18 ~ 2.7 kb) to each of the lambda fragments to yield: 25.8, 12.0, 9.0, 7.0, 5.0, 4.7, 3.3, 2.8, 2.7 (the last of these is the pUC18 vector). The gel profile below is a guide to indicate these plasmids.

White colonies indicate that the plasmid took up DNA. Compare plasmid size to the supercoiled ladder to determine which fragment of HindIII/Lambda DNA (0.13-23.1) was inserted in pUC18.



Note: The 2.8 and 2.68 kb plasmids are difficult to discriminate in the gel. However, the 2.81 plasmid with the smallest insert (0.13 kb) comes from only white clones and the vector 2.68 kb plasmid from only blue clones and this information was recorded on Day 2.

RESULT SHEET: EXPERIMENT 6

Numbers of colonies (cumulative totals for many years)

Ligation 1		Ligation 2		Ligation 3	
blue	white	blue	white	blue	white
46,887	4,401	1,833	11,915	173	0

Interpretation

The trend of results is generally accurate, with anomalies able to be explained by incomplete enzymatic reactions or failure to fully inactivate alkaline phosphatase.

Ligation 3 (double digest with EcoRI and BamHI)

The cumulative multi-year total of blue colonies from Ligation 3 was 173. They are derived from incomplete digestion reactions, since doubly-digested plasmids are unable to relegate, due to incompatible ends. No white colonies should ever be obtained.

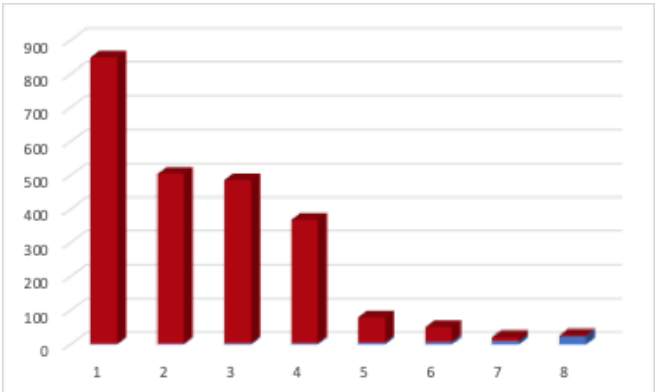
Plasmid numbers and sizes

	Plasmid Size (kb)	Plasmid Numbers
	2.7 (vector)	n/a*
1	2.8	846
2	3.3	502
3	4.7	481
4	5.0	294
5	7.0	72
6	9.0	41
7	12.0	10
8	25.9	0

*(n/a) The number of ~2.7 kb pUC18 plasmids is unimportant and is used only as 'diagnostic' proof that blue colonies contained the original, religated vector.

The lambda DNA HindIII fragment sizes are: 23.1, 9.4, 6.6, 4.4, 2.3, 2.0, 0.56 and 0.13 kb.

The results in the table are plotted below as recombinant plasmid sizes (x axis) versus cumulative plasmid numbers (y axis).



Experiment 6 Report Questions

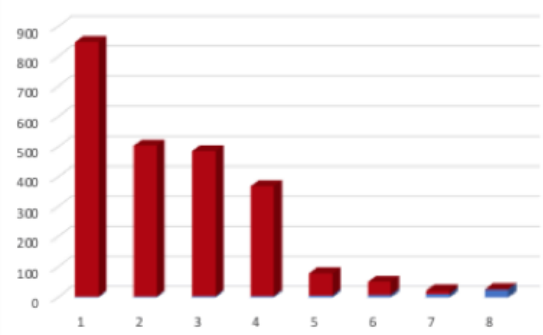
The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

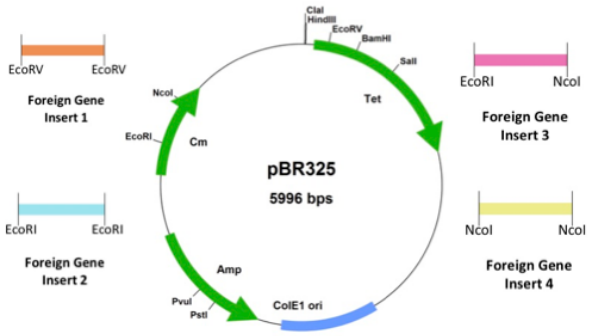
Q1. Consider Ligation 1 on the Result Sheet that shows about a 10:1 ratio of blue to white colonies. Why is this ratio so tilted in favour of blue colonies? Give two reasons. [2 marks]
[Clue: (a) Vector ends are 'open'; (b) Ligation theory about plasmid sizes.]

Q2. A research student's supervisor is on vacation out of mobile phone range. One more experiment to do before a lab deadline in two days' time. There is a blunt ended insert to be cloned into a special vector that only has one cloning site and that is PstI. The problem is that the insert has an internal PstI site. A blunt-end strategy is out of the question as the reagents/enzymes are not available in time to modify the vector PstI ends. However, there is a PstI linker kit in the refrigerator and another kit that the student realizes may be used to begin the cloning task successfully! What is the student's successful strategy using the two kits? [2 marks]
[Clue: Begin with the insert and the requirement to block the internal PstI site. Then it's gung-ho for the attachment of sites to the blunt ends.]

Q3. The graph below is reproduced from the Result Sheet. As we know from theory, transformation efficiency decreases with increasing plasmid size, but this trend doesn't seem to hold for these data, as there is a shoulder around the middle of the graph, namely the bars at abscissa points 3 and 4, which represent the recombinants at 4.7 kb and 5.0 kb (from the inserts 2.0 and 2.3 kb). Can you explain this result? [2 marks]
[Clue: Consider ligation theory relevant to the sizes of the recombinant 'parts'.]

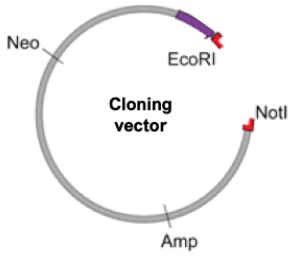


Q4. The scheme below depicts the cloning vector pBR325 and four potential foreign gene inserts. The pBR series of plasmids use differential antibiotic insertional activation to discriminate between religated vectors and true recombinants. Which of the inserts would you ligate into pBR325 to generate recombinants that are resistant to tetracycline and ampicillin and susceptible to chloramphenicol? Explain your choice(s) [2 marks]
[Clue: (a) Choosing the correct insert(s) should be a visual given, if you select the correct sites and antibiotic resistance gene.]



Q5. A list of recombinant plasmid numbers is given near the bottom of the Results Sheet. You will notice that every recombinant was obtained, bar one, namely the largest recombinant, at 25.8 kb, was missing. Can you devise a simple strategy to obtain this elusive largest recombinant? [2 marks]
[Clue: How can I get the 25.8 kb fragment onto a desert island, before the ligation?]

Q6. The plasmid drawing below is a vector ready for cloning. One student uses it as shown, while a second student treats it with alkaline phosphatase before using it for cloning. (a) What difference, if any, would there be in the two outcomes vis-à-vis the transformants? (b) If you were using genomic DNA for the source of inserts for this vector, how would you prepare the genomic DNA inserts? (c) Why is the library of recombinants obtained from (b) only a partial representation of the entire genome? [4 marks]
[Clue: (a) Look closely at the sites. No clue needed, I hope! (b) Look at the sites. (c) A bit tricky – I'd draw a line and put in randomly spaced multiple cut sites for both enzymes, then look at the fragments generated.]



Q7. Two students are setting up a cloning experiment using the vector pUC18 (~2.7 kb). One student is provided with an insert of 2.4 kb and the other student is given a 12.3 kb insert. Would both students perform identical ligation reactions, or would they use a slightly different set-up? Explain. [2 marks]
[Clue: Mass to length differences.]

Q8. This experiment makes use of the ability of hydroxylated silica to bind plasmid DNA from a cell lysate. Other cellular components, such as carbohydrates and proteins do not bind and are found in the collection tube after the spin step. Can you explain why (a) carbohydrates and (b) proteins, do not bind to the silica disc? [2 marks]
[Clue: (a) Carbohydrates are polar only. (b) Proteins have +ve and -ve surface charges, but what happens to proteins in buffered salt lysates Vs the hydroxylated silica discs?]

Q9. Why does the blue colour develop for vector transformants and not the recombinant plasmids? [2 marks]
[Clue: Answer is on the relevant slide/notes in Lecture 6.]

EXPERIMENT 7

AMPLIFICATION OF A GENE IN A RECOMBINANT PLASMID BY THE POLYMERASE CHAIN REACTION

Background

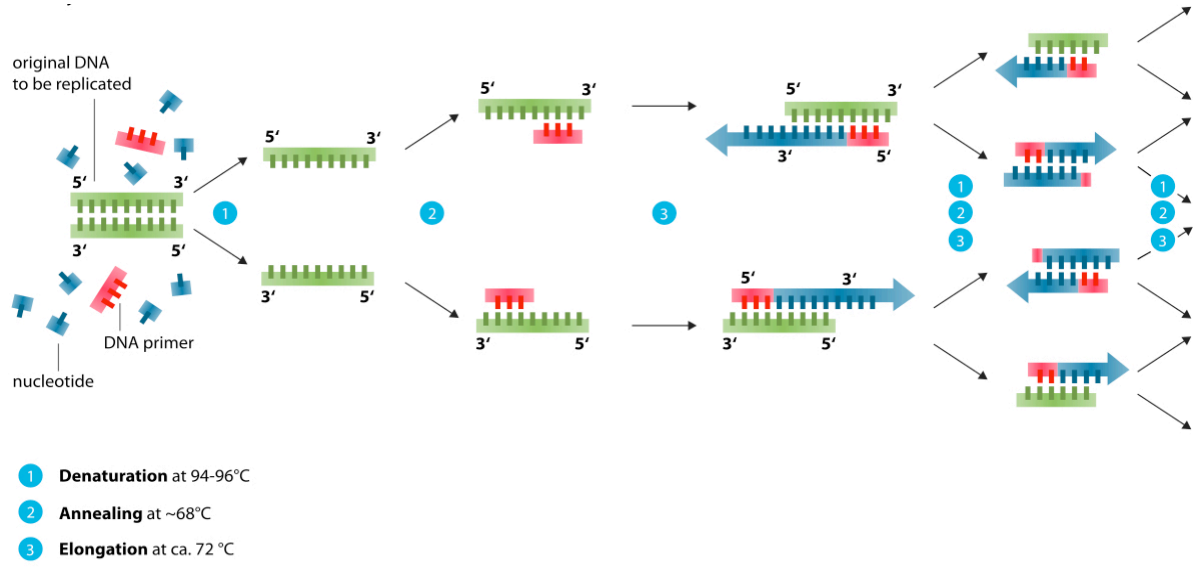
In this experiment you will be introduced to the Polymerase Chain Reaction (PCR), which is a molecular biology tool designed to amplify sequences of DNA of lengths between about 200 base pairs (0.2 kb) up to as long as 30,000 bps (30 kb). The target source DNA to be amplified can come from any of the following: Plasmids, viruses, individual chromosomes, complete genomes, or fragmented DNA from these and other sources. Common modern applications include: Medical (genetic) diagnostics; DNA from forensic, genealogical, anthropological, or archeological sources; including blood stains, fingerprints or thumbnail scrapings, hair roots, skin or tissue, sputum, mouth (gum) scrapings etc. Even degraded DNA can still usually be amplified if two or more short sequences, or single loci, are preserved and available for primer binding and amplification.

The principle of PCR is based around the need to obtain large amounts of amplified DNA from small amounts of DNA, of which the latter is known as the *template DNA* and may be derived from any source, such as chromosomes of any organism, plasmids, historical and ancient samples (e.g. fossils), forensic DNA (hair; fingerprint; semen; blood or bloodstains etc). The means of amplifying the template DNA has been enabled since the discovery of thermostable DNA polymerases through the 1990s. Thermostable polymerases are isolated from bacteria that live in high temperature environments such as hot springs or near deep ocean volcanic trenches. In this experiment, we will use Taq polymerase, named after the thermophilic bacterium *Thermus aquaticus* from which it was first isolated in 1976.

The enzymes are resistant to very high temperatures (~90-98^{o}C), which enable DNA to be denatured (H-bonds between bases break and the DNA strands separate). After an annealing step in which added short oligonucleotide primers bind to their respective matching (complementary) sequences, the temperature is raised to the optimum temperature for polymerase activity, typically 68-74^{o}C, depending on the type of thermostable polymerase. This third step is known as the extension step during which the enzyme adds nucleotides to each primer and extends the primers along the template DNA. This cycling protocol of denaturation, annealing, and extension

is repeated many times (15-40 cycles) to generate large amounts of amplified DNA from the primer/template substrate. PCR will be covered in one of the lectures and you may watch 'You Tube' videos, available as links in the UTS Online MB1 'Helpful Videos' menu.

The rationale behind each PCR amplification experiment is based on the need to obtain DNA sequence or identification information, or to amplify DNA for identification or cloning purposes. In all cases, the starting or source DNA material is present in very small amounts, usually from 10 to about 200 ng. This limiting amount of target DNA is the reason for amplifying the DNA over many repeat cycles until an amount of product is obtained that can easily be seen in an agarose gel. The early stages of the reaction are depicted in the figure below. Typically, PCR will generate µg to possibly mg amounts of DNA so that even if only about 5-10% of the reaction product is run in an agarose gel, the product will be seen easily with Gel-Red staining under UV light. The original amount of DNA template in the reaction is either too little to detect in the stained gel or may be just visible. The amplified product should be of very high intensity in the stained gel, which is indicative of the amplification process.



In some cases, for very limited source material, e.g. in forensics, the success of the amplification process is gauged by obtaining PCR products in the gel, but in these cases, since only single or double bands (loci) are required for the identification of individuals, the products are run in thin polyacrylamide gels and the banding pattern may be identified with a sensitive silver stain. Modern DNA fingerprinting (or profiling as it is now more commonly known) uses primer sets to amplify DNA. When the amplified DNA is run in a gel, the DNA bands are easily detected in Gel-Red/UV. A variation of the PCR technique is able to detect vanishingly small template DNA may be amplified by a modified PCR method known as Reverse Transcriptase PCR (RT-PCR)

(covered in more detail in the lectures), is commonly used to detect gene expression by amplifying messenger RNA sequences and copying them into complimentary DNA (cDNA), which may also be used as a cloning scheme.

The main parameters in PCR amplifications are defined by the specific protocol, which is varied according to the requirements of the particular experiment. In general though, a standard protocol requires: DNA template (typically 10-200 ng); a "bookended" primer pair of 12-50 nucleotides; buffer for pH optimization and containing Mg^{2+}; a dNTP mix for DNA extension; a thermostable DNA polymerase; and water to make up to a designated final volume (usually 50 ml). The PCR mixture is placed in a Thermal Cycler that is programmed to carry out a set of repeat cycles.

A typical cycling programme is:

First cycle:

Single step only - denaturation of template DNA 95°C for 30 - 60 sec.

Second cycle: (repeated 15-40 times depending on the specific application):

Step 1 – denaturation at 95°C for 5 – 60 sec

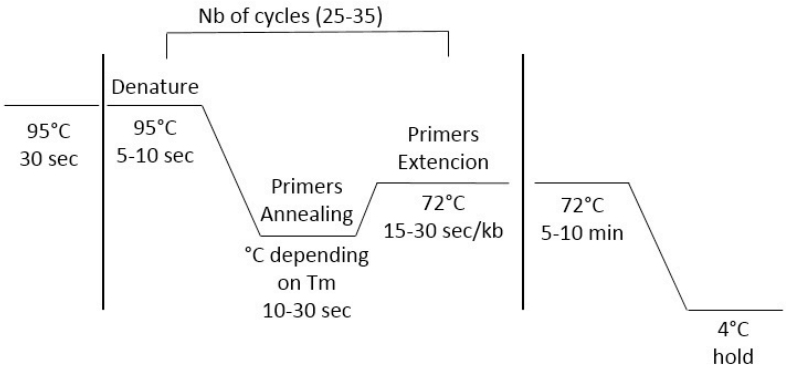
Step 2 – annealing of primers at 55-72°C for 10-60 sec

Step 3 – extension of primers along template strands at 68-72°C for 1 min/kb.

Final cycle:

Single step of "polishing" (finishing) the DNA ends at 72°C for 5-10 mins.

A complete cycling protocol will take 2 to 6 hrs; and during this time evaporation of the reaction mix must be avoided. This is achieved in thermal cycling machines with heated lids that prevent evaporation and condensation. At the completion of the cycling program, an aliquot of the reaction mix (~10%) is run in an agarose gel to check for a visible (amplified) product. Meanwhile, the rest of the mix is kept at 4°C or frozen until required for further processing (DNA sequencing, cloning, etc). An example of a cycling protocol is given below.



In designing a PCR protocol, the most important consideration is choice and design of the primer pair. Fortunately, many software programs are now available for doing this. Briefly, a primer pair needs to be about same length; must have melting points (T_m) that are close together; must not contain internal hairpin (secondary) structure; ideally should have one to three 'Cs' and/or 'Gs' at the 3'-ends, which will improve the reaction rate. Some primer sets will be self-complementary (for example in 'site-directed mutagenesis), but opposite in polarity (just like double-stranded DNA) and will self-dimerise even as the template DNA is being amplified, reducing the yield of product moderately to substantially. These "primer-dimers" are readily seen in the lower part of an agarose gel.

Amplifying the *lmrA* gene in a recombinant plasmid

This experiment has been modified from a research project in my laboratory at UTS. ABC transporters are integral membrane pumps that are responsible for the import or export of a diverse range of molecules across cell membranes. ABC transporters have been implicated in many phenomena of medical importance, including cystic fibrosis and multidrug resistance in bacteria, and higher organisms including humans. The *lmrA* gene, which expresses the multidrug resistance protein LmrA, was isolated from the chromosome of the bacterium *Lactococcus lactis* and cloned into *E. coli*, initially on the shuttle vector pUC18 (modified as pAMG25), then into the expression vector pPOWB2 as the recombinant pAMG28 (Jones & George, 2015). The isolation procedure was made more difficult by the fact that only the open reading frame (ORF) DNA sequence of the *lmrA* gene was ever published. Therefore, the cloning scheme required the use of perfectly bookended primers at the ends of the chromosomal lmrA sequence. The subsequent PCR amplification and cloning protocol worked and the resulting expressed LmrA protein was demonstrated to be functional as an MDR protein in *E. coli*. A site-directed mutagenesis PCR method was then employed to set up a sequence of experiments to demonstrate aspects of the structure and function of this ABC transporter (Jones & George, 2015).

In this experiment, the same two original bookended primers will be used to amplify the *lmrA* sequence using PCR and the primary recombinant plasmid pAMG25. The primers are:

Forward: 5'-ATGGAAAGAGGTCCACAAATGGCCAAT-3' (T_m = 70ºC)
Reverse: 5'-TTATTGACCAACAGTCAATTGTTCTGAAAC-3' (T_m = 66ºC)

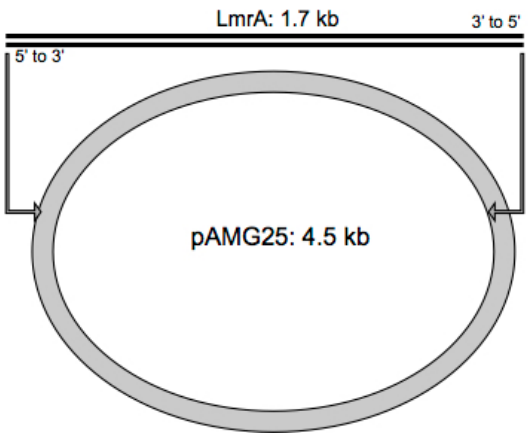
Even though software is now available for designing primers and estimating the melting temperature (T_m), a useful formula for calculating the approximate T_m is the Wallace Rule:

T_m (ºC) = 2 x (A + T) + 4 x (G + C).

This equation is fine for shorter primers (10-20 nucleotides), but more elaborate and accurate equations are available. Promega recommend a quite sophisticated calculator, which can be found on Promega's BioMath calculator page (www.promega.com/biomath/), which also has a number of other useful conversion equations.

Unlike original PCR set-ups, here we will use a newer scheme for the PCR, namely Promega's GoTaq® Green Master Mix (GTGMM), a premixed ready-to-use solution containing Taq DNA polymerase, dNTPs, $MgCl_2$ and reaction buffers at optimal concentrations for efficient PCR amplification of DNA templates. GTGMM is supplied as a 2X concentrate comprising reaction buffer (pH 8.5), 400µM of each dNTP, and 3mM $MgCl_2$. The reaction buffer contains a compound that increases sample density, and yellow and blue dyes, which function as loading dyes when reaction products are analyzed by agarose gel electrophoresis. The blue dye migrates at the same rate as 3–5 kb DNA fragments, and the yellow dye migrates at a rate faster than primers (<50bp), in a 1% agarose gel.

The PCR set-up requires, in addition to 2X GTGMM, template DNA, forward and reverse primers, and nuclease-free water. The *lmrA* primers (see above) will anneal (bind) to either end of the *lmrA* gene in pAMG25. The pAMG25 template for this experiment is ~4.5 kb (comprising 2.68 kb for pUC18 and ~1.7 kb for *lmrA*). This illustration depicts the amplification of the *lmrA* gene within the recombinant plasmid. The primers face each other at either end of the gene (arrowheads on separate strands) and the DNA in between is copied many times over many cycles. At the end of the reaction, the amplified product will



be in excessive abundance to the starting template, and should appear at a different position in the gel. Take note of the fact that the template is a (circular) plasmid, but the product (*lmrA*) will be a linear, double-stranded DNA fragment.

Experimental protocol

Students at each bench will set up PCR mixes that will examine a different reaction parameter according to your bench number as indicated on the day of the experiment.

Day 1: (Work in pairs)
Note: The annealing temperature will be varied for different labs at: 50°C, 55°C, 60°C, 65°C, 70°C, or 75°C. Refer to the LmrA primer details in the introductory notes above to see how these annealing temperature choices straddle the T_{ms} of the primers.

Setting up the PCR mixes.

DNA template (pAMG25) (10 ng/µl)	4 µl
Pre-mixed primers (10 µM)	5 µl
GoTaq Green Master Mix (2X)	12.5 µl
*H_2O (to 25 µl)	3.5 µl

PCR Protocol:
Cycle 1: Denature at 95°C, 2 mins. Cycle 2 (repeated 30 times): Denature at 95°C, 1 min; anneal at *X°C (varied per lab at 50° to 75°C), 40 sec; extend at 72°C, 2.5 mins. Cycle 3: Polish at 72°C, 5 mins. Cycle 4: Hold at 4°C.

1. Into a 0.5 ml thin-wall reaction tube add the components listed above, at room temperature (only the reagents need to be kept on ice). Use separate tips for each addition.

5. Bring all components to the bottom of the tube with a brief (5 sec) spin in a benchfuge. Alternatively, pipette all components together at the bottom of the tube, flick to mix.

6. Put into the rack provided. Staff will add these into the thermal cycler at the end of the sessions. A thermal cycler instrument will be on the side bench for viewing.

Day 2:

Running the PCR products in agarose gels.

[Note: the GoTaq Green Master Mix is sufficiently viscous for gel loading; and it already contains running dyes in the mix (blue + yellow = green).]

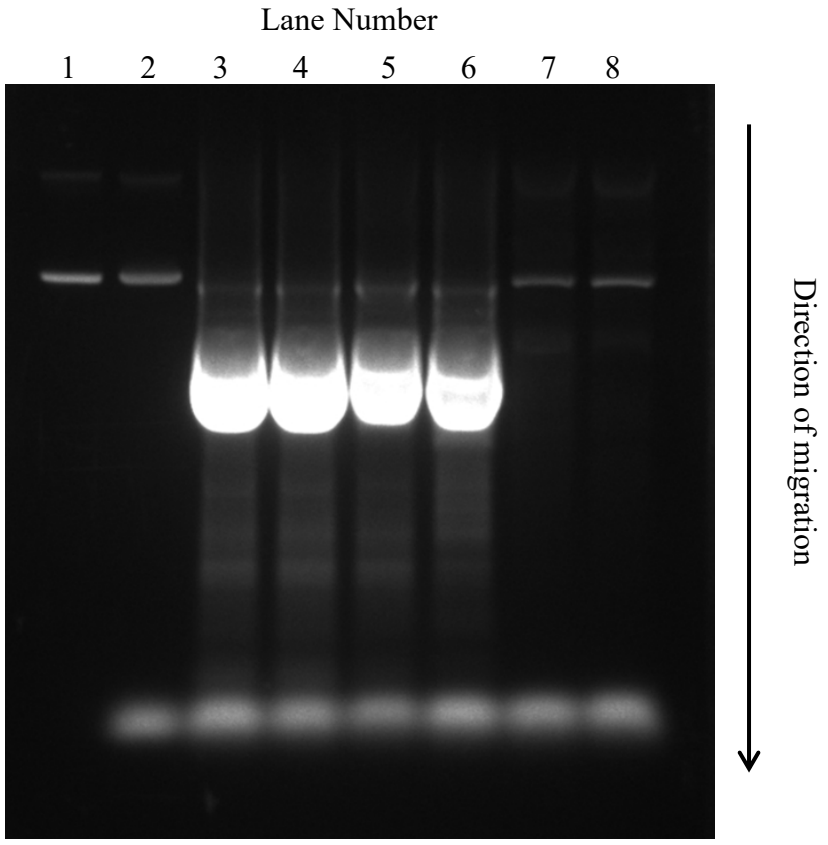
1. Add ~15-20 µl of the PCR reaction mix to one of the gel lanes.

2. Run the gels for about one hour at 120 volts.

3. Observe the agarose gel DNA band pattern on an UV transilluminator. Control reaction sample should be in lane 1.

During the gel run, there will be a PCR tutorial and some revision.

RESULT SHEET: EXPERIMENT 7



Lane Number	Content
1.	pAMG25 Template DNA
2.	Control reaction

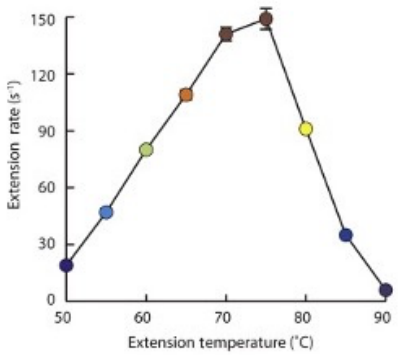
3. Annealing temperature 50°C
4. Annealing temperature 55°C
5. Annealing temperature 60°C
6. Annealing temperature 65°C
7. Annealing temperature 70°C
8. Annealing temperature 75°C

Experiment 7 Report Questions

The breakdown of marks and helpful clues are indicated at the end of each question.

Your report for this experiment is due one week later than the end of the experiment, uploaded into 'Assignments' in Canvas.

Q1. Thermostable DNA polymerases exhibit activity profiles such as the one shown below for a Taq polymerase. The PCR tubes for Experiment 7 are, for convenience, set up at room temperature (RT) until being loaded into the thermal cycler for the amplification process, in which the protocol specifies that each extension step in the cycles be carried out at 72°C for 2.5 min. In one of the PCR cyclers, the technician accidently set the extension temperature at 65°C. Would the product yield be the same as that from the 72°C extended product? Explain your answer briefly. [2 marks]
[Clue: Answer is in the graph below and extension time.]



Q2. Two honours students have related projects that use the *lmrA* gene. Their supervisor asks them to begin their projects by producing a large amount of lmrA DNA, using PCR and the same bookended primers as used in this experiment. In a slightly confusing situation, one student uses the pAMG25 plasmid as the template, but the second student finds a small amount of the actual *lmrA* linear double-stranded DNA. Both proceed independently with their PCRs. Has one of the students acted incorrectly? What do you expect the two PCR outcomes to be when the students view their products in an agarose gel? [2 marks]
[Clue: Check the source of the primers.]

Q3. You are planning a PCR experiment for which you have designed forward and reverse primers as shown here:

Forward: 5'-ATGGCTTAAAGCTACATGAAAGAGGTCCACAAATGGCCAAT-3'
Reverse: 5'-TTGGAAGGCTAAGAATTGACCAACAGTCAATTGTTCTGAAAC-3'

You now need to calculate the Tms of these primers to see if they are suitable for your PCR. A useful online tool for calculating primer Tms is the Promega Biomath Oligo site:
https://www.promega.com.au/resources/tools/biomath/tm-calculator/

When you enter the site all you need to do is enter the forward oligo sequence in Step 1 and choose the first GoTaq Green buffer system in Step 3 and hit the Calculate button. Repeat for the reverse primer. From the calculated Tms, are your primers suitable or unsuitable for your PCR? Explain. [3 marks]

[Clue: Primer Tms calculated from the Biomath site should enable you to answer this question.]

Q4. This question concerns the gel photo on the Result Sheet. The PCR results in lanes 3-6, where the annealing temperatures were 50ºC-65ºC seem to have worked equally well. (a) What would you expect if we had used an annealing temperature of say 40ºC? (b) Can you explain the results in lanes 7 and 8, given that Taq polymerase is still very active at these higher temperatures? [3 marks]
[Clue: (a) Too low an annealing temperature will cause the primers to make mismatched bindings on the DNA template. (b) All about primer Tms.]

Q5. Calculate the theoretical product yield at the end of 15 cycles, that is halfway to the 30 cycles used in this PCR experiment. [1.5 marks]
[Clue: A calculator will do this for you.]

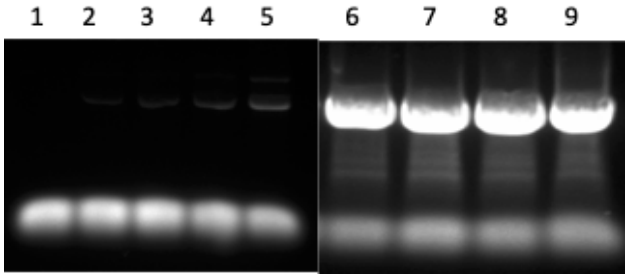
Q6. In this experiment, the extension time was 2.5 mins, comfortably long enough to copy the entire *lmrA* sequence, at ~1.7 kb, bookended by the primers, give the rule-of-thumb of ~1 kb/min. A student who was sceptical of this PCR extension rate, wanted to do the same PCR but by doubling the extension time to 5 mins. The amplified product yields for the two different PCRs were examined in an agarose gel. What differences, if any, would you expect to see in the gel lanes? Explain your answer. [2 marks]
[Clue: Shouldn't need a clue, as the answer is more or less in the question.]

Q7. You are planning a PCR experiment and you are supplied with the set of primers shown below. What is wrong with this primer set? [1.5 marks]

5'-AACTGGATACACTAGCGACATGCCCAGAGACATACTCATGCGG-3'
5'-GAATCTACACACTGGGGGGGGGATCCCCCCCCCCCTAAACTGCC-3'

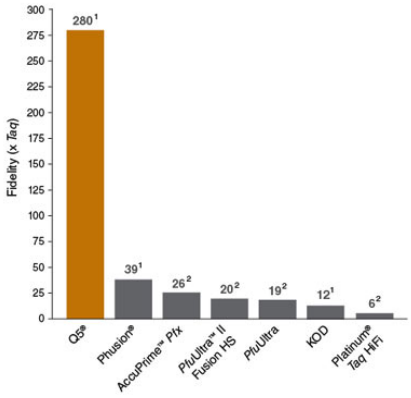
[Clue: Check the primer sequences.]

Q8. The picture below is a composite of separate sets of PCRs. Lane 1 is the template control. Lanes 2 to 5 and lanes 6 to 9 are from PCRs that used 1, 2, 5, and 10 ng of template, respectively, in otherwise identical reaction mixtures. You can just make out small amounts of product in lanes 4 and 5, but lanes 6 to 10 have copious amounts of amplified DNA. (a) Can you say which PCR parameter is different between these two sets of PCRs, that is, in lanes 2-5 versus 6-9? (b) Why are there seemingly equivalent amounts of amplified DNA in lanes 6 to 9, given that the starting template masses were so different? [3 marks]
[Clue: (a) The left set is low yield and the right set is high yield, yet both begin the PCRs with the same set of template masses. Only one parameter can cause the vast increase in yield seen in the second set of products (lanes 6-9)]



Q9. You are about to embark on a research project in which you begin by using PCR to amplify a gene of interest from a eukaryotic chromosomal DNA source. There are two types of thermostable DNA polymerases available in the laboratory, namely Phusion and Platinum Taq

DNA polymerase and Pfu Ultra DNA polymerase. Check the graph below for the fidelity of thermostable DNA polymerases. Which one should you use and why? [2 marks]
[Clue: Answer is in the graph.]



GLOSSARY OF COMMONLY ENCOUNTERED TERMS IN MOLECULAR BIOLOGY

Adaptor. A synthetic, double stranded oligonucleotide containing two adjacent restriction enzyme sites, one of which is compatible with the end of a molecule that is to be adapted to now contain a new terminal restriction site (the second site in the adaptor).

Alkaline phosphatase. An enzyme that catalyses the removal of 5' phosphate groups from linear DNA molecules. In molecular cloning, the vector is usually dephosphorylated to prevent its re-ligation during the ligation reaction that produces recombinant DNA molecules.

Alleles. Alternative forms of a gene at the same locus.

Bioinformatics. Using computer databases and software to obtain and analyse DNA or protein sequences.

Blunt ends. DNA termini without overhanging 3' or 5' ends. Also known as flush ends.

cDNA. Complimentary DNA made from mRNA and dNTPs by reverse transcriptase. A cDNA library is a collection of cloned molecules representing the entire mRNA population of a specific cell or tissue type.

Chimeric/hybrid plasmid. A plasmid constructed from two or more other plasmids and designed to have specific properties for gene and cloning.

Cleared lysate. The contents of cells (nucleic acids) that have been disrupted and separated from membranes and soluble proteins and polysaccharides. Also known as a cell-free extract.

Clone. A group of identical cells derived from a single ancestral cell. Cloning is the process of creating a group of identical cells from a single ancestor. [See 'molecular cloning' below]. Used as a verb to describe the generation of recombinant molecules.

Cohesive ends. Also called sticky ends. Single-stranded, complementary, overhanging 5' or 3' ends.

Competence. A transient physiological state that renders cells capable of absorbing transforming DNA, usually in the form of small recombinant plasmids.

Concatamer. An end-to-end (tandem) array of identical DNA sequences, common throughout the genome of higher organisms, but the function of which is unknown.

Consensus sequence. A sequence for which there may be several variations amounting to single base changes, but for which there is a majority of one type of sequence.

Copy number. The number of copies of a particular plasmid in a bacterial cell.

Denaturation. Disruption of secondary or tertiary structure of proteins or nucleic acids. In DNA, it involves the breakage of hydrogen bonds between base-paired strands, by the action of heat or alkali.

DNA fingerprinting. A method that uses Southern hybridisation and a radioactive DNA probe, or PCR amplification to produce a specific bar code pattern of different sized DNA fragments from the genome of an individual.

DNA ligase. An enzyme that catalyses the joining of two restricted blunt, or compatible cohesive ends of DNA by reforming phosphodiester bonds.

DNA probe. A radioactively or chemically labelled small DNA fragment which will hybridise with, and thereby detect and locate, complementary (related) DNA sequences from a different source.

DNA sequencing. A method of determining the exact order of DNA nucleotide bases a contiguous piece of DNA. If the sequence is that of a gene, it can be “read” into the triplet codon amino acid sequence of the protein product of the gene.

Downstream. Refers to the sequence on the 3’ side of a given site in DNA or RNA.

Duplex. A double-stranded nucleic acid molecule with inter- or intra-complementary base pairing.

Dyad symmetry. Property of a dsDNA sequence that has a two-fold symmetry about a central vertical axis. Typical of restriction sites.

Electroporation. An efficient method of transformation of DNA molecules into cells by applying short high voltages pulses to an electrocompetent suspension.

Endonuclease. An enzyme that digests a molecule of DNA by making internal breaks in a DNA molecule.

Ethidium bromide. An intercalating dye that binds to DNA and fluoresces in UV light (254 to 340nm) producing a red-orange emission spectrum.

Exons. Coding regions of eukaryotic genes interspersed between non-coding intervening sequences (introns).

Exonuclease. An enzyme that digests a molecule of DNA by removing successive nucleotide bases from one or other end of the molecule.

Expression vector. A cloning vector that contains a strong but regulated promoter and ribosome binding site just upstream from where a promoter-less gene is inserted, and its transcription controlled by the level of induction of the promoter.

Frameshift. Insertions or deletions of bases that cause a translation shift in the triplet codons so as to place the sequence out-of-frame.

Fusion protein. The joining of the N-terminus of part of one protein to the C-terminus of part of a second protein as a result of the fusion (ligation) of parts of two genes with codons in phase. The fusion protein is usually created to allow for the detection of an otherwise promoter-less or expression-silent gene, by using the N-terminus of the lacZ gene, for example, from which ß-galactosidase produces indole blue from an X-gal reagent.

Gene expression. The transcription of messenger RNA molecules from a DNA template, and the translation of the "message" into functional proteins and enzymes.

Gene regulation. The "turning-off" or "turning-on" of messenger RNA transcription by regulatory proteins or smaller metabolite molecules - repressors and inducers - which bind to, or are released from, gene promoter sites.

Gene therapy. The replacement of a defective or cancer-causing gene by using "gene shears" to delete the defective gene and insert the correct normal gene in its place; or simply inserting a missing but important gene using the same technology; or inserting recombinant plasmids carrying genes which either over-ride or supplement existing genes and the proteins or enzymes produced from them. This type of therapy will be most useful in correcting genetic and hereditary disorders, perhaps as early as the embryonic stage of development.

Genetic engineering. The construction of recombinant DNA molecules in which an important gene is cloned into a small plasmid, and thereby is able to be grown in large numbers in bacterial batch cultures, enabling the relatively inexpensive mass production of hormones, vaccines, antibodies, and molecules that enhance the resistance of agricultural crops to disease and frost, or that enable the breakdown of pollutants such as oil slicks or plastics. Genes cloned in this way are also used as diagnostic "probes' to identify genetic abnormalities (example, diabetes) and a host of diseases and infections.

Genome. The total DNA genetic content of an organism.

Genotype. The complement of genes of an organism. Designations follow the convention of three letter lower case mnemonics or abbreviations followed by an upper case letter if more than one of a series of related genes exists; and by a number for more than allele. For example: *leuA*, *leuB*, *leuC* are genes, and *leuA1*, *leuA2*, *leuA3* are alleles of one gene.

Guanidinium isothiocyanate. An agent that aids in the disruption of cell membranes. It is also a powerful inhibitor of ribonucleases.

Hogness box. Also called the TATA box (consensus TATAAT). Part of the eukaryotic promoter sequence at about minus 25 nucleotides from the start codon. The second part of the promoter is the CAAT box (consensus GG{T/C}CAATCT) at about minus 75 from the start codon. RNA polymerase binds to these regions in order to initiate transcription.

Homology. The degree to which nucleic acid or amino acid sequences from different sources are related.

Host. A cell used to propagate recombinant DNA molecules.

Hybridisation. The joining together (re-associating or reannealing) of single-stranded DNA molecules by complementary hydrogen bonding.

***In vitro*.** Literally "in glass", meaning outside the cell.

***In vivo*.** Literally "in life", meaning within a cell or organism.

Leader sequence. A region of mRNA or protein that precedes the information region, and which contains ribosome binding sites or transcriptional control elements in RNA, or signal peptides in proteins that enable them to cross membranes and are then processed (removed) to produce the mature protein.

Ligation. The joining of two molecules or pieces of DNA that have been cut by the same restriction endonuclease, using the enzyme DNA ligase which reforms the phosphodiester bonds between the DNA nucleotide bases.

Linker. A synthetic oligonucleotide that contains a single, internal restriction site for an enzyme that generates cohesive termini. It is used to add cohesive ends to blunt-ended molecules.

Locus. The site at which a gene is located on a chromosome.

Molecular cloning. The production of large numbers of small genetically identical recombinant DNA plasmids or viruses bearing one or more genes that were previously present in single copies in very large chromosomes. This technology thus delivers multiple copies of cloned genes for study and manipulation, from sites that were hitherto relatively inaccessible.

Monoclonal antibodies. Antibodies produced from a hybridoma cell line against a specific epitope on an antigen molecule.

Multiple cloning site (MCS). Also called a polylinker region. A short region of DNA in a vector that has recognition sites for several restriction enzymes.

Mutation. A specific genetic change that alters a property of the cell. A suppressor mutation is a second site mutation which restores function (that is, the phenotype) to the gene product. A polar mutation affects all genes downstream in the same operon.

Nick translation. A method of radiolabelling DNA probes using DNA polymerase I (Klenow fragment), ^{32}P-ATP, and the other three dNTPs.

Northern blotting. Autoradiographic or chemiluminescent detection of membrane-bound mRNA by DNA probes.

Nuclease. An enzyme that hydrolyses phosphodiester bonds either internally (endonuclease) or inward from one or both ends (exonuclease).

Operator. Region of an operon, close to the promoter, to which a repressor protein binds.

Operon. Two or more genes in tandem are expressed from a single promoter at the head of the operon, a short sequence with internal repeat sequences (palindromes). A single long polycistronic mRNA is transcribed. Operons are regulated by a repressor gene that produces a protein, which binds strongly to the operator region preventing transcription. Derepression (induction) is effected by the presence of a substrate or catabolite for one of the pathway enzymes produced by the operon.

Palindrome. A word, sentence, or verse that reads the same from right to left as it does from left to right (derived from the Greek *palindromos*, running back again). In molecular biology, it is defined as a sequence of DNA nucleotide bases which are identical in the two DNA strands but in opposite directions, and which are usually specific recognition sequences (restriction sites) for restriction enzymes. For example,
TTTAAA
AAATTT

Phenotype. The observable traits that result from the genotype of an organism. Designation by convention is a three-letter abbreviation with the first letter capitalised and an additional

capital letter for more than one phenotype from a series of related genes. For example: LeuA, LeuB, LeuC are phenotypic symbols for three enzymes in the leucine biosynthetic pathway.

Plasmid. An extrachromosomal circular molecule of double-stranded DNA carrying non-essential genetic information and replicating independently of the host chromosome (relaxed control), or in cycle with the chromosome (stringent control). These may take the form of large natural plasmids conferring, for example, antibiotic or heavy metal resistance, or degradation of complex organic molecules. Small chimeric versions (e.g. pUC18 and pBR322) have been artificially constructed for use in molecular cloning and genetic engineering in biotechnology.

Polycistronic mRNA. An mRNA molecule containing coding information from a bacterial operon of several genes in tandem, with a single promoter/operator region at the 5' end of the first cistron (gene).

Polylinker. Also known as a Multi-Cloning Site (MCS), is an engineered short segment of DNA within a cloning vector that contains several unique restriction sites in tandem.

Polymerase Chain Reaction (PCR). A series of alternating primer annealing and DNA polymerase synthesising reactions during which primer-framed regions of DNA strands are amplified (copied) by geometric progression (1,2,4,8,16,32.....).

Pribnow box. One of two sequences found in prokaryotic promoters required for the binding of RNA polymerase (transcription initiation). The consensus sequence is TATAAT and it is positioned at about minus 10 nucleotides from the start codon. The second part of the promoter is the minus 35 region (consensus TTGACA).

Primary transcript. The initial or precursor mRNA transcript of a eukaryotic gene that is processed or spliced in the nucleus to produce the mature mRNA with collinear exons.

Primer. A short sequence of DNA or RNA that binds to its complementary region on large molecules and provides a priming or starting site for DNA polymerase.

Promoter. The DNA sequence just in front (“upstream”) of the coding region of a gene and to which repressors, inducers, and RNA polymerase bind. RNA polymerase is the enzyme that enables the transcription of a gene into a messenger RNA molecule.

Reading frame. A nucleotide sequence beginning with an initiation codon (almost always ATG) and read as a series of triplet codons specifying consecutive amino acids in a polypeptide. There are potentially three reading frames in any given sequence of DNA.

Recombinant DNA. A small circular vector plasmid molecule into which has been inserted a foreign piece of DNA, usually one or more genes, required to be studied in finer detail, or used for large-scale production of important proteins, enzymes, vaccines or antibodies.

Replicon. Any DNA molecule able to initiate its own replication, includes, plasmids, viruses and chromosomes of bacterial, fungal, plant and animal cells.

Repressor. A protein that binds to the operator region of a gene preventing transcription.

Restriction endonuclease. Bacterial enzymes of the type II class, of which there are many hundreds that recognise short DNA sequences in prokaryotic and eukaryotic DNA, and cleave

DNA in a way that enables it to be rejoined to another piece of DNA cut with the same type of enzyme, producing a hybrid, recombinant molecule. An isoschizomer is a restriction endonuclease that recognises the same palindromic sequence as another enzyme.

Restriction fragment length polymorphism (RFLP). The existence of genetic diversity in a population is termed polymorphism. When polymorphisms can be detected in restriction fragment patterns, as indicated by changes in the length of one or more fragments that should otherwise be identical in normal individuals, the resultant RFLPs are indicative of mutation by base changes, insertions or deletions of DNA within genes, that are thus rendered abnormal.

Restriction mapping. A technique used to determine the location of selected restriction enzyme sites in a DNA molecule.

RNA splicing. An RNA-catalysed processing of pre-mRNAs into mature mRNAs by the removal of introns and the colinear attachment of exons.

Shine-Dalgarno sequence (SD sequence). Also, Ribosome Binding Site (RBS). A short sequence (consensus AGGAGGU) near the AUG initiation codon in prokaryotic mRNAs that base pairs with a complementary sequence near the 3' terminus of the 16S rRNA of the small ribosome particle (30S).

Shotgun cloning. The cloning of the entire genome of an organism in the form of randomly generated DNA fragments, into plasmid or viral genomic libraries.

Southern hybridisation. A method that uses a small, labelled DNA probe to detect like, or nearly like, membrane-immobilised DNA sequences from other sources.

Tandem repeat. A double-stranded sequence (of variable length depending on the location in different DNA molecules) repeated two or more times contiguously in the same orientation.

Topoisomerases (gyrases). Enzymes that break and rejoin DNA strands in such a way that it changes the number of times one strand crosses the other. They can tie and untie DNA knots, add and subtract twists (supercoils) in the DNA, and link and unlink DNA circles.

Transcript. The messenger RNA product of a gene.

Transformation. The process of absorption of plasmid DNA by competent cells. A transformant is a cell that has stably inherited a transformed plasmid.

Transfection. The process of absorption of viral DNA by cells, during which the DNA is "injected" across the cell membrane from the protein coat (capsid) of the viral particle.

Upstream. The region that lies to the 5' side of a given point on a DNA molecule.

Vector. An independent replicon, usually a small plasmid or virus, used in genetic engineering to produce recombinant DNA molecules with a foreign gene, then introduced into a host cell.

Western blotting. Electrophoretic transfer of polyacrylamide gel separated proteins to a membrane for probing with antibody.

Wild-type. The form of a gene (allele) normally found in nature.